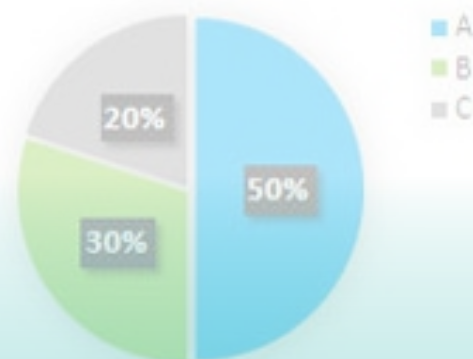
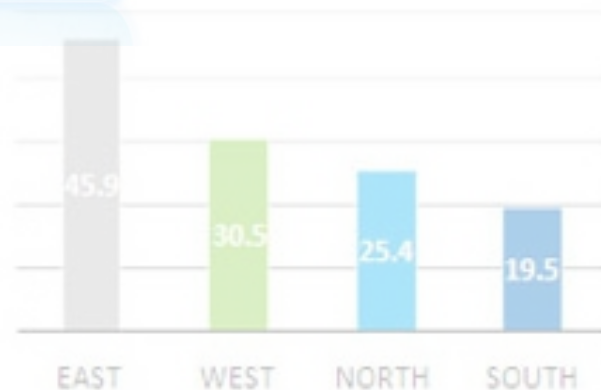
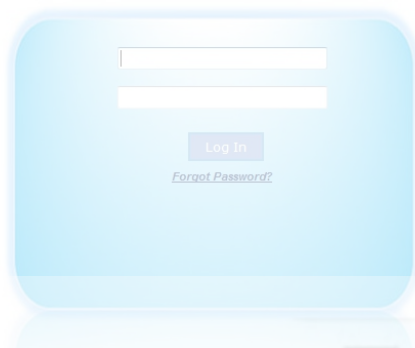




About the Editors

Navanita Gogoi is the Director (Statistics), Ministry of Health and Family Welfare, Government of India

P. Mohanachandran Nair is Honorary Director, Population Research Centre, and Professor and Head, Department of Demography, University of Kerala Thiruvananthapuram

Sajini B Nair is Social Scientist at Population Research Centre, University of Kerala, Thiruvananthapuram



Re-accredited by NAAC with A-Grade

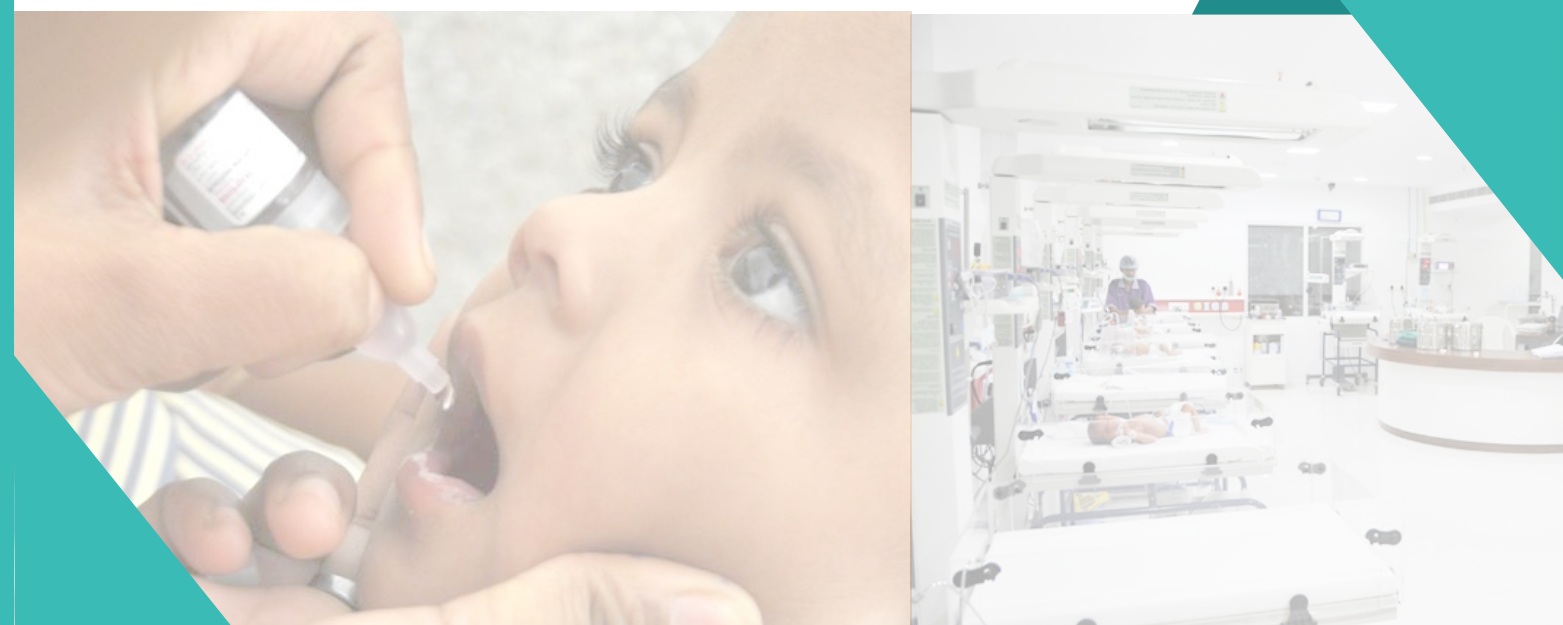
Population Research Centre
University of Kerala, Karyavattom
Senate House Campus, Palayam
Thiruvananthapuram – 34, Kerala, India
E-mail: prc.kerala@nic.in, ku.release@gmail.com
Web: <https://www.keralauniversity.ac.in>

Printed at : Kerala University Press



Critical Issues in Maternal and Child Health in India

**A Compendium of Studies Conducted by
the Population Research Centres 2015-16**



Edited by

Navanita Gogoi
P. Mohanachandran Nair
Sajini B Nair

Ministry of Health and Family Welfare
Government of India
Nirman Bhavan, New Delhi

Critical Issues in Maternal and Child Health in India

**A Compendium of Studies Conducted by the
Population Research Centres 2015-16**

Edited by

**Navanita Gogoi
P. Mohanachandran Nair
Sajini B Nair**



**Ministry of Health and Family Welfare
Government of India
Nirman Bhavan, New Delhi**

2017



जगत प्रकाश नड्डा
Jagat Prakash Nadda



सत्यमेव जयते



स्वास्थ्य एवं परिवार कल्याण मंत्री
भारत सरकार
Minister of Health & Family Welfare
Government of India

MESSAGE

I am delighted to note that the various Population Research Centres in the Country are continuously engaged in carrying out research work on critical issues of maternal and child health. The initiative for disseminating the findings of their studies during 2015-16 in the form of a Compendium of research work is laudable. The second Compendium, like the first one in 2014-15 will, no doubt, guide Policy makers and Health Professionals in formulating new plans as well as recalibrating the ongoing schemes. By contributing significantly to the understanding of the nuances of the functioning of the health sector, the Population Research Centres, located at different parts of the country, are strengthening the developmental initiatives of the Government.

I am confident that this work will help the overall progress of the Health and Family Welfare programmes. I take this opportunity to congratulate all professionals behind this noble initiative. My best wishes to all Population Research Centres in the country in their future efforts for all your future endeavors.

March, 2017
New Delhi

(Jagat Prakash Nadda)

348, ए-स्कंध, निर्माण भवन, नई दिल्ली-110011
348, A- Wing, Nirman Bhawan, New Delhi-110011
Tele: (O) : +91-11-23061661, 23063513, Telefax : 23062358, 23061648
Email: hfwminister@gov.in



सी.के.मिश्रा
सचिव
C.K.Mishra
Secretary



भारत सरकार
स्वास्थ्य एवं परिवार कल्याण विभाग
स्वास्थ्य एवं परिवार कल्याण मंत्रालय
Government of India
Department of Health and Family Welfare
Ministry of Health & Family Welfare

MESSAGE

With the launch of National Rural Health Mission (NRHM) in the year 2005, the Government of India has renewed its commitment in achieving health goals in line with the national targets and international obligations. The NRHM and subsequent National Health Mission give priority to the achievement of Maternal and Child health goals. The reproductive health continues to be an important priority both in the recently concluded Millennium Development Goals and the Sustainable Development Goals now launched.

Undoubtedly, the maternal and child health indicators have substantially improved in the country over the last several decades. The pace of improvement has accelerated in recent times. The rapid decline in infant and maternal mortality in recent years clearly conveys the effectiveness of the several programmes launched for augmenting the reproductive and child health scenario in the country. At the same time, there are still large variations in achievements across regions. Therefore, it is important that these issues surrounding the maternal and child health are examined in detail. To measure the progress and to find out the reasons for backwardness or progress, it is important to have up-to-date information that directly feed into the programme implementation and policy formulation. The studies carried out by various Population Research Centres across the country are of immense value to the Ministry in making course corrections and implementing new programmes and schemes.

Towards the goal of making critical research based input available to the policy makers, stakeholders, researchers and other such users, the Ministry of Health and Family Welfare has started the initiative last year for bringing together the important research work of various Population Research Centres conducted during a particular year. The First issue of the Compendium, titled "Maternal and Child Health", a compilation of selected research work by Population Research Centres during 2014-15 was published last year. I am happy that on similar lines, a Compendium of important studies carried out by various population research centres in the year 2015-16 is being brought out this year. These research papers provide useful information on various programmes and policies carried out in the field of health sector in the country.

I would also like to convey with pleasure that the Ministry of Health and Family Welfare has supported the initiative of screening the research works undertaken by the PRCs. A workshop was organized by the Ministry at Thiruvananthapuram, Kerala where the articles of this compendium were presented and evaluated by a team of experts before selection for this compendium.

I am happy that the Statistics Wing of the Ministry of Health and Family Welfare has taken this initiative of bringing out selected research work of various Population Research Centres. Population Research Centre, University of Kerala under the guidance of its Director-in-Charge Dr. P. Mohanachandran Nair deserves full appreciation for taking up the complete task of collecting, evaluating and disseminating the research studies conducted by PRCs during 2015-16 and bringing out the Compendium.

I am certain that this document will be useful to policy makers both at the Central and State level as well as many other stakeholders like non-governmental organisations, academic institutions, etc.


(C.K. Mishra)

Tele : (O) 011-23061863, Fax : 011-23061252, E-mail : secyhw@nic.in
Room No. 156, A-Wing, Nirman Bhavan, New Delhi-110 011



Dr. SHEELA PRASAD

Economic Adviser

Telefax : 23061730

E-mail : sheela.prasad@nic.in



भारत सरकार
स्वास्थ्य एवं परिवार कल्याण मंत्रालय
निर्माण भवन, नई दिल्ली - 110011
Government of India
Ministry of Health & Family Welfare
Nirman Bhavan, New Delhi - 110011

FOREWORD

The Ministry of Health and Family Welfare, Government of India established 18 Population Research Centres with the mandate to provide critical research based input in the field of population and health. The studies carried out by PRCs include both field based and secondary data based and such analytical studies are of great relevance to planning and policy formulation. The Ministry is earnestly pursuing its mandate for achieving the targets set by the Sustainable Development Goals by 2030. Monitoring, evaluation and a strong data base are vital for the successful implementation of programmes and the role of PRCs in this regard is crucial.

It is indeed a great pleasure to note the initiatives taken by the Statistics Wing of the Ministry of Health and Family Welfare for a wider dissemination of the studies carried out by the Population Research Centres in India in the form of a Compendium, which started with the Compendium of studies conducted by PRCs during 2014-15. The present Compendium is the second issue of the publication. The research papers included in this compendium span a wide range of issues including health care service delivery and quality of health data. The suggestions and recommendations are of great relevance to the successful implementation of programmes.

The papers for the Compendium for this year were selected through a dissemination workshop, among a large number of studies carried out by the PRCs. Prof. K.S. James, Professor of Demography, Jawaharlal Nehru University, New Delhi and Dr. Hukum Chandra, National Fellow & Principal Scientist, ICAR, New Delhi served as experts for the two day Workshop. Moreover, there was a first round of screening of papers preceding the two-day workshop carried out by Shri. C.R.K Nair, Prof. K.S. James, Prof. P. Sankara Sarma, Dr. N Sreedhar, Dr. P.S. Nair, Dr. A.N Rajeev and Dr. P. Mohanachandran Nair. I place on record our deep appreciation for their efforts and valuable inputs.

Healthy Village, Healthy Nation



एड्स - जानकारी ही बचाव है
Talking about AIDS is taking care of each other

: 2 :

The role played by Population Research Centre Kerala in collecting the studies from the PRCs, screening, coordinating the Two-Day Workshop and bringing out the compendium is indeed praise worthy. I am sure that this compendium stands testimony to the useful work done by the PRCs in India in better planning and implementation of health and family welfare programmes.

February 2017
New Delhi


(Sheela Prasad)



Dr. Vishnu Kant Srivastava

Chief Director

Telefax : 23062288

E-mail : cdstat@nic.in



भारत सरकार
स्वास्थ्य एवं परिवार कल्याण मंत्रालय
निर्माण भवन, नई दिल्ली - 110011
Government of India
Ministry of Health & Family Welfare
Nirman Bhavan, New Delhi - 110011

PREFACE

The Population Research Centres were set up by the Government of India with an objective of conducting research in population and health areas and to assist the Ministry of Health and Family Welfare, with policy input and feedback on population and health issues. There are altogether 18 Population Research Centres (PRCs) across the country, carrying out various policy and programme related studies.

The PRCs have been involved in providing critical research based inputs in the field of population and health since its inception. Although these studies have been extremely useful for both Central and State Governments, many of these study reports were not widely disseminated. Last year, the Ministry of Health and Family Welfare has started the initiative to bring out a Compendium of important studies conducted during the year 2014-15 with an objective of wider dissemination of the studies carried out by PRCs. The first Compendium, titled "Maternal and Child Health in India" was released by Hon'ble Minister of Health and Family Welfare on 29th March 2016.

This year, the Ministry of Health & Family Welfare has initiated the work of Compendium in the form of a Dissemination Workshop held at Thiruvananthapuram, Kerala to disseminate the research studies conducted by these PRCs during 2015-16. Altogether, 29 papers were presented in the two days Dissemination workshop, out of the 68 papers received from various PRCs. The first round of short listing was carried out by 6 reviewers; Shri. C.R.K Nair, Prof. K.S. James, Prof. P. Sankara Sarma, Dr. N Sreedhar, Dr. P.S. Nair, Dr. A.N Rajeev and Dr. P. Mohanachandran Nair. Prof. K.S. James, Professor of Demography, Jawaharlal Nehru University, New Delhi and Dr. Hukum Chandra, National Fellow & Principal Scientist, ICAR, New Delhi were invited as experts in the two days Workshop to help the Ministry in finalizing the papers for the Compendium by their valued critical input as well as providing guidance to the PRCs for improving the quality of the papers. I would like to place on record our deep appreciation for the efforts of all the reviewers and experts in bringing out the Compendium.

The present issue of the Compendium, titled "Critical Issues in Maternal and Child Health in India", consists of 22 papers, organized into four Sections. The Section I consists of six papers on 'Maternal and Child Health in India'; Section II includes eight papers on 'Health Care Service: Status and Issues'; Section III contain three papers on 'Health Information: Assessment of Data Quality' and Section IV have five papers on 'National Health Mission: Programme implementation'.

Healthy Village, Healthy Nation



एड्स - जानकारी ही बचाव है

Talking about AIDS is taking care of each other

The introduction of National Rural Health Mission in 2005 and subsequent launch of National Urban Health Mission and the amalgamated programme of National Health Mission aims to provide comprehensive health care for maternal and child health services in the country. However, success of any Governmental initiative depends upon availability of good quality data and its in-depth analysis aimed at addressing real programme specific and context specific issues. The research work compiled in this publication contributes immensely towards this goal.

Overall, there is lot to be achieved in ensuring 'health for all' in the country, despite the fact that there has been significant progress in recent years. It is important that the policies and programmes are backed by accurate data as well as well analytical studies on the reasons for successes and failures. The Population Research Centres, undoubtedly, play a critical role in this direction. The studies in this volume indicate the type of work currently underway in many PRCs. This will definitely help in designing better health programmes and evolve policies in the country.



(Dr Vishnu Kant Srivastava)

March 2017
New Delhi

ACKNOWLEDGEMENT

The compendium of studies conducted by various Population Research Centres in India during 2015-16 seeks to publish good quality research work with great relevance to planning and policy making in India. It covers a wide range of topics covering health care service delivery, maternal and child health care, data quality appraisal of web based health data and evaluation of certain programmes under National Health Mission. Our sincere gratitude goes to the Ministry of Health and Family Welfare (MoHFW), Government of India for taking the great initiative of publishing the work of PRCs in India to the second year. The screening of research work in a Workshop mode by bringing all the PRCs together with two experts to review the work has benefitted the PRC personnel immensely and on behalf of the PRC fraternity we place on record our sincere thanks to Dr. Vishnu Kant Srivastava, Chief Director (Statistics Division), for spearheading the whole initiative. We are indeed grateful to Smt. Navanita Gogoi, Director (Stats) for her continuing support and untiring efforts in planning and organizing the Two Day Dissemination Workshop at Thiruvananthapuram, facilitating the screening activities and publishing the compendium. We sincerely place on record our gratitude to Shri. C.R.K Nair, ADG Stats (Retd.), Ministry of Health and Family Welfare for his valuable contribution in screening the research works and providing constructive suggestions at the Workshop.

We are grateful to the MoHFW for entrusting us with the task of bringing out the second issue of PRC Compendium. Several steps were involved in finalization of this Compendium. Research studies from all 18 PRCs were invited, reviewed by six eminent reviewers in the first phase of screening and 35 papers out of the 68 papers received were selected. We thank our panel of reviewers for scrutinizing and short listing these papers and also providing detailed comments about each of these 68 papers, viz ,Dr. N Sreedhar, Shri. C.R.K Nair, Dr. Sankara Sarma P, Shri. A.N Rajeev, Dr. K.S.James, Dr. P.S.Nair and Dr. P. Mohanachandran Nair. The papers short listed by the reviewers were presented in a Dissemination Workshop. We are extremely thankful to Prof. K.S.James, Professor in Demography, Centre for the Study of Regional Development, School of Social Sciences, JNU, New Delhi and Dr. Hukum Chandra, National Fellow & Principal Scientist, ICAR, New Delhi for their contribution as experts at the two day Workshop and helping in the final selection of papers with their critical input. Their suggestions and detailed instructions as well as comments have contributed greatly in improving every research paper presented in this compendium. After detailed deliberation, a total of 22 papers have been included in this Compendium.

Finally we thank all the PRC personnel for their contribution in carrying our excellent research work on topics of relevance to planning and development of the country. The contributors of the present compendium deserve great appreciation. We place on record our sincere gratitude to the Kerala University Press for printing the compendium. Once again, we sincerely thank the Ministry for bestowing confidence in us in assigning the task of publication of Compendium.

Dr. P. Mohanachandran Nair
Director-in Charge
and
Staff
Population Research Centre
University of Kerala

CONTENTS

Contents		
	<i>Message from Hon'ble HFM</i>	iii
	<i>Message from Hon'ble Secretary, HFW</i>	v
	<i>Foreword</i>	vii
	<i>Preface</i>	ix
	<i>Acknowledgements</i>	xi
	<i>List of Tables</i>	xv
	<i>List of Figures</i>	xxii
	<i>Introduction</i>	xxvii
Chapter No.	Section I: Maternal and Child Health in India	
1	Utilization of Maternal Health Care Services in Karnataka and its Regional Variations <i>T.S.Syamala, N Kavitha, SupriyaVerma</i>	3
2	Household Expenditure on Maternal Care Services in Rural North Karnataka, <i>Jyoti S. Hallad, J. A. Golandaj, ArinKar, Krishnamurthy J, B. M. Ramesh, R. V. Deshpande and B. I. Pundappanavar</i>	19
3	Role of Living Conditions and Hygienic Practices on Childhood Diarrhoea in Rural Areas of Koppal District, Karnataka <i>Dr. Rajarama K. E. T, Mr. C. N. Noolvi</i>	33
4	To Assess the Role of Family Planning on Fertility in Jharkhand State: Based on NFHS Data <i>Dilip Kumar</i>	43
5	Maternal morbidity profile of women in Tamil Nadu, <i>N. Dhanambhagyam, N.Rajakumar, M.Murugesan and B.Karthi</i>	49
6	A comparative analysis of malnutrition among children in Maharashtra <i>Vini Sivanandan, Vandana Shivnekar, Raj Pol, Arun Pisal, A.P. Prashik, Akram Khan</i>	65
Section II: Health Care Service: Status and Issues		
7	A Mixed-Method Study on Knowledge, Awareness and Practices regarding Tuberculosis: Insights from Community and RNTCP Personnel in Delhi and Muzaffarnagar <i>William Joe, AakshiKalra, Abhishek and Suresh Sharma</i>	83
8	Comparative Analysis of Institutional Deliveries in Kanyakumari District of Tamil Nadu <i>S. Ravichandran, V. Saravanakumar, N. Kala, M. Senthil Kumar</i>	103
9	Status of Community Health Centres (CHCs) in Madhya Pradesh: An Assessment based on HMIS Infrastructure Data <i>Nikhilesh Parchure and ReenaBasu</i>	113

10	Assessment of Availability of Service Readiness and Beneficiary Satisfaction in UPHCs of Bhopal and Raipur under NUHM <i>ReenaBasu, NikhileshParchure, Niklesh Kumar, Jyoti Tiwari</i>	133
11	Patient Satisfaction with Health Care Services in Punjab <i>Pawan Kumar Sharma and PoonamSandhir</i>	149
12	Treatment Seeking Behaviour of Chronic and Other Diseases <i>Sajini B Nair, Suresh Kumar S, Rajesh J Nair, Mathew M.C</i>	167
13	Assessment of Home Based Newborn Care in Uttar Pradesh <i>Rajesh K Chauhan and Pradeep Mishra</i>	187
14	Knowledge, Attitude and Practice regarding Non-Communicable diseases among Medical Officers in Haryana <i>Sukhbir Singh and Reena Singh</i>	207
Section III: Health Information : Assessment of Data Quality		
15	Analysis of sub district level HMIS data in three high focus districts of Gujarat, Dahod and Panchmahals <i>Gayatri Desai, Rajnikant Patel</i>	221
16	HMIS Data quality issues in Subcentres and PHCs of Ganderbal district of Jammu and Kashmir <i>Bashir Ahmad Bhat, Farida Qadri and Muneer Ahmad</i>	233
17	Data capture in MCTS in Kerala – An assessment of accuracy of data <i>OommenP.Mathew, Anil Kumar K, Mathew MC, Sachin K.V</i>	249
Section IV: National Health Mission: Programme implementation		
18	AYUSH Health Care Centres under NHM in Punjab: An Appraisal <i>Pawan Kumar Sharma, Poonam Sandhir and Shaik Iftikhar Ahmed</i>	267
19	Status of AYUSH in Co-located Public Health Facilities in Karnatak <i>R.V. Deshpande and H.R. Channakki</i>	283
20	An Overview of the Status of Determinant Inputs in SNCUs under Public Health System in selected Districts of North Karnataka <i>Shriprasad H and M. S. Kampli</i>	295
21	An assessment of Janani Shishu Suraksha Karyakram in Kerala <i>Sreeranjini A, Oommen P Mathew, AnilkumarK & Sachin K V</i>	305
22	Ageing in Assam: A district-wise comparative study based on Census data 1991-2011 <i>Dr. Dilip Kr. Kalita and Arpita Basak</i>	319

LIST OF TABLES

<i>Table No.</i>	<i>Title</i>	<i>Page No</i>
Chapter 1		
1.1	Crude Birth Rate and Total Fertility Rate for the Districts of Karnataka	6
1.2	Distribution of Sample Women across Regions of Karnataka	7
1.3	Background Characteristics of Women in Karnataka	8
1.4	Utilization of Maternal Health Care Services in Districts of Karnataka	10
1.5	Utilization of Maternal Health Care Services by Background Characteristics of Women	11
1.6	Logistic Regression Results of Utilization of Maternal Health Care Services in Karnataka	12
A.1	Number of Health Sub-Centres (HSCs) in the Districts of Karnataka	16
A.2	Number of Primary Health Centres (PHCs) in the Districts of Karnataka	17
A.3	Number of Community Health Centres (CHCs) in the Districts of Karnataka	18
21Chapter		
2.1	Out of pocket expenditure per delivery in the survey districts of Karnataka State, DLHS4-2012-13.	21
2.2	Amount spent for ANC services (in INR) by Type of facility and Caste/Tribe	23
2.3	Proportion of women availed ANC checkups, Sonogram services and treatment for pregnancy complications at private facilities.	24
2.4.a	Expenditure on delivery (in INR) by Type of Facility	25
2.4.b	Expenditure on delivery (in INR) by Type of delivery and Type of Facility	25
2.5	Amount spent on delivery services by the women delivered during previous 3 years (in INR) by selected background characteristics and Type of facility	26
2.6	Proportion of women experienced delivery complications, approached private facilities and prop. Of C-section deliveries at private facilities	27
2.7	Amount spent for PNC services (in INRs) by Caste and Type of facility	28
2.8	Proportion of women received PNC care services and treatment for complications at private facility	28
2.9	Amount spent (in INRs) for getting maternal care services (ANC+Delivery+PNC) by caste/Tribe and Wealth index	28
Chapter – 3		
3.1	Percent distribution of mothers by knowledge of symptoms & danger signs of diarrhoea, according to socioeconomic characteristics	36

3.2	Percent distribution of mothers by knowledge of care & feeding of child during diarrhoea, according to socioeconomic characteristics	37
3.3	Socio-economic Differential of Childhood diarrhoea	37
3.4	Housing Characteristics by prevalence of childhood diarrhoea	38
3.5	Source and treatment of drinking water by prevalence of childhood diarrhoea	38
3.6	Hygienic Practice by Prevalence of Childhood Diarrhoea	39
3.7	Immunization Coverage by Prevalence of Diarrhoea	39
Chapter 4		
4.1	Prevalence of Programme and non-programme contraception, observed fertility rates and estimated natural and gross fertility rates by age-group in Jharkhand State, 2005-06	46
4.2	Gross fertility and gross birth averted by the programme and non-programme contraception by age group in Jharkhand State, 2005-06	47
4.3	Estimated birth averted by programme and non-programme contraception in Jharkhand State, 2005-06	47
Chapter 5		
5.1	Classification of Maternal Morbidity	52
5.2	Maternal Morbidity Incidence Rate in Tamil Nadu	53
5.3	Maternal Morbidity According to Classification of Diseases in Tamil Nadu	54
5.4	Maternal Morbidity due to Pregnancy Morbidity / Diseases in Tamil Nadu	57
5.5	Maternal Morbidity Due to Pregnancy Complications / Complicated Pregnancy in Tamil Nadu	58
5.6	Maternal Morbidity Due to Obstetric Morbidity / Diseases in Tamil Nadu	60
5.7	Maternal Morbidity Due to Obstetric Complications in Tamil Nadu	61
5.8	Maternal Morbidity Due to Post Natal Complications in Tamil Nadu	62
Chapter 6		
6.1	Risk Ratios for Moderately Underweight Children in age group 0-5 years, Maharashtra, 2013-14	66
6.2	Risk Ratios for Severely Underweight Children in age group 0-5 years, Maharashtra, 2013-14	67
6.3	Prevalence of Risk Ratios for Moderately and severely Underweight Children in age group 0-5 years, by districts of Maharashtra	68
6.4	Prevalence of Risk Ratios within Districts for Moderately and Severely Underweight Children in age group 0-5 years, by districts of Maharashtra	69
6.5	Prevalence of Risk Ratios for Moderately and Severely Underweight Children in age group 0-5 years, across districts of Maharashtra	70
6.6	Distribution of Moderately and Severely Malnourished Children by gender and areas, Maharashtra, 2013-14	71

Chapter 7		
7.1	Distribution of households by caste, religion and household size, North-East Delhi and Muzaffarnagar, 2016	88
7.2	Age and education level of respondents by gender, North-East Delhi and Muzaffarnagar, 2016	89
7.3	Awareness about Tuberculosis in North-East Delhi and Muzaffarnagar, 2016	89
7.4	Knowledge about symptoms, source of transmission, prevention and risk of Tuberculosis as per respondents (percent), North-East Delhi and Muzaffarnagar, 2016	90
7.5	Respondent's source of information about Tuberculosis (percent), North-East Delhi and Muzaffarnagar, 2016	91
7.6	Respondents' attitude and practice on TB (percent), North-East Delhi and Muzaffarnagar, 2016	92
7.7	Multivariate analysis of characteristics associated with a high knowledge score among community members, North-East Delhi and Muzaffarnagar, 2016	93
Chapter 8		
8.1	Social profile of women by place of delivery	105
8.2	Economic profile of women by place of delivery	106
8.3	Demographic Characteristics of women	107
8.4	Percent distribution of women availed Antenatal care services	108
8.5	AN Care availed and suggestion for improvement of services	108
8.6	Reasons for preferring a particular place for delivery	109
8.7	Type of delivery and expenses on delivery	110
8.8	Awareness about Government health programmes	111
Chapter 9		
9.1	Growth of Rural Public Health Institutions, MP	116
9.2	Number of Public Health Facilities Mapped in HMIS, MP	117
9.3	Ratio of PHCs and SHCs to CHCs by Districts, M.P., 2015	117
9.4	Essential Infrastructure available in CHCs, MP (HMIS-2015	122
9.5	Correlation coefficient matrix of CHC facility indices and CHC Infrastructure Index, 2015-16	128
9.6	Comparison of Selected Infrastructure Reported in HMIS and as Observed during Visit to Selected CHCs during 2015-16, MP	128
9.7	Distribution of CHCs according to infrastructure index by star rating, MP (2015-16)	129
Chapter 10		
10.1	Coverage by UPHCs of Bhopal and Raipur	137
10.2	Human Resource for UPHCs in Bhopal and Raipur	139
10.3	Availability of Trained Persons in UPHCs in Bhopal and Raipur	139
10.4	Proportion of Availability of Infrastructure, Services and other Public amenities in UPHCs in Bhopal and Raipur	140
10.5	Background Characteristics of Beneficiaries of UPHCs in Bhopal and Raipur	141

10.6	Distribution of Households of Beneficiaries by Standard of Living Scores in Bhopal & Raipur.	142
10.7	Type of Services Received by Beneficiaries at UPHCs in Bhopal and Raipur	142
10.8	Beneficiaries' Satisfaction with Services Received in Bhopal (MP) and Raipur (CG)	143
10.9	Advice received on ANC and Delivery Care by Beneficiaries at UPHCs in Raipur	143
10.10	Proportion of Beneficiaries who Received ANC Services at UPHCs in Raipur (CG)	143
10.11	Suggestions of Beneficiaries who availed UPHC services in Bhopal and Raipur	144
10.12	Health and Social Activities of MAS in Bhopal and Raipur	146
Chapter 11		
11.1	Satisfaction with location of the health institution and facilities availed at arrival (in per cent)	152
11.2	Satisfaction with the services at reception and registration counter	154
11.3	Satisfaction with the facilities at the waiting area in the OPDs	155
11.4	Satisfaction with the facilities at different wards in the IPDs	156
11.5	Satisfaction level of patients in the OPD with the services of doctors	156
11.6	Satisfaction level of patients in the IPD with the services of doctors	158
11.7	Satisfaction level of OPD patients with diagnostic facilities	159
11.8	Satisfaction with finance related matters	162
11.9	Satisfaction with overall experience at the public and private health institution	163
11.10	Patient's views on different aspects of a hospital	164
Chapter 12		
12.1	Percentage distribution of households by background characteristics	170
12.2	Percentage distribution of households by housing characteristics	170
12.3	Percentage distribution population by chronic diseases	171
12.4	Type of Chronic diseases among total diseases reported	172
12.5	Chronic diseases by Age and Sex	172
12.6	Chronic diseases by Socio economic characteristics	173
12.7	Percentage Distribution of population reporting 'any other disease'	173
12.8	Percentage distribution of population by type of chronic Diseases	174
12.9	Geographical distribution of population seeking treatment of chronic disease	175
12.10	Percentage distribution of population seeking treatment for chronic disease	176
12.11	Place of treatment of chronic disease by area	176
12.12	Reason for not choosing Government facility for treatment of chronic disease	177
12.13	Duration of chronic disease and follow up treatment	177

12.14	Total cost of Treatment of Chronic Diseases	178
12.15	Expenditure for chronic illnesses (in Rs)	178
12.16	Expenditure for treatment of Chronic disease	179
12.17	Expenditure for treatment of Chronic disease at Government Health Facilities	179
12.18	Registration in Health Insurance	180
12.19	Cost of treatment for Chronic disease and Insurance Coverage	180
12.20	Source of money for treatment	180
12.21	Any other disease' by Age and Sex	181
12.22	Percentage distribution of population seeking treatment of other diseases	181
12.23	Place of treatment of 'Any other disease' by area	182
12.24	Reason for not choosing Government facility for treatment of 'Any other disease'	182
12.25	Total cost of Treatment of Any other diseases	183
12.26	Expenditure for treatment of 'Any other disease'	183
12.27	Treatment cost of 'Any other disease' (in Rs)	183
12.28	Binary Logistic Regression coefficients showing the odds ratios of choosing Government or Private Health Facilities for treatment of chronic and 'any other disease'	184
Chapter 13		
13.1	Distribution of RDWs by newborn health check-ups by ASHA during home visits	197
13.2	Percent RDWs by health advices given by ASHA related to the newborn during her postnatal home visits	198
13.3	Proportion of RDWs with good NBC practices by ASHA background	199
13.4	Proportion of RDWs with good NBC practices by RDW background	202
Chapter 14		
14.1	Knowledge About Non- Communicable Diseases	210
14.2	Attitude About Non-Communicable Diseases	212
14.3	Practice about Non Communicable diseases	213
Chapter 15		
15.1A	Performance of study districts based on HMIS index for April-June, 2014-15 and 2015-16	224
15.1 B	Performance of study districts based on HMIS index for, 2013-15	224
15. 2.A	Talukas level performance of Dahod District based on HMIS Index, 2013-15	225
15. 2.B	Talukas level performance of Panchmahals District based on HMIS Index, 2013-15	227
15. 2.C	Talukas level performance of Valsad District based on HMIS Index, 2013-15	228
15.3	Performance as per HMIS Index among the talukas of 3 Districts during 2013-15 (based on index value for all talukas of 3 districts)	229

Chapter 16		
16.1	Details of Selected Health Facilities in Ganderbal District	235
16.2	ANC Indicators of Ganderbal District and their comparison with Jammu and Kashmir During 2015-2016.	237
16.3	Matching of HMIS records for ANC indicators and Child Immunization reported by PHC Shuhama and SC Gulab Bagh and SC Khullmulla During Oct-Dec, 2015.	239
16.4	Matching of HMIS records for ANC indicators and child immunization reported by PHC Kachhan and its SCs During Oct-Dec, 2015.	244
16.5	Overall Differences in Recording, Reporting and Uploading in selected HMIS Indicators in Ganderbal during Oct-Dec, 2015.	245
Chapter 17		
17.1	Distribution of women selected for antenatal care services	252
17.2	Distribution according to accuracy of TT1b& TT2 date recorded in MCTS data and MCH register	253
17.3	Distribution according to Institution at which TT was taken and mode of collecting information for MCTS entry	254
17.4	Distribution according to MCTS-reported ANC visit by JPHN and reported by beneficiary	255
17.5	Distribution according to disparity in Polio – 0 doze, Hepatitis B and BCG date recorded in MCTS	256
17.6	Distribution according to place of administration and mode of collecting data for MCTS of Polio- 0 doze, Hepatitis B and BCG	257
17.7	Distribution according to disparity in Pentavalent and Measles date recorded in MCTS and MCH register	258
17.8	Distribution according to place of administration of Pentavalent and measles and mode of collecting data for MCTS	259
Chapter 18`		
18.1	Type of allopathic and AYUSH health care facilities in Punjab	269
18.2	Workforce at AYUSH Health care centres as per IPHS Guidelines	271
18.3	Physical condition of AYUSH health centres buildings	272
18.4	Type of IEC activities in the sampled AYUSH health care centres	273
18.5	Availability of furniture items in the sampled AYUSH centres	275
18.6:	Availability of equipments of routine use in the sampled AYUSH health care centres	276
18.7	Status of medicines in sampled AYUSH health care centres	278
18.8	Average number of OPDs per day in an AYUSH health care centre	279
Chapter - 19		
19.1	Background characteristics of clients availed AYUSH services, Udupi district	290
19.2	Percent distribution of clients by type, duration of illness, stage availed treatment, distance travelled, person motivated and reason for preference, according to type of AYUSH system, Udupi district	291

19.3	Percentage distribution of clients by their opinion on quality of services , Udupi district	292
19.4	Percent distribution of clients reported suggestions for improving AYUSH , Udupi district	294
Chapter - 20		
20.1	Admission Profile	297
20.2	Status of Bed Availability	297
20.3	Radiant Warmer	298
20.4	Phototherapy Unit	298
20.5	Status of Stethoscope	299
20.6	Status of Pulse Oximeter	299
20.7	Multi Channel Parameter	299
20.8	Staff Nurse	300
20.9	Nurse Bed Ratio Vs Nurse Baby Ratio	300
20.10	Doctors	301
20.11	Doctor Baby Ratio	301
Chapter 21		
21.1	Descriptive statistics for no of checkup based on place of treatment during pregnancy	309
21.2	Comparison of Antenatal cost based on place of antenatal checkup	312
21.3	Comparison of type of delivery based on place of delivery	312
21.4	Comparison of cost of delivery based on type of facility	313
21.5	Descriptive statistics showing other costs (bribe amount)of government hospital delivery	314
Chapter 22		
22.1	Percentage of Population by broad age group in Assam, 1991-2029 (Total)	322
22.2	Percentage of Population by broad age group in Assam, 1991-2029 (Rural)	322
22.3	Percentage of Population by broad age group in Assam, 1991-2029 (Urban)	323
22.4	District wise population by broad age group in Assam, 1991-2029	324
22.5	District wise Dependency Ratio of Assam, 1991-2029	325
22.6	District wise Dependency Ratio of Assam, 1991-2029	326
22.7	Sex Ratio by Broad Age Group in Assam	327
22.8	District-wise composition of 'young old' (60-69 years) and 'old old' (70 years and older) among elderly population (60 years and older) in Assam, 1991-2011	327
22.9	District-wise index of ageing in Assam, 1991-2029	328
22.10	District-wise index of ageing in Assam, 1991-2029	329
22.11	Trend in overall growth rate (OGR in percent) of the total population and growth rate of elderly (GRE) in India and Assam	329
22.12	District-wise Median Age in Assam, 1991-2029	329

LIST OF FIGURES

<i>Figures No.</i>	<i>Title</i>	<i>Page No</i>
Chapter 2		
2.1	Study area	21
Chapter – 3		
3.1	Number of children who had diarrhoea last time by treatment & care received	40
Chapter 5		
5.1	Maternal Morbidity Incidence Rate in Tamil Nadu	53
5.2	Incidence of Maternal Morbidity in Tamil Nadu	54
5.3	Incidence of Pregnancy Complication in Tamil Nadu by districts	56
5.4	Incidence of Pregnancy Morbidity in Tamil Nadu by districts	56
5.5	Incidence of Obstetric Morbidity in Tamil Nadu by districts	56
5.6	Incidence of Obstetric Complication in Tamil Nadu by districts	56
5.7	Incidence of Other Maternal Morbidity in Tamil Nadu by districts	56
5.8	Incidence of Post Natal Morbidity in Tamil Nadu by districts	56
Chapter 6		
6.1	Percent Distribution of Moderately and Severely Malnourished Children by Social Groups, Maharashtra, 2013-14	71
6.2	Percent distribution of Moderately and Severely Malnourished Children by Social Groups, Maharashtra, 2013-14	71
6.3	Distribution of Moderately and Severely Malnourished Children in urban areas by Social Groups, Maharashtra, 2013-14	72
6.4	Percent distribution by causes of Infant Deaths in Vidarbha -Amravati region	72
6.5	Percent distribution by causes of Infant Deaths in Konkan-region	72
6.6	Percent distribution by causes of Infant Deaths in Marathwada region	73
6.7	Percent distribution by causes of Infant Deaths in Vidarbha-Nagpur region	73
6.8	Percent distribution by causes of Infant Deaths in Kandesh region	73
6.9	Percent distribution by causes of Infant Deaths in Desh -Pune region	73
6.10	Percent distribution by Neonate deaths by time and In Vidarbha region of Maharashtra, 2013-14	73
6.11	Percent distribution by causes of Neonatal deaths by time and In Konkan region of Maharashtra, 2013-14	74
6.12	Percent distribution by causes Neonate deaths by time and In Vidarbha – Nagpur region of Maharashtra, 2013-14	74
6.13	Percent wise causes of Neonate deaths by time and In Kandesh region of Maharashtra, 2013-14	74
6.14	Percent distribution by causes of Neonate deaths by time and In Desh-Pune region of Maharashtra, 2013-14	74

6.15	Percent distribution by causes of Neonate deaths by time and In Marathwada region of Maharashtra, 2013-14	74
6.16	Percent distribution by Neonate deaths by time and In Vidarbha-Amravati region of Maharashtra, 2013-14	75
6.17	Percent distribution by of Neonate deaths by time and In Konkan region of Maharashtra, 2013-14	75
6.18	Percent distribution by of Neonate deaths by time and In Marathwada region of Maharashtra, 2013-14	75
6.19	Percentage of Neonatal deaths by time and In Vidarbha region of Maharashtra, 2013-14	75
6.20	Percent distribution by of Neonate deaths by time and In Kandesh region of Maharashtra, 2013-14	76
6.21	Percent distribution by of Neonate deaths by time and In desh-pune region of Maharashtra, 2013-14	76
Chapter 9		
9.1	District-wise number of CHCs in 2007 and 2015, M.P.	117
9.2	Distribution of Rural Health Facilities according to Population size in Non High Priority Districts, Madhya Pradesh, 2015	118
9.3	Distribution of Rural Health Facilities according to Population size in High Priority Districts, Madhya Pradesh, 2015	118
9.4	Average Population Covered by CHCs in the Districts, MP (HMIS-2015)	119
9.5	Status of Exterior and Interior in CHCs, MP, (HMIS-2015)	119
9.6	Essential Communication and Support Infrastructure in CHCs, MP (HMIS-2015)	120
9.7	Essential Communication and Support Infrastructure in CHCs, MP (HMIS-2015)	121
9.8	Average number of Health Human Resource available at CHC per 100,000 population by districts, M.P. (HMIS-2015)	121
9.9	Availability of Specialists' Services in CHCs, MP (HMIS-2015)	123
9.10	Distribution of CHCs on Infrastructure Index, M.P. (HMIS-2015)	125
9.11	Distribution of CHCs on Infrastructure Index, M.P. (HMIS-2015)	127
Chapter 10		
10.1	Functioning of UPHCs under NUHM in Bhopal & Raipur	136
10.2	No. of infrastructure amenities	138
10.3	No. of Services	138
10.4	Background characteristics: Economic status	141
10.5	No. of ANC Checkups	143
10.6	ANC Services received	144
10.7	Out of pocket Expenses	144
Chapter 11		
11.1	Satisfaction Level	163
Chapter 12		
12.1	Types of Chronic Diseases among total population	172

12.2	Type of Other Diseases to total cases	174
12.3	Utilization of Health Care facilities for Other Diseases	181
Chapter 13		
13.1	ASHAs aware of danger signs in newborns during first month	191
13.2	Knowledge scores of ASHAs	191
13.3	Age distribution of recently delivered women	193
13.4	Educational Status of RDWs	193
13.5	Mean household income of RDWs	194
13.6	Mothers' Knowledge on Various Aspects of Newborn Care	194-195
13.7	Distribution of Mothers by Knowledge Scores	196
13.8	Babies having health problems within one and half months of birth	197
13.9	Health advices given by ASHA related to the newborn during her postnatal home visits	198
13.10	Breastfeeding status of the children by present age of newborn	200
13.11	Something applied on the cord after cutting and until it fell	200
13.12	Age of baby at first bathe after birth	201
13.13	Proportion of RDWs with all six good NBC practices by ASHA background	201
13.14	Proportion of RDWs with all six good NBC practices by RDW background	201
Chapter 17		
17.1	Distribution according to agreement of antenatal MCTS data with actual data	255
17.2	Distribution according to agreement of postnatal MCTS data with actual data	259
17.3	Distribution according to agreement of postnatal MCTS data with MCH register data	260
Chapter - 20		
20.1	Percentage of Fully functional equipments	300
Chapter 21		
21.1	Awareness regarding JSSK	308
21.2	Awareness regarding JSSK	309
21.3	Source of knowledge	309
21.4	Type of services	310
21.5	Box plot showing the various antenatal cost based on place of antenatal check-up	310-311
21.6	Free entitlements	312
Chapter 22		
22.1	Population Pyramid of Assam (broad age –group), 1991-2029	322-323

INTRODUCTION

A vital index of a healthy society is to have healthy mother and healthy baby. India is at the forefront of ensuring maternal and child health services to all women and children through various policies and programmes. Despite several socio-economic and cultural barriers, the country has made laudable progress in ensuring health care services and thereby bringing down maternal and child mortality. Globally, however, the share of maternal mortality in India remains high at nearly 15 per cent. Therefore renewed efforts are essential for ensuring safe motherhood to all in the country.

The Millennium Development Goals (MDG) have now paved the way to Sustainable Development Goals (SDG) in 2016. Maternal and child health was accorded great emphasis under MDG5. Strategies adopted targeted at reducing the number of maternal deaths and providing women good-quality reproductive health care. At the same time, MDG6 strived towards combating HIV/AIDS, malaria and other diseases. Although the overall achievements under the MDG towards improving maternal and child health have been impressive, the country still faces challenges. Maternal Mortality Ratio (MMR) in India is 167 per 100,000 live births during 2011-13 (RGI-SRS, 2014) a drop from 212 per 100,000 live births during 2007-09. Fertility is nearing replacement levels (TFR of 2.3 in 2014), infant mortality is 39 infant deaths per 1000 live births a decline from 53 in 2008, U5MR is relatively higher at 45 (2014) although it declined from 69 (2008) and neo-natal mortality rate is 26 (2014) as against 35 in 2008. Present challenges pertain to achieving the targets set by the SDG by the year 2030. Reducing maternal mortality ratio to less than 70 per 100,000 live births, ending preventable deaths of newborns and children under age 5 years, reducing neonatal mortality to 12 per 1000 live births, ending AIDS

epidemic, reducing premature deaths due to NCDs and achieving UHC etc are some of the Goal 3 targets. Further improvements in the health of the population are expected through a much broader approach inclusive of 17 goals and 169 targets of the SDG. All the development goals underpin the critical development challenges faced by India. Embracing Universal Health Coverage (UHC), the third goal “Good Health and Well-being” targets to ensure healthy lives and promote well-being for all. Non Communicable Diseases and risk factors, ending AIDS epidemic, tuberculosis, malaria, environmental hazards and strengthening health systems are also accorded priority in the SDG.

India has, over the last ten years, witnessed greater integration of health aspects with factors that by and large directly influence the health outcomes like nutrition, sanitation and hygiene, safe drinking water etc and indirect factors like gender based violence, education etc under the NHM. NHM support in infrastructural development of Government health institutions, improved supply of medicines, equipments, supply of substantial contractual work force have resulted in providing quality health care services. To fill the gap in service delivery to urban poor the National Urban Health Mission (NUHM) has been providing better health care facilities to urban poor since 2013. “National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke (NPCDCS)” helped to address the rising NCDs in the country. The programmes like JSY and JSSK increased institutional deliveries and new born care through the Special New Born Care Units (SNCU) reduced the case fatality among sick new borns. Web-based HMIS has led to the regular reporting of data pertaining to health indicators, and the MCTS initiative effectively tracked mothers

for ensuring timely ANC, PNC, infant and child care.

To track the progress towards Goal 3 and others including achievement of gender equality, clean water and sanitation, ending poverty and hunger which are interlinked to the health and well being goals, one major challenge would be the availability of data. India has so far invested greatly in creating data base for planning, the Management Information System being the latest initiative. Censuses, national level household surveys, coverage of basic information from small scale surveys constitute the cornerstone of data base for planning. Here lies the critical role of research studies utilising these data sources for providing inputs for policy and planning .

Population Research Centres sponsored by the Ministry of Health and Family Welfare, Government of India have been engaged in carrying out national surveys, evaluation studies, projects and analytical research studies on various issues relating to population, health and development which have policy relevance in the context of planning and development. The year 2015-16 has been a landmark in the PRCs' existence when the Ministry published selected good quality research works of PRCs all over the country as 'Maternal and Child Health in India: Compendium of studies conducted by PRCs in India 2014-15'. Ministry took the whole journey one step forward by organizing a workshop to disseminate the research works conducted by PRCs during 2015-16 at Thiruvananthapuram, Kerala towards its efforts to bring out a compendium. Population Research Centre, University of Kerala coordinated the work.

The compendium includes 22 papers classified in four sections. The first out of the five papers in the section 'Maternal and Child Health in India' examines the regional differences in the utilization of maternal health care services in Karnataka by using District Level Household Survey (DLHS)-4 data for Karnataka. The findings of this paper calls

for an urgent need to improve the MCH services in the State by providing full range of ANC services to women and bringing them to the institutions for their delivery which is a major challenge. The second paper is on household expenditure incurred on maternal care services in rural North Karnataka. It estimated the household expenditure on ANC, delivery and post-natal care by categorizing the type of facility broadly as public and private as well as by type of public health facilities. The results clearly demonstrate high burden of maternal health care expenditure which negates the utilization of Governmental free maternal healthcare services. The third paper captures the level of knowledge of mothers on childhood diarrhoea and its management and also attempts to understand the extent of diarrhoea among children and factors associated with it. The study results show that prevalence of diarrhoea was high among children whose household lagged behind in hygienic practice.

Mother's knowledge about child care during diarrhoea was better than the care after the illness. The fourth paper on family planning and fertility in Jharkhand captures the proximate determinants through the programme and non-programme contraception mainly to assess the role of family planning on fertility control. The study concludes that programme contraception has a dominant role in controlling fertility but the non-programme contraception/natural methods use should also be enhanced at the places where accessibility of programme contraception is poor. The fifth paper looks at the overall incidence of maternal morbidity in Tamil Nadu and the incidence of the diseases/complication among women. In the last paper of this section, the authors make a comparative analysis of malnutrition among children in Maharashtra and looks at the regional variation in terms of prevalence of malnourished children and the factors that predict the observed causes of child death. Risk ratio techniques, are applied which shows regional variation in terms of prevalence of moderately and severely malnourished children. The study points out the prevalence of moderate-

ly underweight children to normal weight to be higher in rural areas than in urban areas

Section II of the book is on 'Health Care Service: Status and Issues' which includes eight papers. The first paper applies a mixed methodology to assess the knowledge, awareness and practices regarding Tuberculosis and documents the status and scope of IEC activities at district level from DOTS providers in North-East Delhi and Muzaffarnagar, Uttar Pradesh. The paper highlights the need for broadening the focus of IEC from just pulmonary TB to extrapulmonary, pediatric and drug resistant TB. The study shows that the community was aware of TB disease but neglects health unless symptoms worsen and concludes that the IEC activities need repeated sensitization of the general population, given the limited knowledge and awareness about different aspects of TB. The second paper makes a comparative analysis of institutional deliveries in Kanyakumari District of Tamil Nadu and tries to find out factors influencing decision on place of delivery. The study arrives at the conclusion that doctors' availability, facilities for caesarean section delivery and scan facility are some of the improvements expected by women in public institutions. For bringing the facilities to the domain of the public utility the authors point out the necessity to ensure that newly constructed facilities are made functional as soon as the construction and other works are completed which is the major lacunae in successful implementation of programmes. An assessment of status of Community Health Centres (CHCs) in Madhya Pradesh using information from HMIS infrastructure data is embarked upon in the third paper. It describes the growth and distribution of different types of rural public health institutions, its coverage, infrastructure and amenities, availability of human resource, and computes the estimated facility infrastructure index ascertains the gap in reported and actual available infrastructure at the selected CHCs. The fourth paper is also in the same line but looks at the Urban Primary Health Centres of Bhopal and Raipur Cities un-

der NUHM. The paper also gauges the opinion of beneficiaries' regarding services received and their level of satisfaction. The UPHCs of Raipur are found to be providing services more effectively due to adequate infrastructure facilities, human resources and other public amenities in comparison to those at Bhopal where only OPD services are being rendered. The authors stress the need for creation of more capacity in the health system in terms of manpower, infrastructure and diagnostics to match growing demand for services in the community from UPHCs. Next paper of the section finds out the patient satisfaction with health care services in Punjab based on patient's experience with a variety of attributes at the allopathic public health institutions providing secondary health care services. The study found the mismatch between the huge amount of money spent and the quality of services received to have impelled many patients to seek services of a Government hospital although the satisfaction level was better among the patients utilizing private health care OPD services than those availing public health care OPD services. Sixth paper here examines the treatment seeking behaviour of persons suffering from chronic diseases and also 'other' diseases and the level of utilization of Government health facilities for chronic and 'other illnesses'. The higher prevalence of hypertension and diabetes and the absence of rich-poor divides in prevalence of chronic diseases are the major observations. Out-of-pocket expenditure on tests, medicine and travel is pointed out as a gap in service delivery. But the Governmental efforts, including health insurance coverage, for supporting the treatment costs of chronic and other diseases is pointed out to have succeeded in higher utilization of Government health facilities which reflects the initial success of UHC programme in the State. The seventh paper is on assessment of home based newborn care in Uttar Pradesh and examines the preparedness of ASHAs for conducting home visits, the training status of ASHAs on HBNC and tries to capture their performance and the challenges in implementing this scheme. The authors find message retention on crucial NBC

practices among ASHAs to be quite low and the need for addressing the tendency among ASHAs to follow shortcuts without adhering to the protocol through quality vigil and continuous monitoring and supportive supervision of ASHAs by ASHA coordinators and ANMs. The last paper of the section encapsulates the knowledge, attitude and practice of NCD risk factors among medical officers working at all three levels of health care facilities in Haryana. The study recommends improving knowledge and overcoming identified barriers in order to promote Medical Officers in taking up NCD prevention in community.

The third section “Health Information: Assessment of Data quality” includes three papers with settings in three States of Gujarat, Jammu and Kashmir and Kerala respectively. The first one analyses the performance of all the talukas in Dahod, Panchmahals and Valsad districts of Gujarat using HMIS index measuring various maternal and child service indicators. The study concludes that variations exist in performance across districts, across talukas, as well as differences in performance of indices, when performance of individual districts is assessed by percentage of poor performing talukas. But the contradictory results depicted by combined index calls for addressing the weak indices in poor performing talukas. HMIS data quality issues in Sub Centres and Primary Health Centres of Ganderbal District of Jammu and Kashmir are examined in the second paper. The paper identifies whether all service providing units are reporting for HMIS and examines whether reporting units are regular in reporting or not on all maternal and child health indicators as per HMIS. The study finds regular data uploading to have improved to a great extent but still finds some data quality issues especially in under-reporting of infant and maternal deaths and immunization coverage, that remain to be addressed. The third paper gauges the accuracy of data capture in MCTS in Kerala by measuring the drop out rate in antenatal and birth registration, understanding the level of maintaining MCH reg-

ister etc. The study finds the accuracy of MCTS data regarding BCG and Polio 0 dose, Hepatitis etc to be poor and that on ANC services to be satisfactory. The authors conclude that higher target population, workload assigned other than MCH activities and non-availability of high speed internet connection for HMIS/MCTS are some of the major problems experienced by JPHNs which in turn affect the quality of MCTS data.

The fourth and final section ‘National Health Mission: Programme implementation’ includes four papers on different programmes under NHM and one paper on Ageing of population in Assam. The first paper endeavors to assess the availability of physical infrastructure and health staff, and the overall functioning of AYUSH centres under NHM in Punjab relative to the Indian Public Health Standard Guidelines (IPHS). The study finds major gaps in service delivery based on IPHS standards but the co-location of AYUSH centre has lead to higher satisfaction level of patients in terms of providing choice of treatment. The study concludes that strengthening the functioning of AYUSH centers based on IPHS standards need to be prioritised. The second paper on AYUSH in co-located public health facilities in Udupi district of Karnataka too has objectives similar to the previous paper. In addition it assesses the utilization patterns of clients in terms of socio-economic and demographic characteristics choosing AYUSH services and reasons for preferring the system. The study clearly brings out the increasing demand for AYUSH due to effectiveness and lesser side effects of medicines and calls for creating increased awareness about AYUSH through mass media and other channels and improving the service conditions of AYUSH staff at co-located facilities. The third paper provides an overview of the status of SNCUs under public health system in the selected districts of North Karnataka by addressing the availability of infrastructure, equipments and human resources. The inappropriate location of basic infrastructure, lack of equipments, shortfall of doctors and staff nurses, high-

er average nurse-baby ratio than nurse-bed ratio, higher average baby-bed ratio than doctor-bed ratio due to higher bed occupancy rate and substantial average fatality rate are the issues highlighted in the study. The penultimate paper throws light into the implementation of Janani Shishu Suraksha Karyakram (JSSK) in two districts of Kerala. The awareness of pregnant women on the entitlements of JSSK is found to be satisfactory but the out of pocket expenses at both private and public facilities is pointed out as a major concern. Highest mean OOPs for the diagnostics test and scans were seen at private practicing Government doctors whereas for medicine and OP tickets, private hospitals were charging higher amounts. ANC cost is almost double in the private hospital or private practicing Government doctors compared to that in Government hospitals. The study recommends reducing the gaps by liberalizing the policy about the funding of the components of the JSSK entitlements. The final paper on Ageing placed in this section makes a district-wise comparison of age composition of population of Assam by broad age groups 0-14, 15-59 and 60+ years based on Census data 1991-2011. The levels and trends of aged population in Assam, dependency ratio, sex ratio, median age and ageing index are examined. The author points out that the increasing median age, increasing old dependency ratio, higher ageing index among females than males etc necessitates social security measures for the older persons, especially in rural areas and special policies for women and rural people.

By and large, the studies carried out by the PRCs that are included in the compendium encompass diverse aspects with main focus on maternal and child health and health service delivery in the country. A substantial proportion of studies are based on information collected in the beneficiary perspective which throws light into mismatch between expectation, demand and supply, information about which is a big step in improving the implementation of various health programmes. The studies based on data collected from health institutions carried out in an evaluation mode reflect the

standards maintained by hospitals and the suggestions necessarily could lead to improved quality of services. On the other hand the wealth of information provided by the secondary data sources like the Census, NFHS and DLHS are tapped by the research studies that mirror the various aspects of population, health and family welfare indicators in India. Data quality of HMIS and MCTS have been dealt with in few studies which show that there has been improvement on the data quality and the indicators that need further focus have also been highlighted. So these data quality issues could be dealt with separately. There are studies that focus on the implementation of programmes under NHM like the JSSK, SNCU, AYUSH etc. The findings of such studies are useful in having a deeper understanding of the shortfalls in implementation. Overall the studies included in the compendium provide useful insights into various aspects of health care service delivery, maternal and child health and data quality of web based health data for policy and planning initiatives. The studies have portrayed the lacunae in service delivery both in the beneficiary and provider perspective. Access to affordable and good quality health care remains the basic requirement among the people. Various initiatives in place in this regard are found to have not fully met the requirements in terms of human resources and facilities as per guidelines although there have been considerable changes. Few studies have raised the issue of out of pocket expenditure involved in access to services. Suggestions in this context have been placed in these studies. With regard to data quality issues, it can be seen that although the coverage of HMIS data have improved, data quality issues still remain a challenge. Underreporting of infant deaths, maternal deaths and immunization status have been pointed out. The completeness and accuracy of information provided in MCTS too requires strict monitoring to overcome the dropout rate for instance. So the findings of these research papers could contribute to the designing of measures in the country's long run towards realization of the goals laid by the SDG.

Section I

Maternal and Child Health in India

CHAPTER - 1

Utilization of Maternal Health Care Services in Karnataka and its Regional Variations

T.S.Syamala¹, N Kavitha² and Supriya Verma³

*¹Associate Professor, ²Senior Research Officer (Projects), ³Assistant Professor (Projects),
⁴Population Research Centre, ISEC, Bangalore*

Abstract

Care during pregnancy and childbirth plays a crucial role in reducing maternal, infant and child mortality. Despite the launch of National Rural Health Mission, Karnataka continues to have high burden of maternal, infant and child mortality presumably due to poor utilization of some of the MCH services. Further, some of the demographic and health indicators also exhibit wide regional variations. Although there are studies on interstate differentials in various health indicators, studies discussing intrastate differentials are rather limited. This paper therefore examines the regional differences in the utilization of maternal health care services in Karnataka by using District Level Household Survey (DLHS)-4 data.

The findings indicate that there is an urgent need to improve the MCH services in the State. Providing full range of ANC services to the women and bringing them to the institutions for their delivery are the major challenges. In spite of the introduction of several schemes such as JSY and JSSK under NRHM, still 14 percent of the deliveries are non-institutional ones in the State. Home deliveries are particularly high (more than 30 percent in Raichur and Koppal) in the districts of northern inland region. In addition to this, glaring inequalities are also visible in the provision of various maternal health services, causing the northern inland region to lag behind and necessitating special attention. Overall, the multivariate analysis brings out clear disadvantages faced by women living in the inland northern regions. Utilization of maternal health care services appears to be poor among women belonging to inland northern region even after regulating the impact of other background factors.

INTRODUCTION

Significant number of women in developing countries experience life-threatening and complicated health problems during pregnancy and childbirth. In developing countries more deaths occur due to complications in pregnancy and delivery than any other reproductive health problems (EC/UNFPA, 2000). Use of timely and adequate maternal health care services plays a crucial role in reducing maternal and infant mortality. Estimates of World Health Organisation (WHO) show that 22 percent of the world's maternal deaths occur annually in India (Kenneth et al., 2007). Utilization of health care services depends on several factors. Ensor and Cooper (2004) framed a demand-supply framework in order to understand these factors. Demand side factors consist of individual, household and community characteristics whereas the supply side factors consist of health facilities, drugs, equipment, human resources, geographic distance etc.

In order to improve the maternal health care services and thereby reducing the maternal and child mortality, Government of India has initiated several programmes since the inception of its Five Year Plans (GOI, 1952). Efforts further intensified by introducing programmes such as Child Survival and Safe Motherhood (CSSM) Programme during 1992–1997, Reproductive and Child Health (RCH) Programme phase 1 and 2 during 1997–2004, and more recently, National Rural Health Programme in 2005. The CSSM programme was aimed at training medical officers and traditional birth attendants in safe delivery practices, providing aseptic delivery kits, expanding rural health services, including facilities for institutional deliveries, and ensuring other essential things for the emergency obstetric care. This programme was also aimed at improving the antenatal coverage and intended to increase proportion of deliveries attended by a skilled medical attendant.

The RCH Programme widened the scope and in-

cluded other reproductive and child health services (Dadhich and Paul, 2004). The second phase of RCH Programme contained a comprehensive newborn health strategy that included promotion of institutional deliveries with cash subsidies for poor families and compensation for traditional birth attendants. (Pinel and Islam, 2007). Further, if the villages are far off from the nearest health facility, improved home-based newborn care was envisaged (Kim and Chen, 2006). Utilization of maternal health care is a vital component in reducing maternal and child mortality, as complications can occur anytime during pregnancy and childbirth. Hence, preventing complications through use of health care services is a better option. National Rural Health Mission (NRHM) was launched in order to provide equitable, affordable and quality health care to the rural population. During the NRHM period, to increase the institutional delivery, the facilities were further strengthened with Emoc facilities to handle complicated deliveries while ASHAs were introduced to improve the ANC services. Conditional cash transfer to mothers to encourage institutional delivery through JSY schemes and JSSK schemes to improve newborn care were also introduced. As the focus of NRHM was mainly on the rural population, National Urban Health Mission was launched to provide adequate health care services to urban population. All these programmes were introduced primarily to improve the health of the mother and the children and to reduce maternal deaths. These are the supply side improvements that have been made.

Although the maternal and child health situation in India has improved over the years, still maternal mortality and under-five mortality are quite high in the country. This is mainly because of the poor utilization of MCH services in the country particularly among certain sections of society as utilization of health care services depends on many factors. Several studies have shown that utilization of health care depends not only on the provision of services but also on social, economic, demographic, cultural and regional factors. In India, several

studies have brought out state-wise variation in the utilization of health services and studies have also discussed rural-urban differentials in the utilization of health services (Desai et al., 2006; Baru et al., 2010; Purohit, 2004). However, studies focussing on intrastate differentials in utilization of health services are rather limited.

The present study therefore concentrates on the intrastate differentials in MCH services in the state of Karnataka. Karnataka possesses a very rich cultural heritage and rulers of different dynasties have influenced social, cultural and economic settings of the State. It is evident from the history of Karnataka that north Karnataka was more developed politically, economically and culturally as this region was ruled by some of the most of the popular dynasties such as Kadamba, Rashtrakuta, Chalukya, Kalachuri, Vijaya Nagar etc.. In spite of all these, north Karnataka remained underdeveloped due to the collapse of the Vijaya Nagar Empire and the consequent changes in the political system. In contrast, in South Karnataka, several social welfare schemes and policies were introduced by the Mysore Kings for the development (Shiddalingeswari and Raghavendra, 2010). It is also evident that districts of North Karnataka have lower utilization of health care services compared to other districts (GoK, 2006). In Karnataka, the studies have focussed more on the regional imbalances in economic and developmental factors and less on utilization of health services. In view of this background, the present study focuses on the intrastate imbalances in utilizing the maternal health care at the district and regional level.

Objectives of the Study

1. To study the variations in the utilisation of MCH services such as antenatal, natal and postnatal care services in different regions of Karnataka.
2. To analyse the influence of regions on the use of maternal health care services in

Karnataka.

Data and Methods

Data for this study are drawn from the District Level Household and Facility Survey fourth round (DLHS-4) for Karnataka conducted during 2012-13. The major objective of the DLHS-4 was to provide information on family planning, maternal and child health, reproductive health of ever-married women and adolescent girls, and utilization of maternal and child healthcare services at the district level. In addition to these, DLHS-4 also provides information on newborn care, postnatal care within 48 hours, role of ASHAs in enhancing reproductive and child health care and coverage of Janani Suraksha Yojana (JSY).

An important component of DLHS-4 is the integration of Facility Survey of health institutions (Sub-centres, Primary Health Centres, Community Health Centres and District Hospitals) accessible to the sampled villages. The focus of DLHS-4 is to provide health care and utilization indicators at the district level for the enhancement of activities under National Rural Health Mission (NRHM). The International Institute for Population Science (IIPS) was the nodal agency for the survey at the national level.

In this study the utilization of MCH services is assessed only for the last birth. MCH services such as any or full antenatal care (At least three visits for antenatal check-up, at least one TT injection received and 100+ IFA tablets/ syrup consumed), institutional delivery, and use of postnatal care (Care received within 45 days of delivery) were analysed to understand the regional differentials. In order to understand the regional variations, districts of Karnataka have been categorized into four regions- namely, Coastal & Ghats, Inland Eastern, Inland Southern and Inland Northern (as followed in National Sample Survey Organization).

Coastal & Ghats region comprises districts Uttara

Kannada, Dakshina Kannada and Udupi. The districts coming under Inland Eastern region are Shimoga, Chikmagalur, Hassan and Kodagu. Districts such as Tumkur, Kolar, Bangalore, Bangalore Rural, Mandya, Mysore and Chamrajnagar come under the region Inland Southern. The region Inland Northern consists of Belgaum, Bagalkot, Bajapur, Gulbarga, Bidar, Raichur, Koppal, Gadag, Dharwad, Haveri, Bellary, Chitradurga and Davanagere districts. Both bivariate and multivariate analysis is carried out to understand the influence of each region on maternal health services.

Other socio- economic characteristics considered in this analysis are place of residence, religion, caste, educational status of the women and household wealth index which can influence the utilization of MCH services. Apart from these variables, women's age at first birth, birth order and time of ANC registration have also been considered as these variables can also influence the utilization of MCH services.

Religion is categorized into Hindu and others while caste is categorized into three viz., Scheduled Caste (SC)/ Scheduled Tribe (ST), Other Backward Castes (OBC) and others. Educational level of women is categorized as follows: Illiterate, <5 years of education, 5-7 years of education and 8 or more years of education. Age of women at the time of first birth is categorised as births occurred during adolescence (age ≤ 19) or not (20 years and above), and birth order have been cate-

gorised as order 1, 2, 3 and, 4 and above.

Dependent variables used in the analysis are dichotomous in nature and are categorized as follows: utilization of any ANC care services (whether utilized any of the antenatal services or not), utilization of full ANC services (yes or No), institutional delivery (whether delivery was in institution or not) and postnatal check-up (Whether received any postnatal check-up within 45 days of delivery or not).

Before discussing the variations in maternal health services across regions, an attempt has been made to present the variations in select fertility indicators such as Crude Birth Rate (CBR) and Total Fertility Rate (TFR) across different districts of Karnataka. Data on CBR and TFR show wide variations across the districts of Karnataka. The CBR and TFR for Karnataka are 17.8 and 2 respectively.

The CBR is lowest in Udupi (12.8) and Mandya (13.7) and highest in Yadgir (25.9), Bijapur (22.6) and Bellary (22.6). Similarly, TFR is also lowest in Udupi (1.2) and highest in Yadgir (3.5). It is clear that TFR is below replacement level in (many?) of the districts of Coastal& Ghats, Inland Eastern and Inland Southern regions of Karnataka. However, the districts belonging to Inland Northern have higher levels of fertility indicating clear regional differentials in fertility indicators (Table 1.1).

Table 1.1: Crude Birth Rate and Total Fertility Rate for the Districts of Karnataka

Districts	CBR(2011)	TFR (2011)		CBR (2011)	TFR(2011)
Coastal & Ghats			Inland Northern		
Dakshina Kannada	14.8	1.5	Bagalkot	22.4	2.7
Udupi	12.8	1.2	Belgaum	19.9	2.4
Uttara Kannada	15.5	1.7	Bellary	22.6	2.7
Inland Eastern			Bidar	20.0	2.7
Chikmagalur	13.4	1.4	Bijapur	22.6	3.0
Hassan	13.2	1.5	Chitradurga	17.0	2.0
Kodagu	14.1	1.5	Davanagere	16.6	1.9

Shimoga	15.4	1.7	Dharwad	18.0	2.3
Inland Southern			Gadag	18.8	2.3
Bangalore Rural	16.3	1.9	Gulbarga	22.1	3.0
Bengaluru	17.5	1.7	Haveri	18.6	2.2
Chamarajanagar	14.5	1.6	Koppal	22.8	2.9
Kolar	16.5	1.9	Raichur	22.5	2.9
Mandya	13.7	1.5	Yadgir	25.9	3.5
Mysore	15.2	1.7	Karnataka	17.8	2.0
Tumkur	14.6	1.7			
Chikkaballapura	15.5	1.8			
Ramanagara	14.3	1.6			

Source:- S Irudaya Rajan, (2013) *Fertility at the District Level in India, Lessons from the 2011 Census, Economic & Political Weekly, No.23, vol xviii, June 8.*

Analysis and Discussion

This section provides the discussion on the background characteristics of the sample women in Karnataka. The second part of the section covers different aspect of the utilization of maternal health services such as utilization of antenatal, natal and postnatal care services across religions of Karnataka. Finally, logistic regression technique is employed to assess the magnitude of effect of different regions of Karnataka on the use of maternal health care services (slightly confusing sentence)

Background Characteristics of Sample Women in Karnataka

Table 1.2 presents the percentage distribution of women across different regions of Karnataka. The table shows that majority of the women (52.3) belongs to Inland Northern region. Further, 9 percent of sample women belong to Coastal& Ghats region, about 12 percent belong to Inland Eastern region and 27 percent belong to Inland Southern regions.

The age distribution of women in the sample show that 18 percent are below 24; 35 percent are in the age range of 25-29 and 47 percent are 35 years and above. The sample consists of 59 percent rural women and 41 percent urban women. Majority of

Table 1.2: Distribution of Sample Women across Regions of Karnataka

Background Characteristics	Frequency	Total no. of women
Region		
Coastal & Ghats	9.1	3971
Inland Eastern	11.8	5210
Inland Southern	26.8	11795
Inland Northern	52.3	23052

sample women belongs to Hindu religion. About half of the women belong to ‘Other Background Caste (OBC)’. One-fifth of the women belong to ‘Others’ and slightly less than one-fifth of women are from SC. Thirty four percent of women are illiterate and 32 percent have eight or more years of education. Around 28 percent of women belong to the poorest category and 9 percent belong to the richest wealth quintile and the remaining was in the middle income group. It should be noted that as high as 49 percent of sample women became mothers when they were adolescents. Majority of women registered their pregnancy during the first trimester. Socio-economic profile of women across different regions also show more or less similar pattern except small variations. The regional variations were more pronounced in case of education where proportion of illiterate women (43.2 percent) was much higher in the Inland

Northern region as compared to other regions. demographic and health indicators are concerned, Percent of women who belong to poorer wealth quintile appear to be more in the Inland Northern region compared to other regions. As far as the percent of adolescent pregnancies and higher order births are higher among women living in Inland Northern region than women in other regions.

Table 1.3: Background Characteristics of Women in Karnataka

Background Characteristics	Coastal & Ghats	Inland Eastern	Inland Southern	Inland Northern	Total Karnataka
Age of women					
<=24 years	9.9	12	17.1	22	18.4
25-34 years	36.5	34.3	34.4	35.4	35.1
35+ years	53.6	53.7	48.5	42.6	46.5
Residence					
Rural	57.4	56	57.8	61.4	59.4
Urban	42.6	44	42.2	38.6	40.6
Religion					
Hindu	77	83.1	86	83	83.3
Others	23	16.9	14	17	16.7
Caste					
SC	10.8	18.6	22.2	19.4	19.3
ST	5.3	5.7	9.1	11.9	9.8
OBC	56.4	45	46.3	53.6	50.9
Others	27.6	30.7	22.4	15.1	20
Education					
Illiterate	15.5	21.4	29.4	43.2	34.4
<5 years	33.4	22.7	18.7	19.7	21
5-7 years	12.3	15.2	13.7	10.8	12.2
8+ years	38.8	40.6	38.2	26.3	32.3
Wealth quintile					
Poorest	17.5	17.2	20	35.8	27.7
Second	20.8	22.2	23.5	25.7	24.2
Middle	24.4	27.6	29.1	19.2	23.3
Fourth	20.8	20	18.1	13.1	15.9
Richest	16.5	13.1	9.3	6.4	8.9
Age at first birth					
Adolescent age	19.4	37.5	50.2	56.9	49.3
Non Adolescent age	80.6	62.5	49.8	43.2	50.7
Time of registration of last pregnancy					
First trimester	94.3	82.1	85.6	73.3	79.1
After first trimester	5.8	18	14.4	26.7	20.9
Birth Order					
1	47.3	44.3	40.3	31.3	36.2
2	36.9	42.1	43.9	33.3	37

3	10	10.1	12	22	17.4
4+	5.8	3.5	3.8	13.3	9.4
Total	100	100	100	100	100
No. of women	3,971	5,210	11,795	23,052	44,028

Utilisation of Maternal Health Services

One of the major objectives of the National Rural Health Mission is to strengthen maternal and child health services. Several programmes have been initiated under NRHM to improve the MCH services in the country. Maternal health services included here are: antenatal care, delivery care and postnatal care. In DLHS-4 women who delivered since January 1, 2008 were asked about the details of various maternal health services received by them during their last delivery. Components of maternal health services such as antenatal care and institutional delivery and postnatal care have been analysed here.

Utilisation of Antenatal Care Services

Table 1.4 presents the utilization of different maternal health services across districts of Karnataka. The table shows that the proportion of women who received any antenatal care is 90 percent in Karnataka. The coverage of any ANC varies from 81 percent in Haveri to 100 percent in Hassan and Mandya.

Although 94 percent of the women received some ANC services, the percentage of women who received full ANC is as low as 43 percent. The DLHS-4 defines full ANC as at least three visits for ANC check-up, administration of at least one TT injection and consumption of 100 IFA tablets/syrup. Wide variations in the coverage of full ANC have also been observed across the districts of Karnataka. The percentage of women who received full ANC ranges between 24.6 percent in Dharwad and 72 percent in Tumkur. In the districts of Raichur, Dharwad, Bellary and Davanagere belonging to Northern Inland region, the coverage of full ANC is less than 30 percent. Overall,

full ANC coverage is much lower in the State as compared to any ANC coverage.

It is a matter of great concern that although women are coming into contact with the health system for receiving antenatal care services, they are not receiving all the ANC services required for safe delivery, and instead, dropping out after receiving a few services. Since all the ANC services are essential for ensuring the health of the mother and the children, concerted efforts are required to investigate the reasons for women dropping out without receiving full ANC services. Special focus is also required particularly on Inland Northern region where ANC coverage is very low.

Institutional Delivery

In Karnataka, about 86 percent of deliveries have taken place in medical institutions. This implies that still 14 percent of women deliver at home. Large regional variations are also observed in the proportions of institutional delivery across districts. Among the districts of Karnataka, the highest proportion of women opting for institutional delivery is from Dakshina Kannada, Udupi, Mysore and Hassan (98 percent). The proportion of institutional delivery is 65.8 percent in Koppal. Overall, the data show that institutional delivery is far from universal particularly in the Northern districts of Karnataka.

Postnatal Care

In order to ensure good health of the mother and the newborn, all mothers must receive care soon after the delivery. Postpartum check-ups within two months of the delivery are particularly important for home deliveries, and three postpartum check-ups are generally recommended for better

health outcomes. The data presented in table 1.4 shows that majority of women in Karnataka received postnatal check-ups within 45 days of delivery. Here again the districts belonging to Inland Northern lag behind compared to other regions.

Table 1.4: Utilization of Maternal Health Care Services in Districts of Karnataka

Districts	Any ANC	Full ANC	Institutional delivery	Post Natal Care
Coastal & Ghats				
Uttara Kannada	98.5	54.8	94.2	96.2
Dakshina Kannada	99.7	61.9	98.5	92.5
Udupi	99.8	49.1	98.3	96.7
Inland Eastern				
Shimoga	99.4	68.7	96.6	92.6
Chikmagalur	99.1	52.6	95.9	85.7
Hassan	100.0	55.5	98.0	90.2
Kodagu	99.4	49.0	92.9	85.0
Inland Southern				
Tumkur	98.5	71.8	94.3	97.9
Kolar	93.2	45.0	91.6	87.2
Bangalore	99.7	65.1	94.0	92.8
Bangalore Rural	99.0	61.5	95.5	91.7
Mandya	100.0	71.9	96.1	95.2
Mysore	98.8	49.0	98.4	87.9
Chamarajanagar	98.2	59.3	95.1	96.3
Ramanagara	98.9	46.1	97.7	80.1
Chikkaballarpura	97.9	42.4	86.6	85.2
Inland Northern				
Belgaum	99.0	49.5	87.5	91.6
Bagalkot	93.8	31.2	84.1	88.9
Bijapur	89.9	32.2	77.1	68.8
Gulbarga	93.1	52.4	73.4	82.7
Bidar	91.3	38.5	91.5	91.6
Raichur	86.4	25.5	68.5	73.0
Koppal	93.6	35.4	65.8	72.8
Gadag	97.2	31.3	79.8	84.0
Dharwad	97.2	24.6	92.6	89.1

Haveri	81.0	39.2	87.9	82.1
Bellary	89.6	18.7	77.4	79.3
Chitradurga	92.9	32.3	88.5	92.0
Davanagere	94.8	27.2	93.3	92.6
Yadgir	84.8	34.5	73.9	74.9
Karnataka	94.4	43.4	85.8	85.8
N	12,510	12,510	9,764	12,504

This section clearly shows that in spite of the implementation of NRHM, the State needs to concentrate more on the improvement of maternal health indicators. Providing full range of ANC services to the women and bringing women to the institutions for delivery pose a major challenge. In spite of the introduction of several schemes such as JSY and JSSK under NRHM, still considerable proportion of the deliveries are non-institutional deliveries particularly in the Inland Northern region. Apart from this, glaring inequality is also visible in the provision of various maternal health services in the Northern Inland region which, in turn, demands special attention.

Socio-Economic Factors and Utilization of Maternal Health Care Services

The previous section clearly shows that maternal health services need further strengthening in the State and focus should be more towards reducing regional inequalities. Several studies have also indicated that the utilization of MCH services is highly linked to the socio-demographic background of women. This section therefore discusses the socio-economic variations in the utilization of maternal health services in Karnataka. Further, the paper also discusses whether the existing regional variations are due to independent regional effect or due to the differentials in population composition.

Table 1.5 provides the utilization of maternal health services such as the percentage of women receiving any kind of ANC, full ANC, percentage of women going for institutional delivery and per-

centage of women receiving postnatal check-ups according to various socio-economic factors. The percentage of women receiving partial or full ANC declines with an increase in age and birth order. However, throughout the age group the proportion of women receiving full ANC remained very low. First and second order births received better ANC services compared to higher order births. As expected, the ANC service is also better for better educated and richer sections of society. Although, sharp religious and caste differentials are not observed, ANC services are poorer for Hindu women and women from SC/ST groups.

Percentage of women opting for institutional delivery also shows sharp differentials across various categories. As in the case of ANC services, percentage of institutional delivery also declines sharply with age and birth order. Glaring differentials can also be observed across educational groups and household wealth quintile. Percent of institutional deliveries is the least among illiterate women and also among those who belong to poorer wealth quintile. Percentage of institutional delivery also varies among caste groups, with the lowest percentage among Schedule Tribes. Similar pattern is observed with respect to variations in postnatal care also.

Table 1.5: Utilization of Maternal Health Care Services by Background Characteristics of Women

Background characteristics	Any ANC	Full ANC	Institutional delivery	Post Natal Care
Age of women				
<20 years	95.8	42.7	86.8	89.3
20-29 years	94.9	43.4	86.8	86.1
30-39 years	92.8	44.2	82.5	84.4
40+ years	84.7	34.0	74.5	77.0
Residence				
Rural	92.9	39.4	82.8	83.7

Urban	96.7	49.7	91.4	89.1
Religion				
Hindu	93.9	42.8	84.8	85.6
Others	96.1	45.7	90.3	86.6
Caste				
SC	92.8	39.1	81.6	82.0
ST	88.9	33.5	78.8	79.4
OBC	95.3	44.6	87.6	87.3
Others	96.7	50.1	90.5	89.1
Education				
Illiterate	84.6	26.8	70.9	72.9
<5 years	95.0	40.0	84.6	85.7
5-7 years	97.0	41.6	90.2	87.4
8+ years	98.4	54.3	95.3	92.1
Wealth quintile				
Poorest	89.1	31.0	77.2	78.6
Second	94.2	38.8	84.6	84.8
Middle	97.0	50.0	92.0	89.2
Fourth	97.9	53.6	93.6	91.3
Richest	99.2	62.2	98.2	94.0
Age at first birth				
Adolescent age	91.8	35.7	79.6	81.3
Non-Adolescent age	96.0	48.3	90.7	88.8
Time of registration of last pregnancy				
First trimester	97.1	45.1	88.9	87.8
After first trimester	93.6	41.9	82.2	84.9
Birth Order				
1	97.6	51.1	94.2	91.2
2	95.9	44.25	89.3	88.2
3	89.7	33.72	75.8	78.4
4+	84.9	28.9	68.8	70.7
Total	94.4	43.4	85.8	85.8

Overall, the analysis shows that the utilization of maternal health services in the State needs further improvement. Women receiving full ANC coverage are quite low and 14 percent of women preferred to deliver at home. Sharp regional variations are also observed with low utilization of maternal health services in inland Northern regions.

Variations are not only with regard to the regions, but the socio-economic differences in health indicators are also huge. Utilization is better among wealthier and better educated women as compared to their counterparts.

Logistic Regression Results

Bivariate results clearly showed that the utilization of maternal health care services vary widely across regions and also across socio-economic characteristics. However, these are gross differences without controlling for the effect of background variables. Further, the bivariate analysis does not provide a clear answer to the questions on the reasons for the existing regional variations. Whether it is due to compositional differences across regions or is there any independent effect of regions on health outcomes. For this purpose, binary logistic regression analysis is carried out. Three different regressions, the first one with full ANC service as dependent variable and the second one with women who had institutional delivery or did not have one, and the third one with women received postnatal care or not have been carried out to understand the net effect of a region on these maternal services. The explanatory variables included are age of the woman, region (Coastal and Ghats region as the reference category), birth order (1 as reference), education (illiterate is reference category), wealth quintile (lowest is reference category), caste (Scheduled Caste as reference category), age at first birth (first birth at adolescent age as the reference category) and timing of registration of last pregnancy (registered during first trimester as reference category) are the explanatory variables.

Logistic regression analysis confirms the fact that even after controlling other socio-economic variables, region emerges as a strong variable that determines access to maternal health services. Women from Inland Northern region particularly are at

a disadvantageous position in receiving maternal health services even after regulating other socio-economic variables. The odds of receiving full ANC care, institutional delivery and postnatal care are the least for women belonging to Northern Inland region. It is also noted that use of postnatal care is lower in Inland Eastern and Inland Southern regions as well when compared to the Coastal and Ghats region. Similarly, education of women and household economic status are the other significant variables that determine MCH services. The odds ratio of women going for institutional delivery is 1.8 times higher among literate women compared to illiterate women. Similarly, the odds of women going for institutional delivery are almost double when woman moves from the poorest to the richest quintile.

Table 1.6: Logistic Regression Results of Utilization of Maternal Health Care Services in Karnataka

Background characteristics	Full ANC	Institutional Delivery	Postnatal Care
	OR (p value)	OR (p value)	OR (p value)
Region			
Coastal & Ghats(ref)			
Inland Eastern	1.097 (0.281)	0.956 (0.879)	0.377 (0.000)
Inland Southern	1.158 (0.046)	0.618 (0.042)	0.512 (0.000)
Inland Northern	0.587 (0.000)	0.283 (0.000)	0.402 (0.000)
Residence			
Rural (ref)			
Urban	1.176 (0.000)	1.405 (0.000)	1.198 (0.008)
Religion			
Hindu (ref)			
Others	0.915 (0.092)	1.207 (0.072)	0.859 (0.059)

Caste SC/ST (ref)			
OBC	1.087 (0.082)	1.136 (0.100)	1.245 (0.001)
Others	1.145 (0.026)	1.060 (0.605)	1.205 (0.040)
Education Illiterate (ref)			
Literate	1.339 (0.000)	1.771 (0.000)	1.753 (0.000)
Wealth quintile Poorest (ref)			
Second	1.122 (0.037)	1.194 (0.029)	1.223 (0.006)
Middle	1.471 (0.000)	1.729 (0.000)	1.317 (0.001)
Fourth	1.613 (0.000)	1.783 (0.000)	1.539 (0.000)
Richest	2.024 (0.000)	5.236 (0.000)	1.767 (0.000)
Age at first birth Adolescent age (ref)			
Non-Adolescent age	1.139 (0.002)	1.354 (0.000)	1.127 (0.048)
Time of registration of last pregnancy First trimester (ref)			
After first trimester	1.133 (0.010)	0.818 (0.007)	1.026 (0.708)
Birth Order 1 (ref)			
2	0.792 (0.000)	0.551 (0.000)	0.817 (0.006)
3	0.695 (0.000)	0.295 (0.000)	0.519 (0.000)
4+	0.682 (0.000)	0.274 (0.000)	0.443 (0.000)
Constant	0.609 (0.000)	15.533 (0.000)	8.852 (0.000)
N	11,828	9,174	11,824
-2 Log likelihood	-7663 .166	-2934.395	-4208 .6324
Significance	0	0	0

Overall, the logistic regression analysis brings out clear disadvantages in the utilization of prenatal, natal and postnatal health care services for women living in the Inland Northern region. Even after

controlling the socio-economic variables, utilization of maternal health services are poor for women belonging to Inland Northern region. This confirms the independent effect of region on maternal health indicators. Although interstate differentials can be attributed to a greater extent for the existing differences in the health system administration and state health expenditure, the reasons for intrastate differentials need further scrutiny. One of the possible reasons could be the low health infrastructure and low density of health staff in the northern Karnataka as utilization of health care services can directly be linked to the availability of health infrastructure and human resource. As mentioned earlier, North Karnataka also remains underdeveloped because of political and other reasons (Shiddalingaswami and Raghavendra, 2010). Women's education and household economic status are the other two important variables which determine the utilization of maternal care services. Therefore, concerted efforts are needed to minimise the inequalities across regions of Karnataka. Further, focus on women's education and improvement in household economic status should also help in improving the MCH services.

Conclusion

The study shows an urgent need to improve the MCH services in the State. Providing full range of ANC services to women and bringing them to the institutions for delivery are the major challenges. In spite of the introduction of several schemes such as JSY and JSSK under NRHM, 14 percent of the deliveries are non-institutional ones in the State. In addition to this, glaring inequality is also visible in the provision of various maternal health services owing to which the Northern Inland region lags behind and needs special attention.

Overall, the logistic regression analysis brings out clear disadvantages for women living in the Inland Northern region. Even after controlling the socio-economic variables, utilisation of maternal health services remain poor for women belonging

to the Inland Northern region. This confirms the independent effect of region on maternal health indicators.

In recent years, several schemes and programmes have been rolled out by the Centre to improve the maternal and child health care services. In addition to these schemes, the Government of Karnataka has also implemented additional schemes for improving maternal and child health situation in the State.

Schemes such as Prasooti Araiike (care for the pregnant in backward districts viz., Gulbarga, Bidar, Raichur, Koppal, Bijapur and Bagalkot), MADILU (Caring for the mother and the child) and THAYI BHAGYA (Public-private partnership in maternal health care) have been implemented not only to improve maternal and child health services but also to reduce inequality in health indicators. These programmes primarily aim at improving the MCH situations in the underdeveloped northern region of Karnataka. Prasooti Araiike is particularly focussing on the backward districts. Despite these initiatives, the northern districts of Karnataka still lag behind in the utilization of MCH services. Apart from all these programmes which specifically address these districts, it is also found that number of government health facilities such as Health Sub-centres, Primary Health Centres (PHCs) and Community Health Centres (CHCs) is less in the northern districts of Karnataka compared to other districts (Government of India, 2015).

It is also estimated that in the districts of Inland Northern region, there is a lack of required number of sub-centres. In most of these districts, though number of PHCs does not appear to be a problem according to the district population, there is a significant shortfall in the case of Health Sub-centres and community health centres to some extent (Appendix-Table 1.A1, Table 1.A2 and Table 1.A3). This information brings out the fact that health workers have additional pressure at the grass-root level of covering more population. Such kind of lack of health infrastructure and health workers

may also be a major cause for the lack of use of maternal health care services in Inland Northern districts.

While the achievements in the demographic and health fronts made by the State are commendable, there are significant challenges also in improving the MCH situation in the State. First, in spite of concerted efforts, significant proportion of the deliveries in the Inland Northern region of the State still takes place at home and renewed efforts are necessary to universalize the institutional delivery. Second, Karnataka is also a State with considerable regional variation in health indicators. Therefore, strategies to reduce regional disparities in health indicators are important. Thirdly, just like the region-wise disparities in health indicators,

socio-economic differences in health indicators are also huge. Health also has close connection with several other social determinants and it is also important that the general developmental indicators such as increasing education especially women's education, income, basic amenities etc. are also effectively addressed. Despite several efforts through NRHM to make MCH services cashless, the out-of-pocket expenditure on health continues to be a major health care barrier to many households. Therefore, it is important to continue the measures with renewed vigour for achieving the desired health goals.

References

- Baru, Rama, Arab Acharya, Sanghmitra Acharya, A.K. Shiva Kumar and K. Nagaraj. (2010). Inequities in access to health services in India: Caste, Class and Religion. *Economic and Political Weekly*, Special Article, XLV (38).
- Dadhich J and Paul V. (2004). *State of India's newborns*, New Delhi. National Neonatology Forum and Washington DC, Save the Children US; 2004.
- Desai, Sonalde, Lijuan Wu and B. L. Joshi. (2006).

- Utilization of maternal health care services in India: Understanding the regional differences. Indian Human Development Survey, Working Paper No.1.
- EC/UNFPA (2000). *Making Pregnancy and Childbirth Safer* [fact sheet]. EC/UNFPA Initiative for Reproductive Health in Asia in cooperation with the German Foundation for World Population.
- Ensor, Tim and Stephanie Cooper. (2004). Overcoming Barriers to Health Service Access: Influencing the Demand Side. *Health Policy and Planning*, 19(2): 69-79.
- Government of India (1952). First Five Year Plan. New Delhi, Planning Commission.
- Government of India (2015). Rural Health Statistics. New Delhi: Ministry of Health and Family Welfare, Statistics Division.
- Government of Karnataka (2006). Karnataka Human Development Report. Investing in Human Development. Published by Planning and Statistics Department, Karnataka. <http://bjpkarnataka.org/wp-content/uploads/2013/10/Health-and-Family-Welfare.pdf>
- Kenneth, Hill, Kevin Thomas, Carla AbouZahr, Neff Walker, Lale Say, Mie Inoue, Emi Suzuki (2007). Estimates of maternal mortality worldwide between 1990 and 2005: an assessment of available data. *The Lancet*, 370 (9595): 1311-1319.
- Kim Choe M, Chen J (2006). Potential for Reducing Child and Maternal Mortality through Reproductive and Child Health Intervention Programmes: An Illustrative case study from India. *Asia pacific Population Journal*, 21(1):13-7.
- Pinel, A., Q. M. Islam, P. Villeneuve and R. Lakshminarayanan (2007). Strategies for maternal mortality reduction. *The Lancet*, 369(9561):557-8.
- Purohit, B.C. 2004. Inter-state disparities in health care and financial burden on the poor in India. *Journal of Health and Social Policy* 18(3): 37-60.
- Shiddalingeswari, H. and V. K. Raghavendra (2010). Regional disparities in Karnataka: A district analysis of growth and development. CMDR Monograph Series No. 60. Centre for Multi-disciplinary Development Research.

APPENDIX

Table 1.A1. Number of Health Sub-Centres (HSCs) in the Districts of Karnataka

Coastal Ghats	Existing	Required	Shortfall
DAKSHINA KANNADA	440	219	-221
UDUPI	301	169	-132
UTTARA KANNADA	343	204	-139
Inland Eastern			
CHIKMAGALUR	375	180	-195
HASSAN	456	280	-176
KODAGU	206	95	-111
SHIMOGA	305	226	-79
Inland Southern			
BANGALORE RURAL	195	144	-51
BANGALORE	167	174	7
CHAMARAJANAGAR	245	169	-76
KOLAR	275	211	-64
MANDYA	385	299	-86
MYSORE	438	351	-87
TUMKUR	487	416	-71
CHIKKABALLAPURA	199	195	-4
RAMANAGARA	230	163	-67
Inland Northern			
BAGALKOT	234	258	24
BELGAUM	549	714	165
BELLARY	272	306	34
BIDAR	280	255	-25
BIJAPUR	309	335	26
CHITRADURGA	283	266	-17
DAVANAGERE	301	263	-38
DHARWAD	194	159	-35
GADAG	168	137	-31
GULBARGA	347	346	-1
HAVERI	303	248	-55
KOPPAL	185	231	46
RAICHUR	223	288	65
YADGIR	176	191	15
KARNATAKA	8871	7494	-1377

Source: Government of India (2015). Rural Health Statistics. New Delhi: Ministry of Health and Family Welfare, Statistics Division.

Table 1.A2. Number of Primary Health Centres (PHCs) in the Districts of Karnataka

Coastal Ghats	Existing	Required	Shortfall
DAKSHINA KANNADA	71	36	-35
UDUPI	73	28	-45
UTTARA KANNADA	80	34	-46
Inland Eastern			
CHIKMAGALUR	80	34	-46
HASSAN	137	47	-90
KODAGU	32	16	-16
SHIMOGA	91	38	-53
Inland Southern			
BANGALORE RURAL	75	24	-51
BANGALORE	47	29	-18
CHAMARAJANAGAR	60	28	-32
KOLAR	62	35	-27
MANDYA	114	50	-64
MYSORE	140	59	-81
TUMKUR	143	69	-74
CHIKKABALLAPURA	60	32	-28
RAMANAGARA	60	27	-33
Inland Northern			
BAGALKOT	50	43	-7
BELGAUM	154	119	-35
BELLARY	75	51	-24
BIDAR	53	43	-10
BIJAPUR	68	56	-12
CHITRADURGA	89	44	-45
DAVANAGERE	108	44	-64
DHARWAD	32	27	-5
GADAG	37	23	-14
GULBARGA	89	58	-31
HAVERI	72	41	-31
KOPPAL	47	39	-8
RAICHUR	53	48	-5
YADGIR	46	32	-14
KARNATAKA	2310	1249	-1061

Source: Government of India (2015). Rural Health Statistics. New Delhi: Ministry of Health and Family Welfare, Statistics Division.

Table 1.A3. Number of Community Health Centres (CHCs) in the Districts of Karnataka

Coastal Ghats	Existing	Required	Shortfall
DAKSHINA KANNADA	6	9	3
UDUPI	7	7	0
UTTARA KANNADA	3	8	5
Inland Eastern			
CHIKMAGALUR	4	7	3
HASSAN	14	12	-2
KODAGU	6	4	-2
SHIMOGA	5	9	4
Inland Southern			
BANGALORE RURAL	3	6	3
BANGALORE	1	7	6
CHAMARAJANAGAR	3	7	4
KOLAR	2	9	7
MANDYA	7	12	5
MYSORE	9	15	6
TUMKUR	4	17	13
CHIKKABALLAPURA	2	8	6
RAMANAGARA	3	7	4
Inland Northern			
BAGALKOT	7	11	4
BELGAUM	16	30	14
BELLARY	8	13	5
BIDAR	6	11	5
BIJAPUR	8	14	6
CHITRADURGA	10	11	1
DAVANAGERE	5	11	6
DHARWAD	0	7	7
GADAG	2	6	4
GULBARGA	16	14	-2
HAVERI	5	10	5
KOPPAL	8	10	2
RAICHUR	5	12	7
YADGIR	5	8	3
KARNATAKA	180	312	132

Source: Government of India (2015). Rural Health Statistics. New Delhi: Ministry of Health and Family Welfare, Statistics Division.

CHAPTER - 2

Household Expenditure on Maternal Care Services in Rural North Karnataka

Jyoti S. Hallad^{1}, J. A. Golandaj¹, ArinKar², Krishnamurthy J.^{2,3}, B. M. Ramesh³, R. V. Deshpande¹ and B. I. Pundappanavar¹*

¹Population Research Centre (PRC), Dharwad, India; ²Karnataka Health Promotion Trust (KHPT), Bangalore, India ³University of Manitoba, Winnipeg, Canada

Abstract

This paper aims to estimate the household expenditure made on maternal care services- ANC, Delivery and PNC. Endline Household Survey data on Maternal, Neonatal and Child Health (MNCH) in rural Karnataka, India, has been used for the analysis. The survey was conducted in the year 2015 among 5154 ever-married women in the age group 15-34, of which 2337 women had experienced at least one pregnancy outcome during preceding 3 years. Expenditure made by the women for getting ANC, delivery and PNC services were analysed here by considering different categories like consultation charges, diagnostics, medicine and room rent. In addition, expenses towards food, transportation and money given to Health staff, if any, were also considered, so as to understand the total expenditure. Univariate and Bivariate analyses are used as methods for the study.

Findings reveal that most of the women reported household economic burden for availing maternal healthcare services. On an average, altogether each household had spent 9831- INRs for getting ANC, delivery and PNC services. Even women who belong to Scheduled Caste and Scheduled Tribe, poor households report the cost of maternal health care to be higher than 6500INRs. However, expenses made at public health facilities were considerably low at INR 649, 2057, and 142 compared with that at private health facilities INR 5455, 17549 and 1419 respectively for getting ANC, delivery and PNC services. Experience of complications during pregnancy, delivery or postnatal period as well as non-availability of diagnostic services like ultrasound/scanning at public health facilities are the major compelling factors for seeking maternal care services from private sector.

Results clearly demonstrate high burden of maternal health care expenditure which negates the utilization of government free maternal healthcare services. Attempt should be made to create awareness about the available free maternal healthcare services. Reducing cost of obtaining maternal care, particularly for the socioeconomically deprived, should be an appropriate measure to increase the utilization of skilled care. Extreme poor households should continue to get financial support

INTRODUCTION

India has avoided significant maternal and neonatal deaths in the last two decades. Maternal Mortality Ratio has declined from 398 to 212 (per 100,000 live births), and the Neonatal Mortality Rate has declined from 45 to 34 (per 1000 live births) during 1997-98 to 2007-09 (ORGI 2006; ORGI 2009; ORGI 2011a; ORGI 2011b). Despite this recent achievement, still India is one of the five countries in Asia that accounted for half of the maternal deaths worldwide (Hillet al. 2007), and it is one of the countries having high MMR in the world, having 19 percent of maternal deaths worldwide (WHO 2005; WHO 2012a; Hogan et al. 2010).

Skilled attendance at birth is widely recognized as a single most effective strategy for reducing maternal mortality and morbidity (UNFPA, 2007), and this is being used as an indicator to measure progress toward the fifth Millennium Development Goal of improving maternal health (MDG 5) (United Nations, 2007). However, around half of the deliveries, in the developing world, are not attended by skilled professionals. Safe motherhood still remains a distant dream for many around the world, especially in developing countries (Starrs, 2001), leading to most of the maternal deaths occur in poor countries (Costello et al. 2004) suggesting that, most of these can be attributed to low level of supply and utilization of skilled maternal health services (Rahman and Sarkar, 2009).

A range of social, economic and geographic factors that act as barriers to low use of skilled maternity care during childbirth around the world had been documented in literature (Leslie and Gupta 1989; Thaddeus and Maine 1994; Bloom et al. 2001; Hotchkiss et al. 2005; Haddad et al. 2006; Parkhurst et al. 2006; Gage 2007). As India is concerned, maternal education, economic status of household, caste and religion (Bhatia and Cleland 1995; Singh and Singh 2007; Mohanty and Pathak 2009; Kesterton et al. 2010; Mohanty 2011); the

availability of and accessibility to a health facility, presence of a lady medical doctor, availability of drugs (RamaRao 2001; Navaneetham and Dharamalingam 2002; IIPS 2010); community attributes and programme-related factors play a critical role in the utilization of facility-based maternity care (Stephenson and Tsui 2002). Latest rounds of National Family Health Survey (NFHS 3 and 4) indicated that the financial burden on the families was one of the key reasons for poor utilization of maternal health care services (IIPS, 2007 & 2015). Though, Costs - both direct and indirect - have also been shown to be an important barrier to women's use of facility-based maternity care (Stanton and Clemens 1989; Nahar and Costello 1998; Gwatkin 2004; McIntyre et al. 2005; Koenig et al. 2007).

According to the World Health Statistics 2012 report, India ranks third in the out of pocket (OOP) expenditure on health (60 percent) in the south-east Asia region, only after Myanmar (82 percent) and Bangladesh (65 percent). In comparison, the proportion of OOP expenditure to the total health expenditure of Nepal (49 percent), Sri Lanka (44 percent), Indonesia (41 percent), Maldives (28 percent), Thailand (15 percent) and Bhutan (13 percent) are better in same region (WHO, 2012b). The Indian Ministry of Health and Family Welfare (MoHFW) too accepts that, unlike developed countries, in India more than 70 percent of the health care expenditure is paid by individuals and government bears only 20 percent, rest from businesses and external flows (MoHFW, 2009).

Similarly, a national survey analysis by Leone et al. (2013) indicated that 80 percent of the households reported out of pocket expenditure for maternal health care services. Another study done by Mohanty and Srivastava (2013) documented that out of pocket expenditure on institutional delivery in India was 39 USD in public sector facilities in 2004-08. In many situation, health care expenditure which also includes maternal and neonatal care, is catastrophic, this in turn can push the households into poverty. Especially rural households, the less educated, slum dwellers and poor

households, because these households in India are more likely to finance health care through borrowing and sale of assets than well off population groups (Peters, 2001; Bonuet al. 2009; Skordis-Worrall et al. 2011). The analysis of NSSO, 2004 data also reveals that only less than half of the women receive ANC care at public facilities without incurring any direct costs and it is only 15 percent while getting delivery services. The major reason for household expenditure despite many health schemes is unavailability of medicines and diagnostics within public facilities (Leone, 2013). Expenditure on maternal care services was mainly influenced by the type of delivery, delivery with or without complications and place of ANC (Govilet al., 2013). DLHS3 analysis also indicated that OOP expenditure for delivery is more at private facilities and expenditure increases by six times for C-section deliveries (IIPS, 2010). Government of India initiated several programs to improve the institutional deliveries with the goal of reducing maternal mortality. Janani Suraksha Yojana (JSY), a conditional cash transfer scheme under National Health Mission, has led to increase in the institutional deliveries (Paulet al. 2011; Moduguet al. 2012). Further, Government of India has launched Janani Shishu Suraksha Karyakram (JSSK) scheme in June 2011 to further enhance the institutional delivery and reduce the out of pocket expenses for families of all pregnant women in the public sector facilities. JSSK provides free and cashless delivery in the public sector institutions (MoHFW, 2011). Tripathi, et al. (2014) observed significant reduction in the health expenditure for ANC and delivery services before and after implementation of JSSK scheme in Chandigarh. Even though efforts are being made for free and cashless MCH services, recently conducted District Level Household Survey (DLHS4, 2012-13) has pointed out that even at the public health facilities a household has to spend on an average 3130 INRs for each delivery and it is 3000 INRs in rural areas of Karnataka. Further, mean delivery cost for all the deliveries (public and private) combined is found to be INR 9646 in Karnataka (IIPS, 2014).

Table 2.1: Out of pocket expenditure per delivery in the survey districts of Karnataka State, DLHS4-2012-13.

Name of the District	Mean Institutional delivery cost in Public Health facility		Mean Delivery cost (combined*)
	Rural	Combined (Rural and Urban)	
Bagalkot	1500	1590	7858
Bellary	3850	3690	9964
Bidar	1590	1730	8706
Bijapur	1790	1750	9355
Gulbarga	1780	1770	10184
Koppal	2710	2160	8907
Raichur	2180	1820	9916
Yadgir	3120	2980	8964
Karnataka	3000	3130	9646

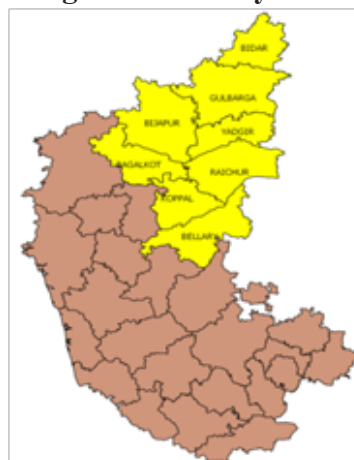
**Includes deliveries at both, Public and Private facilities.*

It indicates that, still women have to spend quite a significant amount while getting various maternal care services. In this background, present paper has been taken up with an objective to examine the extent of expenditure towards ANC, delivery and PNC services and identifies its socio-economic differentials and causing factors among rural women in North Karnataka.

Data Source

The paper utilizes the data from the Maternal, Neonatal and Child Health (MNCH) Endline Household survey on project 'Sukshema' carried out during 2015.

Figure 2.1: Study area



This is a cross sectional survey conducted by PRC, Dharwad in a representative sample of households in rural areas of 8 districts of North Karnataka viz., Bidar, Gulbarga, Yadgir, Bagalkot, Bijapur, Bellary, Raichur and Koppal (Figure 1).

The survey covers the representative sample of 4882 households and 5154 Ever Married Women (EMWs) residing in these households in the age group 15-34. Survey covers both usual residents and visitors who stayed in the household the night before the survey. The survey was basically designed to provide reliable information about awareness regarding the MCH services as well as service utilization and cost of each service. Two different study tools like Household schedule for getting household information and an eligible woman schedule were used for the data collection. Response rate for household questionnaire was 99 percent and that among eligible women was 83 percent. Here maternal and child health services included are Ante Natal Care (ANC), delivery services and Post Natal Care (PNC) services.

Sampling Method

A multistage stratified sampling method was used to create a sample representing individuals from rural areas of 8 districts of North Karnataka. The sample was selected in two stages, with the selection of Primary Sampling Units (PSUs) which are villages, with probability proportional to population size (PPS) at the first stage followed by the selection of households within each PSU in the second stage. The list of villages as per 2011 Census was the sampling frame and 167 villages were selected out of the 5177 villages, applying PPS sampling method. Villages were stratified geographically within each district and taluka. Village size was also considered for implicit stratification.

In all, the selected villages mapping and household

listing operation was carried out separately for SC/ST groups and non-SC/ST groups to prepare a list of households with atleast one eligible woman. In each village, a total of 30 households with atleast one Ever Married Woman (EMW) aged 15-34 years were selected – 15 SC/ST households and 15 non-SC/ST households – with equal probability and without replacement using Systematic Random Sampling in the second stage. SC/ST households were over-sampled to be able to measure extent of differentials by caste and poverty levels. Depending upon the size of the village and its caste composition either a complete listing was done in the village or listing was done in selected segments only, which was randomly selected (for more details on sampling, see Hallad et. al. 2015).

Analysis Approach

As mentioned earlier, a total of 5154 women between the age group 15-34 years were included for the study. Among them 2337 women had experienced atleast one pregnancy outcome in the 3 years period preceding the survey. Hence, the main focus of analysis on extent of expenditure on MCH services is focused on these women only. Univariate and bivariate analyses were conducted to calculate total expenditure associated with utilization of maternal health care services. Here an attempt has been made to understand the expenditure pattern by type of facility by broader categories like public vs. private facilities as well as by type of public health facilities like Sub Health Centre (SHC), Primary Health Centre (PHC), Community Health Centre (CHC), Sub Divisional Hospital/Taluka Hospital (TH) and District Hospital (DH). To make sample representative of the population, proper weights have been used during analysis; hence, the results presented in the study are weighted percentages. Cost amounts are presented here in Indian Rupees (US Dollars – Ru-

1. Funded by the Bill & Melinda Gates Foundation

2. Conventionally, women aged 15-49 years are defined as women of reproductive age in health surveys in India. However, in this survey we decided to include only women age 15-34 years so as to cover maximum births.

pees average exchange rate at mid survey period (May 2015) price was 1 USD = 63.5INR).

Defining Variables

The main objective of the paper is to assess the Household expenditure on each of the maternal health care services received by the women. Information on expenditure on ANC services, delivery services and PNC services were collected from women for each of the health facilities visited by them. Women were requested to report the expenditure on different categories like consultation, diagnostics, medicine charges and tips to the health staff. Expenditure on food, room rent (while stayed in the facility) and expenses towards transportation to reach to the facility and while returning back home at the time of delivery were also collected. If at all, women were unable to provide the cost by category, atleast the total cost was collected. Sum of expenditure towards ANC, delivery and PNC is considered here as total cost for getting maternal services. Predictor variables considered here are caste/tribe, wealth index, experience of complications (during pregnancy, delivery and postpartum period), type of facility and type of delivery.

Expenditure Towards Ante Natal Care (ANC) Services

As mentioned earlier, expenditure towards ANC services was estimated by asking the women to provide the detailed expenditure made by them towards consultation, diagnostics and medicine charges for each of the visits and facilities attended by them for getting ANC check-ups separately. If the women were not able to provide the exact expenditure according to the different categories mentioned above, they were probed to provide at least the total cost of ANC check-up. Expenditure towards ANC services are analyzed in two ways. Firstly, in terms of proportion of women received completely free ANC services at public facilities and secondly, by calculating the average amount spent for ANC services. As presented in Table 2.2, among 2337 women experienced pregnancy outcome during previous 3 years, only eight percent mentioned that all the ANC services were free of cost.

On an average, women had to spend Rs. 3421 (95 percent CI - 3199 and 3642) for getting ANC check-up during their entire period of pregnancy.

Table 2.2: Amount spent for ANC services (in INR) by Type of facility and Caste/Tribe

Type of facility	Number of women	Prop. Received free ANC services	Mean cost	95 percent conf. interval	
ALL					
SHC	954	91.2	21	6	35
PHC	981	74.7	115	49	180
CHC	254	60.2	646	-113	1405
TH	321	49.4	334	242	427
DH	147	42.0	563	375	751
PVT	1844	NA	3918	3666	4171
Combined	2337	7.6	3421	3199	3642
SC/ST					
SHC	495	89.8	7	2	12
PHC	543	73.4	78	52	105
CHC	147	57.4	191	104	278

However, type of the facility visited by the women for getting ANC services makes significant influence on the expenditure. The average cost at SHC was estimated to be only Rs. 21 which is increased to Rs. 115 at PHCs, to Rs. 646 at CHCs, to Rs. 334 at TH and to Rs. 563 at DH. On the other hand, women had to spend Rs.3918 if they visit private hospitals for getting ANC check-up. Even at government facilities not all women who received the ANC services have availed them free of cost, around 10 percent of the women who received ANC service at SHCs had spent some amount. Similarly, one-fourth of pregnant women who received ANC services at PHCs, two-fifths at CHCs and half of the women at TH& DH had spent some amount for their ANC check-ups.

Variation in terms of caste/tribe clearly indicates that 12 percent of the SC/ST women had received all ANC services free of cost compared to five percent of women belonging to Non-SC/ST category. Average expenditure among non-SC/ST women was Rs. 4089 whereas it was Rs. 2784 among SC/ST women for getting ANC services. However, in both the categories, 6-8 fold increase is observed between the expenses at DH and Private sector.

The above discussion clearly indicates that women have to spend significant amount for getting ANC services but it is not much of an economic burden as long as they approach public health facilities. However, as high as 80 percent of the women vis-

it private hospitals at least once for getting ANC services during their pregnancy period (Table 2.3) and 84 percent of the women who had underwent Ultrasound/Scanning had got it done from Private facilities. This clearly reflects the fact that under normal circumstances also most of the pregnant women have to visit private facilities at least once. Further, experience of complications during pregnancy increases the tendency of approaching private health facilities. Half of the women reported one or the other complications during pregnancy and one-third of them had sought treatment from private hospitals for these complications.

Table 2.3: Proportion of women availed ANC checkups, Sonogram services and treatment for pregnancy complications at private facilities.

Number of women experienced pregnancy outcome during previous 3 years	2337
Prop. Visited private facility at least once for ANC checkup (in percent)	78.8
Proportion received Sonogram or Ultrasound at private (among received this service) (in percent)	84.0
Prop. Experienced any pregnancy complications (in percent)	46.6
Prop. Of women sought treatment for pregnancy complications at private facilities (among exp. Pregnancy complications) (in percent)	62.2

The above discussion clearly indicates that women have to spend some amount though less, for getting ANC services even at public health facilities. As claimed in many health schemes, services at public health facilities are not completely cashless. The expenditure increases to some extent while getting basic diagnostic services at private hospitals, but increases drastically if they experience any complications during pregnancy and visit private hospitals for treatment.

EXPENDITURE FOR DELIVERY SERVICES

Further, all the women delivered during 3 years prior to the survey were asked to give the expen-

3. *The standard of living index in this study has been used as a proxy for economic status of the household (Montgomery et al., 2000; Filmer and Pritchett, 2001; Vyas and Kumaranayake, 2006; O'Donnell et al., 2008; Rutstein, 2008; Howe, Hargreaves and Gabrysch, 2009). The standard of living index is based on variables related to household amenities. The variables included are – availability of electricity, mattress, pressure cooker, chair, sofa set, cot, table, fan, radio, black & white television, colour television, sewing machine, mobile phone, telephone, refrigerator, washing machine, bicycle, motorcycle or scooter, cart, car, tractor, water pump, thresher, computer and internet. We use the Principal Component Analysis (PCA) to construct the index. The index is divided into three categories – Poor, Middle and High.*

diture made by them for getting delivery services. Eighty seven percent of the women had normal delivery and for 13 percent of them it was a C-section / Assisted delivery. As far as place of delivery is concerned, as high as 25 percent of the deliveries had occurred at private facilities and 18 percent of the deliveries were home deliveries.

As mentioned earlier, an attempt was made here also to get the information on total expenditure incurred by the women during delivery by asking them to provide the information on amount spent towards consultation, diagnostics, medicine and room rent separately. If at all, women were not able to categorize the expenditure into the above

mentioned categories, they were requested to provide at least the total cost. JSSK scheme provides free food, free transportation services for women to reach the facility and return back. Further, all the women were asked to give whether they had to make any expenditure towards food during their stay at hospital, transportation to reach the Health facility from home and while returning back from facility to home, as well as any amount given to the health staff. All the expenses reported by the women under these categories were added up to calculate the total expenditure made for getting delivery services which are presented below in Table 2.4a and 2.4b by Type of Health Facility and Type of delivery respectively.

Table 2.4a: Expenditure on delivery (in INR) by Type of Facility

Type of Facility	Number of cases	Mean cost	95 percent Conf. Interval		Prop. Received free delivery services
PHC	535	1201	1055	1346	15.5
CHC	184	1539	1258	1820	9.2
TH	349	2344	1865	2824	12.7
DH	202	3712	2965	4459	10.5
PVT	523	17549	16230	18867	NA
Combined	1816	6656	6113	7199	NA
Mean cost is excluding Home deliveries					

Table 2.4b: Expenditure on delivery (in INR) by Type of delivery and Type of Facility

Type of Facility	Number of cases	Mean cost	95 percent Conf. Interval	Prop. Received free delivery services
Normal/Assisted				
ALL	1551	3510	3194	3825
PHC	532	1179	1038	1321
CHC	180	1528	1244	1812
TH	326	1953	1537	2369
DH	164	3207	2539	3875
PVT	329	9999	8954	11045
Caesarean				
ALL	265	24497	22443	26551
TH	23	7860	4297	11422
DH	38	5895	3244	8546
PVT	194	30037	27992	32083

*Mean cost is excluding Home deliveries

On an average, women had to spend Rs. 6656 (95 percent CI- 6113 and 7199) for getting the delivery services. The amount spent is between Rs. 1201 to 3712 at various levels of Public Health facilities. As observed in Table 2.5, an average cost of getting delivery services in all the public health facilities combined is Rs 2057 and it is as high as Rs 17,549 at private hospitals. As expected, significant difference has been observed between Public and Private sector, for both normal and C-section deliveries. The cost of normal deliveries increased by 3 times (Rs 3207 to Rs. 9999) and that of C-section deliveries increased by 5 times (from Rs 5895 to Rs. 30,037) from DHs to Private Hospitals. Among public health facilities, DH deliveries accounted higher expenditure at Rs. 3712 while it was comparatively less at PHCs with Rs. 1201. The drastic difference (nearly 8 times more) between the public and private sector has been observed among all the socio-economic categories. Further, a normal delivery accounts for an expenditure of Rs. 3510 (95 percent CI-3194 and 3825) for rural women; and is increased almost by 8 times to Rs. 24,497 (95 percent CI-22443 and

26551) if it happens to be a C-section.

Above discussion clearly indicates that as such economic burden of delivery on rural women is minimum, as long as it is a normal delivery and occurs at a public health facility. But it is not completely cashless as mentioned by NHM and other Health schemes including JSSK. As we can notice in Table 2.4a, only for around 9 percent to 16 percent of the total delivered women, all the delivery services were completely free of cost. This clearly indicates that, even at public health facilities, quite a large proportion of women have to spend some amount towards consultation, diagnostics, medicine and room rent though the amount is meager.

However, if women experience some complications and if they had to undergo C- section delivery, most likely they go to private facilities and it drastically increases the expenditure towards getting delivery services. As shown in Table 2.6, one-fourth of the women had experienced some pregnancy complications and 40 percent of them had approached private hospitals for seeking treatment

Table 2.5: Amount spent on delivery services by the women delivered during previous 3 years (in INR) by selected background characteristics and Type of facility

	Combined*		Public		Private	
	Number of cases	Mean cost	Number of cases	Mean cost	Number of cases	Mean cost
ALL	1816	6656	1293	2057	523	17549
Caste/tribe						
SC/ST	928	5377	722	1942	206	17126
Non SC/ST	888	8002	571	2206	317	17820
Wealth Index						
Low	494	4313	402	1760	92	15272
Middle	613	6450	457	1972	156	18848
High	702	8558	432	2451	270	17695
Exp. of delivery comp.						
Experienced Del. comp.	517	10485	322	3303	195	21615
Not Experienced comp.	1299	5094	971	1641	328	15027
* Mean cost is excluding Home deliveries						

for these complications. Very high proportion of C-section deliveries at Private hospitals (38 percent compared to 7 percent at TH and 18 percent at DH) clearly supports the fact again that women with some complications usually approach Private hospitals and higher proportion of C-section deliveries at private health facilities leads to higher average amount.

Table 2.6: Proportion of women experienced delivery complications, approached private facilities and prop. Of C-section deliveries at private facilities

Number of women experienced Live birth/ Still birth during last 3 years	2181
Prop. Of women experienced any delivery complications	25.8
Prop. Of women approached private facility (among experienced delivery comp.)	39.9
Proportion of Caesarean delivery at Taluka Hospital	6.8
Proportion of Caesarean delivery at District Hospital	18.2
Proportion of Caesarean delivery at Private Facilities	37.9

EXPENDITURE FOR POST-NATAL CARE (PNC) SERVICES

Post-Natal Care (PNC) services are equally important as that of ANC services to reduce maternal and neo-natal mortality. Expenditure made by the women towards PNC services was also collected as that of ANC services and delivery. Here also expenditure made by the women was categorized into consultation, diagnostics and medicine charges. If at all the women were not able to categorize the cost into these categories, an attempt was made to find out atleast the total cost towards PNC services. Further, information on any money given to the staff of health facilities while getting PNC services was also collected and added to the total cost of PNC.

As presented in Table 2.7, among those women

experienced pregnancy outcome during previous 3 years, 44 percent mentioned that their entire PNC services were free of cost and the women had spent on an average Rs. 550 (95 percent CI - 461 and 639). The expenditure towards PNC services did not show much variation at different public health facilities as it varied between Rs. 50-100 at SHC, PHC, CHC and TH hospitals and Rs. 410 at DH. However, if women approach private facilities, they had to spend as high as Rs. 1419 to get the PNC services. For around three-fourth of the women, all the PNC services were completely free of cost at public sector. As expected, average cost of PNC services was slightly more among non-SC/ST women than that of SC/ST women. But, difference between the two categories was not wide as observed in the cost of ANC and delivery services.

Thus, we can say that expenditure towards PNC services increases mainly when the women approach private hospitals. As shown in Table 2.8 around 80 percent of the women had received PNC care and only 30 percent of them had received this care at private hospitals. On the other hand, nearly half of the women with post-partum complications and three-fourths of the women with complications to their newborns had sought treatment from private hospitals. Hence, as long as everything is normal after the delivery, women rarely go to private hospitals for PNC check-ups. But if any complications arise either for themselves or for their newborns, they are more likely to visit private sector and it leads to some expenditure.

TOTALEXPENDITURE FOR MCH SERVICES (ANC, DELIVERY AND PNC COMBINED)

After discussing the expenditure made by the women specifically towards ANC services, delivery and PNC services, total amount spent by the women for getting all these services combined has been analyzed. Here amount spent towards ANC, delivery and PNC have been added and the amount includes consultation, diagnostics, medicine charges and room rent and expenditure towards food, transportation and any mon-

Table 2.7: Amount spent for PNC services (in INRs) by Caste and Type of facility

Type of Facility	Number of cases	Prop. Received free services	Mean cost	95 percent Conf. Interval	
ALL					
SHC	21	63.6	50	-26	126
PHC	475	80.3	100	30	170
CHC	164	78.9	112	26	199
TH	321	78.4	82	18	147
DH	168	65.3	410	163	658
Private	641	NA	1419	1179	1659
Combined	2337	43.8	550	461	639
SC/ST					
SHC	9	55.6	13	-12	37
PHC	265	83.1	54	16	92
CHC	100	76.8	149	10	288
TH	196	79.2	84	9	160
DH	85	76.9	373	40	706
Private	276	NA	1368	1023	1714
Combined	1205	45.6	465	356	575
Non SC/ST					
SHC	12	69.2	80	-57	217
PHC	210	78.1	159	8	310
CHC	64	81.8	59	-4	121
TH	125	78.1	79	-35	194
DH	83	57.8	455	83	827
Private	365	NA	1458	1127	1789
Combined	1132	42.8	644	502	786

Table 2.8: Proportion of women received PNC care services and treatment for complications at private facility

Number of women	2181
Prop. Received PNC care	79.6
Prop. Received PNC care at private facility (among received PNC care) (in percent)	30.1
Prop. Of women experienced any PP complications	5.7

Prop. Sought treatment at private facilities (among experienced) (in percent)	43.5
Prop. Of newborns experienced any complications	11.7
Prop. Sought treatment at private facilities (among experienced) (in percent)	73.3

Table 2.9: Amount spent (in INRs) for getting maternal care services (ANC+Delivery+PNC) by caste/Tribe and Wealth index

Particulars	Number of cases	Mean cost	95 percent Conf. Interval	
All	2337	9831	9056	10607
Caste/Tribe				
SC/ST	1205	7415	6511	8319
Non SC/ST	1132	12447	11196	13698
Wealth index				
Low	688	6546	5460	7631
Middle	776	9886	8672	11100
High	864	12440	10882	13997

ey paid to staff.

The average cost of MCH services was found to be Rs. 9831 (95 percent CI – 9056 and 10607) (Table 2.9). Amount spent for getting maternal services is Rs. 12,447 (95 percent CI-11196 and 13698) among non-SC/ST women and Rs. 7415 (95 percent CI - 6511 and 8319) among SC/ST women. Higher the wealth index, higher is the expenditure for getting maternal services.

CONCLUSION AND RECOMMENDATIONS

Among all the three maternal services considered here, expenditure towards delivery takes the major share of expenditure (around two-thirds of the total expenditure) and type of delivery and place of delivery are the major contributing factors which decide the cost of delivery. During pregnancy and PP period, if women experience any complications either to themselves or to their newborn, usually they seek treatment from private facilities which leads to higher expenditure for getting ANC and PNC services. Approaching private facilities for getting diagnostics related services, mainly the scanning, leads to still

higher expenditure.

Under normal circumstances during pregnancy and post-partum period and if it is a normal delivery, as such will not lead to higher expenditure among rural women in North Karnataka. Further, still the services like consultation, diagnostics, medicine and room rent are not completely free even at public health facilities. Still women have to spend some nominal amount towards food during their stay and for transportation during delivery.

To avoid expenditure for getting MCH services, specialized services are to be made available in all the public health facilities which may prevent the women from approaching private health facilities. Availability of specialized doctors atleast at SDH and DH level give an assurance for the women to reach these health facilities under emergency and complications. National schemes like JSSK are to be made available strictly and awareness has to be created among rural women.

Thus, still maternal services are not cashless at public health facilities and all the components of JSSK are not fully implemented in rural North Karnataka. Still many rural women are compelled to go to private hospitals during pregnancy, delivery or after the delivery. These factors lead to higher economic burden on poorest and poor households and socio-economically deprived women while getting maternal and child health services in rural North Karnataka.

References

Bhatia J., Cleland J., 1995,” Determinants of maternal care in a region of South India”. *Health Transition Review*, 5: 127–42.

Bloom S. S., Wypij D., Das Gupta M., 2001, “Dimensions of women’s autonomy and the influences on maternal health care utilization in a North Indian City”. *Demography*, 38: 67–78.

Bonu S., Bhushan I., Rani M., Anderson I., 2009, “Incidence and correlates of ‘catastrophic’ maternal health care expenditure in India”. *Health Policy and Planning*, 24: 445–56.

Costello A., Osrin D., Manandhar D., 2004, “Reducing maternal and neonatal mortality in the poorest communities”. *BMJ*, 329: 1166–68.

Gage A. J., 2007, “Barriers to the utilization of maternal health care in rural Mali”. *Social Science and Medicine*, 65: 666–82.

Gwatkin D. R., 2004, “Beyond the averages, countdown 2015”. New York: Family Care International, Population Action International, and International Planned Parenthood Federation, pp. 35–37.

Haddad S., Nougara A., Fournier P., 2006, “Learning from health system reforms: lessons from Burkina Faso”. *Tropical Medicine and International Health*, 11: 1889–97.

Hallad Jyoti, R. V Deshapande, B. I Pundappanavar, J. A Golandaj, 2015, “Baseline and Endline Household Survey for Project Sukshema; Technical assistance to improve maternal, neonatal and child health (MNCH) outcome through the National Health Mission in Karnataka, India”. PRC Report No.164.

Hogan, M. C., Foreman, K. J., Naghavi, M., et al., 2010, “Maternal mortality for 181 countries, 1980–2008: A systematic analysis of progress towards Millennium Development Goal 5”. *The Lancet*, 375: 1609–1623, doi: 10.1016/S0140-6736(10)60518-1.

Hotchkiss D. R., Krasovec K., El-Idrissi M. D. et al., 2005, “The role of user charges and structural attributes of quality on the use of maternal health services in Morocco”. *International Journal of Health Planning and Management*, 20: 113–35.

International Institute for Population Sciences (IIPS), 2014, *District Level Household and Facility Survey (DLHS-4)*, 2012–13: India. Karnataka: Mumbai: IIPS.

- International Institute for Population Sciences (IIPS), 2010. *District Level Household and Facility Survey (DLHS-3)*, 2007–08: India. Mumbai: IIPS.
- International Institute for Population Sciences and Macro International, 2007, *National Family Health Survey (NFHS-3)*, 2005-06. India. Mumbai: International Institute for Population Sciences and Macro International; p. 210.
- Kesterson J. A., Cleland J., Sloggett A., Ronsmans C., 2010, “Institutional delivery in rural India: the relative importance of accessibility and economic status” *BMC Pregnancy and Childbirth*, 10: 30.
- Koenig M., Jamil P., Streatfield T. S. et al., 2007, “Maternal health and care seeking behaviour in Bangladesh: findings from a national survey”. *International Family Planning Perspectives*, 33: 75–82.
- Leone T, James K, Padmadas S. S., 2013, “The burden of maternal health care expenditure in India: Multilevel analysis of national data”. *Maternal Child Health Journal*;17:1622-30.
- Leone, Tiziana, et al., “Does maternal health care expenditure in Indian households vary by source of care?” <http://www.esrc.ac.uk/my-esrc/grants/RES-238-25-0009/outputs/Read/302817ab-d149-4713-82012f8f3869f536> [Accessed on 22/08/2014].
- Leslie J., Gupta G. R., 1989, “Utilization of formal services for maternal nutrition and health care”. *Washington DC: International Center for Research on Women*.
- McIntyre D., Thiede M., Dahlgren G., Whitehead M., 2005, “What are the economic consequences for households of illness and of paying for health care in low and middle-income country contexts?” *Social Science and Medicine*, 62: 858–65.
- Modugu HR, Kumar M, Kumar A, Millett C., 2012, “State and socio-demographic group variation in out-of-pocket expenditure, borrowings and Janani Suraksha Yojana (JSY) programme use for birth deliveries in India”. *BMC Public Health*;12:1048.
- Mohanty S. K., 2011, “Multidimensional poverty and child survival in India”. *PLoS ONE*, 6: e26857.
- Mohanty S. K., Pathak P. K., 2009, “Rich–poor gap in utilization of reproductive and child health care services in India”, 1992–2005. *Journal of Biosocial Sciences*, 41: 381–98.
- Mohanty SK, Srivastava A., 2013, “Out-of-pocket expenditure on institutional delivery in India”. *Health Policy Plan*;28:247-62.
- MoHFW, 2011, “Janani Shishu Suraksha Karyakram (JSSK), National Health Mission”, Ministry of Health and Family Welfare, Government of India, New Delhi. <http://nrhm.gov.in/janani-shishu-suraksha-karyakram.html>. [Accessed on 30/04/2016].
- MoHFW, 2009. “National Health Accounts India 2004–05”. New Delhi: Ministry of Health and Family Welfare, Government of India: http://mohfw.nic.in/NHA_percent202004-05_percent20Final_percent20Report.pdf [Accessed on 22/08/2014].
- Nahar S., Costello A., 1998, “The hidden costs of ‘free’ maternity care in Dhaka, Bangladesh”. *Health Policy and Planning*, 13: 417–22.
- Navaneetham K., Dharamalingam A., 2002, “Utilisation of maternal health care, services in Southern India”. *Social Science and Medicine*, 55: 1849–69.
- Office of the Registrar General of India (ORGI), 2006, “Maternal Mortality in India 1997–2003: Trends, causes and Risk Factors”. New Delhi: Office of the Registrar General, India.
- Parkhurst J. O., Rahman S. A., Ssengooba F., 2006, “Overcoming access barriers for facility-based delivery in low income settings: insights from Bangladesh and Uganda”. *Journal of Health Population and Nutrition*, 24: 438–45.

- Paul VK, Sachdev HS, Mavalankar D, Ramachandran P, Sankar MJ, Bhandari N, et al., 2011, "Reproductive health, and child health and nutrition in India: Meeting the challenge". *Lancet*, 377:332-49.
- Peters D., Yazbeck A., Sharma and R. et al., 2001, "Better Health Systems for India's Poor". *Washington, DC: The World Bank*.
- Rahman KMM, and Sarkar P., 2009, "Level of differentials of maternal healthcare utilisation in Bangladesh". *Research Journal of Medical Sciences*, 3(4): 163-9.
- Ramarao S., Caleb L., Khan M. E., Townsend J. W., 2001, "Safer maternal health in rural Uttar Pradesh: do primary health services contribute?" *Health Policy and Planning*, 16: 256-63.
- Singh L., Singh C. H., 2007, "Rich-poor gap in maternal care: The case of northeast India". *Asian Population Studies*, 3: 79-94.
- Skordis-Worrall J., Pace N., Bapat U. et al., 2011, "Maternal and neonatal health expenditure in Mumbai slums (India): a cross sectional study". *BMC Public Health* 11: 150.
- Stanton D., Clemens J., 1989, "User fees for health care in developing countries: a case study in Bangladesh". *Social Science and Medicine*, 29: 1199-205.
- Starrs, A., 2001, "Skilled attendance at delivery: A review of the evidence". *The Interagency Group for Safe Motherhood, Family Care International*, New York.
- Stephenson R., Tsui A. O., 2002, "Contextual influences on reproductive health services use in Uttar Pradesh, India". *Studies in Family Planning*, 33: 309-20.
- Thaddeus S., Maine D., 1994, "Too far to walk: maternal mortality in context". *Social Science and Medicine*, 38: 1091-100.
- UNFPA, 2007, "Skilled attendance at birth". UNFPA, Geneva. <http://www.unfpa.org/public/mothers/pid/4383> [Accessed on 22/08/2014].
- United Nations, 2007, *UN Millennium Development Goals Report*. New York: United Nations, pp. 16-17.
- World Health Organization (WHO), 2005, "The WHO report: Make every mother and child count". *World Health Organization, Geneva*.
- World Health Organization (WHO), 2012a, "Trends in Maternal Mortality" 1990-2010, WHO, UNICEF, UNFPA and the World Bank Estimates. *World Health Organization, Geneva*.
- World Health Organization (WHO), 2012b, *World Health Statistics 2012 report*. *World Health Organization, Geneva*.

CHAPTER - 3

Role of Living Conditions and Hygienic Practices on Childhood Diarrhoea in Rural Areas of Koppal District, Karnataka

Dr. Rajarama K. E. T¹, Mr. C. N. Noolvi²

¹ Assistant Directors, ² Data Assistant Population Research Centre, Dharwad

Abstract

An attempt has been made in this paper to assess the mothers' knowledge on diarrhoea, management of diarrhoea, prevalence of diarrhoea by living condition and hygiene practices. Although many macro and micro-level studies in India have been conducted on childhood diarrhoea, management of diarrhoea, maternal knowledge on diarrhoea and practice etc. there is a paucity of studies to assess the level of environmental sanitation, extent of household hygienic practices and childhood diarrhoea. The study was taken up in 10 villages of Koppal district of Karnataka. The villages were selected using probability proportional to population size (PPS) sampling method. Considering the budget constraint, it was decided to select 300 mothers from the selected villages. The mothers who delivered in 2012 were selected using systematic random sample method. In all, 282 mothers were contacted and interviewed of the 301 selected. Semi-structured schedule was used to collect information from the respondents. Univariate and bivariate techniques of data analysis were used. The wealth index was constructed using the household assets and economic housing characteristics.

Findings of the study reveal that majority of mothers are aware of symptoms of childhood diarrhoea whereas knowledge on danger signs of the illness is known to less number of respondents. The child care during diarrhoea is known better than the care after the illness. This knowledge is found to be low among the socio-economically backward mothers. The prevalence of diarrhoea among young children is 12 percent. It is more common among the children belonging to Muslim religion, OBC, medium wealth index, household residing at two or less than two room house, drinking public tap water or bore well water, following poor hygienic practices. There is need for a great deal of education regarding importance of hygienic practice, household and environmental cleanliness to mothers, household members and community.

INTRODUCTION

Children below 5 years of age are the most vulnerable population to infectious diseases in general and diarrhoea sickness in particular although it is common among all the age groups. Though the disease looks simple, the consequences of the illness are more severe among the growing children as it may lead to death if we fail to manage it properly. Besides, acute and persistent diarrhoea leads to retardation in growth, inadequate weight gain or loss of weight etc. (Agarwal, 2014). Besides, malnutrition is the most common among the children who suffer from diarrhoea due to inadequate care during and after the illness. Diarrhoea is a disease caused by poor personal, household and environmental hygiene. Eighty percent of diarrhoea is attributable to these causes. (World Bank, 2008). Death due to diarrhoea is largely preventable by following simple hygienic practices such as washing hands with soap before serving and consuming food, washing hands with soaps after using toilet, using toilets for defecation and using safe drinking water and proper management during illness (UNDP, 2003).

Diarrhoea is one of the major causes of mortality and constitutes 24 percent of deaths in India among 1-4 years children (UNICEF, 2006). Laxinarayana recently estimated that deaths caused by diarrhoea constitute approximately 13 percent of all children's deaths (Ranjan, 2015). India is one among the 10 countries of the world which has high mortality rate due to diarrhoea (Liu et.al., 2012).

According to one estimation, annual burden of water borne disease in India is 38 million and 1.5 million children die due to diarrhoea alone. Added to this is the loss of 73 million working days due to water borne diseases. In terms of economic burden, it is \$ 600 million in a year (Khuran, 2008). Many studies have found that better sanitation, safe drinking water and hygiene interventions have positive impact on reduction in diarrhoea illness which is responsible for one-fifth of mortal-

ity among children below 5 years in developing countries (Kosek et. al. 2003).

The estimated number of diarrhoea cases in India between October 2013 and November 2014 was 199 million whereas HMIS reported only 4.8 million such cases during the same period. The large gap between estimation and HMIS data set might be due to under reporting of diarrhoea cases in HMIS (Ranjan, 2015). According to NFHS-4, the prevalence of diarrhoea among children below 5 years is 4.5 percent and 4.3 percent for Karnataka state and rural Karnataka respectively. Little more than half of the infected children (53 percent) received ORS in the state and corresponding proportion for rural children was 59 percent. Almost the same proportions of children (around 70 percent) were taken to facility for treatment both in rural Karnataka and in Karnataka state as a whole.

With respect to prevalence of childhood diarrhoea in the study district (Koppal), it is lower in rural areas compared to urban areas (3.9 percent against 4.3 percent) according to NFHS-4. Although many macro and micro-level studies in India have been conducted on childhood diarrhoea, management of diarrhoea, maternal knowledge on diarrhoea and practice etc. there is a paucity of studies to assess the level of environmental sanitation, extent of household hygienic practices and childhood diarrhoea.

The present study was planned to estimate differential prevalence of diarrhoea by environmental sanitation and household hygienic practice in rural settings.

Objectives:

1. To assess the level of knowledge of mothers on childhood diarrhoea and its management;
2. To know the extent of diarrhoea among children and its governing factors.

Methodology

Study Design And Sample Size

The present study was taken up in Koppal district of Karnataka. Ten villages were selected using probability proportional to population size (PPS) sampling method from all the four taluks. It was decided to interview 300 mothers from the selected villages. In the selected villages, a list of usual resident mothers who delivered in January-December 2012 was obtained from the ASHAs and it was used for selecting the mothers for interview.

Respondents were selected using the systematic random sampling method. In all, 282 mothers were contacted and interviewed of the total selected 301 women. The field work was carried out from October 26th to 6th November 2015. Semi-structured schedule was used to collect information from the mothers. The main focus of the study was to assess the mothers' knowledge and practice on child care, but information on childhood morbidity, including diarrhoea and hygienic practice were collected.

Information on diarrhoea was collected on the index children who suffered from diarrhoea illness during 30 days prior to the survey. Data was entered using Statistical Package for Social Sciences (SPSS) version 19. Univariate and bivariate techniques of data analysis were used. The wealth index was constructed using the household assets and economic housing characteristics. For each household possessions and household economic characteristics, score 0 to 4 were assigned depending on the type of characteristics. Maximum score a household can obtain is 40 and the minimum score a household can get is 0. The score obtained by the each household was summated and all the households were arranged in ascending order. After arranging the households in this manner, it was made into equal three parts according to the score obtained.

The households which fall under the first part scored the least and named as 'low', the households which come under the second part scored medium and named as 'middle' and the households which fall under third part scored the highest and named as 'high'. Minimum and maximum points secured by the households were 3 and 34 respectively.

Limitation Of The Study

The number of episodes of diarrhoea was based on lay report of mothers for the last 30 days prior to the survey and hence, there could be recall bias. As diarrhoea is a seasonal disease and survey conducted during October-November 2015 this does not reflect annual load of childhood diarrhoea.

Results

Mothers' Knowledge On Diarrhoea

"Diarrhoea is passing of watery stools more than three times a day. It causes loss of body water and electrolytes, dehydration sets in soon in children if fluids and electrolytes lost are not replaced timely and adequately. Increased thirst, skin pinch goes back slowly, dry mucus membrane, sunken eyes, lack of tears, decrease in urine output, frequent episodes of loose motion within one hour or mucus or blood in stool etc. are the dangerous signs of diarrhoea" (Ghosh, 1997). Table 3.1 shows percent distribution of mothers by level of knowledge on symptoms of childhood diarrhoea and danger signs of diarrhoea according to socioeconomic characteristics. Majority of mothers are aware of symptoms of diarrhoea (65 percent). Regarding the awareness on danger signs of childhood diarrhoea, 67 per cent of mothers are aware of 'restlessness' as the danger sign of diarrhoea. The other danger signs such as 'skin pinch goes back slowly' (20 percent) and 'frequent loose motion within an hour or blood in stools' (34 percent) are less known to the study subjects. Difference in the level of knowledge with respect to socioeconomic

groups is significant. Awareness on danger signs of the illness is low among the socially and economically unprivileged groups.

Table 3.1: Percent distribution of mothers by knowledge of symptoms & danger signs of diarrhoea, according to socioeconomic characteristics

Characteristics	Watery stools more than 3 times a day	Restlessness	Skin pinch goes back very slowly	Frequent loose motions in an hour or blood in stools	Number of mothers
Education					
Illiterate	52.68	52.68	14.29	28.57	112
Primary	68.89	68.89	16.67	40.00	90
Secondary & above	78.75	83.75	36.25	36.25	80
Social group					
SC & S T	53.49	52.33	11.63	26.74	86
OBC	70.99	73.28	23.66	39.69	131
General	69.23	72.31	29.23	33.85	65
Wealth index					
Low	59.57	59.57	10.64	25.53	94
Medium	67.02	70.21	21.28	36.17	94
High	69.15	70.21	31.91	41.49	94
Total	65.53	67.03	21.73	34.26	282

Source: Survey

Mothers' Knowledge On Care During Diarrhoea

Childhood diarrhoea can be managed at home by giving the child Oral Rehydration Solution (ORS) and home available fluids such as salted butter-milk, ganjee, dal water, weak tea with sugar etc. Breast feeding should be continued if the child is on breast milk. When the child has recovered from the illness, the child may be given a little more quantity of food than normal quantity for 2 weeks to regain the weight lost. If mother knows these cares, worsening condition of the disease can be prevented and further deterioration in the

child's health can be averted by taking appropriate measures. Table 3.2 shows percent distribution of mothers by knowledge on care of child during diarrhoea, according to socioeconomic characteristics. Majority of mothers (60 percent) have knowledge on feeding of ORS after every episodes of loose motion. Only half of the total mothers are aware of giving more fluids along with normal food. Need for continued breast feeding of sick child is known to 53 per cent respondents. Feeding of extra food at least for 15 days is known by a lesser proportion of mothers (5 percent). Disparity in knowledge among education groups is glaring. Illiterate mothers have the lowest knowledge of all other educational groups. Knowledge gap is moderate among social and economic groups. Further it is observed that the SC & ST and the poor mothers have the least knowledge in four of the five care indicators.

Socio - Economic Differential In Prevalence Of Childhood Diarrhoea

Information on childhood diarrhoea and hygienic practice was collected from mothers whose reference child is surviving (273 mothers). Nine reference children died at various point of time after birth. Table 3.3 shows that around 12 percent of children below 5 years of age have suffered from diarrhoea during one month prior to the survey. This rate is around three times higher than that of NFHS-4 rates. This might be due to season in which the NFHS - 4 survey was conducted and partly due to under-reporting. Childhood diarrhoea was more common among children belonging to Muslim religion than their counterparts among Hindus.

Nearly 21 percent of children suffered from diarrhoea in the last one month before the survey, whereas only 11 percent of Hindu children were reported to have such problem in the same reference period. With respect to the illness among the different social groups, it was the highest among children who belonged to OBC category (15 per-

Table 3.2: Percent distribution of mothers by knowledge of care & feeding of child during diarrhoea, according to socioeconomic characteristics

Character-istics	Continue breast feed-ing	More flu-ids along with usual food to be given	ORS to be given after every episode of diarrhoea	Extra food to be given after recovery of illness	Extra food to be given at least for 15 days after recovery of illness	Num of mothers
Education						
Illiterate	40.18	33.04	46.43	21.43	2.68	112
Primary	53.33	53.33	62.22	25.56	2.22	90
Secondary & above	68.75	68.75	76.25	37.50	12.50	80
Social group						
SC & S T	45.35	39.53	43.02	24.42	4.65	86
OBC	57.25	54.96	68.70	30.53	6.87	131
General	52.31	52.31	64.62	24.62	3.08	65
Wealth index						
Low	51.06	43.62	50.00	22.34	5.32	94
Medium	57.45	48.94	63.83	26.60	3.19	94
High	50.00	56.38	65.96	32.98	7.45	94
Total	52.85	50.10	60.11	27.33	5.33	282

Source: Survey

cent) and the lowest among children of SC & ST category. Further, it is observed from the table that prevalence of diarrhoea was low among children of high wealth group and vice versa.

Table 3.3: Socio-economic Differential of Childhood diarrhoea

Socioeconomic Characteristics	Percent	N
Religion		
Hindu	10.65	244
Muslim	20.69	29
Social group		
SC & ST	8.43	83
OBC	14.84	128
General	9.68	62
Wealth Index		
Low	12.22	90
Medium	14.13	92
High	8.97	91
Total	11.72	273

Source: Survey

Diarrhoea And Living Condition

There exists a link between housing characteristics such as type of house, number of rooms in house, number of persons per room etc. and infectious diseases. A study conducted in slum area found that Kachcha and Semi-Kachcha houses are the major causes for communicable illness among children due to its unhygienic type of structure.

Prevalence of diarrhoea among children below five years of age according to housing characteristics is given in Table 3.4. Results show that prevalence of diarrhoea does not vary much according to type of house. The prevalence was close to 12 percent among children who are residing in the pucca and semi-pucca structures and slightly less among children who were staying in the kachcha structures (10 percent). Further, it is noticed that there exist a large amount of difference regarding number of rooms in the houses and diarrhoea prevalence. The prevalence rate is almost 15 percent

among children who reside in the houses which possess ≤ 2 rooms whereas, it is 8 percent among children who are staying in the houses which are having 3 or more than 3 rooms.

Table 3.4: Housing Characteristics by prevalence of childhood diarrhoea

Characteristics	Percent	N
Type of house		
Pucca	12.22	90
Semi-pucca	11.56	173
Kachcha	10.00	10
No. rooms		
2 or less	14.79	142
3+	8.40	131

Source: survey

Source And Treatment Of Water And Childhood Diarrhoea

Studies conducted in various countries reveal that both morbidity and mortality due to diarrhoea could be reduced by 22 to 30 percent by providing improved source of water supply and safe disposal of human solid waste. (Nayar, 1997).

Table 3.5: Source and treatment of drinking water by prevalence of childhood diarrhoea

Source & Treatment of Water	(percent)	N
Source of drinking water		
Own tap	10.10	99
Public tap/bore well	15.38	91
Open well	13.79	29
Panchayat filtering unit	7.41	54
Treatment of water		
Yes	9.31	204
No	18.84	69
Type of treatment*		
Electronic	7.46	67
By using cloth	9.57	94
Other	9.30	43
* percent computed for 204 cases		

Source: Survey

Safe drinking water has been defined as water free from microorganisms, harmful chemical substances, colour and odour and usable for domestic purposes” (UNDP, 2003). Diarrhoea, cholera, typhoid, hepatitis, polio and gastroenteritis etc. are the result of water contamination from these things. Table 3.5 presents source and type of drinking water and prevalence of diarrhoea among children below 5 years of age. The table reveals that prevalence of diarrhoea is less common in the households having own tap connection (10.1 percent) and getting water from Panchayat filtering unit (7.4 percent). The prevalence rate is 14 to 15 percent in households using bore well or open well water. The chances of being contaminated are usually high among these sources. Further, it is noticed that diarrhoea among children is more than two times higher in the households where water treatment is not being done before consumption when compared to the households which are drinking treated water. Degree of purification varies according to the method of purification. Diarrhoea was less common in the households using electronic device for water purification (7.5 percent) compared to other methods of purification (9.5 percent).

Hygienic Practice And Childhood Diarrhoea

Table 3.6 shows prevalence of diarrhoea by practice of personal and household hygiene. It is revealed that childhood diarrhoea was found highest where the mothers not practiced hygiene such as ‘hand washing with soap after cleaning child’s bottom (13 percent against 11 percent), preserved the eatables safe from insects and house flies (25 percent against 12 percent) and household members using toilet for defecation (13 percent against 10 percent). The prevalence of the disease was higher (13 percent) where the mother washed hands before serving food to the child compared to mother who did not follow such practice.

Table 3.6: Hygienic Practice by Prevalence of Childhood Diarrhoea

Hygienic practice	Percent	N
Use of toilet		
Yes	10.08	129
No	13.19	144
Safe keeping of food		
Yes	11.52	269
No	25.00	4
Hand washing with soap after cleaning the child's bottom		
Yes	11.31	221
No	13.46	52
Hand washing with soap before serving food to child		
Yes	12.85	179
No	9.57	94

Source: Survey

Immunization Coverage And Prevalence Of Childhood Diarrhoea

Immunization coverage plays a vital role in preventing many communicable diseases among children. Immunization coverage is universal among the study population for all the immunizations, including booster doses. Table 3.7 present data on immunization coverage of children by prevalence of childhood diarrhoea. The prevalence of diarrhoea is around 12 percent among the immunized children. It is 4-5 percent high among non-immunized children, but their number is negligible.

Table 3.7: Immunization Coverage by Prevalence of Diarrhoea

Immunization	Percent	N
BCG		
Yes	11.81	271
No	0.0	2
Polio3		
Yes	11.61	267
No	16.67	6
DPT3		
Yes	11.61	267
No	16.67	6

HB 3		
Yes	11.65	266
No	14.26	7
Measles		
Yes	11.99	267
No	0.0	6
Polio Buster		
Yes	11.76	255
No	11.11	18
DPT booster		
Yes	11.72	256
No	11.76	17
Vitamin A		
Yes	13.24	219
No	5.88	54
Disinfestation drug given		
Yes	9.86	142
No	13.74	131

Source: Survey

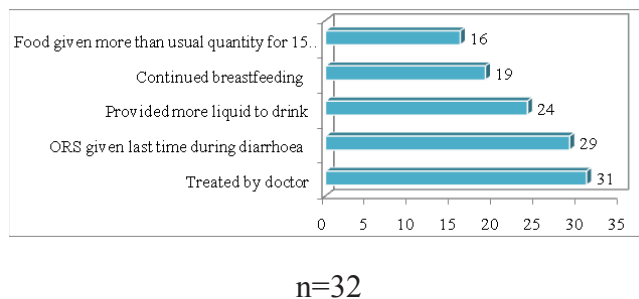
Worm infestations are common among children. It leads to malnutrition and reduces learning ability among them. Therefore, disinfestation drugs to be administered to children once in 6 month to keep them healthy. The government recently started worm disinfestation day in which all children below 14 years of age are being administered disinfestation drugs. The disease is found less among the children who ever received worm disinfestation drug (10 percent). The prevalence rate is five points more among children who never received such medication (15 percent).

Type Of Treatment Received By The Diarrhoeal Children

Figure 3.1 present numbers of children below 5 years of age who had diarrhoea during the last 1 month before commencing the survey, by types of treatment and care received. Overwhelming majority of sick children (31 out of 32 sick children) was treated by doctor and number of children received ORS was 29. Children who received breast milk and received extra food for 15 days after re-

covery of illness are only 19 and 16 respectively. The reason for not giving more fluids, extra food and breast milk are 'child was too tired and was reluctant to drink milk/fluids', 'loose motion would increase' and 'child may get stomach pain'. These reasons indicate that mothers do not have a correct knowledge of care and diet of diarrhoea sick children.

Figure 3.1: Number of children who had diarrhoea last time by treatment & care received



Conclusion And Policy Implimentations

The study portrays a mixed picture on mothers' knowledge as some diarrhoea-related information is well known to mothers while some other facts are least known to them. The danger signs 'Skin pinch goes back slowly' and 'frequent loose motion within an hour or blood in stools' are least known to mothers. If mothers are aware of these danger signs they can take immediate measures. Mothers lack correct knowledge on child care during diarrhoea and care after recovery from the illness. Mothers usually provide less food and less fluid, including breast milk to diarrhoeal child due to the wrong impression that illness may turn worst if they provide more fluid and food. Usual food and more fluids are recommended for preventing the deteriorating condition of diarrhoea. Most of the mothers are not aware of the fact that the child needs extra food at least for 15 days to replenish the weight lost. The ASHA, ANM and Anganwadi workers should make the mothers aware of these issues in their household visits and during Village Health and Nutrition Day. They should focus on socio-economically backward groups as their level of knowledge is low.

The study results revealed that prevalence of diarrhoea was high among children whose household is lagging behind in hygienic practice, such as use of toilet, hand washing practice before serving the food and after using the toilet etc. Type of house does not make much difference in the prevalence rate of diarrhoea whereas number of rooms in the house matters a lot in the disease prevalence. The childhood diarrhoea found more in the households which are drinking untreated water and getting water from open well and bore well. In spite of wide publicity made through the electronic and print media on environmental and personal hygiene, at the ground level its practice seems to be low. We have noticed during the survey that toilets have been constructed at household level and it is being used for dumping crop residuals and fire wood. Therefore, serious effort is required to change the mindset of the people in this direction. Village Health Sanitation and Nutrition Committee along with frontline workers should work in these lines. Water treatment plants should be installed in each village to ensure safe drinking water.

Acknowledgement

We are would like to express our heartfelt thanks to Ministry of Health and Family Welfare, Government of India, New Delhi for their funding and permission to conduct the study.

We would like to record our sincere thanks to all the paper revivers for their constructive comments in improving the quality of the paper.

Reference:

- Agarwal, N, 2014, "Immunization status and childhood morbidities as determinants of PEM among under 5 children in slum of Khanpur", *Indian Journal of Community Health*, Vol. 26, Issue No.4.
- Fewtrell, L. R. B, Kaufman, D, N, W. Enanoria, C. Haller and J. M. Cilford.Jr, 2005, "Water sanitation and hygiene interventions to reduce diarrhoea

- in less developed countries: A Systematic Review and Meta Analysis". *Lancet*, Vol 5.P 42-52
- Ghosh, Shanti. 1997, *Nutrition and Child Care: A practical Guide*. Jaypee Brothers, Medical Publishers (P) Ltd, New Delhi 155 – 168.
- Gubhaju, B. K. and Majumdar, A. K, 1991, "Socioeconomic, demographic and environmental determinants of infant mortality in Nepal", *Journal of Biosocial*, Vol. 23, No.4, pp425-35
- Hughes, G. K. L and Dunleavy. M, 2001, "Environmental Health in India: Priorities in Andhra Pradesh", Washington DC. Environment and Social Development Unit, South Asia Region, The World Bank.
- International Institute for Population Science, 2016, *National Family Health Survey (NFHS-4) 2015-16*, State and District Fact Sheets, Karnataka.
- Jatrana, S, 2001, "Household environmental factors and their effects on infant mortality in Meowat region of Haryana State". *Demography India*, Vol. 30, No. 1, p31-47
- Khurana, Indira and Romit.Sen, 2008, "Drinking Water Quality in Rural India: Issues and Approaches", Background Paper Water Aid. www.wateraid.org
- Kosek, M. C., Bern and Guerrant R. L, 2003, "The magnitude of global burden of diseases from studies published 1992-2000". *Bulletin of the World Health Organization*, Vol. 81, p197-204.
- Liu, L, Johnson, et. al. 2012, "Global, regional and national causes of child mortality: An updated systematic analysis for 2010 with time trends since 2000". *Lancet*, 379, p 2151-2161.
- Nayar. R. K, 1997, "Housing amenities and health improvement: some findings". *Economic and Political Weekly*, Vol. XXXII, No. 2, p 1276
- Prakasam, C. P, 2012, *Influence of Environmental Factors on Child Survival Status: Evidences from the NFHS-3*, in N. Audinarayana, et. al. (eds), *Health, Environment and Sustainable Development: Perspectives and Issues*, Serial Publications, New Delhi.
- Ranjan, R, 2015, "Heavy Burden of Diarrhoea under Five Years of Age in Bihar" in Alok. Ranjan et. al. (eds). *India 2015: Population Human Development*, MLC Foundation, and Shyam Institute, Bhopal, p 111-124.
- UNICEF, 2006, *Water, Sanitation and Hygiene - Annual Report*. New York, UNICEF
- United Nations Development Programmes, 2003, *Population and Environmental Nexus*. UNDP, New Delhi
- WHO/UNICEF, 2013, *Ending Preventable Child Deaths from Pneumonia and Diarrhoea by 2025. The Integrated Global Action Plan for Pneumonia and Diarrhoea (GAPPD)*, Geneva, World Health Organization.
- http://www.unicef.org/media/files/final-GAPPD-main-Report_EN-8_April 2013.pdf

CHAPTER - 4

To Assess The Role Of Family Planning On Fertility In Jharkhand State based On NFHS Data

Dilip Kumar

Joint Director, Population Research Centre, Patna University,

Abstract

In the present study an attempt has been made to assess the role of family planning on fertility in Jharkhand State through the Prevalence Model. If prevalence levels of both programme and non-programme contraception are known, this technique permits the estimation of gross natural and potential fertility for assessing births averted. With the emergence of the National Family Health Survey (NFHS: 2005-06) to monitor family planning and health activities, this method becomes a useful tool. Of special interest is the ability of the procedure to yield estimates by age group as well as by type of contraceptive methods used.

In the study, the standard method-specific use-effectiveness levels weight observed use and prevalence level by method have been applied. Of the total births averted in Jharkhand State by programme contraception, 80 percent of births were averted by Sterilization users in 2005-06 while the spacing methods users contributed to about 20 per cent of the birth prevention. The spacing methods need to be strengthened for the greater use. With regard to the births averted by non-programme contraception/natural methods, the main contribution was made by the users of Rhythm of about 45 percent, which was followed by the users of Withdrawal of 42 percent and by the other methods of 13 percent. Of the total birth averted in Jharkhand State, the contribution of programme contraception and non-programme contraception/natural methods is about 85 percent and 15 percent respectively in 2005-06. The programme contraception has the dominance role to control fertility, however, the non-programme contraception/natural methods use should also be enhanced at the places where accessibility of programme contraception is poor.

Keywords: Births Averted Prevalence Model, Potential Fertility Rate, Natural Fertility Rate, Programme Contraception, Non-Programme Contraception / Natural Method.

INTRODUCTION

With the rapid growth of national family planning programme in 1990 onward, family planning administrators and policy makers felt an increasing need to evaluate the impact of the programme yearly. For such, probably the most widely used measure of family planning impact is the number of births averted by a programme in a given year. An estimation of the number of births averted is typically obtained by subtracting observed fertility (i.e. fertility with programme) from potential fertility (i.e. fertility without the programme) and multiplying the differences by the appropriate base of the population. Since potential fertility is an unobservable quantity, it has to be estimated indirectly. Gross and net potential fertility are two major types of potential fertility used in the study. The gross potential fertility is the fertility that would prevail if all use of programme contraception were eliminated, without switching to the non-programme contraception. The net potential fertility is the fertility that actually would be observed if there had never been a programme. In that case, many who would have been programme users would have obtained supplies from non-programme sources. This substitution would tend to make net potential fertility lower than the gross potential fertility.

The use of contraception greatly reduces the chances of conceiving, but except in the case of sterilization, the chance is not zero. To take this contraceptive failure into account, some methods introduce a penalty for accidental failure. The simplest way to do this consists of multiplying potential fertility by 'e', the contraceptive use-effectiveness. In the present study, Prevalence Model has been used in the State of Jharkhand to evaluate the number of births averted by programme and non-programme contraception efforts. This paper has captured the proximate determinants through the programme and non-programme contraception only to assess the role of family planning on fertility control as the focus of the study.

2. Prevalence Model

The prevalence model is based on age-specific and method-specific prevalence rates of both programme and non-programme contraceptives. This information helps in getting the gross potential fertility and subsequently, the number of births averted by programme and non-programme contraceptive efforts.

Age-specific model: The age-specific model requires information on prevalence of programme and non-programme contraception by age, age-specific fertility rates which have been taken from the NFHS data of Jharkhand State. The Census figures have been used for the female population of reproductive age. Bongaarts, J (1993) provides new estimates of gross and net impact on fertility reductions from family planning (FP) programs for 31 developing countries in Africa, Latin America, and Asia. A comparison is made of net and gross measures, and the interaction with the level of development is identified. The conclusion is reached that FP has been crucial in reducing fertility in many countries. Without FP, the total annual number of births in the late 1980s would have been 164 million instead of 120 million. There is no agreed upon measure for determining the impact of FP on fertility and estimates has ranged from 3-40 percent. Discrepancies in results are due to the use of multiple methodologies, of which some are unsuitable or unreliable for normal evaluation due to difference research objectives and due to conceptual differences in measurement of gross versus net impact. Gross impact refers to the reduction due to the use of contraception available from programme sources. Net impact measures the reduction achieved by the presence of the programme. Net and gross impact varies within each country, with net impact the smaller of the two. Gross impact usually was measured with statistics on acceptors. The measures of gross potential fertility, gross natural fertility and births averted are obtained as follows;

$$NAFa = AFa / \{1 - Ca (ua' + ua'')\} (1)$$

$$PAFa = AFa \{1 - Ca (ua'')\} / \{1 - Ca (ua' + ua'')\} (2)$$

$$BAa = (PAFa - AFa) POPa (3)$$

$$BANa = (NAFa - PAFa) POPa (4)$$

Where;

A = age group of women, a
= 15-19

ua' = prevalence of programme contraception,
by age.

ua'' = prevalence of non-programme
contraception, by age.

AFa = age-specific fertility rate.

PAFa = potential age-specific fertility rate.

NAFa = natural age-specific fertility rate.

BAa = birth averted by programme contraception,
by age.

BANa = birth averted by non-programme
contraception, by age.

POPa = number of women in age group a.

Ca = elasticity coefficient by age.

In order to estimate the gross potential fertility and natural fertility, information on elasticity coefficient of sterility and use-effectiveness by age of women is utilized.

$$C (15 - 19) = 0.620$$

$$C (20 - 24) = 0.620$$

$$C (25 - 29) = 0.823$$

$$C (30 - 34) = 0.940$$

$$C (35 - 39) = 1.022$$

$$C (40 - 44) = 1.309$$

$$C (45 - 49) = 1.898$$

The method-specific model: It derives the number of births averted by each programme and non-programme method through the data on prevalence and use-effectiveness of contraception for both sectors. Estimates of births averted are obtained by the following equations:

$$BA_m = BA \cdot um' \cdot em' / (u' \cdot e') (5)$$

$$BAN_m = BAN \cdot um'' \cdot em'' / (u'' \cdot em'') (6)$$

Where;

um' = prevalence of programme method 'm'.

um'' = prevalence of non-programme method 'm'.

em' = use-effectiveness of programme method 'm'.

em'' = use-effectiveness of non-programme method 'm'.

$$u' = \sum_m um'$$

$$u'' = \sum_m um''$$

$$e' = \sum_m um' \cdot em' / u'$$

$$em'' = \sum_m um'' \cdot em'' / um''$$

3. Data Estimates

National Family Health Survey (NFHS-3) is the third in the NFHS series of surveys in 2005-06. The first NFHS was conducted in 1992-93, and the second (NFHS-2) was conducted in 1998-99. All three NFHS surveys were conducted under the stewardship of the Ministry of Health and Family Welfare (MoHFW), Government of India, New Delhi.

In Jharkhand, NFHS-3 was based on a sample of 2,483 households that is representative at the State level, and within the State at the urban and rural levels. Women aged 15-49 were eligible for interview in all NFHS-3 sample households, while men aged 15-54 were eligible for interview in a subsample of households. NFHS-3 interviewed 2,983 women aged 15-49 and 996 men aged 15-54 in Jharkhand State to obtain information on popu-

lation, health, and nutrition.

The data of the National Family Health Survey (NFHS) in Jharkhand State during 2005-06 were utilized to estimate the number of births averted by programme and non-programme sources separately. The prevalence information by age and method was obtained for currently married females who are currently using and contraceptive method. The female population of reproductive ages 15-49 years was estimated from the 2001 population census figures.

The data problem arose when programme and non-programme methods were to be sorted out. Some of the programme methods and contraceptive services, such as condoms and pills, are available outside the programme at the private clinic. However, the data of such services are not available. It was thus assumed that all modern methods were offered by the programme and were termed programme methods. All traditional methods were considered non-programme methods. This assumption is fairly reasonable because modern

contraceptives, such as condoms and pills, are widely distributed through the programme.

4. Results

The application of the formulae yielded the estimates of gross natural fertility and gross potential fertility of Jharkhand State (Table 4.1). The difference between gross potential and gross natural fertility, on the one hand, and observed fertility, on the other hand, provided the basis for estimating births averted by programme and non-programme contraception for the State (Table 4.2). The results show that young fertile women avert the majority of births by non-programme methods compared to the programme methods. The births averted by non-programme contraception/ natural methods are concentrated among women of ages 20-29 years also in the State. The findings confirm an earlier study where mean age of use was found to be high in the early thirties (Kumar, D; 1990). The method specific birth averted by both programme and non-programme contraception was evaluated for the Jharkhand State (Tables 4.3). It is not-

Table 4.1: Prevalence of Programme and non-programme contraception, observed fertility rates and estimated natural and gross fertility rates by age-group in Jharkhand State, 2005-06

Age group	Prevalence of		Observed fertility, 2005	Elasticity coefficient	$1-C_a$ ($u_a' + u_a''$)	$1-C_a$ (u_a'')	Natural fertility rate	Gross potential fertility rate
	Programme contra-ception	Non-pro-gramme contra-ception						
a	u_a'	u_a''	AF_a	C_a			NAF_a	PAF_a
15-19	4.3	2.4	53.7	0.620	0.95846	0.98512	56.0	55.2
20-24	17.4	3.6	249.3	0.620	0.8698	0.97768	286.6	280.2
25-29	31.8	6	181.2	0.823	0.688906	0.95062	263.0	250.0
30-34	19.1	2.6	116.6	0.940	0.79602	0.97556	146.5	142.9
35-39	24.8	3.4	53.2	1.022	0.711796	0.96525	74.7	72.1
40-44	22.9	1.1	35.2	1.309	0.68584	0.98560	51.3	50.6
45-49	20.9	3.3	18.9	1.898	0.540684	0.93737	35.0	32.8

ed that, in general, the effectiveness or impact of the programme resembles an oval-shaped curve, i.e., in the initial phases pregnancy reduction increased to reach a plateau and then declined in the remaining phases. This may represent a cyclical

occurrence and pregnancy reduction may again increase. Continual follow-up is necessary for an extended time period to analyze any additional trends in fertility reduction.

Table 4.2: Gross fertility and gross birth averted by the programme and non-programme contraception by age group in Jharkhand State, 2005-06

Age group	Female Population, 2006	Gross fertility effect of:		Birth averted by			
		Programme Contraception	Non-programme contraception	Programme methods		Non-programme methods	
a	POP _a	PAF _a - AF _a	NAF _a - PAF _a	BA _a	Percent	BAN _a	Percent
15-19	1586000	0.00149	0.00083	2369	1.3	1322	4.3
20-24	1238000	0.03092	0.00640	38279	21.4	7920	25.5
25-29	1055000	0.06884	0.01299	72624	40.6	13703	44.2
30-34	997000	0.02630	0.00358	26220	14.6	3569	11.5
35-39	929000	0.01894	0.00260	17598	9.8	2413	7.8
40-44	818000	0.01538	0.00074	12585	7.1	605	1.9
45-49	678000	0.01387	0.00219	9401	5.2	1484	4.8
Total	7301000			179076	100.0	31016	100.0

Table 4.3: Estimated birth averted by programme and non-programme contraception in Jharkhand State, 2005-06

Methods	Prevalence of		Use effectiveness of		Estimated birth averted by			
	Programme contra-ception	Non-programme contra-ception	Programme contra-ception	Non-programme contra-ception	Programme contraception		Non-programme contraception	
	u _m '	u _m ''	e _m '	e _m ''	BA _m	Percent	BAN _m	Percent
Oral pills	3.8		0.9		20635	11.5		
IUD	0.6		0.95		3439	1.9		
Condom	2.7		0.7		11403	6.4		
Tubectomy	23.4		1.0		141185	78.8		
Vasectomy	0.4		1.0		2413	1.3		
Rhythm		2.1		0.5			13858	44.6
Withdrawal		2		0.5			13199	42.6
Other methods		0.6		0.5			3959	12.8
Total	30.9	4.7			179076	100.0	31016	100.0

5. Conclusion

The salient feature of the study is that it is not based on the service statistics and the data are obtained from the State-level survey. In the study, the standard method-specific use-effectiveness levels weight observed use and prevalence level by method. Of the 17,9076 births averted in Jharkhand State by programme contraception 14,3598 (80.1 percent) births were averted by sterilization users in 2005-06 while the spacing methods users contributed to only 19.9 per cent of the birth prevention. The spacing methods need to be strengthened for the greater use. With regard to the 31,016 births averted by non-programme contraception, the main contribution was made by the users of Rhythm (44.6 percent) that was followed by the users of Withdrawal (42.6 percent) and by the other methods (12.8 percent). Of the total birth averted in Jharkhand State, the contribution of programme contraception and non-programme contraception is about 85 percent and 15 percent in 2005-06.

6. Recommendation

The programme contraception has the dominance role to control fertility. however, the non-programme contraception/natural methods use should also be enhanced at the places where accessibility of programme contraception is poor.

Acknowledgement: An earlier version of the paper was presented in the International Conference on Recent Advances in Mathematics, Statistics and Computer Science (ICRAMSCS-2015) at Central University of South Bihar, Patna during 29-31 May 2015 and the author is thankful to the participants for their feedback in the first dissemination workshop of research studies conducted by PRCs

during 2015-16 during 6-7th October 2016 at Thiruvananthapuram, Kerala also.

References

Bongaarts, John , 1985, "A Prevalence Model for Evaluating the Fertility Effect of Family Planning Programmes: Age-specific and Method specific Results" *Studies to Enhance the Evaluation of Family Planning Programmes*, United Nations, New York, ST/ESA/SER.A/87, pp.246.

_____ (1993), "The fertility impact of family planning programs". New York, Population Council, Research Division Working Papers No. 47, pp.35.

Census of India, 2001, "Age Level Data of the States of India". India Level (In CD).

International Institute for Population Sciences (IIPS) and ORC Macro. 2008. *National Family Health Survey (NFHS-3), India, 2005-06: Jharkhand*. Mumbai: IIPS.

Kumar, Dilip, 1990, "Evaluation of Family welfare and MCH Programmes in Some PHCs of Patna district". Population Research Centre, Patna University, Patna, PRC Mimeograph Series No. 118, pp.91.

Ministry of Health and Family Welfare, Government of India, New Delhi 'Family Welfare Statistics in India, 2009.

Potter, Robert G. 1969. Estimating births averted in a family planning program, in S. J. Behrman, Leslie Corsa, and Ronald Freedman (eds.), *Fertility and Family Planning: A World View*. Ann Arbor: University of Michigan Press, pp. 413-434.

CHAPTER - 5

Maternal Morbidity Profile of Women in Tamil Nadu

*N. Dhanabaghyam**, *N. Rajakumar***, *M. Murugesan*** and *B. Karthi***, **Assistant Chief, **Data Assistant*

Population Research Centre, Gandhigram, Tamil Nadu

ABSTRACT

Maternal Morbidity is a challenging social and health issue worldwide. Attention to maternal morbidity is considered as a proper approach to prevent maternal mortality. Millennium Development Goals (MDG)-5, spells out an international commitment to reduce maternal mortality by two-thirds between 1990 and 2015. While in Tamil Nadu, concerted efforts were taken up by the Government, through its interventions, to reduce maternal morbidity and mortality, certain districts are far behind. Attention to maternal morbidity has triggered a growing interest to frame a proper approach to prevent maternal and child deaths in these districts. The two major community-level studies (DLHS and NFHS) are the main sources of maternal morbidity in Tamil Nadu. They used women's self-reported morbidity to identify cases, but subsequent work has shown self-reporting to be more unreliable than medical records, with evidence of both over- and under-diagnosis. To adequately monitor maternal morbidity and launch an appropriate programmatic response, it is necessary to identify and measure the maternal conditions based on the evidence, their frequency, burden and its impact. Data for the present study was obtained from the Mother Child Tracking System (MCTS). The study covered 11,17,140 women who had registered and received ANC with an EDD period from 01.04.2014 to 31.03.2015. Maternal morbidity incidence rate in the Tamil Nadu state is 6.56. Perambalur recorded the highest maternal morbidity incident rate (MMIR) followed by Thiruchirapalli and Thiruvallur and the lowest recorded at Chennai. Nagapattinam, Karur, The Nilgiris, Thanjavur, Thiruvallur and Ariyalur districts recorded high prevalence (incident rate ≥ 7.88). Chennai, Pudukkottai, Thoothukudi, Ramanathapuram, Erode, Dindigul, Vellore and Coimbatore districts recorded low prevalence (incident rate ≤ 4.57) of maternal morbidity. Obstetric morbidity plays a significant role in the maternal morbidity followed by pregnancy complications, pregnancy diseases/morbidity, obstetric complications and Postnatal morbidity. Anemia/severe anemia, PIH and GDM were the mostly reported pregnancy morbidities/diseases followed by heart disease, APH, HIV. Jaundice and Vesicular mole. Teenage pregnancy, elderly primi i.e., conception at the age of 35 and above years, RH incompatibility, Multipara, multiple pregnancy, Short primi and hydramnios are the important pregnancy complications/ complicated pregnancies. Obstetric problem includes previous LSCS/assisted delivery, long period of infertility, previous bad obstetric history, pregnancy due to contraceptive failure and handicapped mother. Obstetric complications include CPD and post dated pregnancy, IUGR, Congenital malformation and ectopic pregnancy.

INTRODUCTION

Pregnancy and childbirth-related complications are the leading cause of disability and death among women of reproductive age worldwide. It is a challenging social and health issue in developing countries. All countries have been trying to identify and resolve maternal morbidities to promote better maternal health and prevent maternal mortality. Mother and children were given increased attention from the Alma Ata Conference in 1978, with an initiation to cherish the primary health care through Healthy People by 2000. The Safe Motherhood Conference in Nairobi in 1987, the International Conference on Population and Development in Cairo in 1994, and the 1995 Fourth World Conference on Women in Beijing also insist the same. The focus on women's health was further reinforced in the 2000 Millennium Development Goals (MDG) in particular MDG 5, spells out an international commitment to reduce maternal mortality by two-thirds between 1990 and 2015.

Global estimates of maternal morbidity or disabilities range from 15-20 million. About 20 million pregnant women suffer from pregnancy complications and 2,75,000 women are estimated to die each year in pregnancy and child birth (UN, 2010). Maternal morbidities and disabilities are considered a leading contributor to the burden of disease among women. Maternal conditions contributed to 2.7 percent of the deaths world wide and 12 percentage of deaths among women aged 15-44 years (Koblinsky.et.al., 2012).

1.1. MAGNITUDE OF MATERNAL MORBIDITY

Maternal morbidity is very difficult to measure due to its various definitions. The magnitude of the maternal morbidity varies according to the criteria/classification for identifying the cases. The true burden of maternal morbidity is still not known. Prevalence varies between 0.80 and 8.23 in studies using disease-specific criteria, while the

range is 0.38 to 1.09 in the group that use failure of organ/dysfunction of system. Rates are within the range of 0.01 and 2.99 in studies using management criteria (Mantel, et.al., 1998).

Worldwide it is estimated that about 1.4 million women experience acute obstetric morbidity. There is a large difference existing between developed (UK, 1:118, France 1:222) and developing (India, 1:11, Niger 1:5) countries (Minkauskiene. et.al, 2004). In India, self-reported symptoms of DLHS-4 show pregnancy complication variation from 57.8 percent to 80.2 percent. The highest proportion of pregnancy complication was reported in Bihar(75.4) and the highest complication in delivery was reported in Jharkhand (83.5). In India, Severe acute maternal morbidity (SAMM) was found to be 7.1 percent and case fatality ratio was 13.8 percent. Incidence of SAAM ranged from 0.07 to 8.2 percent and the case fatality ratio 0.02 to 37 percent. Pregnancy-induced hypertension, including eclampsia contributes maximum (35 to 77 percent) to severe acute morbidity cases. Hemorrhage, including resultant shock was observed in six to 35 percent of cases. Puerperal sepsis was reported by one to 19 percent of cases (Doke, 2014). In Tamil Nadu, there is a substantial decline in maternal morbidity over the past few years due to the concerted efforts taken up by the Government through its interventions. Results of the two large-scale community-level studies in Tamil Nadu show a substantial decline in the maternal morbidity from 2008 to 2013. Pregnancy complications were reduced from 47.8 in percent to 45.5 percent. Delivery complications reduced from 37.6 percent to 11.1percent and postnatal complications reduced from 18.8 percent to 13.9 percent (IIPS, 2010, IIPS, 2014). Pregnancy anemia (<11.0g/dl) reduced from 54.9 percent to 44.4 percent (IIPS, 2016).

Even though Tamil Nadu reached its goal, through substantial decline in maternal morbidity over the past few decades, attention is needed to frame a proper approach to decrease maternal morbidity in

some districts to prevent maternal and child deaths in these districts.

The two major community-level studies in Tamil Nadu used women's self-reported morbidity to identify cases, but subsequent work has shown self-reporting to be more unreliable than medical records, with evidence of both over- and under-diagnosis. To adequately monitor maternal morbidity and launch an appropriate programmatic response, it is necessary to identify and measure the maternal conditions based on the evidence, their frequency, burden and its impact. Community-level studies based on self-reported symptoms are not on account of the information. In this situation, assessment of maternal morbidity based on community as well as clinical level is very important (i) to assess the size of the problem in terms of incidence and prevalence for planning, which involve priority setting and allocation of resources in the form of financial and human resources, health facilities, supplies, etc. and assessment of morbidity at the baseline is necessary to monitor progress, (ii) assess the effectiveness of interventions and programmes targeting the public health problem and (iii) identifying the underlying causes and risk factors of maternal morbidity in a community (Doke, 2014).

1.2. OBJECTIVES

The overall objective of the study is to assess the magnitude and pattern of maternal morbidity in Tamil Nadu

1.2.1. SPECIFIC OBJECTIVES

- i. To assess the overall incidence of maternal morbidity in Tamil Nadu
- ii. To identify the spectrum of maternal morbidity and the incidence of the diseases/complication.
- iii. To identify possible discrepancies in maternal morbidity among the districts and in different complication/morbidity conditions.

1.3. IDENTIFICATION AND CLASSIFICATION OF MATERNAL MORBIDITY

Current literature includes a wide range of methods for identifying and classifying maternal morbidity. Maternal morbidities have been grouped under various categories such as direct obstetric morbidity, indirect obstetric morbidity and psychological obstetric morbidity or obstetric complications, pre-existing medical conditions, or categories of obstetric morbidity that occur during pregnancy, during delivery, or after delivery. Another way to categorize potentially life-threatening maternal conditions has been by type of complication, such as hemorrhagic complications, hypertensive disorders, or management indicators of severity. Yet another example of the variety of categorization is the grouping of selected obstetric complications into "non-severe conditions", "severe conditions", and "procedures".

In general, maternal morbidity is defined as "any health condition attributed to, and/or aggravated by, pregnancy and childbirth that has a negative impact on the woman's wellbeing" (WHO, 2013). The Maternal Morbidity Working Group (MMWG) defined a framework for developing criteria by which to identify maternal morbidity. The development of the framework followed several guiding principles: (i) identification and measurement of the selected maternal conditions should be feasible and evidence-based; (ii) maternal morbidity should not be viewed as consisting only of the conditions themselves, but also their complications; and (iii) morbid conditions should be prioritized on the basis of their frequency and impact. The aim is to measure maternal morbidity rather than the burden of each condition.

In line with these principles and the newly-developed definition, the maternal morbidity framework includes the symptoms, signs, clinical tests and management strategies of relevant maternal conditions – similar to the approach used to identify morbidity in a case of maternal near miss-linked

with the direct, indirect and incidental causes of maternal morbidity.

Attention was paid to the similarities and discrepancies of the maternal morbidity definitions across the studies. Maternal morbidity is defined as “morbidity in a woman who has been pregnant (regardless of site or duration of the pregnancy) resulting from any cause related to or aggravated by the pregnancy or its management but not from accidental or incidental causes (WHO, 1989). Maternal Morbidity Working group (MMWG) defined maternal morbidity as “any health condition attributed to and/or aggravated by pregnancy and childbirth that has a negative impact on the woman’s wellbeing”. This new definition of maternal morbidity was proposed for inclusion in the 11th revision of the International statistical classification of diseases and related health problems (ICD) also.

II. MATERIALS AND METHODS

2.1. DATA

The present study is based on the data obtained from the Mother Child Tracking System (MCTS) for the year 2014-15. The study covered women who had registered and received ANC with an EDD period from 01.04.2014 to 31.03.2015. Totally 11,17,140 women were registered for ANC during the reference period.

2.2. MEASURING MATERNAL MORBIDITY

The measures of disease most often used in epidemiology are prevalence, incident and fatality rates. The point of prevalence is the proportion of women /men/child who have the complication, at the specified time regardless of the length of time they had the complication per the total population. On the contrary incidence refers the number of new cases of the complication that occur in the population at risk within a specified period of time. Therefore, in the present study incident rate may be suitable and was used to assess the magnitude

of the maternal morbidity.

The incidence rate of maternal morbidity was calculated as

Number of existing cases of particular maternal complication at a specified point in time

----- X 100

Population at risk

2.3. CLASSIFICATION OF DISEASES

As per the available literature and time frame of the recorded morbidities, the maternal morbidity was classified into five groups as pregnancy complication, pregnancy morbidity, obstetric morbidity, obstetric complication, postnatal morbidity and others (Vanderkruik. et al., 2013). Table 5.1 explained the classification of the diseases according to the symptoms, prevalence and clinical examination.

Table 5.1: Classification of Maternal Morbidity

Type of Morbidity	Name of the Morbidity
Pregnancy Morbidity	Anemia, Pregnancy-induced hypertension (PIH), Eclampsia, Ante partum haemorrhage (APH), Jaundice, Hyper tension / Heart diseases, Fever, Gestational diabetes mellitus (GDM), HIV, Vesicular mole
Pregnancy Complication	Teenage pregnancy (< 18 years), Elderly primi (> 35 years) Short primi (Height below 145cms), Multipara, Hydraminous (Poly/Oliga), Multiple pregnancy, Weight below 35 kg, RH incompatibility
Obstetric Morbidity	Previous LSCS /Assisted, Long period of infertility, Previous bad obstetric history, Pregnancy due to contraceptive failure, Handicapped mother

Obstetric Complication	Cephalopelvic disproportion (CPD), Intra Uterine Death (IUD), Intra Uterine Growth Restriction (IUGR), Congenital malformation, Malpresentation, Ectopic pregnancy, Post-dated pregnancy
-------------------------------	--

III. RESULTS AND DISCUSSION

3.1. MATERNAL MORBIDITY INCIDENCE

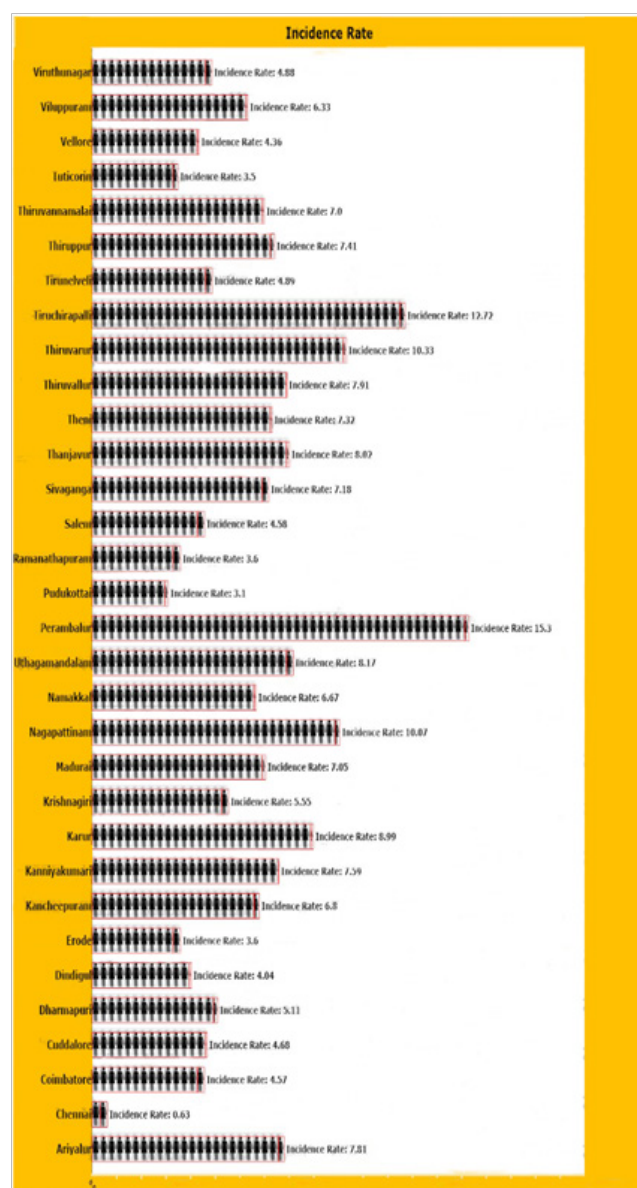
Maternal morbidity incidence rate in the Tamil Nadu state is 6.56 (Table.5.2, Figure-5.1). Perambalur (15.30) recorded the highest maternal morbidity incident rate followed by Thiruchirapalli (12.72) Thiruvavur (10.33) and Nagapattinam (10.07) districts. Chennai recorded the lowest incident rate of 0.63.

Table 5.2: Maternal Morbidity Incidence Rate in Tamil Nadu

S.No	Name of the Districts	Maternal Morbidity
	TAMIL NADU	6.56
1	Ariyalur	7.81
2	Chennai	0.63
3	Coimbatore	4.57
4	Cuddalore	4.68
5	Dharmapuri	5.11
6	Dindigul	4.04
7	Erode	3.60
8	Kancheepuram	6.80
9	Kanyakumari	7.59
10	Karur	8.99
11	Krishnagiri	5.55
12	Madurai	7.05
13	Nagapattinam	10.07
14	Namakkal	6.67
15	The Nilgiris	8.17
16	Perambalur	15.30
17	Pudukottai	3.10
18	Ramanathapuram	3.60
19	Salem	4.58
20	Sivagangai	7.18

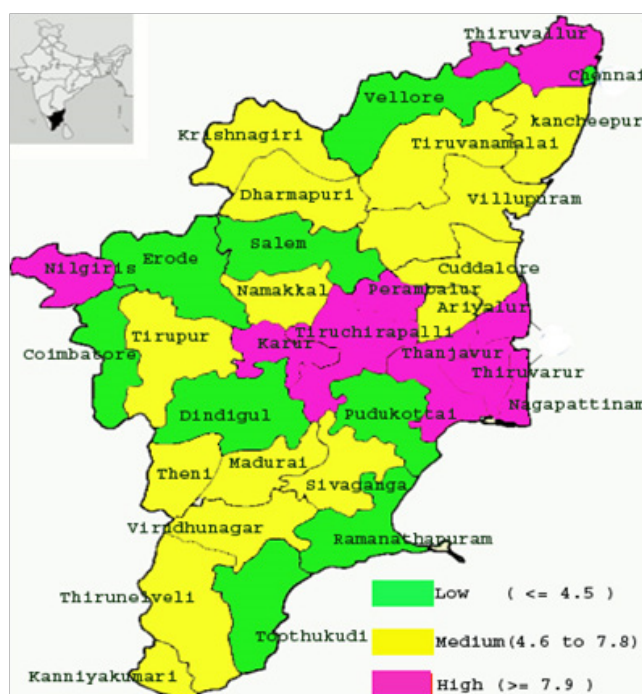
21	Thanjavur	8.02
22	Theni	7.32
23	Thiruvallur	7.91
24	Thiruvavur	10.33
25	Thiruchirapalli	12.72
26	Tirunelveli	4.89
27	Tirupur	7.41
28	Thiruvannamalai	7.00
29	Thoothukudi	3.50
30	Vellore	4.36
31	Villupuram	6.33
32	Virudhunagar	4.88

Figure 5.1: Maternal Morbidity Incidence Rate in Tamil Nadu



Based on the maternal morbidity incidence rate districts were classified into three groups (Figure-5.2) as low incidence districts (incidence rate ≤ 4.5), medium incidence districts (incidence rate 4.6 to 7.8) and high incidence districts (incidence rate ≥ 7.9). Perambalur, Thiruchirapalli, Thiruvavur, Nagapattinam, Karur, The Nilgiris, Thanjavur, Thiruvallur and Ariyalur districts recorded high incidence rate (incident rate ≥ 7.88) and Chennai, Pudukkottai, Thoothukudi, Ramanathapuram, Erode, Dindigul, Vellore and Coimbatore districts were recorded as low incidence districts (incident rate ≤ 4.57) of maternal morbidity. Cuddalore (4.68), Virudhunagar (4.88), Tirunelveli (4.89), Dharmapur (5.11), Krishnagiri (6.55), Namakkal (6.67), Kancheepuram (6.80), Thiruvannamalai (7.00), Madurai (7.05), Sivagangai (7.18), Theni (7.32), Thirupur (7.41), Kannayakumari (7.59) and Ariyalur (7.81) districts recorded a medium incidence rate.

Figure 5.2: Incidence of Maternal Morbidity in Tamil Nadu



3.2. MATERNAL MORBIDITY ACCORDING TO CLASSIFICATION OF DISEASES

In Tamil Nadu, incidence of pregnancy-related disease such as anemia, PIH, Eclampsia, APH, Jaundice, heart disease, fever, GDM, HIV and Vesicular mole was recorded as 13.6. Pregnancy disease/morbidity was higher in Chennai (28.8), Virudhunagar (18.0), Krishnagiri (17.8), Vellore (17.6), Dindigul (16.60), Thiruchirapalli (16.4), The Nilgiris (16.4), Erode (15.8) and Coimbatore (14.9) districts (Table 5.3, Figure 5.4).

Eighteen percent of the pregnancies were recorded as complicated either due to age factor (teenage pregnancy/elderly or short primi), low weight, RH incompatibility, multipara, multiple pregnancy or pregnancy or Hydraminous. Complicated pregnancies were more reported in Perambalur (26.7), Dindigul (24.3), Theni (23.5), Thanjavur (23.3), Pudukkottai (22.2), Thiruvavur (22.0), Thiruchirapalli (21.7), Nagapattinam (21.5), Sivagangai (20.7), and Madurai (20.5) (Table 5.3, Figure-5.3).

Table 5.3: Maternal Morbidity According to Classification of Diseases in Tamil Nadu

Districts	Percentage of Women experienced					
	Pregnancy Diseases/ Morbidity	Pregnancy Complica- tion	Obstetric Morbidity	Obstetric Complication	Others	Post Natal Morbidity
Tamil Nadu	13.6	18.1	47.2	3.8	11.8	5.5
Ariyalur	12.1	19.1	46.5	3.3	13.8	5.2
Chennai	28.8	9.9	34.8	3.4	15.2	7.9
Coimbatore	14.9	16.5	46.6	2.9	10.2	8.9
Cudalore	13.7	16.0	49.5	3.6	8.6	8.6
Dharmapuri	12.8	15.6	42.8	4.7	15.7	8.4
Dindigul	16.6	24.3	38.2	3.0	12.4	5.5
Erode	15.8	14.6	39.3	7.6	11.0	11.7

Kancheepuram	12.4	12.0	53.4	3.8	12.6	5.8
Kanyakumari	8.9	10.2	71.6	1.2	5.1	3.0
Karur	13.8	19.1	45.4	2.9	14.8	4.0
Krishnagiri	17.8	19.2	38.5	7.9	10.4	6.2
Madurai	14.0	20.5	50.3	2.0	9.4	3.8
Nagapattinam	13.7	21.5	50.8	3.5	8.9	1.6
Namakkal	10.3	17.4	48.5	5.8	13.5	4.5
The Nilgiris	16.4	15.2	36.9	5.5	18.2	7.8
Perambalur	10.9	26.7	48.5	2.2	9.7	2.0
Pudukkottai	11.8	22.2	44.8	4.2	6.4	10.6
Ramanathapuram	14.3	15.2	53.8	2.1	9.7	4.9
Salem	13.9	19.9	45.3	4.1	11.6	5.2
Sivagangai	7.3	20.7	51.1	5.0	11.5	4.4
Thanjavur	12.3	23.3	52.3	1.5	8.5	2.1
Theni	12.3	23.5	47.3	1.6	11.5	3.8
Thiruvallur	11.0	11.8	44.3	6.9	19.8	6.2
Thiruvarur	12.7	22.0	55.2	1.9	6.2	2
Thiruchirappalli	16.4	21.7	47.2	2.1	8.9	3.7
Tirunelveli	11.5	15.8	60.2	1.7	7.8	3
Tirupur	12.7	16.9	45.4	4.9	14.7	5.4
Thiruvannamalai	11.0	19.7	39.8	5.4	14.3	9.8
Thoothukudi	13.5	14.9	52.7	2.9	6.9	9.1
Vellore	17.6	15.8	37.7	5.5	15.4	8.0
Villupuram	14.8	16.9	34.6	6.2	19.3	8.2
Virudhunagar	18.0	14.7	48.8	3.6	8.6	6.3

Forty-seven percent of the pregnant women experienced pre-pregnancy/obstetric morbidities like previous LSCS, prolonged infertility, previous bad obstetric condition, contraceptive failure and handicapped mother. Obstetric morbidity was more in Kanyakumari (71.6), Tirunelveli (60.2), Thiruvarur (55.2), Ramanathapuram (53.8), Kancheepuram (53.4), Thoothukudi (52.7), Thanjavur (52.3), Sivagangai (51.1), Nagapattinam (50.8), Madurai (50.3) and Cudallore (49.5) districts (Table 5.3, Figure-5.5).

Four percent of the women experienced obstetric complications like CPD, IUD, IUGR, congenital malformation, mal presentation, Ectopic pregnancy and post-dated pregnancy. Higher prevalence of obstetric complications were mostly reported in Krishnagiri (7.9), Erode (7.6), Thiruvallur (6.9), Villupuram (6.2), Namakkal (5.8), The Nilgiris and Vellore (5.5 each), Thiruvannamalai (5.4) and Sivagangai districts (Table 5.3, Figure-5.6).

Eleven percent experienced other morbidities (Table 5.3, Figure-5.7). Women in Villupuram (19.3), The Nilgiris (18.2), Vellore (15.4), Chennai (15.2), Dharmapuri (15.7), Karur (14.8), Tirupur (14.7), Thiruvannamalai (14.3), Ariyalur (13.8) and Namakkal (13.5) districts experienced other complications.

Five percent experienced post natal morbidities. Postnatal complications were more in Erode (11.7), Pudukkottai (10.6), Thiruvannamalai (9.8), Thoothukudi (9.1), Coimbatore (8.9), Cudallore (8.6), Dharmapuri (8.4), Villupuram (8.2) and Vellore (8.0) districts (Table 5.3, Figure-5.8).

Figure 5.3: Incidence of Pregnancy Complication in Tamil Nadu by districts

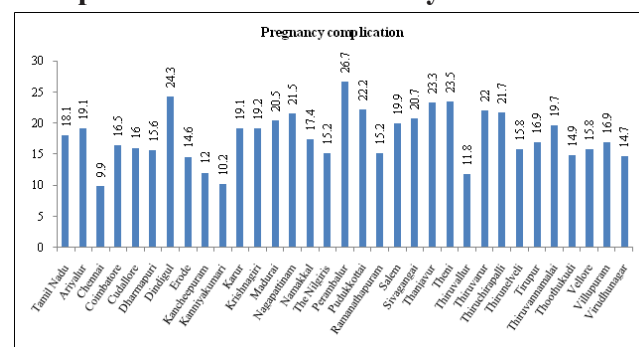


Figure 5.4: Incidence of Pregnancy Morbidity in Tamil Nadu by districts

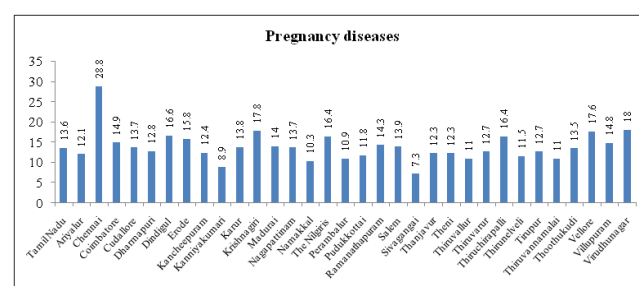
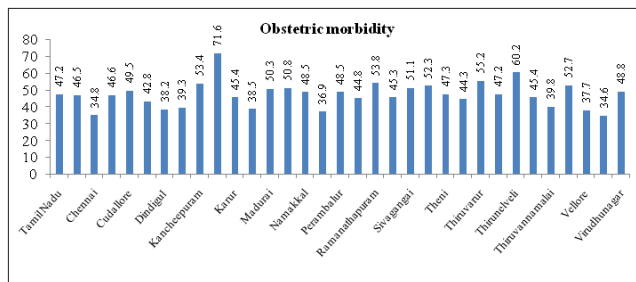
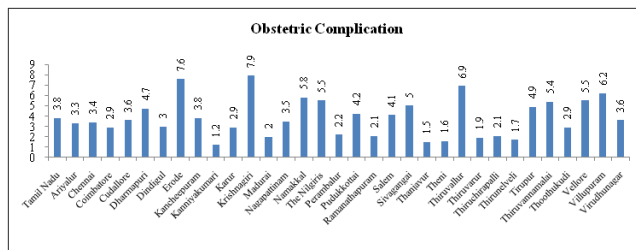
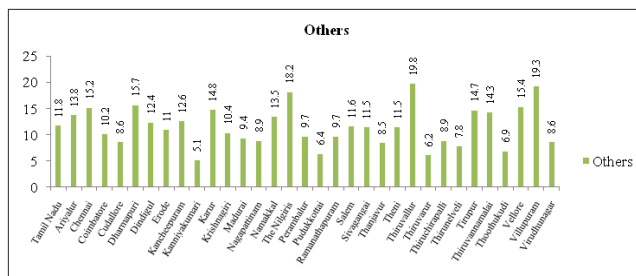
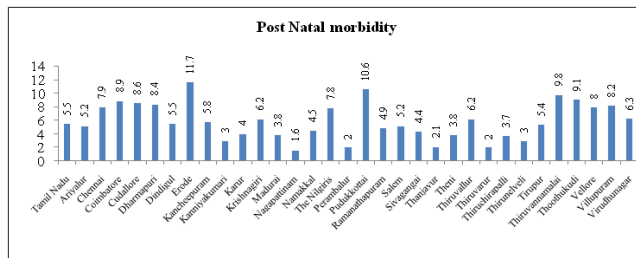


Figure 5.5: Incidence of Obstetric Morbidity in Tamil Nadu by districts**Figure 5.6: Incidence of Obstetric Complication in Tamil Nadu by districts****Figure 5.7: Incidence of Other Maternal Morbidity in Tamil Nadu by districts****Figure 5.8: Incidence of Post Natal Morbidity in Tamil Nadu by districts**

3.2.1. INCIDENCE OF PREGNANCY MORBIDITY / DISEASES

Table 5.4 explains the pregnancy-related diseases according to districts in Tamil Nadu. Disease-wise classification helps to identify the factors behind the morbidity and, in turn, helps the policy makers and programme managers to frame accordingly. In Tamil Nadu, severe anemia (5.67), PIH (3.81 percent) and GDM (1.87 percent) were mostly reported as pregnancy morbidities/diseases followed by heart disease (0.91 percent), APH (0.44 percent), HIV (0.27 percent), Jaundice (0.21 percent) and Vesicular mole (0.13 percent).

Severe anemia was mostly found in Thiruchirapalli (12.90 percent), Villupuram (5.7 percent), Vellore (4 percent), Salem (4.3 percent), Krishnagiri (4.2 percent) and Coimbatore (3.8 percent). PIH was highly reported in Thiruchirapalli (9.4 percent), Vellore and Villupuram (7.7 percent each), Kancheepuram (5.9 percent), Salem (5.30 percent), Madurai (4.8 percent), Thiruvannamalai (4.4 percent), Thiruvallur (4.3 percent), Coimbatore (4.2 percent) and Tirupur (3.8 percent) districts. Eclampsia was highly reported in Coimbatore, Cuddalore, Dharmapuri, Dindigul, Erode, Kancheepuram, Karur, Krishnagiri, Madurai, Nagapattinam, Namakkal, , Sivagangai, Theni, Thiruvallur, Tirupur and Thiruvannamalai districts. APH was mostly reported in Villupuram (14.2 percent), Vellore (7.5 percent), Thiruchirapalli (6.50 percent), Dharmapuri (5.5 percent), Thiruvannamalai (5.5 percent) and Thiruvallur (5.1 percent) districts. Jaundice was mostly reported in Kancheepuram (14.4 percent), Thiruvallur (10.10 percent), Villupuram (10.8 percent) and Vellore (6.5 percent). GDM was reported in Chennai, Coimbatore, Cuddalore, Kanyakumari, Nagapattinam, Thanjavur and Thiruvallur districts. HIV was mostly reported in Coimbatore (3.90 percent), Dharmapuri (3.30 percent), Dindigul (2.8 percent), Karur (3.90percent), Krishnagiri (5 percent), Madurai (6.7 percent), Namakkal, (5 percent) and Thanjavur (5.6 percent) districts.

Table 5.4: Maternal Morbidity due to Pregnancy Morbidity / Diseases in Tamil Nadu

Districts	Percentage of Women Experienced during Pregnancy									
	Severe Anemia	PIH	Eclampsia	APH	Jaundice	Heart Disease	Fever	GDM	HIV	Vesicular mole
Tamil Nadu	5.67	3.81	0.29	0.44	0.21	0.91	0.05	1.87	0.27	0.13
Ariyalur	1.20	1.30	0.00	2.10	2.20	2.70	0.00	1.40	0.60	0.00
Chennai	1.30	1.50	0.50	1.40	3.60	2.20	33.30	3.20	1.70	1.20
Coimbatore	3.80	4.20	5.90	1.00	3.60	4.50	0.00	4.70	3.90	0.00
Cudallore	2.90	2.40	1.10	2.10	2.90	4.50	0.00	3.40	2.20	0.00
Dharmapuri	1.50	2.40	3.70	5.50	0.00	2.30	0.00	1.80	3.30	3.50
Dindigul	3.20	1.80	3.70	4.10	0.00	3.80	0.00	0.90	2.80	2.40
Erode	2.10	2.20	1.10	0.70	0.70	1.30	0.00	1.70	1.70	1.20
Kancheepuram	3.40	5.90	8.00	4.10	14.40	3.80	33.30	8.10	2.20	4.70
Kanyakumari	1.70	0.60	0.00	3.40	0.70	2.70	0.00	4.90	1.10	3.50
Karur	2.10	2.00	3.20	1.70	1.40	3.70	0.00	1.40	3.90	2.40
Krishnagiri	4.20	3.30	2.70	2.40	1.40	1.30	0.00	1.70	5.00	3.50
Madurai	7.40	4.80	4.30	4.80	7.20	4.80	0.00	4.00	6.70	10.60
Nagapattinam	3.60	3.30	2.70	2.10	0.70	4.00	0.00	5.80	3.30	0.00
Namakkal	1.50	1.80	2.70	1.40	1.40	2.50	0.00	1.50	5.00	1.20
The Nilgiris	0.70	1.60	1.60	1.40	2.90	2.50	0.00	1.60	0.60	0.00
Perambalur	2.30	1.30	0.00	1.00	0.70	1.00	0.00	1.60	4.40	1.20
Pudukkottai	1.30	0.90	0.50	2.70	0.70	2.20	0.00	0.40	0.60	1.20
Ramanathapuram	1.70	1.00	1.10	2.10	2.20	1.00	0.00	0.70	0.60	1.20
Salem	4.30	5.30	5.30	1.40	2.90	2.70	0.00	2.20	2.80	2.40
Sivagangai	0.70	1.40	2.70	3.10	0.00	2.00	0.00	1.40	1.10	1.20
Thanjavur	4.00	3.00	2.10	1.40	1.40	4.50	0.00	7.00	5.60	0.00
Theni	1.90	2.00	2.70	1.00	0.70	1.50	0.00	1.90	1.70	1.20
Thiruvallur	2.70	4.30	6.40	5.10	10.10	2.80	33.40	7.20	5.00	7.10
Thiruvarur	2.80	2.80	2.10	1.70	0.00	3.70	0.00	3.90	3.30	4.70
Thiruchirapalli	12.90	9.40	5.30	6.50	5.80	3.70	0.00	8.40	7.80	9.40
Tirunelveli	3.30	2.20	2.70	4.50	1.40	4.30	0.00	2.60	3.30	3.50
Tirupur	3.20	3.80	6.40	2.10	2.20	4.00	0.00	4.20	3.30	1.20
Thiruvannamalai	2.70	4.20	5.30	5.50	8.60	5.10	0.00	3.20	1.70	11.80
Thoothukudi	1.50	1.40	1.10	1.00	0.70	1.80	0.00	1.10	0.60	2.40
Vellore	4.00	7.70	3.70	7.50	6.50	6.10	0.00	3.40	8.10	5.90
Villupuram	5.70	7.70	9.30	14.20	10.80	5.20	0.00	3.90	5.00	10.20
Virudhunagar	4.40	2.50	2.10	1.00	2.20	1.80	0.00	0.80	1.10	1.20

3.2.2. PREGNANCY COMPLICATION / COMPLICATED PREGNANCIES

Pregnancy complications/ complicated pregnancies account for a considerable proportion in maternal morbidity and, in turn, in maternal death (Table 5.5). In Tamil Nadu, maternal age accounts for two percent (elderly primi i.e., conception at the age of 35 and above years) of the pregnancy complications. RH incompatibility accounts for 3.87 percent of the complications. Multipara (3.7 percent) and multiple pregnancies constitute (2.9 percent) a considerable proportion followed by Short primi (3.1 percent) and hydraminos (1.98

percent). Teenage pregnancy is mostly reported in Theni (16.3 percent) followed by Thiruchirapalli (11.1 percent), Coimbatore (6.7 percent) followed by Thiruvallur (5.9 percent), Thiruvannamalai (4 percent), Madurai (3.70 percent), Salem (3 percent), Kancheepuram and Nagapattinam (2.2 percent each). Elder primi (conception >30 years) is reported in Thiruchirapalli (13.5 percent), Tirunelveli (4.6 percent), Thanjavur (5 percent), Villupuram (5.6 percent). Short primi accounts for 3.5 percent of the pregnancy complications and mostly reported in Thiruchirapalli (13.50 percent), Coimbatore (4.30 percent), Kancheepuram (4.7 percent), Madurai (6 percent), Thanjavur (5 per-

Table 5.5: Maternal Morbidity Due to Pregnancy Complications / Complicated Pregnancy in Tamil Nadu

Districts	Percentage of Women who had Complications in Pregnancy							
	Teenage Pregnancy (< 18 Yrs)	Elderly Primi (> 35 Years)	Short Primi (< 145 cms)	Hydraminos (Poly / Oligo)	Multipara	Multiple Pregnancy	Weight < 35 Kg	Rh incompatibility
Tamil Nadu	0.20	1.98	3.15	1.83	3.76	2.90	0.38	3.87
Ariyalur	1.50	1.20	1.70	3.50	1.80	1.50	2.00	0.80
Chennai	1.50	0.60	0.90	0.20	0.20	0.60	0.40	0.10
Coimbatore	6.70	3.10	4.30	3.50	2.10	3.60	2.00	3.60
Cudallore	1.50	1.90	2.40	1.60	3.50	2.30	2.00	2.50
Dharmapuri	1.50	2.80	2.10	1.20	2.10	2.50	2.00	0.90
Dindigul	0.70	1.90	1.90	2.10	2.80	3.80	6.40	3.20
Erode	0.70	1.50	1.70	1.70	1.10	1.20	0.80	1.20
Kancheepuram	2.20	4.70	4.40	5.50	1.60	4.00	5.60	3.40
Kanyakumari	0.00	1.70	1.00	1.60	0.90	1.70	2.00	2.90
Karur	0.70	3.10	1.70	3.90	2.30	2.20	0.00	1.50
Krishnagiri	1.50	1.50	1.60	2.30	4.50	3.00	3.20	2.30
Madurai	3.70	6.00	5.30	6.50	5.00	7.40	3.60	8.90
Nagapattinam	2.20	3.90	2.40	2.10	6.50	3.30	6.40	5.90
Namakkal	3.70	2.10	2.80	1.50	1.70	2.30	1.20	2.50
Nilgiris	0.70	0.90	1.60	1.50	0.50	0.60	0.40	0.70
Perambalur	11.90	1.80	5.40	1.20	3.50	2.60	8.00	2.40

Pudukkottai	0.70	0.80	2.10	1.20	1.50	1.70	1.60	1.80
Ramanathapuram	0.70	1.50	1.30	1.10	0.30	1.00	0.00	1.40
Salem	3.00	3.80	4.20	4.00	5.50	4.60	1.20	3.90
Sivagangai	0.00	1.60	1.80	3.00	1.90	2.00	2.00	5.00
Thanjavur	0.70	5.00	3.90	2.10	8.30	6.40	8.40	6.10
Theni	16.30	2.10	1.90	1.20	3.10	2.20	2.40	3.50
Thiruvallur	5.90	4.10	3.80	7.30	0.90	4.00	0.80	2.60
Thiruvarur	3.00	3.40	2.10	2.20	5.40	4.50	5.20	4.10
Thiruchirapalli	11.10	13.50	8.70	4.40	13.40	7.90	8.00	10.60
Tirunelveli	1.60	4.60	3.30	1.40	3.20	3.00	2.40	2.90
Tirupur	3.00	4.20	4.30	4.40	2.40	2.80	4.00	3.90
Thiruvannamalai	4.40	4.40	6.00	7.80	4.60	4.50	4.00	3.80
Thoothukudi	1.50	1.10	1.10	0.70	1.20	1.40	1.60	1.10
Vellore	4.40	3.80	4.90	4.20	3.50	3.30	2.80	2.60
Villupuram	1.50	5.60	7.40	12.20	3.30	6.10	7.20	2.60
Virudhunagar	1.50	1.80	2.00	2.90	1.40	2.00	2.40	1.30

cent) districts. Three percent of the complicated pregnancies were due to Short primi (<145cms height). Women in Thiruchirapalli (8.7 percent), Villupuram (7 percent), Thiruvannamalai (6 percent), Coimbatore, Salem and Kancheepuram (4 percent each), Vellore, Madurai and Perambalur (5 percent each) districts have prevalence of pregnancy complications due to short primi. Hydraminos (Poly/Oliga) accounts for two percent of the complicated pregnancies, mostly reported in Kancheepuram (5.5 percent), Madurai (6.5 percent), Thiruvallur (7.3 percent), Thiruvannamalai (7.8 percent) and Villupuram (12.2 percent) districts.

Multipara accounts for 3.7 percent of the complicated pregnancies. Higher proportion of women in Krishnagiri and Madurai (5 percent each), Nagapattinam (6.5 percent), Salem and Thiruvarur (5 percent each), Thanjavur (8 percent) and Thiruchirapalli (13.4 percent) districts reported about multipara. Three percent of the complicated pregnancies were multiple pregnancies. Women in Thiruchirapalli (8 percent), Madurai and Villupuram (7 percent each), Thanjavur (6 percent), Salem and Thiruvarur (5 percent each) and Coimbatore, Dindigul, Tirupur, Thiruvannamalai,

Kanchipuram and Thiruvallur (4 percent each) districts experiences higher proportion of multiple pregnancies.

In Tamil Nadu, less than one percent (0.38 percent) of the pregnancy complications were due to underweight of pregnant women. Under weight is recorded in Thiruchirapalli, Perambalur and Thanjavur (8 percent each), Villupuram (7 percent), Dindigul, Nagapattinam and Kancheepuram (6 percent each) and Thiruvarur (5 percent) districts. RH incompatibility accounts for four percent of the pregnancy complications. RH incompatibility is mostly reported in Thiruchirapalli (10 percent), Madurai (9 percent), Nagapattinam and Thanjavur (6 percent), Sivagangai (5 percent) and Tirupur Thiruvannamalai, Thiruvarur, Theni, Salem and Coimbatore districts (4 percent each).

3.2.3. OBSTETRIC MORBIDITY/DISEASES

Previous LSCS/assisted delivery (43 percent) is the main obstetric problem during reported delivery followed by long period of infertility (1.9 percent), previous bad obstetric history (2.78 percent), pregnancy due to contraceptive failure (0.14 percent) and handicapped mother (0.19 percent).

Forty-two percent of the delivery problems were due to previous LSCS/assisted delivery. Thiruchirapalli (8 percent), Coimbatore, Salem, Thiruvallur, Thiruvannamalai, Nagapattinam (4 percent each), Thanjavur, Villupuram, Tirunelveli (5 percent each), Madurai (6.2 percent) and Kancheepuram (7 percent) districts reported more about this. Long period of infertility was reported in Thiruchirapalli and Villupuram (8 percent each), Kancheepuram and Vellore (5 percent each), Madurai and Perambalur, Thiruvallur (6 percent each) districts. Previous bad obstetric history was reported in Thiruchirapalli (13 percent), Tirupur (7 percent), Thiruvallur and Thanjavur (6 percent), Thiruvannamalai and Villupuram (5 percent each) and Nagapattinam, Vellore Coimbatore, Madurai, Kancheepuram (4 percent each) districts. Obstetric morbidity due to failure of contraception was mostly reported in Nagapattinam (13 percent), Villupuram (9 percent), Thiruvannamalai (8 percent), Thiruchirapalli (7 percent), Dindigul, Thiruvallur and Pudukkottai (5 percent each).

Table 5.6: Maternal Morbidity Due to Obstetric Morbidity / Diseases in Tamil Nadu

Districts	Percentage of Women who had Obstetric Complications				
	Previous LSCS / Assisted	Long Period of Infertility	Previous Bad Obstetric History	Pregnancy Due to Contraception Failure	Handicapped Mother
Tamil Nadu	42.9	1.90	2.78	0.14	0.19
Ariyalur	1.50	1.70	2.10	1.10	3.10
Chennai	0.60	0.10	0.40	2.10	0.00
Coimbatore	3.60	2.50	3.70	3.20	1.50
Cudallore	2.90	3.50	3.20	3.20	2.30
Dharmapuri	2.00	1.70	1.40	4.20	1.50
Dindigul	1.60	3.20	1.60	5.30	2.30
Erode	1.40	1.70	0.40	4.20	0.80

Kancheepuram	6.50	5.10	3.80	3.20	6.90
Kanyakumari	4.70	2.50	0.70	3.20	2.30
Karur	2.10	1.10	1.20	2.10	1.50
Krishnagiri	1.90	4.00	3.20	1.10	3.80
Madurai	6.20	6.10	4.00	2.10	5.30
Nagapattinam	4.00	1.80	3.50	13.40	3.80
Namakkal	2.50	1.00	0.90	3.20	2.30
Nilgiris	0.80	0.80	0.60	1.10	0.00
Perambalur	2.10	6.30	2.40	1.10	0.80
Pudukkottai	1.20	1.70	0.80	5.30	1.50
Ramanathapuram	1.40	3.80	0.80	2.10	0.00
Salem	3.80	4.50	3.30	2.10	9.20
Sivagangai	2.50	2.10	2.00	0.00	2.30
Thanjavur	4.80	4.50	6.50	2.10	7.60
Theni	2.10	1.10	2.00	0.00	1.50
Thiruvallur	4.70	6.00	5.90	5.30	6.10
Thiruvallur	3.70	2.50	4.10	1.10	0.80
Thiruchirapalli	8.10	7.80	13.10	7.40	12.20
Tirunelveli	4.60	1.80	2.70	1.10	3.80
Tirupur	3.50	1.50	7.00	0.00	5.30
Thiruvannamalai	3.70	4.30	5.20	8.40	3.10
Thoothukudi	1.60	0.40	1.00	0.00	2.30
Vellore	3.20	5.00	4.40	1.10	0.80
Villupuram	4.50	7.70	5.40	9.10	4.50
Virudhunagar	2.20	2.20	2.70	1.10	0.80

3.2.4. OBSTETRIC / DELIVERY COMPLICATION

Maternal morbidity due to obstetric /delivery complications were reduced considerably due to the active implementation of the RCH programmes in Tamil Nadu. Cephalopelvic disproportion (CPD) cases were observed in Thiruvallur (14 percent), Erode and Krishnagiri (9 percent each), Tirupur (7 percent), Kancheepuram and Villupuram (6 percent each) and Vellore and Thiruvannamalai (5 percent each) districts. Intrauterine deaths account

Table 5.7: Maternal Morbidity Due to Obstetric Complications in Tamil Nadu

Districts	Percentage of women who experienced obstetric Complication						
	CPD	IUD	IUGR	Congenital malformation	Mal presentation	Ectopic Pregnancy	Post dated pregnancy
Tamil Nadu	1.04	0.20	0.30	0.35	0.78	0.09	1.04
Ariyalur	1.20	0.80	0.00	0.90	1.70	0.00	1.90
Chennai	0.10	0.80	1.00	4.30	0.00	5.20	0.10
Coimbatore	1.70	3.80	6.60	6.40	2.50	0.00	1.70
Cudallore	1.20	1.50	3.60	3.40	2.90	3.40	3.60
Dharmapuri	2.30	3.10	1.50	3.80	2.70	3.40	2.90
Dindigul	1.00	1.50	1.50	3.40	2.50	1.70	0.90
Erode	8.60	3.80	0.50	1.70	0.40	1.70	1.50
Kancheepuram	6.10	3.80	8.60	4.70	4.80	1.70	5.50
Kanyakumari	0.40	0.80	0.50	1.70	1.70	3.40	0.40
Karur	1.00	1.50	2.00	1.30	2.10	0.00	1.90
Krishnagiri	8.90	3.10	1.50	2.60	5.60	1.70	4.10
Madurai	1.50	6.90	4.10	3.40	3.70	10.30	1.90
Nagapattinam	4.40	2.30	3.00	1.30	3.90	6.90	2.80
Namakkal	4.50	3.10	1.50	1.70	2.50	3.40	4.50
Nilgiris	1.20	2.30	2.00	1.30	1.60	0.00	1.70
Perambalur	0.60	0.80	2.00	0.40	1.00	3.40	1.90
Pudukkottai	0.40	0.80	1.50	1.70	2.90	1.70	1.30
Ramanathapuram	0.30	0.80	0.00	0.00	1.00	1.70	1.20
Salem	3.80	3.80	1.50	3.00	4.10	1.70	6.30
Sivagangai	1.60	2.30	4.10	2.10	4.30	0.00	3.90
Thanjavur	1.70	2.30	3.60	3.00	0.60	3.40	1.60
Theni	0.10	1.50	1.50	0.00	0.80	1.70	1.60
Thiruvallur	14.10	7.70	6.10	6.00	8.70	3.40	7.70
Thiruvarur	0.40	0.80	2.00	2.10	1.70	6.90	2.00
Thiruchirapalli	3.80	5.40	10.20	3.80	6.20	8.60	2.50
Tirunelveli	1.50	0.00	2.00	2.10	1.40	1.70	1.90
Tirupur	7.40	4.60	1.50	4.70	4.10	3.40	4.10
Thiruvannamalai	5.20	10.00	6.60	9.40	5.20	1.70	7.10
Thoothukudi	0.60	1.50	1.00	0.40	1.00	0.00	1.70
Vellore	5.40	5.40	3.60	5.10	4.50	5.40	8.30
Villupuram	6.40	10.00	11.20	11.50	12.50	10.60	9.60
Virudhunagar	2.60	3.20	3.70	2.80	1.40	1.90	1.90

for only 0.02 percent of obstetric complication. Intrauterine deaths were more in Thiruvannamalai and Villupuram (10 percent each), Thiruvallur (8 percent), Madurai (7 percent), Thiruchirapalli and Tirupur (5 percent each) and Coimbatore, Erode, Kancheepuram (4 percent each).

Intrauterine growth retardation was prevalent in Villupuram (11 percent), Thiruchirapalli (10.20 percent), Coimbatore and Thiruvannamalai (7 percent each), Kancheepuram (9 percent) and Thiruvallur (6 percent). Congenital malformation was reported in Villupuram (11 percent), Thiru-

vannamalai (9 percent) and, Thiruvallur and Coimbatore (6 percent each) districts. Malpresentation of the child was identified in Villupuram (12.5 percent), Thiruvallur (9 percent), Thiruchirapalli and Krishnagiri (6 percent each) and Kancheepuram, Vellore and Thiruvannamalai (5 percent each) districts. One percent of the complications was due to ectopic pregnancy, mostly reported in Villupuram (11 percent), Madurai (10 percent), Thiruchirapalli (9 percent), Nagapattinam and Thiruvarur (7 percent each) and Chennai (5.2 percent). Post-dated pregnancy accounts one percent of the maternal morbidity. Women in Thiruvallur and Vellore (8 percent each), Thiruvannamalai (7 percent), Kancheepuram and Salem (6 percent each) and Namakkal (5 percent) districts reported more about postdated pregnancy (Table 5.7).

3.2.5. POSTNATAL COMPLICATIONS / MORBIDITIES AND OTHER PROBLEMS

Complications other than pregnancy morbidity, pregnancy complications, obstetric morbidity and obstetric complications were also reported. Table 5.8 explains these problems. Women in Vellore (10 percent), Kancheepuram, Thiruvannamalai and Thiruchirapalli (6 percent each), Madurai, Vellore and Tirupur (5 percent each), Thiruvallur (9 percent) reported about other complications

Kancheepuram	6.00	5.80
Kanyakumari	1.30	1.60
Karur	2.60	1.50
Krishnagiri	2.20	2.80
Madurai	4.60	3.90
Nagapattinam	2.80	1.10
Namakkal	2.60	1.90
Nilgiris	1.60	1.50
Perambalur	1.80	0.80
Pudukkottai	0.70	2.50
Ramanathapuram	1.00	1.10
Salem	3.90	3.70
Sivagangai	2.20	1.80
Thanjavur	3.20	1.60
Theni	2.00	1.40
Thiruvallur	8.60	5.70
Thiruvarur	1.70	1.10
Thiruchirapalli	6.30	5.60
Tirunelveli	2.30	1.80
Tirupur	4.80	3.70
Thiruvannamalai	5.50	8.00
Thoothukudi	0.80	2.30
Vellore	5.40	6.00
Villupuram	9.80	8.80
Virudhunagar	1.60	2.40

Table 5.8: Maternal Morbidity Due to Post Natal Complications in Tamil Nadu

District Name	Percentage of women who had experienced other complication and post natal complications.	
	Other Complications	Post Natal Complication
TAMIL NADU	3.1	5.54
Ariyalur	1.80	1.40
Chennai	1.00	1.10
Coimbatore	3.20	5.90
Cudallore	2.10	4.40
Dharmapuri	2.90	3.30
Dindigul	2.20	2.00
Erode	1.50	3.50

Postnatal complications account for five percent of the maternal morbidity. Postnatal morbidity is mostly noticed in Villupuram (9 percent), Thiruvannamalai (8 percent), Coimbatore, Kancheepuram, Thiruvallur, and Thiruchirapalli (6 percent each).

CONCLUSION

Tamil Nadu has considerably achieved its demographic goals particularly the maternal and child health through its programs and policies. But still care should be taken on certain components of maternal health particularly maternal morbidities to reach its goal. Data related to maternal morbidities are available from the two national-level

surveys, the DLHS and NFHS. These community-level surveys are based on the self-reported symptoms pertaining to selected morbidity conditions and also retrospective in nature. The results in the present study exhibit the actual morbidities experienced by women and the clinical symptoms assessed by the health care personnel. Therefore, the incidence of maternal morbidity may be some over and under diagnosis.

Maternal morbidity incidence rate of Tamil Nadu state is 6.56. Perambalur recorded the highest incidence rate and Chennai the lowest. Eastern belt consist of Karur, Perambalur, Thanjavur and the hilly Nilgiris shows high incidence rates. Obstetric morbidity are the major causes followed by pregnancy complications and pregnancy morbidity diseases. The findings of the study have some policy implications. Severe anemia and GDM are noticed as main causes for pregnancy morbidity. There is a need to identify any lacunae in implementing the programme at community level or at facility level and suitable intervention should be taken.

Multi para plays a significant place in pregnancy complication. Efforts ought to be taken for making family planning information and services more accessible to the individual and may be promoted. RH incompatibility which accounts for a considerable proportion in maternal complications, may be focused seriously. Previous LSCS/Assisted delivery and previous bad obstetric history contribute higher proportion in obstetric complications. There is a need to investigate the underlying causes of C-section deliveries and contraceptive failure. Above all, certain district-specific incidences/morbidities were also noticed. Appropriate interventions that reflect the local (district) needs should be framed and implemented according to current district-based best practice, and the progress should be monitored.

REFERENCES

- Doke, P.P, 2014, "Maternal morbidity and Estimates for community studies in India", *Journal of Medical Science*,1(2)56-64.
- International Institute for Population Sciences (IIPS) and Macro International.(2016). *National Family Health Survey-4*.(NFHS-4), Fact Sheet 2015- 16: India, Mumbai.
- International Institute for Population Sciences (IIPS) and Macro International (2010). *District Level Household and Facility Survey-3*.(DLHS-3), 2005- 06: India, Mumbai.
- International Institute for Population Sciences (IIPS) and Macro International.(2016). *District level household and Facility survey*, Fact Sheet, 2012- 13: India, Mumbai.
- Koblinsky.et.al. 2012, "Maternal morbidity and disability and their consequences: Neglected Agenda in Maternal Health" *Journal of Health Population and Nutrition*, (Jun 30 (2): 124-130.
- Mantal.C.D.,Buchman,E.,Rees.H.Pattinson.R.C.1998,"Severe acute maternal morbidity: A Pilot study of definition for a near-miss" *British Journal of Obstetrics and Gynaecology*, 1998.15(9):985-990
- United Nations, 2010, *Monitoring Emergency Obstetric Care: A hand book*, Geneva.
- Vanderkruik,R.C. , Tunçalp.O, Doris .D and Say,L, 2013, "Framing maternal morbidity: WHO scoping exercise", *BMC Pregnancy and Child-birth*2013, 13:213
- World Health Organization, 2013, *The International Classification of Diseases* 2015, Geneva.

APPENDIX – I

Explanation for Maternal morbidities	
Morbidity	Explanation
Teenage Pregnancy (<18 Yrs)	Pregnancy in a woman who is 19 years of age or under
Elderly Primi (>35 Yrs)	Woman who goes into pregnancy for the first time at the age of 35 years or older.
Short Primi (Below 145 cms)	Short stature refers to a height of a women which is below typical (<150 centimeters)
Severe Anaemia	Hemoglobin Anemia level < 7 gm/dl
PIH	Pregnancy-Induced Hypertension which may also be called pre-eclampsia, toxemia, or toxemia of pregnancy.
Eclampsia	Eclampsia is seizures (convulsions) in a pregnant woman.
Hydraminos - (Poly / Oligo)	Chronic polyhydramnios where excess amniotic fluid accumulates gradually.
APH	AntePartum Haemorrhage (APH) / PrePartum Hemorrhage, is genital bleeding during pregnancy.
Multipara	Multipara refers to a woman who has had two or more viable pregnancies.
Multiple Pregnancy	Women carrying multiple babies
Rh incompatibility	Rh incompatibility is a condition that occurs during the pregnancy if a woman has Rh-negative blood and her baby has Rh-positive blood
Previous LSCS / Assisted	Previous Lower Segment Caesarean Section (LSCS) delivery /Assisted delivery
Vesicular mole	Hydatidiform mole is a rare mass or growth that forms inside the womb (uterus) at the beginning of a pregnancy.
Long period of infertility	Infertility means not being able to get pregnant after one year of trying or six months if a woman is 35 or older. Women who can get pregnant but are unable to stay pregnant may also be infertile.
GDM	Gestational Diabetes Mellitus (GDM) is defined as any degree of glucose intolerance with onset or first recognition during pregnancy
Previous bad obstetric history	Occurrence of three or more consecutive pregnancy losses i.e., spontaneous abortions, early neonatal deaths, stillbirths, intrauterine fetal deaths, intrauterine growth retardation and congenital anomalies.
Ectopic Pregnancy	Pregnancy, is a complication of pregnancy in which the embryo attaches outside the uterus i.e., fallopian tube.
Malpresentation	Abnormal positioning of a fetus at the time of delivery.
Congenital malformation	A physical defect present in a baby at birth that can involve many different parts of the body, including the brain, heart, lungs, liver, bones, and intestinal tract.
CPD	Cephalopelvic Disproportion (CPD) occurs when a baby's head or body is too large to fit through the mother's pelvis
HIV	Human Immune Deficiency Virus (HIV)
IUD	Intra Uterine Death (IUD)
Post-dated pregnancy	A pregnancy that lasts more than 42 weeks (294 days since the first day of the last menstrual period) is considered post-term.
IUGR	Intra Uterine Growth Restriction refers to a condition in which an unborn baby is smaller than it should be because it is not growing at a normal rate inside the womb.

CHAPTER - 6

Comparative Analysis of Malnutrition among Children in Maharashtra

*Vini Sivanandan, Vandana Shivnekar, Raj Pol, Arun Pisal,
A.P. Prashikand Akram Khan*

Population Research Centre, GIPE, Pune

Abstract:

Maharashtra is one of the most progressive states in the country and being the second most populous state there exist huge variation in terms of both development and health. Malnutrition is one of the serious health problem and, if not detected and treated in early childhood, it may lead to lifelong health issues and contributes to a major cause of child mortality. In Maharashtra, for every ten children aged three or less, born to illiterate mothers, five children are underweight - too thin for their age. The main objective of this paper is to identify variation in terms of child health by background characteristics. Additionally, we investigate group of factors (measured at district -level) that can best predict the observed causes of child death. Using DLHS4 and ICDS (Integrated Child Development Schemes) MPR data set we compared the height and weight of children across regions. Risk ratio techniques, when applied shows regional variation in terms of prevalence of moderately and severely malnourished children. Prevalence of moderately underweight children to normal weight is much higher in rural areas than in urban areas. However, in rural areas, the prevalence of under weighed children is much higher among tribal groups than among other caste groups. Risk ratio, when comparing tribal groups with other caste groups in urban areas, shows severely malnourished children to be more in tribal areas than in urban areas in districts of Thane, Ahmednagar, Jalgaon, Amravati, and Dhule districts. The highest difference is observed in Amravati and Ahmednagar districts. Comparison of social groups, tribal with other castes in rural areas shows severely malnourished children are more in tribal groups than in other groups in selected districts of Maharashtra except in Chandrapur district. Analysis of prevalence within a district shows, in general, rural areas with below average prevalence as compared to the total average prevalence of severely and moderately underweight except in districts of Raigad, Aurangabad, Satara, Washim, Bhandara and Ratnagiri districts

INTRODUCTION

Millennium Development Goal aims to halve, between 1990 and 2015, the proportion of people who suffer from hunger as measured by the prevalence of underweight among under-5 years children. Child malnutrition is still a persisting issue, mainly in South Asia and sub-Saharan Africa alone, 121 million children are stunted (United Nations Children's Fund 2013). The UN estimates that 2.1 million Indian children die before reaching the age of five every year.

The state of Maharashtra consists of 35 districts, which are grouped into six administrative regions. In Maharashtra for every ten children aged three or less, born to illiterate mothers, five children are underweight - too thin for their age. This is a sign of acute and chronic malnutrition (Compared with: three children born to well-educated mothers)(NFHS III, 2005-06). There is a wide variation among districts in terms of socio-economic development. Out of the six administrative regions, children in Pune division fared better in terms of nutritional and health status whereas children from Amravati and Nashik divisions were worse off (CNSM, 2013). Disparities in terms of socio-economic development lead to differences in maternal and child health. Any future and present programme needs to consider the local needs and address the constrain at the local level for successful implementation of Maternal and Child health schemes. Therefore, it is important to detect malnutrition at early childhood for timely interventions and participation at community level.

The main objective of this paper is to identify regional variation in terms of prevalence of malnourished children. Additionally, we investigate group of factors (measured at district level) that can best predict the observed causes of child death.

Materials and Methods

We use DLHS4 (2012-13), ICDS (Integrated Child Development Schemes) MPR data set to estimate

and compare the height-weight of these children. Analysis was carried out using risk factors in a systematic way to assess the differential in terms of children weighed. This exposure of risk factors and their consequences is an important step in linking the causal determinants (regions in our study). We calculate the risk in terms of moderately underweight and severely underweight children to the total number of children weighed by regions such as tribal, rural and urban regions in all the districts of Maharashtra. Comparison of risk ration was done using relative risk ratios. In addition we use the prevalence method to examine the overall prevalence and to analyze the difference within a district as well as between the districts.

Malnourished Children in Maharashtra

Table 6.1: Risk Ratios for Moderately Underweight Children in age group 0-5 years, Maharashtra, 2013-14

Sr. No.	District	Risk Ratio : Moderately underweight children (0-5 yrs)		
		Urban to Rural	Tribal to Urban	Tribal to Rural
1	Nashik	5.0581	0.4381	2.2160
2	Nanded	3.5483	0.3054	1.0836
3	Thane	3.1132	1.0886	3.3891
4	Yavatmal	3.0977	0.3549	1.0992
5	Ahmednagar	2.5677	1.8437	4.7342
6	Pune	2.4713	0.4583	1.1327
7	Jalgaon	2.2348	0.6681	1.4931
8	Chandrapur	1.8328	0.5741	1.0522
9	Gondia	1.5823	0.7412	1.1727
10	Amaravati	1.3463	2.2684	3.0540
11	Dhule	0.7568	2.3922	1.8104
12	Sangli	5.8859	-	-
13	Kolhapur	4.1658	-	-
14	Jalna	3.6784	-	-
15	Nagpur	3.2497	-	-
16	Parbhani	2.8472	-	-
17	Latur	2.7944	-	-
18	Hingoli	2.5595	-	-
19	Beed	2.5107	-	-

20	Solapur	2.2672	-	-
21	Wardha	1.9121	-	-
22	Osmanabad	1.9078	-	-
23	Akola	1.8928	-	-
24	Ratnagiri	1.6698	-	-
25	Bhandara	1.6199	-	-
26	Washim	1.3373	-	-
27	Satara	1.3074	-	-
28	Aurangabad	1.1511	-	-
29	Buldhana	0.9361	-	-
30	Gadchiroli	-	0.9614	-
31	Raigad	-	-	1.7146

Source: Unit data records from MPR ICDS, Ministry of Women & Child Development 2013-14; India

Table 6.1 presents the relative risk ratio calculated in terms of prevalence of moderately underweight children (0-5 years) by region such as urban to rural; tribal to rural and urban to tribal region. As seen in the table, the prevalence of moderately underweight children to normal weight is much higher in urban areas than in rural areas and is observed in all the districts of Maharashtra except in Buldhana and Dhule districts. The primary reason of prevalence of moderately underweight children in urban areas as compared to rural areas is due to the concentration of children from urban slums.

The highest difference by region is observed in Nashik, Sangli, and Kolhapur districts where the risk ratio of severely underweight children was 5.0, 5.8 and 4.2 times more than rural areas, respectively. Whereas, in Aurangabad, Amravati, Washim and Satara districts the risk ratio was slightly more than one times in urban areas as compared to rural areas. Moderately underweight children were comparatively less in tribal areas as compared to urban areas, in almost all the districts of Maharashtra except in Thane, Ahmednagar, Amravati and Dhule districts. The highest value of risk ratio was from children in Dhule district with risk ratio of moderately underweight children in tribal areas to be almost 2.3 times than children in urban areas.

Comparative analysis of moderately underweight children in rural areas to children in tribal areas shows in selected districts the moderately underweight children are more in tribal areas than in rural areas. In Ahmednagar district, the number of moderately underweight children were 4.7 times more in tribal areas than in rural areas. Overall, urban areas shows more number of moderately underweight children, followed by children in tribal and rural areas.

Table 6.2: Risk Ratios for Severely Underweight Children in age group 0-5 years, Maharashtra, 2013-14

Sr. No.	District	Risk Ratio : Severely underweight children (0-5 yrs.)		
		Urban to Rural	Tribal to Urban	Tribal to Rural
1	Nashik	3.0359	0.9129	2.7716
2	Nanded	1.2121	0.8402	1.0184
3	Thane	1.2556	4.0120	5.0373
4	Yavatmal	2.1265	0.6029	1.2820
5	Ahmednagar	2.0658	4.2683	8.8173
6	Pune	2.4934	0.4497	1.1213
7	Jalgaon	0.3699	3.9399	1.4574
8	Chandrapur	1.5332	0.5828	0.8935
9	Gondia	1.5311	0.7402	1.1333
10	Amaravati	0.8533	6.9265	5.9107
11	Dhule	0.5562	3.5298	1.9632
12	Sangli	6.7000	-	-
13	Kolhapur	3.6077	-	-
14	Jalna	5.2302	-	-
15	Nagpur	2.0605	-	-
16	Parbhani	1.1921	-	-
17	Latur	2.2864	-	-
18	Hingoli	3.6671	-	-
19	Beed	2.1043	-	-
20	Solapur	2.2303	-	-
21	Wardha	0.7203	-	-
22	Osmanabad	2.1569	-	-
23	Akola	1.2874	-	-
24	Ratnagiri	3.2986	-	-

25	Bhandara	1.6690	-	-
26	Washim	1.0582	-	-
27	Satara	0.8511	-	-
28	Aurangabad	0.8748	-	-
29	Buldhana	0.9757	-	-
30	Gadchiroli	-	2.5569	-
31	Raigad	-	-	2.3983

Source: Unit data records from MPR ICDS, Ministry of Women & Child Development 2013-14; India

Table 6.2 depicts the risk ratio of severely underweight children to children with normal weight in all the districts of Maharashtra. As seen in the above table, severely underweight children are more in urban areas than in rural areas and is observed in all districts of Maharashtra except in Satara, Aurangabad, Buldana, Wardha, Dhule, Amravati and Jalgaon districts of Maharashtra. The highest prevalence of severely malnourished children was found in Sangli district followed by Jalna district where the number of severely malnourished children was 6.7 times more in urban areas than in rural areas and 5.2 times more in urban areas than in rural areas respectively. The lowest difference between urban and rural areas was observed in Washim district with presence of severely undernourished children in rural and urban areas to be more or less same.

Risk ratio when compared with tribal to urban areas shows severely malnourished children to be more in tribal areas than in urban areas in districts of Thane, Ahmednagar, Jalgaon, Amravati and Dhule districts. The highest difference was observed in Amravati district where the number of severely malnourished children was 6.9 times more in tribal areas than in urban areas followed by Ahmednagar district where the number of severely malnourished children was 4.2 times more in tribal areas than in urban areas.

Comparison of Tribal to rural areas shows severely malnourished children are more in tribal areas than in rural areas in selected districts of Maharashtra except in Chandrapur district. Tribal to ru-

ral difference, was highest in Ahmednagar district, with number of severely malnourished children, 8.8 times more in tribal areas than in rural areas. This is followed by Amravati district with number of severely malnourished children 5.9 times more in tribal areas than in urban areas, and Thane district, with number of severely malnourished children 5.0 times more in tribal areas than in urban areas. Overall, tribal and urban areas shows more number of severely malnourished children than in rural areas.

Table 6.3: Prevalence of Risk Ratios for Moderately and severely Underweight Children in age group 0-5 years, by districts of Maharashtra

Sr. No.	District	Prevalence of Risk Ratio: Moderately underweight children (0-5 yrs)			
		Rural	Urban	Tribal	Total
1	Nashik	0.06	0.29	0.14	0.12
2	Nanded	0.06	0.18	0.06	0.07
3	Thane	0.07	0.20	0.25	0.18
4	Yavatmal	0.09	0.28	0.11	0.15
5	Ahmednagar	0.04	0.10	0.22	0.05
6	Pune	0.07	0.16	0.08	0.09
7	Jalgaon	0.09	0.18	0.14	0.11
8	Chandrapur	0.11	0.19	0.11	0.12
9	Gondiya	0.08	0.13	0.10	0.09
10	Amravati	0.12	0.15	0.40	0.16
11	Dhule	0.11	0.08	0.20	0.12
12	Sangli	0.03	0.20	-	0.05
13	Kolhapur	0.05	0.20	-	0.07
14	Jalna	0.06	0.25	-	0.07
15	Nagpur	0.06	0.18	0.06	0.07
16	Parbhani	0.06	0.16	-	0.07
17	Latur	0.08	0.25	-	0.13
18	Hingoli	0.09	0.25	-	0.10
19	Beed	0.07	0.17	-	0.08
20	Solapur	0.07	0.16	-	0.08
21	Wardha	0.08	0.14	-	0.09
22	Osmanabad	0.09	0.17	-	0.09
23	Akola	0.07	0.13	-	0.08

24	Ratnagiri	0.08	0.15	-	0.09
25	Bhandara	-	-	-	-
26	Washim	0.09	0.12	-	0.09
27	Satara	0.09	0.11	-	0.09
28	Aurangabad	0.09	0.10	-	0.09
29	Buldhana	0.05	0.08	-	0.05
30	Gadchiroli	-	0.21	0.24	0.22
31	Raigad	0.05	-	0.08	0.04

Source: Unit data records from MPR ICDS, Ministry of Women & Child Development 2013-14; India

Overall, the prevalence of moderately underweight children and severely underweight children shows highest prevalence in urban areas, followed by tribal and rural regions in almost all the districts of Maharashtra except in Gadchiroli, Raigad, Amravati, Dhule, Ahmednagar, and Thane districts which shows prevalence of severely and moderately malnourished children high in tribal areas followed by urban areas and rural areas. The highest prevalence of severely and moderately malnourished children in tribal areas was observed in Amravati district; in urban areas in Nasik district and in rural areas in Amravati district. Overall, the prevalence of severely and moderately malnourished children was highest in Gadchiroli district and lowest in Raigad district.

Table 6.4: Prevalence of Risk Ratios within Districts for Moderately and Severely Underweight Children in age group 0-5 years, by districts of Maharashtra

Sr. No.	District	Prevalence of Risk Ratio: Underweight children (0-5 yrs)			
		Rural	Urban	Tribal	Total
1	Nashik	-0.01	0.11	-0.01	0.12
2	Nanded	-0.01	0.11	-0.01	0.07
3	Thane	-0.11	0.02	-0.07	0.18
4	Yavatmal	-0.06	0.12	-0.05	0.15
5	Ahmednagar	-0.01	0.05	-0.16	0.05
6	Pune	-0.03	0.07	-0.02	0.09

7	Jalgaon	-0.01	0.08	0.03	0.11
8	Chandrapur	-0.01	0.08	0.00	0.12
9	Gondiya	-0.01	0.04	0.01	0.09
10	Amravati	-0.04	-0.01	0.24	0.16
11	Dhule	-0.01	-0.04	0.08	0.12
12	Sangli	-0.01	0.15	-	0.05
13	Kolhapur	-0.02	0.13	-	0.07
14	Jalna	-0.01	0.17	-	0.07
15	Nagpur	-0.05	0.12	-	0.07
16	Parbhani	-0.01	0.09	-	0.07
17	Latur	-	-	-	0.13
18	Hingoli	-0.01	0.15	-	0.10
19	Beed	-0.01	0.09	-	0.08
20	Solapur	-0.01	0.08	-	
21	Wardha	-0.01	0.05	-	
22	Osmanabad	0.00	0.08	-	
23	Akola	0.01	0.05	-	
24	Ratnagiri	0.00	0.06	-	
25	Bhandara	0.00	0.03	-	
26	Washim	0.00	0.03	-	
27	Satara	0.00	0.02	-	
28	Aurangabad	0.00	0.01	-	
29	Buldhana	-	-	-	
30	Gadchiroli	-	-0.01	0.02	
31	Raigad	0.03	0.00	-	

Source: Unit data records from MPR ICDS, Ministry of Women & Child Development 2013-14; India

Table 6.4 shows analysis of within inequality by prevalence of severely and moderately malnourished children. In general, rural areas shows value with below average prevalence of severely and moderately malnourished children as compared to average prevalence of respective districts except in districts of Raigad, Aurangabad, Satara, Washim, Bhandara and Ratnagiri districts. In urban areas only Amravati, Dhule and Gadchiroli districts show prevalence of severely and moderately malnourished children below average as compared to the average total. Nasik, Nanded, Yavatmal and Pune districts show prevalence of severely and moderately malnourished children below average as compared to the average total.

Table 6.5: Prevalence of Risk Ratios for Moderately and Severely Underweight Children in age group 0-5 years, across districts of Maharashtra

Sr. No.	District	Prevalence of Risk Ratio: Moderately underweight children (0-5 yrs)			
		Rural	Urban	Tribal	Total
1	Nashik	0.01	0.01	-0.01	-0.02
2	Nanded	-0.03	-0.01	0.12	-0.05
3	Thane	0.00	0.02	0.06	0.07
4	Yavatmal	0.02	0.10	-0.08	0.05
5	Ahmednagar	-0.03	-0.07	0.03	-0.05
6	Pune	-	-0.01	-0.11	-0.01
7	Jalgaon	0.02	0.01	-0.05	0.00
8	Chandrapur	0.04	0.02	-0.08	0.01
9	Gondiya	0.01	-0.04	-0.09	-0.02
10	Amravati	0.04	-0.03	0.21	0.05
11	Dhule	0.04	-0.10	0.01	0.02
12	Sangli	-0.04	0.03	-	-0.06
13	Kolhapur	-0.02	0.02	-	-0.04
14	Jalna	-0.01	0.07	-	-0.03
15	Nagpur	-0.01	0.01	-	-0.13
16	Parbhani	-0.01	-0.02	-	-0.04
17	Latur	0.01	0.07	-	0.02
18	Hingoli	0.02	0.07	-	-0.01
19	Beed	0.00	0.00	-	-0.03
20	Solapur	0.00	-0.01	-	-0.02
21	Wardha	0.01	-0.03	-	-0.02
22	Osmanabad	0.12	-0.05	-	0.01
23	Akola	0.00	-0.05	-0.03	-
24	Ratnagiri	0.01	-0.02	-	-0.02
25	Bhandara	-0.02	-0.10	-	-0.06
26	Washim	0.02	-0.06	-	-
27	Satara	0.02	-0.07	-	-0.02
28	Aurangabad	0.02	-0.07	-	-0.02
29	Buldhana	-	-	-	-
30	Gadchiroli	-	0.04	0.05	0.11
31	Raigad	-0.01	-	-0.11	-0.01

Source: Unit data records from MPR ICDS, Ministry of Women & Child Development 2013-14

Between analysis of districts shows that prevalence of severely and moderately malnourished children are high in all the districts except in Raigad, Bhandara, Sangli, Kolhapur, Nagpur, Parbhani, Jalna, Ahmednagar and Nanded districts.

In urban areas, Nanded, Ahmednagar, Pune, Gondia, Amravati, Dhule, Solapur, Warda, Osmanabad, Akola, Ratnagiri, Bhandara, Washim, Satara and Aurangabad show below average prevalence of severely and moderately malnourished children as compared to the overall prevalence. In Tribal areas, Nanded, Thane, Dhule, Ahmednagar and Gadchiroli shows above average prevalence of severely and moderately malnourished children as compared to the overall tribal prevalence. Thane, Yavatmal, Gadchiroli, Latur, Dhule, Amravati, Jalgaon and Pune show above average prevalence of severely and moderately malnourished children as compared to the overall total prevalence. In districts of Gadchiroli, Thane and Latur all the area shows above average prevalence of severely and moderately malnourished children as compared to the total prevalence of the respective areas.

Assessment of Malnourished Children

The above analysis assessed the regional differential in terms of severely and moderately malnourished children. Further, to examine the differential in terms of child growth by socio-economic characteristics we used the DLHS4 (2012-13) unit-level data for Maharashtra.

Table 6.6 presents the distribution of severely and moderately malnourished children (males) in the age group of two to five years. Number of moderately and severely malnourished children was estimated by adopting new World Health Organization (WHO) Growth Reference Standards (GRS), which determines the weight for height (2-5 years).

As evident from Table 6, rural-urban divide is striking with more number of malnourished chil-

dren in rural areas than in urban areas. Further, the rural-urban gap is striking among moderately malnourished children as compared to severely malnourished children. It also indicates more number of male children are malnourished as compared to female children.

Table 6.6: Distribution of Moderately and Severely Malnourished Children by gender and areas, Maharashtra, 2013-14

	Severely malnourished		Moderately malnourished	
	Rural	Urban	Rural	Urban
Male	589	327	974	597
Female	269	189	456	321
Total	858	516	1430	918

Source: Unit data records DLHS4, 2013-14

Figure 6.1 Percent Distribution of Moderately and Severely Malnourished Children by Social Groups, Maharashtra, 2013-14

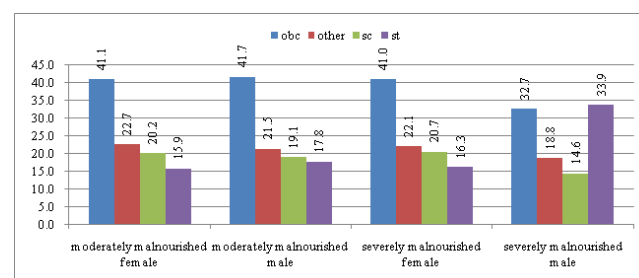


Source: Unit data records DLHS4, 2013-14

Figure 6.1 presents the male-female distribution of moderately and severely malnourished children by social groups. Overall, the percentage of severely and moderately malnourished male child is much higher than female child among all the social groups. However, the male-female difference is striking among the social group STs, whereas the male-female difference was least in case of severely malnourished children among SCs.

Figure 6.2 presents the distribution of moderately and severely malnourished children by social groups. Caste OBCs shows highest percent of se-

Figure 6.2 Percent distribution of Moderately and Severely Malnourished Children by Social Groups, Maharashtra, 2013-14



Source: Unit data records DLHS4, 2013-14

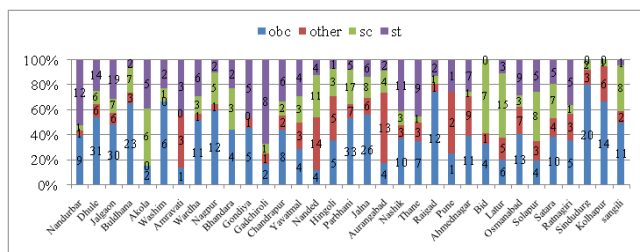
verely and moderately malnourished children for both male and female followed by caste Others and STs. However, it was among social group STs that highest percentage of severely malnourished male child is observed. This indicates, differential by background characteristics in terms of male and female and by social groups.

Figures 6.1-2, presents the overall distribution of malnourished children by social groups and place of residence. The present study also aims to cover the administrative division in Maharashtra and to identify the region with high proportion of malnourished children. Anemia is one of the indicators used to identify severely and moderately malnourished children. Anemic mother and child is of serious concern and if not attended or treated may lead to serious health issues. The CAB file in DLHS4 (12-13) gives results of blood test with severe anemic (HB<7) and Anemic cases (<11).

Figure 6.3 shows distribution of number of children in the age group 0-5 years with severe anemia (Hb< 7) and residing in rural areas by social groups in all the districts of Maharashtra. Overall, the caste OBCs shows proportionately high number of severe anemic children and is highest in Dhule, Jalgaon, and Parbhani districts and least in Akola, Pune and Gadchiroli districts. Among caste Others, the highest proportionate of severe anemic children was observed in Nanded and Aurangabad districts; and among caste SCs the highest propor-

tionate of severe anemic children was observed in Parbhani and Latur districts. Whereas, in case of caste ST, the highest number of severe anemic children was observed in Buldhana, Dhule and Nandurbar districts of Maharashtra.

Figure 6.3: Distribution of Moderately and Severely Malnourished Children in rural areas by Social Groups, Maharashtra, 2013-14



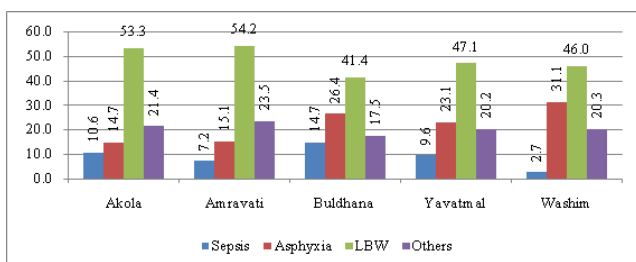
Source: Unit data records DLHS4, 2013-14

In urban areas as shown in figure 3 the number of children with severe anemia was highest among caste OBCs in Nandurbar and Jalgaon districts; and among caste others in Nandurbar and Mumbai districts; Whereas among caste SCs the highest number of severe anemic children was from Thane and Satara districts and among STs in Jalgaon district.

Causes of Infant and Child Deaths

This section examines the causes of Infant and Child deaths across Maharashtra during the year 2013-14. Data from Health monitoring information system was used to examine and identify the main causes of infant and child deaths across various regions in Maharashtra.

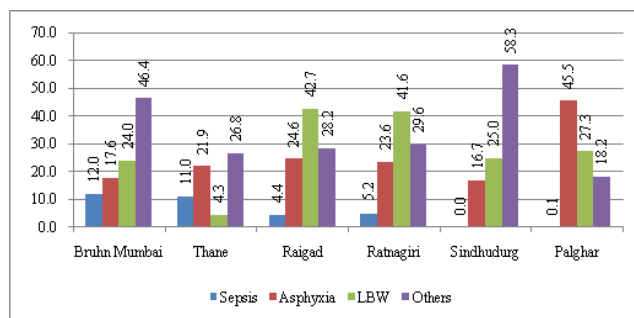
Figure 6.4: Percent distribution by causes of Infant Deaths in Vidarbha -Amravati region



Source: HMIS (2013-14)

Low birth weight was the prominent reason for the deaths of infant in Vidarbha-Amravati region with about half of death due to low birth weight in Amravati and Akola districts. Near about quarter of deaths was also due to asphyxia in Yavatmal and Buldhana districts whereas in Washim district the death due to asphyxia was 30 percent.

Figure 6.5: Percent distribution by causes of Infant Deaths in Konkan-region



Deaths due to other reasons was 46 percent in Brihanmumbai and 58 percent in Sindudurg district. Near about 40 percent of deaths in Raigad and Ratnagiri districts were due to low birth weight and 45 percent of deaths in Palghar district was due to asphyxia. In Thane district, a little more than quarter of deaths was due to other reasons whereas 21 percent of deaths was due to asphyxia. death due to low birth weight was comparatively smaller in Konkan region as compared to other regions and was only 4 percent in Thane district which is the lowest percent of death due to low birth weights as compared to other districts.

Deaths due to low birth weight were highest in Aurangabad and Jalna districts with 60 and 65 percent respectively. Deaths due to other reasons were 55 percent in Beed; 50 percent in Parbhani, and 45 percent in Hingoli districts. Osmanabad district also shows a significantly high number of deaths due to low birth weight with 42 percent and 41 percent of infant deaths Nanded district is due to asphyxia.

Figure 6.6: Percent distribution by causes of Infant Deaths in Marathwada region

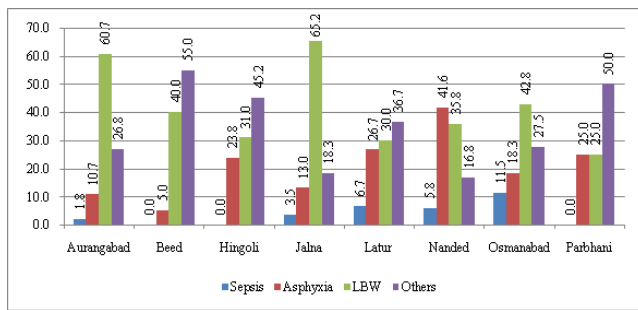
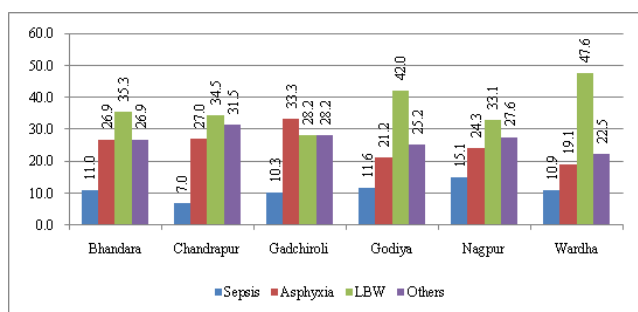
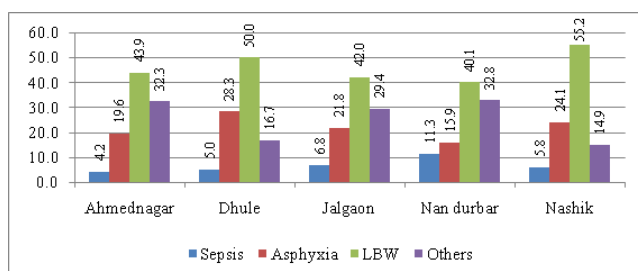


Figure 6.7: Percent distribution by causes of Infant Deaths in Vidarbha-Nagpur region



As seen above in figure 6.7 except Gadchiroli district all the other districts in Vidharba –Nagpur region show high percent of death due to low birth weight which ranges to nearly 35 percent in districts of Bhandara, Chandrapur and Nagpur to 42 percent in Gondiya and 47 percent in Wardha districts. In Gadchiroli district, deaths due to asphyxia with 33 percent was the major cause of infant death.

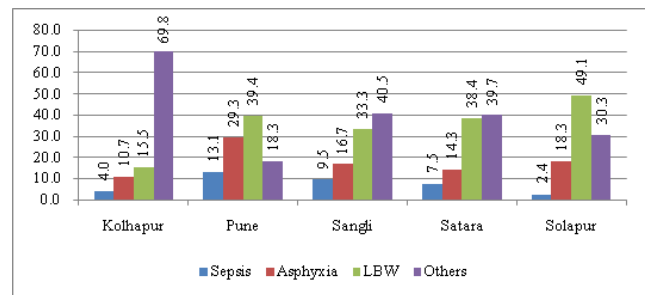
Figure 6.8: Percent distribution by causes of Infant Deaths in Kandesh region



Kandesh region also shows high infant deaths due to low birth weight. Half of infant deaths in Dhule district and 55 percent of deaths in Nashik district were due to low birth weight. Whereas, near about

40 percent of deaths in Ahmednagar, JAalgaon and Nandurbar districts was due to low birth weight.

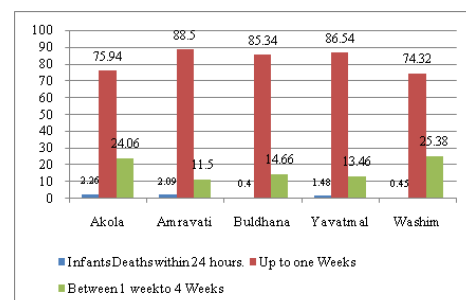
Figure 6.9: Percent distribution by causes of Infant Deaths in Desh -Pune region



In Desh-Pune region deaths due to other reasons were prominent and was highest in Kolhapur district with 69 percent to near about 40 percent in Sangli and Satara districts. Near about 40 percent of deaths in Solapur and Pune districts were due to low birth weight. Significantly, 29 percent of deaths in Pune district was due to asphyxia. Overall, Infant deaths was mainly due to low birth weight, asphyxia and other reasons. With proper ANC care infant deaths due to low birth weight can be reduced.

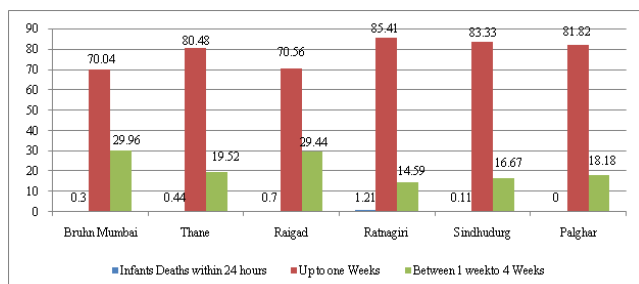
The following figure presents the death of infants in different regions of Maharashtra. Death within 24 hours and within one week after birth is mainly due to genetic reasons. Whereas death between 1 week of birth to one month of birth can be reduced by timely medical intervention.

Figure 6.10: Percent distribution by Neonate deaths by time and In Vidarbha region of Maharashtra, 2013-14



More than three-fourth of deaths among infants is within first week of birth in Washim and Akola districts. Neonatal deaths is much higher in Amravati, Buldana and Yavatmal districts with 88, 85 and 86 percent respectively.

Figure 6.11: Percent distribution by causes of Neonatal deaths by time and In Konkan region of Maharashtra, 2013-14



In Konkan region death of infant within 24 hours of birth is negligible. Death of neonates is highly concentrated in first week of birth. In Brihan Mumbai and Raigad more than quarter of deaths is reported between 1 week and upto 4 weeks of birth. Death upto one week was highest in Thane, Sindhudurg, Ratnagiri and Palghar district with 80, 85, 83 and 81 percent respectively.

Figure 6.12: Percent distribution by causes Neonate deaths by time and In Vidarbha – Nagpur region of Maharashtra, 2013-14

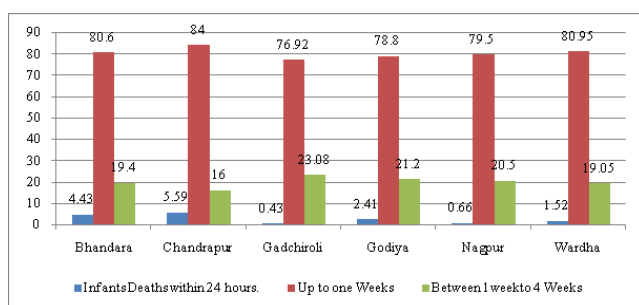


Figure 6.12 shows more than three-fourth of deaths among infants are during the first week of births and is uniformly observed in all the districts and is highest in Bhandara, Wardha and Chandrapur districts with 80 and 84 percent respectively. Infant deaths can be much lower if proper ANC and neonate treatment is provided in a timely manner.

Figure 6.13: Percent wise causes of Neonate deaths by time and In Kandesh region of Maharashtra, 2013-14

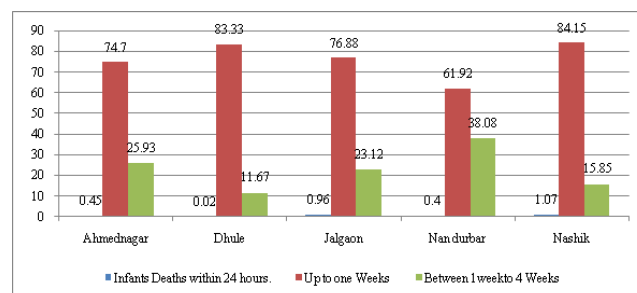


Figure 6.13 shows in Kandesh except for Nandurbar districts more than 75 percent of the neonatal deaths is during first week of birth. In Nandurbar district substantial number of deaths occur between 1 week to 4 weeks of birth.

Figure 6.14: Percent distribution by causes of Neonate deaths by time and In Desh-Pune region of Maharashtra, 2013-14

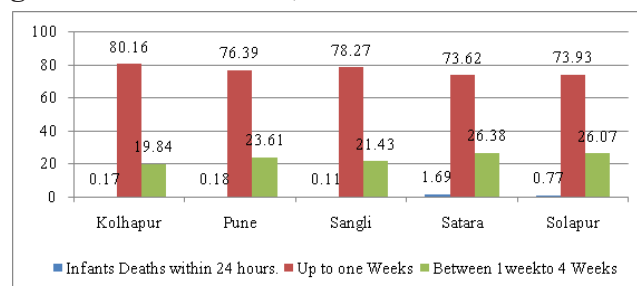


Figure 6.14 shows more than three-fourth of deaths among neonates are during the first week of births. The highest percent of neonatal death was observed in Kolhapur district. More than a quarter of deaths in Satara and Solapur districts are between 1 week to 4 week.

Figure 6.15: Percent distribution by causes of Neonate deaths by time and In Marathwada region of Maharashtra, 2013-14

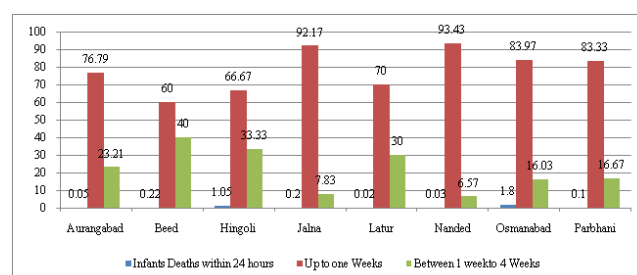


Figure 6.15 shows in Marathwada region in Jalna and Nanded districts nearly 90 percent of the deaths occur during the first week of birth whereas in Hingoli district, 66 percent of neonatal death happens in the first week of birth. Substantial number of deaths with 40 percent in Beed district and 33 percent in Hingoli happen between 1 week to 4 weeks.

Figure 6.16: Percent distribution by Neonate deaths by time and In Vidarbha-Amravati region of Maharashtra, 2013-14

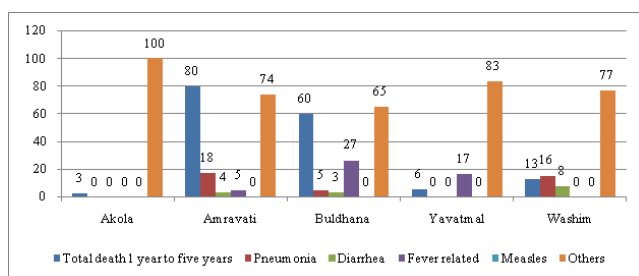
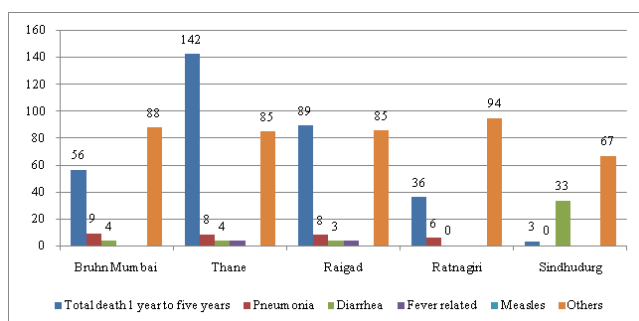


Figure 6.16 presents the causes of deaths among children 0-6 years in Vidarbha-Amravati region. Death of the children 0-6 years was least in Akola district with three deaths. Highest number of deaths of 80 was observed in Amravati district followed by 60 deaths in Buldhana district. In both of these districts, near about 74 percent deaths in Amravati and 65 percent deaths in Buldhana district were due to other reasons.

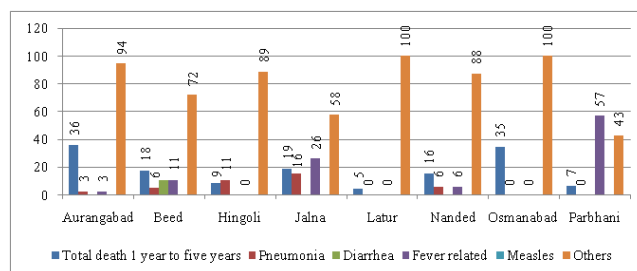
Figure 6.17: Percent distribution by of Neonate deaths by time and In Konkan region of Maharashtra, 2013-14



In Konkan region, the number of child deaths as shown in figure 6.17 above shows highest number of child deaths in Thane district with 142 child

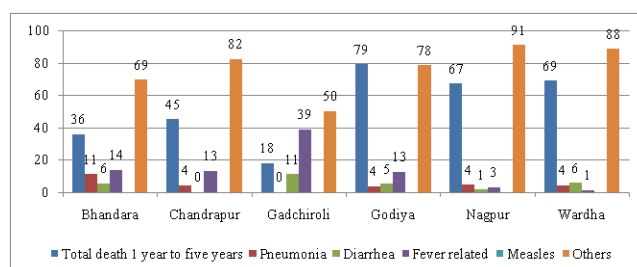
deaths followed by Raigad district with 89 deaths and the lowest number of deaths was observed in Sindudurg district with 3 deaths. Deaths due to other reasons was the major cause and it was observed in all the districts of Konkan region.

Figure 6.18: Percent distribution by of Neonate deaths by time and In Marathwada region of Maharashtra, 2013-14



Deaths in Marathwada region were lesser as compared to other regions and in all the districts the major cause of deaths was other reasons as shown in figure 18 above.

Figure 6.19: Percentage of Neonatal deaths by time and In Vidarbha region of Maharashtra, 2013-14



In Vidarbha – Nagpur region deaths due to other reasons were highest in all the districts. Highest number of deaths was reported in Wardha and Nagpur districts with 69 and 67 deaths respectively and the lowest number of deaths was observed in Gadchiroli district with 18 deaths. Substantial number of deaths was due to fever-related problems in Gadchiroli district.

Figure 6.20 shows Kandesh regions with highest

number of child death of 176 observed in Nandurbar followed by Ahmednagar with 136 deaths and lowest number of deaths was reported in Dhule with 18 deaths. In all the districts the major cause of deaths was other reasons. Significant number of deaths was also observed due to pneumonia with 35 percent of deaths in Nashik and 20 percent of deaths in Nandurbar district.

Figure 6.20: Percent distribution by of Neonate deaths by time and In Kandesh region of Maharashtra, 2013-14

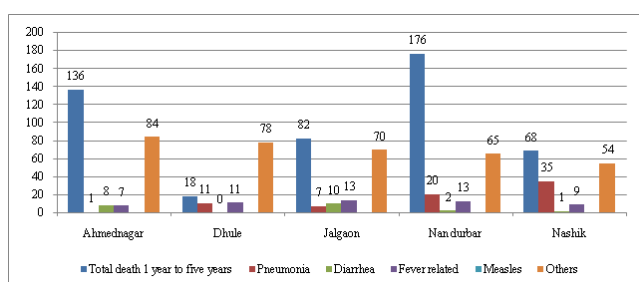


Figure 6.21: Percent distribution by of Neonate deaths by time and In desh-pune region of Maharashtra, 2013-14

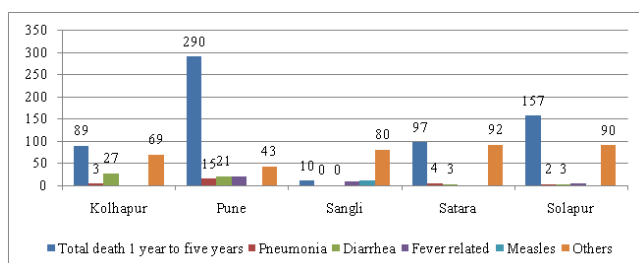


Figure 6.21 shows Pune district of Desh-Pune region with highest number of child deaths of 290 followed by Solapur with 157 deaths. The lowest number of deaths in this region was observed in Sangli district. Death due to other reason was the major cause of child deaths in this region.

Summary of the Findings

Prevalence of moderately underweight children to normal weight is much higher in urban areas than in rural areas and is observed in all the districts of Maharashtra except in Buldhana and Dhule districts. The highest difference by areas shows

in Nashik, Sangli, and Kolhapur districts the risk ratio of severely underweight children in urban areas was 5.0, 5.8 and 4.2 times more than rural areas respectively. In Ahmednagar district, the number of moderately underweight children was 4.7 times more in tribal areas than in rural areas. Similarly, in Thane and Amravati districts it was 3 times more in tribal areas than in rural areas.

Risk ration of severely underweight children to normally weighed children shows severely underweight children are more in urban areas than in rural areas and is observed in all districts of Maharashtra except in Satara, Aurangabad, Buldana, Wardha, Dhule, Amravati and Jalgaon districts. The highest prevalence of severely malnourished children was found in Sangli and Jalna district. The lowest difference between urban and rural areas was observed in Washim district with presence of severely undernourished children in rural and urban areas to be more or less same.

Risk ratio when compared with tribal to urban areas shows severely malnourished children to be more in tribal areas than in urban areas in districts of Thane, Ahmednagar, Jalgaon, Amravati and Dhule districts. The highest difference was observed in Amravati and Ahmednagar districts. Comparison of tribal to rural areas shows severely malnourished children are more in tribal areas than in rural areas in selected districts of Maharashtra except in Chandrapur district. The highest difference was observed in Ahmednagar district where the severely malnourished children was 8.8 times more in tribal areas than in rural areas. Overall, tribal and urban areas shows more number of severely malnourished children than in rural areas.

Analysis of prevalence within a district shows in general rural areas with below average prevalence as compared to the total average prevalence of severely and moderately underweight except in districts of Raigad, Aurangabad, Satara, Washim, Bhandara and Ratnagiri districts. In urban areas, only districts of Amravati, Dhule and Gadchiroli

show prevalence of below average severely and moderately underweight children as compared to the average total of respective districts. Nasik, Nanded, Yavatmal and Pune districts show below average prevalence as compared to the average total. Between analysis of districts shows that prevalence of malnourished children in rural areas are high in all the districts except in Raigad, Bhandara, Sangli, Kolhapur, Nagpur, Parbhani, Jalna, Ahmednagar and Nanded districts. In urban areas Nanded, Ahmednagar, Pune, Gondia, Amravati, Dhule, Solapur, Warda, Osmanabad, Akola, Ratnagiri, Bhandara, Washim, Satara and Aurangabad show below average prevalence as compared to the overall urban prevalence.

In Tribal areas Nanded, Thane, Dhule, Ahmednagar and Gadchiroli shows above average prevalence of severely and moderately underweight children as compared to the overall tribal areas. Thane, Yavatmal, Gadchiroli, Latur, Dhule, Amravati, Jalgaon and Pune show above average prevalence of severely and moderately underweight children as compared to the overall total prevalence. In districts of Gadchiroli, Thane and Latur irrespective of the areas it shows above average prevalence of severely and moderately underweight children as compared to the total prevalence of the respective areas.

Low birth weight was the prominent reason for infant deaths in all the regions of Maharashtra. Deaths due to other reasons and asphyxia were also prominent cause of child deaths. Death due to low birth weight was comparatively smaller in Konkan region as compared to other regions. Deaths due to low birth weight is highest in Aurangabad and Jalna districts with 60 and 65 percent respectively. Except for Gadchiroli district all the other districts in Vidharba –Nagpur region show high percent of death due to low birth weight. In Desh-Pune region deaths due to other reasons were prominent. More than three-fourth of deaths among infants are during the first week of births. Infant deaths can be much lower if proper ANC and neonatal

treatment is provided in a timely manner. In Konkan region, death of infant within 24 hours of birth is negligible. Death of neonates is highly concentrated in first week of birth. In Brihan Mumbai and Raigad, more than a quarter of deaths is reported between 1 week and upto 4 weeks of birth.

To improve the maternal and child health in ICDS health position need to be filled up on priority basis. Once the low birth baby is identified, a routine mechanism should be in place to track the health of child and mother. Awareness programme need to be organised on frequent basis and keeping in mind with the local need. In the tribal areas, some months family move to other places for work. It is not possible to track the sick children identified during that time. ICDS programme need to be further widened and not confined to only children in a particular age group. Local needs and availability need to be taken in providing nutrients to children.

References:

- Greenland S: Bias in Methods for Deriving Standardized Morbidity Ratio and Attributable Fraction Estimates. *Statistics in Medicine* 1984, 3:131–141.
- IIPS. National Family Health Survey (NFHS) – 3 – 2005-2006: national fact sheet, India. International Institute of Population Sciences, Mumbai; 2007<http://www.rchiips.org/NFHS/pdf/India.pdf>.
- Uter W, Pfahlberg A: The Application of Methods to Quantify Attributable Risk in Medical Practice. *Statistical Methods in Medical Research* 2001, 10:231–237.
- WHO. Infant and young child feeding: model chapter for textbooks for medical students, and allied health professionals. WHO. (Ch 1)WHO Press, World Health Organization, ; 2009:22 (20 Avenue Appia, 1211 Geneva 27, Switzerland).

Appendix
Population per Anganwadi (0-6 yrs) by districts and by regions of Maharashtra, 2013-14

Sr. No.	District	Population per Anganwadi (0-6 yrs)			Pregnant & Lactating Women per Anganwadi		
		Rural	Urban	Tribal	Rural	Urban	Tribal
1	Nashik	92	130	77	14	20	12
2	Nanded	77	120	56	13	19	9
	Nandurbar	--	--	71	--	--	11
3	Thane	115	103	65	37	11	10
4	Yavatmal	91	87	59	15	15	10
5	Ahmednagar	73	93	85	11	13	6
6	Pune	74	101	51	11	12	7
7	Jalgaon	88	193	94	13	31	16
8	Chandrapur	52	98	51	8	13	8
9	Gondia	68	89	53	11	12	9
10	Amaravati	58	93	75	10	17	12
11	Dhule	93	126	80	14	19	12
12	Sangli	62	103	--	10	14	--
13	Kolhapur	60	98	--	9	13	--
14	Jalna	95	117	--	15	19	--
15	Nagpur	71	99	--	11	13	--
16	Parbhani	83	136	--	14	20	--
17	Latur	76	151	--	13	22	--
18	Hingoli	102	139	--	16	21	--
19	Beed	93	129	--	15	8	--
20	Solapur	69	100	--	11	12	--
21	Sindudurg	31	--	--	4	--	--
22	Wardha	53	86	--	9	13	--
23	Osmanabad	71	109	--	13	15	--
24	Akola	78	111	--	14	15	--
25	Ratnagiri	36	95	--	5	11	--
26	Bhandara	68	83	--	12	16	--
27	Washim	89	108	--	15	16	--
28	Satara	43	90	--	6	9	--
29	Aurangabad	89	139	--	15	8	--
30	Buldhana	76	112	--	13	17	--
31	Gadchiroli	--	66	41	--	10	7
32	Mumbai	84	--	--	7	--	--
33	Raigad	52	--	54	8	--	9

Sr. No.	District	No. of SNP Beneficiaries (Recipients)			No. of PSE Beneficiaries (3 to 6 years)		
		Rural	Urban	Tribal	Rural	Urban	Tribal
1	Nashik	71	117	62	37	60	32
2	Nanded	65	107	49	39	48	28
	Nandurbar	--	--	59	--	--	32
3	Thane	56	63	48	38	37	29
4	Yavatmal	73	68	51	37	30	28
5	Ahmednagar	58	78	37	31	41	21
6	Pune	51	57	38	24	38	17
7	Jalgaon	69	169	81	36	69	44
8	Chandrapur	44	74	43	24	25	22
9	Gondia	52	31	38	25	28	21
10	Amaravati	43	39	56	19	20	31
11	Dhule	73	107	57	38	54	32
12	Sangli	50	58	--	27	38	--
13	Kolhapur	48	74	--	25	45	--
14	Jalna	83	89	--	44	34	--
15	Nagpur	51	70	--	23	27	--
16	Parbhani	68	116	--	34	62	--
17	Latur	65	91	--	35	45	--
18	Hingoli	88	119	--	50	58	--
19	Beed	72	91	--	38	47	--
20	Solapur	51	65	--	29	39	--
21	Sindudurg	25	--	--	16	--	--
22	Wardha	40	47	--	20	26	--
23	Osmanabad	59	78	--	32	41	--
24	Akola	59	54	--	28	33	--
25	Ratnagiri	28	35	--	16	32	--
26	Bhandara	56	58	--	29	26	--
27	Washim	72	70	--			--
28	Satara	32	46	--	18	32	--
29	Aurangabad	72	112	--	40	59	--
30	Buldhana	63	76	--	30	34	--
31	Gadchiroli	--	60	36	--	20	24
32	Mumbai	57	--	--	32	--	--
33	Raigad	37	--	46	20	--	28

Section II

Health Care Services: Status and Issues

CHAPTER - 7

A Mixed-Method Study on Knowledge, Awareness and Practices regarding Tuberculosis: Insights from Community and RNTCP Personnel in Delhi and Muzaffarnagar

William Joe, Aakshi Kalra, Abhishek and Suresh Sharma

Population Research Centre – Delhi

Abstract

TB still remains a major public health problem in India. In this regard, periodic assessment of IEC effectiveness can open avenues for effective prevention and control of TB. With this motivation, this study aims to assess the knowledge, awareness and practices (KAP) among community members. It also documents the status and scope of IEC activities at district level from DOTS providers in North-East Delhi and Muzaffarnagar, Uttar Pradesh. The study used a mixed methods approach, including both qualitative and quantitative methodologies. The qualitative component used focus group discussions among DOTS providers. The quantitative component consisted of a survey among the community members to assess the KAP with respect to TB. We find that majority of respondents have heard about TB and most of them reported of sharing a friendly and supportive attitude towards a TB patient. However, there is lower awareness regarding DOTS treatment or the fact that TB treatment is provided free at public health facilities. Community members mainly recognized only cough as symptom of TB and believed that TB affects lungs alone. Multivariate analysis shows that those who had prior acquaintance with a TB patient are more likely to know about TB and DOTS compared to their counterparts. Focus group discussions (FGDs) with DOTS providers expressed that the community is aware of TB disease but neglects health unless symptoms worsen. The findings also highlight the need for broadening the focus of IEC from just pulmonary TB to extrapulmonary, pediatric and drug resistant TB. Engagement of social media and celebrities is expected to have greater impact on public awareness. The study concludes that the IEC activities need repeated sensitization of the general population, given the limited knowledge and awareness about different aspects of TB.

INTRODUCTION

Tuberculosis (TB) is an infectious disease caused by the bacterium *Mycobacterium tuberculosis*. It can spread through the air from individuals suffering from TB. Over the last 25 years, one of the major advancements can be visualized in terms of TB-related global mortality which has fallen by 47 percent (WHO, 2015). In fact, an estimated 43 million lives worldwide were saved through collective efforts for TB management between 2000 and 2014. Despite such encouraging evidence and continuing efforts at national and international levels, TB still remains a major public health problem for several low and middle-income countries, including India. It is also a disease with immense effect on social life. Literature suggests that there is fear of social exclusion among individuals infected with TB who deal with stigma and discrimination at their workplaces, within communities, and at times in their households (Singh et al 2002, Mathew and Takalkar 2007, Somma et al 2008, Courtwright and Turner 2010, Dhingra and Khan 2010).

Given such intricate concerns, the World Health Assembly in 2014 introduced a new plan called 'The End TB Strategy' with the goal to end the global TB epidemic (<10 cases per 100,000). The vision is 'A world free of TB - Zero TB deaths, Zero TB disease, and Zero TB suffering' (WHO 2014). As per the estimates for the year 2014, there were 9.6 million individuals who suffered from TB worldwide, however, less than two-thirds of the cases were actually notified. This reiterates the large number of missing cases which remained either un-diagnosed or not reported. It may be noted that India has the highest TB burden accounting for 23 percent of all global TB cases and approximately 220,000 Indians died from TB (WHO

2015, WHO 2016).

As such, The National TB Programme was introduced in 1962 in a phase-wise expansion. In 1992, the Government of India, together with the World Health Organization (WHO) and Swedish International Development Agency, reviewed the programme and concluded that it suffered from various shortcomings. As a result, the RNTCP, based on the internationally recommended Directly Observed Treatment Short-course (DOTS) strategy, was launched in 1997. The programme was rolled out in a phased manner and achieved nationwide coverage by March 2006 (Central TB Division, 2006). Currently, the entire population is covered by RNTCP and the programme has treated and saved millions of TB patients (Central TB Division, 2015). Notwithstanding the huge programmatic efforts, RNTCP has to keep up with national and international strategic expectations that have shifted focus from 'stop TB' (Raviglione and Uplekar 2006) to 'end TB' (Uplekar et al 2015). It is worth noting that in recent years there has been a slow decline in the incidence of TB in India which perhaps makes it challenging to meet Sustainable Development Goals (SDGs) regarding 'end TB'. The issue is further complicated by increasing identification of drug resistant forms of TB in India. In this regard, along with existing clinical and preventive activities, the role of Information, Education and Communication (IEC) would be indispensable for achieving these targets.

The IEC is an integral component of all public health programmes (Verma et al 2013). IEC is a well-accepted concept which ensures better provisioning of care for patients. Operational and technical guidelines have been published in 2016 for the implementation of Advocacy, Communication and Social Mobilization (ACSM) which is a much

4. *Tubercle bacilli enter body through inhalation in aerosolised respiratory droplets causing infection which is, in turn, dependent on duration, probability and proximity of exposure to an infectious case and on the infectiousness of the source (Newton 2008).*

broader and clearer strategy composed of three sub units (Central TB Division 2016). The ACSM was brought in line with 'stop TB' strategy which replaced IEC (Dutta and Das, 2012). It focuses on different areas and multiple stakeholders from policymakers, administrators, health providers, NGOs to community members, and many others (Central TB division 2016). Nonetheless, even after decades of RNTCP implementation, number of studies have reported the need of spreading awareness among patients, community members and providers about TB (Sharma et al 2005, Srivastava et al 2011, Chinnakali et al 2013, Sreeramareddy et al 2013). Importantly, one of the reasons for the delay in TB diagnosis has also been lack of adequate information about TB (Storla 2008, Sreeramareddy 2009).

Assessment of knowledge, attitude, and practice (KAP) is essential to plan, implement and evaluate ACSM work. Also, periodic KAP studies serve as an educational diagnosis of a population or community and are an important way to measure changing beliefs and behaviours over time (WHO 2008). Besides, understanding the personnel perspective can provide a holistic picture of the situation and would aid in further strengthening the IEC efforts. With this motivation, this study aims to assess the KAP related to TB at community level and status of IEC activities at district level from DOTS providers. We attempted to identify the associated challenges and the initiatives undertaken in the districts. Since patients and their family members routinely meet DOTS providers for treatment, it was decided to document the IEC activities status and scope from their perspective. The study was carried out in North-East district of Delhi and Muzaffarnagar district of Uttar Pradesh. It may be reiterated that this study quintessentially aims to examine the KAP of the communities across the two districts and should not be viewed as a comparative study of the two selected districts.

2. Methodology

The study used a mixed methods approach, including both qualitative and quantitative methodologies. The qualitative component used focus group discussions among DOTS providers. The quantitative component consisted of a survey among the community members to assess the KAP with respect to TB. The Focus group discussions utilized a discussion guide while the survey amongst the community members used a structured interview schedule. The survey was undertaken in two blocks each of North-East Delhi and Muzaffarnagar districts (burden determined in consultation with the RNTCP officials). The two blocks were selected to include both high and low burden of TB areas within the particular district to assess the differences in knowledge based on the burden of TB. The FGDs were undertaken amongst DOTS providers to explore the understanding and scope of strengthening IEC.

2.1. Study setting

The study was conducted in Muzaffarnagar district of Uttar Pradesh and North-East district of Delhi. In 2011, Muzaffarnagar had a population of 4,143,512 of which male and female were 2,193,434 and 1,950,078 respectively and the density was 1,034 per square kilometre. The North-East district is having a population of 2,241,624 with a density of 36,155 people per square kilometre (Census of India, 2011). In 2013, the total number of patients registered (includes new sputum smear positive cases, new smear negative cases, new extrapulmonary cases, new others, relapse, failure, treatment after default and retreatment others) for treatment in Muzaffarnagar were 3969 and in the same year, North-East Delhi reported 5934 cases. The study does not intend to compare these two districts. They constitute among the high TB burden areas of the respective states, Delhi and Uttar Pradesh as mentioned. Both dis-

districts have higher number of TB cases registered than the average numbers in the state (Directorate General of Health Services, 2014).

Each district has a district TB Centre, as defined by RNTCP. Accordingly, Tuberculosis Units (TUs), Designated Microscopy Centres (DMCs) and DOTS Centres have been established for effective management of RNTCP. In order to avail treatment services, patients can directly visit these centres to seek care or can be referred by any of the healthcare providers. In addition to the government DOTS Centres, there are non-governmental organisations, community DOTS providers and private providers engaged in treatment delivery effectively to the patients. The high and low TB burden areas selected in Delhi were Shastri Park and Bhajanpura respectively. For Muzaffarnagar, high TB burden area was Laddhawala and low burden area was Gyan Majra (these areas were decided after due diligence and discussions with respective district RNTCP officials).

2.2. Sampling strategy

The study participants were drawn from households in the highest and the lowest TB burden areas of both North-East district of Delhi and Muzaffarnagar district of Uttar Pradesh. The study included only those individuals (18 years and above) who were residents of the sampled areas. A survey sample of 240 eligible individuals was adopted for the study. Since this study used cluster sampling, a sample of 60 households was taken from each area. There were a total of 4 areas which were selected based on TB burden in Delhi and Muzaffarnagar. All those who declined to participate were excluded. Besides, anyone suffering from a debilitating condition and hence, unable to comprehend/

respond to the questions was excluded.

The cluster sampling was adopted for participant selection within the selected areas. The cluster locations determined by the administrative divisions were chosen randomly from the high and low TB burden areas. If there were 6 or more clusters in an area then 6 clusters were chosen randomly and 10 participants were taken from each cluster making a total of 60 from each area. Similarly, if there were less than 6 clusters then 4 clusters were chosen and 15 participants were selected. All four areas had at least 4 clusters. In each cluster the investigator started the survey by visiting the nearest household to the random location. Then every alternate house was selected. Attempt was made to select equal male and female participants alternatively from the households. This continued until the desired sample of 10 or 15 eligible participants as deemed was achieved from the locality. From each household only one individual was interviewed. If more than one eligible participant was present at the time of the survey, the youngest one was interviewed.

2.3. Quantitative and Qualitative Analysis

The analysis was done using STATA version 12 to study the sample characteristics and to estimate the knowledge, attitude and practices (KAP) related to TB among the individuals in the highest and the lowest TB burden areas. The questions were grouped under the categories of Knowledge, Attitude and Practices separately and compared across the different TB burden areas.

For each TB knowledge question, a score of one was given for a correct answer, whereas a zero score was given for incorrect and do not know

5. We restricted to use terminology as IEC than ACSM as we found that ACSM was less understood even by programme personnel.

6. This includes cases reported in 2013 from Karawal Nagar (3034) and SPMH chest clinic (2900), making a total of 5934. According to the administrative division, these 2 centers cater to the major population in North East Delhi.

responses. Questions on the knowledge part were rated and a total score was obtained using Principal Component Analysis (PCA) technique. The median score was then computed. Therefore, those with a total score equal to or below the median were classified as having poor knowledge, whereas those above the median were considered having good knowledge. Multivariate analysis using the logistic regression model was thus computed based on this score.

To corroborate the discussion on the KAP regarding TB, the FGD method was preferred as it contributes to better information collection on a topic compared to individual interviews; it also has the benefit of group synergy. Also, FGD facilitates the exploration of certain topics where views could be put forth openly in a discussion manner. The informal group setting is believed to make participants feel at ease, encouraging them to express freely. A total of 3 FGDs were conducted in the regions of Delhi and Muzaffarnagar. These FGDs were conducted with the DOTS providers mainly. The sample consisted of government and private sector providers serving as DOTS providers. Within the inevitable constraints of time and resources, including those of participants, neither the FGDs nor cross-sectional survey can cover all areas of potential interest. Therefore, the FGDs focused on IEC activities of RNTCP and potential scope for further strengthening them. The IEC component is relevant not only for patients but also for community members and health workers. The questions were purposively open-ended and avoided top-down imposition. This allowed FGD participants to raise those points and issues that were important to them and what they got to know through their experience. The FGD was facilitated around topics related to TB awareness to gain insights into whether IEC-related activities were suitable to the local context. The focus groups were made up of 5 to 13 men and women who were working as DOTS providers with RNTCP, including private

sector DOTS provider. We conducted 2 FGDs in Muzaffarnagar and one in Delhi. There were 2 RNTCP supervisors who participated in the second FGD at Muzaffarnagar.

Following training and familiarization, FGDs were conducted by a team conversant in the language of the area. The team was made up of a three members: an FGD facilitator or the moderator, a note-taker/recorder and an observer. Each FGD lasted for one to two hours. Following introductions and explanations, FGD team facilitated discussions and recorded them in writing and using electronic recorders. Discussions were recorded verbatim as far as possible. The FGD guidelines were developed in English which included questions based on the IEC component of RNTCP with scope for strengthening the awareness. This was converted into Hindi for ease of communication. The selection of respondents was done by taking contact details of all DOTS providers from the district TB office. They were contacted and FGD was scheduled after obtaining prior consent through email or over phone.

Data from the FGDs were stored in specifically designed databases in excel which included fields for recorded transcriptions of the discussions and fields for storing coded data. Although the main value of qualitative data is seen as in the recorded transcriptions, coding of data allowed discussions to be thus summarized. However, some of the information could not be put into codes which mainly included discussion beyond the objective of the study. Coding was completed by those team members who had been involved in data collection. In analyzing results comparisons were made between focus group discussions held in Delhi and Muzaffarnagar, and between focus group discussions held with private sector providers of DOTS and other DOTS providers. This was done in order to bring out differences based on rural and urban sector as well as based on the types of providers.

3. Results

3.1. Key findings from community-level survey

A total of 242 respondents were enrolled in this study, 121 each from Delhi and Muzaffarnagar. There were 59 and 62 respondents from high and low TB burden areas of Delhi respectively. Similarly, in Muzaffarnagar, same numbers of individuals were selected from high burden and low TB burden areas. Of these, 119 (49.2 percent) were male. The mean age of the respondents was 34.4 years. About 74.4 percent were residing in families having five or more members. There were more Hindu respondents (59.9 percent overall) in the sample as well as in both districts. Description of the households as per their caste, religion and household size has been summarized in Table 7.1.

Table 7.1: Distribution of households by caste, religion and household size, North-East Delhi and Muzaffarnagar, 2016

Background characteristics	North-East Delhi	Muzaffarnagar	Total
Social group			
ST	1.7	0.8	1.2
SC	3.3	25.6	14.5
OBC	24.8	53.7	39.3
Others	70.0	20.0	44.0
Religion			
Hindu	65.3	54.6	59.9
Muslim	34.7	45.5	40.1
Household size			
<3	3.3	2.5	2.9
3-4	28.9	16.5	22.7
5-6	43.0	44.6	43.8
7 +	24.8	36.4	30.6
Total (N)	100.0 (121)	100.0 (121)	100.0 (242)

More than 60 percent of the respondents belonged to the age group of 25-49 years. There were no major differences between the two districts in terms of age distribution by gender or otherwise as illus-

trated in table 7.2. A quarter of the sample was not formally educated. Among the female respondents in Muzaffarnagar, around half were illiterate and in Delhi around 53.2 percent were educated up till high school and above.

The knowledge of the respondents was assessed based on a set of questions. These questions mainly focused on general information about TB, its symptoms, causes, mode of acquiring and preventing the disease, its cure, DOTS, cost of treatment, MDR-TB and pediatric TB. Table 7.3 describes a few of the knowledge-based variables. Majority of the respondents had heard about TB irrespective of social group, sex, age and religion. However, within the category of religion, more Hindus had heard about TB in general, pediatric TB and believed that 'TB is a curable disease' than Muslims in both districts individually and collectively. Interestingly, less than 60 percent of the Muslims who had heard about TB had idea about pediatric TB in Delhi. Such differences were evident education and age-wise when asked about TB in children. Within high and low burden areas of Delhi and UP districts, majority knew about TB but had less awareness about pediatric TB. Only two-thirds of those who had heard of TB were aware of pediatric TB in general category in North-East Delhi district. Moreover, approximately three-fourths of SC individuals in Muzaffarnagar sample believed that TB could occur in children.

Overall majority of the respondents held the view that TB could be cured. On the contrary, there were only 24.6 percent respondents who knew that TB could be cured by DOTS regime. Around 64 percent of the sample respondents did not know how TB is cured. The four most commonly recognized symptoms of TB mentioned by respondents were coughing for two weeks and above (55.6 percent), coughing up blood (54.7 percent), weight loss (24.1 percent), and fever (21.6 percent). In Muzaffarnagar, 'coughing up blood' was reported by more than half of the sample, and the most common symptom reported in Delhi was

Table 7.2: Age and education level of respondents by gender, North-East Delhi and Muzaffarnagar, 2016

Background Characteristics		North-East Delhi			Muzaffarnagar			
Total								
Female	Total	Male	Female	Total	Male	Female	Total	Male
Age								
18-24		25.4	22.6	24.0	18.3	14.8	16.5	21.9
18.7	20.3							
25-49		62.7	64.5	63.6	68.3	65.6	66.9	65.6
65.0	65.3							
50 and above		11.9	12.9	12.4	13.3	19.7	16.5	12.6
16.3	14.5							
Education								
No School		10.2	16.1	13.2	20.0	52.5	36.4	15.1
34.2	24.8							
Elementary		11.9	19.4	15.7	36.7	19.7	28.1	24.4
19.5	21.9							
High School and above		76.3	53.2	64.5	41.7	21.3	31.4	58.8
37.4	47.9							

Table 7.3: Awareness about Tuberculosis in North-East Delhi and Muzaffarnagar, 2016

Background characteristics	North-East Delhi			Muzaffarnagar			Total		
	Heard of TB (%)	Among who heard about TB, % who		Heard of TB (%)		Among who heard about TB, %who		Heard of TB (%)	
		Believe TB can occur in children	Believe TB can be cured			Believe TB can occur in children	Believe TB can be cured	Believe TB can occur in children	Be- lieve TB can be cured
Social group									
ST	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
SC	100.0	100.0	100.0	100.0	74.2	90.3	100.0	77.1	91.4
OBC	100.0	83.3	96.7	95.4	80.7	91.9	96.8	81.1	93.7
Others	93.5	66.7	98.6	88.9	100.0	87.5	93.0	72.1	97.7
Religion									
Hindu	97.5	81.8	100.0	97.0	82.8	93.8	97.2	82.8	97.2
Muslim	92.9	59.0	94.9	94.6	73.1	84.6	93.8	68.0	89.7
Age percent									
18-24	96.6	75.0	96.4	95.0	57.9	84.2	95.9	69.4	91.8
25-49	96.1	74.3	100.0	95.1	83.1	90.9	95.6	79.1	95.6
50 and above	93.3	71.4	92.9	100.0	80.0	90.0	97.1	77.1	91.4
Education									
No School	93.8	66.7	93.3	97.7	60.5	90.7	96.7	61.7	91.7

Elementary	100.0	84.2	94.7	94.1	84.4	81.3	96.2	84.9	86.8
High School and above	96.2	76.0	100.0	94.7	94.4	94.4	95.7	82.8	98.3
Gender									
Male	96.6	73.7	100.0	91.7	81.8	85.5	94.1	78.2	93.3
Female	95.2	74.6	96.6	100.0	75.4	93.4	97.6	75.6	95.1
Total	95.9	74.1	98.3	95.9	78.5	89.7	95.9	76.3	94.0

cough for more than two weeks and above. Table 7.4 summarizes knowledge related to symptoms, transmission and prevention. Of the respondents, 64.7 percent believed that TB spreads through air when a person with TB coughs or sneezes. Similar findings were reported in both districts individually. Moreover, 24.1 percent and 23.3 percent thought that TB spreads through sharing of food and eating from same plate respectively. Around

70 percent of the respondents in both districts and overall considered covering their mouth and nose as the most commonly used method for preventing the spread and transmission of TB. Additionally, rest of the respondents mentioned that transmission and spread of TB could be prevented by avoiding sharing of food/utensil (35.8 percent), avoiding hand shake (10.3percent) and other preventive measures.

Table 7.4: Knowledge about symptoms, source of transmission, prevention and risk of Tuberculosis as per respondents (percent), North-East Delhi and Muzaffarnagar, 2016

Knowledge	North-East Delhi	Muzaffarnagar	Total
Symptoms of TB as per respondents			
Cough that lasts longer than 2 weeks	64.7	46.6	55.6
Coughing up blood	54.3	55.2	54.7
Weight loss/loss of appetite	33.6	14.7	24.1
Fever that lasts longer than 2 weeks	26.7	16.4	21.6
Night sweating	1.7	1.7	1.7
Source of transmission of TB as per respondents			
Through handshakes	5.2	8.6	6.9
Through the air when a person with TB coughs or sneezes	69.0	60.3	64.7
Through sharing of food	24.1	24.1	24.1
Through eating from the same plate	25.0	21.6	23.3
Do not know	17.2	18.1	17.7
Prevention of TB as per respondents			
Avoid shaking hands	7.8	12.9	10.3
Cover mouth and nose when coughing or sneezing	75.0	68.1	71.6
Avoid sharing of food/utensil	39.7	31.9	35.8
Close windows at home	2.6	1.7	2.2
By praying	0.9	5.2	3.0
Do not know	14.7	16.4	15.5
Opinion about who can be infected with TB			
Anybody	78.51	85.95	82.23
Only poor people	0.83	9.92	5.37

Only homeless people	2.48	4.13	3.31
Only alcoholics	9.92	1.65	5.79
Only drug users	5.79	0.83	3.31
Only people living with HIV/AIDS	0	0.83	0.41
Only people who have been in prison	0	0.83	0.41

About 50 percent of the respondents reported that TB is caused by contact with other TB patients followed by causes namely bacteria (26.9 percent), 'do not know' (24.4 percent), lack of proper nutrition (14.1 percent) and sin and witchcraft (1.2 percent). Similar kind of reporting was found in both districts but in Delhi, no respondent recognized sin or witchcraft as the cause for TB. Around 82 percent of the sample respondents believed that anybody can acquire the disease. Similar responses were found in both districts except that in Muzaffarnagar around 10 percent felt that only poor people run the risk of getting infected with TB whereas in Delhi, it was believed that only alcoholics (10 percent) could get the disease.

Most study participants (68.5 percent) had heard about TB from family, friends, neighbours or colleagues whereas the role of health professionals was among the least (table 7.5). The second most

common source of information was television followed by newspapers and magazines. Importantly, only 8.6 percent of the sample participants felt that they are well informed about TB whereas around 43 percent believed that they are partly aware. Intervention point was that 74.6 percent of the respondents wished to get more information about TB particularly related to symptoms and prevention of TB. The respondents mentioned that the medium of TV could be most effective for dissemination of health information on TB (59.5 percent overall, 74.1 percent in Delhi and 44.8 percent in Muzaffarnagar) but preferred mode in Muzaffarnagar was information spread by health workers (48.3 percent). Overall, newspapers and magazines, family/friends and brochures/printed material were reported in decreasing preference order. The differences in high and low TB burden areas were evident but overall it reflected similar findings as of districts.

Table 7.5: Respondent's source of information about Tuberculosis (percent), North-East Delhi and Muzaffarnagar, 2016

Source of information about TB	North-East Delhi			Muzaffarnagar		
	Bhajanpura	Shastri Park	Total	Gyan Majra	Laddawala	Total
Newspapers and magazines	25.0	25.0	25.0	21.7	30.4	25.9
Radio	1.7	3.6	2.6	10.0	1.8	6.0
TV	26.7	39.3	32.8	43.3	21.4	32.8
Posters and other printed materials	10.0	10.7	10.3	1.7	8.9	5.2
Health workers	15.0	8.9	12.1	20.0	16.1	18.1
Family, friends etc.	55.0	67.9	61.2	83.3	67.9	75.9

Note: Bhajanpura and Gyan Majra are low burden area in terms of registered cases when compared to Shastri Park and Laddawala which have relatively higher number of registered cases.

In studying the attitudes and practices of respondents, 62.9 percent viewed TB as a very serious disease in general. However, the opinion differed when asked about the seriousness of the disease for the country/region as reported in Table 7.6. About 80 percent of the respondents preferred going to health facility as a choice of treatment for TB symptoms followed by visit to traditional healer; however, 23.3 percent respondents in Muzaffarnagar mentioned about traditional healer compared to only 6.9 percent in Delhi. Community members reported that 70 percent of them would immediately seek care if symptoms emerge. In case of having any doubt related to TB, majority of the respondents (80 percent) would prefer to ask health provider.

Table 7.6: Respondents' attitude and practice on TB (percent), North-East Delhi and Muzaffarnagar, 2016

Opinion of respondents regarding	Delhi	Muzaffarnagar	Total
Seriousness of TB as a disease, in general			
Very serious	52.6	73.3	62.9
Somewhat serious	15.5	3.5	9.5
Not very serious	22.4	20.7	21.6
No idea	9.5	2.6	6.0
Seriousness of TB as a disease, in the country/region			
Very serious	36.2	48.3	42.2
Somewhat serious	19.0	3.5	11.2
Not very serious	12.9	7.8	10.3
No idea	31.9	40.5	36.2
Choice of care if symptoms appear in family members			
Go to health facility	85.3	74.1	79.7
Go to pharmacy	5.2	1.7	3.5
Got to traditional healer	6.9	23.3	15.1
Pursue other self-treatment options (herbs, etc ")	0.9	0.9	0.9

Point of seeking care			
When treatment on my own does not work	6.9	5.2	6.0
wait for symptoms to subside for two three weeks	16.4	14.7	15.5
As soon as I realize that symptoms might be related to TB	73.3	66.4	69.8
Would not go to the doctor	1.7	7.8	4.7
Talk to about doubts related to TB (percent)			
Doctor or other medical worker	74.1	87.1	80.6
Spouse	12.9	5.2	9.1
Parents	11.2	8.6	9.9
Friends	4.3	5.2	4.7

Note: Column percentage may not add up to 100 per cent as there were missing cases.

The perception of the sample regarding cost of TB treatment varied. Only 39.2 percent knew that DOTS treatment is free. Around 20 percent believed that treatment is reasonably priced by the government whereas around 14 percent felt that it is very expensive. There were 141 (60.8 percent) respondents who had known a TB patient. When asked about general behavior of the community towards a person suffering with TB, it was reported that people remain friendly with the patient during interactions but generally try to avoid the patient (41.4 percent). Around 31.5 percent respondents (32.8 percent and 30.2 percent in Delhi and Muzaffarnagar, respectively) believed that hardly anyone avoids the patient rather community supports and helps. There were not many differences within the districts in response to this question.

Table 7.7: Multivariate analysis of characteristics associated with a high knowledge score among community members, North-East Delhi and Muzaffarnagar, 2016

Correlates	High knowledge about TB		Knowledge about DOTS	
	Model 1	Model 2	Model 3	Model 4
Knows TB infected person				
No (Reference)	1.00	1.00	1.00	1.00
Yes	1.77**	1.92**	2.58***	2.93***
	[1.05 - 2.97]	[1.08 - 3.42]	[1.42 - 4.66]	[1.47 - 5.83]
Study Area				
Bhajanpura (Reference)		1.00		1.00
Shastri Park		2.23*		1.92
		[0.9 - 5.55]		[0.75 - 4.9]
Gyan Majra		1.17		0.84
		[0.51 - 2.67]		[0.35 - 2.01]
Laddhawala		1.49		0.62
		[0.48 - 4.63]		[0.17 - 2.23]
Education				
Illiterate (Reference)		1.00		1.00
Up to Primary		1.87		4.95***
		[0.85 - 4.12]		[1.84 - 13.27]
Up to secondary and higher		1.79		3.75***
		[0.81 - 3.96]		[1.38 - 10.19]
Religion				
Hindu (Reference)		1.00		1.00
Muslim		0.57		0.66
		[0.23 - 1.44]		[0.24 - 1.8]
Household has a TV				
No (Reference)		1.00		1.00
Yes		1.81		1
		[0.87 - 3.79]		[0.42 - 2.41]
Gender				
Female (Reference)		1.00		1.00
Male		1.38		2.03**
		[0.79 - 2.41]		[1.09 - 3.79]
Age				
18-24 years (Reference)		1.00		1.00
25 to 49 years		1.14		1.23
		[0.58 - 2.25]		[0.57 - 2.64]
50 years and above		0.97		1.54
		[0.37 - 2.5]		[0.52 - 4.55]

Household size				
Less than 5 (Reference)		1.00		1.00
Greater than equal to 5		1.31		0.83
		[0.69 - 2.5]		[0.41 - 1.65]

Note: *p-value <0.1, **p value <0.05, ***p value < 0.01. [...] Contains 95 percent confidence interval for the odds ratio

Finally, we use logistic regression analysis to understand factors associated with high and low knowledge score (classified as binary variable). Altogether four models were prepared based on high knowledge about TB and knowledge about DOTS regime alone. The results are reported in Table 7.7. The regression estimates shows that, prior acquaintance with a TB patient is independently associated with high knowledge score. For models on knowledge about DOTS, prior acquaintance with a TB patient, formal education and respondent being a male were linked with high knowledge about DOTS. Respondents who knew a TB patient before were around 2 times likely to have high knowledge about TB than those who had no prior acquaintance. Similarly, they were approximately 3 times more likely to have high knowledge score for DOTS information than their counterparts.

3.2. Key findings from FGDs

27 people participated in the FGDs, of these, 11 percent were female. There were no female DOTS providers in Muzaffarnagar district. The participants were in the age group of 24-72 years. There were 9 private sector providers of DOTS and 4 community DOTS providers. None of the private sector providers had allopathic degree. Each participant had attained some level of education. When asked about their education level, 33.3 percent of participants described themselves as educated up to or less than the higher secondary level, 44.4 percent as graduates and above, and the remaining as diploma holders. There were 22.2 percent Muslims and rest belonged to Hindu religion.

A series of topics were discussed with FGD participants looking at their experience in managing TB cases and DOTS delivery. The IEC component included - but not limited to - prevention of disease, awareness regarding symptoms, treatment of disease and support from difference stakeholders for effective communication and mobilization. Questions focused on their perspectives related to current IEC activities, its scope, gaps and recommendations for improvement as per the population needs. FGD questions and prompts facilitated participants to identify and share the issues relating to IEC and discuss measures for improvement thus providing an insight from the programme point of view in adjunct to the community members' perception gained through cross-sectional survey. Overall differences in responses between males and females were negligible. However, attempt was made to capture the differences between two districts and types of DOTS providers.

Community needs and level of awareness related to TB

One of the major points of discussion regarding TB was associated with stigma. As would be expected, people get scared when they hear about screening for TB and tend to neglect unless the symptoms worsen and require medical attention. The univocal response on level of awareness regarding TB among community members was that people are aware about pulmonary TB but are hesitant. Perceptions of government DOTS providers at Muzaffarnagar FGD were summarized by the statement below:

“People do not do “parhez”. Patients eat with family members and create new cases. Stigma is also an issue particularly for young girls who are unmarried. There is one case where patient does

not come to us and we can't give them box so we have given to a community provider still issues are there in taking treatment. People are not ready to accept that they could have TB especially among women (male participants, Muzaffarnagar)''

On comparing Delhi and Muzaffarnagar, issue of migratory population was raised in Delhi FGD whereas there was no such discourse in Muzaffarnagar. Those migratory populations had special needs and therefore, an entirely different approach was needed for these groups and the mechanism of diagnosis also needed to be tailored accordingly. This was expressed by two of the government DOTS providers:

''People are scared of TB, even nobody wants to rent their house to someone with TB. On house visit, we often ask house owners if they will force the person to vacate the room for this reason after we leave (female participant, Delhi)''

''There are lots of migrants who keep shifting. They suffer from TB. They take treatment from here and then go back. It becomes difficult for us to trace them. IEC activities need to focus on explaining these people about harms of leaving treatment (Male participants, Delhi)''

DOTS providers – information and training on TB

Overall, the DOTS providers believed that they were well informed about TB in so far as TB treatment is concerned. With detailed probing, they expressed their misgivings about their ability to deal with drug resistant TB, particularly when patients reported side effects. When they suspected that their knowledge was inadequate to treat/manage a patient, they referred such cases to the District TB Officer. There have been very few training sessions in both districts.

''There has been just 1-2 training sessions since past 5-6 years...first one was at the time of recruit-

ment... We are usually given training sessions by the medical officer, supervisor and all at the district centre. They would discuss the topic with us. Those sessions were useful and kept us informed and updated. We stay in contact with patient till the treatment is complete so we should have knowledge (Government centres DOTS providers, Muzaffarnagar)''

IEC activities and their focus

The participants felt that the IEC activities had reduced over time. Activities in Muzaffarnagar focused on the community. Announcements were made through loudspeakers at religious institutions, informative sessions for students were organised in schools and community meetings were also conducted. In addition to it, the private sector providers had their own way of spreading awareness. They used to refer symptomatic patients to the nearest government TB centre and one of them had written a line at the bottom of his prescription pad about TB. However, such efforts were not made by all private sector providers. In the words of private sector providers during FGD:

''Since we see patients in our clinics and other set-ups so we try to do things which could spread awareness in our own small ways... whenever there is a patient with cough from a long time I refer him to the district centre... Or, if I come to know by seeing somebody that he/she may be having TB, I refer that personal so to the district centre for checkup ... (Private sector provider participants, Muzaffarnagar)''

DOTS providers in Muzaffarnagar described their role in TB awareness as being majorly limited to that of providers of information.

''Supervisors/STS and all do major work in this and we only display banners. We also explain patients particularly sputum positive patients how to manage further. Sometimes during Namaz time in mosques, few of us go with pamphlet and stand

outside mosques. As soon as prayer is over, we distribute the pamphlet on TB awareness to all the people. This all arrangement is made by districts. Some of them or their relatives used to come to us after reading it...we at times go to schools. We tell about TB symptoms and sometimes we get cases. We are in rural area so such information helps. This kind of information is given to students of 7-8th class. (Male participants, Muzaffarnagar) ”

During Delhi FGD, participants mentioned about patient provider meetings which are held at DOTS centre with a few patients fortnightly. They also indicated towards support from NGOs which carry out drug resistant TB IEC activities.

“Whenever activities are planned we are called at the district centre and told that such and such meeting for IEC is to be organized so we plan at any community or required place. MDR-TB related activities are carried by XYZ (NGO name). Else we all do. We conduct patient provider meetings... Usually a few patients gather together and then we explain... this usually happens around 2 times a month... sometimes due to patient load it is difficult to carry out but we still manage somehow... (DOTS providers, Delhi) ”

Difficulties in implementing/managing IEC activities

The nature of difficulties mentioned varied between districts and type of providers. The FGD with the private sector providers indicated that the incentive given by the government to manage or refer a case is negligible. The government centre DOTS providers’ FGD on the other side brought out issues like stigma associated with TB, lack of availability of isoniazid for chemoprophylaxis (relevant in case patient is well-informed about it and asks for treatment for his/her child) and demand by the community members for incentives to participate in TB awareness meetings.

They also pointed out that the community mem-

bers, including symptomatic and/or diseased patients seek early relief and care which is, at times, not available to them at government facilities. In this context, issues of timeliness, cleanliness, adequate attention by the doctor when needed and faith/trust in the private provider were brought as key reasons behind preference for private sector providers. In Delhi, beyond the issues of private sector, there was human resource crunch. Adequate staff is not there to manage the IEC-related activities. As pointed out during Muzaffarnagar FGD about private provider- related issue:

“There are some private providers in the locality who discourage the patient from taking alternate day treatment. They give daily treatment. I will tell you one case... I told the patient to take DOTS for two months and go for sputum test to see whether test is negative? I was able to convince and test came negative after 2 months of treatment. However, it is difficult to catch each patient and make them understood regarding this. Also money wise benefit to doctor is only when he gives medicine by himself. Madam, ghoda ghas se yari karega to khayega kya. There is not much incentive for a private provider from government so patient referral is difficult (Private sector providers, Muzaffarnagar) ”

As explained during Delhi DOTS provider FGD who had been working for more than 5 years:

“Staff is less so it is difficult for us to do IEC activities particularly in communities along with our daily work. I am managing more than 500 patients on DOTS currently. All other providers also manage lot of cases. How would I do other things like patient visits etc.? I work beyond my capacity and capability usually. ”

There were few aspects on which more information needs to be imparted to the community. Majority of the individuals in the community recognize only pulmonary TB and associate the disease with sputum. During discussions in FGDs, pro-

viders were in consensus over negligible focus on extrapulmonary TB by RNTCP programme. They also pointed out that in spite of rising MDR-TB and XDR-TB burden, there is no awareness in the community. However, media coverage by famous personalities has had an impact in spreading information about TB. A similar context was there for pediatric TB as people are able to link TB with adults easily but for children it seemed difficult.

“Community recognizes only lung TB and “ganthwali” TB is not much known to people, madam. It is like story of “do hafte khasi”. We do not talk about “ganth” so awareness is less. You would not believe that when we tell patient that he has “ganth wali” TB then he is unable to accept easily (Government centre DOTS provider, Muzaffarnagar)”

In Delhi FGD, another issue raised was related to health workers being prone to contract TB. They mentioned that along with preventing disease among community members, even of health workers should be considered. The IEC measures have been ignorant of controlling TB among health workers.

“IEC for safeguarding health workers like us from TB... patients do not cover their mouth. When we scold them and explain then they just cover or cover while entering the DOTS center. They think that by asking to cover mouth we are degrading them. Our safety is compromised. Nobody educates anyone for our safety (DOTS providers, Delhi)”

Scope for strengthening IEC measures

There were 3 major themes discussed during FGDs when asked about the scope and suggestions which have been described as sub-codes. The common point brought out was that the efforts should be continuous, method or channel may differ. Repetitive activities alone can bring a desirable change. Nevertheless, there were three key areas which were re-iterated in the FGDs as

priority areas for intervention. The first two are somewhat common to all the FGDs but the issue of migrants was raised in Delhi FGDs.

Sensitization of private sector providers

There was emphasis on increasing awareness among private providers. Usually they are the first level of contact for most people particularly those from the poorer sections of society. People prefer to go to them first for any ailment. However, the treatment they provide can offer only initial relief to TB patients, and may not cure TB. Also, the issues of side-effects of DOTS treatment as discussed above makes them prefer private over government providers. Thus their sensitization for referral and proper management of cases would be useful.

“Private Doctors who may be unqualified sit in local market give daily treatment and for 9 months. Our patients come and argue with us... why difference... we explain our patients but many are missed who don't reach us... such things need attention. Private Doctors are worsening the cases. What Private Doctors do, is they start 2nd line of treatment from the beginning and then resistance so MDR and XDR cases. Awareness is very less which is needed. (Government sector DOTS providers, Muzaffarnagar)”

Information on drug resistant TB-MDR and TB-XDR, extrapulmonary TB and pediatric TB

The current IEC material focuses on pulmonary TB. This results in limited or no knowledge about drug resistant TB or extra pulmonary TB among members of the community. The poor knowledge about these issues renders it all the more difficult for health workers to convince patients and/or their family members on the need to take extra care to avoid drug resistant TB or extra pulmonary TB. Also, there is a need to give special attention to children suffering with TB. The DOTS provid-

ers identified a felt need for improving their own knowledge about these issues.

“MDR, XDR, Extrapulmonary TB all forms are less known to people and to many of us. We are just giving medicine for drug resistant if there is any issue or question we just refer. We cannot manage that (Government sector DOTS providers, Muzaffargarh)”

Migration

In Delhi, completion of TB treatment for patients who are migrants from different states is an issue for the RNTCP as described by the DOTS providers. There should be some provision for these patients as they are likely to be lost to follow-up.

“There are lots of migrants who keep shifting. They suffer from TB. They take treatment from here and then go back. It becomes difficult for us to trace them. IEC activities need to focus on explaining these people about harms of leaving treatment or we can transfer them but they are not aware.”

Proposed IEC activities

The major activities pointed out by private sector DOTS providers during FGD were involving and engaging with NGOs, taking lessons from other programmes, painting walls with information on TB near clinics and keeping a reward for provider who notifies maximum cases or referrals.

“With the help of NGO, earlier we used to put a stamp of “do boond zindagi ki” over every prescription for controlling polio... if such a thing is done for TB then it may be good and useful as a reminder. Printing a Boucher with three/four folded pages like small note book with all the information on TB and distributed to all individuals is also crucial for TB awareness.”

The government centre DOTS providers identified possible ways to address the issue of stigma due to

TB among women patients. They suggested that there should be involvement of women especially to address woman with TB during meetings. Systematic active case finding in areas of high burden was also pointed out along with importance of cleanliness, nutrition and BCG vaccination. For early relief to patient's symptoms, there was a consensus on allowing government centre DOTS provider to give analgesics and antacids. There can be local volunteers, religious organisations, teachers and other NGOs who can extend support for spreading awareness. A proposal of keeping a small amount as caution money which will be returned upon treatment completion to patients was conceived by the group. This was in lieu of the patients not taking government treatment seriously.

“I think we should be allowed to give medication like Paracetamol or antacid... because this will help us retain patient. I also believe we need training for it. Patients are not ready to go and stand in line to receive medication for side effects. This leads to further problems like default and argument over private being better. Also community meetings should have women involvement. Women in general as well as patients do not attend meetings quite often.”

Effective source of information

A special emphasis on ‘nukkadnatak’ was suggested by all providers. They believed that it was difficult to convince people to come for community meetings. People readily relate with something which is demonstrated. Also, printing pamphlets or making announcements at religious institutions at small places are effective sources. Another point of interest discussed by participants in Delhi FGD was TV shows or social media campaigning by famous personalities or celebrities.

“Advertisements by famous people can help... Like Aamir Khan's television show on MDR TB

made people think. Patients asked us questions after that. It had lot of impact. These days, Amitabh Bachchan advertisement is also coming... so such thing help a lot. People are able to connect better with actors (Government centre DOTS providers, Delhi)”

4. Discussions and conclusions

The present study provides insights into the existing knowledge, attitude and practices of the community related to TB; both in high and low TB burden areas. It also documents the perception of DOTS providers on the IEC status and its scope in their respective districts of Delhi and Uttar Pradesh. The study findings suggest that majority of the respondents had heard about TB irrespective of the social group, sex, age and religion they belonged to. However, overall there was less awareness about pediatric TB. The majority of respondents had heard about TB for the first time from their family members or friends/colleagues and neighbours followed by media like TV. A high proportion of respondents were of the opinion that TB as such is a very serious disease but not within their region or country, comparatively. TB was mainly associated with cough either for two weeks or with blood. Based on the present findings, the major source of TB transmission was air when a person with TB coughs or sneezes, and covering mouth and nose was reported as preventive measure. The behavior of the community towards TB patient was friendly and supportive as only 15.5 percent mentioned that TB patient is rejected thus discriminated. Effective means of disseminating information about TB was TV followed by health workers. In continuation, respondents expressed that their first preference would be consulting a doctor or other medical workers for any doubt related to TB. Only one fourth of the sample was aware about DOTS as recommended treatment for TB and only 40 percent knew that treatment was free. Those who had prior acquaintance with a TB patient had high knowledge scores about TB and DOTS compared to their counterparts. Similarly,

those who had received formal education and male respondents had more awareness about DOTS. The differences reported religion-wise should be taken with a caution and cannot be extrapolated given that the multivariate results are inconclusive on the association and there is lower representation of Muslim and other religions in the sample at North-East Delhi and Muzaffarnagar.

The FGD results filled gaps from programmatic point of view, providing a holistic picture of the current levels of understanding about TB. The IEC activities mainly focused on community in Muzaffarnagar along with a few private providers (serving as DOTS providers) initiatives whereas in addition to these, there were patient provider meetings and NGOs' involvement in Delhi district. DOTS providers expressed that community is aware of TB disease but neglects health issues unless symptoms worsen. Also, it was highlighted that the knowledge about TB centers around cough and the organ affected being lungs. There were a few promising strategies mentioned for strengthening IEC activities. There was a felt need for further training on drug resistant forms of TB although overall view was that DOTS providers are well informed about TB. Also broadening the focus of IEC from just pulmonary TB to extrapulmonary, pediatric and drug resistant TB was emphasized. The effect of social media and celebrity involvement for spreading awareness was evident from the findings.

Given the objectives of RNTCP, focus on sputum positive patients has been the mainstay for decades. With evolving strategies and global attention, other forms of TB have also been given due attention. Our hypothesis of differences in high and low TB burden areas was not completely accepted which could be due to small sample size. However, Shastri Park respondents seemed to have high knowledge scores than those from Bhajanpura. It was also concluded that generally, information on TB is lacking beyond awareness about cough for two weeks.

Several studies explained that the majority of respondents heard about TB from health workers and personal experience for the first time (Mushtaq et al 2011). However, our results show that community members acquired information from their acquaintances. This could be due to inadequate IEC activities and/or activities which are not context-specific. The proposed effective sources of disseminating information were TV and health workers.

There is evidence which shows that a greater knowledge level was observed among males (Hoa et al 2009); similar to our study findings. However, studies show age group knowledge differences which was not observed in this study. This might be a result of variation in study population in contrast to other studies. Literature indicates that educational background (Mushtaq et al 2010; Storla et al 2008) is one of the important determinants of TB knowledge. Similar association was found in our study. There seems to be information dissemination majorly in high burden areas and in cases where there is prior acquaintance with a patient. Awareness only after exposure to a patient does not completely satisfies the objective of IEC. Added to these issues is the stigma and discrimination faced at the hands of family and society which was reported in FGDs. There were few misconceptions related to treatment, its cost and other aspects of TB prevailing in the community.

In this regard, the key recommendations emerging from the FGDs are listed as follows: Firstly, there are few lessons to be learnt from other programmes like Pulse Polio Immunization where a stamp was prepared to remind families about polio drops. A similar effort like a line on the prescription pads of general health providers, including those in the private sector, will be helpful. There should be expansion of IEC material on TB and it is also advisable to prepare IEC material on EP-TB, drug resistant forms and pediatric TB. Painting walls with information on TB near private clinics and to plan incentives for provider who notify maximum

cases or referrals can be an effective strategy. As such, audio-visual activities often attract attention and are better understood. A special emphasis on 'nukkad natak' is also suggested by respondents. Importantly, TV shows or social media campaigning by famous personalities or celebrities will help in generating greater public awareness. In addition to these, printing pamphlets or making announcements at community meetings or prayer meetings in villages or other areas is also a feasible strategy for increasing knowledge and awareness regarding such major public health concerns in India.

The programme is undergoing a paradigm shift while focusing on making strategic plans for providing universal access under END TB strategy. In order to achieve the goal of SDG, the role of IEC would be crucial. There is a need to strengthen the existing efforts and chalk out context-specific activities which would cater to the needs of the community, including special population like women, children and migrants. The results also emphasize the need to involve multiple stakeholders, broaden the domain of IEC focus from pulmonary TB to other forms and utilize various media to spread awareness. Importantly, to address the issue of stigma among women TB patients, there should be involvement of women health workers especially to address woman with TB during meetings. Involving local volunteers, religious organisations, teachers and NGOs for spreading awareness is equally important and can help strengthening the existing initiatives.

This study shows that knowledge about TB is limited among the general population which needs to be substantiated by IEC activities through repeated sensitization. It provides baseline information for further action and also opens avenues for evidence-based prioritizing of IEC interventions. It is recommended that such studies are to be conducted across the districts for better implementation of TB programme in the country, particularly in the light of an aim to end TB.

References

- Central TB Division, Directorate General of Health Services, 2016. Technical and Operational Guidelines for TB Control in India 2016.
- Central TB Division, Directorate General of Health Services, 2006. TB India 2006 RNTCP Status Report. MOHFW, New Delhi.
- Chinnakali, P., Gurumurthy, J., Ramakrishnan, J., Vasudevan, K., Upadhyay, R., Panigrahi, K., 2013. Level of awareness about tuberculosis in urban slums: Implications for advocacy and communication strategy planning in the National programme. *Lung India* 30, 139. doi:10.4103/0970-2113.110422
- Courtwright, A., Turner, A., 2010. Tuberculosis and Stigmatization: Pathways and Interventions. *Public Health Reports* 125, 34–42.
- Dhingra, V.K., Khan, S., 2010. A sociological study on stigma among TB patients in Delhi. *Indian J Tuberc* 57, 12–18.
- Dutta, P., Das, J., 2012. An Overview of Information, Education and Communication (IEC) activities in Revised National Tuberculosis Control Programme (RNTCP) And Role Of NGOs. *Indian J. Prev. Soc. Med.* 43, 365–371.
- Esmael, A., Ali, I., Agonafir, M., Desale, A., Yaregal, Z., Desta, K., 2013. Assessment of Patients' Knowledge, Attitude, and Practice Regarding Pulmonary Tuberculosis in Eastern Amhara Regional State, Ethiopia: Cross-Sectional Study. *Am. J. Trop. Med. Hyg.* 88, 785–788. doi:10.4269/ajtmh.12-0312
- Hoa, N.P., Chuc, N.T.K., Thorson, A., 2009. Knowledge, attitudes, and practices about tuberculosis and choice of communication channels in a rural community in Vietnam. *Health Policy* 90, 8–12. doi:10.1016/j.healthpol.2008.08.006
- IHBP, 2014. Operational Handbook on Advocacy, Communication, and Social Mobilization (ACSM) for RNTCP.
- IHBP, 2013. A Status Report on the Advocacy Communication and Social Mobilization (ACSM) in Revised National Tuberculosis Programme.
- Isaac, O., Thiemer, K., 1975. [Biochemical studies on camomile components/III. In vitro studies about the anti-peptic activity of (–)-alpha-bisabolol (author's transl)]. *Arzneimittelforschung.* 25, 1352–1354.
- Mathew, A.S., Takalkar, A.M., 2007. Living with tuberculosis: the myths and the stigma from the Indian perspective. *Clin. Infect. Dis. Off. Publ. Infect. Dis. Soc. Am.* 45, 1247. doi:10.1086/522312
- Mushtaq, M.U., Majrooh, M.A., Ahmad, W., Rizwan, M., Luqman, M.Q., Aslam, M.J., Siddiqui, A.M., Akram, J., Shad, M.A., 2010. Knowledge, attitudes and practices regarding tuberculosis in two districts of Punjab, Pakistan. *Int. J. Tuberc. Lung Dis. Off. J. Int. Union Tuberc. Lung Dis.* 14, 303–310.
- Mushtaq, M.U., Shahid, U., Abdullah, H.M., Saeed, A., Omer, F., Shad, M.A., Siddiqui, A.M., Akram, J., 2011. Urban-rural inequities in knowledge, attitudes and practices regarding tuberculosis in two districts of Pakistan's Punjab province. *Int. J. Equity Health* 10, 8. doi:10.1186/1475-9276-10-8
- Newton, S.M., Brent, A.J., Anderson, S., Whittaker, E., Kampmann, B., 2008. Paediatric tuberculosis. *Lancet Infect. Dis.* 8, 498–510. doi:10.1016/S1473-3099(08)70182-8
- Raviglione, M.C., Uplekar, M.W., 2006. WHO's new Stop TB Strategy. *The Lancet* 367, 952–955. doi:10.1016/S0140-6736(06)68392-X
- Sachdeva, K.S., Kumar, A., Dewan, P., Kumar, A., Satyanarayana, S., 2012. New vision for Re-

- vised National Tuberculosis Control Programme (RNTCP): Universal access - "reaching the unreached." *Indian J. Med. Res.* 135, 690–694.
- Sharma, N., Taneja, D.K., Pagare, D., Saha, R., Vashist, R.P., Ingle, G.K.A., 2005. The impact of an IEC campaign on tuberculosis awareness and health seeking behaviour in Delhi, India. *Int. J. Tuberc. Lung Dis. Off. J. Int. Union Tuberc. Lung Dis.* 9, 1259–1265.
- Singh, M.M., Bano, T., Pagare, D., Sharma, N., Devi, R., Mehra, M., 2002. Knowledge and attitude towards tuberculosis in a slum community of Delhi. *J. Commun. Dis.* 34, 203–214.
- Somma, D., Thomas, B.E., Karim, F., Kemp, J., Arias, N., Auer, C., Gosoniu, G.D., Abouihia, A., Weiss, M.G., 2008. Gender and socio-cultural determinants of TB-related stigma in Bangladesh, India, Malawi and Colombia. *Int. J. Tuberc. Lung Dis. Off. J. Int. Union Tuberc. Lung Dis.* 12, 856–866.
- Sreeramareddy, C.T., Harsha Kumar, H.N., Arokiasamy, J.T., 2013. Prevalence of self-reported tuberculosis, knowledge about tuberculosis transmission and its determinants among adults in India: results from a nation-wide cross-sectional household survey. *BMC Infect. Dis.* 13. doi:10.1186/1471-2334-13-16
- Sreeramareddy, C.T., Panduru, K.V., Menten, J., Van den Ende, J., 2009. Time delays in diagnosis of pulmonary tuberculosis: a systematic review of literature. *BMC Infect. Dis.* 9. doi:10.1186/1471-2334-9-91
- Stop TB Partnership (World Health Organization), 2008. Advocacy, communication and social mobilization for TB control: a guide to developing knowledge, attitude and practice surveys. World Health Organization : Stop TB Partnership, Geneva.
- Storla, D.G., Yimer, S., Bjune, G.A., 2008. A systematic review of delay in the diagnosis and treatment of tuberculosis. *BMC Public Health* 8. doi:10.1186/1471-2458-8-15
- Swaminathan, S., 2014. Tuberculosis in India - can we ensure a test, treatment & cure for all? *Indian J Med Res* 139, 333–336.
- UN, 2016. Sustainable Development Goals [WWW Document]. URL <http://www.un.org/sustainabledevelopment/health/>
- Uplekar, M., Weil, D., Lonnroth, K., Jaramillo, E., Lienhardt, C., Dias, H.M., Falzon, D., Floyd, K., Gargioni, G., Getahun, H., Gilpin, C., Glaziou, P., Grzemska, M., Mirzayev, F., Nakatani, H., Raviglione, M., 2015. WHO's new End TB Strategy. *The Lancet* 385, 1799–1801. doi:10.1016/S0140-6736(15)60570-0
- Verma, R., Khanna, P., Mehta, B., 2013. Revised National Tuberculosis Control Programme in India: The Need to Strengthen. *International Journal of Preventive Medicine* 4, 1–5.
- WHO, 2016. Tuberculosis Fact sheet N°104 [WWW Document]. WHO. URL <http://who.int/mediacentre/factsheets/fs104/en/>
- WHO, 2015. Global tuberculosis report 2015. World Health Organization, Geneva.
- WHO, 2011. *Early detection of tuberculosis An overview of approaches, guidelines and tools.*

CHAPTER - 8

Comparative Analysis of Institutional Deliveries in Kanyakumari District of Tamil Nadu

S.Ravichandran, Chief, V.Saravanakumar, Research Investigator, N.Kala, Research Investigator, M.Senthilkumar, Field Investigator

Population Research Centre, Gandhigram

ABSTRACT

Place chosen for delivery depends on the woman's perception of the quality of services and access. This paper tries to find out factors influencing the decision on place of delivery. Kanyakumari district was chosen due to the high percentage of women (>65 percent) who preferred private health institutions to public institutions for delivery. The interviewed women were from middle level income group. Statistical analysis indicates that educational level of both woman and her husband play an important role in deciding the preference for private institution as place of delivery. Other factors like age of women and gravida also play important roles. It is seen that higher percentage of women who delivered at public institutions have knowledge about government maternity benefit schemes than the women who delivered at private institutions. Among the institutions, Primary Health Centres are preferred for Antenatal Care. Doctors' availability, facilities for caesarean section delivery, scan facility are some of the improvements expected by women in public institutions. In a re-visit to the sampled area it was found that among the private institutions, CSI Hospital was preferred by most of the women for delivery. The charges are nominal and religious affiliations seem to be the reason for even the low income earning families availing private health institution services. Among the public institutions, Government Hospitals are preferred as caesarean section delivery facility, other medical and para-medical attention are always available. Also, the public institutions and PHCs are located in places where transportation and other facilities are lacking. Though huge amounts have been spent on improving the facilities, many of them have not yet become functional. For making the public utilize these facilities it is necessary to ensure that they are made functional as soon as the construction and other works are completed.

INTRODUCTION

Maternal mortality and infant mortality levels are high in India. Among the many factors that cause high mortality levels, domestic delivery is the prime cause. Low use of skilled maternity care is the result of many factors that include absence of quality care, distance and lack of transport to health institutions. Social status of women, education levels and lack of freedom due to cultural norms are the other factors that prevent women from seeking maternity care from appropriate institutions. Direct and indirect costs are also important barriers in seeking health care from the appropriate institutions.

The Union Ministry of Health has launched the National Health Mission to help the states to bridge the gaps in differences in the infant and maternal mortality rates among various states. The objective of the Health Mission is to improve the facilities for maternal and child care with the aim of reducing the highly unacceptable rates with focus on low performing states.

The direct out-of-pocket costs for maternity care include fees charged for delivery care, hospital stay and costs of drugs and supplies. In spite of the high costs at private institutions nearly half of the women delivered in a year sought private health care services. Many studies indicate that, though the services are free of cost at public institutions, other factors like fear of lack of care, absence of medical or nursing staff and other poor facilities at the public institutions are dissuading people from seeking health care services from such facilities. However, many of these issues are addressed by various provisions under National Rural Health Mission.

Tamil Nadu is considered as one of the developed states as far as health facilities and maternal and child health services are concerned. The percentage of women delivering at health institution is as high as 95 percent. It indicates the availability of

health facilities and the health-seeking behaviour of the women in Tamil Nadu. However, more than 50 percent of women in Tamil Nadu deliver at a private health facility every year. Though the maternal care services are free of cost at the public institutions, a majority of women still prefer private care. Particularly in the Kanyakumari district where the percentage of deliveries at private institutions was as high as 65 percent and, in some geographical areas, it goes well beyond this percentage. It is bewildering that in spite of the efforts made by the government to improve the facilities at public institutions the women in this district still prefer private institutions. Whether the commonly quoted 'prestige' - the in-laws insisting on delivery at a private facility - is the reason or the often cited 'inadequate' or 'indifferent attitude of staff' at public facilities is to be ascertained. This study was undertaken to identify the factors influencing the decision to seek delivery service by the family/pregnant woman.

Methodology:

Explorative study method was planned for this study. All the Primary health centres were arranged as per the percentage of deliveries in public health institutions. The PHCs were made into two groups, a group having 65 percent or more deliveries in private institutions and the other group having all the other PHCs. From each group two PHCs were randomly selected. From the Mother and Child Tracking System data, a list of women who delivered a baby during last two years prior to the survey was prepared. From this list, 100 women were randomly selected. Thus a total of 400 women who have delivered in the last two years prior to the study were selected for interviewing.

Results of Analysis

Socio-economic characteristics:

Socio-economic characteristics of households influence women's preference for the place of deliv-

ery. In this section, the socio-economic profile of the households of women interviewed is presented.

Social profile

In Table 8.1 social profile of women is given. Most of the women interviewed were Christians (82 percent) and they belonged to the Backward/ Most Backward Communities (95 percent). In the selected coastal area, there is a concentration of Christians and hence, their percentage is high in the sampled area also.

As the education level of the women increases the percentage of women preferring 'private delivery' also increases. Only 10 percent of the women having an educational qualification upto primary level have preferred the private institutions as place of delivery whereas about three-fourth of the women who are technically qualified or having a degree preferred private institutions for delivery. The trend of choosing private institutions for delivery with the increase in the education level of women is statistically significant. A similar trend pertaining to the husbands' educational qualification and preference for private institution for delivery is also statistically significant.

Type of family has not shown any significant influence on the preference for the place of delivery as there is not much variation in the percentage of place preferred for delivery.

Table.8.1 Social profile of women by place of delivery

Social characteristics	Women delivered at			
	Government (N = 200)		Private (N = 200)	
	No.	Per- cent	No.	Per- cent
Religion				
Hindu	30	15.0	35	17.5
Muslim	6	3.0	0	0
Christian	164	82.0	165	82.5

Caste				
BC / MBC	186	93.0	194	97.0
SC	12	6.0	5	2.5
Others	2	1.0	1	0.5
Education of women				
Primary	9	4.5	1	0.5
High school	84	42.0	55	27.5
Higher secondary	65	32.5	49	24.5
Diploma/ ITI	7	3.5	21	10.5
Degree /Others	35	17.5	74	37.0
Education of husband				
Illiterate	8	4.0	2	1.0
Primary	30	15.0	7	3.5
High school	125	62.5	100	50.0
Higher secondary	20	10.0	40	20.0
Diploma/ ITI	8	4.0	21	10.5
Degree /Others	9	4.5	30	15.0
Type of family				
Nuclear family	72	36.0	77	38.5
Joint family	128	64.0	123	61.5

Economic profile

Economic conditions will have an effect on the preference for place of delivery. For maternal services no fees was collected in public institutions whereas in private institutions the clients have to incur expenses for all services.

The economic profile of the women interviewed is presented in Table 8.2. It is observed that nearly all women have categorized themselves as 'housewives'. There is slightly a higher percentage of housewives preferring public health facility for delivery. It is surprising to note that three-fifth of the women working as 'cooli' preferred private health facility for delivery, however, the number in this category is small (5).

Fishing is the main occupation of the husbands of the women interviewed in the sampled area. About 54 percent of the men are engaged in fishing activities. About 19 percent of the men are working as Cooli and another 12 percent are in service, either in public sector or in private sector. For majority of the families the main occupation is fishing or low income activity of cooli.

Husbands' occupation has a significant bearing on the place of preference for delivery. The fishermen families have preferred public health facilities to private. As expected, higher percentage of women who had their husbands in 'service' have delivered at a private institution. As in the case of women's occupational category, the percentage of women preferring private institution for delivery is higher in the category in which husbands' occupation is shown as 'cooli'. With low incomes why they have preferred private institutions is a question to be probed.

As is the usual trend, the percentage of women preferring private institutions increases as they move from lower monthly income to higher income category. Women from families with an income of less than Rs.5000 have preferred public institutions (82 percent) and for majority of those in the income category of Rs.5000-10000 (55 percent) also preferred public institutions. More than 76 percent of the women in the monthly income category of 'more than Rs. 20000' have preferred private health institution for delivery indicating the 'wealthy' groups' preference for private institutions.

Family income is an important factor that shaped the woman's decision to choose the place of delivery in the study area.

Table.8.2 Economic profile of women by place of delivery

Social characteristics	Women delivered at			
	Government (N = 200)		Private (N = 200)	
	No.	Per-cent	No.	Per-cent
Occupation of women				
Housewife	194	97.0	181	90.5
Cooli	2	1.0	3	1.5
Working in government/ private	4	2.0	13	6.5
Others	0	0.0	3	1.5
Occupation of husband				
Fishing work	132	66.0	82	41.0
Cooli	36	18.0	38	19.0
Working in govt/pvt	7	3.5	40	20.0
Others	25	12.5	40	20.0
Monthly Income				
< 10,000	111	55.5	89	44.5
10,000 to < 20,000	79	39.5	79	39.5
>20,000	10	5.0	32	16.0

Demographic characteristics:

The women's demographic characteristics may have influence on the preference for particular place for delivery. Distribution of women by demographic characteristics and place of delivery is presented in Table 8.3.

More than half of the women aged below 25 years have preferred public institution for delivery. For 44 percent of women aged more than 25, the preference is for private institution delivery. The mean age at marriage of women interviewed was 21 years. Majority of the women got married between 18 and 21. The trend of preference of place of delivery indicates that if a woman gets married late then the preference is for private institution. More than 61 percent of women got married at 22 or later delivered at a private institution. Analysis

shows that as the age advances women prefer private institution as place of delivery.

Table 8.3 Demographic Characteristics of women

Demographic characteristics	Women delivered at			
	Government		Private	
	No.	Percent	No.	Percent
Age of women				
<20	6	3.0	1	0.5
21-25	101	50.5	82	41.0
26-30	72	36.0	91	45.5
31+	21	10.5	26	13.0
Age at marriage of women				
<18	25	12.5	10	5.0
18-21	147	73.5	146	73.0
22+	28	14.0	44	22.0
Family members				
≤3	14	37.8	23	62.2
4 to 6	119	47	134	53
7 to 9	46	54.8	38	45.2
10+	21	80.8	5	19.2
No. of living children				
1	84	42.0	114	57.0
2	101	50.5	78	39.0
3+	15	7.5	8	4.0
No. of gravida				
1	75	37.5	105	52.5
2	97	48.5	71	35.5
3+	28	14.0	24	12.0

Percentage distribution of women by number of family members shows that if they had a small family then the chances are that the woman would deliver the child at a private institution. For woman with seven or more members in the family the preference is for the public institution. A similar trend is observed for number of children too. This indicates that more than 57 percent of women delivering first child have preferred the private institution for delivery. This is confirmed by the distribution by gravida, more than 58 percent of women with first gravida - 'primi' - have delivered at a private institution.

Place of Ante-natal services availed

All the women were asked the place where they got their last pregnancy confirmed. The results are presented in Table 8.4. More than 86 percent of women who delivered at a public institution got their last pregnancy confirmed at the public institution, either at primary health centre (PHC) or at sub-divisional/district hospital or at the Health sub-centre. Nine per cent of the women in this category got the pregnancy confirmed at a private institution. It is interesting to note that five percent of these women confirmed their pregnancy at home and some have used the pregnancy confirmation card.

Among women delivered at a private institution, about 28 percent got their pregnancy confirmed at a public institution. Majority of them (71 percent) sought private institution services. This indicates that even for the women preferring delivery at a private institution the first contact seems to be the public institution, may be due to proximity to services.

As for the registration for receiving Antenatal services is concerned, 96 percent of women who delivered at a public institution have registered with a public health unit for receiving these services. Among women who delivered at a private institution, 37 percent have registered with a public institution for Ante-Natal services. Primary health centre is the most preferred place to get Ante-Natal services with the Health Sub-centre closely following it. This observation is also in agreement with the observation that 'proximity' plays an important role in availing ante-natal services.

Among the Ante-natal services, the essentials viz. 3 ante-natal visits, receiving IFA tablets, TT injection, test for Hemoglobin count, pregnancy-induced diabetics and scan have been availed by majority of women delivered at public institution as well as private institutions. The role of primary health centre (PHC) / block-level PHC is seen as

important for the ante-natal care services as observed from the table.

Table. 8.4 Percent distribution of women availed Antenatal care services

AN CARE	Women delivered at			
	Government (N=200)		Private (N=200)	
	No.	Per-cent	No.	Per-cent
Place of conformation of pregnancy				
HSC	3	1.5	0	0.0
PHC/Block PHC	157	78.5	49	24.5
District/SDH Hospital	12	6.0	5	2.5
Medical col- lege hospital	0	0.0	2	1.0
Private hospital				
Pregnancy card/Home	10	5.0	3	1.5
Place of AN registration				
HSC	43	21.5	23	11.5
PHC / Block PHC	138	69.0	46	23.0
GH	11	5.5	5	2.5
Private hospital	8	4.0	126	63.0
AN Services availed				
3+ AN visits	200	100.0	200	100.0
IFA / TT	170	85.0	159	79.5
Urine/Sugar test	167	83.5	155	77.5
Scan facility	173	86.5	173	86.5

Opinion of women of the services availed

The women were asked to express their opinion about the services and suggest improvements for effective delivery.

The women who delivered at the public institutions expressed that the waiting time for any services is too long. Ninety percent of the women reported that they had to wait for more than one hour to receive any attention or service. Comparatively, 30

percent of women delivered at a private institution reported that they had to wait less than an hour to receive any services at the private institution.

The major improvements expected at the public institutions are providing 'scan facility', facilities for 'Caesarean' section deliveries and doctors' availability.

Table. 8.5 AN Care availed and suggestion for improvement of services

AN CARE	Women delivered at			
	Government (N=200)		Private (N=200)	
	No.	Per-cent	No.	Per-cent
Waiting time for AN Care (advice / services)				
< 1 hr	19	9.5	61	30.5
1-2 hrs	118	59.0	99	49.5
>3 hrs	63	31.5	40	20.0
Total	200	100.0	200	100.0
Suggestions to improve AN services				
Doctors avail- ability	16	8.0	9	4.5
Nurses avail- ability	4	2.0	1	0.5
Scan facility	47	23.5	9	4.5
Staff behaviour				
Caesarean Facility	34	17.0	8	4.0
Cleaning Fa- cility	13	6.5	4	2.0
Child Care Facility / Good care for Deliv- ery	14	7.0	14	7.0
Separate room for delivery	8	4.0	0	0.0
Bathroom Facility				
Staying Facili- ty for Helpers	4	2.0	1	0.5
Total	200	100.0	200	100.0
Urine/Sugar test	167	83.5	155	77.5
Scan facility	173	86.5	173	86.5

Reasons for opting a particular type of institution

All the women were asked why they chose a particular place for delivery. The women delivered at a public institution said that the institution was near their home/ mother's home (42 percent); PHCs refer complicated cases to Government hospitals where facilities are more (24 percent) and the family is poor and cannot bear the expenses at private hospitals (11 percent). The women who delivered at a private institution given varied reasons: Family doctor/good care (19 percent), Previous history of poor health (16 percent), near home/ mother's home (14 percent), not satisfied with public institution services (13 percent) and due to recommendation of relatives (11 percent).

Table 8.6 Reasons for preferring a particular place for delivery

NATAL CARE	Women delivered at			
	Government (N=200)		Private (N=200)	
	No.	Percent	No.	Percent
Reasons for preferring place of delivery				
Doctors' Good care /Family Doctor	0	0	37	18.5
Fear of Going GPHC/GH/not satisfied	0	0	26	13
Good care for Normal Delivery	17	8.5	18	9
Near House/ Mother's House	83	41.5	28	14
First Baby Delivery In private Hospital	0	0	10	5
More Staff's in Private Hospital	0	0	2	1
Caesarean Facility	4	2	2	1
Relation Recommendation	2	1	22	11

Over Pain/BP / complication cases	18	9	32	16
Own Interest	0	0	8	4
VHN's referral	9	4.5	0	0
Money Problem /Poor Family	22	11	3	1.5
PHC's refers to GH	47	23.5	9	4.5
Others	7	3.5	21	10.5
Vehicle used for delivery				
Govt. ambulance/108	39	19.5	0	0
Private vehicle/ Auto	114	57	162	81
Own vehicle	3	1.5	14	7
Bus	19	9.5	19	9.5
By walk	25	12.5	5	2.5
Total	200	100	200	100
Vehicle expenses (Rs.)				
<=100	58	44.6	40	21.4
101-200	10	7.7	23	12.3
201-500	44	33.8	90	48.1
501+	18	13.8	34	18.2
Total	130	100	187	100

The women were asked what type of transportation they used to reach the institution where they delivered the baby and how much do they spent on transportation.

It may be observed from the table that, among the women who delivered at a public institution, 57 percent used private vehicle, 20 percent used government ambulance/108 and 13 percent reached by walk. Sixty-eight per cent of these women spent less than Rs.200 for transportation.

Among women who delivered at a private institution, 81 percent used private vehicle and none used the government ambulance, 108 services and 63 percent of these women spent a minimum of Rs.200 as transportation charges.

Table. 8.7 Type of delivery and expenses on delivery

NATAL CARE	Women delivered at			
	Government		Private	
	No.	Per-cent	No.	Per-cent
Type of delivery				
Normal	116	58	81	40.5
Caesarean	84	42	119	59.5
	200	100	200	100
Total				
Delivery expenses at the institutions				
<10,000	115	89.8	5	2.5
10,000 to 20,000	5	3.9	53	26.9
21,000 to 30,000	4	3.1	51	25.9
30,000+	4	3.1	88	44.7
Total	128	100	197	100
Vehicle used for return after delivery				
Govt. ambulance/108	17	8.5	0	0
Private vehicle/ Auto	178	89	191	95
Own vehicle	2	1	9	5
By walk	3	1.5	0	0
Total	200	100	200	100
Vehicle Expenses (Rs.)				
<=100	49	27.5	26	13.6
101-200	15	8.4	28	14.7
201-500	57	32	90	47.1
501+	57	32	47	24.6
Total	178	100	191	100

For fifty-eight percent of women who delivered at the public health institution, the delivery was normal and 42 percent had to undergo caesarean section. Whereas sixty percent of the women who delivered at the private institution had caesarean section delivery. Ninety per cent of the women who delivered at the public institution did not spend any money.

The women who delivered at a public facility were

asked to list the services they received at the institution and the reasons for preferring the facility. They were also asked to suggest, in their opinion, the things to be improved to make the services better. Seventy-nine per cent of the women received food at the institution while they were in the institution for delivery. They also received the JSY amount of Rs.700. Eighty-seven percent of the women said attendants were allowed to accompany them and eighty per cent of the women were satisfied with the services provided at the hospital.

Sixty- four percent of the women expressed that good care was given at the public hospitals as part of delivery services and 13 percent were hesitant to go to private institutions as they were not satisfied with the services offered there.

The women who delivered at private institution informed that they preferred the institution because, in their opinion, the private institutions offered 'best delivery care' (32 percent); they have a fear that at public institutions the care is not proper (20 percent) and the private institutions were near their house or mother's house (13 percent). Sixty-four percent of the families of women who delivered at private institutions borrowed money to meet the expenses.

Awareness and utilization of Government health programmes

Utilization of the Government health facility depends on the awareness and benefits gained from various health programmes. All the interviewed women were asked whether they were aware of some of the schemes and whether they were the beneficiaries.

Nearly all women who delivered at a Government facility and more than three- fourth in a private facility were aware of the Dr.Muthulakshmi Reddy Maternity Benefit Scheme under which Rs.12,000/- is paid to the pregnant women who register for maternity services. However, less than

50 percent of women who delivered at public institutions received the money. The percentage of women who are aware of Janani Suraksha Yojana (JSY) was very high; 95 percent for women who delivered at a public institution and 44 percent for women who delivered at a private institution.. More than 72 percent of the women who delivered at a public institution and 49 percent of women who delivered at a private institution were aware of the package of maternal and child health services offered by public institutions.

The Government has promoted health insurance for the people living below poverty line. Also, various private and public insurance companies are promoting health insurance. The women were asked whether their family is covered under health insurance. One-fourth of the women who delivered at the public institution and about one-third at private institution replied that their family is having a health insurance. Of the families covered by a health insurance, eighty-eight percent of women delivered at public institution and 54 percent at private institution were covered by the government health insurance scheme.

Table. 8.8 Awareness about Government health programmes

Schemes	Women delivered at			
	Government (N=200)		Private (N=200)	
	No.	Percent	No.	Percent
Government health programmes				
Heard about Dr.Muthu-lakshmi Reddy Scheme	199	99.5	152	76.0
Financial assistance Received	97	48.5	-	-
Aware about JSY	189	94.5	88	44.0
Aware about maternal and child health services of government	145	72.5	98	49.0

Having Health Insurance for family	50	25.0	65.0	32.5
Covered by Government Health insurance	44	88.0	35.0	53.8

Summary and Conclusions

This study was carried out in four Primary Health Centre (PHC) area of Kanyakumari district to identify the factors influencing decision on place of delivery. Kanyakumari district was chosen due to the high percentage of women (>65 percent) preferring private health institutions to public institutions for delivery. Four hundred women who have delivered a baby within two years before survey were interviewed. The sample was selected in such a way that t equal representative respondents were included in both the PHC groups i.e. high percentage of women delivered in private institutions and also high percentage of women delivered in public institutions. Maternal and Child Tracking System records were used for sample selection. The women were from middle level income group and mostly they/their husbands were in occupations like fishing or 'coolie.' Statistical analysis indicate that educational level of both woman and her husband play an important role in deciding the preference for private institution as place of delivery. Other factors like age of women and gravida also play an important role. It is seen that higher percentage of women who delivered at the public institutions have knowledge about government maternity benefit schemes than the women who delivered at private institutions. Among the institutions, Primary Health Centres are preferred for Ante-natal Care. Doctors' availability, facilities for caesarean section delivery and scan facility are some of the improvements expected by women in public institutions.

During our re-visit to the district it was found that the CSI Hospital was the institution visited by most of the women who preferred a private insti-

tution for their delivery. Nominal charges and religious affiliations seem to be the reason for even the low income-earning families availing private health institution services instead of public health facilities in the sampled area. Also it was found that the public institutions like PHCs are located in places where transportation and other facilities are lacking. Though huge amounts have been spent to improve the facilities many of them have not yet been made functional. For example, the 30-bed hospital at Muttom was constructed a year back but it was not yet put into use at the time of the visit. For bringing the facilities to the domain of the public utility it is necessary to ensure that the facilities are made functional as soon as the construction and other works are completed.

REFERENCES

- Bhatia, Jagdish C. and John Cleland, 1995. *Determinants of maternal care in a region of South India* Health Transition Review 5, 1995,
- Kesterton, Amy J, John Cleland, Andy Sloggett and Carine Ronsmans. 2010. *Institutional delivery in rural India: the relative importance of accessibility and economic status* BMC Pregnancy and Childbirth 2010, 10:30
- Kowalewski, Marga, Phare Mujinja and Albrecht Jahn 2002. *Can Mothers Afford Maternal Health-Care Costs? User Costs of Maternity Services in Rural Tanzania*. African Journal of Reproductive Health Vol. 6, No. 1 (Apr., 2002),
- Ladusingh , Laishram & Chungkham Holendro Singh (2007) *Rich Poor Gap in Maternal Care*, Asian Population Studies, 3:1,
- Lim , Stephen S, Lalit Dandona, Joseph A Hoisington, Spencer L James, Margaret C Hogan and Emanuela Gakidou, 2010. *India's Janani Suraksha Yojana, a conditional cash transfer programme to increase births in health facilities: an impact evaluation*. The lancet Vol 375 June 5, 2010
- Nahar, Shamsun and Anthony Costello, 1998. *The hidden cost of 'free' maternity care in Dhaka, Bangladesh*. HEALTH POLICY AND PLANNING; 13(4)
- Skordis-Worrall, Jolene, Noemi Pace, Ujwala Bapat, Sushmita Das, Neena S More, Wasundhara Joshi, Anni-Maria Pulkki-Brannstrom, David Osrin. *Maternal and neonatal health expenditure in Mumbai slums (India): A cross sectional study* BMC Public Health 2011, 11:150

CHAPTER - 9

Status of Community Health Centres (CHCs) in Madhya Pradesh: An Assessment based on HMIS Infrastructure Data

Nikhilesh Parchure¹ and Reena Basu ¹Research Investigator, ² Assistant Director

Population Research Centre, Sagar (MP)

Abstract

Today, the health infrastructure of India for delivery of public health services in general and secondary health care facilities in rural areas in particular, are in a pathetic condition and needs radical reforms to deal with new emerging challenges. Till the launch of HMIS, systematic data regarding existing infrastructure at public health facilities in rural areas was not available for individual facilities. This study is aimed at assessing status of Community Health Centres (CHCs) in Madhya Pradesh in terms of growth, distribution, physical infrastructure, human resources and available services. Community Health Centres are the first referral contact in case of any health emergency in the remote rural areas. Descriptive statistics such as frequencies, cross-tabulation have been used. Infrastructure index has been computed to assess the level of infrastructure available in the CHCs. Results show that after initiation of NRHM, 129 new CHCs have been added to the rural public health institutions in MP. Highest concentration of eight PHCs per CHC is observed in Gwalior and Mandsaur districts. There are more than 30 SHCs per CHC in 12 districts of the State. Concentration of CHCs per 100,000 population is more in high priority districts. Nearly two-thirds of all the CHCs have good interior and exterior. Human Resource availability is highly inadequate in all the CHCs. Only one CHC has all the specialists according to specialty services it provides. Nearly half of all the CHCs have overall poor infrastructure index showing inadequacy in all the dimensions of infrastructure, manpower and services. There is urgent need to upgrade the poor facilities in rural areas to improve the efficacy of secondary level public health care services. The study highlights the need to orient facility-level programme managers in reporting of corrected and authenticated data under HMIS. The infrastructure data for CHCs can be used for better planning and assessment of service delivery vis-à-vis upgradation of infrastructure.

INTRODUCTION

The global health scenario in recent times has indicated that India's health indicators pose greater public health challenges in achieving universal health goals such as Millennium Development Goals (MDGs) and now Sustainable Development Goals (SDGs). It is universally accepted that rural areas have excessive demand for health care services and NRHM has paved the way to cater to rural area's health needs by creating required health infrastructure in terms of manpower, services and linkage with higher health facility. Studies on public health institutions in India, in various Five Year Plan period, show that growth of public health institutions has been a demand-driven initiative under the ever-changing health scenario in India. Given the state- level disparities of health indicators and health being a state subject, every state has evolved its own system of health care service delivery.

Health infrastructure is an important indicator for understanding the health care policy and welfare mechanism in a country. It signifies the investment priority with regards to the creation of health care facilities (Kumar, 2012). Infrastructure has been described as the basic support for the delivery of public health activities. Five components of health infrastructure can be broadly classified as: skilled workforce; integrated electronic information systems; public health organizations, resources and research. From Bhore Committee (1946) to National Commission on Macroeconomics and Health (2005) stressed the need to create and maintain adequate infrastructure for delivering public health care to general masses in a decentralized manner.

In Madhya Pradesh, three types of rural public health institutions have been established and their services and infrastructure vary according to their type. Community Health Centre (CHC) is the larg-

est health care unit in rural area which is usually located at block headquarters. CHC provides most of the secondary care health services, including emergency care. For every 1 lakh population one CHC has been established. Primary Health Centre (PHC) is mainly located in bigger villages and normally caters to a population of 30,000. For every CHC, 5-6 PHCs are established. A sub-health centre (SHC) is the smallest unit among the three health units. It is located at village level and caters to a population of 5000.

Community Health Centres which constitute the secondary level of health care were designed to provide referral as well as specialist health care to the rural population. Indian Public Health Standards (IPHS) for CHCs have been prescribed under National Rural Health Mission (NRHM) as early as 2007 to provide optimal specialized care to the community and achieve and maintain an acceptable standard in quality of care. The First Referral Units (FRUs) constitute of Community Health Centres (CHCs), Sub-district and District Hospitals. The CHCs were designed to provide referral health care for cases from the Primary Health Centres level and for cases in need of specialist care approaching the centre directly. CHC is a 30-bed hospital providing specialist care in Medicine, Obstetrics and Gynecology, Surgery, Paediatrics, Dental and AYUSH. These centres are, however, providing the services only to a limited extent due to limited resources and manpower.

Limited availability of regular data regarding available infrastructure, human resources and its linkages with existing health infrastructure has hampered proper planning for creating adequate, accessible and required health infrastructure in the rural areas (CSO, 2015). Under National Rural Health Mission (NRHM) many architectural corrections in terms of strengthening of rural public health care have been taken up. It gives an opportunity to have a fresh look at the functioning

of CHCs. In this scenario it is necessary to study the growth and distribution of community health centres and to assess the available infrastructure, manpower and functional status of CHCs by analyzing existing Health Management Information System (HMIS) data.

1.Objectives

- a. To study growth and distribution of different types of rural public health institutions.*
- b. To analyze coverage, infrastructure and amenities of community health centres.*
- c. To ascertain the human resource availability in community health centres.*
- d. To study the available services at the community health centres.*
- e. To assess level of infrastructure availability and compute Facility Infrastructure Index.*
- f. To ascertain gap in reported and actual available infrastructure at the selected CHCs.*

Data and Methods

Data for the present paper has been drawn from facility-wise HMIS Infrastructure data for 2015-16 of Madhya Pradesh. HMIS web portal <https://nrhm-mis.nic.in> provides data on health infrastructure for individual public health facilities. This HMIS data is reported annually at the beginning of the financial year i.e. as on 1st April. In HMIS infrastructure details of each public health facility in rural and urban areas are reported.

Number of Data Elements in HMIS Health Infrastructure Data

Health Institution Type	Number of Data Elements
District Hospital	418
Sub District Hospital	332

Community Health Centre	412
Primary Health Centre	231
Sub Health Centre	116

HMIS infrastructure data provides information on ownership of health facility, coverage in term of population and geographic area, availability of basic amenities, including residential facilities, services, human resource, equipment and their functional status etc. In addition to HMIS, in RHS information regarding status of building, residential facility, human resources, clinical facilities etc. are provided. RHS also provides information on health care infrastructure for tribal areas separately. The following table provides number of data items reported in HMIS infrastructure data for each health facility.

Separate formats have been designed in HMIS for reporting of infrastructure details for different types of health facilities; District Hospital (DH), Sub District Hospital (SDH), Community Health Centre (CHC), Primary Health Centre (PHC) and Sub Health Centre (SHC) in HMIS. Altogether, HMIS infrastructure data for all types of public health facilities cover 1509 data items.

Infrastructure data reporting under HMIS has started since 2013-14 for all the existing health facilities on regular and continuous basis. From 2014-15 onwards more than 90 percent of all the health facilities are reporting annual infrastructure data. The study has analyzed status of health care facilities in terms of availability of infrastructure, human resources, basic amenities, services etc.

To accomplish the objectives of the study, primarily HMIS facility-level annual infrastructure data for 2015-16 were used. Most updated and corrected data was downloaded from HMIS portal <https://nrhm-mis.nic.in> as on 8th April, 2016 for 2015-16. The data was then compiled for analysis using excel programming and then transferred to SPSS data files. This paper deals with the status of physi-

cal infrastructure, human resources, amenities and services availability in community health centres of Madhya Pradesh. For analyzing the status of infrastructure facilities and health care services 28 indicators were selected for the study. Proportion, ratios have been computed and results have been shown graphically to analyze the comparative status of health facility infrastructure in each of the 51 districts of MP using bi-variate tables.

Facility Infrastructure index: Facility Infrastructure index has been computed using 41 data items covering six dimensions - Services Availability, Bed Strength, Human Resource, availability of Government Building and its Interior and Exterior, Basic Essential Amenities and Availability of Residential Quarters. These dimensions also cover availability of infrastructure in sub-categories. Including all 41 types of infrastructure facilities, 10 indices were constructed and then overall Facility Infrastructure Index has been computed by averaging all the 10 indices.

Growth of Rural Public Health Institutions

Population of Madhya Pradesh is predominantly rural. As per Census 2011, 72.4 percent population is rural. Catering to the health care needs of rural population is an uphill task where population is sparsely located in small and scattered villages. Population density in rural MP is 175 persons per km². Out of total 51,929 inhabited villages 17,377 villages (34 percent) have a population size of less than 500 persons. During last 15 years, the State has established many health facilities in remote, un-served and under-served rural areas to improve accessibility of health care services at the grassroots level.

Table 9.1 shows number of public health institutions in rural areas. Since the implementation of NRHM in 2005, Madhya Pradesh has upgraded its secondary care public health facilities in rural areas. The number of CHCs have increased by 105, from 229 in 2005 to 334 in 2015. However, there is a decline of 10 sub-health centres since 2005

Table 9.1: Growth of Rural Public Health Institutions, MP

Type of Health Facility	Year					
	2001	2005	2010	2013	2014	2015
CHC	229	229	333	333	334	334
PHC	1194	1192	1155	1156	1157	1171
SHC	8835	8874	8869	8869	8764	8764
Total	10258	10295	10357	10358	10255	10269
<i>Source: RHS, Govt. of India, Ministry of Health & Family Welfare</i>						

Based on the Index values, CHCs were categorized in three groups – Good, Average and Poor. A CHC is categorized as ‘Good’ in terms of infrastructure if index value is in 76-100 range. An index values between 51-75 indicates average infrastructure and index value of 50 or below shows poor infrastructure in the CHC.

Results and Discussion

and presently 8764 SHCs are functional in 2015. Number of Primary Health Centres has decreased from 1192 in 2005 to 1171 in 2015.

To effectively monitor the services being offered at the health care facilities in rural areas, HMIS was introduced in 2008 to capture data on service delivery and available infrastructure in these fa-

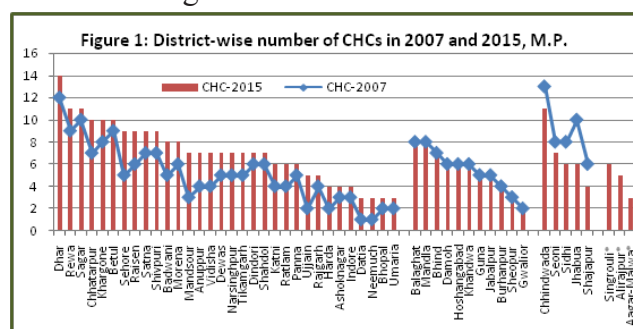
cilities. To enable the data capturing each health facility was mapped in HMIS. During 2008-09, HMIS mapped 10,363 public health facilities in rural areas. Out of these, 331 were CHCs, 1270

were PHCs and 8762 were SHCs. In 2015-16 total number of public health facilities mapped in HMIS was 10,389. In all, 26 more health facilities (4 CHCs, 4 PHCs & 18 SHCs) were added between

Table 9.2: Number of Public Health Facilities Mapped in HMIS, MP

Type of Health Facility	Year							
	2008-09	2009-10	2010-11	2011-12	2012-13	2013-14	2014-15	2015-16
CHC	331	331	331	330	331	331	335	335
PHC	1270	1270	1270	1271	1270	1270	1271	1274
SHC	8762	8762	8762	8762	8762	8762	8762	8780
Total	10363	10363	10363	10363	10363	10363	10365	10389
Source: HMIS Web Portal https://nrhm-mis.nic.in								

2008-09 and 2015-16 (Table 9.2). For providing specialty health care services affordably in rural areas, the state government has set-up nearly 100 new CHCs in rural areas within a span of 3 years after launching of NRHM in 2005.



In all, number of CHCs has increased from 237 to 334 during 2007 and 2015. Distribution of CHCs shows that in Mandla and Sehore districts four new CHCs have been established. Three new CHCs were established in each of the districts of Agar-Malwa, Anuppur, Balwani, Chhatarpur,

Raisen, Ujjain and Vidisha districts. In 14 districts two new CHCs were added while in 10 districts one CHC has been added in the ensuing period. It is noted that 32 districts have added a total of 64 CHCs in this period. Sehore, Mandla, Chhatarpur, Raisen, Balwani, Anuppur, Vidisha and Ujjain have added 26 CHCs. However, in 11 districts, number of CHCs has remained unchanged while in five districts number of CHCs has declined during 2007 and 2015 (Figure 9.1).

Distribution of Rural Public Health Institutions

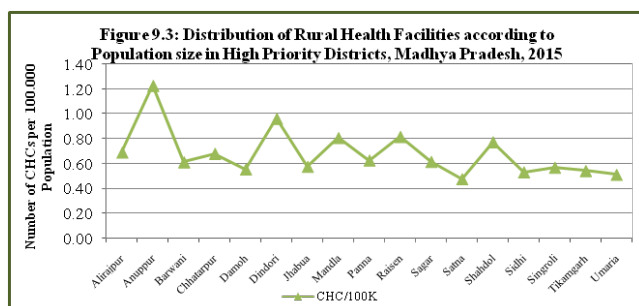
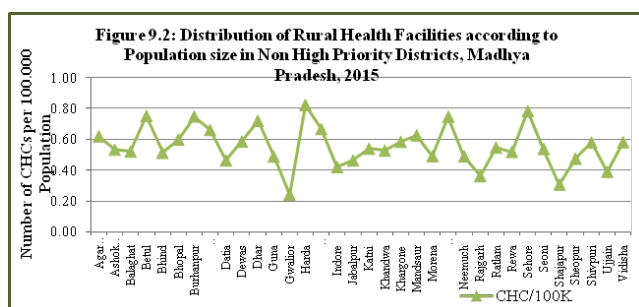
For analyzing the distribution of rural health institutions HMIS facility-level data has been used. A ratio of PHC per CHC and SHCs per CHC has been computed to ascertain the gap in distribution in the district. It is then compared with the state-level ratios in the form of standard distance. During 2015-

Table 9.3: Ratio of PHCs and SHCs to CHCs by Districts, M.P., 2015

District	Ratio PHC-CHC	Ratio SHC-CHC	District	Ratio PHC-CHC	Ratio SHC-CHC
Agar Malwa	1.7	24.0	Khandwa	4.8	32.2
Alirajpur	2.8	31.4	Khandwa	5.2	28.2
Anuppur	2.3	25.0	Mandla	3.2	27.6
Ashok Nagar	2.3	24.3	Mandla	8.4	34.0
Balaghat	4.0	32.0	Morena	2.4	28.7
Barwani	3.6	29.6	Narsinghpur	3.2	24.0
Betul	3.3	26.4	Neemuch	6.0	34.7
Bhind	2.9	27.0	Panna	2.2	24.2

Bhopal	3.0	21.0	Raisen	2.9	26.7
Burhanpur	3.3	24.3	Rajgarh	5.4	33.8
Chhatarpur	3.6	19.1	Ratlam	4.8	32.2
Chhindwada	4.7	22.3	Rewa	2.6	25.7
Damoh	2.3	27.3	Sagar	2.5	22.3
Datia	3.3	32.3	Satna	4.6	29.3
Dewas	3.3	29.8	Sehore	1.7	17.0
Dhar	3.0	27.2	Seoni	3.8	33.4
Dindori	3.1	26.0	Shahdol	3.6	24.0
Guna	3.0	23.6	Shajapur	3.5	27.3
Gwalior	8.0	50.5	Sheopur	3.0	30.0
Harda	1.5	15.8	Shivpuri	1.5	23.9
Hoshangabad	2.1	21.9	Sidhi	4.2	24.2
Indore	6.3	27.5	Singroli	2.3	26.3
Jabalpur	4.8	38.0	Tikamgarh	2.9	22.3
Jhabua	2.4	29.4	Ujjain	4.4	35.8
Katni	2.8	26.3	Umaria	4.0	29.7
M. P.	3.4	26.7	Vidisha	3.0	21.0

16 there were 8943 sub-health centres, 1170 PHCs and 335 CHCs functioning in Madhya Pradesh as per the HMIS facility-level data. In other words, there were 3.5 PHCs per CHC and 26.7 SHCs per CHC are functional in the State. However, the distribution of PHC and SHC per CHC is not uniform in the district (Table 9.3).



Mandsaur and Gwalior districts have eight PHCs per CHC which is highest amongst all the districts.

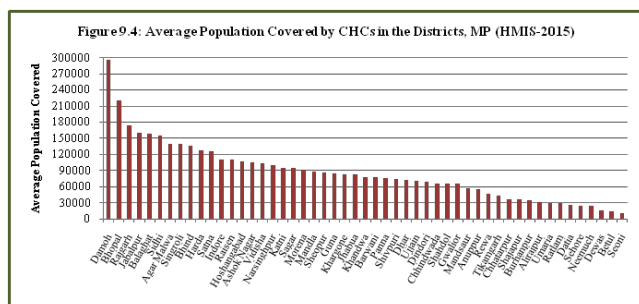
Thirteen other districts have four or more PHCs per CHCs. Harda, Shivpuri, Sehore and Agar-Malwa have less than two PHCs per CHC. The ratio of SHCs per CHC shows a very skewed distribution. In 12 districts, there are more than 30 SHCs functional per CHC. Gwalior has 50 SHCs per CHC followed by Jabalpur with 38 SHCs per CHC. The standard distances of proportion of CHCs in the district to state is highest in Harda district where there are nearly three-fifths more CHCs in the total health institutions than the proportion of CHCs to the total health facilities in the state. Similarly, Sehore, Sagar, Vidisha, Hoshangabad, Chhatarpur and Bhopal districts have higher proportion of CHCs in the district as compared to the state average.

Distribution of CHCs as per population norms indicates that for every 100,000 population there are 0.58 CHCs functional in the State. It is observed that in (among?) non-high priority districts there is nearly one CHC for 100,000 populations in Harda district followed by Sehore and Betul. In Gwalior, Rajgarh and Shajapur districts for every 2-3 lac population only one CHC is functional (Figure 9.2). The situation is somewhat better in high-priority

districts. Anuppur has more than one CHC for every 100,000 population, followed by Dindori and Shahdol. It is further noted that in Mandla, Shahdol and Raisen there is one CHC for every 1.5 lac population. In Satna for every 2 lac population only one CHC is functional (Figure 9.3). Population Coverage

Coverage data of health institutions provides vital information about their catchment areas, population coverage and accessibility. HMIS infrastructure data provide coverage of all the rural public health institutions in terms of population it caters to. This information is vital for planning for transport facility, human resources and services to be offered at the health institutions.

The analysis shows that out of 335 CHCs reporting data under HMIS, 329 CHCs have reported population coverage data in 2015. The average population covered per CHC was 83,919 in the State as a whole. Nearly 30 CHCs in 19 districts have reported less than 10,000 population coverage. All the CHCs in Barwani, Datia, Dewas and Seoni have reported less than 35,000 population coverage. (Figure 9.4).

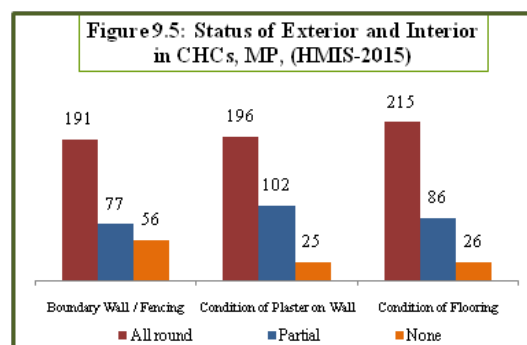


CHC Batiyagarh in Damoh district reported population coverage of 1410000, whereas CHC Chhegaon Makhan in Khandwa reported zero population coverage. It is found that 23 CHCs in 14 districts have reported less than 10,000 population coverage. Population coverage of all the five CHCs in Betul is less than 10,000.

Physical Infrastructure

For every CHC in the rural area availability and status of physical infrastructure is reported in HMIS. For the study purpose availability of building, condition of interior and exterior, essential amenities and infrastructure have been considered.

Building and its Area: In 2015, reporting of ownership of CHC building and its area was done by 330 and 241 CHCs respectively. It is noted that out of 330 CHCs, 324 are functioning from government-owned building while five CHCs in Barwani, Burhanpur, Chhindwara and Dindori districts still have no government-owned building and are functioning from rent-free building. Mean area of the CHC building is 9130 m² in the State. Fifteen CHCs have reported less than 100 m² or 1000 ft² building area. None of the CHCs in Alirajpur, Singroli and Shajapur district have reported their building area. Maximum building area of 1000000 m² is reported by CHC Ashta (Sehore), followed by 100000 m² building area by CHC Khategaon (Dewas) and CHC Jatara (Tikamgarh). There are 35 CHCs which have building area of more than 10000 m². As per IPHS standard, each CHC should have patient services area of 1500 m². It is noted that nearly one-third of CHCs have building area less than the IPHS standard and 23 CHCs have building area of more than 10000 m². It may be mentioned here that HMIS does not specify which area CHCs should report, whether patient services area or total area combined, including open land available with CHC.



Condition of Interior and Exterior: A proper compound wall or fencing all around the CHC premises and proper interior, including plastered walls and flooring are essential, and they must be intact in order to prevent dust, water seepage to maintain contamination-free and hygienic condition in the health institutions. In 2015 number of CHCs reporting about availability of boundary wall or fencing, status of plaster on walls and status of flooring was 324, 323 and 327 respectively. Figure 5 shows that majority of CHCs have reported that exterior and interior were in good condition. All round good flooring is reported by 215 CHCs, all round plastered wall by 196 CHCs and 191 CHCs have reported all around boundary wall. It was found that in 2015-16 all 115 CHCs functioning from government building have boundary wall/fencing all round, interior and exterior walls all plastered and good flooring. CHC at Lalbagh (Burhanpur), Dahi (Dhar), Kailaras (Morena), Mauganj (Rewa) and Mandi Bamora (Sagar) were functioning from a government building but did not have boundary wall, well plastered wall or good flooring.

Basic Amenities: Status of availability of certain basic amenities is also captured in HMIS infrastructure data for each of the CHCs. These are source of water, electricity, and laundry and sewerage system. It is necessary that all the health facilities should have these basic amenities. In 325 CHCs which have reported about source of water supply; 63 have piped water supply and 262 have bore well /tube well / hand pump. Adequate water storage facility in terms of Over Head Tank (OHT) of sufficient capacity and fitted with functioning water pump is available in 262 CHCs while 31 CHCs do not have OHT at all.

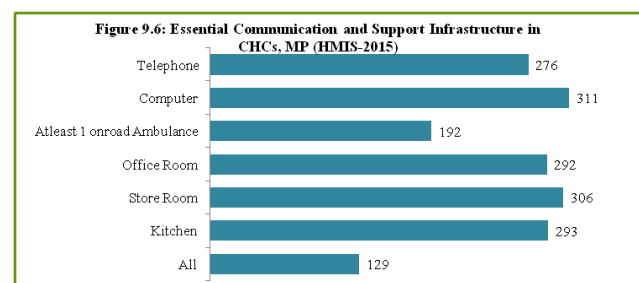
Electricity available in all parts of CHCs was reported by 315 CHCs. Majority of CHCs have generator as standby facility for any power cut or failure. However, availability of regular electricity supply is available in only 165 CHCs. Number of CHCs reporting frequent power cuts and power failure is 163 in the state.

Laundry service is a basic necessity in all the health institution. It ensures basic hygiene for inpatients as well as service providers. For all essential services such as delivery, surgical procedures clean and sterilized linen is a pre-requisite. In-house laundry is available in only 63 CHCs, while all other CHCs have outsourced laundry services. In 2015, all the existing CHCs in 18 districts had outsourced laundry services.

Availability of proper sewerage system is an integral part of any health institution whether providing primary or secondary or tertiary care service. Availability of sewerage system is reported by 310 CHCs. Number of CHCs having soak-pit is 228 and in 92 CHCs waste disposal was connected to municipal sewerage system.

Essential Infrastructure: In conjunction with basic amenities, every CHC is supposed to have essential infrastructure. These include separate male and female wards, adequate number of beds for male, female and children, essential means of communication, infrastructure for administrative set-up of CHC and residential facilities for the staff. It is observed that most of the CHCs are either lacking or are not adequately supported with all essential infrastructures. In CHCs which do have these facilities, these are seldom found in working condition.

HMIS data shows that out of 335 CHCs only 262 have separate male and female wards. Adequate number of beds should be available for patients in the CHC. Normally, all the CHCs are equipped with 30 inpatient beds. It is observed that in MP on an average a CHC has 29 beds. Of these, 11 are for males, 13 for females and 5 for children. It is noted that 91 CHCs in 43 districts have more beds



designated for female patients than male patients. Sixty-four CHCs have less than 15 beds in total. In four districts none of the CHCs has separate paediatric beds.

Figure 9.6 shows that availability of essential communication and support infrastructure vary across different facilities. Availability of kitchen is reported by 293 CHCs which is an integral service with the introduction of JSSK programme.

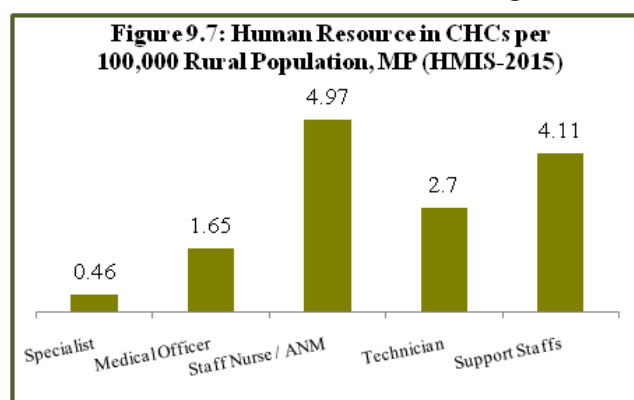
Availability of at least one on road ambulance is lacking in many facilities. Presently, 192 CHCs have this facility. It is noted that there is lot of scope for improving availability of all essential infrastructure in various CHCs. Only 129 out of total 335 CHCs have all the above essential communication and support infrastructure in place.

Residential Quarters: In Madhya Pradesh, CHCs have more than 40 sanctioned posts, including medical, paramedical and support staffs. Ideally, every CHC should have accommodation for all the staff to ensure round-the-clock-services. Status of availability of residential quarters in the CHCs shows a grim picture across the State. In 2015, 307 CHCs have reported about availability of residential quarters for different staff. HMIS data indicates that residential quarters at CHCs are available for a total of 4307 staff in the State. To assess the adequacy of residential quarters at CHCs, number of CHCs having at least four residential quarters for different staff categories – Specialists/ doctors, Paramedical and Support Staff has been computed from HMIS data.

Only 43 CHCs have adequate residential quarters for specialists, in 210 CHCs, four or more quarters are available for paramedical staff and 129 CHCs have four or more residential quarters for support staff (Table 9.4). In half of the districts, none of the CHCs has four or more residential quarters for specialists. In Seoni, Shajapur and Singrauli districts none of the CHCs have four or more quarters for support staff.

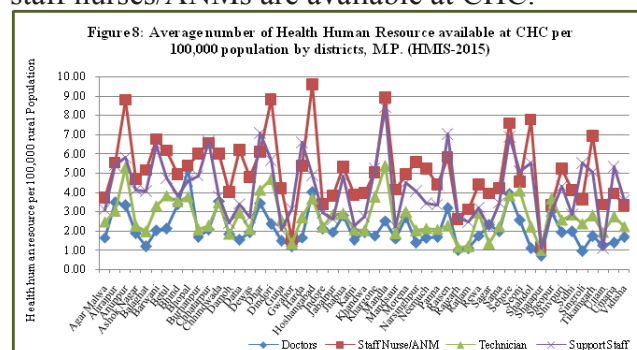
Human Resources

Madhya Pradesh is facing an acute shortage of human resources for health care services. In rural areas, majority of health institutions are functioning without necessary staff. Even contractual staff is not available for placement in rural areas. Despite state government's efforts, doctors and paramedic staff are not willing to join rural health institutions even for higher wages. HMIS provides valuable information on availability of human resources in rural health institutions for 33 staff categories.



Human resources availability in the CHCs of the State is very disappointing. There is a total of 9318 staff posted across all the CHCs. On an average, there are 28 staff posted at each CHC in the State. HMIS data shows that there were 262 specialists and 939 medical officers posted at CHCs in the State. In 7 districts - Anuppur, Damoh, Harda, Shahdol, Sheopur, Singroli and Umaria - not a single specialist is posted in all the 37 CHCs.

Figure 9.7 gives a snapshot of availability of health staff across different categories. It is clear that there is a severe paucity of specialists and medical officers at the CHCs. On an average five staff nurses/ANMs are available at CHC.



Further, district-level variation in the availability of different cadre of health personnel as depicted in Figure 9.8 indicates that CHCs of Alirajpur, Bhopal and Chhindwada have more than two specialists per 100000 rural population. Number of doctors, including specialists per 100000 rural

Table 9.4: Essential Infrastructure available in CHCs, MP (HMIS-2015)

Number of CHCs having Essential Infrastructure															
Districts	Number of CHCs	Separate Male/Female Ward	Average number of Beds for			Telephone	Computer	At least one on road Ambulance	Office Room	Store Room	Kitchen	At least 4 Residential quarters available for			
			Male	Female	Child							Specialist	Paramedics	Support Staff	
Agar Malwa	3	3	17	20	3	3	3	0	3	3	2	1	1	1	
Alirajpur	5	3	7	13	13	5	5	1	4	4	4	2	4	5	
Anuppur	7	5	19	11	4	7	7	4	7	7	7	0	5	4	
Ashok Nagar	4	3	10	16	8	3	4	3	4	4	4	1	4	1	
Balaghat	9	4	15	15	5	9	9	8	9	9	9	0	7	9	
Barwani	8	8	6	7	0	8	8	7	8	8	4	0	8	7	
Betul	10	10	11	13	3	9	10	3	10	10	10	1	5	2	
Bhind	7	7	21	17	11	3	7	3	7	7	7	2	6	4	
Bhopal	3	3	8	15	5	3	3	2	3	3	3	2	3	3	
Burhanpur	4	4	9	9	6	4	4	3	4	4	4	0	4	2	
Chhatarpur	10	10	7	7	2	8	8	6	8	10	10	0	7	2	
Chhindwada	14	9	12	15	5	10	13	11	11	12	14	5	11	6	
Damoh	6	4	10	16	5	5	5	4	5	5	4	0	2	1	
Datia	3	2	10	20	1	3	2	1	3	3	2	1	2	2	
Dewas	6	4	8	15	2	5	5	4	6	6	6	0	5	4	
Dhar	15	13	12	15	2	11	14	9	12	13	13	5	10	7	
Dindori	7	5	12	10	4	6	7	4	5	7	7	0	6	1	
Guna	5	4	13	18	5	3	4	2	5	5	5	0	5	2	
Gwalior	2	1	8	17	8	2	2	0	2	2	2	1	2	2	
Harda	4	4	4	9	10	4	4	2	4	4	3	1	3	2	
Hoshangabad	7	6	11	17	1	7	6	2	7	7	5	0	3	1	
Indore	4	4	11	11	4	4	2	3	2	2	4	0	3	3	
Jabalpur	5	5	11	15	5	5	5	5	5	4	5	0	3	2	
Jhabua	7	7	10	14	1	7	7	3	7	7	7	2	3	2	
Katni	6	5	9	10	7	4	6	4	6	5	5	1	5	2	
Khandwa	6	5	13	14	7	5	6	5	5	5	5	0	4	3	
Khargone	10	7	9	12	2	8	9	7	7	9	6	2	8	7	
Mandla	9	9	11	13	3	9	9	8	9	9	9	0	1	1	
Mandsaur	5	5	12	16	6	5	5	4	5	5	4	1	2	2	
Morena	7	5	9	12	41	6	6	4	5	6	6	3	6	2	
Narsinghpur	6	6	17	12	4	5	5	4	6	6	6	1	3	2	
Neemuch	3	1	15	15	2	3	3	2	3	3	3	0	2	2	
Panna	6	4	13	13	8	6	6	4	5	6	6	0	2	2	
Raisen	7	6	12	13	6	5	7	7	6	5	7	1	6	5	
Rajgarh	5	4	10	13	0	5	5	4	4	4	4	0	3	2	
Ratlam	5	2	12	8	10	4	4	1	3	3	4	1	3	2	
Rewa	11	8	7	9	4	7	11	3	9	8	8	1	6	3	
Sagar	11	8	11	14	8	6	9	4	7	10	10	1	5	2	
Satna	9	5	8	11	1	7	8	4	7	9	7	2	5	3	
Sehore	9	7	16	13	6	9	9	2	9	9	9	2	8	6	
Seoni	8	6	11	12	1	8	7	4	8	8	5	1	5	0	
Shahdol	8	4	11	10	5	4	7	5	6	6	6	0	3	2	
Shajapur	4	4	13	11	0	3	4	0	3	3	2	0	1	0	
Sheopur	3	2	7	13	3	0	3	1	2	3	3	0	2	2	
Shivpuri	8	4	9	12	6	6	5	6	5	6	7	0	4	3	
Sidhi	6	6	13	17	9	6	6	4	6	6	6	0	5	5	
Singroli	6	6	6	6	1	1	6	5	5	6	6	0	0	0	
Tikamgarh	7	4	7	12	7	5	7	6	6	5	6	1	5	4	
Ujjain	5	4	8	18	6	5	5	3	5	5	4	1	3	1	
Umaria	3	3	15	12	5	3	2	1	2	3	3	0	3	3	
Vidisha	7	4	10	12	3	7	7	0	7	7	5	0	5	2	
Madhya Pradesh	33	262	11	13	5	276	311	192	292	306	293	43	217	143	

population is highest in Bhopal (5.2) followed by Sehore and Hoshangabad (4.0). Number of Staff Nurse / ANM per 100000 rural population in CHCs is highest in Hoshangabad (9.6), followed by Mandla, Dindori and Anuppur (8.3). Higher density of technician posted at CHCs was reported from Chhindwada (9.97), Dhar (7.08) and Sidhi (6.35) per 100000 rural population.

A detailed look into the reported data on human resources available at CHCs revealed that in all 41 surgeons were posted in 20 districts, or in other words, 31 districts do not have a single surgeon posted in any of the CHCs. There were only 53 gynaecologists in 26 districts, 52 paediatricians in 32 districts, 63 physicians in 25 districts and 18 anaesthetists in 14 districts. Altogether, 41 districts have at least one specialist posted at CHCs.

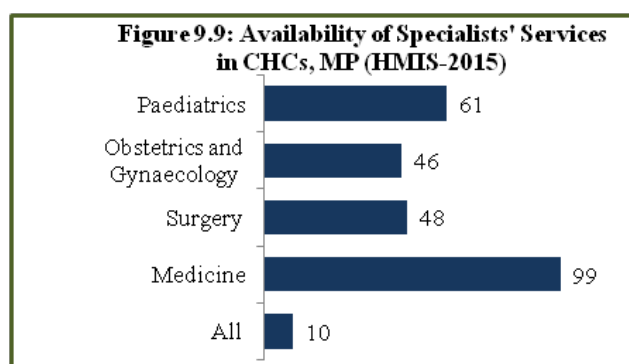
Block-level programme manager and block health officer are posted in 37 and 47 districts respectively at CHCs. There are only 101 block-level programme managers posted at CHCs in the State. There were 696 general duty medical officers and 150 AYUSH medical officers posted at CHCs in the State. Agar Malwa, Damoh, Harda, Mandsaur, Neemuch and Ratlam districts do not have AYUSH medical officers posted at any of the CHCs.

Health Care Service at CHC

HMIS data also captures availability of different services at CHCs. These services can be grouped as - Basic Emergency Services, Specialists Services, Emergency Services and Other Support Services. The analysis of data on Basic Emergency Service shows that round-the-clock Normal / Assisted Delivery is provided by 321 CHCs, newborn care services are provided in 317 CHCs, 248 CHCs having full-range of family planning services, 216 are providing Safe abortion services, 305 are having RTI/STI treatment and 195 are providing counsel-

ing for HIV/AIDS/STD. District-wise comparison shows that in only 102 CHCs across 43 districts all the basic emergency services were available. In Agar Malwa, Balaghat, Betul, Datia, Katni, Panna, Shajapur and Sheopur districts all CHCs were not providing all basic emergency services. Only one CHC has all the basic emergency services across 12 districts.

Availability of specialists' services obviously depends upon availability of specialist doctors. Under this category specialist services for Medicine, Surgery, Obstetrics and Gynaecology and Paediatrics are included. It is observed that this information is not properly reported under HMIS by majority of CHCs. In six districts, Agar Malwa, Ashok Nagar, Barwani, Harda, Rajgarh and Shahdol which comprise 32 CHCs did not report availability of any of the above specialists' services. Further, in 17 districts none of the CHCs had reported about 2-3 out of the five specialists services.



It is found that only 99 CHCs are having specialists services for Medicine, 48 CHCs had Surgical Services, 46 CHCs had Ob&G Services and 61 had Paediatrics services in the State (Figure 9). Data indicates that all specialists services are available in only 10 CHCs covering six districts. These CHCs were Berasia and Kolar (Bhopal), Petalawad and Thandla (Jhabua), Depalpur, Sawyer and Betma (Indore), Hindoria (Damoh), Ashta (Sehore) and Naigarhi (Rewa). A cross verification of availability of services to the corresponding specialists available revealed four CHCs at Kolar, Petalawad, Thandla and Sawyer which had all the

specialty services and specialists in place.

In Madhya Pradesh EmOC services, including Cesarean Section Delivery and other medical intervention is available in 82 CHCs. A maximum of eight CHCs in Barwani are providing EmOC services. It is observed that 17 districts have not reported about availability of EmOC services at CHC. In 2015-16 number of CHCs reported to have emergency care for sick children were 257. In 23 districts, all the CHCs are providing emergency care services for sick children. In Balaghat, Gwalior, Jabalpur and Ujjain only one CHC is providing emergency care services for sick children. Apart from basic and emergency and specialists services, all the CHCs should have some essential services, including diagnostic services, blood storage units, referral transport and ICTC. In MP 320 CHCs are having essential laboratory services, 43 are having blood storage unit, 314 have referral transport facility and 96 CHCs have ICTC / VCTC services. In 22 districts, none of the 127 CHCs reported about availability of blood storage facility.

Facility Infrastructure Index

Services Availability Index: Services availability reported in HMIS CHC infrastructure data pertaining to 14 essential and emergency health care services has been taken for computing using this index. These services are Normal/assisted deliveries (24x7), Newborn care, Family planning, including LTT, Safe abortion, STI/RTI Treatment, HIV/AIDS/STD Counseling, Medicine, Surgery, Obstetrics & Gynaecology, Paediatrics, EmOC, including C Sec and other medical interventions, Emergency care of sick children, Essential Laboratory Services, Blood storage facility, Referral transport service, ICTC / VCTC. For each available service a score of '1' is assigned and total score is obtained by adding all the scores. The index is computed as

Service Availability Index = (No. of Services available at CHC) / 14*100

The index ranges from 0 to 100. Higher index value indicates availability of more services. Index value of 100 indicates availability of all 14 services and a '0' value indicates nil services. A few CHCs have not reported about availability or non-availability of any of the 14 services at the CHC. These were denoted as 'na' i.e. data not reported and index has not been computed for that CHC.

Bed Strength Index: Total number of beds available in CHC has been taken for this index. HMIS data provides number of male and female beds available in each CHC. A CHC generally has 30 beds, which has been taken as upper limit for computing bed strength index. The index ranges between 0-100. Higher index value indicates availability of more beds. Index value of 100 indicates availability of 30 or more beds while a 0 value indicate no beds at CHC. The index value for CHCs having 30 or more beds has been taken as 100.

Bed Strength Index = (Total number of Beds Available in CHC) / 30*100

Human Resource Index: Generally, each CHC has three categories of human resources – Doctors, Paramedics and Support Staff. In HMIS each CHC reported on availability of 33 categories of staff which were grouped in three categories given below. The human resource index for each category is then computed based on the maximum number available in a particular human resource category.

Building Interior and Exterior Index: For each CHC, availability of government building, status of compound wall / fencing, status of plaster on wall and status of flooring has been reported in HMIS. Different scores were given for availability of each of these four items.

Amenity Index: Each CHC reports about availability of certain amenities and support infrastructure in HMIS. For amenity index, 11 amenities and support infrastructure have been considered.

Category	Human Resource	Minimum Value	Maximum Value	Index
Doctor	General Surgeon, Physician, Paediatrician, Gynaecologist, Anaesthetist, General duty medical officer, AYUSH Medical Officer	0	14	Doctor' Index = No. of Doctors / 14*100
Paramedics	Public health nurse, ANM, Staff Nurse, Pharmacists, Lab. Technician, Radiographer, Ophthalmic Assistant	0	41	Paramedics' Index = No. of Paramedics / 41*100
Support Staff	Dresser, Ward boy, Aaya, Compounder, OPD Attendant, OT Attendant, Sweeper, Mali, Peon, Chowkidar, Dhobi	0	24	Support Staff Index = No. of Support Staff / 24 *100
Availability of Building	Government – 2 Other - 0			Any CHC can have maximum score of 8 and minimum score of 0.
Boundary Wall	All around – 2 Partial – 1 None - 0			
Plaster on Interior / Exterior Walls	Plaster all over intact – 2 Plaster coming out on some places – 1 No Plaster - 0			Building Interior and Exterior index = (CHC Interior / Exterior Score – 0) / 8*100
Flooring in the CHC	All around – 2 Partial – 1 None – 0			

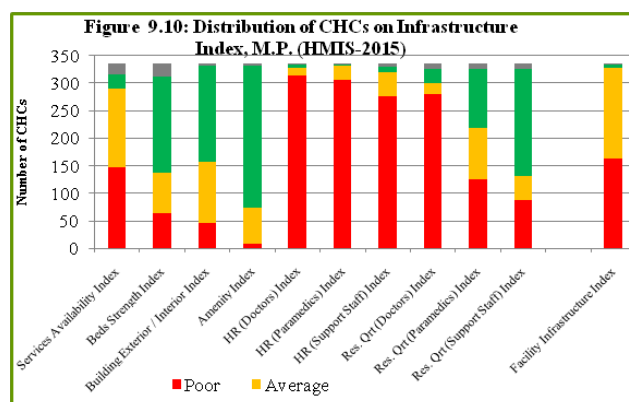
These are water source with over head tank and pump, electricity, regularity of power supply, generator facility, laundry, telephone, computer, at least one ambulance on road, Office Room, Store Room and Kitchen. Different scores were given to amenities and total amenity score is obtained by adding these scores to reflect available amenities in the CHC. The amenity score is then converted into index.

Residential Quarter Index: CHCs have reported about number of residential quarters available for different category of staff. Based on the total residential quarters for different categories of staff - for Doctors, Paramedical Staff and Support Staff three indices were computed.

Doctor Residence Index = (Total No. of Doctor's quarters/6*100) considering availability of at least 6 quarters for doctors at CHC.

Paramedics Residence Index = (Total No. of Paramedic's quarters / 8*100) considering availability of at least 8 quarters for paramedics at CHC.

Support Staff Residence Index = (Total No. of Support Staff's quarters / 3*100) considering availability of at least 3 quarters for support staff at CHC.



Water Source with OHT and Pump	Water Source with Sufficient Capacity OHT with Functioning Pump – 2 Water Source with Sufficient Capacity OHT but no functioning pump – 1, Water Source with insufficient capacity OHT with functioning pump – 1 Water Source without OHT and Pump – 0	Minimum Score of all the amenities - 0 Maximum Score for all the amenities – 14 Amenity Index = (Amenity Score / 14)*100
Electricity Supply line in CHC	All Over in CHC – 2 / Partial – 1 / None – 1	
Regularity of Power Supply	Regular supply / occasional power cut – 2 Regular Power Cut / Power cut in Summer – 1 Frequent Power cut / no supply - 0	
Other amenities	Generator or standby facility of power supply/ Telephone/ Computer / Laundry / At least one ambulance on road / Office Room / Store Room / Kitchen If CHC has above amenities value is given as – 1 If CHC does not have value is given as - 0	

Distribution of number of CHCs according to different constituent indices of facility infrastructure index revealed that majority of CHCs have low index value for human resources particularly for doctors and paramedical staff. Out of 335 CHCs, majority (313 CHCs) have less than 50 percent of required doctors / specialists and paramedical staff compared to the required strength. It is noted that only 174 CHCs (52 percent) have good building interior and exterior index and 258 CHCs have good amenity index. More than four-fifths (280 CHCs) have poor Residential facility index for doctors and two-thirds (213 CHCs) have poor residential facility index for paramedics and support staff (Figure 10). The overall facility infrastructure index shows that 163 CHCs have poor infrastructure, 164 CHCs have average infrastructure

and only 7 CHCs have good infrastructure.

The district-wise status of CHCs according to overall facility infrastructure index revealed that in Betul, Satna, Seoni, Shahdol, Chhatarpur, Rewa, Singroli, Chhindwada, Mandla and Sagar districts more than five CHCs per district have low index value indicating poor infrastructure. In 38 districts out of 244 existing CHCs, 139 have poor infrastructure. Out of 334 CHCs, 226 have index value of 60 or less (Figure 11). The CHCs having good infrastructure with an index value of above 75 are - Ashta (Sehore), Kukshi (Dhar), Pandhurna (Chhindwada), Maheshwar (Khargone), Kolar (Bhopal), Gohad (Bhind) and Sabalgarh (Morena). It is noted that none of CHCs have all required infrastructure i.e. infrastructure index value of 100.

Table 9.5: Correlation coefficient matrix of CHC facility indices and CHC Infrastructure Index, 2015-16

Index	I_Serve	I_Beds	I_DOCHR	I_ParHR	I_supHR	I_in-text	I_ame	I_speqrt	I_par-qrt	I_supqrt	I_CH-CInfr
I_Serve	1.000										
I_Beds	0.332**	1.000									
I_DOCHR	0.536**	0.235**	1.000								
I_ParHR	0.453**	0.212**	0.478**	1.000							
I_supHR	0.259**	0.064	0.268**	0.493**	1.000						
I_intext	0.046	0.250**	0.050	0.076	0.074	1.000					
I_amety	0.137**	0.285**	0.150**	0.188**	0.216**	0.208**	1.000				
I_speqrt	0.561**	0.268**	0.509**	0.401**	0.29**	0.062	0.199**	1.000			
I_parqrt	0.242**	0.192**	0.266**	0.248**	0.184**	0.045	0.158**	0.346**	1.000		
I_supqrt	0.219**	0.151**	0.153**	0.179**	0.139**	-0.024	0.068	0.240**	0.709**	1.000	
I_CH-CInfr	0.611**	0.549**	0.564**	0.574**	0.473**	0.288**	0.428**	0.667**	0.720**	0.655**	1.000
** Correlation is significant at the 0.01 level (1-tailed).											
I_Serve: Index of Services Availability at CHC I_DOCHR: Index of HR-Doctors Availability at CHC I_ParHR: Index of HR-Paramedical Staff Availability at CHC I_supHR: Index of HR-Supporting Staff Availability at CHC I_intext: Index of Interior and Exterior Condition in CHC Building I_speqrt: Index of Specialists Residence availability in CHC						I_Beds: Index of Number of Beds available at CHC I_ame: Index of Amenities available in CHC I_supqrt: Index of Support staff residence availability in CHC I_parqrt: Index of Paramedics residence availability in CHC I_CHCInfr: CHC Infrastructure index					

Table 9.6: Comparison of Selected Infrastructure Reported in HMIS and as Observed during Visit to Selected CHCs during 2015-16, MP

Indicator	CHC, Chanderi (Ashok Nagar)		CHC, Kurwai (Vidisha)		CHC, Kareli (Narsinghpur)	
	HMIS	Field Observation	HMIS	Field Observation	HMIS	Field Observation
Govt. Building	Yes	Yes	Yes	Yes	Yes	Yes
Staff quarter - MO	4	7	1	4	4	1
Staff quarter - Paramedics	5	13	7	2	7	3
Staff quarter - Other	0		4	4	5	2
Electricity with Backup	Yes	Yes	Yes	Yes	Yes	Yes
Running 24x7 Water supply	Yes	Yes	Yes	Yes	Yes	Yes
Separate male / female ward	Yes	No	No	Yes	Yes	Yes
Functioning Blood Storage Unit	No	No	No	No	No	No
Availability of ICTC / PPTCT	Yes	No	No	No	Yes	No
No. of Ambulance	1	2	0	2	1	1
Laundry Facility	Yes	Yes	No	Yes	Yes	Yes
General Surgeon	0	0	0	0	0	0

Gynaecologist	0	0	0	0	1	1
Paediatrician	1	0	0	0	1	1
Physician	0	0	0	0	0	0
Anaesthetist	0	0	0	0	0	0
General Duty Medical Officer	3	4	4	5	1	1
AYUSH MO	0	4	1	0	0	0
Pharmacist	0	1	2	0	1	1
Lab Technician	3	3	1	0	2	0
Radiographer	1	1	0	0	1	1
Ophthalmic Assistant	2	2	1	1	1	1
ANM	1	8	3	4	8	8
Staff Nurse	5	5	5	6	10	10

Table 9.7: Distribution of CHCs according to infrastructure index by star rating, MP (2015-16)

Infra Index	CHC Star Rating						Total
	*	**	***	****	*****	NE	
Poor	1	6	17	49	19	71	163
Average	1	3	17	57	59	27	164
Good	0	0	0	2	5	0	7
Total	2	9	34	108	83	98	334
<i>NE: Not Eligible for Star Rating</i>							

Conclusion

This paper analyzed the HMIS infrastructure data to assess the status of Community Health Centres in MP. The study assessed various aspects of growth, distribution coverage, infrastructure, human resources and availability of services as reported in HMIS.

After implementation of NRHM, number of CHCs increased rapidly. It was necessary, due to emerging challenges of accessibility and affordability, to establish secondary and tertiary care infrastructure in the rural areas. Public health infrastructure has improved during NRHM and at present all the blocks in the State have at least one Community Health Centre. Further allocation of resources for up-gradation of existing services and inclusion of more comprehensive health care services have been provided in the CHCs.

The analysis revealed that out of 51 districts near-

ly half of the districts have added new health facilities in rural areas. For providing specialty health care services affordably in rural areas, the state government has set-up nearly 100 new CHCs in rural areas within a span of 3 years after launching of NRHM in 2005. In total, the number of CHCs has gone up from 237 to 334 during 2007 and 2015 in Madhya Pradesh. The distribution of CHCs is varying across districts. Adequacy of CHCs according to population norm of average one CHC per 100,000 population is not observed in any district. Nearly half of the districts have less than one CHC per 100,000 population. The situation is better in high-priority districts. Alirajpur and Dindori have adequate number of CHCs according to population norms. Number of CHCs with respect to PHC shows higher concentration of PHC per CHC in Mandsaur and Gwalior and higher concentration of SHC per CHC in 12 districts where more than 30 SHC exist per CHC indicating more close linkage between CHC and primary care facilities.

Population coverage reported by CHCs in HMIS data indicates average population catered to by each CHC is less than 100,000 which nearly satisfies the prevailing norms. However, 12 districts have average population coverage of more than 100,000 and ranges between 1.1 lakh for Indore to 3.0 lakh for Damoh indicating tremendous pressure on CHCs.

Adequacy of infrastructure for catering services for secondary care, including emergency care is crucial. It is noted that one-third of all the CHCs have been functioning from the government building having admeasuring area of less than 1500 m², which is against the IPHS standard. Condition of interior and exterior is found to be improved in majority of CHCs where nearly two-thirds of all CHCs have all round compound wall / fencing, all around plastered wall and good flooring. Only one-fifth of CHCs have all kinds of basic amenities, including sufficient water storage facility, electricity in all parts and generator, in-house laundry and proper sewerage system. Likewise, essential infrastructure is available in only one-third of all the CHCs. Only 43 CHCs across the state have at least four residential quarters for specialists and doctors. The gap in availability of all the basic infrastructure amenities needs attention.

Human resources are key to successful implementation of health initiatives and availability of all the required health personnel at CHCs are essential for effective delivery of secondary health care services. It is found that on an average only 2 medical officers, 5 staff nurse/ANM, 3 technicians and 4 support staff are posted in each CHC of the State. It is noted that 31 districts do not have any specialist doctors posted at CHCs. Availability of doctors, including specialists per 100,000 population is lowest in Rajgarh, Shajapur and Singrauli where less than one doctors is available at CHC per 100,000 population.

Availability of a range of basic, emergency, specialist care are other essential services reported under

HMIS for all the CHCs. Data shows that less than one-third of all the CHCs have basic emergency care services, only 10 CHCs in the State have all types of specialty care services, including Paediatrics, Ob&G, Surgery and Medicine. Highest of 99 CHCs in the State have reported about availability of Medicine specialists' service. EmOC services is available in only 82 CHCs. In 22 districts none of the 127 CHCs have blood storage units.

Facility infrastructure index which provides status of CHCs at a glance indicates that nearly half of all the CHCs in the State have poor infrastructure i.e. these CHCs have less than 50 percent of all the required infrastructure, human resources, beds strength, basic amenities and residential facilities in totality. In 11 districts, more than 5 CHCs in each of the districts have poor infrastructure indicating grim situation of secondary care health facilities for rural population in these districts.

The study also found that most of the CHCs have not reported all the infrastructure data and some of the CHCs have reported erroneous data on certain parameters. The study highlights the need to orient facility-level programme managers in reporting corrected and authenticated data under HMIS. Data reported in the infrastructure MIS should be adequately verified for consistency and completeness. From policy point of view infrastructure index can be utilized for accreditation of CHCs according to their available infrastructure. Such accreditation will be useful in mapping CHCs along national and state highways for provide emergency health care in road accidents and for disaster management activities elsewhere. The infrastructure data for CHCs can be used for better planning and assessment of service delivery vis-à-vis upgradation of infrastructure. Based on the available infrastructure and services delivery, accreditation of CHCs can be done for improving efficacy of services using HMIS infrastructure data.

CHCs equipped with necessary infrastructure are essential for providing essential and emergency

health care in rural areas especially the secondary care for maternal and child health. Madhya Pradesh has been witnessing high levels of maternal and infant and child mortality for the last three decades. This itself shows the urgent need for strengthening of CHCs to arrest high levels of maternal and child mortality. Accessibility of emergency comprehensive obstetric (EmOC) and specialized newborn and child health care can be achieved only with adequate infrastructure at CHCs which are the first focal point in the remote rural areas.

References

1. Central Statistical Office (CSO) (2015), *Manual of Health Statistics in India*, Ministry of Statistics and Programme Implementation, Government of India.
2. Kumar, Avanish and Gupta, Saurav, 2012, Health Infra structure in India: Critical Analysis of Policy Gaps in the Indian Healthcare Delivery (Occasional Paper of July, 2012), Viveknanda International Foundation (<http://www.vifindia.org>)
3. MoHFW, 2016. Star Rating Report For CHCs Of Madhya Pradesh (2015-16), https://nrhm-mis.nic.in/hmisreports/frmCustom_reports.aspx?custom=GOC accessed on 24.11.2016
4. MoHFW, <https://nrhm-mis.nic.in> web portal for facility level infrastructure data, Ministry of Health and Family Welfare, Government of India.
5. MoHFW, Rural Health Statistics (Various reports)
6. Novick, Lloyd F., Cynthia B. Morrow, Glen P. Mays, 2008, *Public Health Administration (Principles for Population-based Management)*, 2nd edition, Jones and Bartlett Publications, Massachusetts, p.56

CHAPTER - 10

Assessment of Availability of Services and Beneficiary Satisfaction in UPHCs of Bhopal and Raipur Cities under NUHM

Reena Basu¹, Nikhilesh Parchure², Niklesh Kumar³, Jyoti Tiwari³ ¹*Assistant Director,*
²*Research Investigator,* ³*Field Investigator*

Population Research Centre (MP)

Abstract

National Urban Health Mission (NUHM) endeavours to achieve its goal of providing health services to urban population through institutional mechanism and management systems to meet the health-related challenges of a rapidly growing urban population. Under the framework of NUHM an impetus to urban health programme is envisaged by introducing Urban Primary Health Centres (UPHCs) in cities. The present study focuses on the availability of services of the selected UPHCs in Bhopal and Raipur and their functioning. The study also assessed beneficiary satisfaction for services received. All the visited eight UPHCs in Bhopal and Raipur cities are primarily located in urban slum areas and there is a huge demand for health services in the adjoining areas.

Beneficiaries have expressed satisfaction at services received in UPHCs of both Bhopal and Raipur cities. JSSK services in the UPHCs of Raipur have been well received by beneficiaries who are spared the burden of travelling long distances for delivery services. Free meals through SHGs are being provided at Gudyari and Khokhopara UPHCs which is also proposed for other UPHCs in Raipur.

MAS as a community mobilization group for health activities in urban wards has made inroads in the community in both Bhopal and Raipur, but it has emerged as a strong force which is visible in Raipur because of its presence in the community since the year 2013. MAS has started functioning in Bhopal recently.

The UPHCs of Raipur are providing services more effectively due to adequate infrastructure facilities, human resources and other public amenities. The UPHCs of Bhopal are providing only OPD services and need augmentation in all aspects. More capacity needs to be created in the health system in terms of manpower, infrastructure and diagnostics to match growing demand for services in the community from UPHCs.

INTRODUCTION

National Urban Health Mission (NUHM) endeavours to achieve its goal of providing health services to urban population through institutional mechanism and management systems to meet the health-related challenges of a rapidly growing urban population. As per Census 2011, the population of India has crossed 121 crores with the urban population at 38 crores which is 31 percent of the total population. The process of developing a health care delivery system in urban areas has not as yet received the desired attention. The Tenth Plan Document (2002-07) observed that 'unlike the rural health services there have been no efforts to provide well-planned and organized primary, secondary and tertiary care services in geographically delineated urban areas. As a result, in many areas primary health facilities are not available; some of the existing institutions are underutilized while there is over-crowding in most of the secondary and tertiary centres'. NUHM would cover all State capitals, district headquarters and Class II cities/towns having 50,000 and above population.

Urban Primary Health Centre (UPHC)

UPHCs have been established to cater to need-based city-specific primary urban health care system to meet the diverse health care needs of the urban poor and other vulnerable sections. The urban health was earlier managed by Urban Health Posts, Civil Dispensaries, and Urban Health and Family Welfare Centre (UFWCs) which provided urban health services and were evaluated in 2005. There has been no reorganization and expansion of these schemes for a long period. Based on these evaluations and to serve the growing needs of urban poor, the Urban Primary Health Centres (UPHCs) were created in urban areas of the country in order to ensure access to the urban slum population at convenient timings during the 12th Plan period in the year 2013. Every UPHC is envisaged to cater to sixty thousand population. In areas where the population is sparse it may cater to 25 to 30,000

populations and in densely populated areas the population served by such UPHCs could be upto 75,000 (MoHFW, 2013).

Mahila Arogya Samitis (MAS)

Promotion of access to improved health care at household level has been envisaged through community-based groups at urban ward level. MAS have been visualized on the lines of self-help groups which have provided economic empowerment to women in different parts of the country. MAS as a community-based group would help in meeting urgent cash needs through health insurance scheme for enabling the urban poor to meet medical treatment needs. MAS would also assist in empowering the community to demand improved health services through community participation. Mahila Arogya Samiti (MAS) would comprise a group of around 50-100 households, and would be responsible for health, sanitation and hygiene behaviour change promotion under the leadership of Accredited Social Health Activist (ASHA) and a committee of 5-20 members with an elected Chairperson or Secretary and other elected representative like Treasurer. Partnership with NGOs, profit and non-profit health service providers and other stakeholders is also visualised in planning, implementation, and monitoring of health activities.

Urbanization is increasing rapidly both in Madhya Pradesh (MP) and Chhattisgarh (CG). Proportion of slum households to the proportion of urban household is 28 percent with 5.5 million urban poor in MP and the proportion is 32 percent with 1.5 million urban poor in CG. Both states have carried out situation analysis and initiated the programme. Presently, 19 cities in MP and 11 in CG with population of 50,000 and above have initiated the Urban Health Programme.

Communitization of UPHC services is essential for sustainability and continuity of infrastructure and service augmentation in UPHCs. MAS also

works for communitization of health services by spreading health awareness and establishing its linkage with UPHC. Therefore, to understand the operationalization of UPHCs and their functioning in Bhopal and Raipur cities, it was pertinent to carry out this study.

2. Objectives of the Study

The primary objective of the study was to assess the functioning of UPHCs in Bhopal and Raipur cities. Specifically:

- Assessment of the available infrastructure, human resources and other public amenities in the UPHCs to provide essential services.
- Gauge beneficiaries' opinion regarding services received and their level of satisfaction.
- Explore the role of Mahila Aarogya Samitis in promoting community participation in health in urban areas.

3. Study Design and Methodology

The assessment was carried out in urban wards of Bhopal and Raipur cities where UPHCs were functional. The population of Bhopal is 17.96 lakhs (Census 2011), and more than 50 percent of the population resides in slum areas. Raipur has 10.10 lakh population (Census 2011) and 40 percent of the population reside in slum area.

Both the cities, Bhopal and Raipur, were purposively selected in consultation with the respective state-level programme managers, where Urban Health Programme (UHP) has been implemented by setting up UPHCs with all required infrastructure and other resources. Based on information obtained about the number of OPDs in different UPHCs of Bhopal, and number of deliveries being conducted in the UPHCs of Raipur, four UPHCs from each city which had recorded maximum number of OPDs and deliveries were purposively

selected for the study.

A semi-structured interview schedule was administered to beneficiaries to assess the range of services received and satisfaction from services. Both exit and household-level interviews were conducted for them. Beneficiaries who had received services at the selected UPHCs between April-December 2015 were interviewed. In all, 101 beneficiaries from Bhopal and 108 from Raipur were selected randomly for the study.

Group Discussions with members of MAS were conducted in Bhopal and Raipur cities to understand their role in promoting health in their respective wards.

Availability of services covering 156 indicators across 8 dimensions was assessed and percentages were computed for the following: infrastructure facilities (33), services (36), human resources (4) drugs and consumables (23), equipment (22), laboratory facilities (16), training of staff (7) and public amenities and IEC (15) available in these facilities.

1. In case of categorical variables a score of '1' is assigned to 'yes' if particular service/ amenity/ test is available and '0' if otherwise. The combined score is taken as sum of all the scores for a particular dimension in a facility e.g. for boundary wall with gate existing score of '1' is given to 'yes' and '0' to no.
2. Percentages have been computed for each dimension which indicates the extent of availability of services in terms of the eight dimensions for a particular UPHC on that particular dimension.

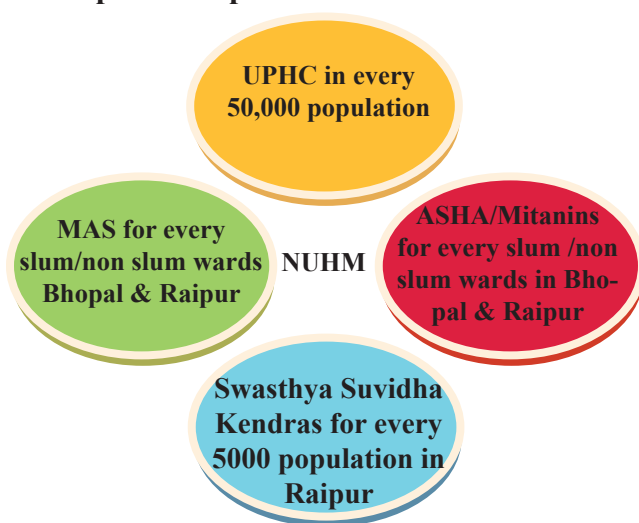
Standard of Living Scores: Standard of Living Scores (SL scores) for beneficiaries was computed based on weights assigned to different household facilities and amenities, and household belongings available, ranging between score zero to four. Maximum score (4) was assigned to pucca house,

floor of cement, tiles and carpet and household items such as washing machine, fridge, computer, car, motorcycle, scooter. Other household items received scores ranging between 1 and 3. Zero was assigned to kuccha house, mud floor etc. After adding all scores of different household amenities, belongings and facilities the household (HH) score was obtained. Total HH score obtained was converted into percent score by adding total of particular HH score/ maximum score*100.

4. Framework of Urban Health Programme under NUHM in MP and CG

The process of strengthening of urban services was envisioned with the initiation of NUHM in different states. UHP was launched in MP in the year 2013 under NUHM but became functional in February 2014. In the year 2015-16, 136 Urban Primary Health Centres (UPHCs) have been operationalized. Earlier, in the year 2007-08, the urban health programme was implemented in 8 districts including, Bhopal, Jabalpur, Indore, Gwalior and Sagar, Ujjain, Ratlam and Satna where the programme was introduced in slum clusters. In MP at the state level there is a separate nodal officer in the rank of a deputy director, who is the overall in charge officer responsible for coordinating

Fig 10.1. Functioning of UPHCs under NUHM in Bhopal & Raipur



and implementing the programme in the State, and there is a consultant for Community Processes, one Monitoring and Evaluation Officer and a data entry operator.

Box 1

Service Package at UPHCs in Bhopal & Raipur

- The UPHCs are expected to provide both IPD and OPD services.
- The UPHCs are expected to promote institutional deliveries.
- The immunization service is available on all 7 days in UPHCs.
- The UPHCs are expected to provide referral services.
- The UPHCs provide preventive and promotive treatments, communicable diseases like malaria, typhoid, leprosy and seasonal diseases are detected and treated.
- Through UPHCs non-communicable diseases like high blood pressure, diabetes etc. are screened and treatment given.
- All the national health programmes are implemented through UPHCs.
- Outreach services are provided every month, particularly in slum area, and daily outreach services/door step health services provided by MMU.

New interventions like UPHCs, Swasthya Suvidha Kendras and Mobile Medical Units in CG to facilitate health services for urban population was envisaged in 2013. In CG, Mukhya Mantri Shahri Swasthya Karyakram (MMSSK) sponsored by the State's own resources, was initiated in June 2012, and subsequently merged with NUHM in May 2013. There is a separate programme management unit (PMU) with State Programme Manager,

one State Finance Manager, two technical experts and one data entry operator to coordinate the programme at state level.

5. Coverage by UPHCs

Table 10.1 below highlights that the average population covered by the eight PHCs which is 65,425

ward by Barkhera Pathani UPHC (Bhopal). The table shows that UPHCs of Raipur cover more wards in comparison to those in Bhopal. All the eight UPHCs cater to slum population but UPHC Saibabanagar in Bhopal city also caters to non-slum population. All the UPHCs are located in government buildings but UPHC Saibabanagar (Bhopal) is functioning from a rented building. UPHCs Bag

Table 10.1: Coverage by UPHCs of Bhopal and Raipur

Coverage	Bhopal*				Raipur#				
	AG*	BS*	BP*	SB*	GY#	KP#	RT#	MW#	Mean
Population covered	150000	34800	23600	30000	100000	100000	40000	45000	65425
Total Coverage Area in sq kms	4	5	6	5	10	5	8	3	5.75
No. of wards the PHC caters to	5	3	1	3	18	11	10	6	7.12
Slum ward/non slum ward	Slum	Slum	Slum	Slum/ N.Slum	Slum	Slum	Slum	Slum	8
Located at an easily accessible area(Yes/No)	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	7
Distance of PHC from District Hospital (in Kms.)	8	6	10	4	10	3	4	6	6.4
PHC located in a designated government building	Yes	Yes	Yes	Rented	Yes	Yes	Yes	Yes	7
Area of the building (Total area in Sq. feet.)	2000	600	3000	600	2000	3000	4800	1250	2156
AG* Ashoka Garden, BS* Bagsevaniya, BP* Barkhera Pathani, SB* Sai Baba Nagar, GY# Gudiyari, KP# Khokho Para, RT# Raja Talab, MW# Mowa									

by the eight UPHCs. Highest population coverage is by UPHC Ashoka Garden (1.5 lakhs) and lowest by UPHC Barkhera Pathani (23,600), both located in Bhopal.

Average coverage area of the UPHCs is 5.75 km² varying between maximum coverage of 10 km² by Gudiyari UPHC and minimum coverage of 3 km² by Mowa UPHC both located in Raipur city. Average number of wards the UPHCs cater to is seven, ranging between maximum of 18 wards covered by Gudiyari UPHC (Raipur) and minimum of one

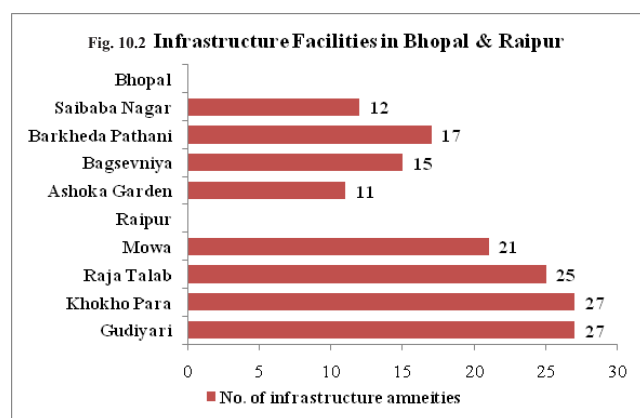
Sevania and Barkhera Pathani (Bhopal) are functioning from civil dispensary buildings.

6. Availability of Services in UPHCs

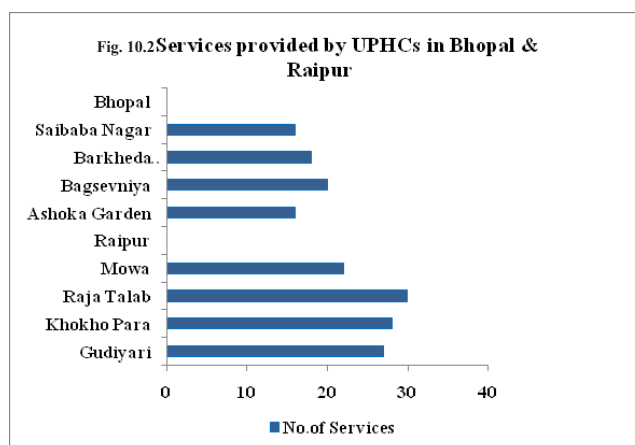
Availability of services of UPHCs on 156 indicators representing eight dimensions is presented in the following section.

Infrastructure Capacity of UPHCs: In all, 33 essential infrastructure facilities have been considered to indicate the capacity of each UPHC for their smooth functioning. The scores obtained for

the infrastructure facilities range between lowest 11 in Ashoka Garden (Bhopal) to highest score of 27 each in UPHCs of Gudyari and Khokhopara (Raipur). These two UPHCs of Raipur have a higher bed capacity than all the other UPHCs. Whereas, UPHCs of Gudyari and Khokhopara (Raipur) have 82 percent infrastructure facilities for services, UPHC Ashoka Garden (Bhopal) has demonstrated only 33 percent on this count. All the UPHCs of Raipur have comparatively better infrastructure facilities than those of Bhopal city as seen in Figure 10.2.



Services Provided by UPHCs: Figure 3 illustrates 36 types of services which should be available in the UPHCs for patients and some of them avoid services emergency services (24 Hours) referral services, in-patient services, antenatal, intranatal and postnatal care, treatment of malaria, primary management of wounds etc. Whereas Gudyari, Khokhopara and Rajatalab UPHCs (Raipur) are providing 82-91 percent of the listed services, including inpatient and delivery services, UPHC Ashoka Garden (Bhopal) has demonstrated only 44 percent service availability, while inpatient and delivery services were not available in the UPHCs of Bhopal. During April-December, 2015 Bag-sevynia UPHC reported 698 average OPDs and Badkheda Pathani highest of 1461 average OPDs per month in Bhopal. Similarly, Gudyari UPHC had the highest average OPD cases (1297) and Mowa the lowest average OPDs (854) per month in Raipur.



Availability of Staff and Trainings: Table 10.2 highlights that there is one medical officer in each of the eight visited UPHCs. Whereas SN, pharmacist and lab technician are in position In all the UPHCs of Raipur, poor staff availability was observed in UPHCs of Bhopal which did not have an SN, pharmacist and lab technician in position. Five to seven staff in different categories is available in the UPHCs of Raipur. The pharmacists and lab technicians of UPHCs of Bhopal have either not joined services or have left owing to pay disparity with their counterparts working in rural areas. Thus staff position in the UPHCs of Bhopal is unsatisfactory and needs immediate attention. Different types of trainings are proposed for the staff of UPHCs like BEmOC, SBA, IMNCI, MTP, IUCD, PPIUCD, immunization and cold chain. Out of the seven types of trainings proposed, in majority of the UPHCs two-three types of trainings have been oriented. In Gudyari the maximum trainings were received by MO (MTP), and SNs in IUCD and PPIUCD. Looking at staff paucity in the UPHCs of Bhopal it is evident that (?) very little training has been conducted. In three UPHCs of Bhopal - Ashoka Garden, Barkhera Pathani and Saibabanagar, not a single staff had received any of the proposed training (Table 10.3).

Availability of Laboratory Facilities: Sixteen types of laboratory tests are part of essential UPHC laboratory services. Routine urine, stool and blood tests, and blood smear examination for malaria parasite tests are mostly available in all the UP-

Table 10.2 : Human Resource for UPHCs in Bhopal and Raipur

Staff Position	Bhopal*				Raipur#			
	AG	BS	BP	SB	GY	KP	RT	MW
Medical Officer Inposition	1	2	1	1	1	2	2	1
Pharmacist Inposition	0	0	0	0	1	1	1	1
Nurse - Midwife (Staff Nurse) Inposition	1	0	0	0	4	1	2	2
Laboratory Technician Inposition	0	0	0	0	1	2	1	1
Total	2	2	1	1	7	6	6	5
AG* Ashoka Garden, BS* Bagsevniya, BP* Barkhera Pathani, SB* Sai Baba Nagar, GY# Gudiyari, KP# Khokho Para, RT# Raja Talab, MW# Mowa								

Table 10.3 : Availability of Trained Persons in UPHCs in Bhopal and Raipur

Training Program	Bhopal*				Raipur#			
	AG	BS	BP	SB	GY	KP	RT	MW
BEmOC	0	0	0	0	0	0	0	1
SBA	0	0	0	0	0	0	1	0
IMNCI	0	1	0	0	0	1	0	0
MTP/MVA	0	0	0	0	0	0	0	0
IUCD	0	0	0	0	1	1	1	1
PPIUCD	0	0	0	0	1	0	0	0
Immunization and Cold Chain	0	1	0	0	1	1	0	0
Total	0	2	0	0	3	3	2	2
AG* Ashoka Garden, BS* Bagsevniya, BP* Barkhera Pathani, SB* Sai Baba Nagar, GY# Gudiyari, KP# Khokho Para, RT# Raja Talab, MW# Mowa								

HCs (Table not given). Laboratories are functional only in three out of eight UPHCs which are Ashoka Garden (Bhopal), Gudiyari and Rajatalab (Raipur). Barkhera Pathani does not have laboratory services and UPHC Bagh Sevania does not have any testing facilities, nor is it conducting any lab test. It was stated by the hospital staff that nearly three-fourths of the laboratory tests are available in Mowa UPHC (Raipur), followed by Rajatalab (Raipur) and Ashoka Garden UPHC (Bhopal).

Availability of Drugs, Consumables and Equipment: The UPHCs are expected to have 23 types of drugs for different services. Maximum drugs were available in Barkhera Pathani (83 percent). Drug availability ranged between 70-78 percent in the four UPHCs of Raipur (Table not given). IFA tablets, iron syrup, Vitamin A, ORS, zinc tablets and paracetamol syrup are the most commonly

available drugs. Two month supply of essential drugs and two month supply of essential vaccines is observed in only half of the UPHCs. Two month supply of essential contraceptives is seen in most of the UPHCs except Ashoka Garden and Rajatalab. Whereas most of the UPHCs of Bhopal have better supply of drugs, UPHCs of Raipur have better supply of essential vaccines. AYUSH drugs are not available in any of the health facilities.

Out of the 22 essential equipment required in UPHCs, Gudiyari and Khokhopara UPHCs have 82-86 percent equipment available followed by Rajatalab UPHC (68 percent) in Raipur. The UPHCs located in Bhopal have equipment in the range of 14-26 percent. All the eight UPHCs have an examination table, while 6-7 UPHCs have thermometer, haemoglobinometer, needle cutter and water purifier. None of the UPHCs have a cradle, and

radiant warmer is available only in Gudiary and Khokhopara UPHCs in Raipur (Table not given).

Public Amenities and IEC: Certain amenities are essential for improving services in the UPHCs. Out of 15 amenities registration counters, pharmacy for drug dispensing and drug storage and OPD rooms / cubicles are available in all the eight UPHCs. Adequate number of windows to let in light and air in each room is observed in all the UPHCs, except Barkhera Pathani. Prominent display boards regarding service availability in local language, education about health and behaviour change communication are available in all the UPHCs of Raipur but not in Ashoka Garden and Barkhera Pathani UPHCs of Bhopal. Maximum public amenities and IEC services were observed

of the visited UPHCs is presented in the table. The table shows that in all respects the UPHCs of Raipur are better placed to provide services than those in Bhopal. One of the reasons for these differences between UPHCs in Bhopal and Raipur may be due to the fact that the UPHCs in Raipur were established in the year 2013 whereas those in Bhopal have become functional only in the year 2015. Moreover, whereas the UPHCs of Raipur are providing OPD, IPD and JSSK services the UPHCs of Bhopal are providing only OPD services.

7. Beneficiary Perspective

In the following section information collected from beneficiaries has been analyzed to understand their view point about the services received and

Table 10.4: Proportion of Availability of Infrastructure, Services and other Public amenities in UPHCs in Bhopal and Raipur

Training Program	Bhopal*				Raipur#			
	AG	BS	BP	SB	GY	KP	RT	MW
Infrastructure (n=33)	33.3	45.5	51.5	36.4	81.8	81.8	75.8	63.6
Services (n=36)	44.4	60.6	54.5	48.5	81.8	84.8	90.9	66.7
Human Resources (n=4)	50.0	25.0	25.0	25.0	100	100	100	100.0
Training (n=7)	0.0	28.6	0.0	0.0	42.9	42.9	28.6	28.6
Laboratory (n=16)	62.5	18.8	0.0	56.3	56.3	50.0	68.8	73.9
Drugs & Consumables (n=23)	52.2	73.9	82.6	47.8	78.3	78.3	69.6	50.0
Equipment (n=22)	27.3	27.3	13.6	27.3	81.8	86.4	68.2	50.0
IEC/Patient Amenities (n=15)	46.7	60.0	26.7	46.7	60.0	66.7	73.3	53.3
AG* Ashoka Garden, BS* Bagsevaniya, BP* Barkhera Pathani, SB* Sai Baba Nagar, GY# Gudiary, KP# Khokho Para, RT# Raja Talab, MW# Mowa								

in UPHC Rajatalab. Emergency room and citizen's charter are the least available amenities (Table not given).

Functionality of UPHCs: Table 10.4 shows the overall functionality of UPHCs. The proportion of availability of infrastructure, services, human resources and other amenities in eight dimensions

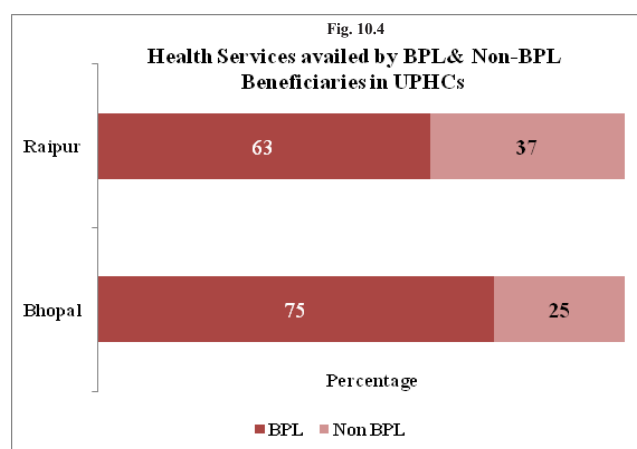
their satisfaction regarding services at the respective UPHCs. A total 209 beneficiaries have been interviewed, 101 in Bhopal and 108 in Raipur city through both household and exit interviews. In all, 146 household (64 in Bhopal & 82 in Raipur) and 63 exit interviews (64 in Bhopal & 82 in Raipur) were conducted in the two cities.

The information about socio-economic and demographic characteristics, including education level, and occupation of the beneficiaries is presented in Table 10.5. More than half of the beneficiaries (56 percent) who availed services at the UPHCs be-

long to the BPL category (Bhopal: 75 percent; Raipur 63 percent). All the respondents belonging to BPL category possessed BPL cards. The proportion of BPL and non-BPL category beneficiaries who availed of services at the UPHCs in Bho-

Table 10.5: Background Characteristics of Beneficiaries of UPHCs in Bhopal and Raipur

Back ground Characteristics	Bhopal*		Raipur#		Total	
	Number	Percent	Number	Percent	Number	Percent
Family Type						
Nuclear	54	53.5	39	36.1	93	44.5
Joint	47	46.5	69	63.9	116	55.5
Religion						
Hindu	78	77.2	98	90.7	176	84.2
Muslim	23	22.8	10	9.3	33	15.8
Caste						
SC	12	31.0	43	11.9	28.7	20.6
ST	4	12.0	16	4.0	11.1	7.7
OBC	52	56.0	108	51.5	51.9	51.7
Other	33	9.0	42	32.7	8.3	20.1
BPL Category						
Yes	76	75.2	68	63.0	144	68.9
No	25	24.8	40	37.0	65	31.1
Total	101	100.0	108	100.0	209	100.0



longed to joint families. (Bhopal: 46 percent; Raipur: 64 percent). Majority of the interviewed beneficiaries are Hindus (84 percent), more than half of them (52 percent) belong to Other Backward Category. More than two-thirds stated that they

pal and Raipur cities is provided in Figure 10.4. More than one-third (37 percent) beneficiaries in Raipur and one-fourth in Bhopal city belonged to non-BPL category.

Table 10.6 shows distribution of households by standard of living scores of beneficiaries in Bhopal and Raipur. Overall, five households of beneficiaries had household scores in the range 20 and below, 38 in the range of 20-40 and more than two-fifths in the range of 60-80, and more than one-third in the range of 80 and above. Forty-eight percent households in Bhopal had SL scores 80 and above and 51 percent households in Raipur had scored SL scores 60 and above, indicating that possession of BPL cards may not always match with standard of living.

Table 10.6: Distribution of Households of Beneficiaries by Standard of Living Scores in Bhopal & Raipur.

SL Scores	Bhopal (N=101)		HHs Raipur(N=101)		HHs Total(N=101)	
	Num	Per-cent	Num	Per-cent	Num	Per-cent
<=20	2	2.0	3	2.8	5	2.4
40	15	14.9	23	21.3	38	18.2
60	36	35.6	55	50.9	91	43.5
>=80	44	47.5	27	25.0	75	35.9
Total	101		108		209	

Different Types of Services Received: Table 10.6 shows that OPD services were availed by majority of the beneficiaries in UPHCs of Bhopal but only by one-fifth beneficiaries in Raipur. In Raipur, visit to UPHCs for ANC, delivery and child immunization services were reported by 81, 69 and 53 percent beneficiaries respectively. Beneficiaries of UPHCs in Bhopal were provided mostly medicines and primary care. A little more than one-third beneficiaries had pathological tests done in the UPHCs at Raipur but not at Bhopal which do not have lab technicians posted in the UPHCs

Table 10.6a: Type of Services Received by Beneficiaries at UPHCs in Bhopal and Raipur

Type of Services at UPHCs	Bhopal (N=101)		HHs Raipur(N=101)		HHs Total(N=101)	
	N =91	Per-cent	N =107	Per-cent	N =198	Per-cent
OPD	86	94.5	24	22.0	110	55.6
ANC	3	3.3	87	81.3	90	45.5
Child Immunization	3	3.3	57	53.3	60	30.3
Pathology	5	5.5	44	41.1	49	24.7
Family Planning	4	4.4	4	3.7	8	4.0
Counseling	24	26.4	4	3.7	28	14.1
Medicine	69	75.8	25	23.4	94	47.5

Primary Care	67	73.6	18	16.8	85	42.9
Institutional Delivery	NA	0.0	74	69.2	74	37.4
DOTS	NA	0.0	2	1.9	2	1.0
Outreach	NA	0.0	18	16.8	18	9.1
NCD	NA	0.0	3	2.8	3	1.5
Referral	2	2.2	2	1.9	4	2.0

Table 10.7 shows that beneficiaries were mostly satisfied with the services received at the respective UPHCs which they visited in Bhopal and Raipur cities. Majority of them said that the concerned doctor heard the problem patiently. Regarding advice for second visit, more beneficiaries (93 percent) received this advice in UPHCs of Raipur because out of total beneficiaries nearly three-fourth of them visited the UPHCs for ANC and or delivery services.

7.1 JSSK services Received by Beneficiaries in Raipur UPHCs

In Raipur city, ANC and delivery care advice and services were received by 79 beneficiaries at the different UPHCs as part of JSSK services which has been analyzed separately.

Advice Received during ANC: Table 10.8 highlights the advice received on delivery care by beneficiaries at different UPHCs of Raipur. It is seen from the table that majority beneficiaries (98 percent) have received the MCP cards for maintaining updating maternal and child health services received. More than three-fourths beneficiaries have reported of 3 or more ANC visits, and health staff suggested about regular ANC checkup to three-fifths beneficiaries and cleanliness during delivery process to more than two-third beneficiaries. Majority beneficiaries (91 percent) received advice on breastfeeding.

Table 10.7 : Beneficiaries' Satisfaction with Services Received in Bhopal (MP) and Raipur (CG)

Satisfaction with Services	Bhopal		Raipur		Total	
	N = 91	Percent	N = 107	Percent	N = 198	Percent
Doctor heard problem patiently	89	97.8	105	98.1	194	98.0
Doctor advised second visit	59	64.8	99	92.5	158	79.8
Satisfied with the service from UPHC	80	87.9	104	97.2	184	92.9
Satisfied with the behaviour of the staff of UPHC	82	90.1	105	98.1	187	94.4

Table 10.8: Advise received on ANC and Delivery Care by Beneficiaries at UPHCs in Raipur

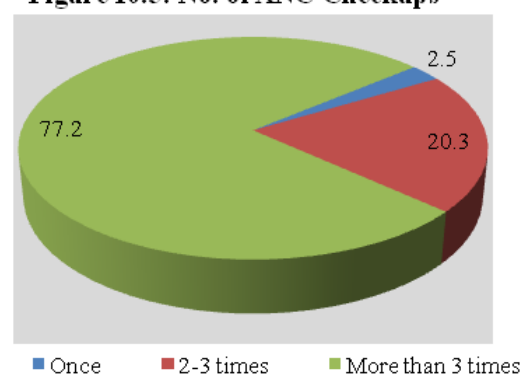
Advise Received on Micro-birth Plan	N=79	Percent
MCP card received	77	97.5
No. of ANC's checkup during Pregnancy		
Once	2	2.5
2-3 times	16	20.3
More than 3 times	61	77.2
Health staff suggested on regular ANC	66	84.6
Health staff suggested on birth preparedness	48	61.5
Health staff suggested on cleanliness during delivery process	53	67.9
Health staff suggested on Kangaroo Care	30	38.5
Health staff suggested on breastfeeding	71	91.0
Health staff suggested on family planning	23	29.5

Table 10.9 : Proportion of Beneficiaries who Received ANC Services at UPHCs in Raipur (CG)

ANC Services	1st Trim. (N=79)		2nd Trim. (N=79)		3rd Trim. (N=79)	
Height	77	97.5	77	97.5	75	94.9
BP	77	97.5	76	96.2	72	91.1
Blood test	61	77.2	49	62.0	47	59.5
Abdominal check-up	67	84.8	68	86.1	74	93.7
EDD	72	91.1	54	68.4	53	67.1
Place of Delivery	67	84.8	53	67.1	50	63.3

Advice on Diet	75	94.9	67	84.8	65	82.3
Vaginal test	17	21.5	17	21.5	29	36.7
TT 1	66	83.5	20	25.3	05	6.3
TT booster	19	24.1	40	50.6	09	11.4
IFA tab/syrup	74	93.7	71	89.9	46	58.2

ANC Services: The UPHCs of Raipur have provided ANC and delivery care services and 79 beneficiaries received these services at the four selected UPHCs. Evidence of ANC services is provided in Table 10.9. Majority beneficiaries received mandatory ANC check-up. Height, BP, abdominal checkup, was done for majority beneficiaries. About 57 percent beneficiaries had at least one check-up within the first trimester. All the available services were provided free of cost which included free check-ups and vaccination to pregnant women and free IFA tablets (Fig 10.5).

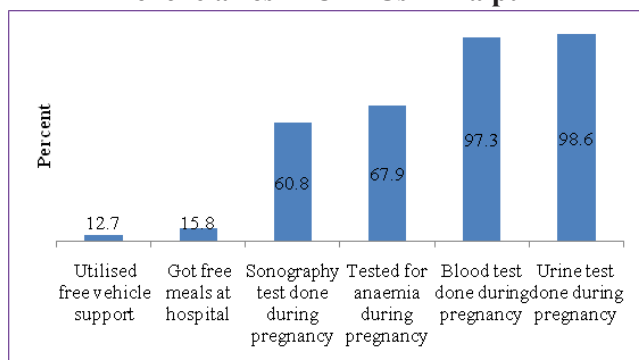
Figure 10.5: No. of ANC Checkups

Pathological Tests: Blood and urine testing services were received free of cost by most beneficiaries in the UPHCs at Raipur. But ultrasonography and HIV test for pregnancy opted by three-fifths of pregnant women on advise of doctors had to be paid for; because these tests were not available at the UPHCs. In general, the beneficiaries were

satisfied with the behaviour of staff who did the lab tests for them at the UPHCs (Table not given).

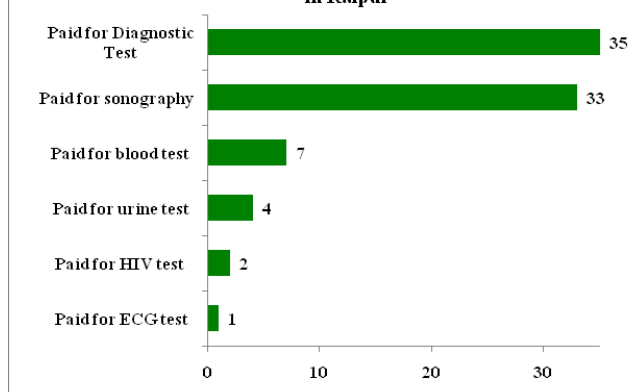
Beneficiary Satisfaction: Satisfaction levels were high among 77 beneficiaries and most of them had received JSY cash incentives. Services of free meals have just started in two UPHCs, Gudiari and Khokhopara, and thus only a few beneficiaries availed free meals. But many beneficiaries staying near the UPHC reported of leaving the health facility within 24 hours after delivery (Fig 10.6).

Figure 10.6: JSSK Services Received by Beneficiaries in UPHCs in Raipur



Out-of-pocket expenditure was reported by beneficiaries in Raipur. Majority of beneficiaries had to pay for ultrasonography test which was not available in the UPHCs (Fig 10.7). Respondents who came to the UPHCs reported high proportion of institutional deliveries in their households in both Bhopal (94 percent) and Raipur (77 percent) when percentage of institutional births in the last 5 years was taken (Table not given). This provides credence to government's efforts to promote institutional delivery.

Fig 7. Beneficiaries Paid for different Tests during ANC in Raipur



7.2 Suggestions given by Beneficiaries

Different suggestions provided by beneficiaries in Bhopal and Raipur for improvements of services are presented in Table 10.10.

A few beneficiaries in Bhopal and Raipur expressed some problems in getting services. In Bhopal, lack of IPD services, laboratory and testing facilities are problems encountered by beneficiaries.

Table 10.10: Suggestions of Beneficiaries who availed UPHC services in Bhopal and Raipur

Bhopal	Raipur
- Facilities for blood and urine test and laboratory facilities in UPHC - IPD facilities - Delivery facilities - Availability of Lady Medical Officer and gynaecological services. - X-Ray and other diagnostic facilities - Diagnostic of diseases like BP, malaria and diabetes - Ultrasonography Facilities - Good Quality of Medicines should be provided. - Sanitation and Cleanliness - Doctor should be available 24*7 - Availability of emergency bed and services - Ambulance services - Adequate waiting space	- Ultrasonography Facilities - Facilities for testing and laboratory - Availability of emergency services - Facilities of Ambulance services - More staff should be available - Sufficiency of Medicines. - Facilities for caesarean section delivery - Availability of Lady Medical Officer - More beds in labour room - Adequate waiting space - Drinking water facilities - X-Ray and other diagnostic facilities - Medicines should be available. - Sanitation and Cleanliness - Doctor should be available 24*7

Non-availability of lady doctor to see female patients, and lack of complete range of services, distance from a bigger hospital and poor cleanliness were some of the concerns expressed by the beneficiaries. In Raipur, non-availability of doctor, poor hygiene in the UPHC OPD room and no facilities for drinking water were some of the problems faced and reported by beneficiaries.

8. Role of Mahila Aarogya Samiti (MAS)

MAS meetings which are an essential part of the communitization of public health programme in states were observed by attending MAS meetings in Bhopal and Raipur. In total, 12 MAS meetings, three in Bhopal and nine in Raipur were attended to understand the community processes adopted by these groups (Table 10.11).

MAS in Bhopal: There were 3000 MAS proposed in MP of which 2394 have been formed. Out of these, 520 MASs are functional in Bhopal city under UHP. Each MAS has a committee of approximately 8-10 members with an elected Chairperson/Secretary and other elected representative like Treasurer. The urban Accredited Social Health Activist (ASHA) has provided the leadership to the Mahila Arogya Samiti in Bhopal. Three MAS meetings were observed in (Amrai Parisar) Bag Sevania, Ward 49, Saibabanagar and (Ward 49, Miranagar) in Bhopal. The MAS members have already received training. Creating awareness on national health issues, cleanliness and sanitation issues were also discussed. Most of the MASs was in the process of opening bank accounts to receive untied funds. Ward/community members showed keen interest in demanding lab technician for their UPHCs

MAS in Raipur: In all, 3245 MAS have been sanctioned but 3699 MAS selected more than sanctioned MAS currently in 19 cities. Total 9 MAS meetings out of 1140 MAS in urban wards of Raipur were observed in Suryanagar, Bajranggharh, Khokhopara, Tatyapara, Bandhuapara, Gopiapara,

Rajatalab, Dhobipara, and Mahtma Gandhi ward. ASHA's/maintains have formed MAS's in their respective wards.

ASHAs, in coordination with the members of the MAS, are promoting proactive community action in partnership with the urban local bodies to take care of issues related to health in their respective area, improved water and environmental sanitation, nutrition and other aspects having a bearing on health. The MAS groups are trained and extremely proactive in their approach to national programmes, public health issues, and have evinced keen interest in a wide range of health, economic and social issues acting as a united force against domestic violence, fighting alcoholism and drugs, seeking micro-finance, providing for nutrition etc. MAS are already receiving untied funds in Raipur.

Strengths of MAS

- Capacity building of MAS through training and orientation workshops regarding activities on local ward issues and significant health subject have been organized in both Bhopal and Raipur cities.
- ASHA/maintains have been successful in mobilizing MAS and are a useful link between these groups and the health system. They have also proven their credentials in resolving several social issues by mobilizing active community participation, as observed in Bhopal and Raipur cities.
- Under NUHM an annual grant of Rs.5000 to the MAS has been provided every year in Raipur and bank accounts of MAS have been opened in Bhopal. This amount is being used for conducting fortnightly/monthly meetings of MAS, sanitation and hygiene, meeting emergency health needs etc.
- In Raipur, MAS has created a mass base for successful implementation and communitization of the national health programme in urban wards.

Table 10.11. Health and Social Activities of MAS in Bhopal and Raipur

Key Activities of MAS in Bhopal	Key Activities of MAS in Raipur
- Child immunization- Baal Surksha Abhiyaan in January, 2016 Vitamin A). - NGO World Vision India has supported child immunization programme. - Removed garbage in front of UPHC Bag Sevania. - Celebrate International Women's Day. - Adolescent Health (Sabla Programme). - Support ANM in National Programmes and community mobilization. - Cleaning of drains and sanitation. - Drinking water issues raised with urban local body. - Celebration of National Deworming Day. - Opening of bank account for untied funds. - Problems of non availability of lab technician in UPHC Saibabanagar discussed. - Health monitoring proforma for children and mothers. - Swachh Bharat Abhiyaan	- Issue of severe malnourishment among children- milk and eggs provided to severely malnourished children through untied - Malaria, awareness about bed nets discussed. - Awareness about institutional delivery. - Child immunization programme. - Street plays organized on social issues. - Untied funds used to purchase medicines for poor and needy in the community. - Celebrate International Women's Day. - Adolescent Health (Sabla Programme). - Health issues like anemia discussed. 27 point MAS monitoring and community alertness - Support ANM in National Programmes and community mobilization. - Cleaning of drains and sanitation. - Drinking water issues raised with ward member and bore well - Fighting alcoholism and drugs. - Micro- financing and SHGs. - Creating awareness on social issues. - Identifying severely malnourished children and sending them to NRC in Medical College. - Wall writing. - Widow pension - Fight against domestic violence. - Family and community counselling - Organizing outreach health camps- jaundice, malaria diarrhoea - Surprise checks in Schools- promote hand washing

Constraints of MAS

- Lack of coordination and cooperation with urban local bodies at times hampers the process of resolving community-level local/ national health, hygiene and sanitation issues like, drainage, piped water, electricity etc.

- In Bhopal, the process of releasing untied funds needs to be expedited to facilitate the process of managing expenditure of MAS meetings and supporting any health emergency.

9. Conclusions

The creation of UPHCs is a laudable programme intervention under NHM in cities like Bhopal and Raipur where there is a tremendous pressure and overload on secondary and tertiary care institutions. All the visited UPHCs in both Bhopal and Raipur cities are primarily located in urban slum areas. There is a huge demand for urban health services in the Bhopal and Raipur cities where services at the doorstep are being provided through the UPHCs. In Bhopal, the process of creating delivery points in the UPHCs is underway with process of recruiting Staff Nurses to initiate delivery. User Charges in OPD, IPD and delivery services are not being levied from the beneficiaries except for nominal registration fees at some of the UPHCs.

Availability of essential drugs and medicines were observed in all the UPHCs in Bhopal and Raipur cities. Delivery kits and all essential medicines for delivery were available at the UPHCs in Raipur and no stock out is reported. Display of IEC material for MCH, FP, available different services, hospital timings and phone numbers were observed in all the visited health facilities.

Beneficiaries have expressed satisfaction at the services received in UPHCs of both Bhopal and

Raipur cities. JSSK services in the UPHCs of Raipur have been well received by beneficiaries who are spared the burden of travelling long distances for delivery services. Free meals through SHGs are being provided at Gudyari and Khokhopara UPHCs which is also proposed for other UPHCs in Raipur.

Certain limitations of this programme are lack of adequate data for health planning based on community needs. Due to limited reporting under HMIS only half of UPHCs services are covered. The trainings for doctors and different category of staff in, BEmOC, MTP, NSV, SBA, F-IMNCI, NSSK, IUCD and PPIUCD have not taken place in Bhopal and MOs have expressed a strong need for training.

MAS as a community mobilization group for health activities in urban wards has made inroads in the community in both Bhopal and Raipur, but it has emerged as a strong force which is visible in Raipur because of its presence in the community since the year 2013. MAS have started functioning in Bhopal only recently.

The UPHCs of Raipur are more effectively providing services due to adequate infrastructure facilities, human resources and other public amenities. The UPHCs of Bhopal are providing only OPD services and need augmentation in all aspects. More capacity needs to be created in the health system in terms of manpower, infrastructure and diagnostics to match rising demand for services in the community from the UPHCs.

There is severe shortage of manpower specifically MOs, SNs, and pharmacists in the UPHCs in Bhopal. There is serious human resource crunch

for augmentation of the programme due to multiple reasons. Lack of pay parity among different categories of staff providing services in rural and urban areas is a cause of dissatisfaction among pharmacists and laboratory technicians who are not joining services or are leaving to join in rural areas a problem persists both in Bhopal and Raipur.

To ensure its sustainability, urban health programme should be made part of the regular health services instead of continuing it in the mission mode. Existing urban dispensaries can be suitably upgraded to urban PHCs by providing required infrastructure. To ensure its continuity, UPHCs may be manned by AYUSH MOs.

A more comprehensive study is required to compare the functioning of PHCs in urban area to those in the rural areas in order to provide a complete picture of primary health care services.

References

- Indian Institute of Population Studies (2005): National Report on Evaluation of functioning of UHPs/UFWCs in India.
- Planning Commission, Government of India (2002-2007): Tenth Plan Document (2002-2007, Volume II).
- Registrar General of India (2011): Primary Census Abstracts 2011, New Delhi.
- Ministry Of Health and Family Welfare: (2013): National Urban Health Mission May 2013 Government Of India, Framework For Implementation.

CHAPTER - 11

Patient Satisfaction with Health Care Services in Punjab

Pawan Kumar Sharma and Poonam Sandhir

PRC - CRRID, Chandigarh

Abstract

The present study explores the satisfaction level of the patients based on their experience with a variety of attributes at the allopathic public health institutions providing secondary health care services in Punjab. Based on the primary data collected from 200 patients, 150 OPD and 50 IPD patients selected at random, in 10 hospitals across five districts of Punjab, it was discovered that contrary to the popular notion, the satisfaction level of the patients with the services delivered at public health institutions providing secondary care health service was significantly high on most of the parameters. Both OPD and IPD patients found the location of the hospitals accessible, diagnostic facilities in order, and quality of medical care satisfactory. There were, of course, some difficulties faced in respect of availability of medicines from the hospitals, availability of ultrasound facilities at these health institutions and owing to laxity in handling emergency cases. The IPD patients were relatively more satisfied than the OPD patients on majority of parameters of satisfaction. Many of the identified areas for quality improvement are related to time factor and non-availability of medicines at the hospitals. The escalating costs involved in the purchase of medicines from the market emerged as a cause of worry for every patient. This is endorsed by a large number of patients who felt that the services offered at the private hospitals were unaffordable and beyond their reach. Moreover, a mismatch between the huge amount of money spent and the quality of services received impelled many of them to seek services of a government hospital. Satisfaction level was better among the patients utilizing private health care OPD services than those availing public health care OPD services. However, there were no differentials in satisfaction level among IPD patients in both types of hospitals. Concerted efforts on the part of state health authorities are required to provide man and machine to offer ultrasound test facilities. Further, to meet the shortage of medicines, in case these are not available free of cost at the hospital pharmacy/dispensary, the state health authorities need to reboot the system of Jan Aushadhi Stores to fulfil the perceived objective of providing generic medicines at cheaper rates. The indispensability of the role of the public sector in health care in Indian context cannot be missed.

INTRODUCTION

Satisfaction, a psychological concept defined in different ways, is considered as a judgment made by individuals regarding any object or event after gathering some experience over time. Satisfaction is a complex concept that is related to a number of factors, including lifestyle, past experiences, future expectations and the value of both individual and society (Verma & Sarma, 2000). Satisfaction refers to the extent of an individual's experience as per his or her expectations. Patient satisfaction is as important as other clinical health measures and is a primary means of measuring the effectiveness of health care delivery.

Patients' satisfaction denotes the extent to which general health care needs of the clients are met to their requirements. Patients carry certain expectations prior to their visit and the resultant satisfaction or dissatisfaction is the outcome of their actual experience. A critical challenge for health service providers in developing countries is to find ways to make them more client-oriented. Indifferent treatment of patients, unofficial payments to providers, lack of patient privacy, and inadequate provision of medicines and supplies are common, yet are rarely acknowledged by traditional quality assessment methods. Assessing patient perspectives give users a voice, which, if given systematic attention, offers the potential to make services more responsive to people's needs and expectations, the important elements of making health systems more effective (Krishna Dipankar Rao et al. 2006).

Patient satisfaction surveys are one of the established yardsticks to measure success of the service delivery system functional at hospitals. The degree of patient satisfaction can be used as a means of assessing the quality of health care and the personnel. It also reflects the ability of the provider to meet the patients' needs. Satisfied patients are more likely than the unsatisfied ones to continue using the health care services, maintaining their

relationships with specific health care providers and complying with the care regimens (Yousef & Mohamed, 2011). Health professionals are also benefited and guided by the outcome of these surveys. The feedback received is likely to help them in identifying potential areas for service improvement. The health expenditure may be optimized through patient-guided evaluation of the system in place.

For assessing the quality of care and for demonstrating that the hospital administrators are interested in quality and in doing things better, the Patient Satisfaction Surveys (PSS) are an important tool as the patients offer a unique perspective for evaluating the non-technical aspects of medical care (Epstein et al. 1996). Such a patient-centered approach is becoming vital in today's competitive healthcare environment. Healthcare consumers demand excellence in care and services from the providers. The providers, especially in the private and semi-public sector, are making concerted efforts to meet the expectations of their clients.

Patient-centered outcomes have taken center stage as the primary means of measuring the effectiveness of health care delivery. It is commonly acknowledged that patients' reports of their satisfaction with the quality of care and services, are as important as many clinical health measures (Ranjeeta et al. 2009). By systematically measuring patient satisfaction and perceptions of medical services in the competitive health care system, the health administrators are making strenuous efforts to increase the effectiveness of care, improve patient outcomes, and control costs (Drain, 2001). Patient's perspective can be highlighted in terms of the quality of care and the services received through adequate, qualitative and quantifiable data. PSS is essentially a tool for self analysis and improvement with the ultimate goal of providing better care and making patients feel that they are getting value out of their investment or effort.

Over the past few years consumer considerations

and perceptions of service quality, which go in for selection of a particular health outlet have been studied by administrators in the public as well as private sector. Among many determinants, the literature points to a negative impact of waiting time on patient's perceptions of health care services, and on patient's subsequent behavior in availing services from a particular hospital (Hill and Joonas, 2006). Depending upon the type of patient, that is, whether availing services in Outdoor Patient Department (OPD) or Indoor Patient Department (IPD), patient satisfaction level may vary due to their different needs, expectations and experiences on parameters such as doctor's behavior, medicine availability, hospital infrastructure, staff behavior, and medical information (Rao et al. 2006). Dissatisfaction amongst the patients, both OPD and IPD needs to be identified through key determinants which can be rectified at a later stage. However, research in the area of "Hospital Administration" is still in its infant stage in India, in general, and Punjab, in particular. Few studies are being carried out for measuring satisfaction of patient with hospital services as a routine process.

Objectives

The present study has been undertaken to fill this gap by exploring the satisfaction level of the patient's experience with a variety of attributes at the allopathic public health institutions providing secondary health care services in Punjab. An attempt is being made to estimate indices of patient satisfaction and to identify the perceived strengths and weaknesses of the public health care services. The findings of the survey are meant to guide the hospital administrators about quality improvement initiatives with respect to services provided to OPD and IPD patients, and also help address the gaps identified.

Materials and Methods

Five districts namely Patiala, Rupnagar, Nawanshahr, Hoshiarpur and Fatehgarh Sahib from Punjab

were selected at random for carrying out the survey. Since the secondary health care services are provided up to the level of Community Health Centre (CHC) in the hierarchy of health institutions in Punjab, two health institutions - one district hospital and one CHC, from each of the sampled five districts were also selected at random. The combined sample of the OPD and IPD patients was fixed at 200; 150 OPD patients and 50 IPD patients. It was decided to cover 15 OPD patients and 5 IPD from each of the two health institutions. OPD patients were selected at random for the exit interviews. IPD patients were also selected at random mainly from four departments, namely General Medicine, Surgery, Gynecology and Orthopedic. While selecting IPD patients, preference was given to interview those patients who were admitted in these health institutions for the last three days or so. However, in case no such patients were available these were replaced with other patients who were recently admitted. Accordingly, 30 OPD patients and 10 IPD patients were covered from each district.

Two sets of interview schedules, one each for OPD and IPD, were used for collection of the data. Five-point scale of highly satisfied, satisfied, somewhat satisfied, not satisfied and very unsatisfied was used to assess the satisfaction level. Face-to-face interview technique was used to collect the quantitative data from the OPD and IPD patients. Qualitative information was also collected through unstructured interviews of the health staff and other members. Weighted averages were calculated on each of the parameter of patient satisfaction for better depiction of the results so as to provide a meaningful comparison. This is calculated as follows:

The patients were classified according to their level of satisfaction with assigned codes (highly satisfied =1, satisfied =2, somewhat satisfied =3, not satisfied =4 and very dissatisfied =5). The responses of the OPD patients on the parameter of easiness to reach different department according

to their satisfaction level were used as follows to calculate its weighted average:

$$39 \times 1 + 93 \times 2 + 14 \times 3 + 3 \times 4 + 1 \times 5 = 284$$

Here the responses as 39, 93, 14, 3 and 1 have been taken arbitrarily for the explanation purpose only. So the weighted average for 150 OPD patients is 1.89 (calculated as 284/150). Similarly, the weighted averages of all the individual parameters were calculated so as to measure extent of the satisfaction level of the OPD as well as IPD patients. The value of weighted average ranged from 1 to 5. Lower the value better is the satisfaction level in the present case.

The study is based as per Indian Public Health Standards (IPHS) Revised Guidelines, 2012 laid by Directorate General of Health Services, MoHFW, GOI. According to this, the Patient Satisfaction Survey for OPD patients should include access to patients, registration and admission procedure, examination, treatment, waiting, toilets and drinking water facilities, complaint and suggestion for OPD patients. Also the satisfaction of IPD patients should be done on the additional counts of linen/beds, staying facilities for relatives, drinking water and toilets.

SATISFACTION LEVEL

A patient's experience within a hospital environment is based on numerous encounters with a wide variety of individuals and locations. The first encounter is with the facility's parking lot, followed by physically accessing the facility, the process of reception and registration counter facility, the admission process, encounters with doctors, nurses, lab personnel, and other service providers and their physical locations. There are a number of factors that could have an impact on the patient's perception of the care.

Location and access to hospital: Long before a patient meets the doctor, the first impression of the hospital and how helpful it would be comes from access-related functions. Difficulty in reaching a destination demotivates a patient from seeking healthcare services from such an institution. Directional signs within the hospital premises facilitate access conditions. The patient's confidence is reinforced by following the signage system to reach the desired destination. Lack of attention towards such matters reflects lack of management on the part of health institution administrators.

Table 11.1: Satisfaction with location of the health institution and facilities availed at arrival (in per cent)

Facilities at arrival	Highly satisfied	Satisfied	Somewhat satisfied	Not satisfied	Very unsatisfied	Weighted average	Total no. of respondents (N)
OPD							
Effectiveness of signage system inside the hospital	26.0	64.7	7.3	1.3	0.7	1.86	150
Easiness to reach different departments	26.0	62.0	9.3	2.0	0.7	1.89	150
Distance between home and hospital	29.3	43.3	22.7	4.7	0.0	2.03	150
Transportation facility to reach the hospital	23.3	48.7	23.3	4.0	0.7	2.10	150
Overall rating of the location and arrival in the hospital	26.0	54.7	19.3	0.0	0.7	1.95	150
IPD							
Easiness to reach different departments	16.0	78.0	6.0	0.0	0.0	1.90	50

Effectiveness of signage system inside the hospital	16.0	76.0	8.0	0.0	0.0	1.92	50
Transportation facility to reach the hospital	24.0	56.0	18.0	2.0	0.0	1.98	50
Distance between home and hospital	22.0	60.0	14.0	4.0	0.0	2.00	50
Overall rating of the location and arrival in the hospital	18.0	72.0	10.0	0.0	0.0	1.92	50

Source: PRC-CRRID, Field Survey, 2015

On the whole, 26 per cent of the OPD patients and 18 per cent IPD patients were highly satisfied with the overall location and access to the hospital. Overall rating of the location and arrival facilities in the health institutions were termed as satisfactory by all IPD patients. Non-availability of convenient transport facilities to reach these health institutions was one reason for the relative dissatisfaction among OPD patients while distance between home and hospital was the prime reason for such dissatisfaction among IPD patients. The weighted averages displayed that the difference in the satisfaction level of OPD and IPD patients was marginal in case of effectiveness of the signage system in hospital and ease of locating the specific departments. The weighted average of 1.92 among the IPD patients and 1.95 among OPD patients reflect that the IPD patients were slightly more satisfied than the OPD patients with the overall location and arrival facilities of the health institutions they were visiting (Table 11.1). The comfort level of patients declined in case the signage system was in English. The health authorities should get all the signage system displayed in Punjabi so as to enhance the satisfaction level of patients.

Reception/registration counter: Once the patient locates and reaches the hospital, his/her first encounter in the hospital is with staff at the reception and registration counter. It is encouraging to know that a large proportion of OPD and IPD patients (88 per cent each) expressed satisfaction regarding their overall experience at reception and

registration counters (Table 11.2). Some variation between OPD and IPD patients emerged as far as their satisfaction levels on parameters, such as co-operation of the staff at the registration counter, ease of the registration process, fairness of queue system and response to enquiries, time taken, and courtesy of reply are concerned. The IPD patients were attended promptly at the reception counter by the hospital staff. The behaviour of the staff at the reception and registration counter was relatively better towards the IPD patients than the OPD patients.

The satisfaction among both the OPD and IPD patients was almost the same on the count of their overall experience at the reception and registration counter as reflected in their weighted average of two in each case. No such complaints as regards the behaviour of the staff and difficulty in getting registered were put forth by the IPD patients.

It was observed during the field survey that the staff at the reception/registration counters of few hospitals was not directing the OPD patients to go to a specific doctor. It was entirely left to the judgment of the patients as to how they reach the doctor whom they wanted to meet. The team came across a number of patients who could not reach a particular specialist doctor whom they wanted to meet but by mistake they kept waiting outside the room of some other doctor. After a long time when their turn came they were again redirected to meet the specialist doctor in another room. This made the patients unhappy with the system. The staff at the reception/registration counters need to

Table 11.2: Satisfaction with the services at reception and registration counter (in per cent)

Services at reception and registration counter	Highly satisfied	Satisfied	Somewhat satisfied	Not satisfied	Very unsatisfied	Weighted average	Total no. of respondents (N)
OPD							
Impartiality kept in queue	24.7	67.3	6.0	2.0	0.0	1.85	150
Time spent in queue at registration and cash counter	13.3	76.7	6.6	2.7	0.7	2.01	150
Assistance of the health staff in helping you to reach at right place and right doctor	10.7	79.3	8.7	1.3	0.0	2.01	150
Behavior of the responding staff	12.7	73.3	12.0	2.0	0.0	2.03	150
Easiness in getting yourself registered	16.0	67.3	14.0	2.0	0.7	2.04	150
Assistance provided by the hospital staff at the registration counter	10.0	71.4	17.3	1.3	0.0	2.10	150
Overall rating of the experience at the reception and registration counter	9.3	78.7	11.3	0.7	0.0	2.03	150
IPD							
Impartiality kept in queue	20.0	78.0	2.0	0.0	0.0	1.82	50
Time spent in queue at registration and cash counter	16.0	80.0	4.0	0.0	0.0	1.88	50
Behavior of the responding staff	14.0	78.0	8.0	0.0	0.0	1.94	50
Assistance of the health staff in helping you to reach at right place and right doctor	12.0	80.0	8.0	0.0	0.0	1.96	50
Easiness in getting yourself registered	14.0	72.0	14.0	0.0	0.0	2.00	50
Assistance provided by the hospital staff at the registration counter	12.0	74.0	14.0	0.0	0.0	2.02	50
Overall rating of the experience at the reception and registration counter	12.0	76.0	12.0	0.0	0.0	2.00	50

Source: PRC-CRRID, Field Survey, 2015

be directed to guide the patients to go to a specific doctor after making preliminary enquiries.

Waiting area facilities for OPD patients: After the registration, the patients are directed to visit different departments to meet the concerned medical officer. Before meeting the medical officer, the OPD patients need a place to sit and wait for their turn. Any lack of facilities, including drinking water, continuous supply of electricity and toilets are likely to make the patients discontented.

Table 11.3 revealed that only a meager proportion of the OPD patients were highly satisfied with the availability of facilities in the waiting area. Of course, a higher proportion of OPD patients were either satisfied (49 per cent) or somewhat satisfied (47 per cent). Highest satisfaction by OPD patients was reported on the count of sitting arrangements. Continuous supply of electricity, with availability of back-up facilities, for use of electrical equipment in the health institution had added to their higher satisfaction levels.

Table 11.3: Satisfaction with the services at reception and registration counter (in per cent)

Facilities at waiting area	Highly satisfied	Satisfied	Somewhat satisfied	Not satisfied	Very unsatisfied	NA	Weighted average	Total no. of respondents (N)
Sitting arrangement	32.7	64.7	1.3	0.7	0.7	0.0	1.72	150
Fan and cooler	4.7	74.0	12.7	0.0	0.0	8.7	2.09	150
Continuous electricity supply	4.7	60.7	22.0	0.0	0.0	12.7	2.20	150
Overall cleanliness	8.0	52.0	37.3	2.7	0.0	0.0	2.35	150
Drinking water	2.0	10.0	12.0	2.7	1.3	72.0	2.69	150
Availability of water in toilet	0.7	8.7	17.3	10.0	12.0	51.3	3.49	150
Toilet	0.7	6.0	18.0	12.0	12.0	51.3	3.59	150
Cleanliness of toilet	0.7	6.0	15.3	14.7	12.0	51.3	3.64	150
Overall rating of the facilities in the waiting area	1.3	49.3	46.7	2.7	0.0	0.0	2.77	150

Source: PRC-CRRID, Field Survey, 2015

Cleanliness of toilets was the prime reason for dissatisfaction among OPD patients. A large number of patients were not using toilets largely because of their bad experience at previous occasions. They knew that the condition of toilet facilities was quite bad. However, one-fourth of the OPD patients who used toilets facilities found these as quite filthy and unhygienic. Not only that, the non-availability of water in these toilets had added to their misery making them to feel highly dissatisfied. Another issue making these patients feel a bit dissatisfied was the non-availability of drinking water. Provision of clean toilets with attached water facility and water suitable for drinking could enhance the satisfaction level of the OPD patients.

Ward Facilities: The IPD patients expressed a higher satisfaction level on the parameters of the easy availability of bed, continuous availability of electricity supply, comfortable bedding, convenient visiting hours and cleanliness of wards. However, poor maintenance of cleanliness of toilets, no seating facility for attendants, non-availability of locker facility near the patient's bed to

keep the personal belongings and no curtain for partition purposes for maintaining the privacy had caused dissatisfaction among a few IPD patients. Further, non-availability of water facility in the toilets had added to the woes of the IPD patients making them feel discontented with these services (Table 11.4).

It was observed during the field survey that the non-availability of sitting facility for the attendants had forced them to sit on the patient's bed. At times more than one attendant was found sitting on the bed of the patient. Non-availability of safe drinking water, mosquitoes and lack of electric plugs for recharging the mobile phones were the other issues observed making the patients feel a bit unhappy. The wards were provided with water filters for making the drinking water safe but these were found non-functional requiring immediate repairs. The concerned health authorities need to be directed to make ad-hoc arrangements for providing the safe drinking water during the period required for repairing these water filters.

Services of Doctor: About 15 per cent of the OPD patients were highly satisfied with the overall ser-

Table 11.4: Satisfaction with the facilities at different wards in the IPDs (in per cent)

Facilities at wards	Highly satisfied	Satisfied	Somewhat satisfied	Not satisfied	Very unsatisfied	NA	Weighted average	Total number of respondents (N)
Availability of bed by first application	70.0	26.0	4.0	0.0	0.0	0.0	1.34	50
Continuous supply of electricity	18.0	80.0	2.0	0.0	0.0	0.0	1.84	50
Availability of fans and coolers	16.0	80.0	4.0	0.0	0.0	0.0	1.88	50
Comfortable bed/bedding	18.0	74.0	8.0	0.0	0.0	0.0	1.90	50
Availability and convenience of visiting hours	8.0	78.0	14.0	0.0	0.0	0.0	2.06	50
Cleanliness of the ward	18.0	56.0	26.0	0.0	0.0	0.0	2.08	50
Cleanliness of sheets and blankets	16.0	48.0	32.0	4.0	0.0	0.0	2.24	50
Availability of requisite medicines in the ward	12.0	20.0	50.0	14.0	4.0	0.0	2.78	50
Curtains for privacy of patients/ availability of partitions	6.0	8.0	54.0	26.0	4.0	0.0	3.14	50
Availability of lockers near beds to keep the personal belongings	6.0	12.0	38.0	36.0	8.0	0.0	3.29	50
Availability of water in toilet	6.0	14.0	34.0	14.0	30.0	2.0	3.49	50
Cleanliness of toilet	6.0	14.0	28.0	16.0	34.0	2.0	3.59	50
Seating facility for the attendant	6.0	10.0	14.0	36.0	34.0	0.0	3.82	50
Overall rating of comfort and facilities in the ward	6.0	32.0	60.0	2.0	0.0	0.0	2.58	50

Source: PRC-CRRID, Field Survey, 2015, Note: NA- Not applicable

Table 11.5: Satisfaction level of patients in the OPD with the services of doctors (in per cent)

Facilities at wards	Highly satisfied	Satisfied	Somewhat satisfied	Not satisfied	Very unsatisfied	NA	Weighted average	Total number of respondents (N)
Impartiality in queue to meet the doctor	25.3	68.0	4.0	2.0	0.7	0.0	1.85	150
Attention given by the doctor in listening to your problems	27.3	60.0	6.0	6.7	0.0	0.0	1.92	150
Treatment going in right direction	14.0	50.0	6.7	2.0	0.0	27.3*	1.95	150
Cordial behavior of the doctor	18.0	72.0	6.7	3.3	0.0	0.0	1.95	150
Time given by the doctor	23.3	62.7	8.7	5.3	0.0	0.0	1.96	150

Doctors competence in understanding your problems in detail	24.7	60.7	7.3	7.3	0.0	0.0	1.97	150
Response of the doctor in answering your all queries	24.7	59.3	8.7	7.3	0.0	0.0	1.99	150
Explanation provided by the doctor to tackle the ailment in future	23.3	56.7	13.3	6.7	0.0	0.0	2.03	150
Privacy maintained by the doctor at time of treatment	9.3	74.0	14.7	2.0	0.0	0.0	2.09	150
Availability of doctor as per your convenience	16.0	60.7	20.0	3.3	0.0	0.0	2.11	150
Availability of the doctor during OPD timings	16.7	60.0	17.3	6.0	0.0	0.0	2.13	150
Waiting time in meeting the doctor after the registration	10.7	52.7	26.7	10.0	0.0	0.0	2.36	150
Enquiry made by the doctor as regards your capacity to purchase medicines from private stores	7.3	32.7	54.7	5.3	0.0	0.0	2.58	150
Overall rating of services provided by the doctor	14.7	61.3	20.7	3.3	0.0	0.0	2.13	150

Source: PRC-CRRID, Field Survey, 2015, Note: NA- Not applicable, *- First timer patients

vices provided by the doctor. In all, about three per cent of the OPD patients were dissatisfied with the services of doctor (Table 11.5). The OPD patients had more or less a higher satisfaction level with the individual parameters related to services provided by the doctor.

The satisfaction level among the OPD patients as regards the parameters on maintenance of impartiality in queue to meet the doctor, treatment provided by the doctor going in right direction and attention given by the doctor in listening to their problems was quite high as reflected in their

weighted averages. However, no enquiry made by the doctor as regards their capacity to purchase medicines from private stores, long waiting hours to meet the doctor after the registration and non-availability of doctors during OPD timings

had made few OPD patients to feel dissatisfied. Long waiting hours to meet the doctor after the registration was the cause of dissatisfaction among 10 per cent of the OPD patients.

Table 11.6: Satisfaction level of patients in the IPD with the services of doctors

Services of doctor	Highly satisfied	Satisfied	Somewhat satisfied	Not satisfied	Very unsatisfied	Weighted average	Total no. of respondents (N)
Time given by the doctor	38.0	56.0	4.0	2.0	0.0	1.70	50
Attention given by the doctor in listening to your problems	34.0	58.0	6.0	2.0	0.0	1.76	50
Response of the doctor in answering your all queries	32.0	62.0	4.0	2.0	0.0	1.76	50
Regular check up by doctor	42.0	32.0	22.0	4.0	0.0	1.88	50
Availability of doctor as per your convenience	36.0	34.0	20.0	10.0	0.0	2.04	50
Doctors competence in understanding your problems in detail	32.0	60.0	6.0	2.0	0.0	1.78	50
Explanation provided by the doctor to tackle the ailment in future	30.0	62.0	6.0	2.0	0.0	1.80	50
Cordial behavior of the doctor	28.0	66.0	4.0	2.0	0.0	1.80	50
Treatment going in right direction	22.0	74.0	2.0	2.0	0.0	1.84	50
Reaching expert doctors in case of specific problems/ diseases	24.0	66.0	8.0	2.0	0.0	1.88	50
Privacy maintained by the doctor at time of treatment	20.0	50.0	28.0	2.0	0.0	2.12	50
Enquiry made by the doctor as regards your capacity to purchase medicines from private stores	20.0	38.0	38.0	4.0	0.0	2.26	50
Overall rating of services provided by the doctor	24.0	62.0	12.0	2.0	0.0	1.92	50

Source: PRC-CRRID, Field Survey, 2015, Note: NA- Not applicable

The satisfaction level among IPD patients on the count of services provided by the doctors was quite high. Barring two per cent of the IPD patients all others had expressed satisfaction, even though with varying degree, at the overall services provided by the doctors (Table 11.6). Non-availability of doctors as per their convenience had caused dissatisfaction among 10 per cent of the IPD patients. The weighted average of IPD patients was 1.9 as compared to 2.1 of OPD patients. Thus, the weighted averages of the overall services provided by the doctors reflect that the satisfaction level among the IPD patients was higher than the OPD patients.

It was observed during the field survey that a number of specialist medical officers were put on duties other than that of attending the patients during the OPD hours. Shortage of staff, medical as well the administrative, was cited to be the main reason for this. In any case, this led to dissatisfaction among a number of the patients who kept on waiting for the doctor with no information about the

whereabouts of the doctors. A large number of patients had come from far off places after spending money on transportation. They were annoyed as they had to return without meeting the doctor. The concerned health authorities need to pay attention to making the specialist doctors available on their scheduled OPD hours. This would minimize the dissatisfaction among the patients visiting these health institutions.

Laboratory tests: The OPD and IPD patients who utilized particular diagnostic facilities were asked to rate their experience. The overall satisfaction level was highest among the OPD patients who had utilized the ECG facility from these health institutions followed by patients using x-ray facility and facility for testing blood, urine etc. The satisfaction was least with the overall experience of using ultrasound facility having weighted average of 2.4 (Table 11.7).

A relatively higher satisfaction level was reported by the OPD patients on the individual parameters

Table 11.7: Satisfaction level of OPD patients with diagnostic facilities (in per cent)

Type of diagnostic facility	Highly satisfied	Satisfied	Somewhat satisfied	Not satisfied	Very unsatisfied	Weighted average	Total number of respondents (N)
BLOOD, URINE AND OTHER RELATED FACILITIES							
Convenience of the timings for the test	17.8	63.0	15.1	4.1	0.0	2.05	73
Politeness of lab technicians	17.8	61.6	15.1	5.5	0.0	2.08	73
Waiting time before test	19.2	58.9	15.1	5.5	1.4	2.11	73
Time taken for the result/reports	13.7	50.7	27.4	6.8	1.4	2.32	73
Overall experience of the blood, urine and other tests facility	13.7	58.9	21.9	5.5	0.0	2.19	73
X-RAY RELATED FACILITIES							
Convenience of the timings for the test	11.1	77.8	5.6	5.6	0.0	2.06	18

Politeness of technicians	11.1	77.8	11.1	0.0	0.0	2.11	18
Time taken for the result/ reports	5.6	77.8	16.7	0.0	0.0	2.11	18
Waiting time before test	5.6	77.8	11.1	5.6	0.0	2.17	18
Overall experience of the X-Ray tests facility	11.1	72.2	16.7	0.0	0.0	2.06	18
ECG FACILITIES							
Convenience of the timings for the test	25.0	75.0	0.0	0.0	0.0	1.75	4
Waiting time before test	25.0	75.0	0.0	0.0	0.0	1.75	4
Politeness of ECG technicians	25.0	75.0	0.0	0.0	0.0	1.75	4
Overall experience of the ECG tests facility	25.0	75.0	0.0	0.0	0.0	1.75	4
ULTRASOUND FACILITIES							
Politeness of doctors concerned	10.0	70	20.0	0.0	0.0	2.10	10
Convenience of the timings for the test	10.0	50	30.0	10.0	0.0	2.40	10
Time taken for the result/ reports	10.0	60	10.0	20.0	0.0	2.40	10
Waiting time before test	10.0	40	30.0	20.0	0.0	2.60	10
Overall experience of the Ultrasound tests facility	10.0	50	30.0	10.0	0.0	2.40	10

Source: PRC-CRRID, Field Survey, 2015

related with blood, urine and other such testing facilities with reference to politeness of laboratory technicians and convenience of the timings for the tests. The OPD patients had more or less a higher satisfaction level with the individual parameters related with X-ray facilities. Again the convenience of the timings for the tests and waiting time before test was one reason for causing a bit of dissatisfaction among the OPD patients. All the OPD patients were satisfied with the convenience of the timings for conducting ECG.

On the count of ultrasound facilities, all the OPD patients were satisfied with the politeness of laboratory technicians. However, one-fifth of the OPD patients were dissatisfied with the waiting time before the test and time taken for the reports. Non-availability of ultrasound machines had made a number of OPD as well as IPD patients unhappy. They were left with no choice but to visit private

clinics for getting ultrasound tests done at an exorbitant price. At times these patients were asked to get repeated ultrasound tests done. On each

occasion these patients had to pay huge amount of money for getting these tests done. Non-availability of radiographer to conduct ultrasound test was stated to be one reason for non-utilization of the machine. Concerted efforts on the part of state health authorities to provide man and machine to provide ultrasound test facilities are required that will augment the satisfaction level of large number of patients visiting these health institutions.

Emergency services: About 10 per cent of the OPD patients and 38 per cent of the IPD patients had utilized the emergency services at some point of time at the health institutions they were visiting. Little more than 40 per cent of the IPD patients had met with an accident because of which they had to

be rushed to emergency care wards of these health institutions, another 10 per cent of them had to be admitted to these health institutions as they were experiencing problems while breathing and almost the same proportion was admitted for childbirth related problems. While three-fourths of OPD patients utilized emergency services related to health related problems and childbirth related services.

Unlike a higher level of satisfaction among OPD and IPD patients on different aspects discussed so far, a lot of variation is found in their experience with the usage of emergency services. The gap in the satisfaction level of the OPD and IPD patients was observed in this case. About 16 per cent of the IPD patients were highly satisfied with the overall experience of emergency services as compared to none of the OPD patients.

The weighted averages reflect that the satisfaction level with the overall experience of emergency services among IPD patients was higher than the OPD patients; the weighted average being 2.0 among IPD patients and 2.6 among OPD patients. On the individual parameters of use of emergency services, the satisfaction level of the IPD patients was higher than that of the OPD patients, barring the availability of wheel chair and of ward boys for shifting patients. About one-fifth of the IPD patients were dissatisfied on the count of non-availability of ward boys for shifting patients from one place to another especially for getting diagnostic tests conducted. The dissatisfaction level was higher among both the OPD and IPD patients as regards the availability of essential drugs and supplies. One-fourth of the OPD patients and 10 per cent of the IPD patients showed their resentment as far as availability of essential drugs and supplies in the hospitals is concerned.

Services at pharmacy/dispensary: Little more than half of the OPD patients were satisfied with the overall experience at the pharmacy/dispensary as compared to 71 per cent of IPD patients. However, the satisfaction level on the count of

availability of prescribed drugs in the pharmacy/dispensary was low among both the OPD and IPD patients. Approximately one-fifths of the OPD as well as IPD patients were dissatisfied with the availability of drugs in the pharmacy/dispensary in these hospitals. About 10 per cent of the OPD patients were dissatisfied with the information imparted by the pharmacists regarding the procedure of taking medicines. Smaller waiting time before being served made almost all the OPD and IPD patients satisfied. They received attention earlier than they expected. The weighted average among the IPD patients was 2.1. This was slightly higher than the weighted average of 2.4 among the OPD patients.

At times, it was discerned that the medicines were available in the store of the district hospital (DH) or CHC but it was not available at the distribution counter. There should not be any time lag between transferring the medicines from store to distribution counter. This would help in minimizing the dissatisfaction level among the patients which arose due to non-availability of medicines at these health institutions.

Financial considerations: Providing affordable treatment to the patients through these hospitals is the prime objective of public health care system. The satisfaction of patients regarding the financial considerations while getting treatment in hospital is one of the indicators for assessing the affordability of the treatment. In general, nearly 54 per cent of the OPD as well as the IPD patients expressed their satisfaction regarding overall financial aspects of getting treatment in these hospitals. A dip in the satisfaction level is reported as far as the distribution of free medicines is concerned. About 20 per cent of the OPD patients and 16 per cent of the IPD patients were dissatisfied as far as availability of free medicines at these hospitals is concerned (Table 11.8). However, an enhancement in the satisfaction level of the 98 per cent of OPD patients and 92 per cent of the IPD patients was evident in respect of affordability of fees to

be paid for diagnostic tests. Unlike a higher level of satisfaction among IPD patients on most of the parameters discussed so far, the satisfaction with the expenditure on getting treatment of ailments was slightly higher among OPD patients.

The dissatisfaction level among the patients because of non-availability of free medicines in the hospital pharmacy/dispensary can be curtailed if these are available at cheaper rates within the hospital. Jan Aushadhi Stores was perceived to be the answer to such problems. To accomplish this task, a campaign was launched in Punjab in December 2008 to provide cheaper, non-branded essential medicines through dedicated outlets named as Jan Aushadhi Stores in all district hospitals. However, the Jan Aushadhi Stores were found almost non-operational in the health institutions across four of the five district hospitals. Further, this Store was as good as non-existent in another hos-

pital as it suffered from a terrible shortage of medicines prescribed by the doctors. In the prevailing scenario, the state health authorities need to reboot the system of Jan Aushadhi Stores to fulfil its perceived objective by maintaining the chain of availability of medicines. This would augment the level of satisfaction among all the patients.

Motivation level to revisit: The higher satisfaction level among OPD and IPD patients is also reflected in their intensity to revisit the same health institution they were visiting at the time of interview. More than 90 per cent of the IPD patients and 81 per cent of the OPD patients were sure that they would revisit the same hospital in future as well in case of need for getting treatment of ailments.

About 17 per cent of the OPD patients were indecisive as regards their intentions to make a revisit to the same hospital in future.

Table 11.8: Satisfaction with finance related matters (in per cent)

Finance related matters	Highly satisfied	Satisfied	Somewhat satisfied	Not satisfied	Very unsatisfied	Weighted average	Total no. of respondents (N)
OPD							
Payment made to get the diagnostic tests conducted	6.7	42.0	48.0	2.0	1.3	2.49	150
Availability of free medicines	6.0	34.7	38.6	20.0	0.7	2.79	150
Overall rating of the expenditure on getting treatment from this hospital	8.0	47.3	40.7	3.3	0.7	2.41	150
IPD							
Payment made to get the diagnostic tests conducted	6.0	58.0	28.0	8.0	0.0	2.38	50
Availability of free medicines	8.0	32.0	40.0	16.0	4.0	2.76	50
Overall rating of the expenditure on getting treatment from this hospital	10.0	44.0	34.0	12.0	0.0	2.48	50

Source: PRC-CRRID, Field Survey, 2015

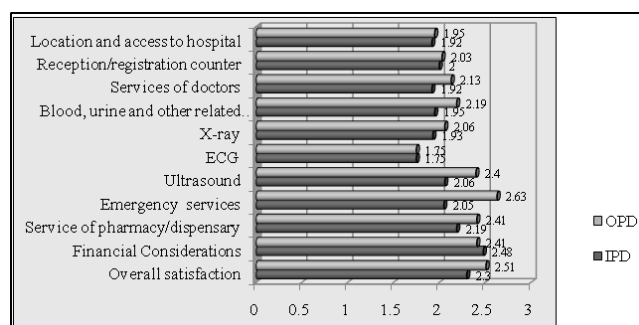
Figure 11.1: Satisfaction level

Figure 11.1 portrays the comparison of weighted average of satisfaction level separately for OPD and IPD patients on different variables. The difference in the weighted averages of OPD and IPD patients with 0.60 points was highest in services related to emergency and lowest with 0.03 points in services related to reception/ registration counter and of location and access to hospital services in favour of OPD patients. There was no difference in the satisfaction level of OPD and IPD patients on the count of ECG facilities as weighted average in both cases was 1.75. It also revealed that satisfaction level among IPD patients was higher than the OPD patients on all individual parameters except on financial considerations. On this parameter, the satisfaction was slightly higher among OPD patients than the IPD patients.

SATISFACTION LEVEL WITH PUBLIC AND PRIVATE HEALTH FACILITIES

Even though the primary aim of the survey was to assess the satisfaction level of OPD and IPD pa-

tients regarding their experiences with utilization of secondary care services at health public institutions, a perusal of their experience with the private sector hospitals were also recorded. Approximately 90 per cent of OPD and IPD patients had been to the private hospitals for seeking treatment. One out of every five OPD patients and one out of every ten IPD patients expressed their higher satisfaction with the services received at the private hospitals (Table 11.9). The dissatisfaction with the private hospitals was higher among IPD patients (six per cent) as compared to OPD patients (three per cent). One reason observed for the lower level of satisfaction with the private hospital services was the gap between the higher expectations formed due to huge amount of money spent and the quality of services received. Both the OPD and IPD patients felt that the services offered at the private hospitals were unaffordable and beyond their reach. Another reason for displeasure with the private hospitals was prescription of expensive medicines which the patients felt was beyond their paying capacity. A higher proportion of patients chose private hospitals for the treatment of specific diseases while public health institutions were selected mostly for minor diseases.

About seven out of every ten IPD patients and five out of every ten OPD patients were satisfied with the overall experience at the public health hospitals. The corresponding satisfaction with the private health institutions was about two out of every

Table 11.9: Satisfaction with overall experience at the public and private health institution (in per cent)

Type of patients	Highly satisfied	Satisfied	Somewhat satisfied	Not satisfied	Very unsatisfied	Weighted average	Total number of respondents (N)
Public hospitals							
OPD	3.3	48.7	41.3	6.7	0.0	2.51	150
IPD	6.0	60.0	32.0	2.0	0.0	2.30	50
Private hospitals							
OPD	22.5	44.4	30.1	3.0	0.0	2.13	133
IPD	11.1	55.6	26.7	4.4	2.2	2.30	45

three IPD as well as OPD (?) patients. Two per cent of the IPD patients and seven per cent of the OPD patients were dissatisfied with services received at public health institutions.

The satisfaction levels among the OPD patients availing services from public health institutions seem to be linked with their lower expectations from these hospitals. The OPD patients were paying minimal amount in the form of registration fee and few other diagnostic tests probably realize that they are getting what they are paying for. The situation was no different in case of IPD patients in this regard. The returns in the form of hospital services were almost matching with their expectations despite the fact that they had made high inputs as hospital charges. Satisfaction level among the patients utilizing private health care OPD services in Punjab was better than those utilizing public health care OPD services. However, in case of IPD services, the satisfaction level of patients was same in private and public health institutions.

PATIENT'S EXPECTATIONS FROM A HOSPITAL ON DIFFERENT PARAMETERS

All the patients were asked to prioritize pre-defined aspects of any hospital, irrespective of it

Table 11.10: Patient's views on different aspects of a hospital

OPD	Rank	IPD
Experience with the doctor – waiting time, time given by the doctor, effectiveness of treatment	1	Experience with the doctor – waiting time, time given by doctor, effectiveness of treatment
Quality of diagnostic facilities at the hospital	2	Quality of diagnostic facilities at the hospital
Provision and services of pharmacy/ dispensary within the hospital	3	Experience with the nurses and other staff

Financial considerations in availing the services of the hospital	4	Comfort of stay in the ward
Location and access to the hospital and departments	5	Quality of emergency services at the hospital
Quality of emergency services at the hospital	6	Financial considerations in availing the services of the hospital
Services at the reception/ registration counter of the hospital	7	Provision and services of pharmacy/dispensary within the hospital
Facilities and experience at the waiting area at the hospital	8	Quality of surgical/ child delivery services at the hospital
Provision and quality of other facilities in the hospital premises like telephone, STD, canteen etc.	9	Location and access to the hospital and departments
-	10	Service at the reception/ registration counter of the hospital
-	11	Provision of quality of other facilities in the hospital premises like telephone STD, canteen , ambulance, blood bank etc.

Source: Field Survey, 2015

being in the private or public sector. The OPD patients were asked to rank nine such aspects of a hospital based on their perceptions and IPD patients on eleven aspects. The basic idea was to understand the expectations of the patients from the hospital administration.

Less waiting time, sufficient time given by the doctor while attending to them, effectiveness of the treatment, and quality of diagnostic facilities were rated as the most critical considerations (Table 11.10). There was no difference in the perception of OPD and IPD patients in this regard. Pres-

ence of pharmacy/dispensary within the hospital was the next priority in order as indicated by OPD patients. In the case of IPD patients, the behaviour of nurses and other staff in the wards was the major consideration. These were followed by the financial consideration of service at an affordable price by the OPD patients and comfort of stay in the ward at these hospitals by the IPD patients.

Location and access of the public hospitals has been a major issue emphasized time and again in the state as well as in the national plans. However, the response to this aspect in the present survey was on different lines. Location and access to the hospital was not the top priority as perceived by all the patients. It was third last priority in case of IPD patients. What does it indicate? Have the increased mobility made the access issues irrelevant? Are people ready to travel to a greater distance for getting the quality services from the medical centres of repute? Are these hospitals already located in an easy accessible distance which made them to perceive it as a non-issue. The achievements of 108 ambulance facility have been remarkable in making distance a non-issue. A detailed exercise, however, need to be taken of this aspect of location and access to get the answers.

CONCLUSIONS

The gap between the expectations and actual experiences with the overall health care services at the public hospitals in Punjab was quite narrow as it caused dissatisfaction only among two per cent of the IPD and seven per cent of the OPD patients. The corresponding figures were almost the same in case patients availing services at private health institutions. Satisfaction level was better among the patients utilizing private health care OPD services than those utilizing public health care OPD services. However, there were no differentials in satisfaction level among IPD patients in both types of hospitals.

On the count of individual parameters such as care

at the reception-cum-registration counter, diagnostic facilities related with blood, urine, X-ray, ultrasound, and emergency services the satisfaction level was also higher among the IPD patients than the OPD patients. Nearly 19 per cent of the OPD and 22 per cent IPD patients were dissatisfied due to the non-availability of essential drugs and supplies in these hospitals.

Even though availability of beds in the ward at first application, continuous supply of electricity, comfortable bedding and convenient visiting hours were the reasons for the satisfaction among all the IPD patients, cleanliness of toilets, non-availability of water in these toilets, lack of seating facility for the attendants, non-availability of lockers near patient's bed to keep the personal belongings and lack of availability of drugs whenever required was a cause of concern among about half of them. Non-availability of water in toilets had caused dissatisfaction among OPD patients as well. Cleanliness of toilets was an issue causing dissatisfaction among a large number of OPD patients. In fact, one out of every four OPD patients using toilet facilities found these filthy and unhygienic. However, satisfaction with the services offered by the medical and other health staff was quite high.

Six out of every ten patients felt that the services offered at the private hospitals were unaffordable and beyond their reach. A mismatch between the huge amount of money spent and the quality of services received impelled many of them to seek services of a government hospital.

On the whole, the experience in these hospitals as characterized by the survey responses of IPD and OPD patients is a positive one. For the OPD and IPD patients, cleanliness of toilets and availability of drinking water was a major concern requiring immediate action. The state health authorities need to direct the concerned district health authorities to take appropriate measures to keep the toilets neat and clean, round the clock in IPDs, and also to make ad-hoc arrangements to provide safe

drinking water during the period required for repairing of the water filters. Apart from this, there was a need to get all the signage system at these hospitals displayed in Punjabi. The availability of specialist doctors, on the scheduled OPD hours, needs to be ensured. Their involvement in other administrative activities can be adjusted in such a way that they are accessible to patients to avoid their revisits from distant places. Many of other identified areas for quality improvement are related to time factor and non-availability of medicines at the hospitals. Concerted efforts on the part of state health authorities are also required to provide man and machine to provide ultrasound test facilities. Further, to meet the shortage of medicines, in case these are not available free of cost at the hospital pharmacy/dispensary, the state health authorities need to reboot the system of Jan Aushadhi Stores to fulfil the perceived objective of providing generic medicines at cheaper rates. The escalating cost involved in the purchase of medicines from the market was a cause of worry for every patient. These remedial measures would minimize the dissatisfaction among the patients visiting these health institutions. These standards set as per the IPHS guidelines need to be repeated for assessing the improvements and shortfall in provision of health services.

REFERENCES

Aldebasi, Yousef Hamoud and Mohamed Issa Ahmed, 2011. "Patients' satisfaction with medical services in the Qassim area". *Journal of Clinical and Diagnostic Research*, 5 (4), 813-817.

Drain M., 2001. "Quality improvement in primary care and the importance of patient perceptions". *Journal of Ambulatory Care Management*, 24(2): 30-46.

Epstein, K. R, C. Laine, N. J. Farber, E.C. Nelson, F. Davidoff, 1996. "Patients' perceptions of office medical practice: judging quality through the patients' eyes". *American Journal of Medical Quality*, 11 (2): 73-80.

Hill C. J. and K. Joonas, 2006. "The impact of unacceptable wait time on health care patients: attitudes and actions". *Health Marketing Quarterly*, 23 (2): 69-87.

Kumari Ranjeeta, M. Z. Idris, V. Bhushan, A. Khanna, M. Agarwal and S. K. Singh, 2009. "Study on patient satisfaction in the government allopathic health facilities of Lucknow district, India". *Indian Journal of Community Medicine*, 34 (1): 35-42.

Ministry of Health and Family Welfare, 2012, *Indian Public Health Standards Revised Guidelines for Community Health Centre*, Directorate General of Health Services, New Delhi, Govt. of India, 2012.

Ministry of Health and Family Welfare, 2012, *Indian Public Health Standards Revised Guidelines for District Hospital*, Directorate General of Health Services, New Delhi, Govt. of India, 2012.

Rao K. D., D. H. Peters and K. B. Roche, 2006. "Towards patient-centered health services in India - A scale to measure patient perceptions of quality". *International Journal for Quality in Health Care*, 18 (6), 414-421.

Verma Aarti and R. K. Sarma, 2000. "Evaluation of the exit proformas in use at special wards of public sector tertiary care centre". *Journal of Academy of Hospital Administration* 2000, 12 (1): 2001-2006.

CHAPTER - 12

Treatment-seeking behaviour of chronic and other diseases

Sajini B Nair, Suresh Kumar S, Rajesh J Nair, Mathew M.C

Population Research Centre. Kerala

Abstract

Chronic diseases are becoming a great burden to the population of Kerala. An assessment of the treatment-seeking behaviour of chronic and 'any other disease' is the aim of the study. The study area is Thiruvananthapuram district. Two coastal and non-coastal areas were selected. Hypertension, which is one of the most important modifiable risk factor for cardiovascular disease, is the most reported chronic disease. Diabetes is the next most reported chronic disease. Rich-poor divide in prevalence of chronic diseases is absent as the BPL category and APL category report the diseases more or less in equal proportions. Hypertension, diabetes and heart disease are found more in the non-coastal areas but cancer/tumour, bone/joint diseases and thyroid are reported more in coastal areas although the difference is marginal. Thyroid is almost double in coastal areas than non-coastal areas. Under Universal Health Care, for providing primary health care, fever is noted as the most reported reason for seeking treatment. So when fever is considered here in the study, one in four cases of 'other disease' is fever which includes viral fever, dengue and chikungunya. Proportion not seeking treatment is very less. Use of healthcare services varies by age and sex of the potential user, and economic situation. Here we find that almost 66 percent of the chronic illness cases are treated in the Government facilities and only less than thirty percent seek treatment in the private sector in the selected areas. Out- of-pocket expenditure is reported on tests, medicine and other expenditure, including travel. Out of the total households covered one in three have no insurance coverage. More than half of the households with insurance coverage are covered by RSBY/CHIS. With regard to any other diseases, significant differentials in prevalence are not evident between coastal and non-coastal regions. Burden of diseases other than chronic are shared more by females. Significant differentials are not discernible in prevalence by background characteristics of the population like religion, social group and the financial background as revealed by the APL/BPL category. Government facilities form a good share of the treatment for 'other diseases'. So various Governmental efforts, including health insurance coverage, for supporting the treatment costs of chronic and other diseases have succeeded in higher utilization of Government health facilities which is the primary objective of providing affordable universal access to health care in the State.

INTRODUCTION

In India, the burden of non-communicable diseases has been increasing. Diabetes, hypertension, cancer and chronic obstructive pulmonary disease are the most reported chronic diseases. Out of the NCD deaths Globally, India's share is over 15 percent which is almost 5.9 million deaths. Also about one third of these deaths occur before 70 years (WHO 2014). A major concern in this regard is the early onset of NCDs. Reddy et. al. (2005) found about 3 – 4 percent of adults (aged 20 and above) in rural areas and 8 – 10 percent in urban areas to be suffering from coronary heart disease in India. Prevalence of most chronic disease also increased with age, and women reported a disproportionately higher burden of chronic illnesses compared to men.

Research studies on morbidity, particularly chronic diseases, among the Indian States, shows higher prevalence of morbidity and also chronic NCDs in the small State of Kerala. Recent statistics from NSSO, 2014 show the Proportion of Ailing Patients (PAP) in Kerala to be the highest (308 per 1000) as against 104 per 1000 persons in India. This is an increase of 60 points for Kerala as against 13 points for India from the year 2004. Rural urban differentials were also evident as urban areas had a higher PAP than rural areas (118 as against 89 per thousand population respectively) (NSSO, 71st Round, 2014). With regard to NCDs, the reported prevalence of any chronic disease at 7 percent, cardiovascular disease at 14.5 percent, hypertension and high blood sugar levels among persons aged 18 years and above at 34.5 percent and 24.7 percent respectively signify the heavy burden (DLHS-4, 2012-13). Such high prevalence has been causing great economic burden to the people in the State.

Health care planning both in India and the State envisages this alarming situation and is committed to address the higher morbidity situation. Studies show that the early recognition of symptoms, pre-

sentation to health care facilities and compliance with effective treatment can reduce morbidity and thereby mortality (WHO, 1995; Hausmann et. al 2003). In every society the public and private health care facilities coexist and health care policies and programmes are designed from information on health care-seeking behavior of the people.

Treatment-seeking behavior is, in simple terms, the measures taken by people to maintain or gain good health in response to the symptoms of illness. Wade et al. (2004) defines health-seeking behaviour as an action undertaken by individuals who perceive themselves as having a health problem or to be ill for the purpose of finding an appropriate remedy. The decisions made encompass all available health care options like visiting a public or private and modern or traditional health facility, self-medication and use of home remedies or not to utilize the available health services etc. Many factors influence health care-seeking behavior. Socio-demographic factors and socio-cultural dimensions (Young 2004), psychological factors, socio-cultural and familial factors, individual factors, situational factors and marketing factors (Atashbahar et. al, 2013) are important among these. Private care providers are usually preferred due to easy accessibility even in the night, quick relief and individual attention, whereas, Public hospitals in developing countries are known for low quality treatment, long waiting period, long distance, inconvenient location, and inadequate facilities (Sudharshanam, 2007).

In Kerala, there has always been a greater tendency to approach private hospitals for treatment. Three fourths of the population usually flows to the private sector. Navaneetham and Kabir (2006) found that inadequate facilities, distance and inconvenient timing were the factors contributing to non-use of public health facilities. KSSP (2006) pointed out the availability of facilities to be the reason for the preference for the private sector, while economic considerations formed the major reason for seeking care from a government hospital. How-

ever, with the launching of the NRHM in 2008 in the State, the public health infrastructure increased tremendously. The subsequent programmes like the Janani Shishu Suraksha Karyakram (JSSK), Rashtriya Bal Swasthya Karyakram (RBSK) or Arogyakiranam etc. accorded free treatment to the mother and children which certainly increased the utilization of Government health facilities. Also the various Governmental Health Insurance Schemes like the Karunya Benevolent Fund, the Comprehensive Health Insurance Scheme PLUS etc. which are unique to Kerala provided financial support to the public suffering from chronic NCDs and reduced the share of household expenditure for treatment of diseases.

With ageing process being rapid in Kerala, the demand for geriatric services has also increased. The setting up of NCD clinics at the PHC level, separate geriatric wards and other Government support in the form of palliative care has certainly set a sound base in the Government health facilities for the population to seek treatment at affordable and good quality care. Among the many factors that potentially influence the healthcare-seeking behavior, living conditions, individual beliefs, gender and age is expected to be the important ones. So it is worthwhile to investigate whether there has been a change in the health-seeking behavior with these innovative programmes both from the Central and State Governments. On these footings, the present study aims at analysing the treatment-seeking pattern of chronic and Other illness in Thiruvananthapuram district.

Objectives

The main objective of the study is

- to examine the treatment-seeking behaviour of persons suffering from chronic illness and
- to assess the level of utilization of Government health facilities for chronic and Other illnesses.

Data and Methodology

Four wards from the Thiruvananthapuram district - two coastal area and non-coastal areas each - were selected at random for the study. The selected wards are Kovilvila and Vengode from non-coastal area and Poonthura and Puthenthope from coastal area. Information pertaining to the household, treatment-seeking behavior and utilization of health care services were gathered using a structured interview schedule.

3.1. Household survey

For the household survey, the sample size was calculated as

$$SS = \frac{Z^2 \times (p) \times (1 - p)}{C^2}$$

SS = Sample Size, Z = Level of Confidence Measure, P = Baseline levels of the indicators

C = Margin of Error (MOE), The Z-value is 1.96 for 95 percent confidence level

$$SS = \frac{(1.96)^2 \times (0.5) \times (1 - 0.5)}{(0.05)^2} = 384.16$$

So the required minimum number of sample is 385. Sample size was initially fixed as 500 to cater to the non response rate also. The final sample households covered was 507. The selection of households was done scientifically beginning with the household in the North-Eastern end of the ward and 125 households from each of the 4 wards were selected randomly.

A responsible adult member of the household was the respondent. Statistical techniques used include both univariate and bivariate analysis. Backward logistic regression technique was employed to find out the covariates of treatment-seeking behavior. The odds of choosing a private health facility given the access to Government health facilities are also examined.

Findings and Discussion

Background Characteristics of Household Head

The survey covered a total of 507 households (247 in coastal and 258 in non-coastal areas). The total population covered is 2365. The average household size of the total sample population, all the four areas taken together, is 4.7.

Table 12.1: Percentage distribution of households by background characteristics

No. of Households	No.	Economic category of HH	
Coastal	249	APL	65.2
Non coastal	258	BPL	34.8
Religion of HH Head	Percent	Social Group of HH	
Hindu	31.2	SC	4.5
Muslim	18.1	OBC	79.7
Christian	50.7	General	15.8
No. of members in HH		Education of Head of HH	Percent
1	3.0	Illiterate	14.8
2	9.1	1-10	71.4
3	15.8	10+2	6.9
4	26.0	≥ 13	6.9
>4	46.2		
Occupation of HH	Percent		
No job	22.5		
Professionals	5.1		
Skilled labourers	4.7		
Unskilled Labourers	52.7		
Clerical/Teacher	3.7		
Business	1.6		
Retired	9.7		

The number of usual residents in the household is an important aspect in any study on disease pattern or prevalence. Here slightly over one quarter of the households have four members and little less than half of the households have more than 4 members. In the study area, Christians have a higher representation (50.7 percent). Muslims form only about 18.1 percent of the total population and 31.2 percent are Hindus. Assessing the distribution of population by social group, one can find that nearly four fifths of the population belongs to OBC category, 4.5 percent are Scheduled Castes and 15.8 percent are from general category. Almost three fourths of the population has had some years of schooling between 1 to 10 years (Table 12.1). About 15 percent of the respondents are illiterate. About 7 percent completed 11 or 12 years of schooling and a similar proportion have completed 13 or more years of formal education.

Among the population covered, over one-fifths are jobless, more than half of the respondents are unskilled labourers mostly fisher folk, coolie labourers etc. Other than these, categorizing the jobs as those with steady income, only one in ten are retired from Government job. Housing Characteristics

The environmental and the housing characteristics have a strong bearing on the incidence of diseases. Prevalence of communicable diseases are directly linked to the hygiene of the surroundings the people live, and non-communicable disease prevalence have been related to life style.

Table 12.2: Percentage distribution of households by housing characteristics

Type of House	Percent	Safe Drinking Water	Percent
Pucca	60.4	Boil	88.4
Semi-Pucca	28.8	Purifying	1.7
Kachha	10.8	Filtering	8.6
		Potassium per manganate / Bleaching powder	1.4

Treat Water		Source of energy for cooking	
Yes	82.8	LPG	83.6
No	17.2	Firewood	16.4
Source of Drinking Water		Sanitation facilities	Percent
Bottled Water	1.2	Septic tank/ Flush	91.9
Piped water -Own	17.8	Pit latrine	2.6
Piped water -Public	14.6	Public latrine	3.4
Well/Bore well	59.2	No facility	1.4
Public Well	2.8	Other	0.8
Other	4.5		

Here we find that, about three-fifths of the population surveyed live in Pucca households. Sanitation facilities are available in 98.6 percent of the households. Out of these, 92 percent use septic tank/flush system. The source of drinking water for almost three-fifths of the households is well or bore well, 17.8 percent use their own piped water facility and 14.6 percent have access to public piped water. Almost 82.8 percent of the households make water safe to drink and out of these boiling is the most widely used method. The source of energy for cooking to about 84 percent of the households is LPG. Firewood is used for cooking by 16.4 percent of the households.

Morbidity

The reported diseases are classified as chronic and 'any other disease' which means diseases other than chronic. Chronic diseases considered here include hypertension, diabetes, tuberculosis, heart disease, cancer, tumour, asthma, bone or joint disease, kidney problems, stroke and thyroid. Any other diseases that last only for a short period here are classified as other diseases. So the reference period for 'any other disease' is taken as 2 months prior to survey.

Chronic Diseases

Demographic transition in Kerala and the subsequent epidemiological transition from communicable to non-communicable diseases in the subsequent years have been associated with an increasing prevalence of chronic degenerative diseases. Hypertension is a major public health problem and is one of the most important modifiable risk factor for cardiovascular diseases (CVDs).

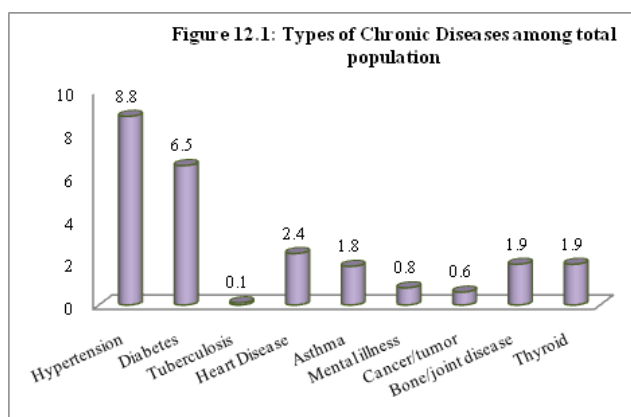
A total of 635 chronic disease cases have been reported among the sample population. Here we consider the cases of diseases because multiple diseases have been reported by most of the respondents. Significant differentials in prevalence of chronic diseases are not evident between coastal and non-coastal regions as about 51 percent of the cases of chronic illness are reported in the non-coastal region and the rest of the cases have been reported from the coastal region of the sampled population.

Table 12.3: Percentage distribution population by chronic diseases

Any Chronic disease among Households	Percent
Yes	75.5
No	24.5
Any Chronic disease among Population	Percent
Yes	22.5
No	77.5

The proportion of households reporting any chronic diseases was 75.5 percent. Out of the total population covered, 22.5 percent reported any chronic diseases during the reference period.

The type of chronic diseases reported is assessed among the total population covered and also among the total cases of chronic diseases reported. Hypertension is the most reported chronic disease (8.8 percent).



Diabetes is the next most reported chronic disease (6.5 percent). Heart Disease (2.4 percent), bone/joint diseases (1.9 percent), thyroid (1.9 percent) and asthma (1.8 percent) are some of the other self reported chronic diseases. Mental illness and cancer/tumour form less than 1 percent each.

The type of disease when assessed among the reported chronic disease cases also shows that hypertension is the most reported chronic disease as one in three cases in the sample population is hypertension. Diabetes is a common disease causing significant morbidity and mortality; thus appropriate health-seeking behaviour may help to plan diabetes care and management upon diagnosis so that complication can be minimized and quality of life can be improved. Here diabetic cases account for 24.2 percent.

Table 12.4 Type of Chronic diseases among total diseases reported

Type of Chronic diseases	Percent
Hypertension	32.9
Diabetes	24.2
Tuberculosis	0.5
Heart Disease	8.8
Asthma	6.6
Mental illness	3.2
Cancer/Tumor	2.4
Bone/Joint Diseases	7.1
Thyroid	7.0
Others	7.4

Any other diseases – epilepsy, paralysis, gastric/peptic ulcer, chronic skin disease, liver disease, kidney problems

Heart Disease (6.6 percent), bone/joint diseases (7.1 percent), Thyroid (7 percent) and asthma (6.6 percent) also are a cause of concern. Mental illness is 3.4 percent of those with chronic disease and 2.4 percent have cancer/tumour.

The common factors attributed to the emergence of chronic diseases are demographic changes, urbanization and lifestyle factors such as diet, exercise, as well as consumption of tobacco and alcohol (Nugent 2008; Upadhyay 2012). So with changes in life style, the age at which a person is at risk of being hypertensive or diabetic has changed drastically. Environmental pollution, exposure to radiations etc. has been projected to be the cause of other non-communicable diseases like cancer and tumour.

Table 12.5: Chronic diseases by Age and Sex

Age of the Patient	Percent	Sex	Percent
<30	4.1	Male	43.5
30-39	9.0	Female	56.5
40-49	14.5		
50-59	26.5		
60+	46.0		

About 4 percent of the cases have occurred at ages below 30 years. Also 9 percent of the total chronic diseases have been reported by those aged 30-39 years and another 15 percent in the 40-49 age group. Older cohorts are naturally at more risk of chronic diseases and slightly more than one fourth of the total number of cases are reported in the age group of 50-59 years and nearly 46 percent of the cases have been reported by those aged above 60 years. As expected, prevalence of most chronic disease increases with age. Chronic diseases are reported more by females. Nearly 56 percent of the patients are females whereas only 44 percent

are males. This finding is in line with the findings of the study by Kowal et al (2012) who showed that women bear a disproportionately higher burden of chronic illnesses compared to men. The study showed that the only exception is diabetes that has no clear age pattern, while a higher percentage of Indian men report diabetes compared to Indian women.

When chronic diseases are assessed by the background characteristics of the population like religion, social group and the financial background as revealed by the APL/BPL category, significant differentials are not discernible by religion.

Table 12.6: Chronic diseases by Socio economic characteristics

Religion		Social Group	
Hindu	25.6	SC	21.0
Muslim	22.0	OBC	21.5
Christian	21.8	General	29.6
Economic category	Percent		
APL	23.0		
BPL	21.8		

About 30 percent of the 'General' category' and 21 percent the SC category have reported of having any chronic disease. One noteworthy finding is the absence of differential by economic category. Some studies even show the risk of chronic diseases such as cardiovascular diseases to be higher among the poorer socio-economic classes (Xavier et al., 2008). The earlier belief that chronic diseases are predominantly diseases of the rich and result from affluent lifestyles is no longer true (Patel et al 2011, Prabhakaran et al 2013; Bhojani et al 2013).

Any other diseases

Those short term diseases other than chronic diseases that require treatment are considered here. Fever is a temporary increase in the body tempera-

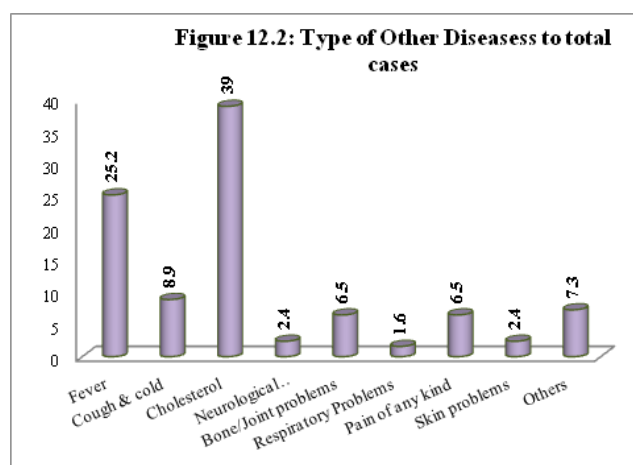
ture, often due to an illness. Having a fever is a sign that something out of the ordinary is going on in the body. Since it is a common sign of some illness it is considered here as acute condition as treatment is usually sought. Similarly, for this survey, 'cholesterol' is also considered as an acute condition since it is often an abnormal condition which leads to some disease. So the self reported diseases other than those which are chronic in nature, are here classified as 'any other disease'.

Table 12.7: Percentage Distribution of population reporting 'any other disease'

Any other disease (2 months) among HH	Percent	
Yes	20.3	
No	79.7	
Any other disease (2 months) among Population	Percent	
Yes	5.0	
No	95.0	
Type of disease (cases)	Percent	Percent (to total population)
Fever	25.2	1.3
Cough & cold	8.9	0.5
Cholesterol	39.0	2.0
Neurological problems	2.4	0.1
Bone/Joint problems	6.5	0.3
Respiratory Problems	1.6	0.1
Pain of any kind	6.5	0.3
Skin problems	2.4	0.1
Others	7.3	0.4

'Others' here include diarrhoea, urinary problem, STI, Jaundice, anemia gynaecological problems, fits and epilepsy as these diseases are reported by very few persons in the sample population

About 20.3 percent of the households report to have some disease other than chronic disease during the 2 months preceding the survey. But the proportion is assessed within the sample population, prevalence is only 5 percent during the two months prior to the survey. Here the reference period is December 2015- January 2016. Studies show that incidence of viral and other infectious diseases are more during the monsoon season - June to October. Thereafter the incidence in communicable diseases usually decreases. So during the reference period in this study one can expect low prevalence of communicable diseases. The study findings also reveal similar findings.



If Cholesterol is considered as an acute condition of the body, it is the most reported disease other than chronic constituting almost two-fifths of the disease. One in four such cases are fever cases which includes viral fever, dengue fever and chikungunya. Cough and cold forms almost 9 percent, and pain in any part of the body like ear, stomach, head, eyes etc. constitutes 6.5 percent. Also bone or joint problems (occurred due to accidents or otherwise) constitute 6.5 percent of the total cases. Neurological problems and skin problems constitute 2.4 percent each.

Chronic Diseases in selected areas

The prevalence of chronic diseases is analysed by area: coastal and non-coastal (Table 12.7). Hypertension and diabetes which form a good

share of the chronic diseases is reported more in non-coastal areas (11 percent) than coastal areas (7 percent). Similarly, there is a difference of almost 2 percentage point in population reporting diabetes in the coastal and non-coastal areas (5.5 as against 7.6 percent respectively). Heart disease, respiratory problems and mental illness are also slightly more in the non-coastal areas. But, cancer/tumour, bone/joint diseases and thyroid are reported more in coastal areas although the difference is marginal.

Table 12.8. Percentage distribution of population by type of chronic Diseases

Type of Chronic Diseases	Percentage to total Popn		Percentage to total Chronic Diseases	
	Coastal	Non-coastal	Coastal	Non-coastal
Hypertension	7.1	10.9	30.1	35.5
Diabetes	5.5	7.6	23.3	25.0
Tuberculosis	0.2	0.1	0.6	0.3
Heart Disease	1.8	3.0	7.8	9.9
Asthma	1.5	2.1	6.5	6.8
Mental illness	0.8	0.9	3.2	3.1
Cancer/Tumor	0.7	0.6	2.9	1.9
Bone/Joint Diseases	2.0	1.8	8.4	5.9
Thyroid	2.1	1.5	9.1	4.9
Others	1.9	2.1	8.1	6.8

If the type of disease is assessed based on the total number of cases of chronic diseases (multiple diseases to a person also included) we find that when 30 percent of the chronic diseases is hypertension in coastal areas, it is 35.5 percent in non-coastal areas. One in 4 cases of chronic disease is a diabetic. When one in ten cases is a case of heart disease in non-coastal areas, it is 2 percentage point less in coastal areas. Thyroid, on the other hand,

is almost double in coastal areas than non-coastal areas (9.1 percent as against 5.9 percent).

Fever is reported more in non-coastal areas compared to coastal areas (28.6 percent as against 21.7 percent). Cough and cold cases reported accounts for 6.7 percent in coastal areas as against 11.1 percent in non-coastal areas. Nearly half the cases of 'Any other diseases' is cholesterol in the coastal areas whereas one in three cases in non-coastal areas is cholesterol.

Treatment-Seeking Behaviour of Chronic diseases

Health care-seeking behaviour is a complex outcome of many factors operating at individual, family, and community-levels. Treatment choices would involve many factors related to illness type and severity, socio-demographic characteristics, pre-existing beliefs about illness causation, accessibility of treatment available and their perceived efficacy and disease profile (Remli and Chan, 2003). Health care-seeking behaviour is influenced by the individual self, diseases, and the availability and accessibility of health services. Treatment-seeking behaviour varies by age and sex also (Hjelm and Atwine, 2011). In India, about 30 percent of the elderly suffering from illnesses seeks treatment and over 55 percent don't receive even minimum care and personal help (Kumar, 1995). However, in Kerala, 90 percent of the elderly, irrespective of sex, consult doctors.

In the present study which covers over 500 households, we find that over 97 percent of the population who had chronic illness sought treatment. Both in the coastal and non-coastal areas only a very few people have not sought treatment. Only for mental illness, asthma and bone/joint diseases around 5 – 7 percent of the patients failed to seek treatment. The reason for not seeking treatment was reported to be due to less severity of disease.

Table 12.9: Geographical distribution of population seeking treatment of chronic disease

Type of disease	Percentage to total Cases			Percentage not seeking treatment
	Total	Coastal	Non Coastal	Total
Hypertension	98.6	97.8	99.1	1.4
Diabetes	98.0	95.8	100.0	2.0
Tuberculosis	100.0	100.0	100.0	0
Heart Disease	100.0	100.0	100.0	0
Asthma	95.2	90.0	100.0	4.8
Mental illness	95.0	90.0	100.0	5.0
Cancer/Tumor	100.0	100.0	100.0	0
Bone/Joint Diseases	93.3	92.3	94.7	6.7
Thyroid	100.0	100.0	100.0	0
Total	98.1	96.8	99.4	1.9

Utilization of Health Care Services

In India, health services utilization statistics indicate that the private sector accounts for around 60 percent of inpatient care use and 80 percent of outpatient care use (NSSO 2006a). In Kerala, studies show that almost 70 percent of the population utilize private sector for treatment. However, with the implementation of various programmes under the NHM which offered free health care services in the Government sector, and the Universal Health Coverage initiative, a slow but steady shift in treatment-seeking behavior towards public sector has been anticipated.

Two-thirds of the chronic illness cases are treated in the Government facilities and only less than thirty percent seek treatment in the private sector

in the selected areas. Although significant differentials by region: coastal/non-coastal is also not evident in the overall treatment-seeking behavior in the Government/ Private facilities, differentials by type of Government or Private facilities are visible between the coastal and non-coastal regions.

Table 12.10: Percentage distribution of population seeking treatment for chronic disease

Type of disease	Percentage to total Cases		Others
	Government Facilities	Private facilities	
Hypertension	68.3	26.3	5.4
Diabetes	65.3	33.3	1.3
Tuberculosis	66.7	0.0	33.3
Heart Disease	71.4	28.6	0.0
Asthma	56.1	34.1	9.8
Mental illness	35.3	58.8	5.9
Cancer/Tumor	100.0	0.0	0.0
Bone/Joint Diseases	59.0	17.9	23.1
Thyroid	61.4	34.1	4.5
Kidney problems	100.0	0.0	0.0
Any other diseases	74.5	21.3	4.3
Total	66.3	28.5	5.2

Generally, a major share of the patients approach Government facilities for treatment of chronic diseases. But this pattern changes in the case of some chronic diseases based on the availability of the health facility, nature of the illness, duration and cost of treatment. In the case of Hypertension, 68 percent of the cases have been treated in Government facilities which are mostly the nearby PHC/CHC and 26 percent of the cases have been treated in private health facilities. Almost a similar pattern is seen with regard to treatment of Diabetes cases with two thirds of the cases of illness being treated in Government facilities (65 percent). Poor quality of medicines available in the Government

facilities prompted them to shift to private hospitals in the case of Diabetes treatment. One-third of the Tuberculosis cases are seen to be treated in health facilities other than Government or private like Ayurveda or traditional healers. Patients with Heart diseases mainly relied on Government facilities (71 percent) and the rest sought treatment in private hospitals.

Patients with mental illness mainly depend on private hospitals (59 percent). Lack of privacy in the public facility created a fear among the respondents and a social stigma against patients with mental illness is the main reason for the patients to opt private facilities for the treatment. Although less in number, all the patients with Cancer/Tumor have sought treatment from the public facilities like Regional Cancer Centre, Thiruvananthapuram or MCH, Thiruvananthapuram. All the Kidney-related problems were treated in public health facilities. This is due to the free treatment available under various health insurance schemes for the Kidney ailments in the State.

Table 12.11: Place of treatment of chronic disease by area

Place of Treatment	Chronic Diseases	
	Coastal	Non Coastal
Not visited	4.5	0.6
Sub centre	0	0.3
PHC	13.9	2.1
CHC	20.1	11.7
SDH	0.3	6.1
DH	8.7	17.5
MCH	20.4	16.6
ESI Hosp	1.0	3.7
CGHS	0.4	8.0
Private Doctor/Clinic	5.5	15.0
Private hospital	19.1	7.7
Private Multi specialty hosp	2.9	3.1
Private MCH	0.6	1.5
Traditional	0	1.9
Other	2.6	4.3

In general, among the Government health facilities, people rely more on the PHC and CHC in the coastal areas whereas in the non-coastal areas we find more dependence on the District Hospital and Sub District/Taluk level hospitals. But since MCH is the ultimate referral point for treatment of diseases, much differentials are not observed by region. Also in the non-coastal region, the proportion of population employed under the Government sector (Central/Defence sector) was relatively more and hence, the proportion of chronic disease cases treated in the ESI/ CGHS is relatively more than in the coastal region.

Among the private sector health facilities, almost 19 percent of the cases are treated in the private hospitals in the coastal regions whereas in the non-coastal region people rely more on the private practice of doctors and clinics. Proportion treated in the private multi- specialty hospitals is more or less the same in the coastal and non-coastal areas.

Table 12.12: Reason for not choosing Government facility for treatment of chronic disease

Reason for not choosing Govt. facility ↓ Disease Type →	Chronic Diseases
Doctor not always available	0.6
Poor facility	15.8
Lack of faith	30.4
Long waiting time	41.5
Lack of transportation facility	0.6
No privacy	2.9
Others	8.2

The main reason mentioned for not choosing a Government facility for treatment of chronic disease is 'long waiting time' (41.5 percent) followed by the reason 'lack of faith' (30.4 percent), 'poor facility' (15.8 percent).

Duration of illness

The duration of illness is suggestive of the disease burden in a society. Chronic diseases are, in fact, highly inequitable with higher risk factors among lower socio-economic groups, and with more adverse financial implications for the poor in India (Thakur et al 2011). The treatment cost is heavily dependent on the duration of illness.

Table 12.13: Duration of chronic disease and follow up treatment

Duration	Chronic Disease
< 1 year	13.4
1-3 years	18.0
3-5 years	19.1
5+ years	47.9
Not known	1.6
Follow up treatment	
Continuing treatment	95.5
Monitoring through Diagnostic tests during fixed intervals	90.2

The respondents who report incidence of chronic disease during the reference period of one year preceding the survey constitute 13.4 percent of those who report chronic disease. Nearly 48 percent of the chronic cases have prolonged for a span of more than 5 years. Around 19 percent of the chronic cases have been reported to be a burden for 3 to 5 years and an almost equal number had been suffering for the past 1-3 years. The reference period for chronic diseases has been one year preceding the survey. Incidence rate here is expressed per 1000 population. So the incidence rate estimated among the population at risk of chronic disease is 37 per 1000 population during the one year preceding the survey. The ratio of prevalence to incidence rate calculated yields duration of 7.2 years which is evidence of the burden of chronic disease in the study area. Almost 96 percent of those suffering from chronic diseases were continuing treatment at the time of survey with 90 percent of them monitoring the disease

through diagnostic tests during fixed intervals.

5.3 Treatment cost

An assessment of treatment cost is important in this context because Government has implemented numerous programmes that extends treatment free of cost on the one hand and Government-sponsored health insurance coverage is provided for various disease treatments on the other hand. So the cost incurred would reflect the out-of-pocket expenditure to a certain extent when government health facility is only considered.

Table 12.14: Total cost of Treatment of Chronic Diseases

Expenditure (in Rs)	Percent
0	20.2
≤100	7.6
101-500	30.7
501-1000	15.9
1001-3000	16.7
3001-5000	2.8
>5000	0.9
Don't know	5.2

The total cost of chronic treatment incurred monthly is calculated as the summary measure of expenditure incurred towards registration, medicine, test and transportation and other expenses. When treatment cost is considered irrespective of type of facility chosen, 21.3 percent of the total chronic cases were treated free of cost. Nearly one-third of the cases had a monthly total expenditure of Rs 100 to 500 and for another 16 percent of the cases, the expenditure ranges between Rs 500 and Rs 1000. About 18 percent of the total chronic cases have a monthly expenditure of about Rs 1000-3000 and 4 percent incurred expenditure above Rs. 3000/-.

As a large portion of the chronic cases have been treated in Government hospitals, the registration

was free of cost for 71 percent of the chronic disease cases. Only 9.8 percent of the total illness cases had incurred more than Rs. 100 for registration in the hospitals (Table 12.15). Expenditure of diagnostics like various blood tests, x-rays were also assessed. Nearly three-fifths of the chronic cases were diagnosed without any costs. For the treatment of about 10.4 percent of the cases Rs. 101 to 500 was spent on monthly diagnosis.

Table 12.15: Expenditure for chronic illnesses (in Rs)

Cost of registration	Percent	Cost of tests	Percent
0	71.5	0	58.7
≤ 100	9.8	≤100	14.3
101-500	6.5	101-500	10.4
Don't know	12.3	501-1000	1.7
Total	635	≥ 1000	1.0
		Don't know	13.9
Cost of medicine	Percent	Other costs	Percent
0	25.5	0	65.2
≤100	7.4	≤100	13.1
101-500	31.2	101-500	6.1
501-1000	14.0	501-1000	0.6
1001-3000	9.4	1001-3000	0.2
3001-5000	1.6	> 3000	0.2
>5000	0.3	Don't know	14.6
Don't know	10.6		

Medicines were free for one-fourth of the chronic disease cases while one in three had spent Rs 101 to 500 per month on medicines. A quarter of cases incurred Rs 501 to 2000 per month for the purchase of medicines for chronic illnesses. The expenditure on monthly medicines exceeds Rs 3000 for about two percent of the chronic disease cases.

Comparing the different purposes for which patients had to spend for treatment, we find that majority incurred expenditure on medicines. All the medicines prescribed by the doctors are not provided from the facility. For diagnostic test also lit-

Table 12.16: Expenditure for treatment of Chronic disease

Type of disease	Amount in Rupees						Total
	0	≤ 100	101-500	501-1000	1001-3000	> 3000	
Hypertension	24.9	10.7	40.6	13.2	10.2	0.5	197
Diabetes	28.6	8.2	27.2	21.8	12.2	2.0	147
Tuberculosis	100	0	0	0	0	0	2
Heart Disease	9.6	7.7	13.5	3.8	46.2	19.2	52
Asthma	24.4	4.9	36.6	22.0	12.2	0	41
Mental illness	5.3	0	31.6	31.6	31.6	0	19
Cancer/Tumor	0	14.3	21.4	14.3	35.7	14.3	14
Bone/Joint Diseases	20.0	7.5	30.0	22.5	17.5	2.5	40
Thyroid	9.3	4.7	48.8	18.6	16.3	2.3	43
Kidney problems	16.7	0	33.3	0	50.0	0	6
Any other diseases	20.0	5.0	20.0	17.5	27.5	15.0	40
Total	21.3	8.0	32.3	16.8	17.6	4.0	601

tle more than one-third of the patients had to spend from their own pocket.

Treatment was free of cost for about one quarter of the cases of hypertension, diabetes, asthma etc. (Table 12.16). Treatment cost was maximum for heart disease as we find that almost 20 percent of those with heart disease spent over Rs. 3000 per month. For cancer/tumour and 'Any other diseases' that included few cases of stroke, almost 15 percent each spent over Rs. 3000 monthly. For asthma, bone diseases, thyroid and mental illness monthly expenditure of Rs 100 to Rs. 1000 was incurred by more than half the number of patients with these diseases. The expenditure was mostly on medicines and diagnostic tests.

Out-of-pocket expenditure is reported on medicine and other expenditure, including travel, food etc. About 12 percent incurred expenditure on registration, 28.5 percent had to spend for tests, over 60 percent incurred expenditure on medicine and over 20 percent on 'other costs' (Table 12.17).

The amount spent on registration is, however,

only less than Rs. 10. In the hospitals they visit, patients have to pay for some diagnostic facilities like x-ray, scanning, blood tests etc. and for those

Table 12.17: Expenditure for treatment of Chronic disease at Government Health Facilities

Cost of registration	Percent	Cost of tests	Percent
0	86.9	0	71.5
< 10	11.8	< 100	15.3
≥ 10	1.3	100-499	10.1
Don't know	0	500-999	1.6
Total	373	≥ 1000	1.4
		Don't know	365
Cost of medicine	Percent	Other costs	Percent
0	36.9	0	78.4
< 100	9.9	< 100	13.7
100-499	31.6	100-499	6.7
500-999	11.2	≥ 500	1.2
1000-2999	7.5	Don't know	0
≥ 3000	2.9	Total	358
Total	374		

diagnostic facilities that are not available, they depend on private clinics in the vicinity.

Health Insurance Coverage

Health insurance has emerged as a key instrument in current health financing reforms in India aimed at achieving universal coverage. The health insurance systems are often characterized by multiple health insurance funds covering different population groups. Some insurance schemes cover the family while others cover individuals.

Table 12.18: Registration in Health Insurance

Registered in Health Insurance scheme	Percent
No Insurance	37.0
RSBY/CHIS	54.7
ESI	1.6
CGHS	1.7
Employer	2.3
Other Policies	2.7

Out of the total households covered, one in three have no insurance coverage. More than half of the households are covered by RSBY/CHIS. Also 2.3 percent report to have some HI policy. Other health insurance includes ESI (1.6 percent), CGHS (1.7 percent) and insurance provided by employer (2.3 percent).

Since a large part of the expenses are covered by Insurance coverage for chronic diseases like heart disease, cancer, tumour, kidney and liver problems etc., the status of insurance coverage by the amount spent is examined. Here it can be noted that among those reporting no expenditure at all on treatment, almost three fourths are due to insurance coverage. For the remaining, quarter proportion, the treatment is perhaps specific to Government facilities when all the medicines, tests etc. are available free of cost. If we look at the other extreme, well over half of the patients reporting various diseases had spent over Rs. 3000 as they

Table 12.19: Cost of treatment for Chronic disease and Insurance Coverage

Cost of Treatment	Chronic Diseases	
	Insurance Coverage	
	Yes	No
0	73.4	26.6
<1000	60.7	39.3
1000-3000	61.0	39.0
>3000	42.4	57.6

Table 12.20: Source of money for treatment

Source of money	Chronic Diseases
Disease Type	
Own income	42.5
Savings of HH	31.8
Insurance	10.2
Loan	1.4
Others help	3.2
Free	10.9

had no insurance coverage or private facilities were chosen for treatment.

With regard to the source of money for treatment of Any other diseases, 42.5 percent of the total amount incurred on chronic diseases treatment was with the patient's own income. Nearly one-third used the savings of the households. Insurance helped 10 percent of the patients and among slightly in over one in ten cases, the treatment was free of cost because the PHC/CHC supplied medicines through the NCD clinics at PHC/CHC level.

Treatment- seeking behaviour of Any other diseases

Although not similar to chronic diseases, certain other diseases require continuous treatment because of repeated occurrence. For instance, people always report cholesterol as an illness although

Table 12.21: ‘Any other disease’ by Age and Sex

Age of the Patient	Percent	Sex	Percent
<30	27.0	Male	38.4
30-39	7.9	Female	61.6
40-49	15.1		
50-59	23.0		
60+	27.0		

it is not considered as a chronic disease. They are often on medication and require repeated monitoring through blood tests as abnormal cholesterol levels are forerunners of many diseases. Table 12.21 provides information on prevalence of other diseases by age and sex in the selected regions.

Here we find that about 27 percent cases have occurred at ages below 30 years. Older cohorts have slightly more share as 23 percent of the total number of cases are reported in the age group of 50-59 years and nearly 27 percent of the cases have been reported by those aged above 60 years. Similar to the sex differentials in chronic diseases, the burden of other diseases’ are also shared more by females. Nearly 61.6 percent of the patients are females. When prevalence of ‘Any other diseases’ are assessed by the background characteristics of the population like religion, social group and the financial background as revealed by the APL/BPL category, significant differentials are absent.

Table 12.22: Percentage distribution of population seeking treatment of other diseases

Type of disease	Percentage to total Cases		Not seeking treatment	
	Total	Coastal	Non Coastal	Total
Fever	85.7	83.3	88.9	14.3
Cough & cold	81.8	75.0	85.7	17.2
Cholesterol	100.0	100.0	100.0	0
Neurological problems	100.0	100.0	100.0	0

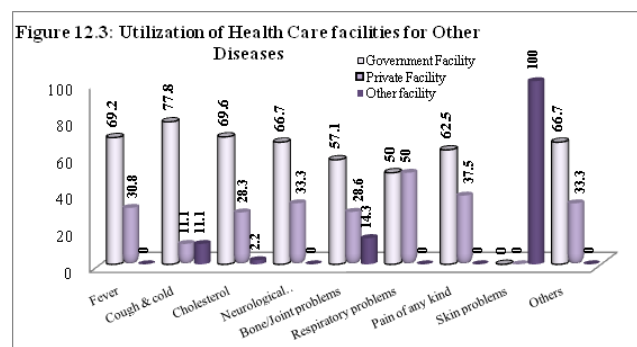
Bone/Joint problems	87.5	75.0	100.0	11.5
Respiratory problems	100.0	100.0	100.0	0
Pain of any kind	100.0	100.0	100.0	0
Skin problems	100.0	100.0	100.0	0
Others	100.0	100.0	100.0	0
Total	93.4	93.1	95.2	6.6

Here we find that for most of the self reported diseases, like neurological problems, respiratory problems etc. treatment has been availed by all patients irrespective of region. Only for fever, cough and cold and cholesterol, some patients did not seek treatment. Overall 93.4 percent of the ‘any other disease’ cases have been treated.

Utilization of Health care facilities

Government facilities form a good share of the treatment for ‘any other diseases’. Over two-thirds of the cases of ‘any other diseases’ were treated in Government facilities whereas only about one-third availed treatment from Private facilities. So Government facility is the place of treatment for 68 percent of the patients, private facility accounts for 28 percent and other facilities, including home remedies and traditional ways of treatment formed 4.2 percent.

Treatment options vary by type of disease (Table 12.23). Seventy percent of the fever cases and also cholesterol, 78 percent of cough and cold cases, 68 percent of fever cases, 68 percent of cough and cold cases, 68 percent of cholesterol cases, 68 percent of neurological cases, 68 percent of bone/joint problems, 68 percent of respiratory problems, 68 percent of pain of any kind, 68 percent of skin problems, 68 percent of others.



percent of the neurological problems are treated in Government facilities. Fifty seven percent of the Bone/Joint diseases were treated in Government facilities whereas 14 percent resorted to Ayurvedic treatment.

If the type of facility is examined, we find that or treatment of Any other diseases, among the Government health facilities, people rely more on the PHC and CHC in the coastal areas whereas in the non-coastal areas we find more reliance on the District Hospital and Medical College Hospital (Table 12.23).

Table 12.23: Place of treatment of 'Any other disease' by area

Place of Treatment	Any other Diseases	
	Coastal	Non Coastal
Not visited	6.7	6.3
Sub centre	1.7	0
PHC	26.7	3.1
CHC	18.3	4.7
SDH	0	6.3
DH	5.0	34.4
MCH	6.7	10.9
ESI Hosp	0	4.7
CGHS	0	3.1
Private Doctor/Clinic	6.7	6.3
Private hospital	21.7	12.5
Private Multi specialty hosp	1.7	1.6
Private MCH	1.7	0
Traditional	1.7	0
Other	1.7	6.3

When one in four cases of 'Any other disease' is treated in PHC in the coastal areas, only 3 percent cases are treated in the PHC in the non-coastal areas. Choice of higher facilities for the treatment can be seen in the treatment-seeking behavior in the non-coastal areas only. More than one-third of the cases were treated in DH and over one in ten cases are treated in Government Medical College. Among the private sector health facilities, almost 22 percent of the cases are treated in the

Table 12.24: Reason for not choosing Government facility fortreatment of 'Any other disease'

Reason for not choosing Govt. facility	Any other diseases
Disease Type	
Doctor not always available	5.6
Poor facility	27.8
Lack of faith	36.1
Long waiting time	27.8
Lack of transportation facility	0
No privacy	0
Others	2.8

private hospitals in the coastal regions and in the non-coastal region also people rely more in the private hospitals although the proportion is only half of that in coastal areas.

The main reason for not choosing a Government facility for treatment of chronic disease is 'lack of faith' (36.1 percent) followed by the reason 'long waiting time' and 'poor facility' (27.8 percent each). About 6 percent opined that doctor was not available.

Treatment Cost

Monthly expenditure on treatment of Any other diseases for 47 percent was less than Rs. 500. Free treatment has been reported by 13 percent of those with Any other diseases. More than Rs. 2000 per month was spent on treatment in 12 percent of the cases. Twenty- seven percent incurred an amount of Rs 500-300 monthly on treatment of Any other diseases.

Expenditure varies by type of disease. Treatment of Neurological problems, bone/joint diseases, skin problems and 'Any other diseases' incurred more than Rs. 3000 for more than one-third of the cases. Only for treatment of fever, cough and cold, respiratory problems, and pain-related issues incurred lesser spending on treatment.

Table 12.25: Total cost of Treatment of Any other diseases

Expenditure (in Rs)	Percent
0	13.1
< 100	14.0
100-499	33.6
500-999	13.1
1000-1999	14.0
≥ 2000	12.1

Table 12.27: Treatment cost of 'Any other disease' (in Rs)

Cost of registration	Percent	Cost of tests	Percent
0	71.1	0	41.3
< 100	16.7	< 100	33.7
≥ 100	12.2	100-499	14.1
		500-999	3.3
		≥ 1000	7.7
Cost of medicine	Percent	Other costs	Percent
0	25.6	0	58.4
< 100	14.4	< 100	22.5
100-499	37.8	100-499	14.6
500-999	12.2	≥ 500	4.4
> 1000	10.0		

Table 12.26: Expenditure for treatment of 'Any other disease'

Type of disease	Amount in Rupees						Total
	0	≤ 100	101-500	501-1000	1001-3000	> 3000	
Fever	8.0	16.0	48.0	4.0	12.0	12.0	100.0
Cough & cold	0.0	25.0	25.0	12.5	37.5	0.0	100.0
Cholesterol	23.8	11.9	40.5	14.3	7.1	2.4	100.0
Neurological problems	0.0	0.0	0.0	33.3	33.3	33.3	100.0
Bone/Joint problems	0.0	0.0	14.3	28.6	28.6	28.6	100.0
Respiratory problems	0.0	0.0	50.0	0.0	50.0	0.0	100.0
Pain of any kind	12.5	25.0	12.5	12.5	25.0	12.5	100.0
Skin problems	0.0	0.0	66.7	0.0	0.0	33.3	100.0
Others	11.1	22.2	0.0	22.2	0.0	44.4	100.0
Total	13.1	14.0	33.6	13.1	14.0	12.1	100.0

Medicines were free for one-fourth of the chronic disease cases while 52 percent had spent Rs 100 to 500 per month on medicines. About 10 percent of the cases required medicines for Rs 1000 per month or more. About 15 percent had to spend about Rs 100 to 500 for meeting other expenses.

Correlates of treatment-seeking behavior

Logistic regression analysis was used to find out

the variables that influenced the treatment-seeking behavior of the respondents for chronic diseases. As more than 98 percent of the respondents received treatment for chronic diseases, the study attempts to find out the factors which influenced the selection of health facilities, Private or Government, for the treatment of chronic diseases. Backward Logistic Regression method was employed for the analysis. The Dependent variable is dichotomous, coded as '0' for choosing a Gov-

ernment health facility and '1' for Private health facility. Independent variables considered for the logistic regression analysis were religion of the respondent, caste, type of house, area of residence (coastal or non-coastal), education of the head of the household and enrolled in Health Insurance scheme.

In the case of treatment-seeking behavior of chronic diseases, the final model was obtained in the fourth step of the analysis and the significant predictor variables in the model are Health insurance of the respondent, type of house, religion and education of the household head (Table 12.28). Compared to the respondents in the pucca households, respondents living in semi-pucca and kuttcha households have 58 percent lesser chance to go to private hospitals for treatment of chronic diseases.

Hindus and Muslims have 33 percent lesser chance to choose private hospitals for treatment of chronic diseases compared to Christian respondents. The odds of choosing private hospitals for treatment of chronic diseases among respondents in households with household heads educated upto secondary level and above is 1.7 times higher than those respondents in households with the household head with education up to primary level. Similarly, respondents with health insurance coverage are at 30 percent lesser odds of choosing private hospitals for treatment compared to those respondents having health insurance coverage.

Table 12.28: Binary Logistic Regression coefficients showing the odds ratios of choosing Government or Private Health Facilities for treatment of chronic and 'any other disease'

<i>Treatment of Chronic Diseases</i>			
Variables in the Model	Categories	β	Exp(β)
Type of House	Pucca®		
	Semi pucca & Kuttcha	-.863***	0.422
Religion	Christian®		
	Hindu & Muslim	-.398**	0.671

Education of head of the HH	Up to Primary®		
	Secondary and above	0.581***	1.788
Health Insurance Coverage	No ®		
	Yes	-.364*	0.695
Constant		0.190	1.209
<i>Treatment of 'Any other disease'</i>			
Type of House	Pucca ®		
	Semi pucca & Kucha	-2.959***	.052
Health Insurance Coverage	No ®		
	Yes	-1.341**	0.261
Constant		3.368***	29.007

In the case of treatment of 'any other disease', the final model was obtained in the sixth step of the analysis. The only two significant predictor variables in the model that explained the choice of private health facilities for treatment of 'any other disease' compared to Government health facilities are enrolment in Health insurance scheme and type of house. Compared to the respondents in the pucca households, respondents living in semi-pucca and kuttcha households are at 95 percent lesser odds of choosing private hospitals for treatment of 'any other disease'. Similar to what was observed in the case of chronic diseases, respondents without health insurance coverage have 74 percent lesser chance to choose private hospitals for treatment of 'any other disease'.

Conclusion

Kerala has taken long strides in the improvement of health care delivery system particularly under the public sector. The State has successfully implemented the Government-level health care programmes that aim at providing universal access to health care. Programmes under NHM like the JSSK, RBSK etc. has been successful in extending free medical care to mother and children.

Health insurance schemes such as the RSBY, Karunya Benevolent Fund, CHIS PLUS catalyzed the health sector reforms as it addressed the heavy burden of non-communicable diseases like heart disease, cancer, tumour etc. The increased health spending and improvement in public health infrastructure, KASH Standards of hospitals under NHM, compulsory rural health services paralleled these initiatives.

So the study finding which portray the treatment-seeking behaviour in favour of the Government sector is a much welcome finding. The higher prevalence of NCDs is, however, a matter of great concern. Hypertension, diabetes and heart disease are found more in the non-coastal areas but cancer/tumour, bone/joint diseases and thyroid is reported more in coastal areas although the difference is marginal. Thyroid, on the other hand, is almost double in coastal areas than non-coastal areas. If Cholesterol is considered as an Other condition of the body, it is the most reported 'any other disease' as almost two-fifths is cholesterol. The absence of any significant differentials by economic category is noteworthy as it implies that the rural poor also bear the great burden of NCDs. So focus on financial stability especially among the rural poor is a much-needed aspect and the programmes need to be monitored so that the benefits of the various schemes reach the poor. The UHC initiative is now a better step to equip the PHCs and CHCs in providing preventive care. The study provides ample evidence to the fact that for the treatment of diseases other than chronic, among the Government health facilities, people rely more on the PHC and CHC in the coastal areas and in the non-coastal areas reliance is more on the District Hospital and Medical College Hospital. Choice of higher facilities for the treatment could also be seen in the treatment-seeking behavior of disease other than chronic cases in the non-coastal areas only.

Further, the evidence provided to the age-sex differential in prevalence of chronic diseases favouring the older cohorts and females argue for

greater investments in health care of the older age groups and women. The supportive role of health insurance schemes in meeting the higher expenditure in treatment of chronic diseases is evident from the lower amount spent on treatment with certain exceptions is a noteworthy aspect. Out-of-pocket expenditure to a certain extent in treatment of chronic and other diseases particularly in the Government sector requires attention. OOPS is evident more in the expenditure on medicines and to some extent on diagnostic tests. Health insurance coverage emerges as a significant variable in the choice of Government health facility both for chronic and other diseases. Majority of the population are covered under RSBY/CHIS. Also, Kerala is the first state to implement the Government's initiative to supply free generic medicine to patients through government hospitals. So the different measure adopted by the Government to reduce the catastrophic expenditure on treatment of diseases has helped the people and the observed pattern of treatment-seeking behavior with more people seeking treatment at the Government health facilities is a much-awaited outcome. Continuous efforts to strengthen the various programmes need to be stressed.

References

1. Atashbahar et. al. 2013, An Examination of Treatment Seeking Behavior Affecting Factors: A Qualitative Study in Iran, *World Applied Sciences Journal* 25 (5): 774-781.
2. Hausmann-Mueala S, Muela Ribera J, Nyamongo I: Health-seeking behaviour and the health system response. In Disease Control Priorities Project (DCPP). Working Paper no.14; 2003. <http://www.dcp2.org/file/29/wp14.pdf>.
3. Hjelm K, Bard K, Nyberg P, Apelqvist J: Religious and cultural distance in beliefs about health and illness in diabetic women of different origin living in Sweden. *Inter national Journal of Nursing Studies*. 2003, 40: 627-643.

4. Hjelm, Katarina and Fortunate Atwine, Healthcare seeking behaviour among persons with diabetes in Uganda: an interview study, BMC International Health and Human Rights, BMC series – open, inclusive and trusted, 2011
5. Kalavathy, M.C, K.R Thankappan, P. Sankara Sarma and R. S Vasan, 2000, Prevalence, awareness, treatment and control of hypertension in an elderly community-based sample in Kerala, India, The National Medical Journal of India Vol. 13, No. 1, 2009.
6. Kumar KV, Sivan YS, Reghu JR, Das R, Kutty YR. Health of the elderly in a community in transition: A survey in Thiruvananthapuram city, Kerala, India. Health Policy Plan 1994;9:331-6
7. Kumar S Vijaya. Growing old in rural India, problems and perspective. In: S. VijayaKumar, editors. Challenges before the elderly: an Indian scenario. New Delhi; MD
8. Navaneetham K, Kabir M. 2006. Health status of Kerala: a life course perspective. Project report submitted to IDPAD, . Thiruvananthapuram, India: Centre for Development Studies.
9. Ramachandran A. Epidemiology of Diabetes in India: Three decades of research. J Assoc Physicians India. 2005;53:34–8
10. Reddy KS, Shah B, Varghese C, Ramadoss A. Responding to the threat of chronic diseases in India. Lancet 2005; 366: 1744-9.
11. Remli, R and S. C. Chan, “Use of complementary medicine among diabetic patient in a public primary care clinic in Ipoh,” Medical Journal of Malaysia, vol. 58, no. 5, pp. 688–693, 2003.
12. Shaw JE, Sicree RA, Zimmet PZ: Global estimates of the prevalence of diabetes for 2010 and 2030. Diabetes Res Clin Pract. 2009, 87: 4-14.
13. Sudharsanam MB RS. Factors determining treatment seeking behaviour for sick children in a fishermen community in Pondicherry. Indian J Community Med. 2007; 32:71-2.
14. Wade DT, Halligan PW: Do biomedical models of illness make for good healthcare systems? BMJ 2004; 329:1398- 401.
15. WHO (2014): Noncommunicable Diseases Country Profile 2014, World Health Organization, <http://www.who.int/nmh/publications/ncdprofiles-2014/en/>, viewed on 7 July 2015
16. World Health Organization: Rapid assessment of health seeking behaviour in relation to sexual transmitted disease: draft protocol. 1995. <http://www.who.int/hiv/topics/en/HealthcareSeeking.pdf>.
17. Young J T,: Illness behaviour: a selective review and synthesis. Sociol Health Illn 2004; 26:1-31.
18. Young JT, Menken J, Williams J, Khan N, Kuhn R. Who receives healthcare? Age and sex differentials in adult use of healthcare services in rural Bangladesh. World health Popul. 2006;8(2):83–100
19. <http://www.idf.org/latest-diabetes-figures-paint-grim-global-picture>

Acknowledgement

We sincerely thank the Ministry of Health and Family Welfare, Govt. of India for providing us the opportunity to present our paper at the Workshop. Our sincere gratitude goes to the Experts for providing critical comments and constructive suggestions which helped us improve immensely.

CHAPTER - 13

Assessment of Home-Based Newborn Care in Uttar Pradesh

Dr. Rajesh K Chauhan, Joint Director, Dr. Pradeep Mishra, Additional Director

Population Research Centre, University of Lucknow

ABSTRACT

Calibrated efforts of Government of India and State Governments reflected in the form of undertaking comprehensive programme of child survival in early ages. The Home Based Newborn Care programme has been initiated primarily to deal with the danger signs in newborns by way of their adequate and timely identification and referral. ASHAs have been identified as key persons in undertaking series of home visits, and understanding, educating and identifying around the issues and challenges faced by newborns and mothers. ASHAs were trained to undertake the home visits and are provided with incentives for the same. To assess the status of its implementation a field-based study was undertaken in one CHC area of Lakhimpur Kheri district where the programme was well implemented. The study sought responses from 271 recently-delivered mothers and all 21 available ASHAs in the villages falling under five randomly selected Sub Centre area. Semi structured questionnaires were used to survey the ASHAs while fully structured schedules were canvassed to recently-delivered women. Under the present study it came to light that message retention on crucial NBC practices among ASHAs is quite low, and at times, they tend to follow shortcuts without adhering to the protocol. Quality vigil and continuous monitoring and supportive supervision of ASHAs by ASHA coordinators and ANMs will prove to be significant for the programme. Training on Module-7 is to be provided to all ASHAs along with regular refresher trainings as contents more specifically deal with the danger signs.

INTRODUCTION

Despite rise in institutional care of deliveries there has not been much improvement in child survival. Infant mortality remains high especially the neonatal mortality. Majority of the neonatal deaths in our country occur at home. Likewise, Uttar Pradesh also has very high infant as well as neonatal mortality rates. Uttar Pradesh accounts for nearly over a quarter of all neonatal deaths in the country. Government of India (GoI) therefore released Home Based Newborn Care (HBNC) guidelines to increase access to newborn care and counselling through ASHAs with the goal of reducing neonatal mortality (NRHM, 2011 and 2014). Govt of India also included relevant aspects of child health in ASHA Module-7 for the training of ASHAs (NRHM, 2010). The guidelines to implement HBNC in Uttar Pradesh were issued in 2012 (Department of Health and Family Welfare, Govt of Uttar Pradesh, 2012).

Box 1: Who is the provider of HBNC

It is important for all peripheral providers of services to be aware of the principles and practices of Home Based Newborn Care. This includes the AWW, the ANM and the Medical Officers. However, as envisaged in XI Plan the main vehicle to provide this is the ASHA. The reasons for this include-

1. She is resident and available in the village
2. She is being equipped with the skills and support to provide such care
3. The findings of a recent evaluation show that ASHA is much more likely to visit the newborn and post-partum mother at home than the ANM or AWW, and is also more likely to be consulted for care of the sick child. The ASHA is emerging as the first port of call for sick newborn and children.
4. The ASHA is supported and guided by the health system which is directly responsible for newborn and child survival. This relationship with the health system is essential for facilitating referral.
(insert from HBNC Guideline)

Under this, ASHAs are expected to make a series of home visits to promote essential newborn care practices, identify signs of illness in young infants and refer infants if needed. In selected districts of Uttar Pradesh, ASHAs had also received 10 days training in HBNC to prepare them for these responsibilities. ASHAs in these districts also receive a performance payment for conducting the recommended home visits. This study aims to understand the status of HBNC implementation through a field-based study of ASHAs and their performance in the field.

REVIEW OF LITERATURE

The review has been clubbed into groups as HBNC field study of Gadchiroli, Maharashtra was the basis of the earlier IMNCI and HBNC programme in India, followed by earlier IMNCI studies, studies comparing skills of ASHAs through trained investigators/doctors, district- based studies and studies based on training modules. Almost all of the reviewed study desired that ASHAs training and its retention should be the prime focus of the programme if ASHAs were to be meaningfully engaged in this programme. The details of reviews are provided as under.

The home or community-based maternal and child health models have been experimented across the globe and deemed successful in low resource settings (Dynes et. al. 2011). In India, a field trial was undertaken in Gadchiroli, Maharashtra for seven years (1996-2003) and it has displayed that effect of home-based neonatal care on management of sepsis and neonatal mortality have been encouraging (Bang et. al. 1999). The study has reiterated its feasibility and effectiveness in Diagnosis and Treatment of Neonatal Sepsis (Bang et. al. 2005a). This study formed the basis for the community-based management in early childhood ages as evidence of mortality decline clearly emerged (Bang et. al. 2005b).

A study of 155 health workers in Purulia district, West Bengal, on implementing the Integrated Management of Neonatal and Childhood Illness (IMNCI) suggested that IMNCI programme offered a scope for capacity-building and infrastructure strengthening of the health system (Biswas et. al 2011).

In a clinic-based study, using the similar chart of symptoms the skills of female health workers were assessed through doctors examining the same infants and it came to light that trained health workers' skills were not satisfactory for assessment of illness severity among young infants, and it was suggested that during training, importance of these signs needs to be emphasized (Bandyopadhyay et. al. 2003). A field-based study using trained investigator and ASHAs in Jhansi district of UP identified that ASHAs misclassified 80 percent of infants on colour codes and ASHAs were not following home-based newborn care formats and skipped critical signs. The study recommended need for improved training, tools, and supportive supervision (Das et. al. 2014) as was done by earlier study. A study of community-based health workers (CHW) and recent mothers identified CHWs' knowledge as one of the crucial aspects for improving the coverage of community-based newborn health care programmes which, in turn, improves the adherence to essential newborn care practices at the household level (Agrawal et. al. 2012).

One study at a CHC of Haryana also revealed that IMNCI-trained workers performed well in all aspects of counselling, except follow-up. The study observed that training without effective implementation plans will not result in long-term skill retention (Shewade et. al. 2013). Similarly, a study of 255 ASHAs in rural Karnataka found that knowledge of key danger signs among ASHAs was associated with repeated, recent and practical training. It suggested of critical need for the implementa-

tion of appropriate training and follow-up supervision of ASHAs within a supportive, functioning and responsive health care system (Kochukuttan et. al. 2013).

One study in rural Haryana demonstrated that 5 days of interrupted training was better placed compared to 8-day training at a stretch (Kumar et. al 2009). However, same was not true when knowledge retention after three years were observed (Venkatachalam et. al 2011). One review study on neonatal health interventions was conducted to understand the status of studies and it identified that family-community care interventions can have a substantial effect on neonatal and perinatal mortality (Schiffman et. al. 2010). A study of 150 ASHAs was conducted at Palghar Taluka in the Thane district of Maharashtra to understand their knowledge about child care. The study found that among the trained ASHAs lacunae in their knowledge regarding various aspects of child health morbidity were observed. It suggested that monthly meetings can be used as a platform for the reinforcement along with periodical refresher training conducted for ASHAs. (Shrivastava & Shrivastava, 2012).

In a study of 33 ASHAs posted at PHC Dayalpur, Faridabad, Haryana were trained for one day which was followed by two refresher trainings. The mean knowledge scores were found to be increasing with stages/time i.e. at pre-training, immediately after training, and after three and six months, respectively. Study concluded that ASHAs could be involved in providing care for newborn. The study warned to ensure a focus on skill development and practical experience for such efforts in community.

OBJECTIVES

The major objective of this study is to understand the preparedness of ASHAs for conducting home

visits and present status of these home visits in selected place. The specific objectives are-

- To examine the training status of ASHAs on HBNC.
- To verify the performance of ASHAs in the field.
- To understand the challenges in implementing this scheme.

METHODOLOGY

This study focuses on the status of implementation and ASHAs preparedness for conducting successful home visits in one of the selected district of Uttar Pradesh where HBNC scheme is well implemented. After discussions with the state programme management unit (SPMU), National Health Mission (NHM), Uttar Pradesh, Palia Kallan Development Block of Lakhimpur Kheri district was selected as the site for the study. Within this block (CHC area), five sub centres were randomly selected. All the villages in these sub centre areas were visited to seek responses of ASHAs and mothers of infants.

This study is aimed at understanding the preparedness of ASHAs and to some extent the impacts of home visits conducted by them. This study is exploratory in nature and does not aim to estimate many rates and ratios. The study covered 271 recently delivered mothers and all the 21 ASHAs. Structured schedules of inquiry were used for recently delivered women (RDW) and ASHAs. Data entry of the questionnaires was carried out on CS-Pro based custom made data entry applications developed in the Centre.

Background of ASHAs

Age and Education: A total of 21 ASHAs were covered under this study. The age of ASHAs ranged between 24 and 48 years, with median age of 30

years and mean age of 30.8 years. On an average, these ASHAs are fairly young and half of them are educated to the 1 high school level or above, fulfilling the precondition of them being educated at least up to the Class eighth. All of these ASHAs reside in the villages where they are engaged as ASHA. All the interviewed ASHAs were Hindu except one of them which was Muslim.

Social group: Among the 21 surveyed ASHAs, seven belonged to 'Scheduled Tribe' category, eight to 'Scheduled Caste' category and six to 'other backward classes'. Majority of ASHAs have served for more than 6 years as only five of them have served for less than 6 years. Mean duration of serving as ASHAs was six years and eleven months.

Population covered and working schedule of ASHAs: Average population covered by ASHAs ranged between 666 and 2200 with mean population of 1128. Mean number of ASHAs in the villages visited was found to be 2.86. While enquired about how many days ASHAs work per week, two thirds responded saying that they work every day and followed by one-sixth working six days a week. ASHAs are engaged in increased number of programmes and activities. It was enquired that how many hours on an average they serve in their village. Responding to this, three ASHAs each informed that they engage for 10 hours, eight hours and six hours every day followed by five each mentioning five and three hours each and remaining two engaging for 2 hours. The mean hours of ASHAs engagement comes out to be five and half hours. Thus an ASHA on an average is putting 35 hours of work in an average week.

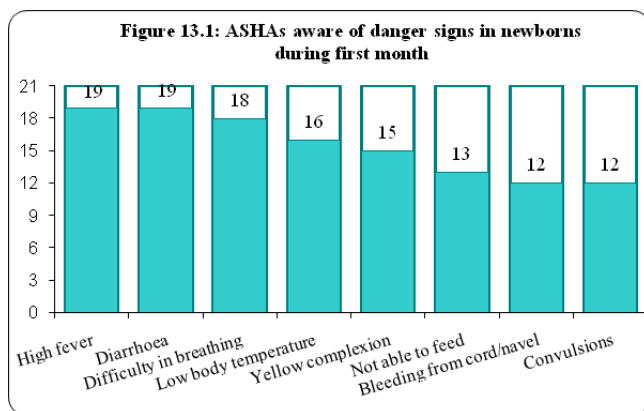
Training on HBNC

HBNC-related training has been received by all 21 surveyed ASHAs and they have received the training on ASHA Module-6. Two of these ASHAs have also received training provided through previous

programme of comprehensive child survival programme (CCSP). Majority of ASHAs were trained at community health centre, Palia. The trainings were organised between January and July 2014. The ANMs positively responded to the training sessions. They were enquired about the themes of training, and looking at the spontaneous responses, highest number of recall was about the 'breast-feeding' as nine ASHAs recalled about it followed by 'cord care' and 'keeping newborn warm', and three ASHAs each could recall spontaneously that they were trained in 'VHIR filling' and 'identification of danger signs and their referral'. Aided recall was attempted to those who failed to recall spontaneously about five fundamental themes. If both the recalls are taken together, all the ASHAs were able to recall 'Breastfeeding', identification of danger signs and their referral' and 'VHIR filling' and 20 of these ASHAs also recalled 'Keeping newborn warm' and 'Cord care'.

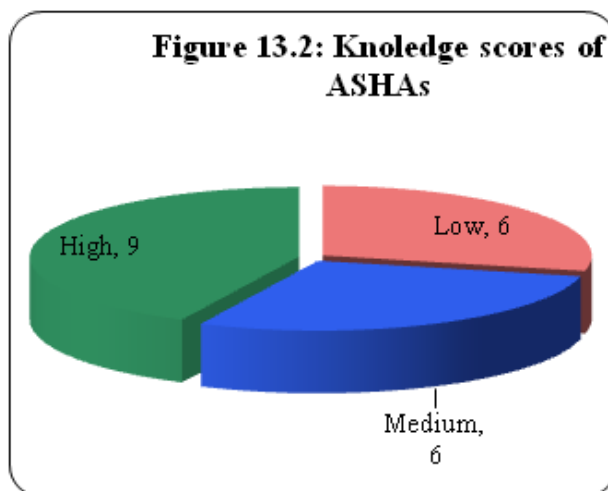
ASHAs knowledge about ANC, PNC and JSY

A section was kept in ASHA schedule to know about their knowledge about antenatal care, post-natal care and JSY.



All the ASHAs were aware of correct number of visits to newborns with normal birth weight, however, five of the ASHAs responded incorrectly to number of visits for the underweight babies. All of the ASHAs were well aware of visit schedule of Day 1, 3, 7, 14, 21, 28, and 42 for home deliveries

and normal birth weight. Likewise, they were also aware of the visit schedule of Day 3, 7, 14, 21, 28, and 42 for institutional deliveries.



About the danger signs newborns have during first month of their lives; 21 of interviewed ASHAs were aware of high fever, diarrhoea, difficulty in breathing and nearly two-thirds were aware of low body temperature, yellow complexion and unable to feed, and majority of them also knew about bleeding from cord/navel and convulsions. Thus eight of the ASHAs were aware of these eight critical danger signs, followed by two each with having knowledge of seven, six and five signs and one third were aware of at least 3 signs. If knowledge of home visits for normal and low birth weight babies, the schedule to undertake home visits and responses about danger signs are accumulated to generate an index of knowledge about newborn danger signs, then 9 ASHAs were found with high knowledge followed by six with medium and remaining six with low knowledge scores.

HOME VISITS TO NEWBORN BY ASHAs

All the 21 interviewed ASHAs reported to have conducted home visits. When asked about number of home visits, one third responded saying that they visit two to four homes followed by nine ASHAs saying five to seven homes and remaining five saying eight or more homes on an average day.

When asked about record keeping, 17 said they maintain register and all except one mentioned to update VHIR.

Records

As per ASHAs record, the mean number of pregnant women registered in the past six months in their area was 14 with a range of 2 to 35. The mean number of women in their third trimester of pregnancy was reported to be five with a range of 0 to 15. It was reported that except one, all these women were visited by ASHAs. In the past six months mean number of institutional and home deliveries per ASHA were reported to be 10.6 and 4.6. Of these, women who have delivered in the health institutions, ASHAs have accompanied most of them (10.5). All the mothers who were present received PNC visits by ASHAs. While asked about the number of visits ASHA pay to the pregnant women during entire duration of pregnancy, nine responded saying that they go to these women nine or more times followed by six saying of six home visits, five mentioning five visits and remaining four saying that they go once only.

When enquired about the specific messages related to ANC they deliver to the women, all of them recalled that they inform women about getting TT injection(s), consuming 100 IFA tablets and getting their blood pressure examined, followed by 20 ASHAs mentioning about getting abdominal examinations, and 17 said of message of completing three ANC visits by a trained health personnel.

When asked about the messages related to birth preparedness, 20 ASHAs mentioned that they suggest women to save money for delivery and arrange for transport to hospital, followed by 19 ASHAs who mentioned of informing about JSY, 18 ASHAs recalling that they suggested to 'identify a place of delivery and/or birth attendant' and 10 ASHAs suggested about identifying a potential blood donor. Thus 10 ASHAs gave all the five messages followed by 7 ASHAs delivering

four messages, two ASHAs giving three messages and one each giving one or two messages on birth planning.

When asked about what messaged you need to give to women who decides to deliver at home two thirds of ASHAs mentioned 'identify a clean place for delivery at home' followed by 13 ASHAs advising to arrange DDK kits; 11 suggesting to identify a skilled birth attendant and 6 each suggesting to 'save money for emergency and arrange transport for emergency.' ASHAs asserted that most likely no one of the women was interested to deliver at home by planning, but by turn of events it so happens. Thus one third of ASHAs could not connect to recall any of the messages, while 4 ASHAs delivered two messages, 2 ASHAS gave 3 messages and eight ASHAs recollected four or all the five messages.

When asked about messages to pregnant about newborn care and suggestions it was mentioned by all the ASHAs that they tell about drying and wrapping newborns immediately after child birth, followed by 20 each mentioning about initiation of breastfeeding within an hour and exclusive breastfeeding up to six months; 19 mentioned about delaying bathing for a week and not applying anything to cord while 18 said that they advised that weight of newborn should be taken and eight mentioned of suggesting to take equal care of girl or boy.

When asked about the number of visits they undertake to the newborns, it was told by 19 ASHAs that for normal birth weight babies they undertake 6/7 home visits while for low birth weight babies only 12 ASHAs mentioned of 11 or more visits. While asked about what do they do during the home visits, almost all mentioned that they 'counsel mother/family members on newborn care' and 'assess the newborn's health by taking temperature, assessing breathing and breastfeeding'. Taking care of mothers and taking additional care of low birth weight babies were also suggested by 17 of the interviewed ASHAs. ASHAs were well versed in

messages about cord care and keeping baby warm. Delay in bathing, initiation of breast feeding and exclusive breast feeding up to six months were recalled to be propagated by 19 of ASHAs. Danger sign identification, taking weight of newborn and identification of danger signs in mothers were mentioned by 15, 14 and 10 ASHAs respectively. In the event of danger signs, 20 ASHAs said that they would refer newborn to a government facility followed by 8 who said they would refer the newborn to ANM while three responded saying that sometimes they would refer the case to a private doctor. Only one ASHA mentioned that she would treat the newborn before referring him/her to health provider.

Job Aids

When inquired about the job aids, only six mentioned of receiving some. Of these, five mentioned of flip book and one each mentioned of calendar, FAQ booklet and poster. Of these six, only three mentioned of taking this material to the homes during home visits and two mentioned of using them during each home visit. These ASHAs mentioned about the benefits of having job aids during home visits, like ease to counselling and complete messaging etc. Those who used job aids did not mention of facing any difficulty in using them. While asked about the difficulties faced during home visits, 12 of these ASHAs mentioned of facing 'no constraints'. Eight of the ASHAs mentioned of non-payment, two about 'scattered work area' and one each of 'not getting monitored' and 'not liked by some families.'

ANM Support to ASHAs

It was asked to ASHAs that in the last three months how many times they met their ANMs, and ASHAs responses account to mean of 14.9 times with a range of 3 to 60 times. The last meeting was just interview day to 23 days with mean of 4 days. Most of them met ANMs on days other than VHND days. Most of the ASHAs told that

ANMs checked their registers and updated names, 15 ASHAs mentioned that ANMs reviewed their home visit record and also demonstrated how to fill register correctly; 13 of the ASHA recollected that ANMs asked about their problems in completing the register while 10 mentioned of making joint home visits, 8 observed of receiving counselling, 6 mentioned about providing feedback and guidance on using job aids and 5 mentioned of making home visits to difficult families.

BACKGROUND AND DETAILS OF RDWs

The mean age of interviewed RDWs was 24.9 years and median age was 25 years with a range of 19 to 37 years. Among the 271 interviewed RDWs, 94 (34.7 percent) were from SC/ST social group followed by 130 (48.0 percent) from other backward classes and 47 (17.3 percent) from 'others'.

Fig 3: Age distribution of Recently Delivered Women

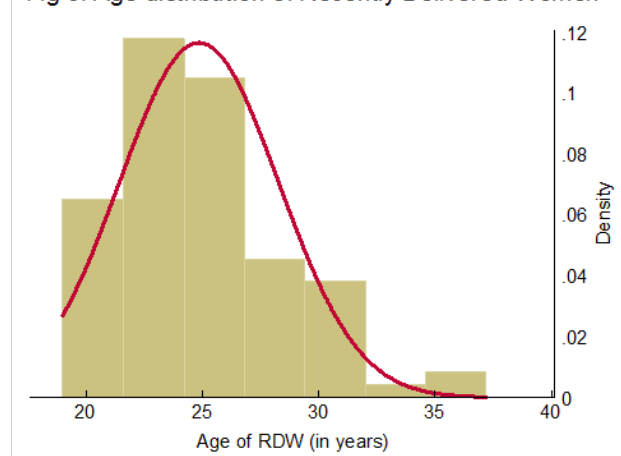
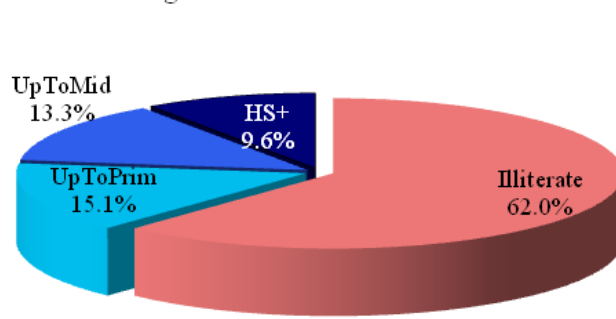
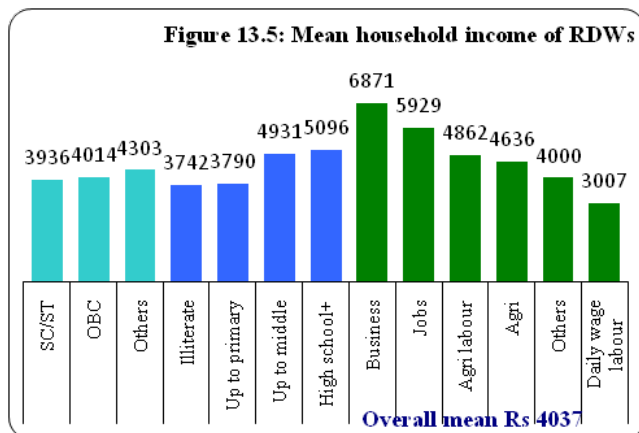


Figure 13.4: Educational Status of RDWs



Looking at the educational status of recently delivered women, it is noticed that 62 percent of these women were illiterate, followed by 15.1 percent who were educated up to primary level, 13.3 percent who were educated more than primary but below middle level (grade VIII) while the remaining 9.6 percent were adequately educated i.e. high school or more. Mean reported income of households of RDW was Rs 4037 with a range of Rs 1000 to Rs 30000. Mean income was the lowest for SC/ST households and highest for others.



Mean income was found positively associated with the level of education of the RDWs. Among the occupational categories highest income was for the households who were drawing income from 'business and the lowest was for the 'wage labour'.

Knowledge of RDWs

Knowledge of RDWs was assessed on eight counts and graphical display of their responses have been provided in figures 6 (a to h) below.

Timing of first bath of newborn: Only 54 percent of RDWs could clearly mention the correct timing of seven days or later followed by 20 percent saying on the sixth day and 15 percent saying on or after third day.

Initiation of breast feeding: Most of the RDWs (92.3 percent) were aware of the correct timing of

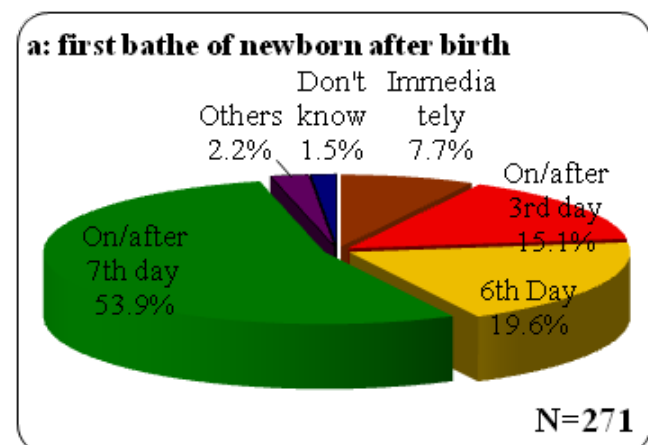
initiation of breast feeding after birth i.e. within an hour of birth.

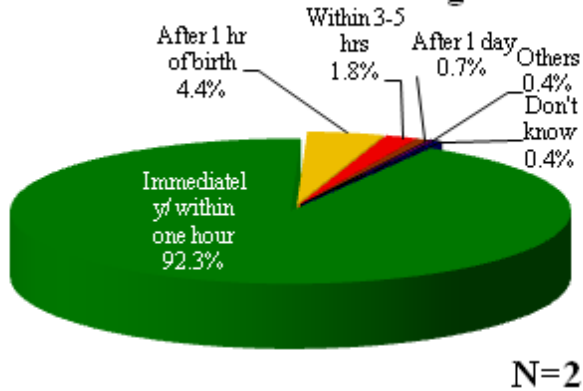
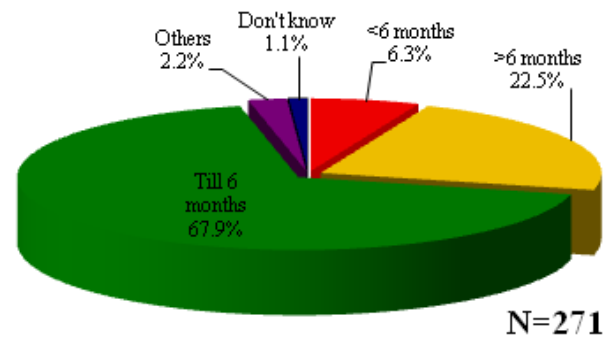
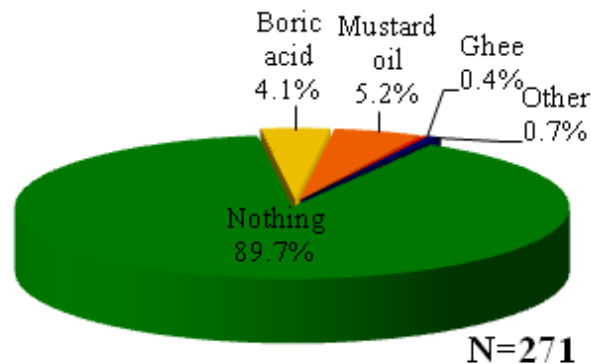
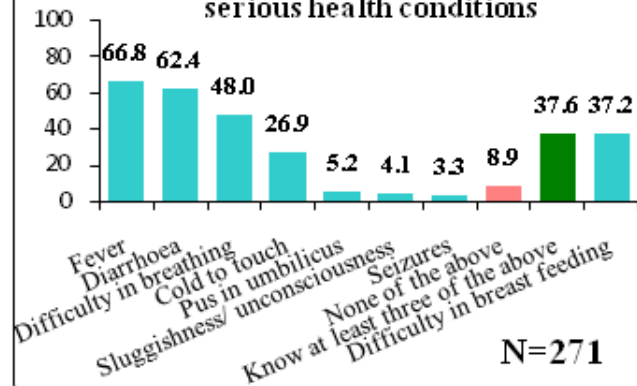
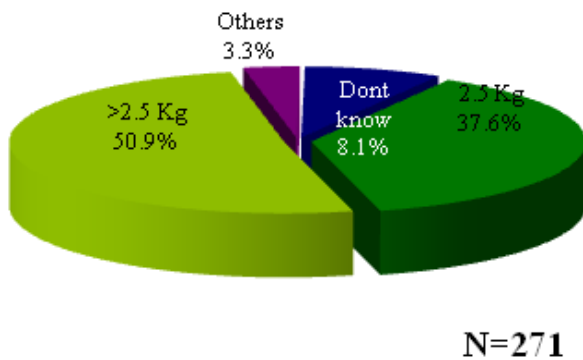
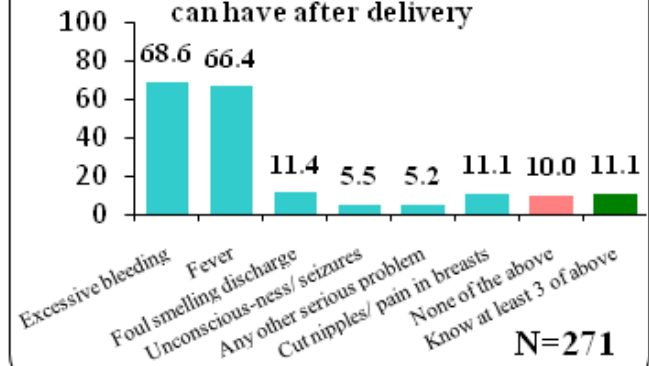
Duration of exclusive breast feeding: A proportion of 68 percent RDWs were aware of the correct duration of exclusive breast feeding (six months) to the newborns followed by 23 percent who mentioned more than six months and 6 percent mentioning it to be less than six months.

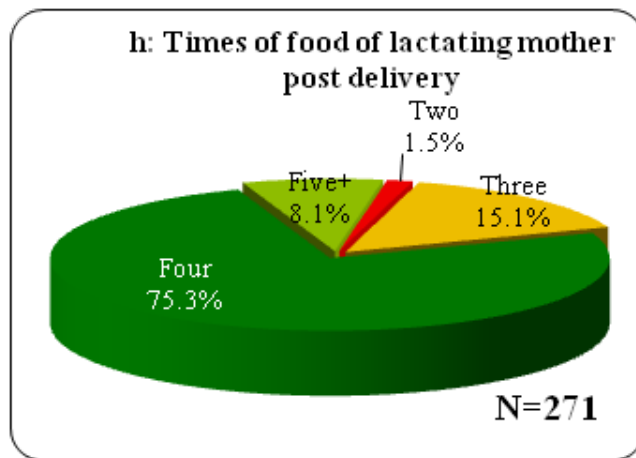
Cord care: Nine out of ten RDWs were aware of the fact that 'nothing' should be applied on cord after the birth of newborn.

Treatment of serious health conditions: Two-third of women mentioned about fever (66.8 percent) followed by diarrhoea (62.4 percent), difficulty in breathing (48 percent), cold to touch (26.9 percent), pus in umbilicus (5.2 percent), sluggishness/ unconsciousness (4.1 percent) and seizures (3.3 percent) as reasons for taking newborns to the doctor or treatment. About 9 percent of women were not aware of any of the above symptoms. Nearly one third of RDWs were aware of at least three of the above (37.6 percent) reasons for taking newborns for treatment. Difficulty in breast feeding as a serious problem was also mentioned by 37 percent women.

Figure 13.6: Mothers' Knowledge on Various Aspects of Newborn Care



b: Initiation of breast feeding**c: Exclusive breast feeding****d: What should be applied on cord after delivery****e: Taking newborn for treatment for serious health conditions****f: Birth weight of newborns****g: Possible health problems mother can have after delivery**



Birth weight of newborns: More than one-third of women (37.6 percent) mentioned '2.5 kilograms' and more than half (50.9 percent) mentioned 'more than 2.5 kilogram' to be birth weight of newborns at the time of birth. Thus 88.5 percent women were aware of the adequate birth weight of the newborns.

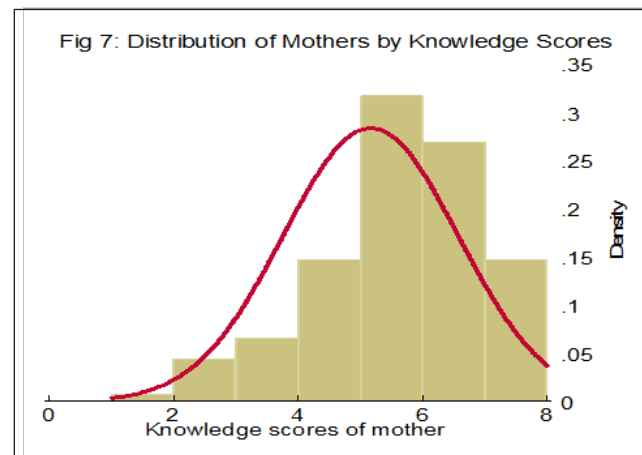
Health issues mother may face after delivery: More than two-third of women were aware of excessive bleeding (68.6 percent) and fever (66.4 percent) followed by a small proportion of 5 to 11 percent of women being aware of foul smelling discharge, unconsciousness/ seizures, cut nipples/ pain in breasts and any other serious problem to the recently delivered women after the child birth. One out of ten women could not mention any of the potential health problems. Only 11 percent women were aware of any three of the specific health issues women may face after delivery.

Times of food to lactating mothers: Three quarters of women were aware that a lactating mother should take 'four meals' a day.

Knowledge scores of RDWs

All the eight items were dichotomised by recoding correct responses to unity and remaining to zero. These eight were summated to generate the knowledge scores of RDWs. Distribution of RDWs as per scores is presented in Fig 7. Mean knowledge score was 5.16 with range of 1 to 8. Most of the

women had a score of five (31.7 percent) followed by six (26.9 percent), four (14.8 percent), three (6.6 percent), two (4.4 percent), eight (4.1 percent) and one (0.7 percent).



An attempt was made to understand the variation in scores by background characteristics of women. The mean scores for various social groups were not much different after controlling for 'number of years of schooling of RDWs, household income and number of children ever born. Mean scores for educational attainments were found to be increasing with the education and these differences were highly statistically significant.

NEWBORN HEALTH CARE PROVIDED BY ASHA

Mothers were inquired about the health check-ups and assessments by the ASHAs on an average visit day to their newborns. It was told by 59 percent mothers spontaneously and 33 percent by probe that ASHAs checked the cord and advised on cord care of newborn. By taking spontaneous and probed responses together 91 percent mothers informed that ASHAs measured weight of newborn, 74 percent confirmed of temperature measurement, 67 percent said ASHAs talked and discussed about difficulties faced while breastfeeding, checked newborn for jaundice and diarrhoea. Assessment of breastfeeding and swelling in eyes were examined by 59 and 53 percent of ASHAs on the visits to newborn. Relatively small proportion of mothers informed that ASHAs as-

sessed newborn for chest in-drawing (40 percent), frequent vomiting (39 percent), pinching skin of stomach (33 percent), boils on skin (29 percent), grunting (28 percent), rashes in skin folds (27 percent), respiratory rate count (24 percent), about sluggishness (23 percent), rash in mouth (21 percent) and flaring of nostrils (6 percent).

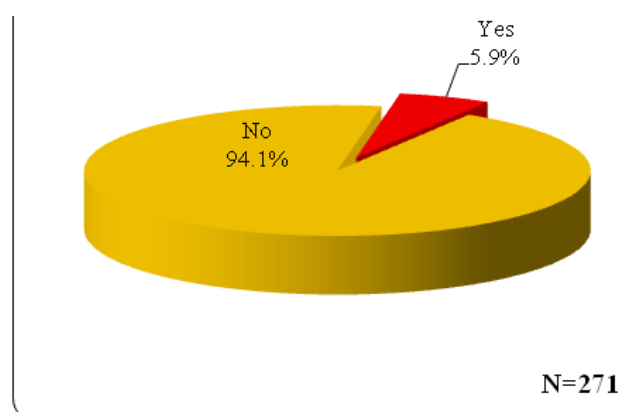
Table 13.1: Distribution of RDWs by newborn health check-ups by ASHA during home visits

Check-up relating to	Yes			No
	Spontaneous	Probe	both	
Flaring of nostrils	0.4	5.2	5.6	94.5
Chest in-drawing	3.0	14.4	17.4	82.7
White patch/ rash in mouth	1.8	18.8	20.6	79.3
Sluggish or unconscious	0.7	22.5	23.2	76.8
Respiratory rate count	10.0	14.4	24.4	75.6
Rash in skin-folds	2.2	24.7	26.9	73.4
Grunting	6.3	21.4	27.7	72.3
Boils on skin	3.7	25.1	28.8	71.2
Pinching skin of stomach	3.3	30.3	33.6	66.4
Distended stomach/ frequent vomiting	8.1	31.4	39.5	60.9
Chest in-drawing	4.8	35.1	39.9	60.1
Swelling in eyes/ pain/ watering	7.4	46.1	53.5	46.5

Assessment of breastfeeding	31.4	28.0	59.4	41.0
Diarrhoea/ dysentery	36.5	29.9	66.4	33.6
Checked for jaundice/ yellowing of eyes	12.5	55.0	67.5	32.8
Difficulty in breastfeeding	37.3	30.6	67.9	32.5
Temperature measurement	44.6	29.5	74.1	26.2
Weight measurement	58.7	23.2	81.9	18.5
Cord care	58.7	33.2	91.9	8.5
No check-up	25.5	7.4	32.9	3.7

A very small proportion of 5.9 percent (only 16 mothers out 271) reported that their babies had health problems during the age birth to one-and-a-half months. Of these 16 mothers, seven reported that child had fever which was followed by five who reported problems in breathing,. While two mentioned that their children were cold to touch, one reported that child was sluggish and another one said that child had diarrhoea. Four mentioned of other problems. Off these 16 mothers, 15 reported that they went for the treatment of children. One child could not be taken to health facility/providers as the family had economic hardship.

Figure 13.8: Babies having health problems within one and half months of birth



Ten of these 15 children were treated at the private hospital and three went to the CHC/PHC/SC and remaining two went to other providers. For three of these children very high costs of treatment such as Rs 80,000, 40,000 and 16,000 were to be spent as they were repeatedly admitted or taken to the doctors. Among them no money was spent for three and for other ten amounts between Rs 10 to Rs 2400 were spent. When inquired if ASHA gave medicines to these ailing newborns, except one all the others said no. Among these 16 ailing newborns only six were referred by ASHAs to doctors/health facilities for seeking medical advice/treatment. One of these newborns was referred to SC, three to PHC/CHC and two to the private doctors by the ASHAs. Of these six mothers, four went to the referred place for the treatment of their newborn. Two women who did not go to the referred place for the treatment did not specify reasons for not doing so. Among the 15 ailing newborns 10 were reported to have improved and five mentioned that there was not much change. Among four women who visited treatment places referred by ASHAs two reported notable improvement in the health of newborns.

Advice provided by ASHAs during home visits

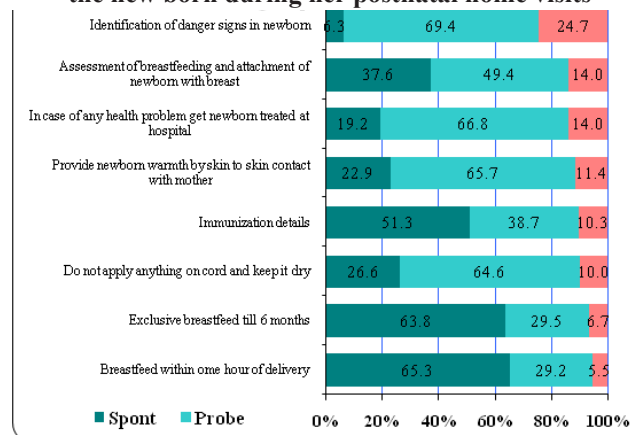
During the home visits ASHAs have been providing messages/ advises regarding the newborns' care in general and health in particular. Most of the mothers either spontaneously or on prompt recollection that ASHAs gave advice regarding the initiation of breastfeeding within one hour of the birth followed by 93 percent recalling of advice on 'exclusive breastfeeding of children up to the age of six months' and ninety percent each about 'cord care' and 'immunization' advices. A very high proportion of women (89 percent) told that they received advice about 'kangaroo mother care', 84 percent each received advice and help on 'taking newborn to health facility in case of health problem' and 'assessment of breastfeeding and attachment of newborn with breast'. Three-fourth of women also confirmed that ASHAs advised and

educated them on 'identification of danger signs in newborns'.

Table-13.2: Percent RDWs by health advices given by ASHA related to the newborn during her postnatal home visits

Health advices	Spontaneous	Probe	Both	No/DK
Breastfeed within one hour of delivery	65.3	29.2	94.5	5.5
Exclusive breastfeed till 6 months	63.8	29.5	93.3	6.7
Do not apply anything on cord and keep it dry	26.6	64.6	91.2	10.0
Immunization details	51.3	38.7	90.0	10.3
Provide newborn warmth by skin to skin contact with mother	22.9	65.7	88.6	11.4
In case of any health problem get newborn treated at hospital	19.2	66.8	86.0	14.0
Assessment of breastfeeding and attachment of newborn with breast	37.6	49.4	87.0	14.0
Identification of danger signs in newborn	6.3	69.4	75.7	24.7

Figure 13.9: Health advices given by ASHA related to the new born during her postnatal home visits



While asked about job aides during ASHAs home visits, it was responded by about three-fourth of mothers that ASHAs did 'mouth conversing' followed by 24 percent mentioning about use of 'chart booklet'. A small proportion of mothers (only 4 percent) mentioned 'demonstrated herself' and 3 percent women said that ASHAs used 'other material' during their home visits.

When asked about the use of paper format, which is to be used for recording the details of all the visits; more than half of the women confirmed that they have not noticed ASHAs using paper formats followed by a quarter saying that they used them but not on each visit. Thus only a quarter of ASHAs are using them in all the visits.

NEWBORN CARE

Newborn care practices

Birth weight of newborn: It is very important to take weight of the newborn immediately after the birth. More than half of the newborns born at home (53.9 percent) and very few (2.6 percent) born at institution were not weighed at the birth making it 17.3 percent of all. Only 6.6 percent newborns were reported to have had weights less than 2.5

kilograms. Nearly one third of mothers mentioned the exact weight of 2.5 kilograms while as many as 43.2 percent mentioned weights which were more than 2.5 kilograms. It does not seem correct that 89 out of 271 newborns would weigh exactly 2.5 kilograms. The cards shown to the teams were showing these weights clearly written as 2.5 kilograms causing some suspicion that it may have mentioned as 2.5 kilograms without weighing the child.

Initiation of breastfeeding: Initiation of breastfeeding soon after the childbirth and up to one hour is very important for making child alert and tendency to engage. Nearly 12.6 percent of newborns were not breastfed within an hour of birth. This proportion was quite high (28.2 percent) for children born at home and reasonably low (6.2 percent) for children born at health institutions. Thus a large proportion (87.5 percent) were initiated with the breastfeeding within the recommended one hour of birth.

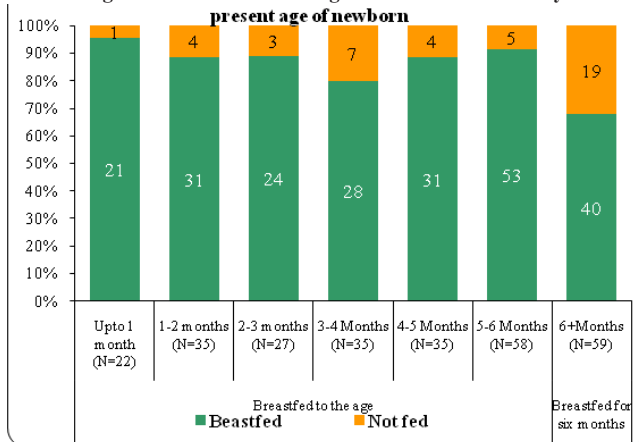
Fed anything before breastfeeding: Only 10 percent of newborns were fed with something before initiation of breastfeeding. Of these 26 children, 15 were given 'other milk' followed by 9 who were given honey and one each were fed with Janam Ghutti and Mustard Oil.

Table13.3: Proportion of RDWs with good NBC practices by ASHA background

Background characteristics of ASHAs	Weighed at birth	Breastfed within an hour	Not given anything before breastfeeding	Given breast milk for six months	Nothing applied on cord after delivery before it fell off	Given first bath after 7 days of birth	All the six good practices	N
Age of ASHA								
<29Yrs	83.7	87.0	90.2	82.9	88.6	52.0	40.7	123
30+Yrs	84.6	88.9	88.0	82.9	91.5	50.4	40.2	117
ASHA not present	71.0	83.9	100.0	93.5	96.8	51.6	32.3	31
Social group of ASHA								
Scheduled caste	68.5	81.5	83.3	74.1	70.4	25.9	16.7	54

Scheduled tribe	96.0	98.0	96.0	92.0	98.0	54.0	52.0	50
Other Backward class	86.0	86.8	89.0	83.1	94.9	60.3	45.6	136
ASHA not present	71.0	83.9	100.0	93.5	96.8	51.6	32.3	31
Work hour in week								
14-21Hrs	71.8	82.4	83.5	78.8	83.5	41.2	30.6	85
22-35Hrs	84.1	88.6	95.5	84.1	97.7	63.6	43.2	44
36+Hrs	93.7	91.9	91.0	85.6	91.9	54.1	46.8	111
ASHA not present	71.0	83.9	100.0	93.5	96.8	51.6	32.3	31
Messages related to ANC, NC, PNC and NBC								
Low	33.3	44.4	55.6	55.6	100.0	66.7	33.3	9
Medium	90.7	93.0	95.3	90.7	95.3	44.2	37.2	43
High	85.1	88.8	89.4	82.4	88.3	52.1	41.5	188
ASHA not present	71.0	83.9	100.0	93.5	96.8	51.6	32.3	31
Identification of danger signs								
Low	81.0	82.8	86.2	82.8	98.3	60.3	48.3	58
Medium	92.3	94.2	98.1	92.3	96.2	44.2	38.5	52
High	82.3	87.7	86.9	79.2	83.8	50.0	37.7	130
ASHA not present	71.0	83.9	100.0	93.5	96.8	51.6	32.3	31
Support of ANMs								
Low	33.3	44.4	55.6	55.6	100.0	66.7	33.3	9
Medium	83.8	89.0	89.0	83.8	83.8	44.1	36.0	136
High	89.5	90.5	92.6	84.2	97.9	60.0	47.4	95
ASHA not present	71.0	83.9	100.0	93.5	96.8	51.6	32.3	31
Total	82.7	87.5	90.4	84.1	90.8	51.3	39.5	271

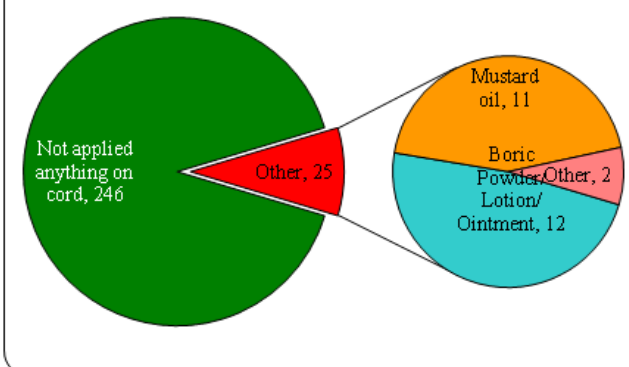
Figure 13.10: Breast feeding status of the children by present age of newborn



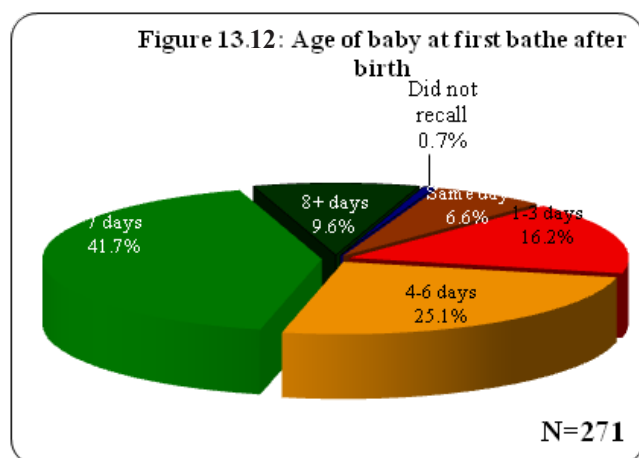
Breastfeeding status of children: It was enquired that whether children were being breastfed up to age six months and, in the case those children who are more than six months, the mothers were asked whether they had breastfed their infants up to the age of six months. Only one child of age less than one month was given something else along with

breast milk followed by four of age '1-2 months', 3 of age '2-3 months', 7 of age '3-4 months', 4 of age '4-5 months', 5 of age '5-6 months' and 19 of age more than six months. Thus 16 percent children were reported to have given something additional along with breast milk.

Figure 13.11: Something applied on the cord after cutting and until it fell



Cord care: Caring for the cord is an important component under newborn care.- Cord has to be kept dry and, to avoid infections, nothing is to be applied. This was enquired from the mothers if they had applied anything on the cord after cutting and until it fell off. Only 25 women responded saying that they had applied something; 12 said they applied boric powder/lotion/ ointment followed by 11 who applied mustard oil and the remaining two applied other things. Rest of the women were aware of applying nothing and did follow that advice of ASHAs.

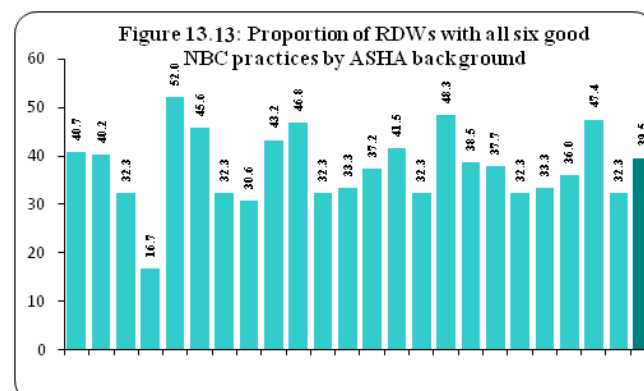


Bathing first time: In order to provide adequate thermal care it is of paramount importance that babies are given first bath on or after the 7th day of the birth. It is notable that nearly half of the newborns were bathed before the age of seven days; seven percent on the same day of birth, 16 percent during first to third day and 25 percent during the age of four to six days. It was also asked to the mothers that if they were aware of bathing on or after the seventh day of birth, most of them responded in assertion but added that elderly women of household or nearby households were keen on bathing of child.

Correct practices

All the six practices discussed in the earlier section are very important for child survival and healthy growth. It is expected that they would vary by the background characteristics of ASHAs and RDWs.

RDWs who had followed all the six good practices were computed and their percentage or various background variables of ASHAs were computed.

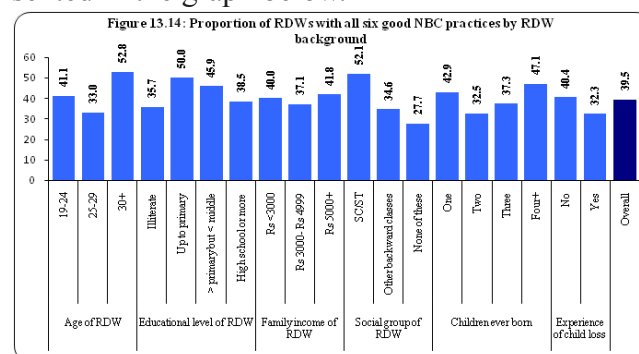


Correct practices by ASHA background

ASHAs are the most important link in the chain of home based newborn care (HBNC). An attempt has been made to understand if ASHAs background explains some of the variation in all good practices of newborn care. Nearly 40 percent of RDWs confirmed of the all six good practices. The women whose ASHAs were ST had highest (52 percent) practices followed by those who worked more than 36 hours a week, who delivered more messages and were able to receive more support from their respective ANMs. ASHAs age and identification of danger signs were not found to be related to good newborn care practises.

Correct practices by RDW background

Similarly, it is equally important to learn about this outcome from the perspective of RDWs background. Proportion of all six good practices for the RDW background has been computed and are presented in the graph below.



The age of RDW had a unique pattern of following good practices, at one hand women with age 30 years or more displayed highest proportion, and women with younger ages (i.e. 19-24 years) displayed reasonably good leaving while women of prime reproductive ages (i.e. 25-29 years) remained at the lowest level.

This could be due to older women having good knowledge due to maturation by age and past reproductive experiences. This also points out that messages provided by ASHAs are not reaching equally to all ages. Similarly, for educational level of RDWs, a strange pattern of all NBC practices have been observed. It was the highest for

Table-13.4: Proportion of RDWs with good NBC practices by RDW background

Background characteristics of RDWs	Weighed at birth	Breastfed within an hour	Not given anything before breastfeeding	Given breast milk for six months	Nothing applied on cord after delivery before it fell off	Given first bath after 7 days of birth	All the six good practices	N
Age of RDW								
19-24	83.7	85.3	91.5	84.5	92.2	51.9	41.1	129
25-29	81.1	91.5	90.6	84.0	85.8	46.2	33.0	106
30+	83.3	83.3	86.1	83.3	100.0	63.9	52.8	36
Educational level of RDW								
Illiterate	81.5	86.3	89.9	83.9	91.1	49.4	35.7	168
Up to primary	80.0	90.0	97.5	90.0	92.5	57.5	50.0	40
More than primary but less than middle	86.5	91.9	86.5	86.5	81.1	48.6	45.9	37
High school or more	88.5	84.6	88.5	73.1	100.0	57.7	38.5	26
Family income of RDW								
Rs <3000	82.1	92.6	93.7	85.3	94.7	54.7	40.0	95
Rs 3000- Rs 4999	77.3	83.5	90.7	85.6	91.8	47.4	37.1	97
Rs 5000+	89.9	86.1	86.1	81.0	84.8	51.9	41.8	79
Social group of RDW								
Scheduled caste/scheduled tribe	83.0	95.7	92.6	89.4	93.6	61.7	52.1	94
Other backward classes	83.8	84.6	90.0	81.5	90.0	47.7	34.6	130
None of these	78.7	78.7	87.2	80.9	87.2	40.4	27.7	47
Children ever born								
One	89.3	86.9	91.7	81.0	94.0	52.4	42.9	84
Two	80.5	84.4	90.9	85.7	87.0	44.2	32.5	77
Three	78.0	94.9	89.8	86.4	86.4	50.8	37.3	59
Four+	80.4	84.3	88.2	84.3	96.1	60.8	47.1	51
Experience of child loss								
Experienced no child death	83.3	88.8	90.8	83.8	90.4	52.5	40.4	240
Experienced at least one child death	77.4	77.4	87.1	87.1	93.5	41.9	32.3	31
Total	82.7	87.5	90.4	84.1	90.8	51.3	39.5	271

women who were educated up to primary level, and declined with the educational level. Does this indicate that ASHAs connect well with less educated women. As one would expect that family income had very little to do in following these good practices as not much variation across these categories was observed. Women from SC/ST group displayed highest proportion in following good practices followed by OBCs and others. If this finding would not have been place-specific, this would have presented itself as a double edged sword. As the programme has been extremely positive in bringing SC/ST women and including them in propagating the good practices through ASHAs. But at the same time, it has not been that affective in the case of other social groups. Varying pattern in case of women by 'children ever born' have been observed, good outcomes were observed for women with four or more children and with one child. This may be parallel to the age.

CONCLUSIONS

This study of one district of Uttar Pradesh shows that HBNC programme has been very well received by the ASHAs and they received good co-operation from the community. In the process, all the ASHAs have been trained on Module-6 but have not received training on Module-7, which reflects lower percentage of ASHAs and women being aware about danger signs in newborns and danger signs in mothers which are crucial to the success of programme. ASHAs are undertaking mundane home visits to mothers and newborns with many areas of possible improvement, namely undertaking home visits for few minutes resulting in skipping the protocols of check-up and examinations of newborn, not using paper formats on the spot and filling them afterwards, taking too many casual type visits etc. ANMs support is a crucial area for strengthening the ASHAs home visits and their overall performance towards HBNC programme. The study clearly brings out the support ANMs leading for improved outcomes for newborns. ASHAs also brought the issues of irregular

payments, this need to be taken care of strictly as incentives cannot be deferred indefinitely or may hang in balance eg. for the FY 2013-14 continued under CCSP, home visit details have been compiled and payments to the ASHAs have not been materialised. Likewise, periodicity of these payments should be reduced to quarterly than it being an annual/ end-of-the-year affair.

RECOMMENDATIONS

The programme is very well placed and contextual to the ground situation of high neonatal and infant mortality and so needs to be accorded high priority. A quality vigil on this programme is recommended by engaging ASHA coordinators and ANMs. Also, continuous monitoring and supportive supervision of ASHAs may provide impetus to a well planned programme to evolve into a good public programme. Training on Module-7 is to be provided to all ASHAs as their contents deal more specifically with the danger signs. The themes to be covered under HBNC require constant practice and continuous learning so refresher's training assumes very high significance. Message retention on crucial NBC practices is quite low, suggesting the need for providing improved inputs to ASHAs through training, encouraging them to adhere to the guidelines during home visits, following the protocols on using paper formats strictly and ensuring constant scrutiny and support by ANMs. Any slowness in disbursing the incentives needs to be addressed and ensure adherence to instructions by the health officials. Release of funds should be done in accordance with the guidelines of the programme leaving no space for delays in payments to ASHAs, at least quarterly. Earlier years' payments, as promised, should be made to maintain the faith and enthusiasm of ASHAs in the programme.

References

Agrawal PK, Agrawal S, Ahmed S, Darmstadt GL, Williams EK, Rosen HE, Kumar V, Kiran U,

- Ahuja RC, Srivastava VK, Santosham M, Black RE and Baqui AH (2012). Effect of knowledge of community health workers on essential newborn health care: a study from rural India. *Health Policy and Planning*; 27:115–126.
- Bandyopadhyay S, Kumar R, Singhi S and Aggarwal AK (2003). Are primary health workers skilled enough to assess the severity of illness among young infants? *Indian Pediatr* ;40: 713–8.
- Bang AT, Bang RA, Baitule SB, Reddy MH and Deshmukh MD (1999). Effect of home-based neonatal care and management of sepsis on neonatal mortality: Field trial in rural India. *Lancet*; 354:1955-61.
- Bang AT, Bang RA, Stoll BJ, Baitule SB, Reddy HM and Deshmukh MD (2005a). Is Home-Based Diagnosis and Treatment of Neonatal Sepsis Feasible and Effective? Seven Years of Intervention in the Gadchiroli Field Trial (1996 to 2003). *Journal of Perinatology*; 25:S62–S71.
- Bang AT, Reddy HM, Deshmukh MD, Baitule BS and Bang RA (2005b). Neonatal and Infant Mortality in the Ten Years (1993 to 2003) of the Gadchiroli Field Trial: Effect of Home-Based Neonatal Care. *Journal of Perinatology*; 25:S92–S107.
- Biswas AB, Mukhopadhyay DK, Mandal NK, Panja TK, Sinha N and Mitra K (2011). Skill of Frontline Workers Implementing Integrated Management of Neonatal and Childhood Illness: Experience from a District of West Bengal, India. *Journal of tropical Pediatrics*, 57 (5).
- Das E, Panwar DS, Fischer EA, Bora G and Carlough, MC (2012). Performance of Accredited Social Health Activists to Provide Home-based Newborn Care: A Situational Analysis (2014). *Indian Pediatr*; 51: 142-144.
- Department of Health and Family Welfare, Government of Uttar Pradesh. Government order for roll out of home based newborn care. Lucknow; 2012.
- Dynes M, Rahman A, Beck D, Moran A, Rahman A, Pervin J, Yunus M, Rashid MH, Gazi T, Biswas KK, Buffington S, Patterson JM and Sibley L (2011). Home-based life saving skills in Matlab, Bangladesh: a process evaluation of a community-based maternal child health programme. *Midwifery*; 27:15–22.
- Kochukuttan S, Ravindran TKS and Krishnan S (2013). Evaluating Birth Preparedness and Pregnancy Complications Readiness Knowledge and Skills of Accredited Social Health Activists in India, *International Journal of MCH and AIDS*; 2 (1): 121-128.
- Kumar D, Aggarwal AK and Kumar R (2009). The effect of interrupted 5-day training on Integrated Management of Neonatal and Childhood Illness on the knowledge and skills of primary health care workers. *Health Policy and Planning*; 24:94–100.
- National Rural Health Mission (NRHM), Ministry of Health and Family Welfare, Government of India (2011). Home based newborn care: Operational guidelines. Nirman Bhavan, New Delhi.
- National Rural Health Mission (NRHM), Ministry of Health and Family Welfare, Government of India (2014). Home based newborn care: Operational guidelines (Revised 2014). Nirman Bhavan, New Delhi.
- National Rural Health Mission (NRHM), National Health System Resource Centre, Govt of India. (2010). ASHA Module 7: Skills that Save Lives-Focus on Child Health and Nutrition. New Delhi.
- Schiffman J, Darmstadt GL, Agarwal S and Baqui AH, (2010). Community-Based Intervention Packages for Improving Perinatal Health in Developing Countries: A Review of the Evidence. *Semin Perinatol*; 34:462-476.
- Shewade HD, Aggarwal AK, Bharti B (2013). Integrated management of neonatal and childhood illness (IMNCI): Skill assessment of health

- and integrated child development scheme (ICDS) workers to classify sick under-five children. *Indian J Pediatr*; 80:448-54.
- Shrivastava SR and Shrivastava PS (2012). Evaluation of trained Accredited Social Health Activist (ASHA) workers regarding their knowledge, attitude and practices about child health. *Rural and Remote Health* 12: 2099. (Online).
- Stalin P, Krishnan A, Rai SK, Agarwal RK (2011). ASHAs' involvement in newborn care: A feasibility study. *Indian Pediatr*; 48:897-9.
- Venkatachalam J, Kumar D, Gupta M, Aggarwal AK (2011). Knowledge and skills of primary health care workers trained on integrated management of neonatal and childhood illness: Follow-up assessment 3 years after the training. *Indian J Public Health*; 55:298–302.

CHAPTER - 14

Knowledge, Attitude and Practice regarding Non-Communicable Diseases Among Medical Officers in Haryana

Sukhbir Singh and Reena Singh

Population Research Centre, Punjab University Chandigarh

Abstract

Non-communicable diseases (NCDs) are defined as diseases of long duration, and are generally slow in progression. Non-communicable diseases (NCD) are a significant health problem in India. The national health policy recommends building the capacity of Medical Officers (MO) in the national health system to address this issue. This study is aimed at assessing the knowledge, attitudes and practices of Medical officers in government health facilities of Haryana state. It was a cross-sectional study which was carried out to assess the knowledge, attitudes and practices of medical officers working in government health facilities. A total of 98 respondents from various health facilities (DH, CHC and PHC) were randomly selected from 7 districts of Haryana. A 5-point Likert scale was used to assess knowledge, attitudes and practice of the respondents. A scoring system (1.00 to 5.00) was devised with low scores representing unfavorable attitudes with 3.00 as the cut off to classify respondents. Nearly one-fourth (24.5percent) of the study population were in the age group of 35 to 39 years followed by 17.3 per cent in the age group of 40 to 44 years. The mean age of the respondents was 38 years. The average work experience among the doctors was between 5 to 6 years. Nearly 23 percent of respondents responded that diabetes is the most common NCD found in India followed by hypertension (19.1 percent), cancer (14 percent), cardiovascular disease (11.7 percent) and stroke (9.9 percent). Nearly 80 percent of the respondents were able to identify four modifiable risk factors for NCDs, while 20 percent of medical officers were able to identify three modifiable risk factors. In relation to non-modifiable risk factor, nearly one-third of the respondents (30 percent) were not aware about the non-modifiable risk factors. Among the study population, 76.5 percent were able to identify the correct formula for BMI calculation. Nearly 42 percent of doctors were aware that the screening services are rendered by team of health workers at the time of large-scale surveys in the country. Knowledge and practices of NCD prevention strategies that are applicable in community settings was good among the majority of MOs while attitudes were mostly favourable. The study recommends improving knowledge and overcoming identified barriers in order to promote MOs to take up NCD prevention in community.

INTRODUCTION

Non - communicable diseases (NCDs) are defined as diseases of long duration, and are generally slow in progression. NCDs were responsible for 38 million (68 percent) of the world's 56 million deaths in 2012. More than 40 percent of them (16 million) were premature deaths under age 70 years. Almost three quarters of all NCD deaths (28 million), and the majority of premature deaths (82 percent), occur in low- and middle-income countries (WHO, 2014). The majority of these deaths (36 million) were attributed to cardiovascular diseases (48 percent), cancers (21 percent), chronic respiratory diseases (12 percent) and diabetes (3 percent) (WHO, 2010).

The global burden of NCDs is increasing and it is a major barrier to development and achievement of Millennium Development Goals (MDGs) (Beaglehole et al, 2011). NCDs are more prevalent in developing nations and India is not exempted from it. According to the recent statistics on NCDs, 53 percent of the deaths were due to NCDs in India. Cardiovascular disease (CVDs) alone account for 24 percent of all deaths. Chronic respiratory diseases (CRDs), cancers and diabetes accounted for 11, 6 and 2 percent of all deaths, respectively (WHO, 2011).

Lifestyle modifications, such as smoking cessation, dietary fat and cholesterol reduction, exercise, and weight control are important components of treatment for reducing mortality rates due to Non-communicable diseases (Al-Omran, 2007; Chobanian et al., 2003).

Secondary as well as primary prevention, including lifestyle modification is an essential part of primary care physician's work in accordance with the current recommendations (NPCDCS, 2009). The primary care physician is in a unique position of influence, which may lead to the adoption of healthy lifestyles and prevention of obesity (Lawlor et al, 1999). Advice from the family physician

(FP) may prompt weight loss attempts (Colvin & Olson, 1983) and may encourage other health-promoting behaviors (Price et al, 1987). FPs' awareness and diagnosis of the obese and overweight is low and may contribute to its under management (Heath et al, 1993). There is also reluctance to treat patients without co-morbidities (Campbell et al, 2000) as there is to treat the overweight as opposed to the obese (Kristeller & Hoerr, 1997). In an era of rapidly growing prevalence of NCDs in India, it is important to explore the current knowledge, attitudes and practices of primary care physicians.

OBJECTIVE

To assess the knowledge, attitude and practice of NCD risk factors among medical officers working at all three levels of health care facilities in Haryana.

METHODOLOGY

This cross sectional study was conducted at health facilities (district hospitals, community health centre, and primary health centre) of the state of Haryana on the sample size of 100 respondents. Medical officers (a qualified medical officer having an MBBS or an equivalent degree and who is not an intern or a specialist) working in the selected health facilities were recruited as the study respondents.

A structured, pre-tested self-administered questionnaire (5 point likert scale) was used to collect data. Whenever possible, participants were requested to fill up the questionnaire in the presence of the investigator to avoid cross-contamination. Data was analyzed using SPSS version 16 statistical software. A marking scheme was formulated and a cut off mark that was based on possessing knowledge about the questions on 'must know' areas of the theme was used to categorize respondents as 'good' and others as 'poor' in knowledge

on health promotion.

A scoring system (1.00 to 5.00) was devised with low scores representing unfavorable attitudes with 3.00 as the cut off to classify respondents. Level of knowledge and overall attitude were cross analyzed with selected factors to assess any association. Specific practices on applying NCD prevention activities were also analyzed.

FINDINGS

Out of the total 100 sample size, 98 participants (Medical Officers) agreed to take part in the study and were included in the analyses. Nearly one-fourth (24.5 percent) of the study population were in the age group of 35 to 39 years followed by 17.3 per cent in the age group of 40 to 44 years. The mean age of the respondents was 38 years. About two-third (66.3 percent) of the medical officers were working in OPD and the remaining were on duty at SNCU or emergency. Doctors having work experience of 1-10 years comprised the largest group (51 percent). The average work experience among the doctors was between 5 to 6 years.

KNOWLEDGE ON NCD AND ITS PREVENTION

To assess the knowledge pertaining to NCDs, a set of eleven statements were used. Nearly sixty percent (57.1 percent) of the respondents strongly disagreed with the statement 'Non-communicable diseases are those which can be easily communicated from one person to other' and 35.7 percent respondents strongly agreed with the statement. Nearly 47 percent (46.9 percent) of the respondents were aware that 'lack of regular exercise, poor eating habits,' and 'tobacco consumption' (45.9 percent) lead to NCDs. Respondents were aware that cancer and diabetes fall under NCDs and cannot be communicated from one person to another person. For 42.9 per cent of the respondents, 'HIV/AIDS is not an NCD and it can be transmitted from one person to another person.'

Nearly half (47 percent) of the respondents agreed with the statement that 'Obesity can cause NCD.' For one-third of the respondents If a man is having BMI more than expected then there are high chances of NCDs.' Nearly two-third of the respondents agreed with the statement that 'Children can be affected if their parents are diabetic'. Half of the respondents agreed with the statement that 'NCDs are not associated with habits. People, nowadays are not bothered to change themselves, for which nothing can be done'. When the medical officers were categorized based on their overall knowledge related to NCD, it was found that 51.5 percent of doctors were having poor knowledge and 48.5 percent having good knowledge about Non-communicable diseases (Table 14.1).

Nearly 23 percent respondents responded that diabetes is the most common NCD found in India followed by hypertension (19.1percent), cancer (14 percent), cardiovascular disease (11.7percent) and stroke (9.9 percent). One-fifth (20.4percent) of the respondents mentioned that diseases like dengue, typhoid, seizure, allergy and defects of vision also fall in the category of non-communicable diseases.

The four NCDs that contribute the most to the global NCD mortality (cardiovascular diseases, cancers, chronic lung diseases and diabetes), share four modifiable risk factors related to lifestyle; poor diet or poor nutritional intake, low levels of physical activity, tobacco and alcohol use. Nearly 80 percent of the respondents were able to identify four modifiable risk factors for NCDs, while 20 percent of medical officers were able to identify three modifiable risk factors. Other modifiable risk factors of NCDs mentioned by the respondents were lifestyle of the individual (22.6 percent) followed by dietary habits (20.2 percent), smoking (16.9 percent), and stress (15.9 percent). Apart from the modifiable risk factors, there are important NCD non-modifiable risk factors, such as genetics, sex and age. Nearly one-third of the respondents (30 percent) were not aware about

non-modifiable risk factors. Among those who were aware about the non-modifiable risk factors, 28.4 percent mentioned heredity/genetic factor, followed by age (14 percent), sex (8.5 percent) and environment.

be consumed in lesser amounts to prevent NCDs. Over three-fourths of the study population were able to identify the correct type of fat in sunflower oil, 62.3 percent respondents were able to correct-

TABLE 14.1: KNOWLEDGE ABOUT NON-COMMUNICABLE DISEASES

Statements	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
Non-Communicable diseases are those which can be easily communicated from one person to other.	56 (57.1)	6(6.1)	1 (1.0)	-	35 (35.7)
Lack of regular exercise and poor eating habits may cause NCDs	26(26.5)	9(9.2)	1(1.0)	16(16.3)	46(46.9)
Tobacco consumption may lead to NCDs	25(25.6)	10(10.2)	1(1.0)	17(17.3)	45(45.9)
Cancer is not an NCD, cancer can be communicated from one person to another	50(51.0)	9(9.2)	5(5.1)	6(6.1)	28(28.6)
Diabetes falls under NCD category	27(27.6)	5(5.1)	1(1.0)	14(14.3)	51(52.0)
Stress can leads to NCD	24(24.5)	10(10.2)	1(1.0)	23(23.5)	40(40.8)
AIDS is a non-communicable disease	42(42.9)	6(6.1)	1(1.0)	8(8.2)	41(41.8)
Obesity can cause NCDs	27(27.6)	7(7.1)	1(1.0)	17(17.3)	46(46.9)
If a man is having BMI more than expected then there are high changes of NCDs.	22(22.4)	11(11.2)	4(4.1)	24(24.5)	37(37.7)
To what extent you agree with statement that “the risk of heart disease is due to lack of physical activity”?	15(15.3)	16(16.3)	3(3.1)	36(36.7)	28(28.6)
Children can be affected if their parents are diabetic	10(10.2)	22(22.4)	3(3.1)	36(36.7)	27(27.5)
Diabetes is more among rich people	14(14.3)	25(25.5)	20(20.4)	30(30.6)	9(9.2)
Obesity is found more amongst adolescent for which parents are responsible	8(8.2)	15(15.3)	21(21.4)	38(38.8)	16(16.3)
NCDs are associated with habits. People, now a days are not bothered to change themselves, for which nothing can be done.	17(17.3)	29(29.6)	17(17.3)	22(22.4)	13(13.3)

A large majority (90.8 percent) of the doctors were aware of the correct age of NCD screening. According to respondents, the mean age for NCDs screening is 30 years. Nearly 42 percent of doctors were aware that the screening services are rendered by team of health workers at the time of large-scale surveys in the country.

It is advisable to choose unsaturated fats, instead of saturated fats. More than two-third of the doctors (68.4 percent) correctly knew the type of fat to

ly tell the type of fat in fish, 84.7 percent in butter and 77.2 percent in pork. About 70 percent of the respondents were able to identify the correct type of fat mostly seen in coconut oil.

WHO recommends intake of 5 grams of salt per person per day (equivalent to 1 teaspoonful) and sugar intake of not more than 25 grams or 5 teaspoonfuls a day for an adult with normal BMI and six servings of fruits and vegetables in a day. Respondents' knowledge on recommended intake of

salt, sugar and serving of fruits and vegetables was assessed. Recommended daily amounts of salt, sugar and servings of fruits and vegetables per day per adult to prevent NCD was known correctly to 35.7 percent, 41.8 percent and 15.3 percent of the respondents respectively. This shows that less number of medical officers knew the correct number of servings of fruits and vegetables to be consumed per day. Thus the findings reveal that medical officers knowledge on intake of salt, sugar and servings of fruits and vegetables needs to be updated.

WHO has recommended that an adult aged 18-64 years should do at least 150 minutes of moderate-intensity aerobic physical activity throughout the week (minimum 30 minutes/day for at least 5 days a week) or do at least 75 minutes of vigorous-intensity aerobic physical activity throughout the week. Aerobic exercises should be performed in bouts of at least 10 minutes duration. Reduction in the risk of CVD, diabetes, colon cancer and breast cancer is achieved with at least 30 minutes of regular, moderate-intensity physical activity on most days. The respondents were assessed for their knowledge on the recommended level of physical activity for an adult to prevent NCDs. Only 22.7 percent of the medical doctors knew that moderate level physical activities have to be performed 30 minutes per day and for at least five days a week to prevent NCDs while the remaining were not aware of it. Thus most of the respondents lacked knowledge regarding the recommended physical activity to be performed by an individual per day.

Regarding the knowledge about smoking and alcohol, approximately half of the study population (46.9 percent) incorrectly identified the statement 'consumption of a small quantity of alcohol per day is recommended to prevent ischemic heart

disease', as correct. The statement indicating that 'advice to smokers should be to gradually reduce smoking and then stop smoking cessation should not be a gradual process' (slightly confusing?) was identified correctly as true by 57.1 percent.

The word "obesity" means too much body fat. It is usually based on your body mass index (BMI), which one can check using a BMI calculator. BMI compares one's weight to one's height. The formula for calculating BMI is weight (kgs)/height (m²). Among the study population, 76.5 percent were able to identify the correct formula for BMI calculation. Using the range of values of the latest WHO classification on BMI, underweight category (<18.50) was correctly known by 39.8 percent, normal (18.50 - 24.99) by 40.8 percent, overweight (≥ 25.00) by 37.8 percent and obesity category (≥ 30.00) was correctly known by 40.8 percent. Thus the medical officers' knowledge of range of values of BMI was average. NCDs can be reduced by common prevention strategies. For instance, research has shown the benefits of avoiding tobacco use, regular physical activity, and having a healthy diet reduced the burden of non-communicable diseases.

Attitudes of the respondents regarding NCD preventive activities were assessed using a set of eleven attitudinal statements. Nearly two-thirds of the respondents (65.3 percent) agreed and strongly agreed to the statement that 'Determinants of health can reduce the burden of illness and enhance the quality of life' and the same proportion of respondents also agreed and strongly agreed that 'health of individuals and community also depends greatly on the access to quality health care and health information.' The response rate related to attitude of study respondents is given in Table 14.2.

TABLE 14.2: ATTITUDE ABOUT NON-COMMUNICABLE DISEASES

Statements	Strongly disagrees	Disagree	Neutral	Agree	Strongly agree
Determinants of health can reduce the burden of illness and enhance the quality of life.	17(17.3)	16(16.3)	1(1.0)	30(30.6)	34(34.7)
Health of individuals and community also depends greatly on the access to quality health care and health information.	20(20.4)	11(11.2)	4(4.1)	25(25.5)	38(38.8)
Health promotion activities in field are a time consuming process and it is very difficult to achieve desired outcome.	19(19.4)	31(31.6)	14(14.3)	21(21.4)	13(13.3)
I don't think we need to counsel people on NCDs and associated risk factors because people are already aware of it.	33(33.7)	29(29.6)	-	16(16.3)	20(20.4)
I don't mind including risk factors of NCDs in my current programme for the betterment of client.	15(15.3)	23(23.5)	3(3.1)	31(31.6)	26(26.5)
Doctors can efficiently contribute to reduce NCD burden in the country if they perform preventive action in the wards	15(15.3)	17(17.3)	8(8.2)	28(28.6)	30(30.6)
Doctors in hospitals should not implement NCD prevention activities as it is not part of curative services	44(44.9)	20(20.4)	3(3.1)	13(13.3)	18(18.4)
A doctor trying to advice patients regarding NCD prevention is a waste of time	39(39.8)	21(21.4)	2(2.0)	9(9.2)	27(27.6)
Doctors cannot contribute towards NCD prevention because they are overburdened with other ward work	26(26.5)	27(27.6)	11(11.2)	19(19.4)	15(15.3)
I personally feel health promotion activities may improve the community's life style	21(21.4)	13(13.3)	1(1.0)	27(27.6)	36(36.7)
It is possible for a counsellor to counsel on integrated health programmes.	18(18.4)	16(16.3)	1(1.0)	32(32.7)	31(31.6)

Nearly 50 per cent of the respondents disagreed and strongly disagreed with the statement that 'health promotion activities in field are a time-consuming process and it is very difficult to achieve desired outcome.' Thus in the opinion of the doctors there is need to counsel the community member regarding NCDs and associated risk factors as people are not aware about the preventive measures. About 60 per cent (58.1 percent) of the respondents wanted to include the risk factor guidelines in their current programme for the betterment of client.' Nearly 50 percent agreed and strongly agreed to the statement that 'doctors can efficiently contribute to reduce NCD burden in the country if they perform preventive action in the wards'. Nearly two-third (65.3 percent,) of the doctors disagreed and strongly disagreed to the statement that 'doctors in

hospitals should not implement NCD prevention activities as it is not part of curative services.' Similarly, 61.2 percent of the respondents disagreed and strongly disagreed with the statement that 'A doctor trying to advice patients regarding NCD prevention is a waste of time.' There is positive attitude of the doctors towards prevention of NCDs as more than half (54.3percent) of the respondents disagreed and strongly disagreed with the statement 'Doctors cannot contribute towards NCD prevention because they are overburdened with other ward work'. Respondents felt that 'health promotion activities may improve the community's lifestyle and counselling on integrated health programmes.' Overall, there is positive attitude (87.8 percent) of the medical officers towards the prevention strategies used for prevention of non

communicable diseases.

To assess the practice of the medical staff for the prevention and control of non-communicable diseases a set of twelve statements were used in the present study. Of the study respondents, nearly 80 percent of the respondents 'referred clients of NCDs detected during counselling to the respective departments for proper testing and treatment.' A large majority (85.7 percent) of the respondents 'advised all in-patients with NCDs on changing their habits/lifestyles so that existing NCDs are controlled and the future complications are avoided.' A large majority of doctors (80.6 per cent) 'advised family members of the inpatients with NCD about measures that the family can take to control existing NCDs/avoid the patient getting

other NCDs in the future'. The practice of advising in-patients who are smokers on stopping smoking was done by 86.7 percent of the study participants for all the patients. Inquiring into the practice of the study population on advising ward staff members on changing their lifestyles/habits to prevent the development of NCDs, it was found that only 72.4 percent had ever (?) advised staff members in this regard. Approximately 29.6 percent had taken a personal effort to ensure that health education materials were available in the wards to provide information on NCD prevention. Nearly half of the respondents had ever (?) done group health education sessions in the ward targeted at patients in the ward in relation to NCD prevention (Table 14.3).

Table 14.3: Practice about Non Communicable diseases

Statements	Yes I do it for all patients	Yes I do it sometime	No I do it occasionally	No I not practice it
I am referring clients of NCDs detected during counselling to the respective departments for proper testing and treatment.	80 (81.6)	8 (7.8)	6(6.1)	4(3.9)
IEC materials on NCDs should be provided to me for distribution and counselling	78(79.6)	7(6.9)	5(5.1)	7(6.9)
NCDs are linked with habits and to bring change in habit one need continuous counselling.	86(87.8)	4(3.9)	2(2.0)	6(5.9)
Do you advise patients already been diagnosed to have NCDs on changing their habits /life styles so that the existing NCDs are controlled and the future complications are avoided?	84(85.7)	4(3.9)	3(3.1)	2(2.0)
Do you advise the family members of patients who have already been diagnosed to have NCDs about measures that the family can take to control existing NCDs/ avoid the patient getting other NCDs in the future?	79(80.6)	7(6.9)	8(8.2)	4(3.9)
Do you advise the family members about measures that they should take to avoid getting NCDs in the future?	81(82.7)	6(5.9)	5(5.1)	6(5.9)
Do you inquire and advise patients admitted to ward other than NCDs but having risk factors on changing their habits /life styles so that the risk of getting NCDs in the future minimized?	65(66.3)	6(5.9)	9(9.2)	18(17.6)
Do you inquire and advise patients on stopping smoking so that the risk of getting NCDs in the future is minimized	85(86.7)	6(5.9)	3(3.1)	4(3.9)

Do you inquire and advise patients on stopping alcohol consumption so that the risk of getting NCDs in the future is minimized?	61(62.2)	29(28.4)	5(5.1)	2(2.0)
Do you talk to patients who are not diagnosed with NCD in the ward and advised them on NCD screening services?	58(59.2)	13(12.7)	15(15.3)	12(11.8)
Have you personally taken any effort to ensure that health education materials (Handbill, leaflet, posters and video) are available in the ward to provide information on NCD prevention?	29(29.6)	12(11.8)	19(19.4)	35(34.3)
Have you ever done any group health education sessions in the ward targeted at patients in the ward/unit related to NCD prevention?	30(17.6)	18(17.6)	28(28.6)	32(31.4)

DISCUSSION

Very limited data is available on the level of knowledge, attitude and practices of non-communicable diseases among medical professionals in developing countries like India. However, such type of data is extremely important to plan the public health policies as increased urbanization and industrialization has resulted in drastic changes in lifestyles globally but the impact is felt more in developing countries because of their more rapid pace of growth. One of the consequences of this transition is a change in disease patterns with communicable diseases being replaced by non-communicable or lifestyle-related diseases like diabetes, obesity, cardiovascular disease and cancer. Knowledge about the about non-communicable diseases is the first step in formulating a prevention programme for NCDs. The review of literature on knowledge, attitude and practices about NCDs in developing countries yielded very few studies which actually deal with the NCDs among medical professionals. Even in other developing countries, such studies have mainly focused on NCDs among patients and studies are mostly clinic-based which introduce referral bias.

In the present study, 51.5 percent of the doctors were found to have poor knowledge about non-communicable diseases. A study among 394 internal medical physicians in 10 hospitals in Guatemala found that their knowledge on NCD and its prevention was limited (Corral, 2012). A study by

Wang et al, 2001 was done in Beijing to assess the practices of cardiologists, neurologists and other physicians in advising about lifestyle modifications in the management of hypertension (Wang et al,2001) and it was found that 48.6 percent of cardiologists and 28.3 percent of other physicians prescribe lifestyle modifications to control hypertension. Another study in China among 837 physicians attached to cardiology units of 35 tertiary care hospitals to assess the influence of education and working background on knowledge of secondary prevention guidelines for coronary heart disease found that only 53.9 percent fulfilled the expected standard of knowledge (Gong, 2011). Few more studies comply with the findings of the present study (Wellapuli & Gunawardena, 2015).

As there has been a shift from communicable to non-communicable diseases, the Ministry of Health and Family Welfare, Govt. of India has set up a dedicated NCD division that acts as the focal point for coordinating the NCD control programmes in the country. Currently, the division is under reorganization. The National Rural Health Mission (NRHM) provides an overarching umbrella, subsuming the existing programmes of the Ministry of Health, including all NCD control programmes. Indian Council for Medical Research (ICMR), the National Institute of Communicable Diseases, the All India Institute of Medical Sciences (AIIMS), and the Public Health Foundation of India (PHFI) are the key institutions dealing with NCDs. Ministry of Health and Family Welfare has

launched National Cancer Control Programme, National Tobacco Control Programme and National Programme for Prevention and Control of Cancer, Diabetes, CVD and Stroke (NPCDCS) to address the NCDs such as cancer, CVD, diabetes and stroke that are the major factors reducing potentially productive years of human life and resulting in huge economic loss.

For prevention and control of Non-Communicable Diseases (NCD) in the state of Haryana, MoHFW, GOI has launched National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke (NPCDCS) & National Programme for the Health care of the Elderly (NPHCE) in 2010-12 across 100 district of the country. It is an independent programme with 80:20 centres: state share. These two programmes are integrated at various levels for optimum utilization of resources and are managed/supervised by State NCD cell at State Head Quarter (SHQ) & district NCD cell at the district headquarter. State NCD cell is established in the office DGHS, at Panchkula. In Haryana, the programme is implemented in 4 District Hospitals of Mewat, Ambala, Kurukshetra and Yamunanagar and 17 Community Health Centres (CHCs), 52 Primary Health Centres (PHCs) & 406 Sub Centers (SC) of the above said districts.

CONCLUSIONS

More than one-fifth of the study population agreed that diabetes is the common NCD found in India followed by hypertension (19.1 percent), cancer (14.0 percent), cardiovascular disease (11.7 percent) and stroke (9.9 percent). The sampled population showed a high level of knowledge regarding the modifiable risk factors. A large proportion of respondents were able to identify four modifiable risk factors for NCDs but only few were able to identified non-modifiable risk factors.

Nearly 70 percent of doctors correctly knew the type of fat to be consumed in lesser amounts to

prevent NCDs. Small proportion of doctors knew correct amount of servings of fruits and vegetables to be consumed per day. Similarly, small proportion of doctors knew that moderate level physical activities have to be performed 30 minutes per day and for at least five days a week to prevent NCDs.

For diet and physical activity, respondents were not aware of the health consequences and the health impact related to an insufficient amount of fruits and vegetables, high salt and fat diet, and being physically inactive. Additionally, the benefits of eating healthy and being physically active are not always clear to the medical officers.

IMPLICATIONS

Respondents' knowledge and attitudes related to risky behaviors, risk perceptions and the awareness of potential consequences of unhealthy behaviors are not sufficient to motivate changes in individuals to take action for their own health. In line with the WHO recommendation on establishing a programme for promotion of health across the life course and prevention and control of NCDs, fundamentals of a healthy lifestyle should be included in the curriculum of trainings of medical officers.

Emphasis should be given to the harms of alcohol and tobacco use and more knowledge related to a healthy diet and the increase of physical activity should be included. The diversity in the general knowledge and attitudes of the population to NCD behavioral risk factors demand different and tailored approaches.

REFERENCES

Al-Omran, M. 2007, "Knowledge and attitude of physicians in a major teaching hospital towards atherosclerotic risk reduction therapy in patients with peripheral arterial disease". *Vascular Health and Risk Management*, 3, 1019–1027.

- Alwan A, Maclean DR, Riley L.M, d'Espaignet E.T., Mathers C.D., Stevens G.A., Bettcher D., 2010, "Monitoring and surveillance of chronic non-communicable diseases: progress and capacity in high-burden countries". *Lancet*. 376(9755):1861-68.
- Beaglehole et al, 2011, "Priority actions for the non-communicable disease crises". *The Lancet* 377:9775; 1438-1447.
- Campbell K, Engel H, Timperio A, Cooper C, Crawford D. 2000, "Obesity management: Australian general practitioners' attitudes and practices". *Obes Res.*;8:459–66. [PubMed]
- Chobanian, A.V., Bakris, G.L., Black, H.R.,ushman, W.C., Green, L.A., Izzo, J.L. Jr., Roccella, E. J. 2003, "The seventh report of the Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure: The JNC 7 report". *JAMA*, 289, 2560–2571.
- Colvin RH, Olson SB. 1983, "A descriptive study of men and women who have lost a significant weight and are highly successful at maintaining the loss". *Addict Behav.*;8:287–95. [PubMed].
- Corral JE, Arnold LD, Argueta EE, Ganju A, Barnoya J. 2012, "Clinical preventive services in Guatemala: a cross-sectional survey of internal medicine physicians". *PloSone* [Internet]. Pmc3485332; 7(10):[e48640 p.]. Available from: *PLOS ONE* | www.plosone.org.
- Gong YJ, Hong T, Jiang J, Yu RH, Zhang Y, Liu ZP, et al. 2012, "Influence of education and working background on physicians' knowledge of secondary prevention guidelines for coronary heart disease: results from a survey in China." *Journal Zhejiang University Science- B (Bio medicine and Biotechnology)*.;13(3):231-8.
- Heath C, Grant W, Marcheni P, Kamps C. 1993, "Do family physicians treat obese patients?" *Fam Med*.993; 25:401–2. [PubMed]
- Hosseinpoor A.R, Bergen N., Kunst A., Harper S., Guthold R., Rekve D., D'Espaignet E., Naidoo N., Chatterji S., 2012, "Socio-economic inequalities in risk factors for non-communicable diseases in low-income and middle-income countries: results from the World Health Survey". *BMC Public Health* 12:912-20.
- Khan F.S., Lotia-Farrukh I., Khan A.J., Siddiqui S.T., Sajun S.Z., Malik A.A., Burfat A., Arshad M.H., Codlin A.J., Reiningger B.M., .2013, "The burden of non-communicable disease in transition communities in an Asian megacity: baseline finding from a cohort study in Pakistan.
- Kristeller JL, Hoerr RA. 1997. Physician attitudes toward managing obesity: Differences among six specialty groups. *Prev Med.*;26:542–9. [
- Lawlor DA, Keen S, Neal RD. 1999, "Increasing population levels of physical activity through primary care: GPs' knowledge, attitudes and self-reported practice". *Fam Pract.*; 16:250–4. [
- National program for prevention and control of diabetes, cardiovascular diseases and stroke: A guide for health care workers, directorate general of health services, ministry of health and family welfare, government of India. 2009. Apr, [Last accessed on 2014 Oct 7]. Available from: http://www.searo.who.int/india/topics/cardiovascular_diseases/NCD_Resources_Training_module_for_NPDCS_for_health_workers.pdf?ua=1 .
- Price JH, Desmond SM, Krol RA, Snyder FF, O'Connell JK. 1987, "Family practice physicians' beliefs, attitudes, and practices regarding obesity". *Am J Prev Med.*;3:339–45. [PubMed]
- Wang Z, Cao L, Wu Y. 2001, "Differences in knowledge, attitude and behavior with respect to hypertension among cardiologists, neurologists and other physicians in internal medicine".

Hypertension Res. Available from: https://www.jstage.jst.go.jp/article/hypres/24/4/24_4_459/_article.

Wellapuli, N. T. and N.S. Gunawardena. 2015, "Knowledge, attitudes and practices of medical officers in two selected teaching hospitals in the colombo district in the application of non-communicable disease prevention activities in the ward setting". *Journal of the postgraduate institute of medicine* 2:e12:1-14

World Health Organization. 2010. Global Status Report on non -communicable diseases.

World Health Organization. 2011. Non-communicable Diseases Country Profile.

World Health Organization. Deaths from NCDs.

http://www.who.int/gho/ncd/mortality_morbidity/ncd_total/en/index.html. [Last assessed on 2016 February]

World Health Organization. 2014. Global Status Report on Non-communicable diseases

Section III

Health Information : Assessment of Data Quality

CHAPTER - 15

Analysis of sub district level HMIS data in three high focus districts of Gujarat, Dahod and Panchmahals

Gayatri Desai, Rajnikant Patel

Population Research Centre, Baroda

Abstract

The study was undertaken for 3 high focus and tribal districts of Gujarat. The specific objectives were to analyse the performance of all talukas in Valsad, Dahod and Panchmahals districts; to study the differences, if any, between talukas of study districts; to examine the trend in performance of talukas of study districts based on the HMIS index. The study has analysed the district-level performance as well as talukas of study districts. For such analysis, the composite index as available on the portal was utilised to calculate index values. Based on this index value, the talukas are divided into 4 categories using quartiles to assess their performance. To examine trend in performance of districts and talukas, analysis was carried out for 2013-14 and 2014-15 by i) examining taluka-level performance for each districts individually and ii) assessing taluka-level performance on the basis of index values calculated from combined values of all talukas of 3 districts. The overall index showed that in 2014-15 Dahod and Panchmahals had two talukas each in good and very low performance categories whereas Valsad had one in good and two in very low performance categories. As per combined analysis, overall index for 2014-15 showed that, in Dahod, one taluka fall under good and three under very low performance categories. In Panchmahals, two talukas fall in good and three in very low performance levels. In Valsad two talukas fall in good and none fall in very low category. Furthermore, out of 20 talukas, only 5 have shown good performance, including 1 from Dahod, 2 each from Panch Mahals and Valsad. Half of the talukas are poor performing - 6 from Dahod and 4 from Panchmahals. Thus Valsad ranked highest as all talukas fall under good or promising levels of performance. However, none of the talukas has shown good performance across all four indices despite having good performance in overall index. Certain talukas have shown poor performance in all indices indicating need for special focus. To conclude, the study has highlighted that despite variations in performance across districts, across talukas, as well as differences in performance of indices, when performance of individual districts is assessed by percentage of poor performing talukas, Valsad has highest percent followed by Panchmahals and Dahod. However, in combined index, results are reversed. The programmers need to address weak indices in poor performing talukas which will ultimately improve performance of respective districts.

Introduction

The government of India introduced the National Rural Health Mission in the year 2005 to ensure architectural corrections in the areas of health care system. The Mission has now been termed National Health Mission to indicate the inclusion of urban along with rural areas of the country.

As a result of such programmatic endeavors, India has made important strides in various areas of health service delivery as indicated by the decline in maternal and infant mortality levels during the past few decades. Needless to say, however, that the MMR and IMR of the country are far from the desired level.

The interventions introduced under NRHM led to an increase in the demand for data on population and health for planning and programme implementation. Apart from this, aspects such as understanding the synergy between service availability, cost involved in public health service delivery, expenditure and pattern of utilization among different sections of society also influence decisions pertaining to public health. The demand for a continuous flow of good quality data on input indicators, output indicators, outcome indicators etc. has given rise to a nationwide web portal –HMIS.

The Health Management Information System (HMIS) has been an important initiative of MoHFW, specially created to assist the health departments in monitoring of the objectives of NRHM. HMIS was designed with an aim to utilize the data generated through this programme for improvement of service delivery and health outcomes.

Subsequently, the demand for disaggregated data for micro-level planning further led to facility-based data reporting system under HMIS. In Gujarat, facility-based reporting has been adopted in all the districts of the State since the past few years. This system facilitates tracking of errors or

discrepancies in data up to a health facility at the grassroots level and helps in monitoring and improvement of the data.

Certain state and district-level studies have been conducted to assess data quality under HMIS. A state-level study conducted in Haryana showed significant differences in the reported and evaluated coverage of MCH services (Sharma et. al., 2016). Another study conducted for all districts of Karnataka state highlighted four types of errors in data reporting i.e. procedural, lack of clarity, lack of understanding and deliberate over/under reporting (James and Prabhuswamy, 2016).

Similarly, a district-level study carried out in Jammu and Kashmir showed trends in ANC registration as well as issue of duplication in ANC registration (Bhat 2016). A taluka-level study carried out in Narmada district of Gujarat showed varied level of errors in HMIS data of the 2 selected talukas (Desai and Shah, 2016).

Thus very few attempts have been made to study the status of health service delivery at sub-district level using HMIS data. In this context it becomes pertinent to note that the composite health index formulated by MoHFW, New Delhi available on HMIS web portal helps in judging the overall health performance at district as well as below district level

An analysis carried out using this index facilitates assessment of the reach of health services and provide valuable inputs for the health programmers to strengthen the poor performing units in the districts. The present study is an attempt in this direction. The study was undertaken for the 3 high focus and tribal districts of Gujarat viz. Dahod, Panchmahals and Valsad.

Objectives

The specific objectives of the study were:

1. To analyse the performance of all the talukas in Dahod, Panchmahals and Valsad districts of Gujarat.
2. To study the differences, if any, between the talukas of the 3 study districts.
3. To examine the trend in performance of the talukas of the study districts based on the HMIS index.

Data and Methodology

The study has analysed the district-level performance as well as the sub-district-level performance i.e. the talukas of the 3 study districts. For the district-level as well as sub-district-level analysis, the composite index as available on the portal was utilised to calculate index values. (MoHFW, New Delhi). Based on this index value, the study units i.e. talukas are divided in to 4 categories

Understanding the computation of health index (For details see Annexure-I)

Sl.No	Stages of life cycle	Indicators	Association
		Proportion of:	
1		Post-partum sterilization against total female sterilization	Positive
2	Pre Pregnancy / Reproductive age	Male sterilization to total sterilization conducted	Positive
3		IUCD insertions to all family planning methods (IUCD plus permanent)	Positive
4		1st Trimester registration to total ANC registration	Positive
5		Pregnant women received 3 ANC check-ups to total ANC registration	Positive
6	Pregnancy care	Pregnant women given 100 IFA to total ANC registration	Positive
7		Cases of pregnant women with Obstetric Complications and attended to reported deliveries	Positive
8		Pregnant women receiving TT2 or Booster to total number of ANC registered	Positive
9		SBA attended home deliveries to total reported home deliveries	Positive
10	Childbirth	Institutional deliveries to total ANC registration	Positive
11		C-Section to reported deliveries	Positive
12		Newborns breast fed within 1 hour to live births	Positive
13	Postnatal maternal & new born care	Women discharged under 48 hours of delivery in public institutions to total no. of deliveries in public institutions	Negative
14		Newborns weighing less than 2.5 kg to newborns weighed at birth	Negative
15		Newborns visited within 24hrs of home delivery to total reported home deliveries	Positive
16		Infants 0 to 11 months old who received Measles to reported live births	Positive

using quartiles to assess the performance of the study units. The talukas falling in 1st quartile indicated with red colour represents very low performance. Similarly, talukas belonging to 2nd quartile marked with pink, 3rd quartile marked with yellow and 4th quartile marked with green colour represent low, promising and good performance levels respectively.

To examine the trend in performance of districts and of each talukas, for all the 3 study districts,

analysis was carried out for the year 2013-14 and 2014-15. This was done in two ways, i) by examining the taluka-level performance for each districts individually i.e. finding the index values based on the total value of all talukas of a district and ii) by assessing the taluka-level performance of all the 3 districts on the basis of index values calculated from the combined values of all the talukas of all the 3 districts.

As regards the performance of the 3 districts in the

Table:15.1A Performance of study districts based on HMIS index for April-June, 2014-15 and 2015-16

Indicators		Composite index: 2015-16					Composite index: 2014-15				
		2015-16					2014-15				
		Pre-pregnancy/ Reproductive age	Pregnancy Care	Child Birth	Postnatal Maternal & new born care	Overall Index	Pre-pregnancy/ Reproductive age	Pregnancy Care	Child Birth	Postnatal Maternal & new born care	Overall Index
1	Dahod	0.3228	0.4163	0.0559	0.5235	0.3647	0.4396	0.3985	0.3833	0.4310	0.4135
2	Panch Mahals	0.3887	0.5400	0.2019	0.5600	0.4545	0.3417	0.4930	0.3632	0.3907	0.4083
3	Valsad	0.1272	0.6614	0.2594	0.6094	0.4696	0.0749	0.4898	0.5312	0.4701	0.4136
	Q1	0.2686	0.5144	0.2019	0.5213	0.4261	0.2423	0.4513	0.4017	0.4090	0.4000
	Q2	0.2999	0.6361	0.2337	0.5439	0.4696	0.2844	0.5196	0.4885	0.4589	0.4387
	Q3	0.3228	0.6912	0.3033	0.5999	0.5037	0.3212	0.6230	0.5495	0.4975	0.5014
Note: The lowest ranking (lower quartile) talukas colour coded as Red(D) - depict very low performance, Pink (C) - Low performing, Yellow (B) - promising and Green (A) – Good performance											

Table:15.1B Performance of study districts based on HMIS index for, 2013-15

Indicators		Composite index: 2014-15					Composite index: 2013-14				
		2014-15					2013-14				
		Pre-pregnancy/ Reproductive age	Pregnancy Care	Child Birth	Postnatal Maternal & new born care	Overall Index	Pre-pregnancy/ Reproductive age	Pregnancy Care	Child Birth	Postnatal Maternal & new born care	Overall Index
1	Dahod	0.1903	0.4513	0.3288	0.5777	0.4189	0.1816	0.2530	0.3840	0.4946	0.3397
2	Panch Mahals	0.2306	0.4980	0.3779	0.4718	0.4172	0.2547	0.5154	0.3564	0.4292	0.4098
3	Valsad	0.0858	0.6062	0.4665	0.6876	0.5079	0.0459	0.5506	0.4422	0.5370	0.4314
	Q1	0.2145	0.4980	0.4077	0.5319	0.4460	0.1997	0.4358	0.3891	0.4577	0.4093
	Q2	0.2654	0.6084	0.4834	0.5904	0.5027	0.2752	0.4806	0.4512	0.5063	0.4481
	Q3	0.3201	0.6623	0.5670	0.6540	0.5580	0.3287	0.6305	0.5793	0.5556	0.5212

Note: The lowest ranking (lower quartile) talukas colour coded as Red(D) - depict very low performance, Pink (C) - Low performing, Yellow (B) - promising and Green (A) – Good performance

first quarter i.e. April – June, of 2015-16 the bifurcation by talukas was not available on the portal hence, the overall district-level performance had been analysed and compared with that of the corresponding quarter of 2014-15.

Results and Discussion

The following section first discusses the performance level of the three study districts for the first quarter of the year 2015-16 and corresponding quarter of 2014-15 as well as for previous two years [Table:15.1A and 15.1B]. This is followed by a discussion of the taluka-level performance as

Performance of Dahod, Panchmahals and Valsad

Looking to the district-wise performance scenario, it is discouraging to note that the overall performance of all the three high focus study districts was poor i.e. low during first quarter of 2014-15 (Table: 15.1A). This status remained the same for Panchmahals and Valsad whereas for Dahod it worsened during the first quarter of 2015-16.

For the first quarter of 2015-16, the indicator wise status of the study districts showed that, in Valsad district, three of the four sets of indicators i.e. those

Table 15. 2A: Talukas level performance of Dahod District based on HMIS Index, 2013-15

Indicators (DAHOD DISTRICT)		Composite index: 2014-15					Composite index: 2013-14				
		2014-15					2013-14				
		Pre-pregnancy/ Reproductive age	Pregnancy Care	Child Birth	Postnatal Maternal & new born care	Overall Index	Pre-pregnancy/ Reproductive age	Pregnancy Care	Child Birth	Postnatal Maternal & new born care	Overall Index
1	Dahod	0.3923	0.2285	0.5239	0.3057	0.3387	0.4691	0.4519	0.5863	0.3423	0.4461
2	Devgadbaria	0.0000	0.2981	0.3884	0.2614	0.2477	0.2750	0.6687	0.6176	0.2786	0.4634
3	Dhanpur	0.3217	0.5701	0.3383	0.5302	0.4676	0.3333	0.6676	0.3245	0.4201	0.4632
4	Fatepura	0.1333	0.3331	0.5300	0.4790	0.3781	0.1902	0.5313	0.4107	0.4997	0.4348
5	Garbada	0.4675	0.3104	0.3397	0.4904	0.4016	0.3810	0.0401	0.2527	0.5876	0.3150
6	Jhalod	0.0294	0.5588	0.4242	0.5651	0.4363	0.0609	0.3453	0.7245	0.3477	0.3638
7	Limkheda	0.0417	0.5825	0.5711	0.3430	0.4041	0.0614	0.6609	0.2724	0.4088	0.3969
8	Sanjeli	0.3333	0.0702	0.3333	0.3993	0.2717	0.3004	0.3607	0.3333	0.4689	0.3781
	Q1	0.0386	0.2807	0.3394	0.3337	0.3220	0.1580	0.3569	0.3114	0.3463	0.3745
	Q2	0.2275	0.3217	0.4063	0.4392	0.3899	0.2877	0.4916	0.3720	0.4145	0.4159
	Q3	0.3481	0.5616	0.5254	0.5003	0.4122	0.3453	0.6626	0.5941	0.4766	0.4504

Note: The lowest ranking (lower quartile) talukas colour coded as Red(D) - depict very low performance, Pink (C) - Low performing, Yellow (B) - promising and Green (A) – Good performance

also the trend in their performance for each of the three study districts viz. Dahod, Panchmahals and Valsad (Table 15.2A, 15.2B and 15.2C). There after there is a discussion on comparative picture of the performance of index across all the talukas of the three study districts (Table 15.33).

pertaining to ‘pregnancy care’, ‘child birth’ and ‘postnatal, maternal and newborn care’ showed promising to good performance level. Similarly, in Panchmahals ‘pre pregnancy/reproductive age’ and ‘postnatal, maternal and newborn care’ showed promising to good performance level. The

Dahod district showed promising performance only in one set of indicators viz. 'pre pregnancy/reproductive age'. For rest of the indicators the districts had showed low or very low performance level.

For the years 2013-14 and 2014-15 (Table 15.1B) the results of the overall performance of the health index of the 3 study districts which are tribal and high focus in nature, were in the expected direction. For the year 2013-14 the districts of Panchmahals and Valsad were in the low performance category whereas Dahod district was in very low performance category. The overall performance of health index in the year 2014-15 remained constant for Dahod, improved for Valsad and deteriorated for Panchmahals.

For the year 2014-15, the relative overall performance of Dahod district was poor due to the low or very low performance reported in all the four sets of indicators. For Valsad district, the performance of only one set of indicator i.e. 'postnatal, maternal and newborn care' was good and for Panchmahals none of the sets of indicators showed good performance.

Performance of the districts by their respective talukas during the past two years

The following section discusses the taluka-level scenario for the three study districts as regards their performance during the years 2013-14 and 2014-15.

Dahod

The taluka-level scenario in Dahod district by overall health index as well as its four sets of indicators are presented in Table 15.2A. For the year 2013-14, as it is clear from the table, out of 8 talukas 2 each fall in good, promising, low and very low categories. Similar pattern can be observed for each set of indicators i.e. two talukas each falling in each of the four performance categories.

For the year 2014-15, a similar pattern of performance was observed. Thus out of 8 talukas 2 each fall in good, promising, low and very low categories of performance for overall health index as well as the four sets of indicators. It should be noted that there was some improvement in health index values of certain talukas viz. Dhanpur, Garbada, Zalod and Limkheda.

Furthermore, for the year 2014-15 four talukas viz. Dhanpur, Jhalod, Garbada and Limkheda despite showing good or promising level of performance, variations in the performance in the four sets of indicators was observed. Thus Dhanpur showed very low performance in child birth, Jhalod in pre pregnancy/reproductive age, Garbada in childbirth and pregnancy care whereas Limkheda was poor in pre pregnancy/reproductive age and in postnatal maternal and newborn care. Thus programmers need to address the weak indicators in these talukas.

The talukas which require more attention of the programmers are Dahod, Devgadbaria, Fatepura and Sanjeli as they fall in low or very low performance category. The taluka-wise need for specific services to be strengthened, as highlighted by the study, include: in Dahod - pregnancy care and postnatal maternal and newborn care; in Devgadbaria - services under all four sets of indicators; in Fatepura - pre pregnancy/reproductive age and in Sanjeli - pregnancy care, child birth and postnatal maternal and newborn care.

Panchmahals

Looking to the taluka-level performance of the Panchmahals district for the two years, as it is evident from Table 15.2B, the overall performance and indicator-wise performance shows similar pattern. For the year 2013-14, out of the 7 talukas of Panchmahals district the overall health index of two talukas each fall in good, low and very low categories of performance and one taluka is under

the category of promising performance. Similar pattern is observed for the four sets of indicators.

The analysis for the year 2014-15 also showed repeat pattern. However, the overall health index values showed improvement in performance of certain talukas viz. Shahera, Kalol and Morwa (Hadaf).

lol - services under all four sets of indicators; in Morwa (Hadaf) - pregnancy care and child birth.

Valsad

The taluka-level performance of Valsad district for the year 2013-14 as viewed from table 15.2C revealed that for overall health index, one talu-

Table 15.2B: Talukas level performance of Panchmahals District based on HMIS Index, 2013-15

Indicators (The Panchmahals District)		Composite index: 2014-15					Composite index: 2013-14				
		2014-15					2013-14				
		Pre-pregnancy/ Reproductive age	Pregnancy Care	Child Birth	Postnatal Maternal & new born care	Overall Index	Pre-pregnancy/ Reproductive age	Pregnancy Care	Child Birth	Postnatal Maternal & new born care	Overall Index
1	Ghoghamba	0.2843	0.3171	0.5338	0.1117	0.2874	0.2572	0.4307	0.3155	0.2320	0.3145
2	Godhra	1.0000	0.4979	0.4837	0.4669	0.5797	1.0000	0.5828	0.3582	0.4625	0.5813
3	Halol	0.0000	0.3743	0.4348	0.4113	0.3270	0.2754	0.7235	0.3079	0.5213	0.4984
4	Jambughoda	0.0945	0.2205	0.0000	0.1668	0.1387	0.0000	0.0386	0.3260	0.3631	0.1867
5	Kalol	0.1095	0.6738	0.7773	0.5318	0.5430	0.0404	0.5827	0.7890	0.4871	0.4898
6	Morwa (Hadaf)	0.2829	0.2284	0.4166	0.5461	0.3732	0.0400	0.3600	0.3647	0.5105	0.3479
7	Shehera	0.1663	0.7025	0.6790	0.7832	0.6228	0.0817	0.6024	0.5331	0.8175	0.5590
	Q1	0.1020	0.2728	0.4257	0.2891	0.3072	0.0402	0.3953	0.3208	0.4128	0.3312
	Q2	0.1663	0.3743	0.4837	0.4669	0.3732	0.0817	0.5827	0.3582	0.4871	0.4898
	Q3	0.2836	0.5858	0.6064	0.5390	0.5614	0.2663	0.5926	0.4489	0.5159	0.5287

Note: The lowest ranking (lower quartile) talukas colour coded as Red(D) - depict very low performance, Pink (C) - Low performing, Yellow (B) - promising and Green (A) – Good performance

For the year 2014-15, the table showed that Godhara, Shahera and Kalol fall in overall good performance category. However, all the sets of indicator did not show uniformity in performance level. Thus Godhara was weak in services of childbirth and postnatal maternal and newborn care whereas the other two talukas were weak in pre-pregnancy/reproductive age. Therefore, the weak indices in these talukas need to be strengthened.

The table further showed overall poor performance in remaining four talukas viz. Ghoghamba, Jambughoda, Halol and Morwa (Hadaf).

Therefore, the talukawise need for specific services to be strengthened in Panchmahals include: in Ghoghamba - pregnancy care and postnatal maternal and newborn care; in Jambughoda and Ha-

ka each belong to first three categories i.e. good, promising and low performance levels whereas two talukas belonged to the fourth category i.e. very low performance. As in case of the other two districts, this pattern gets repeated for the indicator wise performance of talukas as well.

Again, for the year 2014-15 the pattern of taluka level performance remained same for overall index and the sets of indicators.

It may be mentioned that two talukas with overall good performance viz. Dharmpur and Umbergaon, however, did not show good performance in all four sets of indicators. In fact, Dharmpur did not show good performance in all the indicators except 'pregnancy care' and Umbergaon showed good performance in 'pregnancy care' and 'post-

Table 15.2C: Talukas level performance of Valsad District based on HMIS Index, 2013-15

Indicators (Valsad District)		Composite index: 2014-15					Composite index: 2013-14				
		2014-15					2013-14				
		Pre-pregnancy/ Reproductive age	Pregnancy Care	Child Birth	Postnatal Maternal & new born care	Overall Index	Pre-pregnancy/ Reproductive age	Pregnancy Care	Child Birth	Postnatal Maternal & new born care	Overall Index
1	Dharampur	0.6097	0.7572	0.4087	0.4670	0.5735	0.6667	0.8546	0.5019	0.5004	0.6426
2	Kaprada	0.7330	0.5477	0.0269	0.6492	0.5165	0.6157	0.3080	0.0394	0.6528	0.4231
3	Pardi	0.5145	0.4907	0.4798	0.6115	0.5309	0.1750	0.2362	0.8220	0.6496	0.4638
4	Umbergaon	0.1437	0.7119	0.3040	0.7833	0.5512	0.2122	0.8078	0.6624	0.7245	0.6428
5	Valsad	0.6474	0.0000	0.8952	0.3604	0.4019	0.5333	0.3403	0.8180	0.4221	0.4916
	Q1	0.5145	0.4907	0.3040	0.4670	0.5165	0.2122	0.3080	0.5019	0.5004	0.4638
	Q2	0.6097	0.5477	0.4087	0.6115	0.5309	0.5333	0.3403	0.6624	0.6496	0.4916
	Q3	0.6474	0.7119	0.4798	0.6492	0.5512	0.6157	0.8078	0.8180	0.6528	0.6426

Note: The lowest ranking (lower quartile) talukas colour coded as Red(D) - depict very low performance,

Pink (C) - Low performing, Yellow (B) - promising and Green (A) – Good performance

natal maternal and newborn care' but not in the other two sets. Therefore, there is a need to improve weak indicators in these talukas.

The analysis showed that three talukas viz. Kaprada, Valsad and Pardi need more attention in terms of improvement of health services.

Looking to the taluka-wise need for specific services to be strengthened in Valsad district include: in Kaprada taluka - pregnancy care and child birth; in Valsad and Pardi talukas - pregnancy care as well as postnatal maternal and newborn care.

Comparison of talukas based on index value for all the talukas

Table 15.3 shows the performance of all five indices, including overall index across all the talukas of three study districts. This analysis was done to attain a broader view regarding position of each taluka among all talukas of 3 districts. For such an exercise, the values of quartiles have been calculated based on values of indices of all the talukas of 3 districts. Such calculations are carried out for the two study periods i.e. 2013-14 and 2014-15, which revealed interesting variations.

For the overall index of the year 2013-14, as evident from table 15.3, in Dahod out of eight none of the talukas showed good performance, two belonged to promising performance category and three each showed low and very low performance. In Panchmahals, out of seven talukas two each showed good, promising and very low levels of performance and one showed low performance. In case of Valsad, three out of five fall under good and one each under promising and low performance categories.

Thus, out of the total twenty talukas, three talukas from Valsad, two from Panchmahals and none from Dahod showed good performance in overall index in 2013-14.

On examining the overall performance of all the talukas for the year 2014-15, it was observed that out of eight talukas of Dahod, one each fall under good and promising category and three each fall under low and very low level. In Panchmahals, two talukas each showed good, low and very low performance levels whereas only one taluka showed promising level. Out of five talukas of Valsad two fall in good and the remaining three in

promising level category.

Furthermore, out of the 20 talukas, only 5 have shown good performance of which 1 belongs to Dahod viz. Dhanpur, 2 belong to Panch Mahals viz. Godhara, and Shahera and remaining 2 fall in Valsad district viz. Dharampur and Kaparada. The talukas showing promising level is Limkheda of Dahod district, Kalol of Panchmahals and Pardi, Umargaon and Valsad talukas of Valsad district. It should be noted that in terms of combined index, half of the talukas are in the poor performing zone i.e. low or very low performance levels. These

include 6 from Dahod, 4 from Panchmahals and none from Valsad district.

Thus as per the combined analysis of all the talukas, Valsad district ranked highest as all the talukas fall under good or promising levels of performance.

In this context it needs to be noted that none of the talukas has shown good performance across all four indices despite having good performance in overall index. In fact, certain talukas have shown poor performance in all the 4 indices which has resulted in the overall poor performance of those talukas. These talukas are Devgadbaria of Da-

Table 15.3:		Performance as per HMIS Index among the talukas of 3 Districts during 2013-15 (based on index value for all talukas of 3 districts)									
Indicators		Composite index: 2014-15					Composite index: 2013-14				
		2014-15					2013-14				
		Pre-pregnancy/ Reproductive age	Pregnancy Care	Child Birth	Postnatal Ma- ternal & new born care	Overall Index	Pre-pregnancy/ Reproductive age	Pregnancy Care	Child Birth	Postnatal Ma- ternal & new born care	Overall Index
1	Dahod	0.3714	0.3015	0.2473	0.3739	0.3271	0.2289	0.2456	0.3131	0.4429	0.3168
2	Devgadbaria	0.1527	0.3747	0.2775	0.3600	0.3102	0.2959	0.4063	0.4287	0.3311	0.3663
3	Dhanpur	0.3211	0.5949	0.2521	0.5540	0.4665	0.3333	0.4454	0.3058	0.4193	0.3901
4	Fatepura	0.2225	0.3867	0.3806	0.4485	0.3741	0.2414	0.3491	0.2918	0.5190	0.3712
5	Garbada	0.5563	0.3348	0.3199	0.4486	0.4091	0.2009	0.0971	0.2393	0.5585	0.2874
6	Jhalod	0.1694	0.4731	0.3814	0.4865	0.4032	0.1493	0.1708	0.4881	0.4140	0.3022
7	Limkheda	0.1745	0.5684	0.4626	0.4063	0.4241	0.1571	0.4380	0.2724	0.4461	0.3568
8	Sanjeli	0.3272	0.1983	0.1863	0.5222	0.3215	0.3121	0.2447	0.2631	0.6154	0.3766
1	Ghoghamba	0.3032	0.3457	0.3740	0.2051	0.2991	0.2887	0.3985	0.2879	0.2644	0.3153
2	Godhra	0.6710	0.4962	0.3326	0.3731	0.4599	0.8308	0.5294	0.3143	0.3585	0.4922
3	Halol	0.1287	0.3667	0.3247	0.4269	0.3330	0.3009	0.6387	0.2972	0.3891	0.4333
4	Jambughoda	0.1867	0.2760	0.0478	0.2502	0.2084	0.1839	0.0829	0.3251	0.3494	0.2305
5	Kalol	0.1959	0.5486	0.4396	0.3911	0.4128	0.2018	0.4725	0.5759	0.3906	0.4155
6	Morwa (Hadaf)	0.3060	0.3063	0.2979	0.4735	0.3569	0.2036	0.3215	0.3311	0.4287	0.3347
7	Shehera	0.2308	0.5862	0.4802	0.6028	0.5049	0.2172	0.5033	0.4584	0.6137	0.4757
1	Dharampur	0.4727	0.8108	0.4800	0.4181	0.5626	0.3783	0.6676	0.5135	0.3945	0.4991
2	Kaparada	0.5654	0.6322	0.2434	0.5101	0.5086	0.3776	0.4069	0.2478	0.3679	0.3594
3	Pardi	0.3020	0.5655	0.4117	0.4502	0.4512	0.1178	0.4257	0.7977	0.3446	0.4124
4	Umbergaon	0.0860	0.7382	0.2701	0.5073	0.4560	0.1127	0.6613	0.6944	0.4983	0.5137
5	Valsad	0.3864	0.2266	0.8560	0.4313	0.4386	0.2768	0.5072	0.7610	0.3786	0.4714
	Q1	0.1837	0.3277	0.2656	0.3868	0.3315	0.1966	0.3025	0.2908	0.3655	0.3302
	Q2	0.3026	0.4299	0.3286	0.4399	0.4109	0.2351	0.4163	0.3197	0.4042	0.3739
	Q3	0.3751	0.5729	0.4186	0.4917	0.4569	0.3037	0.5043	0.4945	0.4591	0.4428

Note: The lowest ranking (lower quartile) talukas colour coded as Red(D) - depict very low performance, Pink (C) - Low performing, Yellow (B) - promising and Green (A) – Good performance

hod district as well as Halol and Jambughoda of Panchmahals district. Therefore, the present study has clearly indicated that these talukas need special focus of the programmers.

Furthermore, the possible reasons affecting performance of a particular set of indicators as suggested by the study are as follows:

i) For certain data items under a particular index, no data/values were reported. For e.g. the low performance in the index “pre-pregnancy/reproductive age” of Devgadhbharia taluka (in Dahod district) was due to ‘0’ value being reported for 2 out of 3 data items in this set. Furthermore, in the third item i.e. IUD insertions included under this index, the value was at the lowest.

ii) Similarly, over-reporting of data for an indicator in a taluka also affects performance of other talukas in a district. For e.g. against the item, ‘percent of newborns visited within 24 hours of home delivery to total reported home deliveries’, the value reported by one taluka (Morva (Hada) of Panch Mahals district) was 144. Such over reporting of data has relative negative effect in terms of performance value (low) of other talukas.

iii) Furthermore, the presence or absence of a higher level health facility in a taluka of a district may get reflected in the performance of a certain indicator in that taluka. For e.g. in Dahod taluka of Dahod district, the performance of the indicator ‘pre-pregnancy/reproductive age’ was good. On examining the performance of different items under this set of indicator it was found that, for the two out of three items i.e. percent of male sterilisation and percent of post-partum female sterilisation, the performance of Dahod taluka was relatively better which was possibly due to the presence of DH in this taluka where in sterilisation services are available. In the case of the other two districts too good or promising performance of the indicator ‘pre-pregnancy/reproductive age’ in the headquarter taluka of the respective districts could

possibly be attributed to the presence of the higher level health facility in that taluka.

Summary and Conclusions

The study was undertaken for the 3 high focus and tribal districts of Gujarat. The specific objectives of the study were to analyse the performance of all the talukas in Valsad, Dahod and Panchmahals districts of Gujarat; to study the differences, if any, between the talukas of the 3 study districts; to examine the trend in performance of the talukas of the study districts based on the HMIS index.

The study has analysed the district-level performance as well as the sub-district-level performance i.e. the talukas of the 3 study districts. For such analysis, the composite index as available on the portal was utilised to calculate index values. (MoHFW, New Delhi). Based on this index value, the study units i.e. talukas are divided into 4 categories using quartiles to assess the performance of the study units.

To examine the trend in performance of districts and of each talukas, analysis was carried out for the year 2013-14 and 2014-15. This was done in two ways, i) by examining the taluka-level performance for each districts individually i.e. finding the index values based on the total value of all talukas of a district and ii) by assessing the taluka-level performance of all the 3 districts on the basis of index values calculated from the combined values of all the talukas of all the 3 districts.

Looking to the district-wise performance scenario, the overall performance of all the three high focus study districts was poor i.e. low during first quarter of 2014-15. This status remained the same for Panchmahals and Valsad whereas for Dahod it worsened during the first quarter of 2015-16. For the years 2013-14 and 2014-15 the results of the overall performance of the health index of the 3 study districts were in the expected direction. For the year 2013-14 the districts of Panch-

mahals and Valsad were in the low performance category whereas Dahod district was in very low performance category. The overall performance of health index in the year 2014-15 remained constant for Dahod, improved for Valsad and deteriorated for Panchmahals. In Dahod district the overall health index for the year 2013-14 showed that out of 8 talukas 2 each fall in good, promising, low and very low categories. Similar pattern was observed for each set of indicators i.e. two talukas each falling in each of the four performance categories. For the year 2014-15, a similar pattern of performance was observed. There was some improvement in health index values of certain talukas viz. Dhanpur, Garbada, Zalod and Limkheda. However, for these four talukas despite their good or promising level of performance, variations in the performance in the four sets of indicators were observed. Thus certain sets of indicators need to be strengthened even in these talukas. The talukas which require more attention of the programmers are Dahod, Devgadhbharia, Fatepura and Sanjeli as they fall in low or very low performance category.

In Panchmahals district for the two years, the overall performance and indicator-wise performance shows similar pattern. For the year 2013-14, out of the 7 talukas the overall health index of two talukas each fall in good, low and very low categories of performance and one taluka is under the category of promising performance. Similar pattern is observed for the four sets of indicators. For the year 2014-15 also the result showed repeat pattern. However, the overall health index values showed improvement in performance of certain talukas viz. Shahera, Kalol and Morwa (Hadaf).

Godhara, Shahera and Kalol fall in overall good performance category. However, all the sets of indicator did not show uniformity in performance level therefore requiring strengthening of weak indicators. There was poor performance in overall index in remaining four talukas viz. Ghoghamba, Jambughoda, Halol and Morwa (Hadaf). These talukas therefore need more attention.

In Valsad district for the year 2013-14, with regard to overall health index, one taluka each belong to good, promising and low performance levels whereas two talukas showed very low performance. As in case of the other two districts, this pattern gets repeated for the indicator wise performance of talukas as well. Again, for the year 2014-15 the pattern of taluka level performance remained same for overall index and the sets of indicators. Talukas of Dharmpur and Umbergaon with overall good performance did not show good performance in all four sets of indicators thus requiring attention for weak indicators. Further, talukas of Kaprada, Valsad and Pardi need more attention in terms of improvement of health services.

Comparison of talukas based on index value across all the talukas of study districts showed that in the overall index of the year 2013-14, in Dahod none out of eight talukas showed good performance, two belonged to promising performance category and three each showed low and very low performance. In Panchmahals, out of seven talukas two each showed good, promising and very low level of performance and one showed low performance. In Valsad, three out of five fall under good and one each under promising and low performance category. Thus only five talukas i.e. three from Valsad and two from Panchmahals showed good performance in overall index in 2013-14.

The overall performance of all talukas for 2014-15, showed that in Dahod, one taluka each fall under good and promising category and three each fall under low and very low level. In Panchmahals, two talukas each showed good, low and very low performance levels whereas only one taluka showed promising level. In Valsad two talukas fall in good and three fall in promising level category. Furthermore, out of the 20 talukas, only 5 have shown good performance of which 1 belongs to Dahod, 2 belong to Panch Mahals and 2 to Valsad district; 5 talukas have shown promising level - one each from Dahod and Panchmahals and

3 from Valsad district. It should be noted that in terms of combined index, half of the talukas are in the poor performing zone i.e. low or very low performance levels. These include 6 from Dahod, 4 from Panchmahals and none from Valsad district. Thus in the combined analysis Valsad district ranked highest as all the talukas fall under good or promising levels of performance. However, it needs to be noted that none of the talukas has shown good performance across all four indices despite having good performance in overall index. In fact, certain talukas have shown poor performance in all the 4 indices. Therefore, the present study has clearly indicated that these talukas need special focus of the programmers.

To conclude, the study has brought to light that despite variations in performance across districts, across talukas of the districts, as well as differences in performance of indices, when performance of individual districts is assessed by percentage of poor performing talukas, Valsad has the highest percent followed by Panchmahals and Dahod district. However, in combined index, the results are reversed. The possible reasons for good or poor performance of a taluka and/or of a particular set of indicators, as highlighted by the study should also be looked into. Nonetheless, the programmers need to address the weak indices in the poor performing talukas which will ultimately improve the performance of the respective districts.

REFERENCES

- Bashir Ahmad Bhat, 2016, "Trends and Patterns of Antenatal and Delivery Care in Jammu and Kashmir – An Analysis of Health Management and Information System". Maternal and Child Health in India – A Compendium of Studies Conducted by the Population Research Centres, edited by C.R.K. Nair and K.S. James, MoHFW, New Delhi
- Gayatri S. Desai and Urvi Shah, 2016, "An In-depth Analysis of HMIS Data in Narmada District of Gujarat". Maternal and Child Health in India – A Compendium of Studies Conducted by the Population Research Centres, edited by C.R.K. Nair and K.S. James, MoHFW, New Delhi
- K. S. James and P. Prabhuswamy, 2016, "Quality of HMIS Data in Karnataka: A District Level Analysis". Maternal and Child Health in India – A Compendium of Studies Conducted by the Population Research Centres, edited by C.R.K. Nair and K.S. James, MoHFW, New Delhi
- Sharma A, Rana SK, Prinja S, Kumar R, 2016, "Quality of Health Management Information System for Maternal & Child Health Care in Haryana State, India". PLoS ONE 11(2): e0148449. doi:10.1371/journal.pone.0148449

CHAPTER - 16

HMIS Data Quality Issues in Sub Centres and Primary Health Centres of Ganderbal District of Jammu and Kashmir

Bashir Ahmad Bhat, Associate Professor, Farida Qadri, Research Assistant, Muneer Ahmad, Research Investigator,

Population Research Centre, University of Kashmir,

Abstract

Various interventions have been launched under NRHM to carry out necessary architectural correction in the basic health care delivery system. These interventions have increased the demand for data on population and health for use in both micro-level planning and programme implementation. Moreover, continuous flow of good quality information on inputs, outputs and outcome indicators facilitates monitoring of the objectives of NRHM. To meet this demand of good quality information under NRHM, the Ministry of Health and Family Welfare has devoted a specially designed web portal namely “NRHM Health Statistics Information Portal” where all the districts of various States are uploading consolidated information on a monthly basis especially to monitor, evaluate and understand various health-related situations. Currently, various health facilities located in 643 districts in the country are gathering information on more than 250 indicators on monthly basis. A look at the data uploaded on the HMIS web portal since 2008 shows that regular data uploading has improved to a great extent but there are still some data quality issues that remain to be addressed. Besides, one is not sure as to what extent the data uploaded on the web portal depicts the grassroots level reality so far as reproductive child health indicators are concerned. The present study was conducted in Ganderbal district of Jammu and Kashmir to understand the mechanism of record keeping, reporting and uploading of information, so that data quality issues are identified and are shared with the State and necessary steps are taken by them to address them. We selected 2 PHCs and under each PHC, 2 SCs were selected to match the information contained in the registers with the information contained in HMIS formats and also with the information available on HMIS web portal. It was found that HMIS reporting has improved to a great extent in case of ANC registration, TT, and BCG immunization. Duplication of reporting of ANC cases and BCG immunization has stopped. 100 IFA distribution, measles, full immunization is over-reported. Birth and infant and maternal deaths, spacing methods of family planning and HB<11 is under-reported. There is a need to sensitize all the FMPHWs the difference between MCTS and HMIS. It was found that ANMS generally report from MCTS registers which contain information about services received by the women and children from other facilities also. This results in over reporting of services (duplication). The study recommends reorientation of workers involved with recording and reporting of HMIS information at SCs and PHCs.

INTRODUCTION

The Government of India launched the National Rural Health Mission (NRHM) on 10th May, 2005 under the Chairmanship of Union Minister for Health and Family Welfare especially to carry out necessary architectural correction in the basic health care delivery system after recognizing the importance of health in the process of economic and social development and improving the quality of life of its citizens. NRHM follows a synergistic approach by relating health to determinants of good health like segments of nutrition, sanitation, hygiene and safe drinking water. The main goal is to improve the availability of and access to quality health care by people, especially for those residing in rural areas, the poor, women and children. The plan of action of National Rural Health Mission includes increasing public expenditure on health, reducing regional imbalance in health infrastructure, pooling resources, integration of organizational structures, optimization of health manpower, decentralization and district management of health programmes, community participation and ownership of assets, induction of management and financial personnel into district health systems, and operationalizing Community Health Centers (CHC) into functional hospitals in each block across the country that meet Indian Public Health Standards (IPHS).

These interventions have increased the demand for data on population and health for use in both micro-level planning and programme implementation. At the same time, understanding the synergy between availability of services, cost involved in provision of public health care services, expenditure and pattern of utilization among various sections of population, including vulnerable sections of society, are important aspects that influence decision making. A continuous flow of good quality information on inputs, outputs and outcome indicators facilitates monitoring of the objectives of NRHM. For reasons such as these, efficient Health Management Information System

(HMIS) is required for effective implementation of NRHM and to meet the continuous demand for quality information on inputs, outputs and outcome indicators mentioned in the plan of action for the involvement of data in micro-level planning and programme, the Ministry of Health and Family Welfare has devoted a specially designed web portal namely “NRHM Health Statistics Information Portal” where all the districts of various States are uploading monthly consolidated HMIS report on a monthly basis especially to monitor, evaluate and to understand various health-related situations. This web portal has been specially designed to assist health programme managers, at all levels, in managing and planning health programmes.

Currently, various health facilities located in 643 districts in the country are gathering information on more than 250 indicators on monthly basis. A look at the data uploaded on the HMIS web portal since 2008 shows that regular data uploading has improved to a large extent and all the districts all over the country are now regularly uploading monthly reports. Not only the timely uploading of reports has improved but the quality of data being uploaded on the website also seems to have improved. However, there are still some data quality issues that remain to be addressed. For example, it is very difficult to gauge from consolidated district reports whether the information pertains to all health facilities or not and whether all facilities have reported on a particular indicator or not. Besides, one is not sure as to what extent the data uploaded on the web portal depicts the grassroots level reality so far as reproductive child health indicators are concerned.

In order to examine the quality of data being uploaded and address some of the data quality issues, Population Research Centre, University of Kashmir has conducted a series of studies to analyze the record keeping, reporting and uploading mechanism, so that data quality issues are identified and are shared with concerned Programme Manage-

ment Units and necessary steps are taken by them to address them. We conducted one such study in Ganderbal district of Jammu and Kashmir.

OBJECTIVES

The objectives of the study are:

- To identify whether all service providing units are reporting for HMIS and examine whether reporting units are regular in reporting or not.
- Whether the reporting units are providing data on all maternal and child health indicators as per HMIS.
- To make field visits to understand the mechanism of data collection and reporting and trace the source/ reason for data error.
- To match information reported with information recorded and information uploaded on HMIS website.
- To identify the problems in the system of record keeping, reporting and uploading of services rendered by selected Sub Centres (SCs) and Primary Health Centres (PHCs).
- To suggest measures to improve quality of HMIS data.

loading the data on the website. The number of validation errors has declined in all the districts and there are only a few indicators which are now showing inconsistencies. But if we look at the number of expected deliveries and actual number of women registered for ANC services in the State during 2014-15, ANC registration is 124 percent to estimated deliveries. While analyzing the HMIS data quality in various districts of Kashmir Valley, Ganderbal district has witnessed considerable improvement in HMIS data quality. Further, Ganderbal is very close to Srinagar and significant proportion of women from Ganderbal district also visit Srinagar-based health care facilities for antenatal, natal and postnatal care and thereby, minimizing the work load of some PHCs and SCs located in the vicinity of Srinagar District and it would be interesting to examine how the HMIS reflects this pattern of service delivery. We therefore decided to analyse the HMIS data quality in Ganderbal districts so that we are in a position to understand how Ganderbal has improved its HMIS so that other districts can also benefit. We did not try to match HMIS information with Mother and Child Tracking System (MCTS) because the two types of information are conceptually different. The HMIS reports the services which are provided by a facility irrespective of the fact that whether the beneficiary belongs to facility catchment area

Table:16.1 Details of Selected Health Facilities in Ganderbal District

District	Block	Primary Health Centres			Primary Health Centres		
		Total	Selected	Name of selected PHC	No of SCs		Name of selected sc
					Total	Selected	
Ganderbal	Ganderbal	6	2	1. Shuhama	5	2	1. Gulab Bagh 2. Khulmulla
				2. Kachhan	4	2	1. Gund Rehman 2. Harran

METHODOLOGY

A look at the HMIS data pertaining to Jammu and Kashmir uploaded on the HMIS website during the last five years shows that almost all public health institutions in the State are regularly up-

or not. The MCTS, on the other is a community-based reporting system and information about all women belonging to a village/facility are reported irrespective of the fact whether the women have received the services from the local facility or from any other facility outside the catchment area

of the facility, including private facilities/clinics.

Ganderbal district consists of 4 medical blocks, namely Ganderbal, Lar, Kangan, and Wakura. We analysed block-wise ANC registration and among these four blocks, Ganderbal block has registered ANC registration which is in accordance with its population, so we thought it is prudent to select Ganderbal block for our in-depth analysis. Ganderbal district has 6 PHCs and we did a PHC-wise desk review of ANC indicators of Ganderbal block for the year 2014-2015 and identified errors (validation and inconsistency errors) and selected 2 PHCs (PHC Shuhama and PHC Kachhan) which have the least validation and inconsistency errors.

Once the PHCs were identified, we trickled down to SCs and identified two SCs in each PHC which had minimum validation errors. Once the SCs and PHCs were identified, we visited these PHCs and SCs and made an in-depth analysis of recording, maintaining and reporting of services provided by these facilities. The detailed list of facilities selected is presented in Table 16.1.

Once the facilities were selected, we visited the offices of Block Medical Officer and collected detailed information about the reporting system, formats used and regularity of reporting for all the health facilities. We collected the reporting formats for the months of October, November and December 2015 of the selected facilities and also downloaded the uploaded HMIS formats from web portal for these facilities. Finally, we visited the selected health facilities and collected information from the registers maintained by these facilities about ANC, HB, immunization, family planning methods, births and deaths. We matched the reported figures with the information uploaded HMIS web portal as well as with the records maintained at the health facilities. If there were any inconsistencies in the recording, reporting and uploading, reasons for same were explored from the officials responsible for recording and reporting of such information.

STUDY AREA

Ganderbal District is located at 34.23°N 74.78°E. It has an average elevation of 1,619 meters (5,312 feet). It is bordered by the Srinagar district in the south, Bandipora to the north, Kargil in the north-east, Anantnag to the southeast and Baramulla in the southwest. It is divided into four blocks viz, Ganderbal, Kangan, Lar and Wakura. According to 2011 Census, Ganderbal had a population of 297,446 which constituted 2.4 percent of total population of the State. The decadal growth rate during 2001-2011 was about 36.3 percent which is a higher than the decadal growth rate of 23.7 percent for the State as a whole. The sex ratio of the population in the district according to 2011 Census is 874, which is much lower than for the State as a whole (883). Though the child sex ratio in the district is very low (863) but it is slightly higher compared to the child sex ratio at the State level (859). Seventeen percent of the total population of the district lives in urban areas as compared to 35 percent in the State. Around 20 percent of the population of Ganderbal district is from Schedules Tribe category. Ganderbal district has a very low literacy rate. As per the latest Census, the literacy rate among population age 7 and above is 58 percent as compared to 67 percent in the State. Female literacy (46 percent) continues to be lower than the male literacy (69 percent). The health services in the public sector are delivered through a network of 1 District Hospital, 1 CHC, 1 Trauma Hospital, 21 PHCs (15 PHCs+6 NTPHCs) and 43 SCs.

During 2015-2016, Ganderbal has registered a total number of 3900 pregnant women for ANC services. This number is lower than the 4500 expected number of pregnancies in the district. Thus ANC registration is 14 percent lower than the estimated number of pregnancies in the district during April, 2015-January, 2016. This indicates that 14 percent of women do not get registered for ANC services in health facilities located in Ganderbal but at the same time some of these women of Ganderbal are registered for ANC in health facilities

Table:16. 2 ANC Indicators of Ganderbal District and their comparison with Jammu and Kashmir During 2015-2016.

Indicator	Ganderbal	J&K
Expected pregnancies	5972	251800
ANC Registration	3900	252978
percent ANC Registration against expected pregnancies	85	100.4
percent 1 st trimester Registration to ANC Registration	94.2	54
percent JSY Registration to ANC Registration	97.1	81
percent 3ANC checkups to ANC Registration	142	91
percent TT2/Booster to ANC Registration	81.2	55.1
percent 100 IFA to ANC Registration	44	29
percent Women with HB < 11 to ANC Registration	42	
percent having HB <7 treated at institutions	3	16
percent Institutional deliveries to reported deliveries	97	90
percent Newborns weighted at birth to live births	95	77
percent Newborns having weight <2.5 kg to newborns weighted at birth	0.4	6
percent Newborns breastfed within 1 hour of birth to total live births	10.4	69
percent Children less than 1 year Fully Immunized	113.3	125
Sex Ratio at Birth	980	938

located in Srinagar. ANC registration in the first trimester is 94 percent as compared to State average of 54 percent. JSY registration is 98 percent as against 81 percent in the State. The percentage of women receiving 3 ANC check-ups is very high in the district, indicating over reporting of 3 ANC under HMIS. TT coverage again in the district is far higher (81 percent) than in the State (55 percent). IFA coverage in the district is 44 percent as against 29 percent in the State.

Percentage of pregnant women having severe anaemia (Hb <7) treated at institution to women having HB level <11 of women in the district is lower (3 percent) as compared to State average of 16 percent. The district has good coverage of institutional deliveries (97 percent) against 90 percent in the State. Twenty percent of the reported institutional deliveries are conducted through C-section in the district as compared to 35 percent at the State level. Higher proportion of women in the district receives post-partum services as compared to most of the districts. The district has recorded a very good sex ratio (F/M) at birth of 980 which is higher than the State Sex ratio at birth of 938

females per thousand males. However, the births underweight (0.6 percent) is also lower than the State (6 percent) average. Percentage of newborns breast fed within 1 hour in the district is very high (93 percent) as compared to 69 percent in the State. The district also shows a low rate of abortions (3 percent) than the State (7 percent) average.

HMIS Reporting

The District Programme Management Units (DPMU) in J&K was established in November 2008 and the Block Programme Management Units (BPMU) was put in place in 2009-2010. Almost all the positions in the DPMU and BPMU in Ganderbal district are in place. The districts started reporting its monthly performance from April, 2008 initially in the form of hard copies and later on when the DHIS portal was put in place, the blocks started uploading their data on this website. From July, 2011 district started facility-based uploading of information on HMIS portal. An analysis of information on HMIS portal shows that there are a total 67 health institutions in Ganderbal which include 1 District Hospital, 1 Communi-

ty Health Centre, 1 Mother and Child Hospital, 19 Primary Health Centres, 43 Sub Centres and 2 Urban Family Welfare Centres. During the last two years (2014-16), all the facilities had submitted their monthly performance reports to the concerned BPMU. The BPMUs have examined these reporters and upload them on HMIS web portal. It needs to be mentioned that although Kashmir valley experienced worst floods during September, 2014 but it needs to be appreciated that all the facilities in the district had timely submitted their monthly progress reports and were also uploaded on HMIS web portal.

Formats Used for Reporting

A set of formats to be used by various health facilities for monthly reporting have been developed by the Ministry and the health facilities are required to use the appropriate format for monthly reporting. An examination of the formats used for reporting shows that all the facilities use the latest HMIS formats for reporting of information.

We were informed by the officials of the PMU Unit that reporting has improved considerably since June 2012 and all the health facilities are now regularly reporting. However, we thought it prudent to verify the authenticity of this statement by physically checking the availability of these formats at the selected BMO offices for the month of October, November and December, 2015. An examination of the formats made available to us showed that all the health facilities in the district during the last three months (October-December 2015) had regularly submitted their Monthly Progress Reports on HMIS format to BPMU for uploading.

Training

Trainings for HMIS were organised in the district during 2012-13. The Block Monitoring and Evaluation Officers have participated in HMIS train and are fully conversant with HMIS terminologies,

uploading of data and various consistency and validation checks. Almost all the FMPHWs and other staff involved with HMIS have also received HMIS training. During our visit to the selected health facilities, we found all the FMPHWs had clear understanding of various HMIS indicators and were also fully conversant with preparations of HMIS monthly formats. To gauge the level of training among the staff who is involved with reporting, we asked a question regarding the difference between HMIS and MCTS. Surprisingly, none of the workers except BME&O could clarify the difference between HMIS and MCTS. Since most the FMPHW prepare the reports from MCTS registers, they have a tendency to report all the services irrespective of the fact whether the beneficiary has received the services from the concerned facility or from any other facility. Thus, when HMIS reports are prepared from MCTS registers, violates the concept of facility-based reporting under HMIS.

DATA QUALITY ISSUES

Primary Health Centre Shuhama

The HMIS monthly reporting formats of the PHC for the 3 selected months were collected from the PHCs and the web portal information of the PHC for the same period was downloaded from HMIS website. These two types of information were compared with the information contained in the registers maintained by the PHC. At the first instance it has been seen that there is no variation between web portal information and information reported on HMIS format so far as ANC registration is concerned. Both web portal and HMIS format submitted by the PHC shows that 34 women have been registered for ANC at the PHC during the last 3 months (Table 16.3). But while checking the registers the total number of pregnant women who have been registered for ANC at the PHC during the last 3 months is 21. Thus 13 women could not be traced from the ANC registers maintained at the PHC.

This variation between the portal and registers at the PHC can be observed for all the 3 months. Similarly, if we look at the information about 1st trimester registration, TT1 Injection, TT2 Booster and IFA, web portal and HMIS format matches completely but the actual information in the registers does not match with HMIS format. HMIS portal shows higher number of services delivered than registers for almost all ANC indicators with

for all women currently registered for ANC services but HMIS portal and format shows 5 women have been provided IFA from the PHC. FMPHW could not give any reason for this mismatch. The FMPHW mentioned she does not properly consult the registers at the time of preparation of the report. She generally looks at the previous report and modifies the figures according to her perception. If we look at the HB information, web portal and HMIS format shows that 44 women have a

Table: 16.3 Matching of HMIS records for ANC indicators and Child Immunization reported by PHC Shuhama and SC Gulab Bagh and SC Khullmulla During Oct-Dec, 2015.

Indicators	PHC Shuhama			SC Gulab Bagh			SC Khullmulla		
	Web Portal	HMIS Format	Records	Web Portal	HMIS Format	Records	Web Portal	HMIS Format	Records
No of Women registered for ANC registration	34	34	21	9	8	6	15	15	10
1 st Trimester registration	34	34	20	9	8	4	15	15	6
TT1 Injection	34	34	17	9	7	6	15	15	11
TT2 /Booster Injection	25	25	24	0	0	4	17	17	15
100 IFA Tablets	5	5	10	2	2	4	0	0	17
HB<11 gms	44	44	0	0	0	4	0	0	0
HB<7 gms	0	0	0	0	0	0	0	0	0
BCG	0	0	3	0	4	4	0	0	0
Measles-1	12	12	12	15	22	18	9	8	8
Measles-2	21	21	22	17	17	9	7	7	7
Births	0	0	0	0	0	0	0	0	0
Deaths	0	0	0	0	0	0	0	0	0
IUD insertions	4	4	NA	0	0	NA	0	0	NA
Condoms distributes	310	210	NA	360	360	NA	0	0	NA
Oral Pills distributed	13	7	NA	16	15	NA	0	0	NA
ECs	0	0	NA	0	0	NA	0	0	NA

the exception of IFA. While 17 women have received TT1 injection but both portal as well as the format shows that 34 women have received TT1 from the PHC.

IFA is not available at the PHC for the last four months and the IFA column in this register is blank

HB less than 11, but there is hardly any information available with the PHC about HB. PHC has a laboratory which is handled by one lady technician. Some blood and urine tests are conducted at the PHC. The laboratory maintains a register of tests conducted at the PHC but only the name of the patient and the nature of tests are entered

in the register. The lab does not maintain any information about the results of the investigations conducted. But if we see the ANC register, all women registered at the PHC are shown to have normal HB and most of these women have actually visited a private clinic for HB testing. Thus while the information for HB should have come from Lab Register, facility reports HB information from ANC/MCTS register. This is the reason why there is a lot of variation between the information on web portal and actual information available at the PHC.

Facility of routine immunization and BCG is available at the PHC. This centre is maintaining information about routine immunization in immunization register and MCTS register. There is also a third register which is used to record information about immunization given to children who do not belong to PHC area. While preparing the HMIS report, children who have received immunization from PHC but are outside the catchment area of PHC are not included. ANM mentioned such children who do not belong to the PHC area but receive BCG from the PHC will be reported by the concerned SC. This step has been taken to minimize the duplication of immunization reporting. Thus the information available on web portal and HMIS format does not match with the information contained in the register. For example, PHC has given BCG vaccination to 3 children during the reference period but has not reported them in HMIS format or HMIS portal. Medical Officer reported since these 3 children are outside the PHC area and they do not report any service that has been provided to any person not belonging to the PHC area as the same will be reported by the facilities serving them. This is a serious issue as it has the potential to under-report the performance of the facilities. There are no data reporting issues with other doses of child immunization and measles as the number of children to have received these doses is same in web portal, format and register.

The total number of children aged 0-11 months who have received full immunization is another indicator where there is some confusion. As the registers do not have any column for reporting full immunization, the reporting person generally does not take into account the age of child and various doses given to the child while calculating full immunization status and he/she usually takes measles vaccination as a proxy indicator for reporting full vaccination; this is the reason that the number of children fully immunized and the number of children who have received measles vaccine is same.

Information about IUDs inserted is available at the PHC. There is no difference in the number of IUDs inserted between format and web portal. However, the PHC does not maintain the information about Oral Pills (OPs) and condoms. HMIS formats show that 210 OPs have been distributed but at the time of uploading, 310 have been uploaded. The FMPHW mentioned that this mismatch is due to oversight/typing error. Similarly, there are some discrepancies in the number of condoms distributed by the PHC. In the absence of any records about contraceptives, PHC reports the information about contraceptives from stock register.

PHC is supposed to record all births that take place at the PHC. Births that take place at home in the PHC areas are also to be reported by the PHC. Similarly, PHC is also supposed to register and report all deaths that take place at the PHC or at home in its catchment area. But it was found that PHC neither maintains any register for home deliveries nor reports the home births. Similarly, it does not have a register to maintain information about deaths that take place at the PHC or deaths taking place at home in PHC's own catchment area. It was also found that the HMIS format available with the PHC is without the line listing of deaths section. They are not aware that they have to report all births and deaths that take place at home in PHC catchment area.

Sub Centre Gulab Bagh

Gulab Bagh Sub Centre is located on the premises of Madri Meharbaan School which is a private school. While examining the ANC indicators uploaded by SC Gulab Bagh on HMIS portal with HMIS formats, it can be seen that web portal and HMIS Format figures for 3 ANC indicators namely ANC registration, 1st trimester registration and TT1 do not match. Web portal figures are somewhat higher than the figures recorded in HMIS format. For example, HMIS format shows 7 women have received TT1 from SC but HMIS portal shows 9 women have received it from the SC. But if we look at the ANC register, only 6 women have received TT1 (16. 3) from the SC. Thus it appears that data fudging not only takes place at the SC level but it also takes place when the SC data is uploaded by the Data Entry Operator at the BPMU level.

As per the records maintained by the SC, 4 women have received 100 IFA Tablets from SC during the last 3 months, but both the web portal and HMIS format shows only 2 women have been provided IFA. When the FMPHW was asked to mentioned why they have not reported these 2 women in progress report, they mentioned that these two women do not belong to the SC area and they are reporting the services provided to the population of the SC area. Thus services given by the SC to population belonging to outside area are not reported. This results in under-reporting of some services. Thus, under-reporting of IFA is one of the causes that have resulted in low IFA distribution in the district.

HB testing kit has been provided to the SC and the FMPHWs are trained to use this kit, but it was found FMPHWs posted at the SC are unable to use the HB kit due to the non-availability of HB testing strips. Pregnant women generally visit the local DH or a private diagnostic facility for HB testing and therefore, they do not count such wom-

en when reporting HB status under HMIS. However, SC keeps a record of HB of the women who are registered with it.

Although BCG facility is not available at the SC even then it has recorded 4 cases in the register and HMIS format. But web portal does not show any BCG cases (Table 16.3). FMPHW reported these 4 cases had received BCG from DH and they were not aware that they have to report only those cases which have received the services from the SC. The BPMU, however, clarified that a decision has been taken that BCG will only be reported by the PHCs and CHC and SCs will not report BCG as the BCG facility is not available at the SC. He also mentioned that if any facility where BCG facility is not available reports BCG in HMIS format, he does not upload it to avoid duplication.

There are some serious data quality issues with Measles-1 and Measles-2. As per the register SC has provided measles-1 to 18 children but has reported 22 children in HMIS format and web portal shows 15 cases. Similar data quality issues can also be observed with Measles-2. There are only 9 children who have received Measles-2 from SC but 17 cases have been uploaded on web portal. Thus it can be concluded that HMIS data of SC Gulab Bagh has serious data quality issues and BPMU needs to visit this SC to monitor the data recording and data reporting issues and take necessary steps to rectify the HMIS issues at the facility.

There are no data quality issues so far as OPs and condoms are concerned. SC maintains a births and deaths register. The births register contains information about all births that have taken place in the SC area. FMPHWs mentioned that home deliveries generally do not take place in the SC area. Therefore, they have not reported any birth during the last 3 months. SC also maintains a death register of SC area. But the HMIS format provided to the SCs does not have any Death Reporting Section; with the result it does not report deaths that take place in the SC Village.

Sub Centre Khullmulla

SC Khullmulla is located in a single-storey government building. Two FMPHWs are posted in the SC. SC is responsible for registration of women for ANC registration and providing ANC services. It also provides ANC services and counselling related to pregnancy care, delivery care and child care. SC also provides information about various family planning methods, breast feeding and immunization. Immunization services are also provided by the SC. SC is also supposed to conduct HB and pregnancy detection tests.

SC maintains a number of registers to record the services provided by them. Two important registers maintained by the SC are ANC register and Immunization register. SC also maintains MCTS registers for tracking of mothers and children. HMIS formats have been provided to the SC for reporting their monthly progress reports. SCs generally compile the monthly progress reports and submit the HMIS report by 20th of each month to the Block Programme Management Units (BPMU). BPMU uploads the facility-wise HMIS reports.

While examining the ANC indicators uploaded by SC Khullmulla on HMIS portal with HMIS formats, it can be seen that web portal and HMIS format figures are exactly matching for almost all indicators. But when we compare the HMIS format figures with registers maintained at the SC, they do not match for any of the ANC indicators. HMIS format figures are higher than the figures recorded in the registers. For example, while only 10 women have been registered for ANC services in the registers, but both portal and formats have entered 15 women for this indicator. The SC also has over reported other ANC indicators like 1st trimester registration, TT1 injection and TT2/booster (Table 16.3).

So far as IFA is concerned, there is a lot of vari-

ation in the number of IFA beneficiaries recorded in the registers and web portal. HMIS portal and formats have not reported any IFA beneficiary, but ANC register shows 17 women have received 100 IFA from the SC. However, it was observed that there was no IFA supply at the SC for the last so many months. FMPHW has reported IFA from MCTS register and these women had obtained IFA Tables/Syrup from some other facility or from a medical shop.

HB testing kit has been provided to the SC and the FMPHWs are trained to use this kit, but it was found FMPHWs posted at the SC are not using HB testing kit. The anaemia column in the ANC register/MCTS register was found to be blank. HMIS format and the web portal shows that there are no anaemic women in the SC. FMPHWs mentioned that women generally visit DH or a private facility for HB testing and therefore, they do not count such women when reporting HB status under HMIS.

BCG facility is not available at the SC and SC has not reported any BCG case in HMIS format and web portal. FMPHW reported that if any of their children receive BCG from any facility, they just keep a record of such doses but do not report it to avoid multiple reporting. Thus there is confusion between PHC and SC on the issue of BCG recording and reporting. There are no serious data quality issues with Measles-1 and Measles-2. But SC has reported 1 case more in the web portal than is reported in the HMIS format. FMPHWs mentioned that this is probably a typing error.

Oral Pills and Condoms are available at the SC. FMPHWs posted at the Centre mentioned that they provide OPs and other contraceptives but they are not maintaining any information about the distribution of Oral Pills and Condoms. Consequently, it has not reported the information about Condoms and Pills and web portal also is silent on these two indicators.

PHC Kachhan

PHC Kachhan is housed in a government building. Like other PHCs, it offers basic health care services, ANC, immunization services. AYUSH services are also available at the PHC. An analysis of information available on web portal and that contained in HMIS format shows that there are no differences in ANC indicators between the two sources. But information about IFA, BCG and Measles-2 does not match between these two sources. For example, web portal has reported 7 women less than reported in HMIS format. Similarly, PHC has reported 16 BCG cases but web portal has not reported any BCG case. However, if we match the information contained in the register with web portal, most of the ANC indicators except IFA recorded in the register are higher than reported on web portal. PHC has not distributed IFA to any of the ANC cases as per register, but 16 cases have been reported in HMIS format and 9 have been reported on web portal (Table 16.4). No justification was given by the FMPHW for this mismatch.

HB testing facility is available at the PHC and the lab maintains a register of the lab tests conducted. It was found that it contains information about 5 women who have been tested for HB. But this register maintains the name of the women and the test conducted and does not mention the result of the tests conducted. Therefore, with this register it is difficult to ascertain the number of women who have a HB level of < 11 or women who have a HB level of < 7 . However, if we look at the HMIS format and web portal information, 18 women are reported to have a HB < 11 . When the FMPHW who has prepared the HMIS format was asked to explain the reason for this mismatch, she explained that they do not collect the information about HB from the lab and instead report it from MCTS register. It needs to be mentioned that the information about HB status in MCTS register includes the tests done at other facilities and private clinics. Thus information about HB status on web

portal is over reported.

Facility of routine child immunization is available at the PHC. Since deliveries do not take place at the PHC, therefore, BCG facility is not available at this PHC. PHC is maintaining information about immunization in immunization register and MCTS register. There is also a third register which is used to record information about immunization given to children who do not belong to PHC area. Sixteen children from the PHC area had received BCG from other health facilities at the time of delivery but this PHC had included these 16 in its MCTS register and also included them in the HMIS format but Data Entry Operator has dropped all these 16 cases from the PHC at the time of uploading to avoid duplication. Similarly PHC has recorded higher number of children who have received Measles-2 in the HMIS format and in its immunization/MCTS register but the DEO reported that he included only those children in the web portal who actually have received Measles-2 from this PHC and this was done in consultation with the FMPHW. While it appears that the FMPHW does not have clear understanding what is to be recorded and what is to be reported but at the same time fudging of data takes place at the PHC level as well as at the time of uploading of information.

Facility for IUD insertion, oral pills and condoms are available at the PHC. Information about IUD insertions is available at the PHC. But information about OPs and Condoms is not properly maintained. There is no difference between the figures reported and the figures uploaded so far as family planning methods are concerned. But there is a need to properly maintain such information in the PHC. The PHC does not report deaths as the HMIS format available with the PHC does not have a Deaths section. FMPHW mentioned that they do not report any home deliveries as they are not aware that PHC has to report the home deliveries of its catchment area.

Table: 16.4 Matching of HMIS records for ANC indicators and child immunization reported by PHC Kachhan and its SCs During Oct-Dec, 2015.

	PHC Kachhan			SC Gund Rehman			SC Harran		
Indicators	Web Portal	HMIS Format	Records	Web Portal	HMIS Format	Records	Web Portal	HMIS Format	Records
No of Women registered for ANC registration	18	18	17	15	15	12	4	4	4
1 st Trimester registration	18	18	16	14	14	9	4	4	4
TT1 Injection	16	16	20	7	9	11	5	5	5
TT2 /Booster Injection	12	12	19	0	0	0	6	6	6
100 IFA Tablets	9	16	0	9	0	5	15	0	15
HB<11 gms	18	18	5	0	0	1	0	0	3
HB<7 gms	0	0	0	0	0	0	0	0	0
BCG	0	16	16	0	0	0	0	0	0
Measles-1	18	18	16	2	0	2	8	8	8
Measles-2	8	14	14	0	0	2	7	7	7
Births	0	0	0	0	0	0	0	0	0
Deaths	0	0	0	0	0	0	0	0	1
IUD insertions	3	3	3	0	0		0	0	
Condoms distributes	983	983		50	50		46	0	
Oral Pills distributed	41	41		0	0		0	0	
ECs	0	0		0	0		0	0	

Sub Centre Gund Rehman

SC Gund Rehman is housed in a government building. While examining the ANC indicators reported by this SC on HMIS format with registers, it can be seen that the information contained in the registers does not match with the information contained in HMIS Format and HMIS Format figures do not match with the information uploaded on HMIS web portal for any of the ANC or immunization indicator. For example, against 12 ANC registrations in the register, there are 15 on HMIS format. Similarly, for 9 1st trimester registrations, SC has reported 14. It appears that while the SC is deliberately over reporting its performance, it also does not count properly the services rendered

by it. This is substantiated by the fact that it has reported 9 women who have received TT1 but if we look at the register 11 women have received TT1 from the SC. Similarly, SC records show that 5 women have received IFA from SC but IFA column in HMIS format is blank and web portal has reported 9 women to have received IFA from SC.

Table 16.4 shows that there is a complete mismatch in the number of children receiving vaccination as reported by the SC in 3 sources. Immunization register maintained by the SC has information about 2 children who have received Measles-1 and Mesles-2 from the SC, but the HMIS format is blank in both cases and web portal shows 2 children have received measles-1 and none has

received mesles-2. Thus it appears that data fudging not only takes place at the SC level but it also takes place when the SC data is uploaded by the Data Entry Operator at the Block level.

HB testing facility is not available at the SC. Therefore, all columns pertaining to HB are blank in HMIS format and website has entered '0' against these columns. But a look at the MCTS register shows that all women registered with the SC for ANC during the last 1 year have a normal HB. It was observed that FMPHW fills up this column in the register without looking at the HB tests conducted by the women at a private clinic or any other facility.

IUD insertion facility is not available at the SC. However, condoms and oral pills are available. SC has no records about condoms and oral pills. But HMIS format has reported 50 OP cycles and web portal also reports 50 OP cycles.

SC maintains a births and deaths register. The births register contains information about all births that have taken place in the SC area. FMPHWs mentioned that home deliveries generally do not take place in the SC area. Therefore, they have not reported any birth during the last 3 months. SC also maintains a death register of SC area. But the HMIS format provided to the SCs does not have any Death Reporting Section; with the result it does not report deaths that take place in the SC Village. Surprisingly, BPMU also has never asked the SC to report the deaths that take place at home in the SC catchment area. FMPHW has a notion that she has to report the deaths that take place only at SC (institutional deaths).

Sub Centre Harran

SC Harran has comparatively a better HMIS. Adequate copies of HMIS formats are available at the SC. ANC and immunization registers are properly maintained. This is the only SC where there is no mismatch in figures among the three sources of information so far as ANC registration, 1st trimes-

Table:16. 5 Overall Differences in Recording, Reporting and Uploading in selected HMIS Indicators in Ganderbal during Oct-Dec, 2015.

	Record	HMIS Format	Web Portal
No of Women register for ANC registration	70	94	95
1 st Trimester Registration	59	93	94
TT1 Injection	70	86	86
TT2 /Booster Injection	68	60	60
100 IFA Tablets	51	23	40
HB<11 gms	13	62	62
HB<7 gms	0	0	0
BCG	23	20	0
Measles-1	64	68	64
Measles-2	61	66	60
Births	0	0	0
Deaths	1	0	0
IUD insertions	3	7	7
Condoms distributed	NA	1603	1749
Oral Pills distributed	NA	63	70
ECs	NA	0	0

ter registration, TT1 and TT2 is concerned (Table 16.4). Immunization information is also fairly matching in all the 3 sources.

Oral Pills and Condoms are distributed at the SC but records are not maintained at the SC. SC has not reported any performance against Condoms and Oral Pills. While web portal also has not reported any performance against condoms but 46 OP cycles are reported by web portal.

The SC maintains a birth and death register. The death register is well maintained. One death has taken place in the SC area at home during the last three months. SC has maintained all relevant details about this death in the death register but has

not reported it in HMIS format. FMPHW reported that they have never been asked to report the deaths in HMIS that have taken place at home in the SC catchment area.

Combined Analysis

When we combine the recording reporting and uploading of all facilities visited by us in Ganderbal district, it can be found that the mismatch found between the figures reported in HMIS format and that of uploaded on web portal for most of the ANC indicators gets wiped out (Table 16.5). The figures reported in HMIS format and web portal exactly match for 4 selected indicators namely number of women register for ANC registration, 1st trimester registration, TT1 registration and TT2 /Booster injection. IFA supply seems to be over reported by web portal. Thus overall, we can conclude that information pertaining to 4 selected indicators namely no of women register for ANC registration, 1st trimester registration, TT1 registration and TT2 /Booster injection is matching in HMIS formats and Web portal. But IFA reporting and uploading is not matching with each other indicating data fudging in IFA takes place at the uploading stage.

The figures reported in HMIS format and web portal also match in case of number of women with HB<11, but portal figures are generally highly over reported as compared to registers. This indicates that some women from Ganderbal who have been tested for HB in facilities located in Srinagar-based facilities have been reported by Ganderbal-based facilities. BCG seems to have a serious problem in Ganderbal as no performance has been reported on web portal, despite the fact that about 20-23 BCG doses have been recorded and reported by the selected facilities.

Since Ganderbal district does not have much variation in records and web portal information pertaining to various doses of immunization, the explanation given by the FMPHWs that some

children who get immunizations from SC/PHC but are outside the attachment area of these facilities are reported on web portal by the respective facilities is not substantiated by these figures.

So far as IUD figures are concerned, there are no data reporting and uploading issues in Ganderbal but it was seen that most of the facilities do not maintain information about temporary methods of family planning.

7. DISCUSSION WITH HEALTH FUNCTIONARIES

All the problems identified above have been discussed with the concerned health officials. The health officials have assured that they will address all these issues. Medical Officers and BPMUs were requested to bring these data issues into notice of all officials concerned with recording, reporting, uploading and monitoring and supervision of HMIS data so that remedial measures can be taken to rectify the data quality issues. They were also requested to issue uniform directions about reporting of services delivered to clients who do not belong to their service area.

We requested the District and Block Programme Management Units to regularly monitor the HMIS reports and visit the SCs and PHCs in a phased manner and match the HMIS reported information with actual records and try to help the ANMs in improving reporting.

The Lab Technicians posted at the PHCs mentioned that they are not provided any stationery like registers for maintaining information about the tests that are conducted in the laboratory. MOs were requested to provide the requisite stationery to Lab Technicians and Lab Technicians were requested to record the results of the investigations conducted in the lab and also to maintain a separate register for ANC cases, so that it is easy to count and report the HB indicators for HMIS.

8 CONCLUSION AND RECOMMENDATIONS

Our physical examination of information contained in various registers and information reported in HMIS formats and the discussion held with FMPHWs, BPMUs and Medical Officers and Lab technicians reveals the following facts about recording, reporting and uploading of HMIS at SC and PHC level in Ganderbal district.

1. Almost all FMPHWs have no clarity between HMIS and MCTS. When HMIS reports are prepared from MCTS registers, this completely violates the concept of facility based reporting in HMIS.

2. MCTS and ANC Registers are available in all facilities. FMPHWs generally use MCTS register for preparing HMIS format and thus include some services which the women have received from other health facilities including private health facilities. This results in over reporting and duplication of services.

3. Recording of information pertaining to IUD, Condoms and Oral Pills seems to be a neglected area. Registers are generally not available or even if they are available they are not updated. Figures pertaining to condoms and Oral Pills are hypothetically reported.

4. Generally, services are not denied if the clients do not belong to the catchment area of SC/PHC. But facilities generally do not report such services because it is assumed that such services will be reported by the facility with which the client is actually linked. This step has been taken to stop the duplication, but it is resulting in under reporting of services

5. IFA was not available in most of the facilities visited by us and, even if it was available, its supplies are irregular and erratic. Since the FMPHWs have to fill up the IFA column in MCTS reg-

ister, they generally resort to fudging of this indicator both on registers as well as while preparing HMIS report. There is a need to streamline the IFA supply and FMPHWs need to be sensitized about proper reporting of this indicator.

6. HB testing equipment was available in all SCs but HB tests are not conducted due to lack of training or non-availability of strip. Most FMPHWs report HB level from MCTS register, despite the fact most of these tests have been conducted at PHCs/ CHCs/DH. Since the facilities where such investigations are conducted are also reporting it under HMIS, therefore HB indicators have the potential to get reported twice under HMIS.

7. All the PHCs have laboratories. But Lab Technicians are not fully aware about the importance of HMIS. They generally do not record the results of investigations conducted and there is therefore no record available with the PHC which they can use to fill up the HB indicators in HMIS format. Therefore, HB reporting (HB<11 and <7) is far from grassroots reality. There is a need to provide them requisite stationery and develop a uniform register for recording the results of investigations, so that they can report the HB indicators properly.

8. SCs and PHCs have maintained registers for recording of deaths. But FMPHWs at SC and PHCs have a confusion that they have to report only those deaths that have taken place at the facility and therefore home deaths are not reported. The HMIS format available with them does not have deaths section. Therefore, proper HMIS formats be provided to the facilities for reporting of deaths.

9. It was found that there are misconceptions regarding various HMIS reporting terminologies. Therefore, it is suggested that training of the health workers should become a continuous and integral part of the monthly meetings.

10. All the health facilities need to be encouraged to use HMIS data in planning, implementation and supervision of public health services. This will make the workers to realize the importance of good data quality.
11. There is a need to sensitize all the FMPHWs on the difference between MCTS and HMIS. Workers generally report from MCTS which contains information about services received by the women and children from other facilities. This results in over reporting of services (duplication). Clear cut instructions need to be given to all the health workers to report only those services which they have provided at their facility and not the services which clients have received from other facilities.
12. A lot of variations were observed in the figures submitted by the PHCs and SC and those available on web portal. Block Monitoring and Evaluation Officers are to be directed not to fudge the figures and instead return the formats back to the facilities for rectification of errors. There is also a need to investigate why these differences are there and who takes this decision to change the figures submitted by the facilities. BM&EOs should regularly visit the facilities for monitoring of HMIS. Block Medical Officers need to facilitate and cooperate in undertaking such monitoring visit.

CHAPTER - 17

Data capture in MCTS in Kerala – An assessment of accuracy of data

Oommen P.Mathew, Anil Kumar K, Mathew MC, Sachin K.V

Population Research Centre Kerala

Abstract

Mother and Child Tracking System (MCTS) is formulated to facilitate monitoring of maternal and child health services received by all pregnant women and children. The objective of the study is to assess the dropout rate in antenatal and birth registration, accuracy of MCTS data, level of maintaining MCH register, mode and frequency of contact by the JPHN and ASHA, mode of collecting information from beneficiaries, guidance provided on ANC, PNC and immunization of infant by JPHN. Kottayam district was selected for the study. From each of the five Taluks of Kottayam, one Block is selected at random. One CHC/ PHC was selected from each of the selected blocks and one non-vacant sub centre from the selected CHC /PHC at random was selected. Twenty antenatal women or mothers of child born or after 1st January 2014 from each sub centre were selected by house-to-house visit. A total of 88 antenatal women and 150 postnatal women were interviewed. It was found that 12.5 percent pregnant women were dropped out of MCTS. When considering the accuracy of major components of ANC services, the accuracy is not satisfactory for the reporting of TT1 (57 percent) and TT2 (45 percent). Accuracy for the other parameters are LMP - 82 percent and level of anemia - 60 percent and the timely maintenance of MCP card - 63 percent. The accuracy of data maintenance in MCH register is slightly better than MCTS data. Information from about half of beneficiaries regarding administration of TT1 and TT2 were found collected through phone. The level of guidance provided by JPHN on care taken during antenatal period, about JSY, JSSK etc. is not satisfactory. In MCTS data capture, 8.7 percent children were dropped from MCTS on immunization. Nearly eighty percent accuracy was found for the three doses of pentavalent and measles. The accuracy of MCTS data regarding BCG and Polio 0 dose is around 50 percent and the accuracy is poor for the reporting of Hepatitis B (49 percent). Guidance provided by JPHN on breast feeding, initiation of CF, management of diarrhoea and danger signs of Pneumonia were not satisfactory. MCP card is not maintained timely for 19 percent postnatal women. The proportion of collecting information for MCTS through phone from the beneficiaries is rather high for postnatal care services. Comparatively high target population, workload assigned other than MCH activities and non-availability of dedicated computer with high speed internet connection for HMIS/MCTS are some of the major problems experienced by JPHN which, in turn, affect the quality of MCTS data.

INTRODUCTION

Mother and Child Tracking System (MCTS) is formulated to facilitate monitoring of maternal and child health services received by all pregnant women and children. It is developed jointly by the Ministry of Health and Family Welfare and National Informatics Centre and it was launched in collaboration with States/UTs in December 2009. Data quality challenges in Health Information Systems may hamper the usefulness of such systems in achieving better health outcomes². There has been a dearth of systematic evaluations to gauge the data quality and implementation effectiveness of Health Information Systems³⁻⁴. The few Health Information Systems assessments that have been carried out in the developing world have demonstrated usefulness in not only highlighting Health Information System implementation challenges, but also in pointing the way towards implementation improvement measures⁵⁻⁸. The Mother and Child Tracking System is a beneficiary-specific database for MCH services delivered through the Indian public health system. It was launched in 2009 as part of a global trend towards harnessing e-health innovations in improving service delivery, and also because India's existing Health Management Information System (HMIS) was not meeting the service delivery needs of FHWs¹. As a Mission Mode Project (MMP) under the Government of India's National e-Governance Plan (NeGP), MCTS has clearly defined "objectives, scopes, and implementation timelines and milestones, as well as measurable outcomes and service levels"⁹. The MCTS is designed to capture and track all pregnant women (from conception up to 42 days post-partum) and all new-born children (up to 5 years of age). Its objectives are to ensure that¹⁰: (i) All pregnant women receive their full Antenatal Care (ANC) and Postnatal Care (PNC) services at the due times; (ii) Institutional deliveries for pregnant women, particularly for high risk mothers, are encouraged; and (iii) all children receive the full immunization schedule at the due times.

MCTS facilitates the service provider at the grass-roots level in delivering services to women and children according to their specific needs and supports health and family welfare managers and policy makers in measuring and monitoring the efficiency of the maternal and child health services. MCTS is a tracking mechanism based on a unique key provided to each mother and child which strengthens the monitoring mechanism. In the realization of policies of the government in reducing maternal and child mortality and morbidity, MCTS is perceived to be an important intervention. Ultimately, MCTS will act as a tool in enhancing the effectiveness of the public healthcare delivery system. MCTS made it possible to health providers to generate the services due to women and children and to track every pregnant woman during pregnancy and every newborn during the first five years of life so that basic maternal and child health services can be delivered to them.

The ASHA workers and JPHNs are assigned the duty of identifying women and children in their area and assigning them a unique key. This must be done effectively to facilitate services to all pregnant women and children. The vacant positions existing at the health department for JPHNs remain a hindrance in monitoring ASHA workers for the effective coverage of their area and correctness of the collected information.

From the MCTS uploaded data, it is not feasible to assess how much antenatal women and postnatal women skipped from MCTs registration. If a pregnant woman or child is excluded from MCTS, the service for the child for the rest of the period was omitted from the MCTS. There is no other relevant mechanism in the State to identify the dropout from MCT. Thus, one of the intentions is to find out the dropout rate in registration. For antenatal women, the date of LMP, dates of TT1, TT2, HB level and number of antenatal visits have been uploaded in MCTS server. For postnatal women, the date of delivery, place of delivery, dates regarding the immunization of Polio, Hepatitis, BCG, Pen-

tavalent and measles are uploaded in the MCTS site. The agreement between actual dates of immunization service and the MCTS uploaded dates is a measure of quality of MCTS data. Hence, this study is intended to find out how much is the discrepancy of MCTS data with that of actual dates of service and that of MCH register.

Objectives, Data and Methodology

The objectives of the study are

1. To assess the background characteristics of beneficiaries of MCT services.
2. To assess the dropout rate in MCTS registration of pregnant women and children
3. To assess the accuracy of data uploaded in the MCTS
4. To find out discrepancy between MCTS data and MCH register

The analysis of the MCTS data carried out by the State revealed that Kottayam district is one of the districts showing less error in MCTS data. State Health department asked to conduct an assessment of the quality of MCTS data in Kottayam district. Hence, Kottayam district was selected for the study to assess how much is the agreement of the finding of the secondary analysis with the field-level data.

The quality of MCTS services provided by the sub centre was assessed using a structured questionnaire. The dates of actual service on ANC registration, TT1 & TT2 injection, and HB level assessed during antenatal period, date of delivery, actual dates of child immunization were collected from the records kept by the beneficiaries. The accuracy of MCTS data was assessed later by comparing the actual dates of service with the MCTS portal data. Later, the information regarding all the above was collected from MCH register kept at sub centre to

find out the discrepancy between MCTS data. The dropout in antenatal registration and child registration were identified from the data collected from the household survey and MCTS data. The quality of service rendered by JPHN and ASHA workers on antenatal and child immunization services such as mode and frequency of contact by the JPHN and ASHA, mode collecting information from beneficiaries were also collected from the beneficiaries. The background characteristics of the beneficiaries were also collected during the survey.

Population, Sample size and sampling: The population of the study is defined as all antenatal women and mothers of child born or after 1st January 2014 in Kottayam district. The inclusion criteria are that the beneficiary should be either a usual resident or residing temporary for more than 6 months of period.

Sample size was calculated by the formula

where, p is the proportion of dropouts in MCTS

$$n = \frac{Z_{\alpha}^2 p(1-p)}{d^2}$$

registration and d is the Level of precision. From PRC Report series 2015-2, the proportion of dropouts in MCTS in Palakkad is 0.15, the Level of precision is taken as 30 percent. Hence, the sample size for the assessment of dropouts in MCTS is 240.

The Kottayam district has five taluks, and in order to get the adequate representation, all the five taluks were selected for the study. The analysis of uploaded figures of the MCTS data shows that there is not much variation in the quality of data between block PHCs. Hence, the chance of errors is equally likely to be among all the PHC. Hence, on simple random method, one block is selected from each taluk, two CHCs or PHCs selected from each of the selected blocks and one non-vacant sub

centre selected at random from the selected CHC/PHC. Each sub centre is further divided into four regions based on geographical boundaries and two regions selected at random. Using consecutive sampling, twenty-five antenatal/postnatal women selected through house-to-house visit from selected regions constitute the sample of the study. The period of data collection is June – July 2015.

Limitations : If an eligible woman household visited is not available in the household at the time of visit, due to shifting from her own house for delivery, the case was avoided from the sample.

Background characteristics: The background variables at the facilities and beneficiaries are described in this chapter. The details of selection of facilities and socio-economic and demographic variables of individuals selected for the study are analysed.

Background characteristics of antenatal women : Majority of the antenatal women are in the age group of 24-33 which constitute about 77 percent of the total sample, 12.5 percent of the antenatal women belong to the 19-23 age group and the remaining 10.2 percent belong to the age group of greater than 33. The other variables considered are religion, social class and type of residence, 59.1 percent of women belong to the Hindu religion and 36.4 belong to the Christian. The caste-wise break-up shows that about 65 percent of women fall in backward or SC/ST category, 52.3 percent coming under the BPL category. Nearly 30 percent of the women have degree or higher education compared to 20.5 among their spouses. About 93 percent of the women are house wives; occupation of husband shows that only a minor percentage (2.3) is employed in government sectors while 45.5 percent are employed in menial types of jobs named Coolie.

Table 17.1 Distribution of women selected for antenatal care services

Taluk	PHC	Sub Centre	ANC	PNC
Changanassery	PHC Madapally	Palamattom	13	9
	PHC Thrikodithanam	Amara	8	21
Kottayam	CHC Kumarakam	Kumarakam (N)	11	16
	PHC Thiruvappu	Chengalam (S)	5	18
Meenachil	CHC Edamaruk	Melukavu	8	12
	PHC Thidanadu	Chemmalamattom	6	17
Vaikom	CHC Edayazham	Pariyaram	13	14
	PHC Thalayazham	Thottakam	8	16
Kanjirappally	PHC Manimala	Karikkattoor	9	14
	PHC Parathode	Pazhoothadam	7	13
	Total		88	150

Table 17.1 discusses the selection of sub centre and antenatal and post-natal women from the district. A total of 88 antenatal women and 150 post-natal women were interviewed. Among the 88 ANC women studied, Palamattom and Pariyaram sub-centres have the highest numbers enumerated with 13 and Amara sub centre which comes under Thrikodithanam PHC has the highest number of postnatal women.

The obstetric background of antenatal women shows that, about 56 percent of them are in the second order pregnancy and 36.4 percentage are in their first order pregnancy, more than half of the women utilized government facility for their last delivery.

Background characteristics of postnatal women: About 38 percentage of the beneficiaries belong to

the 24-28 age group followed by 30 percentage in the age group 29-33; about 55 percent of them belong to the Hindu religion followed by Christian. Considering the ownership of ration card, it can observe that majority are APL card holders. Most of the women studied are residing as permanent residents and only 2.6 percentages has rented status for their residence.

Nearly 40 percentage of postnatal women and 32.5 percentage of their husbands have higher secondary education: 27.3 percentage of women are degree or above educated and half of the husbands are secondary educated. About 93 percentage of beneficiaries are house wives (92.7 percent) and most of their husbands (66.9 percent) are engaged in unorganized sector.

The delivery characteristics shows that, more than half of the beneficiaries utilized government facility for their delivery, 44.7 percent of the women are having first order birth and 39.3 percent were caesarean deliveries. The male-female gap in birth is admissible as the sex ratio is 1307 favouring female births and about 64 percent and 85 percent of women delivered at government facility utilized JSY and JSSK benefits respectively.

Accuracy of MCTS data on antenatal care services : The quality of MCTs service provided

by the JPHN and ASHA workers at sub centre-level in terms of the antenatal registration and server uploaded data for the parameters such as LMP, antenatal visits, TT1, TT2 and anemia are presented in this section. Eighty-eight antenatal women were interviewed for the study and the findings are explained below.

Antenatal dropouts : Out of the 88 pregnant women interviewed, 77 women were found registered in MCTS server. It means that 12.5 percent (95 percent CI – 5.6 – 19.4) antenatal women dropped from the MCTs. Among the non-registered 11 cases, 10 antenatal women were found excluded from the MCTS server and one women was reported as abortion in the server but the gestation age is at third trimester at the time of interview.

Discrepancy in LMP, TT taken and anaemia level : Out of the 78 women interviewed, the actual LMP date is found matched with the server figure of 82.1 percent. For 9 women, the date of LMP registered in MCTS server is later than the actual date and for 5 cases, the server date is before actual LMP date. The disparity in LMP date with the MCH register is found in 13 cases. For 8 women, the date of LMP entered in MCH register is later than the actual date and for 5 case, the register date

Table 17.2 Distribution according to accuracy of TT1b& TT2 date recorded in MCTS data and MCH register

	Difference	TT1	TT2
Difference in actual TT and MCTS data	No difference	42 (56.8)	28 (45.2)
	1 week before actual date	2 (2.7)	1 (1.6)
	1 week - 1 month before actual date	3 (4.1)	4 (6.5)
	>1 month before actual date	6 (8.1)	10 (16.1)
	Before actual date	11 (14.9)	15 (24.2)
	1 week after actual date	1 (1.4)	0 (0)
	1 week - 1 month after actual date	5 (6.8)	8 (12.9)
	>1 month after actual date	9 (12.2)	4 (6.5)
	After actual date	15 (20.3)	12 (19.4)
	Not uploaded in MCTC	6 (8.1)	7 (11.3)
	Total	74 (100)	62 (100)

Difference in actual TT and MCH register data	No difference	48 (64.9)	34 (54.8)
	1 week before actual date	2 (2.7)	2 (3.2)
	1 week - 1 month before actual date	2 (2.7)	4 (6.5)
	>1 month before actual date	3 (4.1)	5 (8.1)
	Before actual date	7 (9.5)	11 (17.7)
	1 week after actual date	2 (2.7)	1 (1.6)
	1 week - 1 month after actual date	5 (6.8)	5 (8.1)
	>1 month after actual date	5 (6.8)	3 (4.8)
	After actual date	12 (16.2)	9 (14.5)
	Not recorded in Register	7 (9.5)	8 (12.9)
	Total	74 (100)	62 (100)

is before actual LMP date.

Out of the 74 women who have taken TT1, the actual dates of TT taken matched with server data in only 56.8 percent cases. For 15 cases, the MCTS dates are later dates than actual dates and for 11 cases, MCTS dates are earlier than the actual dates. Six women who have already taken TT1 were not uploaded in MCTS server till the date of interview.

Table 17.3 Distribution according to Institution at which TT was taken and mode of collecting information for MCTS entry

	Institution	TT1	TT2
Institution from which TT taken	DH /GH/ MC	22 (29.7)	18 (29)
	CHC / PHC	14 (18.9)	11 (17.7)
	SC /AW	14 (18.9)	14 (22.6)
	Pvt Hospital	24 (32.4)	19 (30.6)
	Total	74 (100)	62 (100)
If Not SC, mode of collecting information	Personal contact by JPHN	25 (41.7)	17 (34.7)
	Personal contact by ASHA	11 (18.3)	7 (14.3)
	Through Phone	24 (40)	25 (51)
	Total	60 (100)	49 (100)

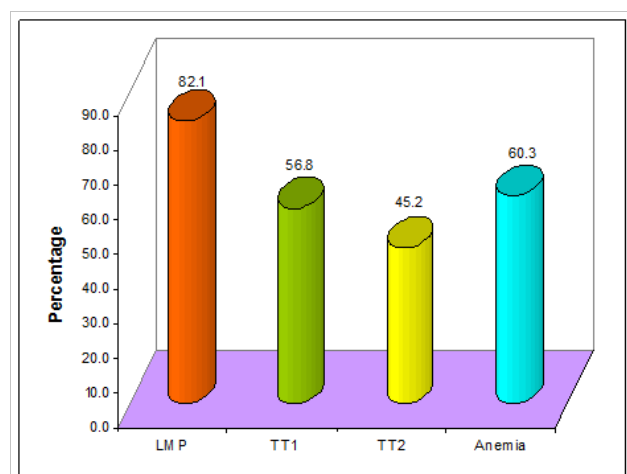
Out of the 62 women who have taken TT2, the actual dates matched with server data is only in 45.2 percent cases. For 12 cases, the MCTS dates are later dates than actual dates and for 15 cases MCTS dates are earlier than the actual dates. Seven women who have already taken TT2 were not uploaded in MCTS server till the date of interview. The mode of collection of information regarding date and place from which TT taken was sought from women who took TT from institutions other than sub centre. The information was collected by the JPHN through personal meet in 25 cases and ASHA workers collected information from the beneficiaries in 11 cases. The details collected by JPHN or ASHA workers through phone for 40 percent cases.

It shows that JPHN/ASHA collected the information regarding TT2 administration through phone from more than half of the sample women. In MCTS it can be observed as; Out of the 78 women, 59 were recorded as normal and nothing marked for 19 cases and no moderate anemia reported. As per lab report kept by the beneficiaries, 59 were normal and 19 were moderately anemic. Among the 19 moderately anemic cases 12 were reported as normal in MCTS while the remaining 7 were not recorded in MCTS. From the table it can be assessed that the agreement of level of anemia reported in MCTS is only 60 percent.

Graphical presentation of accuracy of MCTS server data on antenatal care services with actual data reported by beneficiary is presented in Fig. 1.

From the figure the agreement is high for the reporting of LMP date. The disparity is high for the reporting of TT2 vaccination.

Figure 17.1 Distribution according to agreement of antenatal MCTS data with actual data



Antenatal visits by JPHN : The dates of four antenatal visits made by JPHN have to be uploaded in the MCTS server. The server figures are gathered for the 88 women interviewed for the antenatal care services. The server figure shows that for 53 percent of women, JPHN carried out four antenatal visits, JPHN visited 3 times in the case of 20.8 percent women and two visits completed in 10.4 percent cases. For detecting the disparity in number of antenatal visits reported by JPHN in MCTS and actual visits made by JPHN, details of visits made by the JPHN during the ANC period were collected from the beneficiary.

Table 17.4 Distribution according to MCTS-reported ANC visit by JPHN and reported by beneficiary

No. of visit (as per MCTS)	MCTS Reported	Actual visit as reported by beneficiary				
		Not visited	One	Two	Three	Four or more
One	12	6	6			
Two	8	3	4	1		
Three	16	5	1	4	6	
Four	41	5	7	6	4	19

From Table 17.4, it can be seen that there is disparity in actual number of visit by JPHN and uploaded figures. MCTS figure shows that one ANC visit was carried out for 12 women, but the beneficiary survey revealed only 6 women were visited by the JPHN. Eight women were visited twice, according to MCTS report, but as per beneficiary report, one woman was visited twice, four women once while 3 women were not visited at all. While sixteen women were visited thrice as per MCTS, the beneficiary report suggests three visits to six women and no visit to 5 women. Among the 41, who were reportedly visited four times as per MCTS figures, only 19 were visited four times or more by JPHN as per the beneficiary report.

Accuracy of MCTS data on Child Immunization care services: The quality of MCTS service provided by the JPHN and ASHA workers for data uploading at sub centre-level in terms of the birth registration, immunization of infant at birth, after 4 week, 10 weeks, a 14 weeks and after 9 months are assessed and presented in this section. One hundred and fifty postnatal women were interviewed for the study and the findings are explained below.

Birth registration and maintenance of MCP cards: The registration of birth and uploading details of delivery such as date of birth of child, place of delivery and gender of infant etc. is the first stage of uploading in MCTS. The maintenance of MCP card kept by the beneficiary is also expected from the JPHN.

It is found that the date of birth of infant was correctly entered in MCTS in 92.7 percent cases. Gathering information from relatives through phone is the observed reason for disparity in the entry of date of birth. The MCP card is maintained up to date for about 81 percent of women interviewed. For 17 percent cases, the MCP card were not issued till the time of interview.

Postnatal dropout : MCTS has been devised as a name-based tracking system to ensure that all pregnant women receive their Antenatal Care Check-ups (ANCs), postnatal care (PNCs) and children receive their immunization full. It is not possible to ensure that all children receive the immunization full if dropouts occur in birth registration in MCTS. Out of the 150 postnatal women interviewed, 13 are found out of MCTS. The 95 percent CI for the dropout is (4.0 – 13.4). Among the 13 dropouts, the antenatal and postnatal data (of 8) were found missing from MCTS service. Thus it can be inferred that the tracking of antenatal care services were not possible for these 8 women and a higher chance for out of tracking of the immunization coverage of child in the coming years. For the remaining five cases out of 13 dropouts, antenatal care services were uploaded but birth registration and immunization details were not uploaded so far. Since all the five are within two months of postpartum, there is high chance of locating the women by the JPHN in the near future and upload the data regarding birth and immunization taken so far.

Discrepancy in immunization at birth: Polio zero dose, BCG and Hepatitis B are the three immunization administered to children at birth. The discrepancy in the uploaded figures of immunization at birth with dates of actual services are assessed and presented in this section. Among the 137 children whose birth was registered in MCTS, the dates of Polio 0 dose administration was found matching with the MCTs data in only 75 cases.

Table 17.5 Distribution according to disparity in Polio – 0 dose, Hepatitis B and BCG date recorded in MCTS

	Polio – 0 dose	Hepatitis B	BCG
No difference	75 (54.7)	68 (49.6)	81 (59.1)
1 week before actual date	27 (19.7)	24 (17.5)	29 (21.2)

1 week - 1 month before actual date	2 (1.5)	2 (1.5)	4 (2.9)
>1 month before actual date	1 (0.7)	1 (0.7)	1 (0.7)
Before actual date	30 (21.9)	27 (19.7)	34 (24.8)
1 week after actual date	29 (21.2)	28 (20.4)	19 (13.9)
1 week - 1 month after actual date	3 (2.2)	3 (2.2)	3 (2.2)
After actual date	32 (23.4)	31 (22.6)	22 (16.1)
Total	137 (100)	137 (100)	137 (100)

For 30 cases, the MCTs dates are before the actual dates of administration, and for 32 cases, the MCTS dates are after the dates of actual administration of vaccine. For 20 percent cases, the MCTS entry regarding Hepatitis B is on an earlier date than the actual administration date and for 22.6 percent cases, MCTS shows later date than the actual date. For 8 cases, Hepatitis B administered but not uploaded in MCTS and for 3 cases, the entry was wrongly made in MCTS while actually those children have not been given Hepatitis vaccine. For nearly 60 percent of cases, the MCTS figure for BCG exactly matched with the actual dates and for 25 percentage cases the dates found in MCTS is earlier than the actual date of administration.

The JPHNs are not in a position to collect the information personally during the time, they collect the information either through phone or from relatives. This is the main reason observed for disparity. In certain cases, the Hepatitis vaccine was not administered along with BCG or Polio because of lack of vaccine or some other reason, JPHN assumes all the vaccine were taken on the same date and enter the same dates for BCG, Polio zero dose and Hepatitis B. This is the reason for the wrong entry regarding administration of vaccines, while actually the vaccines were not administered.

The place of administration of Polio zero dose,

Hepatitis B and BCG, and mode of collection of information regarding place of administration for MCTS is presented in Table 17.6. The table shows that about half of the samples taken these immunizations from government facility. The information for MCTS collected from the beneficiary or relatives through phone for about 50 percent cases by the JPHN.

Table 17.6 Distribution according to place of administration and mode of collecting data for MCTS of Polio- 0 doze, Hepatitis B and BCG

	Difference	Polio – 0 doze	Hepatitis B	BCG
Place of administration	DH/GH/MC	79 (53)	77 (53.1)	83 (55.7)
	Private Hospital	70 (47)	68 (46.9)	66 (44.3)
	Total	149 (100)	145 (100)	149 (100)
Mode of collection	Personal contact by JPHN	46 (33.6)	45 (33.8)	45 (32.8)
	Personal contact by ASHA	20 (14.6)	19 (14.3)	25 (18.2)
	Through Phone	71 (51.8)	69 (51.9)	67 (48.9)
	Total	137 (100)	133 (100)	137 (100)

Discrepancy in immunization at 4th week, 10th week, 14th week and 9th month after birth: First, second and third doses of Pentavalent and Polio are scheduled for 4th week, 10th and 14th week after birth of children. In most of the cases polio is given along with pentavalent. The discrepancy of MCTS data with the actual administration of Pentavalent and measles was observed and presented in Table 17.7.

As compared to the accuracy observed in the administration of immunization at birth, the accuracy was much better in the management of data regarding administration of Pentavalent and measles vaccines. The MCTS data exactly match with these actual dates of immunization administration in about 75 percent cases. For 11 percentage cases of Penta 1st doze, MCTS shows later dates compared to actual date of administration. In 4 cases, Pentavalent was administered but not uploaded in MCTS.

The agreement between MCTS data and actual dates of administration of Pentavalent 2 is around 80 percent. For 10 percent cases, the MCTS-entered dates are before the actual dates of administration while for 8 percent cases, the entry dates are later dates as compared to actual dates of administration. The agreement between MCTS data with actual dates of administration of Pentavalent 3 is observed as 79 percent. It is observed that there is no specific error pattern in MCTS. The early and late entry in MCTS observed regarding Pentavalent 3 is 10 percent.

For 52 cases out of 69 children who administered measles vaccine, MCTS data perfectly agree with the actual dates of administration of measles vaccine. It is found that 5 children actually took measles but the information was not uploaded in MCTS till the date of interview and for 7 cases the dates uploaded in MCTS are later dates than the actual dates of administration.

The accuracy of maintaining information in MCH register is much better than MCTS figures. For 83 percent cases, the date of actual administration of Penta I and MCH register figures exactly matched. For Penta II, 86 percent cases in the MCH register information exactly agree with the actual dates of service. For Penta III, MCH registers shows that the information is correct in 80 percentage cases. For Measles, the agreement between actual dates of service and the information in MCHs register is only for 67 percent cases, which is less than the

Table 17.7 Distribution according to disparity in Pentavalent and Measles date recorded in MCTS and MCH register

	Difference	Penta I	Penta II	Penta III	Measles
Difference in actual date and MCTS date	No difference	103 (78)	99 (79.8)	90 (78.9)	52 (75.4)
	1 week before actual date	7 (5.3)	5 (4)	3 (2.6)	1 (1.4)
	1 week - 1 month before actual date	2 (1.5)	5 (4)	6 (5.3)	3 (4.3)
	>1 month before actual date	1 (0.8)	2 (1.6)	2 (1.8)	1 (1.4)
	Before actual date	10 (7.6)	12 (9.7)	11 (9.6)	5 (7.2)
	1 week after actual date	7 (5.3)	1 (0.8)	6 (5.3)	1 (1.4)
	1 week - 1 month after actual date	6 (4.5)	8 (6.5)	3 (2.6)	3 (4.3)
	>1 month after actual date	2 (1.5)	1 (0.8)	2 (1.8)	3 (4.3)
	After actual date	15 (11.4)	10 (8.1)	11 (9.6)	7 (10.1)
	Not uploaded in MCTC	4 (3)	3 (2.4)	2 (1.8)	5 (7.2)
	Total	132 (100)	124 (100)	114 (100)	69 (100)
Difference in actual date and register date	No difference	110 (83.3)	106 (85.5)	91 (79.8)	46 (66.7)
	1 week before actual date	3 (2.3)	4 (3.2)	6 (5.3)	2 (2.9)
	1 week - 1 month before actual date	1 (0.8)	2 (1.6)	3 (2.6)	5 (7.2)
	>1 month before actual date	1 (0.8)	2 (1.6)	2 (1.8)	3 (4.3)
	Before actual date	5 (3.8)	8 (6.5)	11 (9.6)	10 (14.5)
	1 week after actual date	6 (4.5)	2 (1.6)	6 (5.3)	2 (2.9)
	1 week - 1 month after actual date	5 (3.8)	4 (3.2)	2 (1.8)	5 (7.2)
	>1 month after actual date	3 (2.3)	2 (1.6)	2 (1.8)	2 (2.9)
	After actual date	14 (10.6)	8 (6.5)	10 (8.8)	9 (13)
	Not maintained in Register	3 (2.3)	2 (1.6)	2 (1.8)	4 (5.8)
	Total	132 (100)	124 (100)	114 (100)	69 (100)
Difference in MCTS date and register date	No difference	114 (86.4)	110 (88.7)	93 (81.6)	54 (78.3)
	Later date in MCTS than Register	5 (3.8)	2 (1.6)	7 (6.1)	7 (10.1)
	Later date in Register than MCTS	10 (7.6)	7 (5.6)	10 (8.8)	3 (4.3)
	Date not in Register but in MCTS	1 (0.8)	2 (1.6)	2 (1.8)	2 (2.9)
	Date not in MCTS but in Register	2 (1.5)	3 (2.4)	2 (1.8)	3 (4.3)
	Total	132 (100)	124 (100)	114 (100)	69 (100)

accuracy observed in the case of data maintained in MCTS server.

The comparison with MCTS information and MCH register information shows that for 14 per-

cent cases the dates are not matched for Penta I. The agreement of information between MCH register and MCTs server is almost 90 percent and 82 percentage for Penta II and Penta III. For 22 percentage cases, the information regarding Measles

Table 17.8 Distribution according to place of administration of Pentavalent and measles and mode of collecting data for MCTs

		Penta I	Penta II	Penta III	Measles
Place of administration	DH/GH/MC	37 (26.6)	31 (23.5)	20 (16.5)	5 (7.2)
	CHC/PHC	47 (33.8)	48 (36.4)	49 (40.5)	28 (40.6)
	SC/AW	29 (20.9)	31 (23.5)	34 (28.1)	24 (34.8)
	Private Hospital	26 (18.7)	22 (16.7)	18 (14.9)	12 (17.4)
	Total	139 (100)	132 (100)	121 (100)	69 (100)
Mode of collection	Personal contact by JPHN	40 (38.8)	31 (33.3)	28 (34.6)	11 (24.4)
	Personal contact by ASHA	21 (20.4)	24 (25.8)	17 (21)	9 (20)
	Through Phone	42 (40.8)	38 (40.9)	36 (44.4)	25 (55.6)
	Total	103 (100)	93 (100)	81 (100)	45 (100)

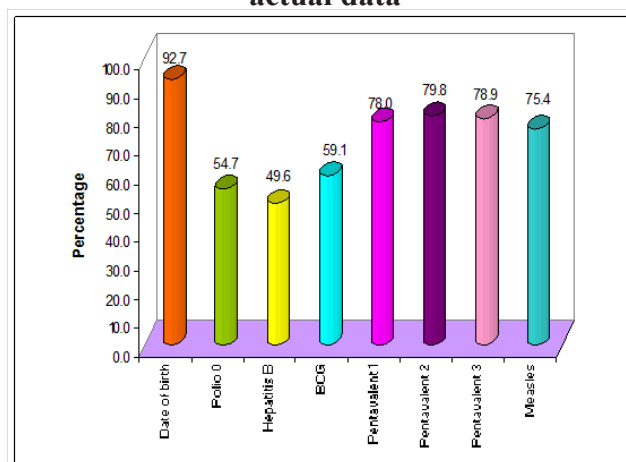
data of MCTS did not match with that in the MCH register.

The place of administration of immunization and mode of collecting data for MCTs were presented in Table 17.8. It can be observed that about 20 percent women utilize Sub centre for Pentavalent 1 administration. The utilization of SC is 24 percent, 28 percent and 35 percent for Penta II, Penta III and measles vaccines respectively.

Overall utilization of government facility for immunization after 4th week onwards is more than 80 percent. A major share of women opted private facilities for their delivery, but depend government facility for immunization.

The information regarding administration of vaccine was personally collected by JPHN in 39 percent cases while ASHA workers collected information personally in 20 percent cases. The information from the remaining 40 percent cases were collected over phone. The rate of collecting information over phone increases thereafter. The information collected over phone regarding Penta II, Penta III and Measles are 41 percent, 44 percent and 56 percent respectively. The higher percentage of collection of information over phone is may be the major reason for the disparity in actual service details and MCTS-entered figures.

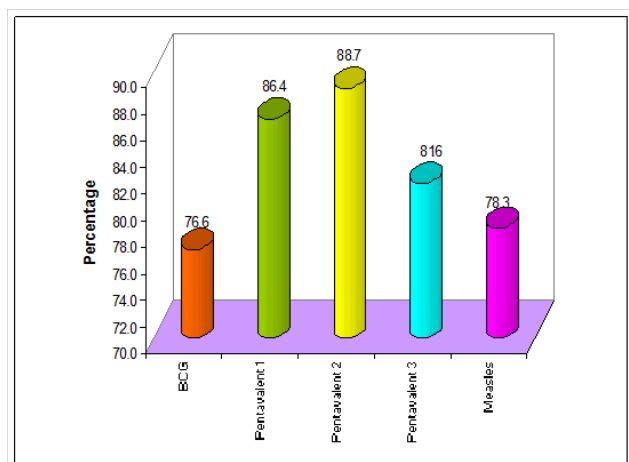
Graphical presentation of accuracy of MCTS server data on postnatal care services with actual data reported by beneficiary is presented in Fig. 2.

Figure 17.2 Distribution according to agreement of postnatal MCTS data with actual data

From the figure it is found that the agreement is high for the entry of date of birth. The accuracy in the reporting of Pentavalent 1,2,3 and measles vaccines is almost similar. The disparity is found more in the reporting of Hepatitis B.

Graphical presentation of agreement of MCTS figures with MCH register is presented in Fig. 3, it can be observed that the agreement of MCTS data with MCH register data is high for the reporting of Pentavalent 2. The disparity is found more regarding BCG and measles.

Figure 17.3 Distribution according to agreement of postnatal MCTS data with MCH register data



Summary : Kottayam district is selected for the study. Five Taluks constitute this district. To get the adequate representation of district, data was collected from all the five Taluks. From each Taluk, one Block is selected at random. One CHC or PHC was selected at random from each of the selected blocks and one non-vacant sub centre was selected at random from the selected CHC /PHC. The population of the study is defined as all antenatal women and mothers of child born or after 1st January 2014. Twenty antenatal/postnatal women selected by house-to-house visit from different geographical part of each sub centre constitute the sample of the study. The dropout rate in antenatal and birth registration, accuracy of MCTS data, level of maintaining MCH register, mode and frequency of contact by the JPHN and ASHA, mode of collecting information from beneficiaries, guidance provided on ANC, PNC and immunization of infant etc. were collected from the beneficiaries using a structured interview schedule.

It was observed that only 62.5 percent women had timely maintained MCP card. According to beneficiary report, only 73 percent knew the JPHN of their area and 92 percent knew the ASHA of their area. It was found that 12.5 percent pregnant women were dropped out of MCTS. The validation of recording of LMP dates revealed that MCTS date

agreed with actual data only in 82 percent cases. From the MCTS data regarding TT1, the actual dates of service have agreement for 57 percent cases, for TT2 the agreement is only for 45 percent cases. The agreement of MCH register data with actual service is slightly better than MCTS data. The institutions from which the service of TT taken shows, about 70 percent utilize Government facilities. For 40 percent cases, the dates of TT1 taken collected by JPHN over phone and for TT2, its share is about 51 percent. There is significant variation in the reporting of MCTS visits carried out as reported in MCTS and the actual ANC visits as reported by beneficiaries. Variation observed in MCTS reporting of anemia; among the 19 moderately anemic pregnant women, MCTS reported only 12 cases as moderately anemic .

The mode and frequency of visit by JPHN during antenatal period show that only 70 percent women were visited by JPHN, and among the personal visits, only 28 percent visited four or more times. It is found that ASHAs personally visited 87 percent cases, while there are four or more visits only in 42 percent cases. When considering the service rendered by the JPHN/ASHA during antenatal period, 72 percent women reported that the SC has not arranged any class on antenatal care services. The level of guidance on antenatal care services by the JPHN is found poor. Around 40 percent women received guidance on diet, exercise and sleep management and caring of newborns. The percentage of women received guidance on JSY and JSSK is below 25 percent.

MCTS data regarding child immunization revealed that, 8.7 percent children were dropped from MCTS. Among those found dropped out from MCTS, it is found that the details of antenatal registration of 8 women were not uploaded. The status of MCP card revealed that it was correctly maintained for 81 percent cases. When considering the accuracy of major components of child immunization services at birth such as BCG, Hepatitis B and Polio 0 doze, the exact agreement

of server data with actual service is around 50 percent only. The accuracy of MCTS date is more than 75 percent for immunization services carried out at 10th week, 14th week and 9th month after birth. The accuracy in maintaining MCH registers is more than that observed in maintaining MCTS data. The place of administration of immunization services shows that nearly 50 percent of immunization at birth were carried out at private hospitals. More than 80 percent postnatal women utilize government facilities for the immunization of their child charted for 4th week, 10th week, 14th week and 9th month after birth. When considering the way of collecting information regarding the administration of immunization by the JPHN for MCTS, it is found that in more than 40 percent cases JPHN collected information over phone.

When considering the mode and frequency of contact by JPHN and ASHA, for 70 percent cases JPHN personally visited the women at postpartum after delivery. The frequency of contact revealed that for 46 percent cases, JPHN visited three or more times. When considering the mode of contact by ASHA after delivery, 80 percent postnatal women were visited by ASHA. Frequency of visits shows, nearly 50 percent cases are visited four or more times.

Findings :

- MCP card not provided to 29.5 percent antenatal women
- Out of the 88 antenatal cases 10 were found dropped out from MCTS and one current pregnant case was reported as abortion in MCTS.
- MCTS dates regarding LMP, actual date matched for only for 82 percent cases.
- For TT1, the actual date matched with MCTS date only for 57 percent cases; for 65 percent cases, the MCH register date matched with actual dates, for 40 percent cases, the dates of

TT1 were gathered from beneficiaries over phone.

- For TT2, the actual TT2 dates matched for 45 percent cases, for 51 percent cases the dates of TT2 gathered from beneficiary over phone.
- There is high level of disagreement between the antenatal visits reported by JPHN in MCTS with antenatal visits reported by beneficiaries.
- ASHA workers not visited 10 women out of 88 and three or more visits by ASHA workers are for 54 cases.
- The level of guidance provided by JPHN on care taken during antenatal period, about JSY, JSSK etc. is found poor.
- MCP cards correctly maintained for 81 percent postnatal women
- Out of the 150 children, 13 were dropped out from MCTS. For 8 cases, both antenatal registration and child immunization were not uploaded in MCTS.
- Out of the 150 children, actual date of birth did not match with MCTS data for 10 children.
- For polio – 0 dose, date of actual service matched with MCTS data for 54.7 cases. The dates were gathered over phone in 51.8 percent cases
- For Hepatitis B, the date of actual service matched with MCTS data for 53.1 cases. For 8 cases, dates were not uploaded in MCTS and for 3 cases, actually Hepatitis B was not taken but dates were uploaded in MCTS. The dates gathered through phone for 51.9 percent cases

- For BCG, date of actual service matched with MCTs data only for 49.6 percent cases. The dates gathered over phone for 48.9 percent cases.
- For Pentavalent 1, date of actual service matched with MCTS data for 78.6 percent cases. The dates of MCTS and MCH register not matched for 18 cases. The dates gathered over phone for 40.8 percent cases
- For Pentavalent 2, date of actual service matched with MCTs data for 79.84 percent cases. Dates not uploaded for 2 cases. The dates of MCTs and MCH register not matched for 14 cases. The dates gathered over phone for 40.9 percent cases
- For Pentavalent 3, date of actual service matched with MCTS data for 78.9 percent cases. Dates not uploaded for 2 cases. For 21 cases, the dates of MCTS and MCH - registered figures did not match. The dates gathered over phone for 44.4 percent cases
- For Measles, date of actual service matched with MCTs data for 75.4 percent cases. For 15 cases, the dates of MCTS and MCH register figures did not match. The dates gathered over phone for 55.6 percent cases
- Only 105 women out of 150 cases visited by JPHN after delivery. There are 30 cases not visited after delivery by ASHA workers.
- Guidance provided by JPHN on breast feeding, initiation of CF, management of diarrhoea and danger signs of Pneumonia is not satisfactory.

A great share of first order pregnant women and a considerable share of higher pregnant women went to their own home for delivery and returned only after a minimum of two months after delivery. This is one major reason for higher rate of gather-

ing MCTS information over phone. When JPHN/ASHA collect information over phone, usually they do not ensure that the immunization cards are with the beneficiary at the time of call, this is the main reason for high disparity on the MCTS data on immunization at birth, 4th week after birth and even 10th week after birth. Since the workload is high for JPHN, a large number of JPHN blindly depend on ASHA and upload the figures provided by them without further enquiry. For example, one ASHA reported that an antenatal women aborted the pregnancy, without further enquiry the JPHN uploaded the fact in MCTS; during the survey it was found that the women is at 28th week of gestation. It is found that the ASHA workers collect information from the beneficiary without confirming the information with records and JPHN upload the information collected by ASHA directly from beneficiary without further enquiry. This is another reason for disparity between actual dates of service and MCTs data. The time schedules followed in Government hospital for the implementation of TT1 and TT2 during antenatal period are not followed by most of the private hospitals. If women was not given the TT as per the government schedule, there should be a gap in the work plan generated; to avoid such gap, JPHN upload an appropriate date in MCTS. This is the main reason for the high rate of disparity of MCTS data with actual data.

The problems experienced by the JPHN have to be considered while looking into the quality of MCTS data. The major problems experience by JPHNs, which affect the quality of MCTs data are:

- As per norms, the population to be covered by a JPHN is 5000, but in effect, the population covered varies from 6000 to 12,000. When the catchment area of a sub centre is hilly terrain, the geographical area will be vast; in that area also the population assigned is found extremely higher than norms population.
- When the target population is high and the area is vast, the JPHNs are overloaded with usual

MCH services such as conducting antenatal clinics, antenatal visits, immunization sessions, family planning programmes, collection of information for MCTS, action according to the work plan generated for MCTS, timely uploading of HMIs and MCTS, management of several registers etc.

- Apart from the MCH services, the JPHNs are forced to do several other duties such as conducting NCD clinics, staff nurses' duty as part of working arrangement in the absence of staff nurse in PHC/CHC and do even pharmacists' duty as part of working arrangement in their absence and chlorination and source reduction for communicable disease. They are even forced to take part in surveys of other departments - recently disability survey by ICDS

- Non-availability of dedicated computer and uninterrupted internet connection for HMIS and MCTS in some PHCs.

Suggestions : The following suggestions are presented for the betterment of quality of MCTS data. The suggestions were arrived at mainly from the field experience, interviews with JPHNs, medical officers of the selected PHCs and district-level officials. The initiation from the side of JPHN to improve data quality in MCTS is given below

- Provide MCP cards in time and maintain it on regular interval
- Insist ASHAs to collect information such as LMP, dates of TT1, TT2 and HB level from slips or lab reports of antenatal checkup while they visit antenatal women.
- Insist ASHA to collect child immunization details from immunization card supplied from the facility while they visit postnatal women
- JPHN/ASHA should ensure that the women provide information using the hospital records.

- Avoid the practice of uploading false information to justify work plan regarding TT1 and TT2

- Periodic cross verification is required on the data provided by the ASHA worker to ensure the genuineness

- Presently, the house visits made by JPHN is to collect information for MCTS. The level of guidance on antenatal care, national programmes such as JSY, JSSK, breastfeeding etc. is not satisfactory. So steps needed to rectify this gap.

Certain change in policy and some initiative from supervisory level officers are essential to increase the quality of MCH-related activities and MCTS data in particular.

- It is very essential to limit the population of sub centre area to 5000 to improve the quality of MCH-related activities and data quality in MCTS

- Exclude JPHNs from duties such as NCD clinics, Chlorination and source reduction for communicable disease and surveys of other departments

- Provide dedicated computer and uninterrupted internet connection for HMIS and MCTS in all PHCs.

- In majority of states, data entry regarding HMIS and MCTS is carried out by data entry staff at block level or at district-level. Here, all JPHNs upload data regarding HMIS and MCTS using internet café or personal computers. Providing laptops to all JPHN will be appreciated.

- Supervisory level monitoring and field-level spot-check by LHI and other supervisory staff is essential to minimize dropout in MCTS and to enhance the quality of MCTS data

References

1. Garai A, Ganesan R. Role of information and communication technologies in accelerating the adoption of healthy behaviors. *J Fam Welf*. 2010;56:109–18.
2. Piette JD, Lun KC, Moura L a, Fraser HSF, Mechael PN, Powell J, Khoja SR. Impacts of e-health on the outcomes of care in low- and middle-income countries: where do we go from here? *Bull World Health Organ*. 2012;90:365–72.
3. Mair FS, May C, O'Donnell C, Finch T, Sullivan F, Murray E. Factors that promote or inhibit the implementation of e-health systems: an explanatory systematic review. *Bull World Health Organ*. 2012;90:357–64.
4. Rahimi B, Vimarlund V. Methods to evaluate health information systems in health-care settings: a literature review. *J Med Syst*. 2007;31:397–432.
5. Mate KS, Bennett B, Mphatswe W, Barker P, Rollins N. Challenges for routine health system data management in a large public programme to prevent mother-to-child HIV transmission in South Africa. *PLoS One*. 2009;4, e5483.
6. Mphatswe W, Mate KS, Bennett B, Ngidi H, Reddy J, Barker PM, Rollins N. Improving public health information: a data quality intervention in KwaZulu-Natal, South Africa. *Bull World Health Organ*. 2012;90:176–82.
7. Stewart JC, Schroeder DG, Marsh DR, Allhasane S, Kone D. Assessing a computerized routine health information system in Mali using LQAS. *Health Policy Plan*. 2001;16:248–55.
8. Wilms MC, Mbembela O, Prytherch H, Hellmold P, Kuelker R. An in-depth, exploratory assessment of the implementation of the National Health Information System at a district level hospital in Tanzania. *BMC Health Serv Res*. 2014;14:91.
9. Mission Mode Projects. [<http://deity.gov.in/content/mission-mode-projects>].
10. MCTS Cell. Mother and Child Tracking System (MCTS). In: MCTS Cell, editor. Meet between Public Heal Found India MCTS Cell, 7th December 2012. New Delhi, India: MCTS Cell; 2012.

Section IV

National Health Mission: Programme implementation

CHAPTER - 18

AYUSH Health Care Centres under NHM in Punjab: An Appraisal

*Pawan Kumar Sharma, Poonam Sandhir and
Shaik Iftikhar Ahmed*

Population Research Centre -CRRID, Chandigarh

ABSTRACT

The present study endeavors to assess the functioning of AYUSH centres under NHM in Punjab vis-à-vis the laid Indian Public Health Standard Guidelines (IPHS) by Government of India. Most of the guidelines prescribed in the IPHS are based on mainstreaming of AYUSH. Besides few essentials, most of these guidelines are categorized under desirable services. For the purpose, 20 AYUSH health care centres; 16 ayurvedic and 4 homeopathic dispensaries and 62 patients availing treatment from these centres in three districts of Amritsar, Jalandhar and Sangrur were covered during the field survey. Two structured and pre-coded questionnaires: one for AYUSH facilities and the other for exit interview of patients- were used for the collection of the data. The ground realities revealed that the space allotted to AYUSH centres in the parent institution where these were co-located invariably was insufficient. Inconsistent with IPHS, majority of the AYUSH centres were functioning from one room allotted to them. While efforts were put in to spruce up the interiors of buildings, the surroundings were left unattended in majority of cases. The implementation of IPHS desirable guideline that signboard of the PHCs and DHs should mention AYUSH facilities was disappointing. Nine of the 16 ayurvedic dispensaries and two of the four homeopathic dispensaries were functioning without a dispenser exhibiting gaps in the IPHS standard laid in this regard. According to IPHS guidelines, two separate rooms - one for consultation by the AYUSH doctor and the other for dispenser needs-should be provided wherever AYUSH facilities exist. The situation was found wanting on the count of another desirable IPHS guideline that locally available medicinal herbs/plants should be grown around the AYUSH centres. No AYUSH medications were included in the drug kit of ASHAs as envisaged in mainstreaming of NHM. The IPHS guideline that AYUSH doctor shall attend to patients for AYUSH system-specific preventive, promotive and curative health care was not complied with up to the desired level. The overemphasis on accomplishment of OPD targets in the AYUSH health care centres was leading to a number game, low impact, neglect of the quality aspect, insensitivity to client needs and stagnation of the programme. This also displayed that how a well-intentioned policy laid by the Government of India led to negative results at the ground level. However, a happy situation existed on the count of higher satisfaction among the patients visiting AYUSH centres.

INTRODUCTION

India has built up a vast health infrastructure and health personnel base at primary, secondary and tertiary care levels in public, voluntary and private sectors during the last six decades or so. Considerable achievements have been made in the efforts to improve health standards, such as life expectancy, child mortality, infant mortality and maternal mortality since independence. But still the health problems are quite huge. An unacceptable high proportion of population continues to suffer and die from new diseases that are emerging apart from continuing ailments and new threats posed by the existing ones. There is a resurgence of interest in holistic systems of health care, especially, in the prevention and management of chronic lifestyle-related non-communicable diseases and systematic diseases.

Considering the humongous size of country's population and its widespread geographical distribution, health sector trends suggest that no single system of health care has the capacity to solve all of society's health needs. Steered with this realization, it was decided to continue with the policy of having integrative system of medicines being offered by multi-provider systems of medicines, including AYUSH which covers the medical systems that are being practiced in India such as Ayurveda, Yoga & Naturopathy, Unani, Siddha and Homoeopathy (AYUSH).

It took almost three decades for initiating systematic revival of the AYUSH system of medicine in the country. While formulating National Health Policy (NHP) in 1983, it was realized that the vast physical infrastructure and manpower base including AYUSH systems were being underutilized. The NHP- 1983 highlighted the need to regulate AYUSH systems in order to fill the large infrastructural gaps in the healthcare system. This policy laid stress on planned efforts to integrate the AYUSH systems with modern health systems. The Ministry of Health and Family Welfare, Gov-

ernment of India created the Department of Indian Systems of Medicines and Homoeopathy (ISM&H) in 1995 with the objective of bringing lost traditional and Indian systems of medicine into the mainstream. This department was renamed as Department of AYUSH in November 2003 (AYUSH, 2015). The objective was to provide focused attention to development of education and research in AYUSH systems of medicines. A centrally-sponsored scheme for the development of health care institutions was introduced during the 10th Five Year Plan to integrate AYUSH and bring it into the mainstream of health care services. Under the scheme, financial assistance was provided for setting up of AYUSH treatment centres in allopathic health centres and procurement of drugs in these AYUSH health care centres. The 11th Five Year Plan also laid emphasis on mainstreaming of AYUSH by designing strategic interventions for its wider utilization (Sharma and Goyal, 2013). The AYUSH Ministry was established in 2014 to significantly improve the Department's outreach in terms of planning, supervision and monitoring (MOHFW, 2014).

However, it was only after National Rural Health Mission (NRHM renamed as NHM) in 2005 and its mainstreaming policy for integration of AYUSH within the existing health systems, that major changes have been taken place in the transformation of AYUSH system in the country. The rationale behind the mainstreaming of AYUSH was to strengthen the public health system in the country at all levels, by engaging practitioners of alternate medicine, especially in the rural communities. The key strategies that were identified by the Government of India towards the process of mainstreaming were integration and mainstreaming of the Indian systems of medicine and homoeopathy into the existing public healthcare system and the national health programmes; encouragement and establishment of Indian systems of medicine specialty centres; facilitation and strengthening of quality control laboratories for the AYUSH system; strengthening drug standardisation research;

advocacy for AYUSH; and establishing sectoral linkages for AYUSH (MOHFW, 2005).

It was realized that exclusive AYUSH hospitals under the NRHM will not be able to sustain, therefore, NHM planned to integrate AYUSH treatment systems, facilities and faculties into the existing healthcare system. AYUSH practitioners were to be co-located in the existing District Hospitals (DHs), Community Health Centres (CHCs) and Primary Health Centres (PHCs). The Mission seeks to revitalize local health traditions and mainstream AYUSH infrastructure, including manpower and drugs, to strengthen the public health system at all levels. It was planned that AYUSH medication shall be included in the drug kit provided to ASHAs. The additional supply of generic drugs for common ailments at sub-centre/PHC/CHC levels under the Mission shall also include AYUSH formulations. At the CHC level, two rooms shall be provided for AYUSH practitioner and pharmacist under the Indian Public Health System (IPHS), the prescribed quality norms for health care services in India. Under the Mission the first set of standards was released in 2007 which were revised and released in 2012.

The IPHS guideline mentions about AYUSH services under the heading of service delivery (both outpatient department and inpatient department) in CHCs in the document by MoHFW, GOI under the title IPHS Revised Guidelines for CHCs by Directorate General of Health Services, New Delhi, GOI, 2012. The IPHS guidelines mention about AYUSH services under the category of essential health services in a district hospital published by the same office. Bringing standards in AYUSH services would help in two ways (a) the diseases, which are difficult to approach in Allopathic system can be done with AYUSH systems and vice versa and (b) some diseases could be approached in an integrated manner blending different systems of medicine. Most of the guidelines prescribed in

the IPHS are based on mainstreaming of AYUSH and majority of these are categorized under desirable services (Samal, 2014).

In Punjab, considerable efforts have been made to strengthen and upgrade the AYUSH systems. The number of standalone AYUSH health care centres has reached 640 in 2015 in the State. New set up of AYUSH has emerged under mainstreaming policy of NHM where the allopathic and AYUSH services are being provided under the same roof. Under the mainstreaming policy of NHM, the numbers of sanctioned AYUSH health care centres are 322. The details are displayed in Table 18.1.

Table 18. 1: Type of allopathic and AYUSH health care facilities in Punjab

Type of health centres	No. of health centres
Allopathic	
District hospitals	22
Community Health centres	150
Primary Health Centres	428
Sub-centres (SCs)	2950
AYUSH	
Ayurvedic hospital	5
Ayurvedic dispensaries	490*
Homoeopathic dispensaries	111
Unani dispensaries	34
AYUSH under NHM mainstreaming programme	
Specialized clinic	12
Specialized therapy centre	1
AYUSH wing	14
Co-located centres	
Ayurvedic	188
Homoeopathic	104
Unani	3

Source: Department of Health and Family Welfare, Government of Punjab, Chandigarh retrieved from <http://Pbhealth.gov.in>

*Note: * - 473 Government Ayurvedic Dispensaries + 17 Swasthya Kendras.*

No systematic efforts have been made for the state of Punjab to assess the gaps and the deficiencies in the functioning of the AYUSH health care centres under NHM related to infrastructure, equipment and instruments and the nature of services provided. Moreover, how effective these steps taken are and what more can be done for further development of the AYUSH health care centres still need to be established.

Objectives: The present study attempts to explore the functioning of AYUSH health care centres under NHM in Punjab. The performance of AYUSH health care centres in the State has been assessed on a number of associated parameters impacting upon their performance based on IPHS. The specific objectives of this evaluation study are to investigate the availability of physical infrastructure and health staff, analyze their equipment, materials and supply situation, identify gaps for strengthening the functioning of these centers by augmenting the provision of facilities and services; and assess satisfaction level of patients about the quality of services received at AYUSH facilities.

Data and methods: The sampling was done to select districts, AYUSH health care centres and clients. Staff in-position at AYUSH centres under NHM was the criteria adopted for the selection of the districts. The districts with higher number of AYUSH centres without any staff in-position especially the medical officers were excluded from the sample. Accordingly, Amritsar from Majha region, Jalandhar from Doaba region and Sangrur from Malwa with higher number of functional AYUSH centres were selected. Prior to that, information regarding existence of the AYUSH centres and staff in-position in each district were obtained from the Directorates of Ayurveda and Homoeop-

athy, Punjab.

The next stage involved the selection of AYUSH health care centres from each of the selected districts. Under the mainstreaming policy of NHM in the State, the homoeopathic dispensaries are co-located either in district hospitals or CHCs. This forms the basis for the selection of homoeopathic facilities. Homoeopathic facilities co-located in the district hospitals in Amritsar and Sangrur were selected for conducting the survey. However, the position of homoeopathic medical officer was found lying vacant at district hospital in Jalandhar. It was replaced with homoeopathic dispensary at CHC Kartarpur. While, the ayurvedic dispensaries were primarily co-located at PHCs in the State under mainstreaming policy of NHM. This also formed the basis for the selection of ayurvedic dispensaries. Ayurvedic dispensaries mainly co-located at PHCs were selected at random depending upon the availability of medical officer therein. The last stage involved the selection of patients at exit points of AYUSH centres. The patients were selected at random for the purpose of interviewing them.

For the collection of information two different types of structured and pre-coded questionnaires - one for AYUSH facilities and the other for exit interview of patients - were used. The basic objective was to capture ground realities in terms of adherence to IPHS guidelines. In all, 62 patients: 44 at ayurvedic and 18 at homoeopathic dispensaries were interviewed. The authors of this Project Report conducted the survey by themselves. This ensured the reliability and comparability of data.

MANPOWER AND PHYSICAL INFRA-STRUCTURE: COMPARISONS WITH IPHS GUIDELINES

Adequate physical infrastructure and professionally trained staff at the health centres are the two basic prerequisites for providing quality health

care services. Any deviation from the set norms for these parameters renders the system weak. The location of AYUSH health care centres within the parent institutions where these are co-located should be such that it does not prevent people to visit these. Generally speaking, AYUSH centres should be positioned away from toilets, garbage collection centre, water logging areas and mortuaries etc.

Manpower: Provision of one AYUSH doctor and one dispenser has been made in the IPHS guidelines (Table 18.2). All the AYUSH centres had sanctioned posts of the MOs which were duly occupied by them. Nine of the 16 ayurvedic dispensaries and two of the four homeopathic dispensa-

ries were functioning without dispensers causing problems in distribution of medicines. There was a shortage of 11 dispensaries as per the IPHS guidelines set for AYUSH centres. The co-location of AYUSH doctors has taken place as per the IPHS guideline which is not the same with AYUSH dispensers in Punjab.

Physical Infrastructure: All the 20 AYUSH dispensaries surveyed in the State, 16 ayurvedic and four homoeopathic dispensaries, were functional in designated district hospitals/CHCs and PHCs. The ayurvedic dispensaries were mainly co-located in the PHC buildings and the homoeopathic dispensaries in the DH or CHC buildings. This was as per the IPHS guidelines

Table 18.2: Workforce at AYUSH Health care centres as per IPHS Guidelines

Personnel	Essential	Desirable	Essential	Desirable	Qualification
SUB-CENTRE					
No workforce					
PRIMARY HEALTH CENTRE					
Type A (<20 deliveries per month)			Type B (>20 deliveries per month)		
Medical officer (AYUSH)	—	1*	—	1*	Graduate in AYUSH
Pharmacist (AYUSH)	—	1	—	1	
COMMUNITY HEALTH CENTRE					
Graduate medical officer (AYUSH)	1	—	No categorization		Graduate in AYUSH
Pharmacist (AYUSH)	1	—	No categorization		
SUB-DIVISIONAL HOSPITAL					
(31-50 bedded SDH)			(51-100 bedded SDH)		
AYUSH physician	1 [#]	—	1	—	
DISTRICT HOSPITAL [‡]					
	100 beds	200 beds	300 beds	400 beds	500 beds
AYUSH physician [√]	1	1	1	2	2
AYUSH pharmacist	1	1	1	1	1

Source: Ministry of Health and Family Welfare (2012): Indian Public Health Standards, Revised Guidelines for Sub-Centre, Directorate General of Health Services, New Delhi, Government of India, 2012

Note: *To provide choices to people wherever an AYUSH facility is not available in the near vicinity, #Provided there is no AYUSH hospital/dispensary in the district headquarters, one from AYUSH Safai Karamchari is to be outsourced, [√]If more than one AYUSH doctors are available; at least one doctor should have a recognized PG qualification in relevant system under AYUSH, [‡]AYUSH services have proposed in the essential category at district hospital.

Condition of the plaster was found intact in 11 of the 16 ayurvedic dispensaries and in two out of four homoeopathic dispensaries (Table 18.3). In five ayurvedic dispensaries and two homoeopathic dispensaries, the plaster was found to be coming off at some places. The condition of floor was found good in 15 of the 16 ayurvedic dispensaries and three of the four homoeopathic dispensaries. It was found somewhat broken in one ayurvedic dispensary and the other homoeopathic dispensary. Condition of the roof was found completely intact in 15 of the 16 ayurvedic dispensaries and two of the four homoeopathic dispensaries. It was found leaking in one ayurvedic dispensary and two homoeopathic dispensaries. This was causing problems in smooth functioning of the AYUSH dispensaries especially during the monsoons. The survey of AYUSH dispensaries was conducted during the rainy season only. During the field survey, the team got a chance to observe the kind of difficulties faced by the AYUSH staff. The leakages were causing damage to infrastructure facilities besides preventing the smooth conduct of the dispensary work. The qualitative aspects on the physical condition of AYUSH centres needs to be defined in the IPHS guidelines meant for these facilities.

Among other things, the electricity connection at these centres is basic to provide quality of services. The situation was found perfect on this count. All AYUSH centres surveyed were reported as having this facility. Also, the backup facilities in case of power cuts were more or less satisfactory. One ayurvedic and the other homoeopathic dispensary were yet to be connected with electricity back-up facilities. In both dispensaries, the parent health institutions had the electricity back-up facility available. Water supply posed no problem in any of the four homoeopathic dispensaries and 14 of the 16 ayurvedic dispensaries surveyed. All ayurvedic dispensaries barring one in Sangrur district and the other at CHC in Jalandhar district had water facility available within the AYUSH centre building. In case it was not available within the AYUSH centre it was fetched from the parent

building where the AYUSH centre was co-located. There was uninterrupted supply of water in 18 of the 20 AYUSH health care centres surveyed. However, the quality of water was found unfit in six AYUSH health care centres; four ayurvedic and two homoeopathic dispensaries.

Table 18.3: Physical condition of AYUSH health centres buildings

Parameters	Ayurvedic (n=16)	Homoeopathic (n=4)	AYUSH (n=20)
Condition of the plaster			
Intact at all places	11	2	13
Broken at some places	5	2	7
Condition of the floor			
Good	15	3	18
Broken at some places	1	1	2
Condition of the roof			
Completely intact	15	2	17
Leakage at some places	1	2	3

Source: PRC-CRRID Field Survey, 2015.

Toilet facility was found as available at all the 20 AYUSH health care centres surveyed. In 12 of the 16 ayurvedic dispensaries and in all the four homoeopathic dispensaries surveyed, the toilets were located within the space allotted for AYUSH buildings. The toilet facilities available at the parent health institution where AYUSH centres were co-located were used in case these were not provided separately for AYUSH staff. Some centres had a provision of two toilets, one for the staff and the other for public. Out of 20 AYUSH health care centres surveyed, 11 had separate toilet facilities for the staff with the facility of running water. During the visit of these AYUSH centres it was observed that while the toilets reserved for the staff were kept neat and clean, toilets meant for the public were in bad shape. Maintenance and cleanliness of toilets was a big problem. In some cases, the condition of toilets was quite pathetic. It was difficult even to approach the AYUSH centre due to foul smell all around. Seven out of 16 ayurvedic dispensaries did not have hand washing

soap available with them. They purchased this of their own.

Overall assessment by the survey team on the up-keep of AYUSH health care centres was on a more positive note. Of the 20 AYUSH health care centres, 12 were found quite clean while eight others had acceptable level of cleanliness. The surroundings of the AYUSH health care centres were found good or somewhat good in 17 of the total of 20 dispensaries surveyed. The surroundings of the homoeopathic dispensary in Amritsar hospital were found to be quite filthy. This dispensary was located on the back side of the civil hospital, at a place where variety of hospital waste was either disposed off or thrown in open. The surroundings of two ayurvedic dispensaries in Jalandhar district were also not hygienic. The ayurvedic MOs were found to be making personal efforts for making the surroundings of the dispensary clean but at times it was not yielding the desired results. The surroundings of another ayurvedic dispensary were not clean because of its location. There was no boundary wall of PHC where ayurvedic dispensary was co-located. It was located on the outskirts of the village in the vicinity of a place where villagers were throwing their dead animals which led to foul smell all around the PHC as well as in the ayurvedic dispensary.

IEC Activities: The scenario on the display of posters/banners informing about the existence of AYUSH facilities in the DHs/CHCs/PHCs where these were co-located was not very encouraging. One of the desirable services under IPHS guidelines was that the signboard of the CHC and district hospital should mention existence of AYUSH facilities. It was disheartening to note that while the names of DHs/CHCs/PHCs were prominently displayed on the signboard but there was no mention of AYUSH facilities therein. The names of the MOs with their specialization in allopathic medicines were displayed along with designated rooms at different floors. Here also recognition as well as mention of AYUSH doctors was ignored.

AYUSH dispensary name was displayed just on the name plate of AYUSH medical officer outside their rooms or on a board or wall nearby these dispensaries. These names were displayed mostly in Punjabi which have got faded away in due course of time. These names were not much visible even from a very short distance. Names of seven AYUSH dispensaries out of 20 surveyed were not mentioned anywhere. Even citizen charter displayed otherwise at the parent institutions where the AYUSH centres were co-located did not have any mention on the provision of these services at their respective institutions.

Signage marks leading to AYUSH health care centres were absent in 18 of the total of 20 dispensaries surveyed (Table 18.4). At number of places it was very difficult for the patients, who otherwise want to avail these services, to locate the AYUSH dispensary in the DHs/CHCs/PHCs where these were co-located. The survey team also experienced difficulties in finding the exact location of AYUSH dispensaries. At some places, the staff posted at these AYUSH dispensaries escorted the survey team to AYUSH dispensaries from the main gate of the DHs/CHCs/PHCs where these were co-located. One wonders how the patients must be locating these health care centres. Before undertaking the actual process of assessing the functioning of AYUSH centres, one could anticipate the extent to which these centres must be getting utilized.

Table 18.4: Type of IEC activities in the sampled AYUSH health care centres

IEC activity	Ayurvedic (n=16)	Homoeopathic (n=4)	AYUSH (Total) (n=20)
Display of AYUSH centre name			
On board and wall in Punjabi	7	1	8
On board and wall in English	0	1	1
On front wall in Punjabi	1	1	2

Others	1	1	2
Not displayed at all	7	0	7
Signage marks to AYUSH centre			
Displayed in Punjabi	1	0	1
Displayed in English	0	1	1
Not displayed at all	15	3	18
Different sections within AYUSH centre marked with name plates			
Displayed in Punjabi	3	1	4
Not displayed at all	13	3	16

Source: PRC-CRRID Field Survey, 2015.

According to IPHS guidelines, two separate rooms one for consultation by the AYUSH doctor and the other for dispenser needs to be provided wherever AYUSH facilities exist. Contrary to these standards, majority of the AYUSH centres were functioning from one room allotted to them. In few dispensaries both the AYUSH MOs as well as the dispensers were operating from one single room only. All activities, including the registration of the patients, examination of the patients and distribution of medicines were carried out from this room only. In few dispensaries, embarrassing situation was observed. On one side of the table ayurvedic MO was sitting and attending patients and on the other side the allopathic MO was attending patients.

At places where there was no dispenser, the Ayurvedic Medical Officers (AMOs) or Homeopathic Medical Officers (HMOs) were distributing medicines themselves. Nine AYUSH centres - seven ayurvedic and two homoeopathic dispensaries - had the provision of a store room where medicines and equipment were kept. In rest of the dispensaries the space for the purpose of storing medicines and equipment was shared with DHs/

CHC/PHCs where these were co-located.

According to IPHS guidelines, size for each sub-function was to be housed in 20.24 square metres. These included pharmacy-cum-store for AYUSH and other for space for AYUSH doctors. Adherence to the guidelines in the State was not as per the IPHS guidelines. In fact, more than half, 9 of the 16 ayurvedic and two of the four homoeopathic dispensaries, were without a store room facility. However, majority of the AYUSH medical officers were housed as per the IPHS guidelines.

In one of the institutions in Amritsar district, the sitting arrangement of ayurvedic and homoeopathic MO was found quite awkward. For meeting the ayurvedic MO a patient will have to pass through the room of the homoeopathic MO. While both the ayurvedic and homoeopathic dispensers were sitting in one big room. It was felt that some kind of rearrangement needs to be done for optimal space management. The ayurvedic MO and the associated dispenser can be allocated one room while the homoeopathic MO and the dispenser the other room. This would enhance the efficiency of both types of dispensaries.

Furniture: This is being done with a realization that while the optimal location of a health centre within a convenient distance is the prime factor influencing their utilization pattern, availability and adequacy of different items of furniture, equipment and materials required in delivering the health care services are critical to an assured and sustained usage. Non-functional equipment or non-availability of materials may either force the beneficiaries to look for alternatives or go without the service. The availability and supply situation of furniture such as doctor's chair, table, waiting benches, examination table, patient stool etc. and equipment of routine use, such as weighing machine, stethoscope and blood pressure instrument, and of materials including the medicines was examined. It also assesses the functioning status of the furniture, equipment and adequacy of materials.

None of the four homoeopathic dispensaries had waiting benches, dispensing tables, bedside screens, staff chairs and examination tables available with them (Table 18.5). The doctor's chair, doctor table and staff table were available only at homoeopathic dispensary at district hospital in Sangrur. Other homoeopathic dispensaries were devoid of most of the furniture items essential for its optimal functioning. Waiting benches, dispensing tables and staff tables were not available in two-thirds of the ayurvedic dispensaries. More than one-fourth of the ayurvedic dispensaries were functioning without the essential facility of doctor chair, doctor table, patient stool and visitor chairs. These furniture items were borrowed from the parent health institutions where these ayurvedic dispensaries were co-located. However, positive aspect was that the condition of furniture items was quite satisfactory in terms of its quality. These were found in good condition wherever these were available barring two ayurvedic dispensaries where examination tables were lying unutilized as these were broken.

Equipment: An equipment-in-order and in requisite quantity contribute to the community's confidence in efficacy of the health staff. Table 18.6 presents the availability scenario of the listed

equipment. None of the AYUSH centres had all the listed items of equipment available with them. On the whole, the homoeopathic dispensaries were ill equipped with materials such as thermometer, weighing machines, tongue depressor, height and weight measuring instruments. The situation on the count of availability of equipment was relatively better in ayurvedic dispensaries. More than half (9) of the 16 ayurvedic dispensaries had blood pressure measuring instrument and stethoscope available. Thermometer was not available in half of the AYUSH dispensaries covered under the survey.

It was observed during the survey that the ayurvedic MOs in few of the dispensaries had borrowed blood pressure measuring apparatus from the health institutions where these were co-located. These apparatus were lying unutilized in the parent health institution where posts of allopathic MOs were vacant. It was also observed that the ayurvedic MOs in some dispensaries were using digitized blood pressure measuring apparatus without understanding its correct usage. This was even true of the parent health institutions where the allopathic MOs were also found to be using these apparatus incorrectly. In a few cases, the ayurvedic MOs had to purchase this apparatus

Table 18.5: Availability of furniture items in the sampled AYUSH centres

Furniture items	Ayurvedic(n=16)			Homoeopathic(n=4)			AYUSH (Total)(n=20)		
	0	1	>=2	0	1	>=2	0	1	>=2
Waiting bench	12	2	2	4	0	0	16	2	2
Dispensing table	11	5	0	4	0	0	15	5	0
Bedside screen	9	7	0	4	0	0	13	7	0
Staff chair	9	6	1	4	0	0	13	6	1
Staff table	10	6	0	3	1	0	13	7	0
Examination table	6	10	0	4	0	0	10	10	0
Medicine rack	7	1	8	3	0	1	10	1	9
Doctor chair	6	9	1	3	1	0	9	10	1
Doctor table	5	10	1	3	1	0	8	11	1
Almirah	5	3	8	3	0	1	8	3	9
Visitor chair	4	0	12	3	0	1	7	0	13
Patient stool	4	5	7	3	0	1	7	5	8

Source: PRC-CRRID Field Survey, 2015.

Note: The furniture items have been arranged in descending order of their non-availability

from their own pockets. Ironically, while some ayurvedic dispensaries had more than the required number of stethoscopes and blood pressure instruments quite some of them had to do without these two basic items. The availability of items including imam dasta, dai kit bags, measuring spoons, kharal, vials and bottles, ice caps and dispensing cup was quite poor in the sampled AYUSH health care centres.

During the field survey few of the ayurvedic dispensaries were found to be equipped with desktop computers. Neither the survey team nor the AYUSH MOs could find any kind of utility of keeping these computers in these dispensaries. The computers were just placed on a table having no functional value. These were also not at all connected with any internet facility. In fact, none of the ayurvedic MOs required these computers in their dispensaries because of the concerned security issues. The space they were allocated was already short and unsafe as well due to non-availability of lock and key facility. Feeling afraid of the theft of these computers in the given physical space constraints, these were locked in one of the rooms at DHs/CHCs or PHCs. In the given circumstances if these computers cannot be utilized properly by the AYUSH centres these can be adjusted in such a way that these are made operational by the health staff members of any stream

of health system in the State.

The availability status on electrical equipment, including fans, tube lights/bulbs and coolers was more or less satisfactory in the AYUSH dispensaries surveyed in the three districts in the State. Four ayurvedic dispensaries had table lamps available with them. These appliances were more or less in working condition barring desert coolers in two dispensaries.

Type of Services offered at AYUSH Centres: One of the desirable guideline under IPHS is that AYUSH medical officers should support implementation of the national health programmes after requisite training, if required. AYUSH staff in all the dispensaries surveyed was not only involved in the health activities carried out by the AYUSH department but also in the activities undertaken by respective DHs, CHCs and PHCs where these were co-located. All the AYUSH MOs stated that they are also involved in carrying out national health programmes at one or the other stage especially in repeated pulse polio immunization campaigns. The ayurvedic dispensaries were involved in providing larger basket of health services as compared to homoeopathic dispensaries.

However, the involvement of AYUSH medical staff was found wanting on the count of their

Table 18.6: Availability of equipments of routine use in the sampled AYUSH health care centres

Items	Ayurvedic			Homoeopathic			AYUSH (Total)		
	0	1	>=2	0	1	>=2	0	1	>=2
Height measuring instrument	12	4	0	4	0	0	16	4	0
Height measuring tape	12	4	0	4	0	0	16	4	0
X-ray view box	12	4	0	4	0	0	16	4	0
ENT diagnostic set	11	5	0	4	0	0	15	5	0
Tongue depressor	10	5	1	4	0	0	14	5	1
Torch	9	5	2	4	0	0	13	5	2
Hammer	9	7	0	4	0	0	13	7	0
Weighing machine (Adult)	9	7	0	4	0	0	13	7	0
Thermometer	8	4	4	4	0	0	12	4	4
BP apparatus	7	6	3	3	1	0	10	7	3
Stethoscope	7	6	3	3	1	0	10	7	3

Source: PRC-CRRID Field Survey, 2015.

Note: The equipment items have been arranged in descending order of their non-availability.

co-ordination with the allopathic staff especially with multipurpose worker (female) as regards the activities pertaining to women and child health, promotion of institutional deliveries, nutritional health counselling, the supplementation of micro-nutrients deficiencies through AYUSH therapy, education, motivation and counselling to adopt appropriate family planning methods, health counselling; newborn care, coordination and integration of the all national health programmes by providing early screening for diseases and appropriate treatment through AYUSH Therapy.

Half of the MOs in ayurvedic dispensaries were members of Rogi Kalayan Samiti (RKS). On the other hand, none of the MOs in homoeopathic dispensaries was a member of RKS. District AYUSH officials in all three districts were members of District Health Society. However, they opined that they were just part of the district health societies on papers but their involvement in decision making was not up to their own desired levels. Their presence was not considered significant as they were considered from a peripheral system of medicine or less important.

Supply of medicines: The MOs at ayurvedic as well homoeopathic dispensaries highlighted that they always had to contend with acute shortage of medicines required for treating different kind of ailments. Shortage of medicines was reported in 17 of the total 20 AYUSH dispensaries visited; 14 of the 16 ayurvedic and three of the four homoeopathic dispensaries. For identifying the availability status of medicines in the AYUSH health care centres, four groups based on percentage share as displayed in the Table 18.7 were made according to shortage of medicines.

The availability status of medicines was slightly better in homoeopathic dispensaries as compared to ayurvedic dispensaries. One fourth of the ayurvedic MOs reported that they were getting 25 per cent share of the medicines as required at their dispensaries, thereby indicating 75 per cent of

shortage in their requirement. No MO at homoeopathic dispensaries reported shortage of medicines required by them up to this extent.

The MOs at both types of AYUSH centres were asked about their strategies to meet shortfall in medicines. Half of the MOs had given a prescription slip to the patients asking them to get these from the private medical stores. This strategy was followed by the AYUSH MOs despite the fact that they themselves were convinced that these patients will not go to private medical stores to get the prescribed medicines as these stores were located at a distance of more than five kilometers away from the dispensaries. This peculiar scenario was found in half of the AYUSH centres. The other strategy followed by MOs to meet the shortage of medicines was that the patients were directed to take allopathic medicines and revert back to them whenever the requisite medicines come in stock. The MOs in few ayurvedic dispensaries did not hesitate in prescribing allopathic medicines themselves to patients in case requisite ayurvedic medicines were not available in the stock of the dispensary. This is contrary to IPHS guidelines which states that AYUSH doctor shall attend to patients for AYUSH system-specific preventive, promotive and curative health care. Also the AYUSH doctors should take up educational activities which are based on their stream of medicines. The ground reality was not as envisaged in NRHM. The circumstance at few of AYUSH centres were forcing them to prescribe allopathic medicines which was beyond their purview.

The NHM had also envisaged that AYUSH medications shall be included in the drug kit provided at village level to ASHAs. This practice was found completely absent in the 20 AYUSH health care centres surveyed. In fact, the State as well as district health authorities were unaware in this regard. One of the desirable guidelines as laid in the IPHS was that locally available medicinal herbs/plants should be grown around the AYUSH centres. Here also the situation was found wanting. No herbal

Table 18.7: Status of medicines in sampled AYUSH health care centres

Medicine supply aspects	Ayurvedic (n=16)	Homoeopathic (n=4)	AYUSH (Total) (n=20)
Status of medicines			
Sufficiently available	2	1	3
Insufficient	14	3	17
Shortfall in percentage terms			
Less than 25 per cent	1	1	2
25-50 per cent	3	1	4
50 to 75 per cent	6	1	7
More than 75 per cent	4	0	4
Not applicable	2	1	3
Strategy to meet shortfall in medicines			
Ask patients to go to private medical store	8	2	10
Ask patients to come other day	1	1	2
Others	7	1	8
Distance of private medical store for AYUSH medicines			
Less than 1 km	6	1	7
1-5 kms	4	0	4
More than 5 kms	6	3	9
AMO/HMO prescribe allopathic medicines			
Yes	3	0	3
No	13	4	17
AYUSH medicines given in ASHA drug kit			
Provided in AYUSH drug kit	0	0	0
Not provided in AYUSH drug kit	16	4	20

Source: PRC-CRRID Field Survey, 2015.

plantations were grown on the ayurvedic dispensary premises. The community members were not imparted any kind of skill to treat common ailments with the home remedies. The AYUSH health care staff needs to widen their basket of services for making them relevant for the community at large. Early detection, management and referral of lifestyle diseases, including diabetes and hypertension along with others should be done. Their treatment through practice of Yoga and Ayurveda Therapy should be encouraged.

AYUSH OPDs: All the 20 AYUSH dispensaries surveyed were expected to provide OPD service for a total of six hours a day from 8 AM to 2 PM in summers and 9 AM to 3 PM in winters. The benchmark for minimum OPD attendance as set by state AYUSH authorities was reported unequivocally by the AYUSH staff to be about 35 patients per doctor per day. The comparable data on the number of OPD conducted during the last three years prior to the survey could not be collected despite best efforts made by the survey team. The posts of AYUSH MOs were lying vacant for different time periods in a year. Under these circumstances, no OPD services were provided in these centres. Similarly, there were AYUSH centres where one MO was serving two AYUSH dispensaries alternatively for three days a week. It was difficult to assess the overall performance of such AYUSH centres. But still, whatever data the survey team could gather from these AYUSH centres, displayed that the number of OPD per day was higher in homoeopathic dispensaries as compared to ayurvedic dispensaries (Table 18.8). OPDs of AYUSH centres were monitored along with parent institutions in both the ayurvedic and homeopathic dispensaries.

It is imperative to mention here about the quality of OPD data as provided by the ayurvedic as well as homoeopathic dispensaries covered under the survey. The attainment of higher OPDs has become the sole indicator for measuring the success of these AYUSH health care centres in the State. The higher officials at the state level were report-

ed to be obsessed with fixing of minimum number of OPD targets in the AYUSH centres. The MOs at ayurvedic and homoeopathic dispensaries were under the pressure to attain the fixed target of about 35 patients per day per MO.

Table 18. 8: Average number of OPDs per day in an AYUSH health care centre

Type of AYUSH centre	2012-13	2013-14	2014-15
Ayurvedic dispensaries	14	19	28
Homoeopathic dispensaries	24	42	49

Source: PRC-CRRID Field Survey, 2015.

Note: For calculating average number of OPD patients, 280 working days in each of the three years were used.

This is not surprising that this kind of pressure had led to recording of fake entries in the OPD registers. The OPD data in ayurvedic as well as homoeopathic dispensaries was found quite erratic. This was confirmed from the mismatch between the number of entries made in the OPD registers and the actual number of patients attended by the MOs.

Mainstreaming of AYUSH system had envisaged to actively supplement the efforts of the allopathic systems. However, competition between the allopathic and ayurvedic MOs was quite visible as regards the number of OPD patients they were attending. This was against the spirit of mainstreaming of AYUSH as laid by the NHM. The ayurvedic and homoeopathic MOs were blaming the allopathic MOs for low turnout of OPDs in their respective dispensaries. It was stated that the allopathic MOs were treating AYUSH MOs as quite inferior to them. Instead of making referrals to AYUSH health care centres, the allopathic staff members at the registration counter were reported to be discouraging the patients from visiting these centres. The negative competition between the AYUSH and allopathic MOs against each other

was spoiling the whole idea of mainstreaming of AYUSH in the State contrary to what NHM had envisaged. This overemphasis on accomplishment of OPD targets in the AYUSH health care centres was leading to a number game, low impact, neglect of the quality aspect, insensitivity to client needs and stagnation of the programme. This has also displayed that how a well-intentioned policy laid by the Government of India led to negative results at the ground level.

CONCLUSIONS

Mainstreaming of AYUSH has largely come to be perceived as 'Co-location' of AYUSH dispensaries in DHs, CHCs and PHCs in the state of Punjab. The whole process has been primarily restricted to contractual appointment of AYUSH doctors. The activities like specialty clinics, specialized therapy centre and AYUSH wings have been initiated under AYUSH mainstreaming but need to be strengthened yet.

Most of the guidelines prescribed in the IPHS are based on mainstreaming of AYUSH. Most of these are categorized under desirable services. Adherence to the guidelines in the State on the count of physical space requirements in 11 of the 20 AYUSH centres was not as per the IPHS guidelines. However, the medical officers were housed as per the IPHS guidelines. According to IPHS guidelines, two separate rooms - one for consultation by the AYUSH doctor and the other for dispenser - need to be provided wherever AYUSH facilities exist. Contrary to these standards, majority of the AYUSH centres were functioning from one room allotted to them. All activities such as attending the OPDs, testing, placing equipment and storing materials were carried out at one spot. It was disheartening to note that while the names of DHs/CHCs/PHCs were prominently displayed on the signboard there was no mention of existence of AYUSH facilities in 18 of the 20 such centres. This was contrary to IPHS guidelines.

Provision of one AYUSH doctor and one dispens-

er has been made in the IPHS guidelines. All the 20 AYUSH centres had sanctioned posts of the MOs which were duly occupied by them. Nine of the 16 ayurvedic dispensaries and two of the four homeopathic dispensaries were functioning without dispensers causing problems in distribution of medicines. There was a shortage of 11 dispensaries as per the IPHS guidelines set for AYUSH centres. The co-location of AYUSH doctors has taken place as per the IPHS guideline which is not the same with AYUSH dispensers in Punjab.

Contrary to what NHM had envisaged, no AYUSH medications were included in the drug kit at village levels to ASHAs. The IPHS guideline that AYUSH doctor shall attend patients for AYUSH system- specific preventive, promotive and curative health care was not followed as desired. Few of the medical officers were prescribing allopathic medicines. Another desirable guideline as laid in the IPHS was that locally available medicinal herbs/plants will be grown around the AYUSH centres. Here also the situation was found wanting. No herbal plants were grown on the ayurvedic dispensary premises.

The AYUSH medical officers were mainly concentrating on providing OPD services. In fact, attainment of higher OPDs due to pressure exerted by higher officials has become the sole indicator for measuring the success of these AYUSH health care centres in the State. This is not surprising that this kind of pressure had led to recording of fake entries in the OPD registers. This overemphasis on accomplishment of OPD targets in the AYUSH health care centres was leading to a number game, low impact, neglect of the quality aspect, insensitivity to client needs and stagnation of the programme. The involvement of AYUSH staff in national health programmes was not up to the desired levels. This also displayed that how a well-intentioned policy laid by the Government of India led to negative results at the ground level.

A happy situation, however, was visible on the

count of the higher satisfaction among the patients visiting AYUSH centres. Patients with chronic diseases such as arthritis, liver diseases, allergies, diseases related to digestive and respiratory systems were found to be getting greater solace from this system of medicines.

RECOMMENDATIONS

The recommendations as emanating from the discussions with the AYUSH officials at different levels in the State are as follows:

1. **Medicine supply:** The disbursement of AYUSH medicines, both in terms of distribution of type of medicines and their quantity, should be as per the demand of the medical officers rather than one based on a uniform supply. The supply of medicines should be adequate and regular. Medicines which are about to expire should be supplied in time to check their wastages. Low quality of some of the ayurvedic medicines was an issue highlighted by a number of MOs. For ensuring the quality, minimum quality standards should be fixed and followed in such cases.
2. **Manpower:** The need to induct dispensers at both the ayurvedic and homoeopathic dispensaries is underlined. Helpers and other class IV employees posted at the parent health institutions should be directed to provide their services to AYUSH MOs as well. Further, it was suggested that the AYUSH MOs should not be engaged in meetings that are avoidable as their frequent participation in these meetings was hampering their OPD services. The available manpower needs to be optimally utilized at the dispensaries.
3. **Financial aspects:** The staff at majority of the centres opined that amount of contingency needs to be enhanced. This would facilitate meeting the daily requirements of stationary and minor repairs of equipment more effectively. Failing this, they felt that there is every probability of the programme getting weak.

4. **Infrastructure:** Majority of the AYUSH health care centres were operating from one single room. Provision of a separate room for dispenser with a store to keep medicines and equipment especially at Panchkarma centres was described as necessary. A need for additional built space was demanded by virtually every centre. A case was made for the supply of vials and bottles by the homoeopathic MOs which are an essential component to carry the medicines at home. Supply of proper prescription slips and paper pouches for the patients in ayurvedic dispensaries to carry prescribed medicine was considered no less critical.

5. **Referral for testing facility:** Arrangements for referring and conducting basic blood and urine tests were deemed as most necessary. An early identification of ailment would help in rendering timely treatment. This would also check drop out of patients from the AYUSH centres as well. The allopathic MOs should refer patients with ailments which they think could be treated more effectively by the AYUSH system of medicines. This will make the referral system more dependable and effective.

6. **Publicity:** For enhancing the service utilization of AYUSH centres, two measures were deemed as very effective (i) awareness camps for publicity, and (ii) distribution of pamphlets and posters for spreading awareness.

7. **Skill upgradation:** Efforts should be made to upgrade the skills of the AYUSH doctors so that their involvement in the national health programmes becomes more significant. This can be achieved through their participations in training programmes as well through seminar/workshops.

On the whole, it is suggested that different stakeholders be made well-versed with the vitality of the AYUSH system for its smooth functioning under the mainstreaming component of NHM in the

State. All in all, the AYUSH system of medicines should continue in the State with necessary improvements and modifications, as indicated above.

REFERENCES

Department of Health & Family Welfare, 2015, Retrieved from Pbhealth.gov.in Government of Punjab, Chandigarh.

Ministry of Ayurveda, Yoga, Naturopathy, Unani, Siddha and Homeopathy (AYUSH), 2015, Annual Report 2014-15, Government of India, New Delhi.

Ministry of Health & Family Welfare, 2005, Accessed from <http://nrhm.gov.in/nhm/nrhm/nrhm-framework-for-implementation.html>.

Ministry of Health and Family Welfare, 2012, Indian Public Health Standards Revised Guidelines for Community Health Centre, Directorate General of Health Services, New Delhi, Govt. of India, 2012.

Ministry of Health and Family Welfare, 2012, Indian Public Health Standards Revised Guidelines for District Hospital, Directorate General of Health Services, New Delhi, Govt. of India, 2012.

Ministry of Health & Family Welfare, 2014, National AYUSH Mission- Framework for Implementation, (2014), Department of AYUSH, Government of India, New Delhi.

Samal, J. 2014, "Indian Public Health Standards for Ayurveda, Yoga and Naturopathy, Unani, Siddha and Homeopathy Facilities: An Assessment", *International Journal of Medicine and Public Health*, Oct.- Dec. 2014, Vol. 4, Issue 4. Pp. 331-335

Sharma, D. K. and R.C. Goyal, 2013, *Hospital Administration and Human Resource Management*. PHI Learning Pvt. Ltd, New Delhi.

CHAPTER - 19

Status of AYUSH in Co-located Public Health Facilities: A Study in Udupi District of Karnataka

R.V. Deshpande and H.R. Channakki

Population Research Centre, Dharwad

Abstract

Though AYUSH services existed for long, it was felt more since the launch of NHM where AYUSH has received much support with co-location in public health facilities. With an aim to understand the status of AYUSH co-located facilities and use of services, a field survey was carried out in Udupi district of Karnataka state. All the co-located facilities in the district namely DH, 2 SDHs, 4 CHCs were covered and information related to availability of basic amenities, manpower, equipment, drugs and perception of service providers were collected using structured questionnaire. Exit interviews of clients who availed services were also conducted to understand their perception about the system. In all, 141 exit interviews, 11 MOs from 9 co-located facilities and 4 district-level officers, were covered. The analysis showed that infrastructure facilities were good at DH but poor at SDHs. Except Ayurveda section at DH, none of the facilities satisfy the norms of manpower, and DH and SDHs are experiencing severe short supply of medicines. Except MOs at DH, all other MOs have received training in National Health Programmes and are actively involved. However, MOs expressed the need of training in Pan-chakarma and understanding/analyzing allopathic diagnostic reports. Cross referrals from Allopathic to Ayush and vice versa are taking place which is a good sign for the growth of AYUSH system. However, MOs of AYUSH and DAO expressed discrimination by the health system in terms of salary and respect to AYUSH system.

Patterns of utilization of services showed a clear distinction in availing AYUSH by religion; Hindus have availed more of Ayurveda and Muslims have availed more of Unani. More use of Unani by Muslims could be partly due to easiness in understanding dosage and content as script is in Urdu/Arabic. Further, use of the systems for illnesses broadly indicated that Unani and Homeopathy are used more for chronic illnesses while Ayurveda is used for both chronic and acute illnesses. Half of the clients availed treatment after trying allopathic medicines and one-third opted the system since beginning indicate the continuing demand for AYUSH services. Effectiveness of medicines and no/less side effects are the main reasons for preferring the system. Adequate and appropriate supply of medicines as per patients' illness profile was the major suggestion made, both by service providers and clients, to further improve the utilization of AYUSH services.

INTRODUCTION

The Indian Systems of Medicine and Homoeopathy (ISM & H) have been existing since ancient periods. The systems use herbs for preparation of medicines and people have faith in these systems as they are in their culture and social beliefs. Some people entirely depend on these systems and some others begin with these systems before switching over to other system of medicines. Further, some of the medicines of AYUSH are widely used by the people for the diseases which are not curable by the allopathic medicines. Realizing the importance of ISM & H system in strengthening health care delivery system, Government of India has set-up a separate department for ISM and H in 1996 under the Ministry of Health and Family Welfare and it was renamed as AYUSH in 2005 and included in the National Rural Health Mission (NRHM) with an aim to mainstream AYUSH services by promoting and expanding accessibility of AYUSH services. Further, under NRHM, strengthening of AYUSH services was carried out by integrating AYUSH services with national health care delivery systems through co-locating AYUSH services in the public health facilities namely, DHs, SDHs, CHCs and PHCs, and infrastructure, manpower, equipment and drugs have been provided to the co-located facilities. An understanding of the current status of co-located facilities in terms of availability of infrastructure and utilization of AYUSH services is important to improve the services. In this background, the present study aims to assess the status and utilization of AYUSH co-located facilities in DHs, SDHs and CHCs in the state of Karnataka.

REVIEW OF LITERATURE: Studies specifically on AYUSH co-located facilities are rare. However, existing studies related to AYUSH are reviewed.

Analyzing all India survey data of about 18,000 households covering both rural and urban areas, Sundar (1994), found that allopathic system of medicine is dominant over other systems of med-

icines and found that homeopathic treatment was sought by only 2 percent and ayurvedic treatment by only 3.8 percent. In an attempt to understand the utilization patterns of health care services by the households, a question on “preference for source of health care in case of sick” was asked in all the sampled households of NFHS2 and NFHS3 surveys. The surveys revealed that majority of households reported that they prefer allopathic system and AYUSH system was preferred by less than one percent of the households (NFHS2, 1998-99, NFHS3, 2005-06).

The NRHM-PIP monitoring studies conducted in different districts of Karnataka state found that AYUSH system is co-located in all the district hospitals and in few CHCs, and AYUSH co-location is largely not existed in PHCs. The studies also found that due to lack in equipment, lack of space in OPD and shortage of medicines, the services are poorly used by the people (Hallad et al, 2013). Review of literature showed that AYUSH is used by less proportion of people and the facilities lacked basic infrastructure and medicines.

OBJECTIVES:

- To assess the availability of infrastructure, manpower, equipment and drugs in AYUSH co-located facilities
- To understand utilization patterns of clients in terms of socio-economic and demographic characteristics choosing AYUSH services
- To know the choice of AYUSH services for illness and reasons for preferring the system

METHODOLOGY:

Udupi district, which has more AYUSH streams in the state, has been selected, and within the district, all the AYUSH co-located facilities have been selected. Accordingly, District Hospital which has Ayurvedic, Unani and Homeopathy services, two

SDHs, namely Kundapur and Karkala and four CHCs, namely Brahmapura, Byndur, Hebri and Nitte which have co-located with Ayurvedic have been selected.

In each of the selected facility, AYUSH medical officers were interviewed to gather information on the availability of infrastructure, manpower, equipment, drugs, training needs, jobs performing, involvement in national programmes, referral system, problems in providing services and suggestions to improve the utilization of services. Informal discussion was also held with district-level officers of both Allopathy and AYUSH systems on functioning of AYUSH system. Further, exit interviews of clients who availed services were conducted to understand the utilization patterns of clients choosing AYUSH services and satisfaction of services received. The targeted sample size of 150 exit interviews were allocated to Ayurvedic, Unani and Homeopathy streams based on the proportion of AYUSH OPDs to the total AYUSH OPDs in the district. The proportion of AYUSH OPD to the total OPDs worked out to be 60 percent for Ayurvedic, 20 percent each for Unani and Homeopathy. Accordingly, 90 exit interviews were targeted to be covered for Ayurvedic, 30 interviews each for Unani and Homeopathy. Exit interviews were done serially as they come out after consultation till our target is reached. In all, 11 medical officers, four district-level officials and 141 exit interviews were covered from 9 co-located facilities. The data were computerized using SPSS 20.0 version and analyzed.

SURVEY TOOLS:

Two types of schedules were used; one for facility and another one for exit interviews of the clients' who availed AYUSH services. While facility questionnaire aimed to collect information related to availability of infrastructure, manpower, equipment etc., the exit interview schedule gathered information from the client on the type of treatment availed (ayurveda, homeopathy, unani), nature of

illness, duration of illness, reason for preferring the method, perception about the quality of service received and suggestions for improving the utilization of AYUSH services.

FINDINGS:

INFRASTRUCTURE:

Availability of basic amenities like running water, generator, separate consultation room, adequacy of space, ventilation, waiting space, benches/chairs in waiting space, availability of privacy, examination table and telephone facility have been collected. A summary index has been derived to assess the overall status of infrastructure at AYUSH facilities using the information collected on facilities mentioned above. For this, we have taken the presence or absence of above infrastructure in the facility and in case all the infrastructures considered are present, availability is considered as 100 percent. If three items are present, for instance, and the remaining are absent in a facility, then availability index will be 30 percent.

The index derived based on the availability of above facilities showed that all the three AYUSH co-located facilities at district hospital, namely Ayurveda, Unani and Homeopathy have good basic amenities with availability index of 80 percent. The co-located AYUSH facilities at SDHs have availability index ranged between 40-50, indicating half of the considered infrastructure facilities were absent in the facilities. Among the CHCs, Hebri and Byndoor stand top having majority of infrastructural facilities (80-90 percent) but CHCs Brahmapura and Nitte lag behind in infrastructure with availability index of 50 percent each.

With respect to manpower, the sanctioned medical and paramedical posts in the co-located facilities are basically on contract basis. However, Ayurveda and Homeopathy sections have one each regular sanctioned medical officers' posts. Of the sanctioned three medical officers (one regular and two contractual) in the Ayurveda section, one specialist

MO is in position on contract basis and other two posts are vacant. Unani and Homeopathy sections have one sanctioned Medical Officer each and are in position.

All the four sanctioned contractual para medical workers (2 staff nurses and two masseurs) and one Gr-D in Ayurveda section are in position. However, Unani and Homeopathy sections don't have sanctioned posts of paramedical and Gr-D. The staff nurse working in Ayurveda section, besides her job, helps Unani and Homeopathy MOs in record keeping, medicines stock management etc. after OPD hours. One Gr-D post sanctioned for Ayurveda section is helping Unani and Homeopathy sections in sweeping and cleaning.

Both the SDHs have two sanctioned contractual MOs each and CHCs have one sanctioned MO each, and all are in position. However, SDHs and CHCs don't have sanctioned PMWs and Gr-D. Clinical related aspects like OPD, dispensing medicines, preparation of medicines, informing dosage etc. are being managed by the respective MOs and Gr-D of parental facility clean the consultation room.

The manpower situation revealed that Ayurvedic section of the DH has required manpower but SDHs and CHCs are functioning only with Medical Officer without paramedic and Gr-D.

EQUIPMENT AND DRUGS:

In all the co-located facilities, information on availability of common minimum equipment such as availability of BP apparatus, stethoscope and thermometer and system-specific equipment were collected. All the co-located facilities, except Byn-dur, Brahmapur and Nitte CHCs, have functional stethoscope and BP apparatus and only district Ayurvedic and Homeopathy sections have thermometer. However, system-specific equipment to carry out the procedures like Panchakarma (in ayurvedic) and Hijma (in unani) are available with the respective sections.

Source of supply of equipment differ from one co-located facility to the other. While district Ayurvedic section had received equipment from State AYUSH department, the other two sections, namely Unani and Homeopathy had received equipment from the NRHM wing. If co-located facility at Kundapur had got equipment from NRHM, it was the State AYUSH department that had supplied the equipment to the Karkala co-located facility. CHC Hebri had got the equipment from NRHM wing.

Regarding drug supply, all the interviewed MOs reported that they faced shortage of medicines. While co-located facilities at DH and SDHs experienced shortage of medicines to the tune of about 50 percent of total required medicines during 2014-15, the facilities at CHCs experienced shortage of about 70-90 percent. The CHCs Byndoor and Nitte are worst affected as shortage was as high as 90 percent. It is also found from records that facilities have been experiencing the shortage problem since the start of the facility which is being compensated through either providing allopathic medicines or reducing dosages or by prescribing medicines.

Regarding procurement of medicines, Ayush facilities at DH and SDHs receive medicines through indent and direct supply from NRHM while facilities at CHCs procure medicines directly from State AYUSH department. The medicines are supplied once a year but periodicity, and the quantity of medicines being supplied is not fixed. Further, it is reported that of the medicines received, only 30-40 percent are relevant/appropriate as per the illness profile of the people and the remaining are general medicines- neither cures the illness nor lead to side effects.

It is clear from the above that all the sampled facilities are experiencing shortage of medicines and shortage is more acute in CHCs followed by SDHs and DH.

SERVICE PROVIDERS' PERSPECTIVE:

The views of Medical Officers of the AYUSH facilities on their job responsibility, training needs, referrals, monitoring system, involvement in national programmes, problems in delivering services and suggestions to improve the utilization of AYUSH services were collected through a structured questionnaire. In all, 9 Ayurvedic, one each Unani and Homeopathy Medical Officers were interviewed and informal discussions were also held with four district-level officials, namely district health and family officer (DHO), district AYUSH officer (DAO), district surgeon (DS) and district programme management officer (DPMO).

CHARACTERISTICS OF PROVIDERS:

While seven out of nine MOs of Ayurveda were aged less than 35 years, MOs of Unani and Homeopathy were more than 35 years. Except two MOs of Ayurveda, the remaining were females. Except Unani MO who belongs to Muslim community, all the other MOs are Hindus. Almost all MO posts, except Homeopathy and Unani, are on contract basis and Ayurveda MO receive an average monthly salary of Rs.16,000, Homeopathy MO is getting Rs.26,000 and Unani MO is getting Rs. 41,000. While Ayurveda MOs receive salary from NRHM wing, Unani and Homeopathy MOs receive from Centrally Sponsored Scheme (CSS). Regarding their prior experience, four had worked in their established clinic, another four MOs had worked in other's clinic and one MO had worked at government facility.

JOB FUNCTIONS:

As per the guidelines issued jointly by the Mission Director-NRHM and State AYUSH department on works to be done by AYUSH doctors, AYUSH doctors, besides treating patients, should also be actively involved in all the national programmes such as pulse polio, routine immunization, school health programmes, VHNDs, NVBDCP, ASHA trainings and supervision of AWCs.

The jobs MOs performing differ according to the level of facility. At district hospital, the work of MOs of AYUSH is restricted mainly to treating OPD patients and providing awareness on the system during ASHAs and ANMs training programmes. The MOs at SDHs, besides OPD, conduct ANC check-ups, provide injections and administer IV and also do primary management of emergency cases. The MOs at CHCs, besides OPD, perform all types of works as assigned by the chief medical officer such as conducting Allopathic OPDs and attending IPDs in case of emergency, primary management of accident cases, ANC check-ups, monitoring immunization sessions and attending anganwadi meetings and gram sabhas. Further, in addition to clinical works, all the AYUSH MOs reported that they are maintaining records and registers.

TRAINING:

While Ayurvedic and Unani MOs at DH had undergone system-specific in-service trainings such as Panchakarma and Hijma therapies respectively, all MOs at SDHs and CHCs have received training in skill birth attendance (SBA), tuberculosis and HIV, communicable and non-communicable diseases. All the MOs have also received training in many maddu-tips on ayurvedic remedies for common illnesses. On training needs, MOs at DH reported that they require training on new concepts and techniques in handling the system and also in understanding and analysing allopathic diagnostic reports like ECG, MRI, pathology reports etc. which help to continue AYUSH treatment. All the MOs at SDHs and CHCs reported that they need training in Panchakarma procedure.

REFERRALS:

One of the important functions of the AYUSH MOs is to have coordination with medical officers of other systems and to encourage referrals. Interviewed MOs were asked about the referrals within the system i.e. one AYUSH to another AYUSH

and cross referrals i.e from AYUSH to Allopathic. The illnesses for which cases were referred have showed that the cross referrals i.e from AYUSH to allopathy and allopathy to AYUSH do exist. For dog bites, uncontrolled BP, high diabetic, surgery cases and children's diseases, providers refer cases to allopathy and conditions like chronic asthma, skin diseases, arthritis etc. are referred from Allopathy to AYUSH.

Within AYUSH, referrals are more from ayurveda to unani and homeopathy and from unani and homeopathy to Ayurveda. However, referrals from unani to homeopathy or homeopathy to unani don't exist. While chronic allergy and gynec-related problems are referred from ayurveda to homeopathy, warts and blood circulation-related problems are referred from ayurveda to unani. The cases referred from unani and homeopathy to ayurveda are for panchakarma procedure and for physiotherapy. However, both within and cross referrals are done verbally and in an informal way and thereby, not being documented.

SUPERVISION AND MONITORING:

As per the guidelines issued jointly by Mission Director, NRHM and State AYUSH department, administrative affairs of the AYUSH medical officers at co-located facilities are to be looked upon by the health and family welfare department and technical/professional aspects are to be looked upon by the AYUSH department.

All the medical officers were asked about details on mechanism of supervision/ monitoring such as person monitoring, mode and frequency of monitoring. It is found that the works of AYUSH MOs are being monitored by the head of the respective facility. The MOs of co-located facilities at DH are monitored by district surgeon through periodic visits- once in 15 days. However, MOs are not being invited for the monthly staff meetings to review their performance and to provide guidance. Though each MO submits performance reports

monthly to the DS and DAO, very less likely they had got feedback from them. The MOs of facilities at SDHs are being monitored by their respective CMOs through attendance register. However, their performance is reviewed in the monthly meetings where 10-15 minutes is devoted for AYUSH-related aspects. The monitoring system seems to be weak in AYUSH facilities located at CHCs where monitoring is only through attendance registers and they were not called for monthly meetings. The visits by the DHO to the co-located facilities are limited, very casual and lasts a few minutes.

As per the guidelines, the District Ayush Officer (DAO) is also expected to conduct monthly/bi-monthly meetings of MOs of AYUSH of the co-located facilities to review the performance, however, due to TA/DA provision problem, they are not being called but DAO visits the facilities during the routine tour programme to convey new programmes/ new orders/circulars and requirements/ complaints of MOs to the higher authorities. The DAO also expressed that effective supervision by AYUSH department is not possible as co-located staff feel that DHO officials are their real bosses since they get salary from them.

The above analysis showed that though supervision is in place at the facility level by the respective CMOs, it is very casual and superficial. The supervision from the higher level, namely at DHO/DAO level is very poor. This seems to be because of two bosses in the co-located facilities, namely DHO and DAO. The DHO being an administrative head is not being involved in technical aspects whereas DAO, being a technical head supposed to supervise effectively, could not do it because of poor acceptance by the MOs of co-located facilities.

INVOLVEMENT IN NATIONAL PROGRAMMES:

The MOs of AYUSH are expected to involve actively in all the national programmes. The information collected on this revealed that MOs of

AYUSH at DH are not involved in any national programmes but involved themselves in promoting awareness on AYUSH through lectures in schools/colleges and other community meetings and trainings. The MOs at SDHs are involved in supervising pulse polio programme and helping school health programmes. The involvement of MOs of CHCs seems to be more as all MOs are involved in weekly immunization programmes, screening malaria cases and TB cases, providing awareness on family planning methods, supervision of ICDS programmes and attending VHND meetings.

IEC:

All the MOs of AYUSH were asked about the IEC materials they have received, source and adequacy of materials. The DH and SDHs they have received posters, pamphlets and folders on different themes of AYUSH and they are adequate. However, Byndoor and Brahmavar CHCs have not received any IEC materials but the other two CHCs had received IEC materials which are inadequate. All the IEC materials were supplied by the AYUSH department.

PROBLEMS IN DELIVERING SERVICES:

All the AYUSH MOs were asked on the problems facing in delivering services. The responses are categorized into system- related problems and clinical-related problems.

i. **System-related problems:** Except MOs of Unani and Homeopathy systems, all the MOs expressed that they are being discriminated. According to them, discrimination exists in terms of salary, in terms of allocation of space and inviting for monthly meetings. Incidents of showing disrespect to AYUSH system/medicines and asking to quit the meeting in between have also been reported. This discrimination, according to them, had hurt and often made them psychologically depressed. Further, not allowing to display IEC materials in the facility and not sparing Gr-D workers are the

other problems reported by few MOs. The DAO also admitted that there is discrimination in terms of salary, providing manpower and infrastructure.

Admitting the salary disparity between allopathy and AYUSH, the DHO, DS and DPMO opined that they cannot be equivalent to Allopathy doctors as qualification and work responsibilities differ between the systems.

ii. **Clinical-related problems:** All the 11 MOs expressed the severe shortage of essential medicines. They also pointed out that many medicines being supplied are not appropriate for the illnesses prevailing in this area. Further, 10 out of 11 MOs said that Medical Officer alone managing OPD, dispensing medicines, informing on preparation of medicines, advising on dosages, keeping records and doing other related works are posing problems.

iii. **Coordination related problem:** Except, MOs of Unani and Homeopathy systems, all other MOs of AYUSH facilities and DAO expressed that coordination between the two systems is lacking in terms of providing infrastructure, popularizing the system and sparing manpower in case of emergency works. The DHO and DPMO admitted poor understanding and coordination between the two systems and also expressed that they don't want to bring them up and treat them as equivalent to allopathic. The State director of AYUSH also expressed lack of coordination between Allopathic and AYUSH departments and felt that there is a systematic plan to suppress AYUSH system and prejudice for AYUSH from top to bottom is an obstacle for growing AYUSH systems.

UTILIZATION OF AYUSH SERVICES:

In order to understand utilization patterns of services in terms of what are the socio-economic and demographic characteristics of people who sought AYUSH services, type of system chosen for the illness, stage when started availing treatment, rea-

sons for preference, perceived quality of services and suggestions for further improving the utilization of services were sought from the persons availed services through exit interviews. In all, 141 exit interviews comprising 90 interviews from Ayurveda, 25 from Unani and 26 from Homeopathy were conducted.

PATTERNS OF UTILIZATION:

The clients who availed AYUSH services were analyzed by their characteristics. Table 19.1 showed that the proportion of children below age 15 years who used AYUSH is negligible, adults of age 15-59 availed more of Homeopathy and Unani and those aged 60+ used more of Ayurveda. More than 90 percent of Unani clients, 77 percent of Homeopathy clients and 53 percent of Ayurveda clients are females which indicates that unani and homeopathy systems are more used by females and Ayurveda is used almost equally by both males and females.

Table 19.1. Background characteristics of clients availed AYUSH services, Udupi district

Chara cteristics	Ayurveda (n=90)	Unani (n=25)	Homeopathy (n=26)	All (n=141)
Age				
0-14	2.2	0.0	0.0	1.4
15-59	66.7	84.0	88.5	73.8
60+	31.1	16.0	11.5	24.8
Sex				
Male	46.7	8.0	23.1	35.4
Female	53.3	92.0	76.9	64.6
Religion				
Hindu	86.7	0.0	30.8	61.0
Muslim	10.0	96.0	53.8	33.3
Other	3.3	4.0	15.4	5.7
Caste				
SCs/STs	8.9	0.0	3.8	6.4
Other	91.1	100.0	96.2	93.6

Literacy				
Illiterate	16.7	0.0	3.8	11.3
Literate	83.3	100.0	96.1	88.7
Occupation				
Not working (HH Work)	32.2	68.0	38.5	39.7
Working (outside home)	51.1	24.0	50.0	46.1
Other	16.7	8.0	11.5	14.2
HH status				
APL	37.8	56.0	53.8	44.0
BPL	57.8	44.0	42.3	52.5
DK	4.4	0.0	3.8	3.5
Total	100.0	100.0	100.0	100.0

Eighty seven percent of those availed Ayurveda belong to Hindu community, almost all (96 percent) who had availed Unani belong to Muslim community and those availed homeopathy belong to both Hindu (31 percent and Muslim 54 percent) and only 6 percent of interviewed clients who availed AYUSH services belong to SCs/STs. Muslims opting more of unani system could be due to the fact that script related to symptoms of disease, dosage and content of medicine is in Arabic or urdu language which is easy for them to learn and understand about the disease. While almost all clients availed unani and homeopathy are literates, 84 percent of those who availed ayurveda are literates. Majority (68 percent) of those sought treatment of unani are engaged in household works and it is those who work more outside their home who sought homeopathy and Ayurveda treatment. Regarding household status, more than half of clients availed unani and homeopathy belong to APL and more than half (58 percent) of ayurveda belong to BPL.

DETAILS ON ILLNESS:

All the clients were asked about details on illness such as type, duration of illness, stage when started availing services, person motivated and reason for preferring the system. The responses are presented in Table 19.2.

Table 19.2. Percent distribution of clients by type, duration of illness, stage availed treatment, distance travelled, person motivated and reason for preference, according to type of AYUSH system, Udupi district

Item	Ayurveda (n=90)	Unani (n=25)	Homeopathy (n=26)	All (n=141)
Type of illness				
Knee joint pain / Leg pain	28.9	60.0	38.5	36.2
Cough & Fever	21.1	4.0	19.2	17.7
Back Pain	14.4	0.0	15.4	12.1
Stomach pain	12.2	8.0	3.8	9.9
Abdominal Pain	11.1	8.0	0.0	8.5
Other	12.3	20.0	23.1	15.6
Duration of illness				
Less than 12 months	41.1	40.0	23.1	37.6
More than 12 months	58.9	60.0	76.9	62.4
Stage started availing treatment				
After Allopathic treatment	53.3	36.0	50.0	49.6
From the beginning	31.1	28.0	26.9	29.8
After other AYUSH treatment	7.8	12.0	7.7	8.5
In combination	5.6	16.0	3.8	7.1
Other	2.2	8.0	11.5	5.0
Distance travelled				
Less than 5Kms	50.0	24.0	34.6	42.6
6-10	23.3	20.0	15.4	21.3
11+	26.7	56.0	50.0	36.1
Persons Motivated				
Self	38.9	12.0	7.7	28.4
Allopathy doctor	6.7	0.0	0.0	4.3
Relatives/friends/ husband /wife/ children	38.8	72.6	84.6	53.1

Other	15.6	16.0	7.7	14.2
Reason for preference *				
More effective	76.7	96.0	100.0	84.4
No/less side effects	73.3	84.0	80.8	76.6
Less cost	28.9	32.0	30.8	29.8
Cures from the root	20.0	12.0	34.6	21.3
No faith in other medicines	6.7	24.0	30.8	14.2
Other	15.6	24.0	7.7	15.6
Total	100.0	100.0	100.0	100.0
*- Multiple responses				

Type of illness: Thirty six percent of clients availed AYUSH services for knee joint/leg pain followed by cough and fever (18 percent). While majority of cases of knee joint /leg pain sought treatment from Unani and Homeopathy, most of the cough and fever cases availed treatment from Ayurveda.

Duration of illness: The use of the systems for illnesses by clients broadly revealed a clear distinction between the conditions for which Ayurveda, Unani and Homeopathy are being used. Unani and Homeopathy are mostly used for chronic illnesses while Ayurveda is used both for chronic and acute illnesses. Large proportion of clients was suffering from the illness for more than one year (could be defined as chronic) and such cases are more among those availed Homeopathy (77 percent). However, clients availed treatment for illnesses of less than one year (could be defined as acute) were found comparatively more among those availed Ayurveda and Unani.

Stage started availing treatment: All the clients were asked about the stage at which they started availing treatment for the present illness. It is to be noted that half of the clients availed the AYUSH treatment after trying allopathic medicines. However, nearly one-third had opted the system since beginning of the illness and 10 percent had switched over after availing other AYUSH system of medicines.

Distance travelled: Nearly two-fifths (43 percent) of clients had travelled less than 5 Kms to avail the

services. However, more than half of clients who opted Unani and Homeopathy reported to have travelled more than 10 Kms to avail the services as they are not available at periphery but only at district hospital.

Person motivated: Relatives, friends and neighbours were the main motivators of clients for availing services of Unani and Homeopathy. Allopathy doctor motivating to avail ayurveda medicine was also observed.

Reason for preference: Majority of clients expressed that they preferred the system because the “medicines are more effective” and substantial proportion had also told that “no/less side effects” are the main reasons for preferring the system. “Low cost for the treatment” and “medicines cure the illness from the root” are the other reasons quoted for preferring the system.

QUALITY OF SERVICES:

Use of services depends on the experiences and perceptions about the quality of services received. Table 19.3 presents the percent of clients reported on selected quality of service aspects.

Table 19.3. Percentage distribution of clients by their opinion on quality of services, Udupi district

Indicator	Ayurveda (n=90)	Unani (n=25)	Homeopathy (n=26)	All (n=141)
Waiting time (in mts)				
less than 10	31.1	20.0	46.2	31.9
10-20	61.1	60.0	50.0	58.9
more than 20	7.8	20.0	3.8	9.2
Medicines				

Provided medicines	83.3	48.0	100.0	80.1
Partly provided and partly prescribed	13.3	36.0	0.0	14.9
Neither provided nor prescribed	3.3	16.0	0.0	5.0
Type of medicine received				
Ayurveda	85.3	—	—	85.3
Unani	—	100.0	—	—
Homeopathy	—	—	100.0	—
Allopathic	14.7	—	—	—
Behaviour of the doctor				
Satisfactory	96.7	96.0	100.0	97.2
Somewhat satisfactory	3.3	4.0	0.0	2.8
Cleanliness in facility				
Good	93.3	88.0	84.6	90.8
Fair	6.7	12.0	11.5	8.5
Poor	0.0	0.0	3.8	0.7
Total	100.0	100.0	100.0	100.0

Waiting time: All the clients were asked on how much time they had to wait to get services. Waiting time is determined as the time spent between registration and doctor’s call. It is found that one third of clients reported to have waited for less than 10 minutes to meet the doctor and three-fifths had waited for 10-20 minutes.

Medicines received: Majority (79 percent) of clients reported to have received medicines for the illness. However, among clients availed Unani, one-six had not received any medicines and one-third had received medicines only partially. The contradiction that high proportion of clients reported receiving medicines despite shortage of medicines could be explained in the light of the

fact that such facilities usually receive medicines which are not specific to any illness and could be provided to any illness. So, in case of shortage of appropriate/relevant medicines, patients are given general type of medicines instead of sending them empty handed.

Type of medicines received: Type of medicine received is a proxy for quality of service. Ethically, the person has to be treated with the system of medicine which he/she had opted. To understand whether clients had received medicines in the opted system, all the clients who had received medicines were asked the type of medicines received from the doctor. It is found that all the clients who opted Unani and Homeopathy received medicines of the same system. However, about 15 percent of clients who had opted for Ayurveda were reported to have provided with allopathic medicines and this was found in facilities co-located at CHCs.

Behaviour of the doctor and cleanliness in facility: Almost all the clients expressed satisfaction about the behaviour of the doctor and majority (90 percent) reported that co-located facilities are clean and well maintained.

SUGGESTIONS:

i. **From clients:** It can be observed from table 19.3 that supply of medicines was the major suggestion reported by the clients and they opined that availability of medicines for all illnesses throughout the year would help to avail more AYUSH services. Publicity through mass media and display of availability of AYUSH services in the public health facilities were also reported. Establishment of IPD facilities at co-located centres, separate registration counter for AYUSH and establishment of panchakarma facilities are the other suggestions reported by the clients. However, all MOs strongly suggested improving the supply of medicines. Salary hike, regularization of service, in-service training and more publicity about the AYUSH system at ground level through ASHAs, AWWs as well as

through TV/radio are the other suggestions. The DAO suggested that all AYUSH co-located facilities should be brought under the control of DAO, both administratively and technically, which helps to monitor and improve performance of the facilities.

ii. **From Providers:** The views of the district-level allopathic officials on improving the utilization of AYUSH services differ. The DHO, District Surgeon and DPMO felt that utilization of AYUSH could be improved much if AYUSH MOs practice their own system of medicines instead of practicing allopathic system. Many privately practicing AYUSH MOs conduct allopathic OPD and IPD, prescribing high doses of antibiotics, which is a disturbing aspect leading to the decreasing popularity of AYUSH system and spoiling both AYUSH and Allopathic systems.

CONCLUDING REMARKS:

The analysis broadly revealed that infrastructure facilities, manpower and drugs availability are better in co-located facilities existed at DH but SDHs lack infrastructure and also experiencing severe shortage of medicines. The publicity on AYUSH is poor as most of the facilities don't have board either outside or inside the facility. Except MOs at DH, all other MOs have undergone trainings on National Health Programmes and are involved in them. However, MOs expressed the need for training in Panchakarma and in understanding and analysing allopathic diagnostic reports. The encouraging fact is that cross referrals are taking place from AYUSH to Allopathy and vice versa which is good for the growth of AYUSH system. However, reported discriminatory attitude towards AYUSH by the allopathy needs to be taken care of by the higher authorities.

Patterns of utilisation of AYUSH showed children are less likely to use AYUSH, females are more among users of Unani and Homeopathy, and males are more users of Ayurveda. Clear distinction is observed in availing AYUSH by religion. Hindus

Table 19.4. Percent distribution of clients reported suggestions for improving AYUSH , Udupi district

Suggestions *	Ayurveda (n=90)	Unani (n=25)	Homeopathy (n=26)	All (n=141)
Medicines for all illnesses and all the time	33.3	42.3	36.0	35.5
More Publicity	31.1	7.7	64.0	32.6
IPD facilities	25.6	38.5	28.0	28.4
Separate OPD counter	2.2	15.4	12.0	6.4
Panchakarma facilities	4.4	0.0	0.0	2.8
Good service from Staff	4.4	3.8	20.0	7.1
Waiting time should be reduced	1.1	0.0	4.0	1.4
No idea/ Suggestions	15.5	0.0	0.0	9.9
*Percentage may not tally to 100.0 due to multiple responses				

are more likely to avail Ayurveda, Muslims are more likely to avail Unani, but Homeopathy is availed by both. The use of the systems by clients broadly revealed a clear distinction – Unani and Homeopathy are used more for chronic illnesses and Ayurveda is used for both chronic and acute illnesses. The increasing demand for AYUSH is clear from the study as half of the clients started availing treatment for the illness after trying allopathic medicines and one-third opted the system since beginning. Effectiveness and less side effects of medicines are the main reasons reported for preferring the system. Adequate and appropriate supply of medicines as per patient's illness profile was the major suggestion made by both service providers and clients.

8.6. RECOMMENDATIONS;

- Adequate and appropriate medicines as per patients' illness profile should be supplied.
- Training needs to be provided in understanding allopathic diagnostic reports
- Coordination, mutual trust and respect between AYUSH and Allopathic should be improved.
- Awareness about AYUSH should be created through mass media and other channels.
- Service conditions of AYUSH staff at co-located facilities need to be improved.

References:

1. Hallad J.S, S.R. Vatahati and C.N. Noolvi, 2013, Quarterly Monitoring of PIP in Davanagere district, Karnataka, PRC Analytical Report no.8.
2. Hallad J.S. and S.R. Vatahati, 2013, Quarterly Monitoring of PIP in Dharwad district, Karnataka, PRC Analytical Report no.16.
3. International Institute for Population Sciences (IIPS) and ORC, macro international, National Family Health Survey (NFHS2), 1998-99, Mumbai
4. International Institute for Population Sciences (IIPS) and ORC, macro international, National Family Health Survey (NFHS3), 2005-06, Mumbai
5. Sundar Ramamani, 1994, Household Survey of Health Care Utilisation and Expenditure, Margin, Journal of National Council of Applied Economic Research, Vo. 26, No.4 July-September, 1994, New Delhi

CHAPTER - 20

An Overview of the Status of Determinant Inputs in SNCUs under Public Health System in the selected Districts of North Karnataka

*Shriprasad H, Joint Director and M S Kampli,
Data Assistant,*

Population Research Centre, Dharwad, Karnataka

Abstract

It has been estimated that about 70 per cent of neonatal deaths could be prevented if a right intervention is made through newborn care facility. The study basically intended to look into the performance analysis of SNCUs in the four districts of North Karnataka, in terms of infrastructure, equipment and manpower status. The required data was gathered for the period of six months. Birth Asphyxia was the single largest cause under morbidity profile of the admitted neonates in the visited SNCUs with an average admission of 726 for the reference period. Interestingly, none of the visited facilities had the location of basic infrastructure in appropriate place. The ancillary area was very limited. The number of basic functional equipment like radiant warmer, phototherapy units were very much limited and not working to their full potential. The same condition is applicable for monitoring equipment also. The equipment was remaining non-functional for quite long indicating slow repair mechanism of such equipment. There was also problem of human resource in the visited SNCUs. Shortfall of staff nurses was 20 per cent. And average nurse-baby ratio for four districts was greater than nurse-bed ratio because of more bed occupancy rate. There was a shortfall of doctors for four districts by 45 per cent on an average against recommendations. Also, average baby-bed ratio for four districts was greater than doctor-bed ratio due to higher bed occupancy rate. The average fatality rate for the visited SNCUs was 23 per cent indicating little higher than the normal. Providing higher space for SNCU set-up, alternative repair mechanism of equipment, right combination of staff nurses and doctors for number of neonate beds are some of the ways to improve the quality of SNCUs in terms of infrastructure, equipment and human resource. Perhaps, addressing the above issues related to infrastructure, equipment and human resource may yield better results in terms of fatality rate.

Key words: SNCU, bed occupancy, doctor-bed ratio, staff-bed ratio, infrastructure, fatality rate

INTRODUCTION

India has the single largest share of neonatal death in the world. Neonatal deaths constitute two third of infant deaths in India, and 45 per cent of death occur within the first two days of life. It has been estimated that about 70 per cent of neonatal deaths could be prevented if proven interventions are implemented effectively. Depending on the level of health system there is different neonatal health facility. In District hospitals level II neonatal care system is functioning, generally known as Special Neonatal Care Unit (SNCU). A special emphasis is given under NRHM on various inputs of a SNCU like infrastructure, equipment and manpower. In fact, they are considered as pillars of the SNCU set-up. Infrastructure includes number of beds in relation to delivery load and ancillary area to enhance the efficacy while equipment include warmers, phototherapy units, ancillary units, and monitoring units like stethoscope, multi-channel meters etc. The location of SNCU also has a role to play in the outcome of an SNCU. All these are crucial inputs without which an SNCU set-up cannot be functional at all. Very few studies are available in the literature that highlights the importance of appropriate infrastructure and manpower combination for the effective outcome of SNCUs. A study by (Sen et al 2009), found that shortage of trained nurses and doctors in the newborn care system is not likely to be solved easily and soon. Our innovation of creating a group of newborn aides volunteers by training young women with some education from the neighbourhood was found useful in partially mitigating the shortage of nurses. A study by (Neogi et al 2011), conducted to assess the functioning of SNCUs in eight rural districts of India, found that repair and maintenance of equipment were a major concern in all the SNCU set-ups. It's possible to set up and manage quality SNCU and to improve the survival of newborns with LBW and Sepsis in developing countries, although several challenges relating to human resources and maintenance of aseptic practice remain. A study by (EL Sayed et al 2013)

pointed out that availability of right combination of human resources & equipment is the prerequisite for the successful outcome of any SNCU set-up. Further, the quality of nursing care has a direct impact on the survival of neonates. In this paper an attempt has been made to assess the infrastructure and manpower status in the visited districts.

Objective

The basic objective of this paper is to see the status of infrastructure, equipment potential and human resource status of different category of the visited facilities.

Methodology

Four districts, Belagavi, Haveri, Kalaburagi and Bidar were randomly selected from 2 administrative divisions of North Karnataka. To assess the infrastructure, equipment and manpower, a structured questionnaire was administered to MOs and Staff nurses. Service statistics containing admission load, birth weight of the babies, morbidity profile, fatality rate etc. of the visited SNCUs were gathered for the six months i.e. from April 2015 to September 2015. Discussions were held with the doctors to understand their perceptions and to know the hurdles for performance under SNCU. Further, level of knowledge of staff nurse were also assessed under SNCU. Analysis was made with the help of SPSS package and spread sheet. Results are presented in bivariate and univariate tables.

Admission and Morbidity Profile

The average admission for the reference period was 726 for the four districts, (Table 20.1) with highest admission load in Kalaburagi and least in Haveri. Of the total admissions, 66 per cent were inborn cases, with highest in Kalaburagi and least in Haveri. The average number of males per 100 live births of four districts was 55. The average per-

centage of LBW babies was 34 and extreme LBW babies were 12 in the four districts. The extreme LBW babies were the highest in Bidar, higher by 3 per cent than the averages of visited districts. The extreme LBW babies were least in Belagavi and were equal to the averages of 4 districts. Regarding Morbidity profile, Birth Asphyxia is the single largest cause (32 per cent), followed by respiratory distress, (14 per cent) jaundice (14 per cent) and the sepsis (11 per cent) for the admissions in the visited facilities. The Birth Asphyxia cases found highest in Bidar (48 per cent) and least in Haveri (25 per cent)

Table 20.1. Admission Profile

District	Total Adm.(six months)	Admission/100 Delivery	Inborn cases per 100 delivery	No. of males per 100 live births	Perc. of newborn with birth weight of 1.5-2.49 Kgs	Perc. of newborn with birth weight of <1.5 Kgs
Belagavi	702	18	13	54	38	12
Haveri	480	18	12	55	40	11
Bidar	783	21	13	55	28	15
Kalaburagi	941	26	16	56	33	10
Total(Avg)	726	21	14	55	34	12

Physical Infrastructure

Basic infrastructure or physical infrastructure like beds (warmers), location of SNCUs, ancillary areas plays a very important role in ensuring better outcome of an SNCU. Table 20.2 shows the status of beds. The number of beds required under an SNCU level II is based on the number of delivery load in a hospital. The recommended bed for an SNCU is 12, for every 3000 deliveries and this proportion is to be increased by 4 beds for every additional 1000 deliveries.

Table 20.2. Status of Bed Availability

District	Delivery/Yr	Required Beds	Existing	Shortfall
Belagavi	7482	30	30	-
Haveri	5148	20	8	12
Bidar	7250	29	22	07
Kalaburagi	7174	29	14	15

Accordingly, given the delivery load and bed norms, Belagavi district hospital requires 30 beds and has no shortage of beds. In rest of the visited hospitals, there was a shortage of beds, as shown in the table 20.2. The shortage is more acute in Haveri. As a whole, in the four visited districts, there were only half the beds existing against the recommended number of beds.

Location of SNCU plays an important role in ensuring the quality of service in any SNCU. Therefore, an attempt has been made to assess the location of SNCU in the visited SNCUs. As per the protocol (IPHS), SNCU should be near the labour room and should not be located at the top floor. Further, there should be an exclusive space for breastfeeding and its promotion. Apart from these basic requirements, an SNCU level II is expected to have a step-down unit, to treat the recovered babies. Among the visited districts Belagavi and Kalaburagi had SNCU and labour room on the ground floor and also were having designated breastfeeding area. However, SNCU in Haveri and Bidar were not having the location of SNCU and labour room combinations as per the protocol. None of the visited facilities were having the step-down unit.

Ancillary areas are also known as additional support service areas that are expected to prevent the possible infections for neonates and to provide quick reports of investigations. Among the visited districts, provision for hand wash and changing

room were found only in Belagavi and Kalaburagi. However, it is found that in the visited facilities, there was a provision of hand detergent to wash the hands before the doctor/staff nurse attends the neonates. Portable X-ray machine were found in Belagavi and Kalaburagi. In the rest of the districts, facilities were dependent on central system. None of the visited facility had side laboratory, store room and washing autoclaves. For laboratory and washing autoclaves, the facilities were reported to be dependent on the central system of the hospital. None of the visited facilities were having designated store room and nurse workstation. A separate partition was made in all the visited facilities, which was used as nurse work-station and store room. The major reason for poor location protocol is the absence of enough space in most of the visited facility.

Monitoring Equipment

Monitoring equipment is very much crucial to take timely decisions on accurate neonatal treatment. Radiant warmer is one of the equipment for managing the conditions of the neonates. Their full functionality alone will ensure the best outcome of the SNCUs. Based on the delivery load, the number of beds recommended for the four visited district is 108. The average available beds (warmer) against the recommendation were 65 per cent for four districts. However, fully functional radiant warmer was only 42 per cent (Table 20.3).

Haveri district SNCU was found to be performing poor in terms of functional warmers. Belagavi district SNCU was found to be relatively better in terms of performance of warmers. On the whole, the performance of the radiant warmers was not satisfactory. Phototherapy unit is the core equipment in a SNCU that is used for the controlling the Jaundice level (Bilirubin level) of the neonate. Fully functional phototherapy can deliver effective treatment for the control of jaundice. Usual recommendation for the number of phototherapy

Table 20.3 Radiant Warmer

District	No. Recommended	Available	Functioning Fully	Functioning Partially	Available against recommendation percent	Functional fully against recommend percent
Belagavi	30	30	20	10	100	66
Haveri	20	08	06	02	40	30
Bidar	29	22	12	10	75	41
Kalaburagi	29	14	10	04	48	34
Total	108	74	48	26	67	42

unit is half of the beds available under each unit. In the visited districts, according to the guidelines (IPHS), the number of recommended phototherapy units is 37 against 74 beds.

Table 20.4. Phototherapy Unit

District	No. Recommended	Available	Functioning Fully	Functioning Partially	Available against recommendation (%)	Functional fully against recommend(%)
Belagavi	15	08	00	08	53	00
Haveri	04	04	04	00	100	100
Bidar	11	06	00	06	54	00
Kalaburagi	07	06	04	02	85	57
Total	37	24	08	16	64	22

The Table 20.4 shows the status of phototherapy unit in the visited districts in terms of recommendation and availability. As it is witnessed from the above table that the available numbers of phototherapy units are only 64 per cent against the recommendation and the fully functional ones are only 22 per cent. In Haveri district, phototherapy

units are functioning to the extent of 100 per cent. In Belagavi and Bidar none of the units is fully functional. Stethoscope is the equipment which is used to listen to the internal sounds of the body, especially to heart and lung sounds. The recommended number of stethoscope is one each for a bed. The reason is mainly to prevent the infections from one baby to another. The following table shows functionality of stethoscope in the visited districts.

Table 20.5 Status of Stethoscope

District	No. Recommended	Available	Functioning	Available against recommended (%)	Functional against recommend (%)
Belagavi	30	22	04	73	14
Haveri	08	06	04	75	50
Bidar	22	03	03	14	14
Kalaburagi	14	04	04	29	28
Total	74	35	15	48	21

As it is witnessed, the available number of stethoscopes against recommendation is only 50 per cent (Table 20.5) and functioning is very poor, i.e. only 21 per cent against the recommendations. Belagavi and Bidar districts had very less functional stethoscope. Pulse Oximeter is an instrument that is used to measure the oxygen level of the baby and also to measure the BP level.

As the above table reveals, the available pulse Oximeter is 75 per cent (Table 20.6) against recommended. However only, 45 per cent are fully functional. Except in Haveri, none of the visited facility was having fully-functional Pulse Oximeter. As the term indicates, multi channel parameter performs multi-monitoring functions inside a SNCU. It is used to monitor BP, saturation level, heart rate and ECG. The recommended number for each SNCU is half of the beds available in a given unit. As the table indicates the available unit against the recommendation is 60 per cent and a

functional unit against recommendation is 48 per cent.

Table 20.6 Status of Pulse Oximeter

District	No. Recommended	Available	Functioning	Available against recommended (%)	Functional against recommend (%)
Belagavi	15	08	07	53	47
Haveri	04	04	04	100	100
Bidar	11	06	02	54	18
Kalaburagi	07	07	05	100	71
Total	37	27	17	75	45

Table 20.7. Multi Channel Parameter

District	No. Recommended	Available	Functioning	Available against recommended (%)	Functional against recommend (%)
Belagavi	15	12	10	80	66
Haveri	04	03	02	75	50
Bidar	11	02	02	18	18
Kalaburagi	07	05	04	71	57
Total	37	22	18	60	48

None of the visited district has fully functional Multi-Channel units. (Table 20.7) Bidar is poor in terms of functional units and Belagavi is relatively better in terms of functional units. The status of Resuscitation equipment is also not so good in the visited facilities. In fact, they are expected to play a vital role in saving the life of neonates. The timely available oxygen and its accessories have positive impact on the performance outcome of the SNCUs. In the visited SNCUs, there is central oxygen system in all the districts except in Haveri where the facility was not available for 24 hrs. The

status of functionality of oxygen cylinders, oxygen concentrator and Ambu bag were moderate in all the visited SNCUs. Figure 20.1 shows percentage of fully functional equipment against recommendations.

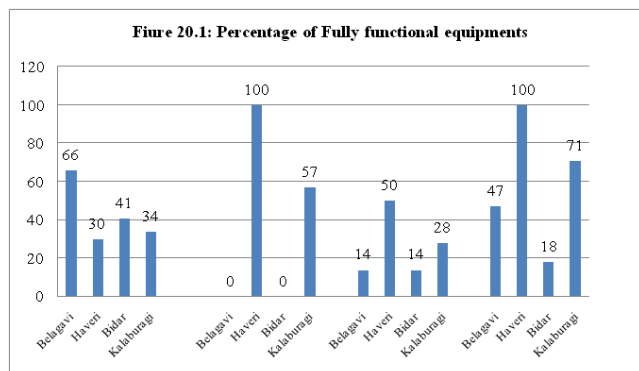


Figure 20.1, above shows the extent of fully functional Radiant warmer, Phototherapy unit, Stethoscope and Pulse Oximeter across the district. In none of the visited facilities these equipment are functional to the extent of hundred per cent. The major reasons identified for the low performance of some of the equipment are poor repair mechanism and complex decision-making system in the SNCUs. In the visited SNCUs, repair and maintenance of equipment were outsourced. On an average, time taken to get it repaired was 4-6 months.

Human Resources

Human resource plays a major role in ensuring optimum utilization of infrastructure, equipment and reduction of neonatal mortality in the SNCU. Among the different category of human resources, staff nurses have a prominent role in the reduction of fatality rate in the SNCU

The recommended nurse-bed ratio for a 12-bed SNCU is 1:4. Table 20.8 shows the status of the nurse-bed ratio in the four districts of North Karnataka. The available beds in the SNCU under four districts is 74, based on which the total staff nurse for 24 hrs is estimated (at 1:4 ratio) to be 57. However, the number of working staff nurse was 46, thus the shortfall is 20 per cent. The short fall

Table 20.8 Staff Nurse

District	Beds Available	Recommended	Working	Shortage
Belagavi	30	23	12	-11
Haveri	08	06	11	+05
Bidar	22	17	12	-05
Kalaburagi	14	11	11	00
Total	74	57	46	11

was more in Belagavi district on the other hand; Kalaburagi district has no shortfall of staff nurse. But Haveri district SNCU, interestingly has excess number of staff nurse than the recommended number.

Though speaking absolutely staff nurse and bed ratio is moderate in the four districts, the actual status of the ratio is revealed by the calculation of nurse-baby ratio. This calculation gives what is the number of babies under each staff nurse. The result can be captured by calculation of bed occupancy rate in the district.

Table 20. 9. Nurse Bed Ratio Vs Nurse Baby Ratio

District	Nurse Bed Ratio per shift	Nurse baby ratio per shift
Belagavi	1:7.5	1:5.4
Haveri	1:2.1	1:3.7
Bidar	1:5.5	1:4.7
Kalaburagi	1:3.5	1:7.1
Average	1:4.6	1:5.2

Table 20.9 shows nurse-bed ratios and the nurse-baby ratios across the visited districts. As it is observed, average Nurse-Baby ratio for 4 districts was greater than Nurse-Bed ratio, because of more bed occupancy rate. The average bed occupancy rate for the four districts is 137 per cent indicating more than one baby is kept on one warmer. This naturally pulls up or down the average nurse-baby ratio under each district. For instance, Nurse-Bed

ratio in Kalaburagi is 3.5, but the nurse-baby ratio is 7.1 due to higher (205) bed occupancy rate. Similarly, in case of Bidar, nurse- bed ratio was 5.5, however, nurse-baby ratio was 4.7 due to lower bed occupancy rate (87 per cent), and this has pulled down the nurse-baby ratios for Bidar.

Doctors expected to play a crucial role in a SNCU set-up. As per the protocol, the expected doctor-bed ratio is 1:12. Thus 74 beds in the four districts require a total of 20 doctors for 24 hrs (assuming that 3 shifts of 8 hrs each a day).

Table 20.10 Doctors

District	Beds	Recommended	Working	Shortfall
Belagavi	30	8	3	5
Haveri	08	3	2	1
Bidar	22	6	3	3
Kalaburagi	14	3	3	-
Total	74	20	11	9

In the visited four districts, as the Table 20.10 shows, the working doctors were 11 against 20 recommended numbers. Thus there was a shortfall of doctors for four districts by 45 per cent. Except Kalaburagi district, rest of the districts were having shortage of doctors in varying degrees. The shortage was more acute in Belagavi than the rest of the districts.

Paediatrician plays an important role in SNCU set-up. Though, a pediatrician is not expected to be in SNCU, he should be made available for 24 hrs on on-call basis. Thus a recommended number of Pediatricians is 'one' for a SNCU. In all the visited districts there was a pediatrician in the hospital.

Table 20.11. Doctor Baby Ratio

District	Doctor Bed Ratio per shift	Doctor baby ratio per shift
Belagavi	1:30	1:22
Haveri	1:08	1:15
Bidar	1:22	1:19
Kalaburagi	1:14	1:29
Average	1:18	1:21

However, none of them had a sanctioned post of pediatrician exclusively for SNCU. In Belagavi, Bidar and Kalaburagi, the Pediatricians were working from the medical college on a rotation basis. Doctor-baby ratio is an important indicator that measures, what is the number of babies actually being monitored by the doctors during the reference period. This is calculated based on the bed occupancy rate and number of doctors under a given SNCU. As it can be observed from the above table that the average doctor-baby ratio for 4 districts was greater (Table 20.11) than the doctor-bed ratio due to higher bed occupancy rate. Based on the number of babies under each warmer/bed the doctor-baby ratio is more or may be less. For instance, in Belagavi district the doctor-bed ratio per shift was 1:30, however, the doctor-baby ratio was 1:22 i.e. less, due to less bed occupancy rate (73 per cent) when compared to the available beds. However, scenario of Kalaburagi is quiet opposite, i.e. its doctor-bed ratio was 1:14. However, its doctor-baby ratio was almost double i.e. 1:29 due to much higher bed occupancy rate (205 per cent) during the reference period. The major cause for lack of specialist was poor attraction towards public health system due to poor remuneration compared to the private sector, excessive bureaucratic control and fear of ill-treatment by the public.

Absence of exclusive D group staff is a major problem in the visited districts. Neonates are highly sensitive for the opportunistic infections. Therefore, recommended times of cleaning the SNCU is three in 24 hrs. None of the visited facilities,

except Bidar SNCU had D group staff. As a result, there was hardly one time cleaning process taking place in such units. It is very surprising to know that in none of the visited SNCUs fumigation takes place regularly. The recommended times of fumigation under the SNCU set-up is at least once in a fortnight. However, in all the SNCUs the fumigation was reported to have not taken place since last one year. This has certainly contributed to the high fatality rate under SNCU due to infections. Another major problem due to the absence of D group staff was the unrestricted entry of the public. This was also a cause for high infections among the neonates.

Conclusion

There were only half of the beds existing against the recommended number of beds in the sample facilities. This calls for necessary steps to be taken in this direction. Even half of the visited facilities were not having SNCU in the recommended location. Very surprisingly, none of the visited facilities had step-down units due to lack of sufficient space. Further, the lack of portable X-ray machine in half of the visited facilities resulted in considerable delay in diagnosis. From the equipment perspective also the functioning of SNCUs is not satisfactory. Hardly 25-40 per cent of the core equipment like phototherapy unit, radiant warmer, monitoring equipment, stethoscope, Pulse Oximeter etc. is fully functional against recommendation.

The major reasons identified for the low performance of some of the equipment are poor repair mechanism and complex decision-making system in the SNCUs. All these issues need to be considered very seriously by the concerned hospital authorities to optimize the functions of the given SNCUs. There is much necessity to focus on issue of human resource also. The average Nurse-Ba-

by ratio for four districts was greater than nurse-bed ratio because of shortage of staff nurse due to higher bed occupancy rate. There was a short fall of doctors for four districts by 45 per cent. The average doctor-baby ratio for 4 districts was greater than doctor-bed ratio due to higher bed occupancy rate. Addressing the above issues related to infrastructure and human resource can only bring about better survival rate inside the SNCUs.

Policy Implications

- a) There is every necessity to enhance the space reserved for SNCUs in every SNCU set-up to accommodate the basic infrastructure and equipment.
- b) Instant repair system, mechanism may be thought about to reduce the duration required to repair the equipment.
- c) Appointment of doctors and staff nurses must be priority task to reduce the doctor-bed and staff-bed ratios that ensure better care of neonates.
- d) Ensuring the required number of 'D' group staff is very essential for the expected outcome from SNCU set-up

References

- A Sen, D Mahalanabis, A K Singh, T K Som and S Bandyopadhyay (2009) 'Impact of a district level sick newborn care unit on neonatal mortality rate': *Journal of Perinatology* (2009) 29, 150-155; doi:10.1038/jp.2008.177; published online 23 October
- Neogi Sutapa Bandyopadhyay, sumit Machotra, Sanjaya Zodpey and Pavitra Mohan (2011) "Assessment of Special Care Newborn Units in India" *Journal of Health, population and Nutrition*, 2011 oct: 29 (5): 500-509 ISSN 1606-0997.

• Rajendra Kumar Mahore, , Sanjay Dixit, SB Bansal, Veena Yesikar, , Nirbhay Mehta, Sachin Parmar (2015) ‘ Overall Outcome of Selected Special Care Newborn Units (SCNUs) of Madhya Pradesh at Different Levels’ *Online Journal of Health and Allied Sciences* ISSN 0972-5997 Volume 14, Issue 1; Jan-Mar 2015

• Sayd Sabah Mohammed EL, Yomn Y Sabry, Hamed M Sharkawy, Ebtisam M elsayed & Thanaa Ali (2013) ‘Establishing Basic standards of

Nursing care protocol at Neonatal Intensive care Unit, Nature & Science 2013; 11(4

Acknowledgement

My Sincere thanks to MoHFW New Delhi, for funding this study, compendium expert committee for their valid suggestions in strengthening this paper and support of our Director in this juncture.

CHAPTER - 21

An Assessment of Janani Shishu Suraksha Karyakram (JSSK) in two districts of Kerala

Sreeranjini A, Oommen P Mathew*, Anilkumar K* & Sachin K V***

**Research Investigator, **Field Investigator
Population Research Centre, Thiruvananthapuram*

Abstract

This community and facility-based observational cross-sectional study evaluates the awareness about the JSSK programme and out-of-pocket expenditure incurred, among the pregnant women in Ernakulam and Alappuzha districts of Kerala. The study analyses the quantum of shift of deliveries from private to public health facilities. The study sample covered 420 mothers, who delivered between July 2015 and the preceding date of survey (January to March 2016). They were interviewed using a schedule and the data was analyzed using the software SPSS. Caesarean deliveries are more than 50 percent which is very high when compared to WHO recommended level. The awareness level about JSSK entitlements was 67.4 percent. Women who delivered at government facility have more knowledge regarding JSSK. The awareness level about the scheme was high at Alappuzha district. Only 17.3 percent pregnant women received free ANC care from the government facility. Scan, diagnostic tests and medicines are the major causes of high OOPs incurred at government facility during ANC. Only 17.3 percent received completely free ANC care from government hospitals. Highest mean OOPs for the diagnostics test and scans were seen at private practicing government doctors whereas for medicine and OP tickets, private hospitals were charging higher amounts. The disparity between government and private hospital is minimum for blood and urine test and maximum for op ticket favoring government hospitals. It can also be seen that the total amount of the ANC cost is almost double in the private hospital or private practicing government doctors compared to that in government hospitals. It is also seen that maximum exploitation of the beneficiaries happens at the time of initial appointment in the private hospitals. Normal delivery is comparatively higher at Government hospital. More than 50 percent of the beneficiaries of the government hospital received complete free delivery care service. Mean cost is around 15 times and 19 times higher in private hospital for normal and caesarian deliveries respectively. The supply of medicines and essential supplies/consumables from KMSCL is insufficient to manage normal and C-section delivery. The tie-up with the medical stores is ineffective forcing the beneficiaries to buy medicines at their own cost. There is no uniformity in providing travelling allowance to beneficiary after delivery. The provision for free diet is almost uniform in all facilities. For successfully utilizing the scheme and for increasing the awareness level among pregnant women, more efforts are to be taken by the service providers. The present study thus points out the fact that there exists OOPs for ANC and delivery services and hence, the gaps are to be reduced by liberalizing the policy about the funding of the components of the JSSK entitlements.

Key Words: JSSK entitlements, pregnant, OOPs, ANC, delivery, beneficiaries, government, private.

INTRODUCTION

Reproductive health and child health are the main apprehensions of any country. According to Ministry of Health and Family Welfare (2011), in India, the existing pace of decline in Maternal Mortality Ratio (MMR) and Infant Mortality Rate (IMR) is deficient to achieve the target of Millennium Development Goals. About 67,000 women die every year due to pregnancy-related complications like hemorrhage, infection, high blood pressure; unsafe deliveries, etc. (Ministry of Health and Family Welfare, 2011). Appropriate medical attention at the time of delivery can prevent 75-80 percentages of total maternal deaths (Paul V. K, 2011 et al). Government of India has introduced a National Policy, Reproductive and Child Health Programme under the National Rural Health Mission with a specific intention to reduce the maternal and infant mortality. It ensures that every pregnant woman gets delivered by a skilled birth attendant. In order to achieve this target, Janani Sureksha Yojana (JSY) was launched which promotes demand for institutional delivery. As a result of launching JSY, the number of institutional deliveries has increased significantly. However, about 27 percent of pregnant woman still hesitate to deliver at health facilities. Those who have selected for institutional delivery are not ready to stay at hospital for 48 hours which impeded the provision of necessary services to both mother and neonates. It is critical for the recognition and management of the complications during the first 48 hours after delivery. Financial burden plays a vital role in accessing health care in general and maternity care in particular in resource-constrained settings (World Health Organization, 2006). This is because of various causes, including high out-of-pocket expenditure on transport and medical care incurred by pregnant women and their families for normal or C-section delivery. This becomes a constraint for the family particularly the poor and BPL families in having access to health facilities. This prevents them from availing essential and emerging (emergency?) care during pregnancy and also in the postpartum peri-

od. Important factors affecting the access of public and private health facilities include high out-of-pocket expense on 1) User charges for OP ticket & IP admission, 2) cost of transportation from home to facility; facility to home and expense in case of reference to higher facility 3) delivery cost especially for caesarian section 4) lab services & diagnostics 5) cost of medicine and other consumables 7) blood 8) food etc. Hence, another programme named Janani Shishu Suraksha Karyakram (JSSK) was launched by Govt. of India on 1st June, 2011, which is aimed at providing better health facilities for every pregnant women and sick neonates up to 30 days. This scheme supplements the cash assistance given to a pregnant woman under Janani Suraksha Yojana and is aimed at alleviating the burden of out-of-pocket expenses incurred by pregnant women and by those women with sick newborns. This scheme envisages the following entitlements to all pregnant women delivering in public health institutions:

- ⇒ Absolutely free and no expense for delivery, including caesarean section.
- ⇒ Free drugs and consumables
- ⇒ Free diet up to 3 days during normal delivery and up to 7 days for C-section
- ⇒ Free diagnostics
- ⇒ Free blood wherever required.
- ⇒ Free transport from home to institution and back.
- ⇒ Exception from all kinds of user charge

The national initiative JSSK has made a great breakthrough in India in reducing the out-of-pocket expenses and increasing the institutional deliveries.

Situation in Kerala

Kerala's maternal mortality rate (MMR) has shown a remarkable drop of 15 points from 2007-09 and it now stands at 66, according to the Sample Registration Survey (SRS) report 2010-12, which is achieved through various innovative programmes initiated by the government. However, the MMR in Kerala is high when compared with other health

indices. The infant mortality rate (IMR) in Kerala is 12 per thousand populations, according to SRS 2013.

JSSK was launched in Kerala on 17th August 2012 - to be known as Mother and Child (Ammayum Kunjum) scheme in Kerala, and would help in bringing down the present MMR and IMR by half within the next five years. The programme has been implemented with the all provisions of JSSK entitlements in all government health facilities that function as delivery points. Every pregnant woman coming to a government hospital would thus be entitled to availing travel expenses to the hospital and back. However, the utilization of any scheme depends on the awareness among the beneficiaries. Awareness regarding JSSK among pregnant mothers is not much studied in Kerala. With this background the present study focused on assessing the level of awareness about JSSK among the pregnant mothers in the districts of Kerala.

Data sources show that Kerala has less than 2 percent home deliveries and only about 30 percent of population use government facilities for delivery. It has been assessed that the implementation of JSSK programme at health facility has reduced the out of-pocket expenses. But a recent community-level assessment of the scheme shows that out-of-pocket expenses mostly during the ANC period has increased to a considerable level. Of late, a discussion at the state level on the expenditure incurred under JSSK programme highlights increased expenditure on ultrasound scan during the ANC period. The increase in the number of caesarian cases raises concern. Based on this, an assessment of the JSSK programme both at the facility and community (?) level is imperative. Hence, the objective of the study is to assess the implementation of JSSK programme and to learn about the quantum of shift in private to public health deliveries that the government-level health facilities claim to have occurred.

Objectives

1. To assess the awareness of pregnant women on the entitlements of JSSK
2. To assess the out- of-pocket expenses of JSSK entitlement
3. To compare the out-of-pocket expense of JSSK with delivery expenses of private hospitals
4. To assess the mode of implementation of JSSK entitlement in selected government hospitals of Ernakulam and Alappuzha

Data and Methodology

For assessing the awareness and out-of-pocket expenditure incurred by pregnant women under JSSK in Kerala, a community and facility-based observational cross-sectional study was conducted during January 2016 to March 2016 at two districts of Kerala i.e., Ernakulam and Alappuzha. The criterion of the district selection is based on the place of delivery. At Ernakulam, delivery at private hospitals is more than in the government hospitals whereas, at Alappuzha, government hospital delivery is higher than in private hospitals. For the representation of district, one urban area and two rural areas were selected from three taluks of each district by simple random sampling method. The study areas selected from Ernakulam district were Vengola (R) Varapetty (R) and Nettoor (U) and from Alappuzha district were Kuppappuram (R) Ambalapuzha (R) and Muhamma (U). A total of 420 mothers, who delivered from July 2015 to preceding the survey date were interviewed, of which 232 women were from Ernakulam and 188 were from Alappuzha. Out of the 232 beneficiaries interviewed from Ernakulam, 73 women were from Vengola area, 81 from Varapetty and 78 from Nettoor. In Alappuzha (188) district, 57 were from Kuppupuram, 61 from Ambalappuzha and 7 from Muhamma areas.

Information was collected by face-to face interview techniques using a schedule (as per JSSK Guidelines) prepared in regional language. Consent from mothers and health professionals were sought before the interview. Public health professionals of the district, taluk hospitals, other delivery points and JPHNs at Sub-Centres were interviewed for knowing the implementation of JSSK entitlements. The information obtained from mothers include Socio-demographic information, awareness and out-of-pocket expenditure under JSSK. The collected data was entered in Cs Pro and was analyzed using the software SPSS. The descriptive analysis of data was depicted in graphs, percentages etc. Chi-square test is used and significance level is considered, at p-value < 0.05 and <0.01.

Results

The socio-demographic profile of the 420 mothers indicated that their mean age of mothers is 26.9 ± 4.3 [$\pm$ standard deviation (SD)] and around 42.6 percent of the beneficiaries fall within the age group 18-25 years. Around 45.2 percent of the study population received a level of education of degree and above and the proportion of women received SSLC or less education is about 18.3 percent. Occupational distribution of the sample population shows that more than 87.0 percent are housewives. More than three-fifth of the sample population belong to APL category (above poverty line).

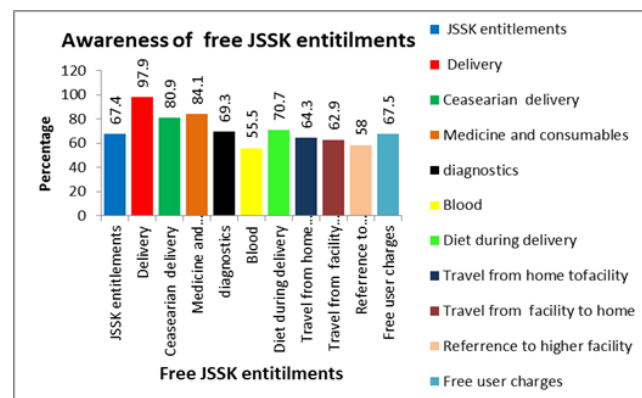
The obstetric characteristics of the sample population revealed that above 49.3 percent and 39.8 percent of the beneficiaries are in the parity 1 and parity 2 respectively. Maximum parity of the study population was 3. The utilization of health services is mainly associated with availability, accessibility, affordability and quality of health services (Abbas et al, 1986). The percentage utilization of private health facility (56.0 percent) is more than that of Government facility (44.0 percent). According to NFHS-1 and DLHS-3, a significant rise is observed in the institutional and C-Caesarean

deliveries in Kerala as well as in India. Of the total beneficiaries, Caesarean deliveries are more than 50 percent which is very high when compared to WHO recommended level.

Awareness on JSSK entitlements

With regard to the knowledge regarding the JSSK entitlements for pregnant women, 67.4 percent are aware of the free entitlements (figure 1). Around 98.0 percent of the eligible women in the study population know about the provision of free delivery and 80.9 percent are aware of free C- section delivery. Of the participants, more than 84.1 percent knows about free medicine and consumables but, only 69.3 percent knows about free lab test, blood test, urine test & scan facility.

Figure 21.1: Awareness regarding JSSK



Only 55.5 percent are aware about the provision of free blood. The availability of free food at government health facilities is known to 70.7 percent. More than three-fifth of the respondents are aware about free transport for mother from home to facility, and back home after delivery. It is also seen that 58.0 percent of the mothers are aware about transport for mother between health facilities in referral cases. Around 67.5 percent of the participants are aware about the free user charges like OP ticket and admission charges.

About 63.8 percent of the eligible participants living in Ernakulam district have awareness re-

garding the free entitlements for pregnant women whereas in Alappuzha district the corresponding percentage is about 71.8 percent (Figure 21.2). More than three-fourth of the pregnant women who delivered in the government hospital knew about free entitlements provided by the government facility, while three-fifth of the women who delivered in the private hospital knew about the JSSK entitlements.

Figure 21.2: Awareness regarding JSSK

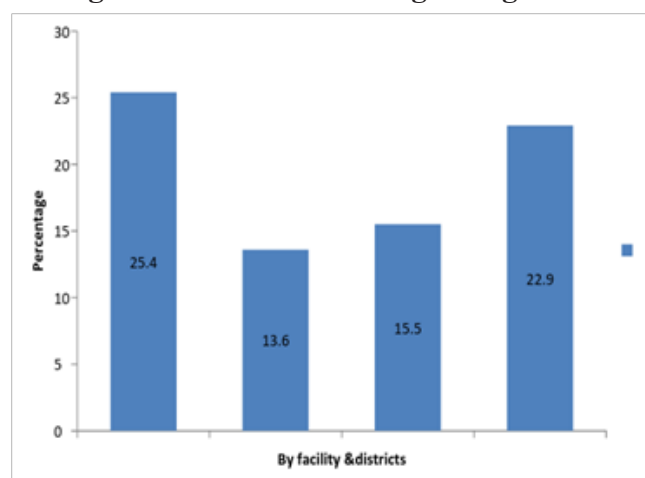
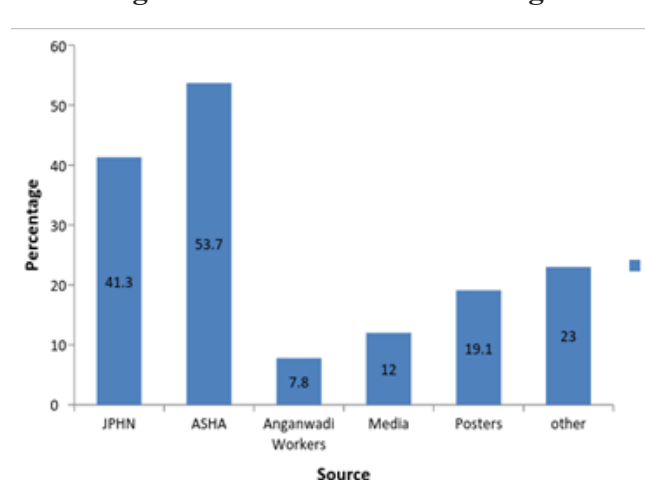


Figure 21.3: Source of knowledge



The main JSSK promoters are ASHA workers, who have imparted the awareness regarding the programme to more than half of the total sample women. The corresponding contribution from the JPHN part is 41.3 percent (Figure 21.3). About 19.1 percent, 12.0 percent and 23.0 percent of the eligible participants have achieved their knowledge

regarding the JSSK entitlements through posters, media and other sources respectively.

Antenatal Care

ANC improves the survival and health of babies directly by reducing still births and neonatal deaths and indirectly by providing an entry point for health contacts with the woman at a key point in the continuum of care (Antenatal care- WHO). A very high percentage (59.0 percent) of women utilized the private health facility for treatment during pregnancy while those who utilized government facility are only 17.9 percent. Around 23.1 percent of the women preferred to go to private practicing government doctor for ANC care. Around three-fourth of the respondents (74.5 percent) received 9 -12 check-ups during pregnancy with the average being 10.7 ± 2.6 and median being 10. Completely free ANC care received from the government facility is only 17.3 percent.

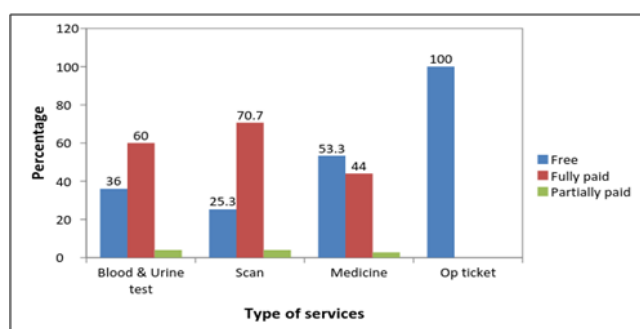
Table 21.1 explains the mean and median of the number of check-ups received during pregnancy based on place of treatment. More number of check-ups [$11.2 (\pm 3.0)$] received by the respondents are from private practicing government doctors followed by private hospitals [$10.8 \pm (2.5)$]. Average number of check-ups received from government facility during pregnancy is $9.9 (\pm 2.4)$ whereas, from private hospital is $10.8 (\pm 2.5)$.

Table 21.1 Descriptive statistics for no of checkup based on place of treatment during pregnancy

Median	Mean $\pm$ SD	Place of treatment during pregnancy
10	9.9 ± 2.4	Govt. hospital
10	10.8 ± 2.5	Private hospital
10	11.2 ± 3	Private practicing Govt. doctor

Classification according to nature of payment and type of service received from government facility is given in Figure 4. Majority of the sample women remitted full payment for their blood/urine test (60.0 percent) and for Ultrasound scan (70.7 percent) at the government facility during the pregnancy period. Full free services received for blood/urine test and ultra sound scan at government facility is only 36.0 percent and 25.3 percent respectively.

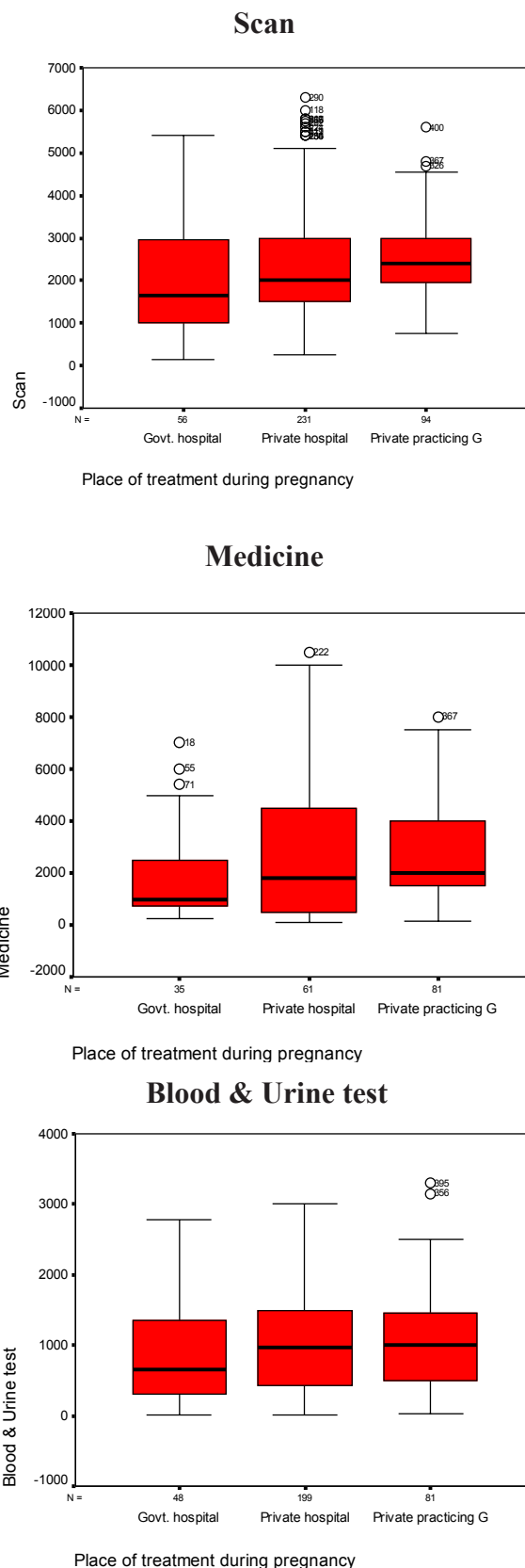
Figure 21.4: Type of services

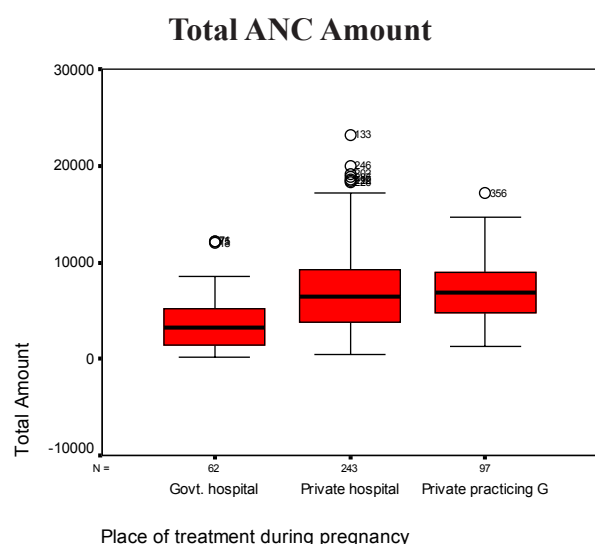


In the case of medicine and consumables, around 53.3 percent got the free services and 44.0 percent paid full amount for getting medicine and consumables. Payment for OP ticket is completely free for all the respondents.

The hospital bills of the beneficiaries were verified and the cost incurred by them with respect to various services were taken and analyzed. Whenever the beneficiaries were unable to produce such type of evidence, the expenses reported by the sample women were included in the data set. In addition, the expenditure might have been high due to some emergency situation that arose during ANC period and at the time of delivery. This may lead to over reporting or under reporting of the cost of ANC and cost of delivery. To eliminate the outliers based on the cost of ANC and delivery services, box plots are constructed. The graphical representation of the box plot showing various costs of antenatal care is given below.

Figure 21.5: Box plot showing the various antenatal cost based on place of antenatal check-up





A comparison of ANC cost with respect to various ANC services based on the place of ANC check-up clearly shows a wide data set indicative of comparatively high SD. The mean is not clustered and skewed to smaller SD, suggesting that there is irrational cost levied on the beneficiaries irrespective of institution or cost. But on comparison, the cost incurred with respect to private practicing government doctors is comparatively higher for blood and urine test, scan and admission at the time of ANC whereas in the case of medicine and OP tickets, the private hospitals are charging higher amounts (Table 21.2). The disparity between government and private hospital is minimum for blood and urine test and maximum for op ticket favoring government hospitals. It can also be seen that the total amount of the ANC cost is almost

double in the private hospital or private practicing government doctors compared to that in government hospitals. It is also seen that maximum exploitation happens to the beneficiaries at the time of initial appointment itself in the private hospitals. This clearly indicates that it is not for the treatment or any tests that the patient spend much but for merely getting an appointment or visit to consult with the doctor for seeking remedy for an ailment. Since the SD was considerably higher, the interval estimate is found out using confidence interval which indicated the extent of uncertainty in our estimate of true mean. It can be seen that a narrower interval estimate is observed for the total cost in each category. A 95 percent confidence limit indicates there is not a high uncertainty in our estimate of the true mean. The P value <0.05 corresponding to scan and OP tickets and p value <0.01 corresponding to total cost are significant.

Delivery Care

Table 21.3 shows the comparison of type of delivery based on place of delivery. At government hospital the percentage of normal delivery (57.8 percent) is higher than the caesarian delivery. By comparing the government and private hospital, caesarian delivery is more in the private hospitals (57.4 percent). The χ^2 value showing the relationship between type of delivery and place of delivery is significant at 0.01 level.

Table 21. 2 Comparison of Antenatal cost based on place of antenatal checkup

Place of ANC checkup by ANC services		Mean	SD	N	95 percent CI	F	p
Blood & Urine test	Govt. hospital	902.7	745.2	48	691.88 - 1113.5	0.78	0.461
	Private hospital	945.5	736.1	199	843.18 - 1047.73		
	Private practicing G. doctors	1046.5	654.8	81	903.89 - 1189.07		
	Total	964.1	717.8	328	886.47 - 1041.83		
Scan	Govt. hospital	1937.1	1207.5	56	1620.87 - 2253.41	3.88*	0.022
	Private hospital	2333.3	1288.6	231	2167.14 - 2499.48		
	Private practicing G. doctors	2495.2	927.3	94	2307.7 - 2682.62		
	Total	2315.0	1206.5	381	2193.86 - 2436.17		

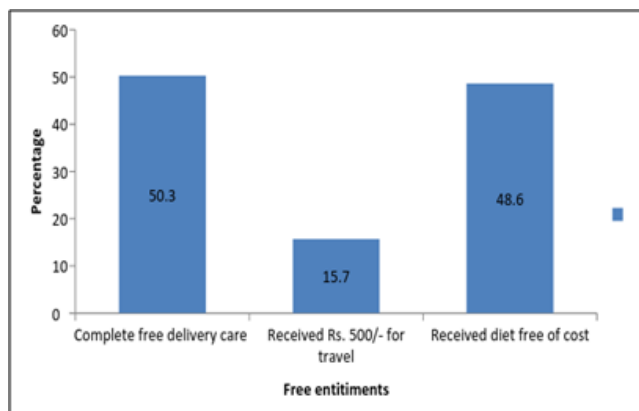
Medicine	Govt. hospital	2001.5	1805.8	35	1403.25 - 2599.78	1.83	0.163
	Private hospital	2867.9	2854.8	61	2151.46 - 3584.28		
	Private practicing G. doctors	2709.6	1746.4	81	2329.3 - 3089.96		
	Total	2624.1	2212.6	177	2298.18 - 2950.11		
Op ticket	Govt. hospital	.	.	0		5.94*	0.016
	Private hospital	3729.1	4068.7	159	3096.66 - 4361.53		
	Private practicing G. doctors	2167.1	3039.5	47	1298.09 - 3036.04		
	Total	3372.7	3906.9	206	2839.18 - 3906.24		
Admission at the time of ANC	Govt. hospital	725.0	671.8	2		0.45	0.637
	Private hospital	1382.9	1000.5	105	1191.54 - 1574.27		
	Private practicing G. doctors	1600.0	.	1			
	Total	1372.7	992.7	108	1185.5 - 1559.96		
Total Amount	Govt. hospital	3653.4	2802.8	62	2955.76 - 4351.08	** 20.04	0.000
	Private hospital	7181.4	4460.8	243	6620.48 - 7742.24		
	Private practicing G. doctors	6951.4	3229.4	97	6308.7 - 7594.04		
	Total	6581.8	4156.6	402	6175.42 - 6988.09		

** Significant at 0.01 level

* Significant at 0.05 level

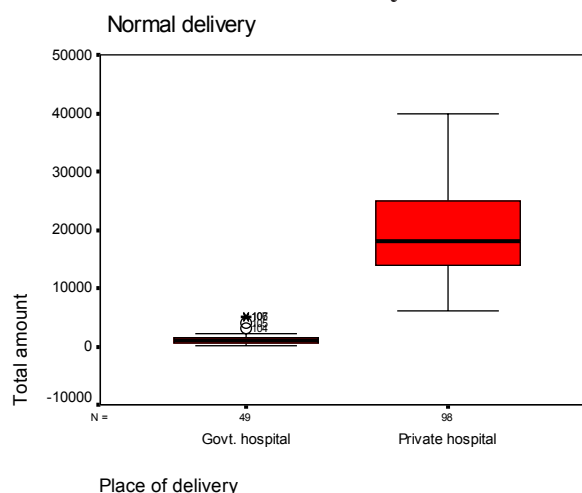
Table 21.3 Comparison of type of delivery based on place of delivery

Type of delivery	Govt. hospital	Private hospital	Total
	Percent	Percent	Percent
Normal	57.8	42.6	49.3
Caesarean	42.2	57.4	50.7
Total	100	100	100

 $\chi^2 = 9.67^{**}$, $p = 0.002$ **: - Significant at 0.01 level**Figure 21.6: Free entitlements**

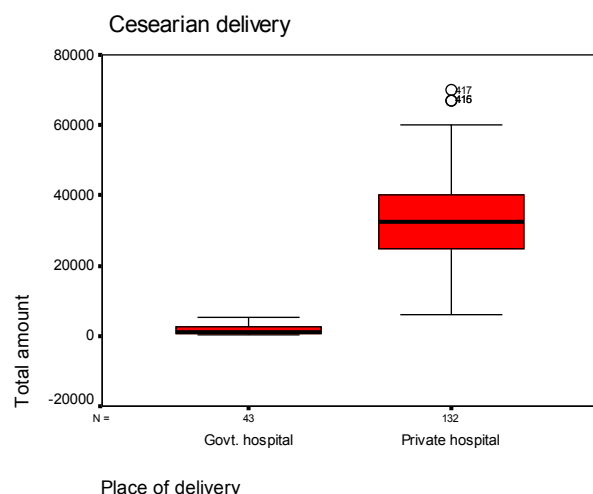
Distribution of free entitlement under JSSK for delivery received by the beneficiaries is shown in figure 21.6. Complete free delivery care service from government facility was received by 50.3

percent of the respondents. Only 15.7 percent of the beneficiaries received cash assistance for travel from government hospital. Around 48.6 percent of the participants have got food at free of cost.

Normal Delivery

Irrespective of the type of delivery, the cost incurred is very high in private hospital (Table 21.4). It is around 15 times and 19 times higher in private hospital for normal and caesarian deliveries respectively. The SD is very high and the 95 percent confidence interval gives a good estimate of the cost of delivery. The P value is <0.01.

Caesarean Delivery



while the remaining 44.9 percent informed that they are satisfied with the services received from the government facility.

Implementation of JSSK entitlement at delivery points

As per government orders all the delivery points of the State had implemented the free entitlements of JSSK by October 2012. Interviews with the key personalities in health department reveals that a major proportion of the deliveries are carried out in DH Aluva, THQH Perumbavoor, THQH Thripunithura, GH Ernakulam, THQH Kothaman-

Table 21. 4 Comparison of cost of delivery based on type of facility

Type of delivery	Place of delivery	Mean	SD	N	95 percent CI	t	p
Normal	Govt. hospital	1231.5	1098.5	49	923.95 - 1539.12	16.25**	0.000
	Private hospital	18980.4	7594.7	98	17476.74 - 20484.08		
Caesarean	Govt. hospital	1728.8	1339.6	43	1328.4 - 2129.18	16.87**	0.000
	Private hospital	33503.6	12302.1	132	31404.95 - 35602.32		

** : - Significant at 0.01 level

It is evident from the sample that only 21.6 percent of the eligible women reported that they have given bribe amount for delivery. The average and median of out-of-pocket expenditure of other costs (amount paid for bribing) for government hospital delivery is revealed in table 21.5. The average out-of-pocket expenditure was Rs.1702.8 ($\pm$ 830) and the median was Rs.1550. The major amount is usually paid to gynecologist for delivery followed by anesthetist and nurses.

Level of satisfaction of beneficiaries

Majority of the respondents (61.1 percent) reported that the food quality is good whereas 37.8 percent of the mothers are satisfied with the quality of the food provided from the facility. Around 55.1 percent of the respondents informed that the services provided from the health facility are good;

galam and THQH Moovattupuzha at Ernakulam district, whereas in Alappuzha, a large number of beneficiaries opted for MCH Vandanam, W&C Hospital Alappuzha and THQH Cherthala for delivery. The findings regarding the implementation of JSSK entitlement in the above delivery points based on interviews with key personalities also point out the following facts. A hospital-based assessment lead to the fact that implementing the entitlements under JSSK varied from hospital to hospital. There are so many factors which hinder the implementation of JSSK entitlements which are different at different hospitals.

The flow of fund is often limited and is not continuous and is thus regarded as the main obstacle raised by the hospital management for the effective implementation of JSSKs. The supply of medicines and consumables essential to manage nor-

Table 21.5 Descriptive statistics showing other costs (bribe amount) of government hospital delivery

Variables	Mean $\pm$ SD	Median	Mini	Maxi	N
Gynecologist	1212.5 $\pm$ 394.8	1000	1000	2000	16
Anesthetist	583.3 $\pm$ 204.1	500	500	1000	6
Other doctors	400 $\pm$ 0	400	400	400	1
Nurses	500 $\pm$ 282.8	450	200	1000	6
Other staffs	334.6 $\pm$ 167.6	300	50	600	13
Total amount	1702.8 $\pm$ 830	1550	50	3300	18

mal delivery and cesarean section is not sufficient from the KMSCL. Some of the needy medicines/ consumables such as measured volume set, disposable enema, and suction drain (Klear site) T. Misoprost etc. are not being supplied by the KMSCL. A few consumables used for cesarean section supplied by the KMSCL are of poor quality and doctors are not ready to use such medicines/ consumables. The facility such as THQH Perumbavoor and DH Aluva take monthly/bi-weekly intend from the wards. Then medicines are purchased from Neethi or Maveli medical stores, after getting non-availability certificate from KMSCL. The procedure for purchasing from outside is cumbersome and the chance of audit objection/ fixing liability later on is higher. Hence, the authorities of certain hospitals are not ready to take any risk in this regard. In Alappuzha district, hospitals have made tie-ups with Karunya medical, Neethi store and Medi bank. The process of getting intent from the ward nurse in the absence of medicine/ consumables and procuring medicine from the tie-up medical shop are time-consuming and often not

feasible. So for getting the medicine immediately, doctors write prescription to patients and ask them to purchase immediately and insist them not to wait for the intent from the ward. In the present system, out-of-pocket expense is incurred for getting medicines/ consumables but this varies with hospitals. The mode of implementation for drugs and consumable is good in DH Aluva and THQH Perumbavoor (missing?). The out-of-pocket expense regarding medicine is high in all the government facilities of Alappuzha district.

When considering the implementation of getting free lab tests to beneficiaries, there are not much differences among the hospitals. In all the hospitals, the tests available in the hospital lab were made available to inpatients free of cost. All the hospitals have tie-ups with outside labs to provide free diagnosis based on the intent provided from the duty nurse signed by doctor. Due to shortage of funds, hospital takes certain steps to limit the costs. The GH restricted the outsourced diagnostic care service to inpatients only. Most of the hospitals restricted the diagnostic tests which are available in the hospital lab to OP antenatal women. The DH Aluva had stopped all the outsourced service for two months in last year due to acute shortage of funds.

Regarding the implementation of getting free scanning to antenatal women, different modes were selected, and it is seen that beneficiaries were not benefitted in most of the hospitals. Since the Antenatal women attending the clinic days were less, almost all women receive three antenatal scanning, including anomaly scan in THQH Perumbavoor. Three free normal scanning were provided at DH Aluva with the scanning machine of Lab and with the outsourced service. THQH also provided free scanning. The GH Ernakulam, Medical College Vandanam had scanning machines, but were not providing scanning to OP antenatal women and

were limited to inpatients only. W&C Alappuzha provides 20 scans per week and free scanning is available to some of the beneficiaries. But THQH Cherthala has limited free scanning due to lack of fund. In Alappuzha medical college, doctors insist antenatal women to get scanning done in private scanning centres and almost all the beneficiaries of MCH Vandanam completely depend on private scanning centres.

Providing free blood under JSSK was implemented in different manners in different hospitals. DH provided free blood to needy from the Regional Blood Bank which is attached to the hospital and medical college provided free blood from its blood bank. THQH Perumbavoor also provided blood from the BSU attached with or made arrangements to get free blood from Aluva. Only thing was that the patient had to arrange it from Aluva at their own cost. The cost for cross matching is provided in GH Ernakulam and THQH Thripunithura, the patient has to bear the cost of blood.

Free diet was provided in the facilities selected. The contract for the diet was given to external agencies and food was provided free for 3 days for normal delivery and 7 days for caesarian cases. In Alappuzha Medical College, free diet was not implemented for a period of 3 months in the last year due to some technical reasons.

Providing the travelling allowance was not uniform. The facilities selected in Ernakulam district, except for GH, provided Rs. 500/- towards travelling expense at the time of discharge. Travelling allowance has not been disbursed in any of the facilities at Alappuzha. Most of the facilities used own ambulance for referral, but if the ambulance was not readily available, patient has to bear the cost of travel for referral to higher facility.

There was no separate Help desk for JSSK, and

no grievance redressal system for the JSSK in all the facilities selected. There was only one PRO in each hospital, who had to monitor the functioning of JSSK, RSBY and 7-8 programmes under NRHM. Taluk hospital managed 10-20 deliveries per month and the PRO alone was attending the help desk and handling the grievances regarding JSSK. But in facilities such as MCH Vandanam where around 300 deliveries are managed per month, the only one PRO had to supervise and monitor JSSK and all other programmes under NRHM. Hence, guidance regarding JSSK and grievance redressal mechanism in such facilities was poor. The IEC regarding JSSK was poor in all the facilities. No boards or hoarding were found in any of the facilities.

The JPHNS who were working in the field played an important role in the success of all health-related programmes. An interview with JPHNS was conducted to find out the merit and demerits of JSSK experienced by the JPHNS. According to JPHNs, usually pregnant women had to purchase medicine from outside. They had to depend on private labs for investigations and private scanning centres. Medicines and consumables were usually prescribed to purchase from outside during delivery. More importantly, free referral transport or free blood was not available in most cases. In fact, they were not imparting awareness on free entitlements of JSSK to the antenatal women. If the beneficiaries are given proper awareness about the free entitlements, and they are denied of these entitlements from the facility, the JPHNs fear that this would lead to aggressive reaction from the part of the beneficiaries. Other out-of-pocket expenses related to JSSK, according JPHNs are: A large share of doctors receives money after delivery, a few of them demand fixed rates for normal delivery and cesarean. If beneficiaries directly approach OP without consultation at their private practicing centre, the behaviour of the doctors was report-

ed to be very rough. According to JPHNs, when considering the doctors' consultation charges, lab and scanning expenses, the antenatal care cost of women approaching private practicing doctors of government facilities is almost equal to those approaching private hospital. According to JPHNs of Alappuzha, number of scanning recommended by the private practicing doctors was higher than the number of scanning done in private hospitals.

Discussion: A significant rise in institutional and C-section delivery is found in the district. The scheme like JSY and JSSK may have influenced the pregnant women on accepting institutional deliveries. Caesarian deliveries at private hospital are high which may lead to the conclusion that place of delivery is becoming very important to regulate the rise of C- section (CS) delivery. An increase in the rates of CS delivery is a burden on family as well as on health system and also lead to many health hazards. Government should caution against the promotion of C-section deliveries at private hospitals through measures like external auditing. Government should provide more enhanced infrastructure facilities with sufficient number of doctors according to the patients' strength in all government hospitals and at all delivery points.

Women who delivered at government facility have more knowledge regarding JSSK. The awareness level about the scheme was high at Alappuzha district. Awareness level of all JSSK entitlements for the pregnant women in our study is not complete whereas awareness on provisions of each component is more than 55 percent. Maximum awareness was noticed on the provision of free normal delivery followed by medicine and consumables and C-section delivery. Further efforts are needed from the part of health providers to increase the overall awareness of various entitlements under JSSK among pregnant women. Despite the fact

that sincere effort has been made in imparting awareness regarding various programmes, many of the facilities have not implemented them properly due to the non-availability of fund at times. For the successful utilization of the scheme, more efforts are to be taken from the part of service providers to increase the awareness level among the pregnant women. Steps are to be taken to ensure effective implementation of the envisaged programmes.

A smaller proportion of the pregnant women received completely free ANC care from the government facility. Scan, diagnostic tests and medicines are the major causes of high OOPs incurred at government facility during ANC. Highest mean OOPs for the diagnostics test and scans were seen at private practicing government doctors whereas in the case of medicine and OP tickets, the private hospitals were charging higher amounts. The disparity between government and private hospitals is minimum for blood and urine test and maximum for op ticket favoring government hospitals. It can also be seen that the total amount of the ANC cost is almost double in the private hospital or private practicing facility of government doctors compared to that in government hospitals. It is also seen that maximum exploitation happens to the beneficiaries at the time of taking appointments in the private hospitals. Normal delivery is comparatively higher at Government hospital. More than 50 percent of the beneficiaries of the government hospitals received complete free delivery care service. Mean cost is around 15 times and 19 times higher in private hospital for normal and caesarian deliveries respectively. The supply of medicines and essential supplies/consumables from KMSCL is insufficient to manage normal and C-section delivery. The tie-up with the medical stores is ineffective forcing the beneficiaries to buy medicines at their own cost. There is no uniformity in providing travelling allowance to beneficiary after deliv-

ery. The provision for free diet is almost uniform in all facilities. For the successful utilization of the scheme and for increasing the awareness level among pregnant women, more efforts are to be taken by the service providers. The present study thus points out the fact that there exists OOPs for ANC and delivery services and hence, the gaps are to be reduced by liberalizing the policy on funding of the components of the JSSK entitlements. Over expenditure incurred for the beneficiaries at public facilities are mainly caused by the complications during pregnancy and delivery period. The major proportion of OOPs in a government health facility was caused by Medicine, Ultrasound scan and diagnostic test. No homogeneity is found in the payment towards travelling allowance. By reducing the gaps found in the out-of-pocket expenditure, the policy decision regarding JSSK has to be modified to remove the financial obstruction.

In general, majority of the beneficiaries visited have complained against the non-availability of sufficient specialist doctors. Hence, they are compelled to visit the private hospitals for ANC check-ups and delivery. In addition, people have a general belief that private hospitals are more equipped to tackle any kind of emergency situations. But it is true that the attitude is slowly changing due to the recent governmental efforts to upgrade infrastructural and related supports at public health facilities.

Reference

1. Abbas, A. A, and Walkern, G.J.A, 1986, "Determinants of the utilization of maternal and child health services in Jordan". *International Journal of Epidemiology*. 15: 404-407.
2. Antenatal Care - World Health Organization. http://www.who.int/pmnch/media/publications/aonsectionIII_2.pdf
3. Ministry of Health and Family Welfare, 2011, *Guidelines for Janani-Shishu Suraksha Karyakram* (JSSK). National Rural Health Mission, Maternal Health Division, Government of India, Nirman Bhavan, New Delhi.
4. Paul V. K, Sachdev H .S, Mavalankar D, Ramachandran P, Shankar M. J, Bhandari N, *et al*. 2011, "Reproductive health, and child health and nutrition in India: meeting the challenge". *Lancet*, 337:332-49.
5. World Health Organization, 2006, Birth and Emergency Preparedness in Antenatal Care: *Integrated Management of Pregnancy and Child-birth (IMPAC)*, Department of Making Pregnancy Safer. Geneva, Switzerland: World Health Organization (WHO).

CHAPTER - 22

Ageing in Assam: A district-wise comparative study based on Census data 1991-2011

Dr. Dilip Kr. Kalita and Arpita Basak

Population Research Centre, Guwahati

Abstract

One of the most critical factors of demographic trends is ageing and this implies that the share of older individuals living in the country increases over time. Ageing is a global phenomenon in the sense that it is experienced by all societies in all generations and all countries but its magnitude and manifestations are not the same everywhere. In the present paper, an attempt has been made to give a comparative perspective of ageing of the population taking place in Assam and its districts. The data has been taken from Office of the Registrar General and Census Commissioner, 1991-2011. The study reveals that in Assam the percentage of child population (0-14) has decreased whereas the percentage of population in the age group of 60 and above has increased continuously during 1991-2011. Among the districts of Assam, Dhubri recorded the highest percentage of child population (0-14 years) which might be due to lowest literacy rate (58.33 percent) (as per 2011 census) in the district. The total dependency ratio of Assam has decreased, but on the other hand, the old dependency ratio has exhibited an increasing trend over the years 1991-2011. Considering the growing elderly population in the State, the study recommends adoption of different policies relating to socio-economic and demographic conditions especially for women and rural population.

INTRODUCTION

Population ageing is perhaps the most significant feature of global demography, with every country projected to experience an increase in the share of people aged 60 and above in the coming decades. Ageing is a global phenomenon in the sense that it is experienced by all societies in all generations and all countries but its magnitude and manifestations are not the same everywhere.

Elderly or old age consists of ages nearing or surpassing the average lifespan of human beings. The boundary of old age cannot be defined exactly because it does not have the same meaning in all societies. Government of India adopted 'National Policy on Older Persons' in January, 1999. The policy defines 'senior citizen' or 'elderly' as a person who is of age 60 years or above.

When the size of the older segments of population increases in relation to that of younger segments the population is said to be ageing. In the ageing process, the youth population not only often undergoes a decrease in absolute terms but also loses its proportion in relation to the older population. As the size of young cohorts has declined and size of old cohorts has increased, the age of national population has been increasing, resulting in gradual population ageing. It has been observed that science and technology, industrialization, modernization and globalization have improved health care and standards of living, making people live much longer than expected and, thereby causing an increase in the growth of elderly population in all societies. The rise in the population of elderly was thought to be the concern of only developed countries, but now it has become an equally important issue even in the developing countries as well. The population aged 60+ is increasing rapidly in our country and it is expected to touch 177

million by 2025.

Increasing number of older persons in the population is leading to a dramatic increase in academic and professional interest in gerontology- a field for the study of ageing, which is still developing its own scientific basis. It is a field concerned primarily with the changes that occur between the attainments of maturity and the death of individual and with the factors that influence these changes.

As per available data from SRS, in India there has been a gradual decline in the share of population in the age group of 0-14 from 36.3 to 29.5 percent during 1991 to 2011, whereas, the proportion of economically active population (15-59 years) has increased from 57.7 to 62.5 per cent during 1991 to 2011. On account of better education, health facilities and increase in life expectancy, the percentage of elderly population (60+) has gone up from 6.0 to 8.0 percent during the period under reference.

In Assam, the percentage of child population (0-14 years) has steadily decreased from 40.37 percent in 1991 to 32.86 percent in 2011 and it is expected to decline to 25.39 percent by the year 2029. On the other hand, there has been a gradual increase in the percentage of elderly population (60+ years) from 5.35 in 1991 to 6.66 percent in 2011 and it is expected to increase upto 8.21 percent by 2029.

A direct consequence of an increasing proportion of older population on the population is an increasing dependency ratio. It is important to note that due to ageing process the total dependency ratio may not change very much, but the old dependency ratio may increase significantly, or we can say that structure of dependency ratio change dramatically. This has significant implications since elderly people need and use much more health and

other services than younger people.

Objectives

The objective of the paper is to study the district-wise age composition of population of Assam by broad age groups of 0-14, 15-59 and 60 years and above. It also aims to gauge the levels and trends of aged population (60 years and above) in Assam. To meet the general objectives, some specific objectives are also planned to assess some variables viz. dependency ratio, sex ratio, median age and ageing index.

Data and Methodology

The paper reports on data drawn from the Registrar General and Census Commissioner, 1991-2011. District-wise population in the five-year age group of Assam have been considered. For the purpose of the paper, the population has been divided into three broad age groups viz. 0-14, 15-59 and 60 and above years. Projected figures for the years 2019 and 2029 have been calculated based on 2011 census and projected population of 2019 respectively using exponential rate of population growth.

Results and Discussion

Broad age structure:

The ageing process can be measured by the broad age structure. The broad age structure in any population reflects changing composition of children (0-14 years), adults of working ages (15-59 years) and the elderly persons (60+). Table 22.1 reveals that the child population (0-14 years) of Assam has declined continuously during last three censuses from 1991 to 2011. It was found declined from 40.37 percent in 1991 to 32.86 percent in 2011. The child population will be 29.41 percent as projected for the year 2019 and 25.39 percent

for the year 2029. On the other hand, the population of working group (15-59 years) has found to be increased from 54.28 percent in 1991 to 60.47 percent in 2011. The population belongs to age group 60 years and above has also found to be increased from 5.35 percent in 1991 to 6.66 percent in 2011 in Assam. The projected figures also exhibit the increasing trend of the population of both the adult groups.

Tables 22.2 , 22.3 and Figure 1 exhibit the declining trend of child population and increasing trend of old population in Assam during the periods 1991-2029. The rate of decline of the child population in urban areas (8.1 percent) is found to be somewhat higher than that of the rural counterparts (7.1 percent) during the period 1991 to 2011. This finding, however, reflects the increase of working population (15-59 years) in rural areas. The population of age group 15-59 years is found to be increased by 5.96 percent in rural Assam from 1991 to 2011 against 4.57 percent increase of their counterparts in urban Assam during the same period. Of course, in contrast to this, the increase of elderly population (60+) in rural areas is found to be somewhat lower (1.13 percent) than that of their urban counterparts (2.53 percent) during the same period.

Table 22.4 depicts the district-wise population of broad age group during 1991-2011 and the projected population for the years 2019 and 2029. The child population is found highest in Dhubri district (38.59 percent) whereas the lowest is observed in Kamrup Metro district (23.50 percent) in 2011. When the district-wise population distribution of the State is considered, it is observed that all districts exhibited a decreasing rate of child population during the reference periods while the working and elderly population has shown an increasing trend.

Table 22.1: Percentage of Population by broad age group in Assam, 1991-2029 (Total)

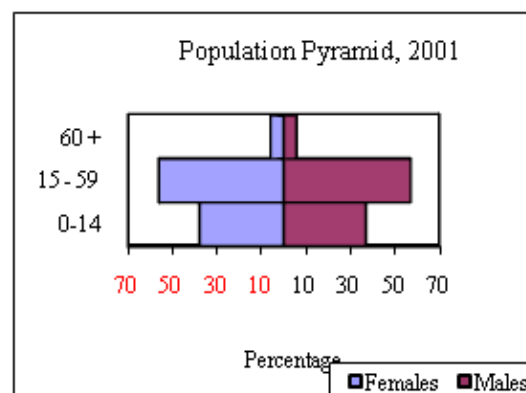
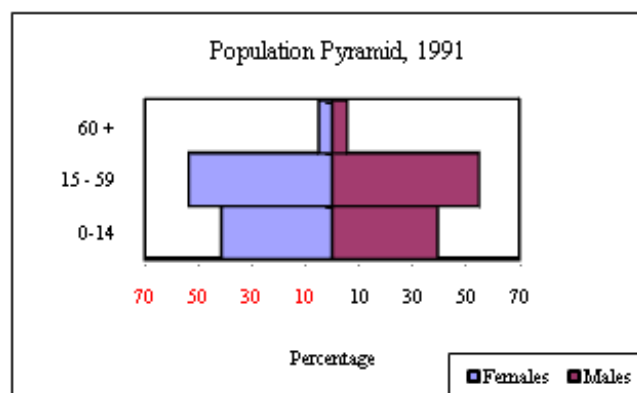
Age Group	1991			2001			2011		
	Male	Female	Total	Male	Female	Total	Male	Female	Total
0-14	39.44	41.38	40.37	37.03	37.89	37.45	32.84	32.89	32.86
15-59	54.89	53.62	54.28	57.15	56.21	56.69	60.54	60.40	60.47
60+	5.68	4.99	5.35	5.82	5.91	5.86	6.62	6.71	6.66
Total	11657989	10756333	22414322	13777037	12878491	26655528	15929549	15257203	31186752
Age Group	2019			2029					
	Male	Female	Total	Male	Female	Total			
0-14	30.0	29.15	29.41	25.94	24.83	25.39			
15-59	63.04	63.48	63.25	65.87	66.95	66.39			
60+	7.30	7.37	7.34	8.2	8.22	8.21			
Total	18010999	17622150	35626968	21140946	21292194	42406040			

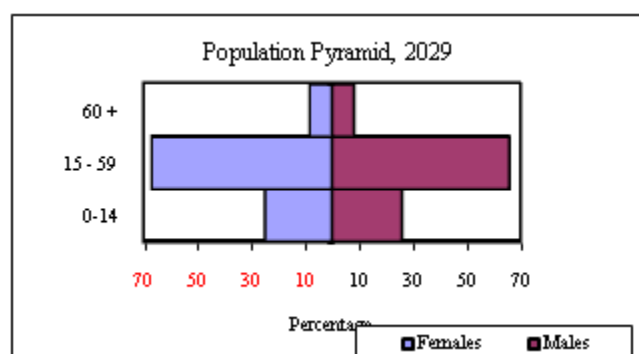
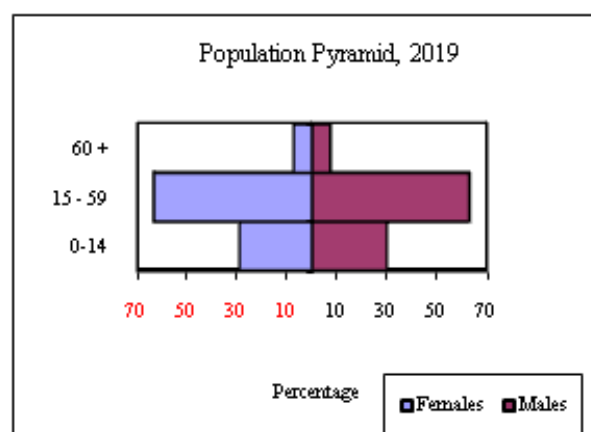
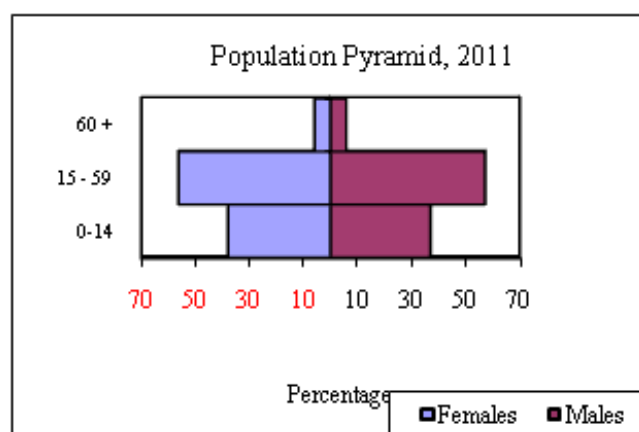
Source: Registrar General and Census Commissioner, 1991-2011

Table 22. 2: Percentage of Population by broad age group in Assam, 1991-2029 (Rural)

Age Group	1991			2001			2011		
	Male	Female	Total	Male	Female	Total	Male	Female	Total
0-14	40.66	42.18	41.39	38.52	39.04	38.77	34.28	34.33	34.31
15-59	53.58	52.82	53.21	55.65	55.06	55.36	59.25	59.09	59.17
60+	5.76	5.00	5.39	5.84	5.90	5.87	6.47	6.58	6.52
Total	10304161	9622366	19926527	11939945	11276343	23216288	13671137	13120878	26792015
Age Group	2019			2029					
	Male	Female	Total	Male	Female	Total			
0-14	31.06	30.76	30.92	27.28	26.61	26.95			
15-59	61.95	62.11	62.03	65.08	65.57	65.32			
60+	6.98	7.13	7.05	7.63	7.81	7.72			
Total	15333672	14922471	30254029	17814799	17666068	35471718			

Source: Registrar General and Census Commissioner, 1991-2011

Figure 22.1: Population Pyramid of Assam (broad age –group), 1991-2029



Source: Registrar General and Census Commissioner, 1991-2011

Dependency Ratio: A related measure of population ageing is the dependency ratio. It is defined as the ratio of population below 15 years of age and above 60 years of age to the population of working ages (15-59 years).

The young and total dependency ratios of Assam have gradually decreased during 1991 to 2011. The young dependency ratio of the State is found to have declined from 74.37 percent in 1991 to 54.34 percent in 2011 and is expected to decrease to 38.26 percent by the year 2029. During this period, the total dependency ratio of Assam has also decreased from 84.22 percent in 1991 to 65.37 percent in 2011. On the other hand, old dependency ratio indicates an increasing trend during the reference period from 9.85 percent in 1991 to 11.02 percent in 2011. This ratio is expected to grow to 11.60 percent by 2019 and 12.37 percent by 2029.

Table 22.3: Percentage of Population by broad age group in Assam, 1991-2029 (Urban)

Age Group	1991			2001			2011		
	Male	Female	Total	Male	Female	Total	Male	Female	Total
0-14	30.15	34.59	32.17	27.38	29.78	28.49	24.07	24.06	24.07
15-59	60.82	60.45	63.83	66.93	64.30	65.71	68.38	68.43	68.40
60+	5.03	4.95	5.00	5.96	5.92	5.80	7.55	7.51	7.53
Total	1353828	1133967	2487795	1837092	1602148	3439240	2258412	2136325	4394737
Age Group	2019			2029					
	Male	Female	Total	Male	Female	Total			
0-14	21.56	20.03	20.83	18.6	15.69	17.18			
15-59	69.04	71.0	69.97	69.18	73.27	71.15			
60+	9.40	8.97	9.20	12.22	11.04	11.67			
Total	2687678	2726867	5404370	3374871	3754349	7081481			

Source: Registrar General and Census Commissioner, 1991-2011

At district level, it is found that old dependency ratio has increased in all districts except in Karbi Anglong where the ratio decreased by one percent during 1991 to 2011. The highest young dependency is found in Dhubri district observed as 70.20 percent in 2011. Dhemaji district shows the rapid decrease in young dependency from 91 percent in 1991 to 57.49 percent in 2011. It is observed that the young dependency would remain highest (52.72 percent) in Dhubri district as projected for 2029.

$$\text{Total Dependency Ratio} = \frac{P_{0-4} + P_{6+}}{P_{5-9}} \times 100$$

$$\text{Young Age Dependency} = \frac{P_{0-4}}{P_{5-9}} \times 100$$

$$\text{Old Age Dependency} = \frac{P_{6+}}{P_{5-9}} \times 100$$

Table 2.3.1 reveals that the overall sex ratio of Assam has increased from 923 in 1991 to 958 in 2011 (per 1000 males). Though sex ratio in the

Table 22.4 District wise population by broad age group in Assam, 1991-2029

State/ District	Percentage of population by Broad age group														
	0-14					15-59					60+				
	1991	2001	2011	2019	2029	1991	2001	2011	2019	2029	1991	2001	2011	2019	2029
Assam	40.37	37.45	32.86	29.41	25.40	54.28	56.69	60.47	63.25	66.39	5.35	5.86	6.66	7.34	8.21
Kokrajhar	40.98	39.92	33.94	-----	-----	54.11	54.96	59.93	-----	-----	4.91	5.12	6.12	---	-----
Dhubri	44.34	41.84	38.59	35.90	32.36	50.31	52.71	55.36	58.09	61.37	5.34	5.38	5.79	6.01	6.27
Goalpara	43.53	41.28	36.13	32.21	27.65	51.48	53.30	58.12	61.81	66.13	4.99	5.42	5.75	5.98	6.22
Barpeta	42.31	40.14	36.44	-----	-----	51.69	53.62	57.07	-----	----	6.00	6.23	6.49	-----	----
Morigaon	43.16	40.65	36.85	33.92	30.42	51.12	53.47	56.67	59.11	61.98	5.72	5.88	6.48	6.97	7.60
Nagaon	41.31	39.35	35.62	32.76	34.29	52.65	54.30	57.60	60.14	56.97	6.04	6.36	6.77	7.10	8.74
Sonitpur	39.74	36.82	32.23	28.76	24.71	55.20	57.58	60.97	63.35	65.86	5.06	5.59	6.80	7.90	9.43
Lakhimpur	41.90	37.86	33.31	29.87	25.86	52.78	56.38	60.06	62.76	65.79	5.32	5.76	6.63	7.37	8.35
Dhemaji	45.28	39.74	34.25	30.15	25.43	49.76	54.89	59.59	63.06	66.97	4.96	5.37	6.15	6.79	7.60
Tinsukia	39.43	35.97	31.05	27.38	23.19	55.86	58.57	62.57	65.45	68.60	4.71	5.46	6.38	7.17	8.21
Dibrugarh	37.34	33.87	28.66	24.86	20.60	57.85	60.56	65.21	68.58	72.33	4.81	5.57	6.13	6.56	7.07
Sivsagar	36.70	33.14	28.05	24.33	20.14	58.04	60.72	64.54	67.15	69.8	5.25	6.14	7.41	8.53	10.06
Jorhat	36.59	32.19	27.05	23.80	20.07	58.28	61.64	65.62	69.76	74.51	5.18	6.17	7.33	6.44	5.42
Golaghat	40.28	35.72	29.66	25.23	20.29	55.20	58.94	66.42	66.36	69.16	4.51	5.34	6.92	8.0	10.55
Karbi-Anglong	42.20	40.42	35.15	31.18	26.58	52.62	54.67	59.62	63.37	67.74	5.18	4.92	5.23	5.45	5.67
Dima-Hasao	38.99	36.63	33.51	31.10	28.22	56.97	58.79	61.21	63.00	65.05	4.04	4.58	5.28	5.89	6.73
Cachar	38.26	35.77	32.17	29.43	26.21	55.47	57.51	60.82	63.34	66.32	6.28	6.72	7.01	7.23	7.47
Karimganj	39.95	38.65	36.00	33.94	31.44	53.34	54.56	56.98	58.87	61.15	6.71	6.79	7.02	7.20	7.41
Hailakandi	40.58	40.07	36.24	33.30	29.79	53.51	53.96	57.24	59.73	62.67	5.92	5.97	6.52	6.97	7.54
Bongaigaon	42.54	39.33	34.29	30.49	26.09	52.51	54.94	59.4	62.75	66.6	4.95	5.72	6.31	6.77	7.32
Chirang	-----	----	34.26	----	----	----	----	59.7	----	---	----	----	6.04	-----	----
Kamrup	37.18	33.06	30.29	23.01	18.92	58.00	61.25	62.51	68.59	73.45	4.83	5.68	7.2	8.39	7.62
Kamrup-Metro	---	-----	23.50	-----	----	-----	-----	69.50	-----	-----	-----	-----	7.00	-----	----
Nalbari	39.71	34.95	28.76	----	-----	54.89	58.49	63.07	-----	-----	5.40	8.31	8.17	-----	----
Baksa	-----	-----	30.40	----	-----	-----	-----	62.60	-----	----	-----	-----	6.99	-----	-----
Darrang	-----	-----	36.46	-----	-----	-----	-----	57.08	-----	-----	-----	-----	6.46	-----	-----
Udalguri	-----	-----	31.45	-----	-----	-----	-----	62.12	-----	-----	-----	-----	6.43	-----	----

age group 0-14 years had decreased from 969 in 1991 to 957 in 2001 but the ratio is found to have increased to 959 in 2011 and it is suppose to increase up to 962 by the year 2019 and 964 by the year 2029. On the other hand, the sex ratio of both

Ageing of the Elderly population:

It is an indicator to study the structural changes among elderly population. It is the proportion of the 'old –old', or the persons 70 years and old-

Table 22. 5: District wise Dependency Ratio of Assam, 1991-2029

State/Districts	Dependency Ratio								
	1991			2001			2011		
	Young	Old	Total	Young	Old	Total	Young	Old	Total
Assam	74.37	9.85	84.22	66.05	10.34	76.39	54.34	11.02	65.37
Kokrajhar	75.74	9.07	84.81	72.62	9.32	81.94	56.63	10.22	66.85
Dhubri	88.14	10.62	98.76	79.37	10.21	89.58	70.20	10.45	80.65
Goalpara	84.57	9.69	94.26	77.46	10.14	87.6	62.15	9.89	72.04
Barpeta	81.85	11.60	93.45	74.86	11.63	86.49	63.86	11.38	75.24
Morigaon	84.43	11.20	95.63	76.02	11.0	87.02	65.02	11.44	76.46
Nagaon	78.45	11.47	89.92	72.46	11.71	84.17	61.84	11.76	73.60
Sonitpur	71.99	9.17	81.16	63.95	9.71	73.66	52.86	11.16	64.02
Lakhimpur	79.38	10.08	89.46	67.16	10.22	77.38	55.47	11.04	66.51
Dhemaji	91.00	9.98	100.98	72.39	9.79	82.18	57.49	10.32	67.81
Tinsukia	70.59	8.44	79.03	61.41	9.32	70.73	49.62	10.19	59.81
Dibrugarh	64.54	8.31	72.85	55.92	9.19	65.11	43.95	9.4	53.34
Sivsagar	63.24	9.05	72.29	54.57	10.11	64.68	43.46	11.48	54.94
Jorhat	62.82	8.89	71.71	52.21	10.01	62.22	41.22	11.17	52.4
Golaghat	72.97	8.18	81.15	60.59	9.07	69.66	46.77	10.92	57.67
Karbi-Anglong	80.19	9.84	90.03	73.93	8.99	82.92	58.96	8.77	67.73
Dima-Hasao	68.45	7.09	75.54	62.32	7.79	70.11	54.75	8.62	63.37
Cachar	68.96	11.32	80.28	62.20	11.68	73.88	52.9	11.53	64.43
Karimganj	74.90	12.59	87.49	70.85	12.44	83.29	63.18	12.32	75.50
Hailakandi	75.83	11.06	86.89	74.26	11.06	85.32	63.32	11.40	74.72
Bongaigaon	81.01	9.43	90.44	71.59	10.41	82.00	57.72	10.62	68.34
Chirang	-----	-----	-----	-----	-----	-----	57.39	10.12	67.51
Kamrup	64.11	8.32	72.43	53.97	9.28	63.25	48.45	11.51	59.96
Kamrup-Metro	-----	----	-----	----	-----	-----	33.81	10.07	43.88
Nalbari	72.35	9.84	82.19	59.75	14.20	73.95	45.6	12.95	58.55
Baksa	-----	----	----	-----	-----	-----	48.56	11.17	59.74
Darrang	-----	----	----	----	-----	-----	63.88	11.31	75.19
Udalguri	-----	----	-----	-----	-----	-----	50.63	10.35	60.98

the groups, 15-59 years and 60+ years, is found to have gradually increased from 1991 to 2029. The study also reveals that as per projected population 2029, the females of these two age groups outnumber their male counterpart in the State.

er, as related to 'young old', or persons aged 60-69 years. From the Table 22.8 it is observed that in Assam as a whole, percentage of 'young old' population is higher than 'old old' population. The proportion of 'young old' population in As-

Table 22.6: District wise Dependency Ratio of Assam, 1991-2029

State/Districts	Dependency Ratio					
	2019			2029		
	Young	Old	Young	Old	Young	Old
Assam	46.49	11.60	46.49	11.60	46.49	11.60
Kokrajhar	-----	-----	-----	-----	-----	-----
Dhubri	61.81	10.35	61.81	10.35	61.81	10.35
Goalpara	52.12	9.67	52.12	9.67	52.12	9.67
Barpeta	----	----	----	----	----	----
Morigaon	57.38	11.80	57.38	11.80	57.38	11.80
Nagaon	54.47	11.80	54.47	11.80	54.47	11.80
Sonitpur	45.39	12.47	45.39	12.47	45.39	12.47
Lakhimpur	47.60	11.75	47.60	11.75	47.60	11.75
Dhemaji	47.81	10.76	47.81	10.76	47.81	10.76
Tinsukia	41.83	10.95	41.83	10.95	41.83	10.95
Dibrugarh	36.25	9.56	36.25	9.56	36.25	9.56
Sivsagar	36.23	12.70	36.23	12.70	36.23	12.70
Jorhat	34.12	9.23	34.12	9.23	34.12	9.23
Golaghat	38.02	12.67	38.02	12.67	38.02	12.67
Karbi-Anglong	49.20	8.59	49.20	8.59	49.20	8.59
Dima-Hasao	49.37	9.35	49.37	9.35	49.37	9.35
Cachar	46.47	11.41	46.47	11.41	46.47	11.41
Karimganj	57.65	12.23	57.65	12.23	57.65	12.23
Hailakandi	55.74	11.68	55.74	11.68	55.74	11.68
Bongaigaon	48.59	10.79	48.59	10.79	48.59	10.79
Chirang	-----	-----	-----	-----	-----	-----
Kamrup	33.55	12.24	33.55	12.24	33.55	12.24
Kamrup-Metro	-----	-----	-----	-----	-----	-----
Nalbari	-----	-----	-----	-----	-----	-----
Baksa	-----	-----	-----	-----	-----	-----
Darrang	-----	-----	-----	-----	-----	-----
Udalguri	-----	-----	-----	-----	-----	-----

Note: Projection is not done in some districts of Assam, namely Barpeta, Baksa, Darrang, Udalguri, Nalbari and Kokrajhar because these districts are either created or divided after 2001 census.

sam has shown a negligible increase from 61.30 percent in 1991 to 61.55 percent in 2011, whereas the ‘old old’ population exhibits a negligible decrease from 38.70 percent in 1991 to 38.45 percent in 2011. Among the districts of Assam, Golaghat district shows the highest percentage of ‘young-old’ population (64.48 percent) whereas lowest percentage is found in Barpeta district (58.18 percent) in 2011.

Sex Ratio: The life expectancy of female is higher than male. In other words, it can be mentioned that women outnumber men in the elderly population. But India is one of the few countries in the world where males outnumber females in some age groups. The sex ratio in India is defined as females per thousand males.

Table 22.7 Sex Ratio by Broad Age Group in Assam

Age group	1991	2001	2011	2019	2029
0-14	969	957	959	962	964
15-59	902	920	956	985	1024
60+	812	949	971	988	1010
All ages	923	935	958	978	1001

Index of Ageing: Another indicator of age structure is the ageing index. It is defined as the number of elderly persons per 100 youths under age 15.

$$\text{Ageing Index} = \frac{\text{Elderly population age 60 and over}}{\text{Child population below age 15}}$$

It is observed from the Table 22.9 that in Assam the proportion of elderly persons has increased

Table 22.8 District-wise composition of 'young old' (60-69 years) and 'old old' (70 years and older) among elderly population (60 years and older) in Assam, 1991-2011

States/Districts	Young Old(60-69)			Old Old (70+)		
	1991	2001	2011	1991	2001	2011
Assam	61.30	61.47	61.55	38.70	38.53	38.45
Kokrajhar	58.66	60.64	59.97	41.35	39.36	40.03
Dhubri	57.57	58.92	58.61	42.43	41.08	41.39
Goalpara	59.92	60.81	60.11	40.08	39.19	39.89
Barpeta	57.50	58.08	58.18	42.50	41.92	41.82
Morigaon	58.39	58.19	59.15	41.61	41.81	40.85
Nagaon	58.94	59.12	59.52	41.06	40.88	40.48
Sonitpur	63.53	63.26	62.82	36.47	36.74	37.18
Lakhimpur	61.28	61.20	61.88	38.72	38.80	38.12
Dhemaji	58.75	61.02	61.91	41.25	38.98	38.09
Tinsukia	65.25	64.47	63.60	34.75	35.53	36.40
Dibrugarh	65.9	64.4	64.14	34.10	35.60	35.86
Sivsagar	65.02	64.32	63.58	34.98	35.68	36.42
Jorhat	65.25	64.36	63.98	34.75	35.64	36.02
Golaghat	65.41	65.24	64.48	34.59	34.76	35.52
Karbi-Anglong	59.38	60.04	61.17	40.62	38.96	38.83
Dima-Hasao	63.61	61.51	62.11	36.39	38.49	37.89
Cachar	60.51	60.02	62.0	39.49	39.98	38.0
Karimganj	59.67	58.67	59.99	40.33	41.33	40.01
Hailakandi	60.11	61.04	62.86	39.19	38.96	37.14
Bongaigaon	61.21	60.07	59.40	38.79	39.93	40.60
Chirang	----	----	60.60	-----	-----	39.40
Kamrup	62.94	63.83	62.44	37.06	36.17	37.56
Kamrup-Metro	-----	-----	64.12	-----	----	35.88
Nalbari	65.19	56.98	61.70	34.81	43.02	38.30
Baksa	-----	-----	62.21	-----	-----	37.79
Darrang	-----	-----	60.32	-----	-----	39.68
Udalguri	-----	-----	62.64	-----	-----	37.36

Table22. 9A: District-wise index of ageing in Assam, 1991-2029

State/ Districts	Index of ageing									
	Male					Female				
	1991	2001	2011	2019	2029	1991	2001	2011	2019	2029
Assam	14.40	15.71	20.17	24.62	31.60	12.07	15.59	20.40	25.30	33.12
Kokrajhar	13.44	13.39	17.42	----	-----	10.48	16.39	18.70	----	
Dhubri	11.94	11.75	13.90	15.9	18.81	12.71	14.04	15.93	17.62	19.99
Goalpara	11.32	12.15	14.84	17.41	21.26	11.61	14.17	17.04	19.75	23.75
Barpeta	14.15	14.40	16.77	-----	-----	14.20	16.73	18.90	-----	-----
Morigaon	14.19	14.29	17.09	19.71	23.56	12.29	14.66	18.12	21.46	26.52
Nagaon	15.59	16.18	18.99	21.59	25.65	13.62	16.14	19.04	21.74	25.65
Sonitpur	14.39	15.88	21.98	28.51	39.46	11.03	14.47	20.21	26.4	36.87
Lakhimpur	13.88	15.73	19.96	24.16	30.67	11.48	14.68	19.84	25.25	34.13
Dhemaji	12.44	14.21	18.20	22.19	28.41	9.45	12.80	17.69	22.90	31.64
Tinsukia	13.86	15.92	20.88	25.93	34.00	9.95	14.39	20.19	26.48	37.15
Dibrugarh	14.69	17.17	23.82	30.95	42.93	11.01	15.67	23.58	32.7	49.22
Sivsagar	16.54	19.61	27.26	35.48	49.33	12.0	17.41	25.52	34.65	50.79
Jorhat	16.04	20.21	30.71	42.91	65.19	12.81	18.07	29.27	43.06	69.74
Golaghat	12.73	15.92	24.04	33.41	50.44	9.64	13.96	22.63	33.30	53.99
Karbi-Anglong	14.65	12.93	15.22	17.35	20.42	9.85	11.37	14.50	17.62	22.48
Dima-Hasao	12.54	13.78	16.64	19.35	23.36	8.10	11.16	14.81	18.58	24.67
Cachar	17.99	18.85	21.87	24.62	28.56	14.83	18.70	21.72	24.49	28.44
Karimganj	17.90	17.68	19.21	19.35	22.29	15.68	17.43	19.81	18.58	24.94
Hailakandi	-----	15.23	18.51	21.63	26.29	-----	14.54	17.48	20.25	24.35
Bongaigaon	12.21	13.83	17.19	20.46	25.43	11.05	15.29	19.65	24.02	30.87
Chirang	-----	-----	17.35	-----	-----	-----	-----	17.94	-----	-----
Kamrup	13.98	17.21	22.90	-----	38.29	11.96	17.17	24.65	-----	
Kamrup-Metro	-----	-----	31.46	-----	-----	-----	-----	28.02	-----	-----
Nalbari	14.03	23.37	27.44	-----	-----	13.14	24.17	29.39	-----	-----
Baksa	-----	-----	22.16	-----	-----	-----	-----	23.88	-----	-----
Darrang	-----	-----	17.74	-----	-----	-----	-----	17.68	-----	-----
Udalguri	-----	-----	20.36	-----	-----	-----	-----	20.51	-----	-----

from 13.25 percent in 1991 to 20.28 percent in 2011 and is projected to increase to 24.95 percent by the year 2019 and 32.34 percent by the year 2029. The table also reveals that the rate of ageing index is found somewhat higher among the females (12.07 percent in 1991 to 20.40 percent in 2011) than that of male counterparts (14.40 percent in 1991 to 20.17 percent in 2011). The highest proportion of elderly persons per 100 youths is found in Kamrup Metro with 29.78 percent in 2011 whereas Karbi Anglong district shows the

lowest percentage (14.87 percent) in this regard.

Growth rates among Elderly Population:

Table 22.11 reveals the growth rate of elderly population. The growth rate of elderly persons has found to be higher than that of the overall growth rate of total population in Assam. However, it is also observed that the growth rate of elderly population has increased from 3.06 percent in 1991-2001 to 3.32 percent in 2001-2011, where-

as overall growth rate of Assam is found to have decreased during the reference period. The same conclusion can be drawn for India also.

Table 22.10: District-wise index of ageing in Assam, 1991-2029

State/ Districts	Index of ageing				
	Total				
	1991	2001	2011	2019	2029
Assam	13.25	15.65	20.28	24.95	32.34
Kokrajhar	11.71	12.83	18.04	-----	-----
Dhubri	11.56	12.86	14.89	16.74	19.37
Goalpara	11.41	13.13	15.91	15.56	22.48
Barpeta	14.89	15.53	17.81	-----	-----
Morigaon	13.34	14.47	17.59	20.56	24.99
Nagaon	14.22	16.16	19.02	21.66	26.00
Sonitpur	12.59	15.19	21.11	27.47	38.18
Lakhimpur	12.71	15.21	19.9	24.68	32.29
Dhemaji	11.03	13.52	17.95	22.52	29.89
Tinsukia	11.16	15.18	20.54	26.17	35.42
Dibrugarh	12.45	16.44	21.38	26.38	34.32
Sivsagar	12.29	18.53	26.41	35.06	49.96
Jorhat	13.89	19.16	27.11	27.06	27.00
Golaghat	11.05	14.96	23.34	33.32	52.00
Karbi-Anglong	12.63	12.17	14.87	17.46	21.35
Dima-Hasao	10.67	12.50	15.75	18.94	23.85
Cachar	15.88	18.78	21.8	24.56	28.51
Karimganj	16.37	17.56	19.5	21.21	23.57
Hailakandi	-----	14.89	18.0	20.95	25.32
Bongaigaon	11.48	14.54	18.4	22.2	31.17
Chirang	-----	-----	17.64	-----	-----
Kamrup	12.78	17.19	23.76	36.47	40.29
Kamrup-Metro	-----	-----	29.78	-----	-----
Nalbari	13.62	23.76	28.4	-----	-----
Baksa	-----	-----	23.0	-----	-----
Darrang	-----	-----	17.71	-----	-----
Udalguri	-----	-----	20.44	-----	-----

Median Age:

The median age is the age above which and below which exactly half of the population falls. This perhaps is the most widely used indicator of population ageing. From the table 22.12 it is observed

Table 22.11: Trend in overall growth rate (OGR in percent) of the total population and growth rate of elderly (GRE) in India and Assam

	1991-2001	2001-2011
India		
OGR	2.27	1.77
GRE	3.52	3.55
Assam		
OGR	1.89	1.71
GRE	3.06	3.32

that in Assam the median age has gradually increased from 20 years in 1991 to 23 years in 2011 and it will be 25 years as projected for the year 2019 and 27 years by the year 2029. This finding, however, reflects the decrease in young population and increase in elderly population in the State. The table reveals that among the districts, Kamrup Metro shows the highest median age (28 years) while lowest median age is found in Dhubri district (21 years) in 2011. This finding obviously reflects the highest percentage of child population in Dhubri district (38.59 percent in 2011) and lowest child population in Kamrup district (23.50 percent) in the same period. This might be due to fact that fertility rate in Dhubri district is high and at same time contraceptive prevalence rate might be low in the district.

Table 22.12: District-wise Median Age in Assam, 1991-2029

State/ District	Years				
	1991	2001	2011	2019	2029
Assam	19.83	21.58	23.13	25.47	27.20
Kokrajhar	19.24	20.12	22.52	-----	-----
Dhubri	18.17	18.96	20.71	21.76	23.00
Goalpara	18.49	19.57	21.57	22.76	25.06

Barpeta	19.02	20.60	21.90	-----	-----
Morigaon	18.55	20.04	21.56	22.60	23.93
Nagaon	19.59	20.77	22.07	23.09	25.45
Sonitpur	20.30	21.93	23.50	24.93	27.68
Lakhimpur	18.65	20.82	22.98	25.85	28.22
Dhemaji	17.34	19.64	22.33	25.30	27.48
Tinsukia	20.43	22.18	23.65	26.04	27.70
Dibrugarh	21.28	23.26	25.75	27.36	30.32
Sivsagar	21.12	23.67	26.28	28.18	31.93
Jorhat	21.13	23.84	26.73	28.81	32.63
Golaghat	19.56	22.09	25.25	27.09	30.17
Karbi-Anglong	18.90	19.67	21.58	22.71	23.80
Dima-Hasao	20.38	21.33	22.36	23.60	25.57
Cachar	21.20	22.40	23.26	25.17	26.53
Karimganj	20.56	21.10	21.7	22.22	23.15
Hailakandi	20.20	20.24	21.52	22.37	23.42
Bongaigaon	18.66	20.62	22.58	23.91	25.88
Chirang	-----	-----	22.42	----	-----
Kamrup	21.25	23.83	25.16	28.39	32.10
Kamrup-Metro	-----	----	27.95	-----	-----
Nalbari	----	----	25.88	-----	-----
Baksa	-----	-----	25.02	----	-----
Darrang	----	----	22.05	----	-----
Udalguri	----	-----	23.64	----	-----

Major Findings:

- The percentage of child population (0-14 years) of Assam has declined continuously during last three censuses from 1991 to 2011. It is found to have declined from 40.37 percent in 1991 to 32.86 percent in 2011.
- The population of working group (15-59 years) has increased from 54.28 percent in 1991 to 60.47 percent in 2011.
- The population of the age group of 60+ years also has increased from 5.35 percent in 1991 to 6.66 percent in 2011.
- The study reveals that the rate of decline of the child population in urban areas (8.1 percent) is found to be somewhat higher than that of the rural counterparts (7.1 percent) during the period 1991 to 2011
- The population of age group 15-59 years is found to be increased by 5.96 percent in rural Assam from 1991 to 2011 against 4.57 percent increase of their counterparts in urban Assam during the same period. Of course, in contrast to this, increase of elderly population (60+) in rural areas found to be somewhat lower (1.13 percent) than that of their urban counterparts (2.53 percent) during the same period.
- The total dependency ratio of Assam has decreased from 84.22 percent in 1991 to 65.37 percent in 2011.
- The young dependency ratio of the State has declined from 74.37 percent in 1991 to 54.34 percent in 2011 and is expected to decrease to 46.49 percent by the year 2019 and to 38.26 percent by the year 2029.
- The old dependency ratio indicates an increasing trend during the reference period from 9.85 percent in 1991 to 11.02 percent in 2011. This ratio is expected to grow to 11.60 percent by 2019 and 12.37 percent by 2029.
- The highest young dependency is found in Dhubri district at 70.02 percent in 2011. Dhemaji district shows the rapid decrease in young dependency from 91 percent in 1991 to 57.49 percent in 2011.
- The overall sex ratio of Assam has increased from 923 in 1991 to 958 in 2011 (per 1000 males).
- Though the sex ratio in the age group 0-14 years had decreased from 969 in 1991 to 957 in 2001 the ratio is again found to have increased to 959 in 2011 and it is expected to increase up to 964 by the year 2029.
- The proportion of 'young old' population

in Assam has shown a negligible increase from 61.30 percent in 1991 to 61.55 percent in 2011 whereas the 'old old' population exhibits a negligible decrease from 38.70 percent in 1991 to 38.45 percent in 2011

- The rate of ageing index in Assam is found somewhat higher among the females (12.07 percent in 1991 to 20.40 percent in 2011) than that of male counterparts (14.40 percent in 1991 to 20.17 percent in 2011). The highest proportion of elderly persons per 100 youths is found in Kamrup Metro district with 29.78 percent in 2011. Karbi Anglong district shows (14.87 percent) the lowest percent age in this regard.

- In Assam, the median age has found to be gradually increased from 20 years in 1991 to 23 years in 2011 and it will be 27 years as projected for the year 2029. Kamrup Metro district shows the highest median age (28 years) while lowest median age is found in Dhubri district (21 years).

Conclusion:

The study has analyzed the levels and trends of aged population (60 years and above) in Assam and its districts during 1991-2011. The study reveals that over the years 1991-2011, the percentage of child population (0-14 years) in Assam has declined continuously. It might be due to increase in the use of different family planning methods among the eligible couples. From NFHS data it is observed that, current contraceptive prevalence rate in Assam has increased significantly from 43 percent in 1992-93 (NFHS-I) to 56.5 percent in 2005-06 (NFHS-III). The highest percentage of child population was observed in Dhubri district (38.59 percent in 2011) and lowest in Kamrup district (23.50 percent) in the same period. The young dependency ratio of the State is found to have declined in 1991-2011 whereas the old dependency ratio has exhibited an increasing trend during the reference period. In Assam, the median

has gradually increased from 20 years in 1991 to 23 years in 2011, and it is expected to increase to 27 years by the year 2029. The National Policy on Older Persons was announced by the Government of India in the year 1999. The policy envisages State support to ensure financial and food security, health care, shelter and other needs of older persons, equitable share in development, protection against abuse and exploitation, and availability of services to improve the quality of their lives. The study recommends adoption of special policies for women and rural people. Problems of the elderly women are exacerbated by lifetime gender-based discrimination often stemming from deep-rooted cultural and social bias. In India, over 50 percent of women above the age of 80 are widows (National Policy for Older Persons, 1999). Again, in India many households in rural areas belong to the below poverty line category. They cannot be expected to participate in long-term saving schemes for old age and they do not. Hence, older persons in rural areas would need social security in large measures.

Reference

1. International Institute for Population Sciences (IIPS), 2004. Dynamics of Age-Sex Structure and Population Projection. Block-1 Dynamic of Age Structure. 4th Edition.
2. International Institute for Population Sciences (IIPS), 2004. Dynamics of Age-Sex Structure and Population Projection. Block-2 Population Estimates and Projection. . 4th Edition.
3. International Institute for Population Sciences (IIPS), 2004. Statistical Methods for Population Studies. Block-1: Some Basic Mathematical Tools. 4th Edition.
4. International Institute for Population Sciences (IIPS), 1995. National Family Health Survey (NFHS-I), 1992-93.
5. International Institute for Population Sci-

- ences (IIPS), 2002. National Family Health Survey (NFHS-II), 1998-99.
6. International Institute for Population Sciences (IIPS), 2007. National Family Health Survey (NFHS-III), 2005-06.
7. Ministry of Social Justice and Empowerment, Government of India, 1999. National Policy for Older Persons, 1999.
8. Office of the Registrar General and Census Commissioner, 1991-2011. Population in five year age group by residence and sex (India/States/UTs/Cities).
9. United Nations, Department of Economic and Social Affairs Population Divisions, 2002. World Population Ageing 1950-2050, New York.
10. United Nations (1991), World Population Prospects, 1990, Sales No. E.91XIII.A
11. United Nations, Department of Economic and Social Development, 1992, Economic and Social Aspects of Population Ageing in Kerala, India. Incorporate

